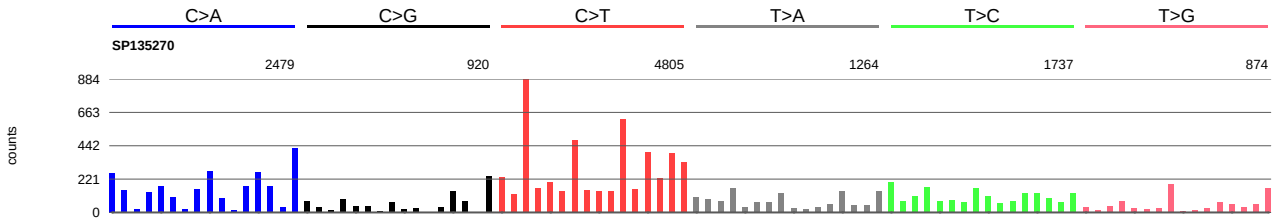
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C>G
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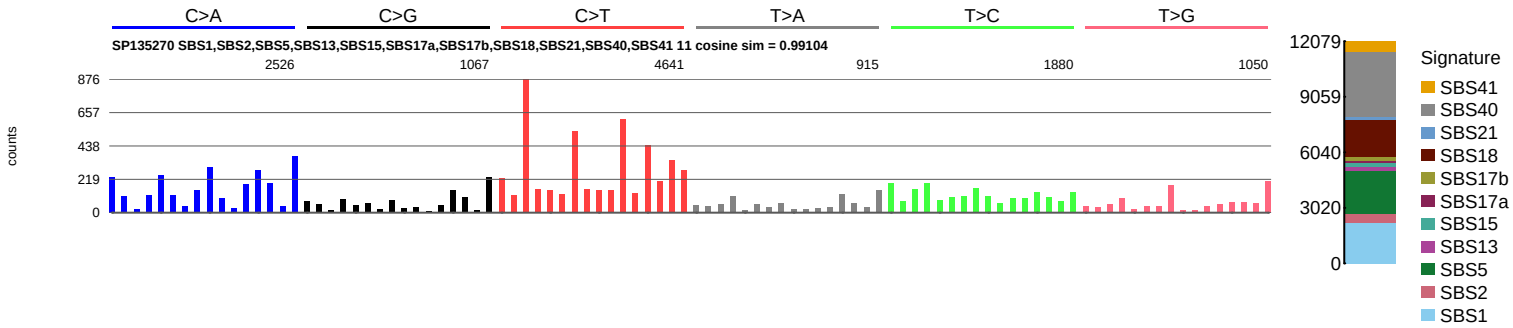
SP135270
2479
920
4805
1264
1737
874
884
663
442
221
0
counts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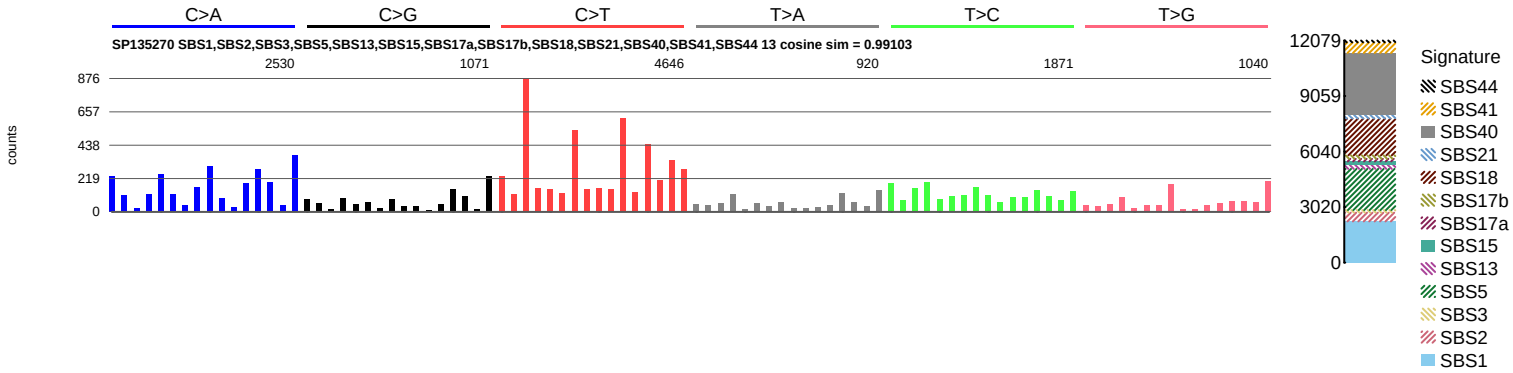
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99104
2526
1067
4641
915
1880
1050
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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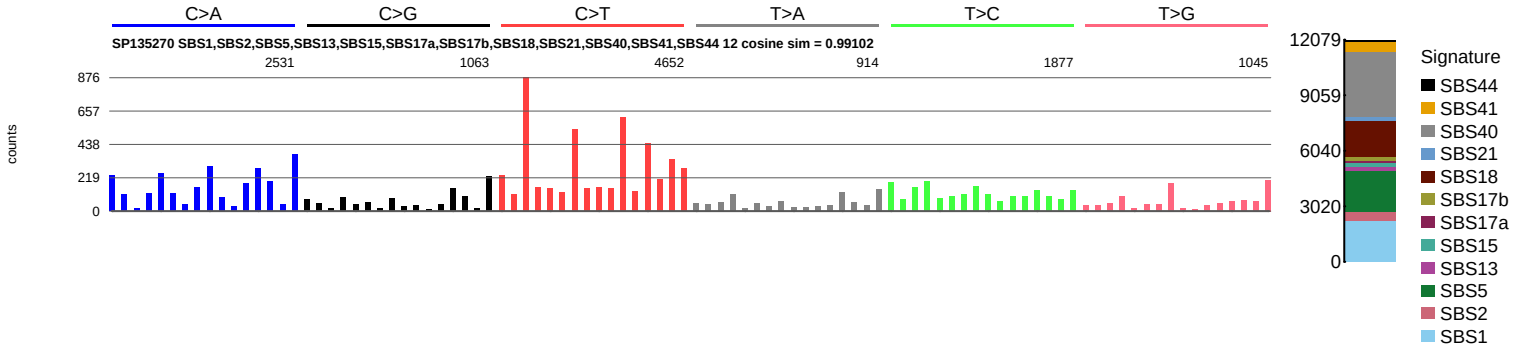
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 13 cosine sim = 0.99103
2530
1071
4646
920
1871
1040
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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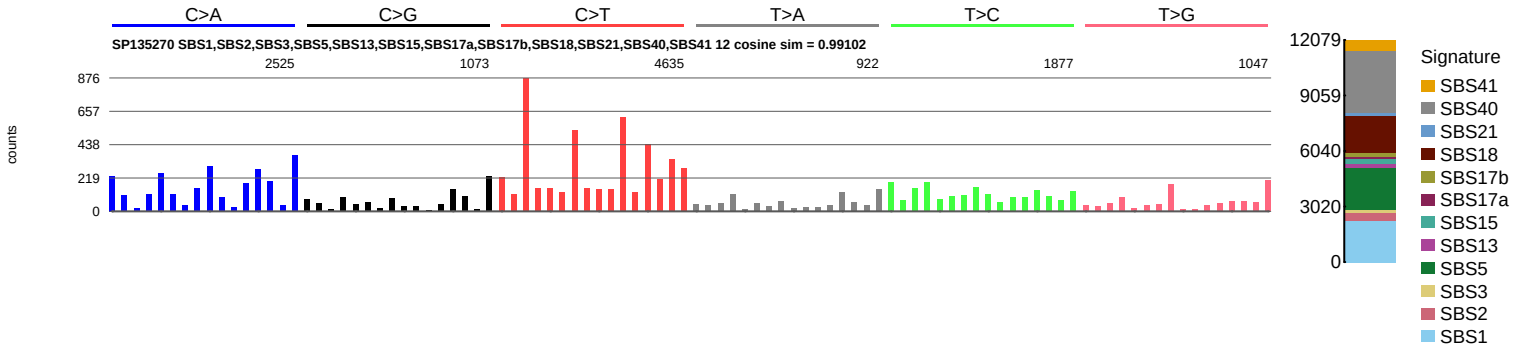
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.99102
2531
1063
4652
914
1877
1045
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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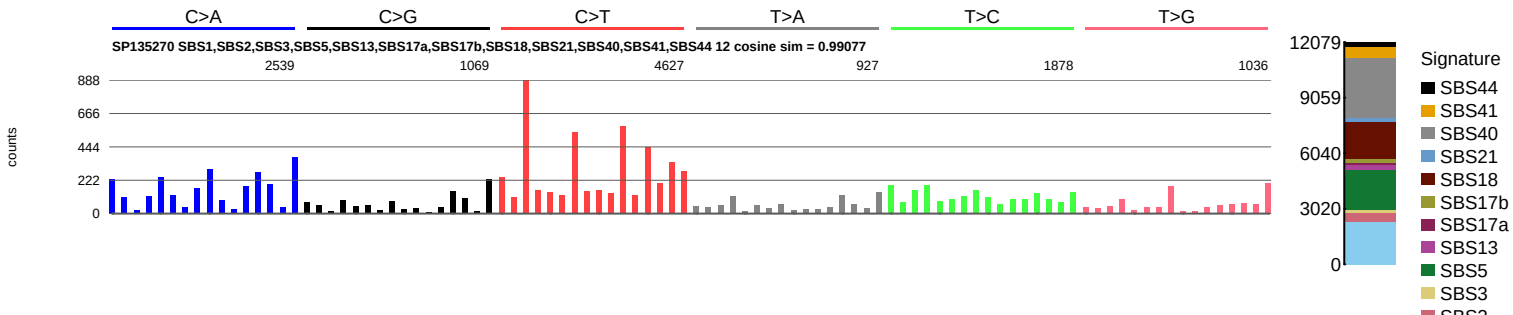
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 12 cosine sim = 0.99102
2525
1073
4635
922
1877
1047
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
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C>T
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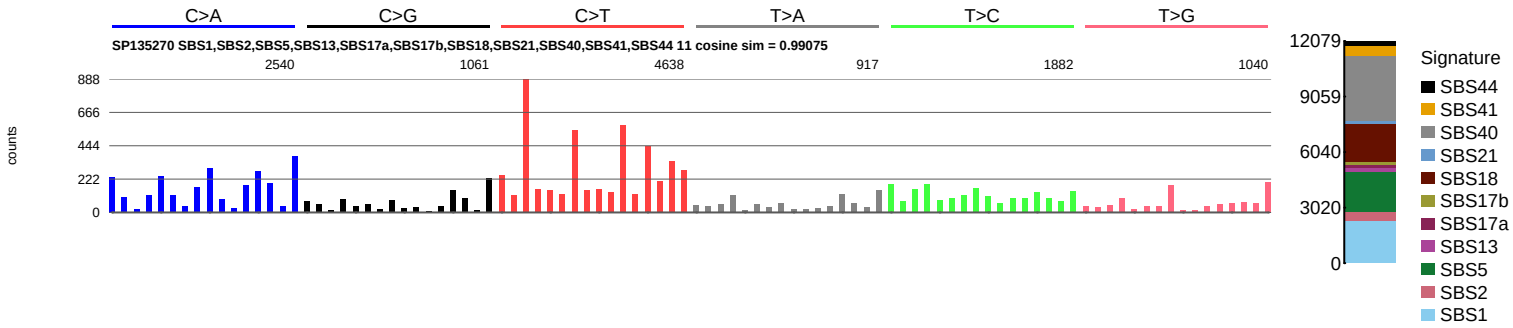
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.99077
2539
1069
4627
927
1878
1036
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3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C>G
C>T
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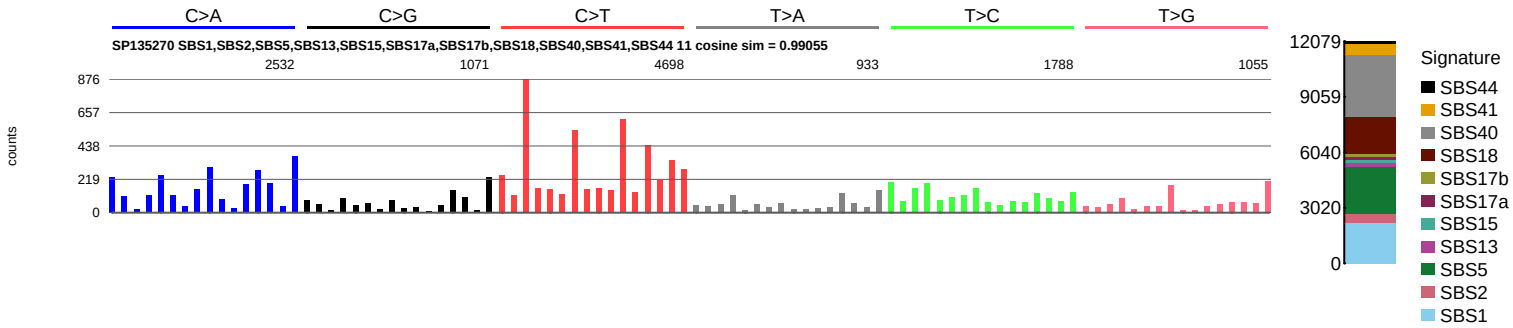
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99075
2540
1061
4638
917
1882
1040
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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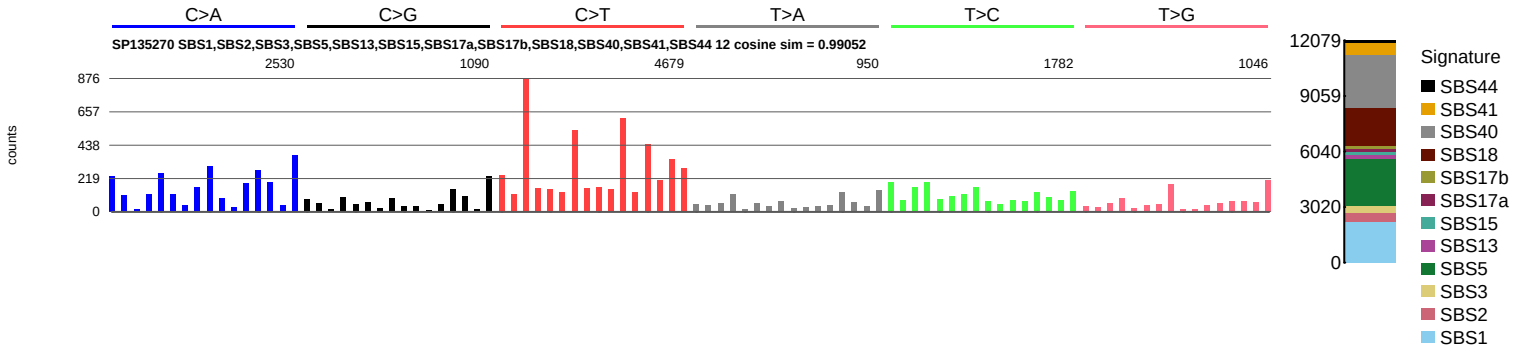
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.99055
2532
1071
4698
933
1788
1055
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
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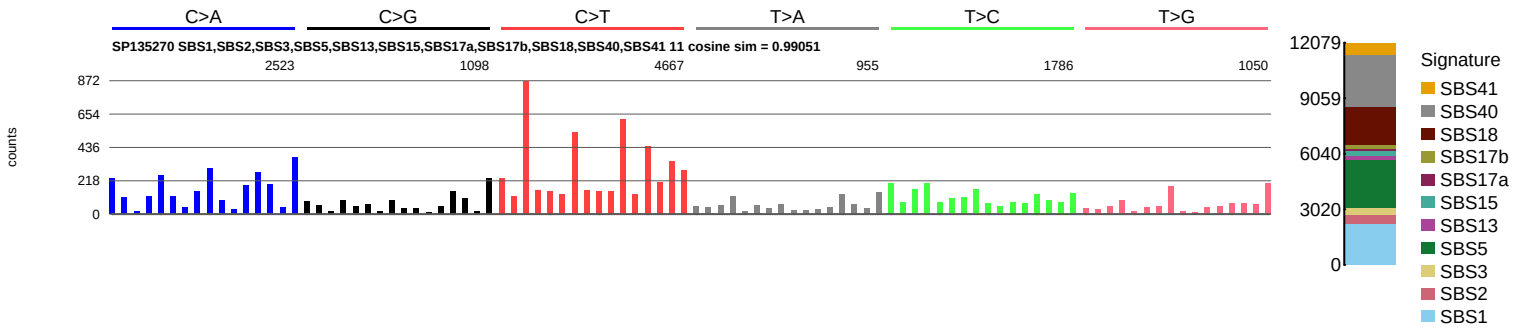
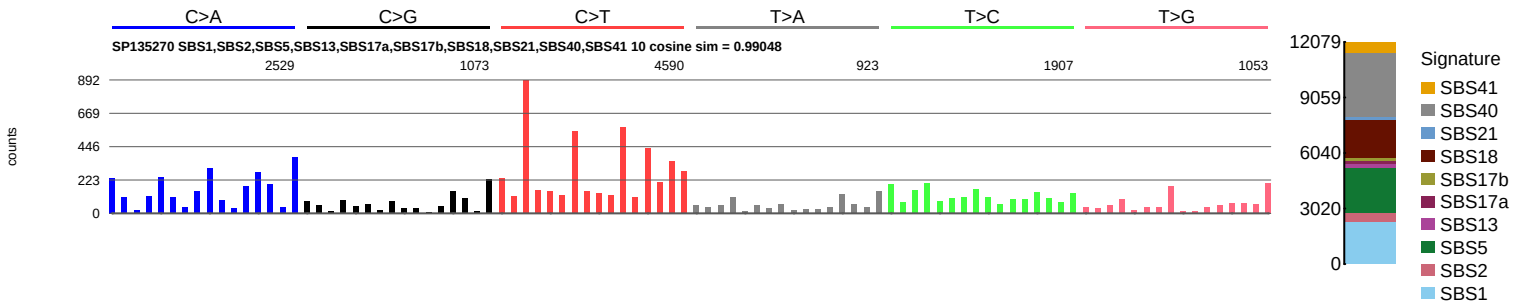
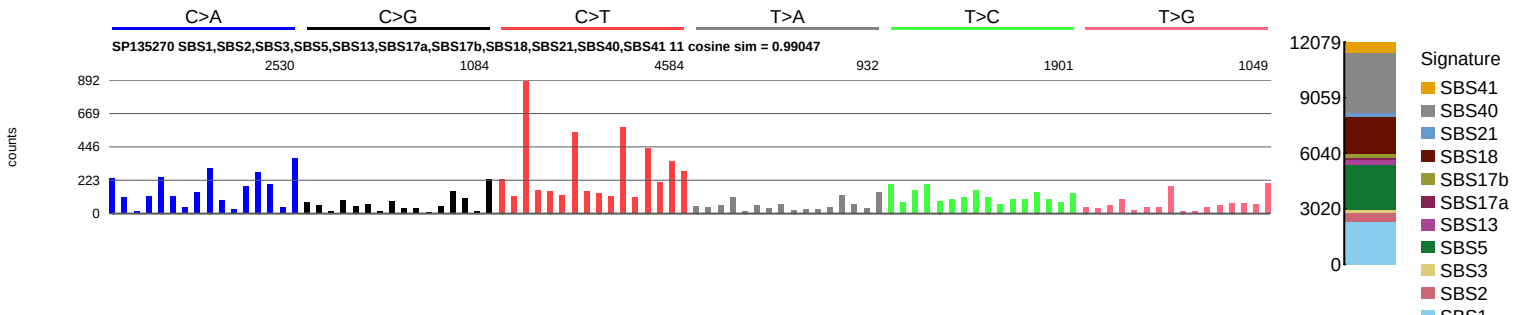
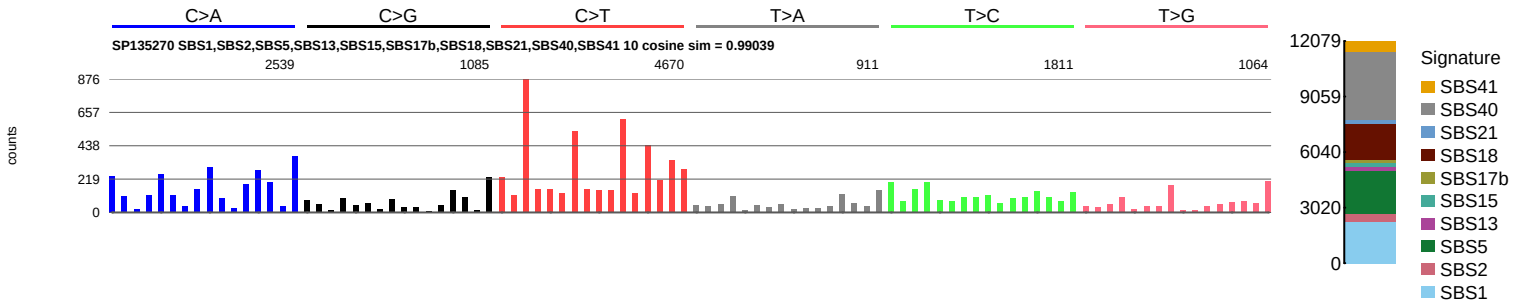
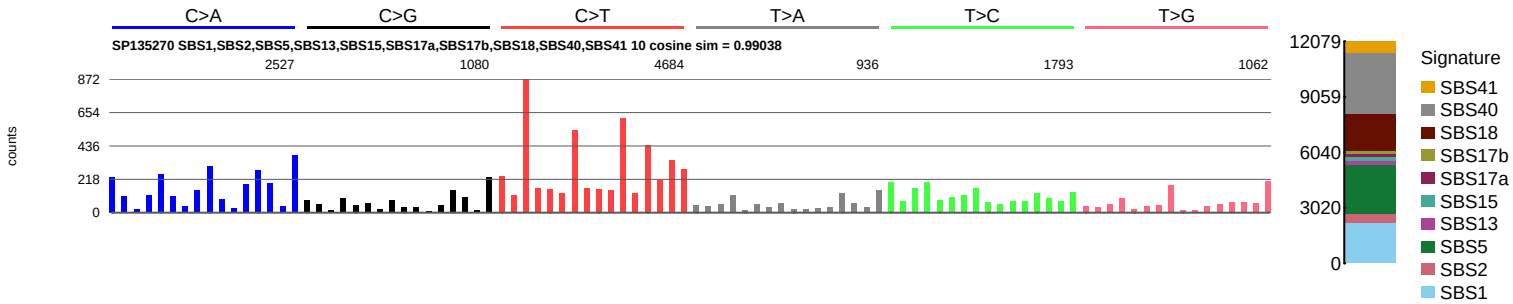
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 12 cosine sim = 0.99052
2530
1090
4679
950
1782
1046
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

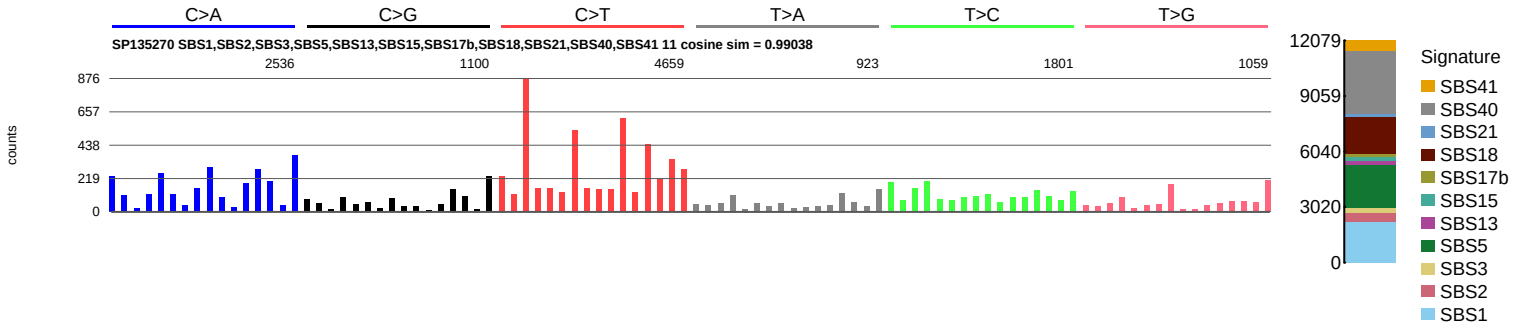

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 11 cosine sim = 0.99051
2523
1098
4667
955
1786
1050
872
654
436
218
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

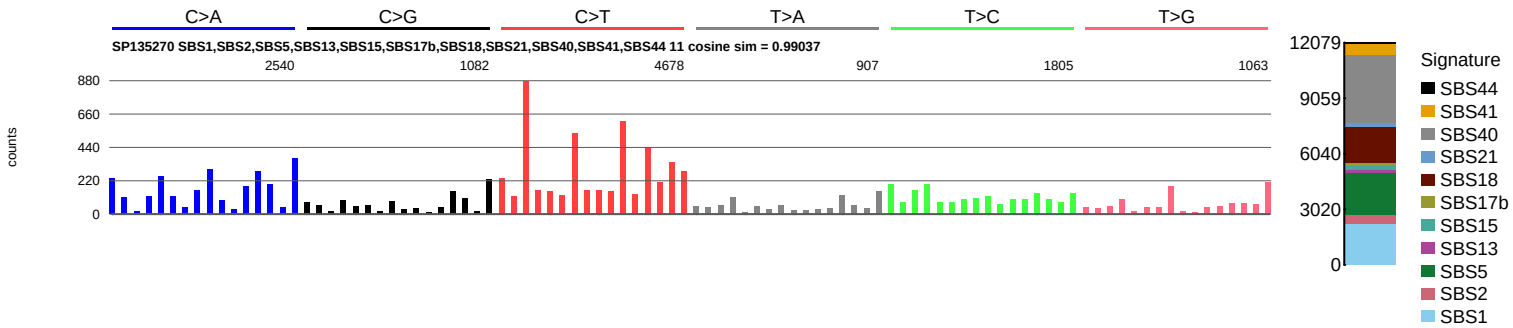

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99048
2529
1073
4590
923
1907
1053
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

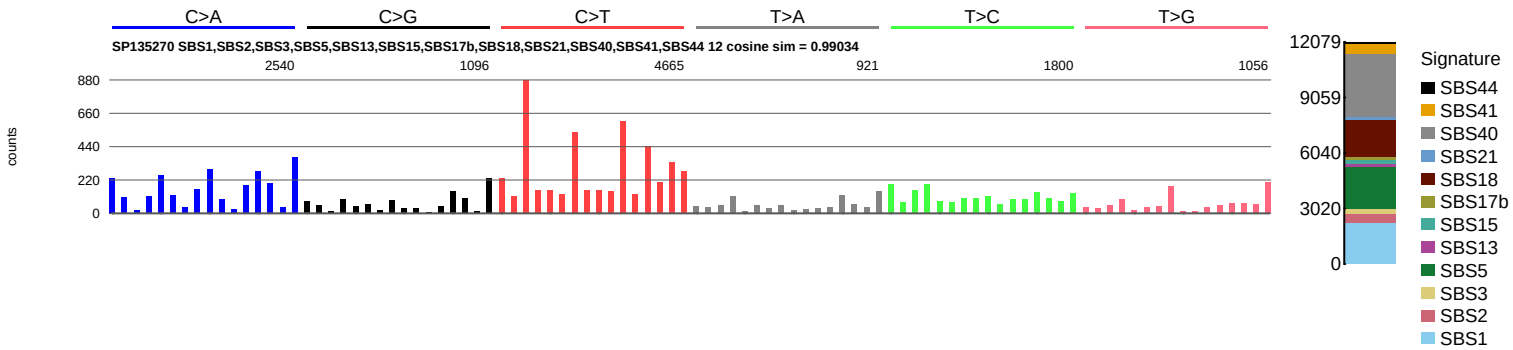

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99047
2530
1084
4584
932
1901
1049
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

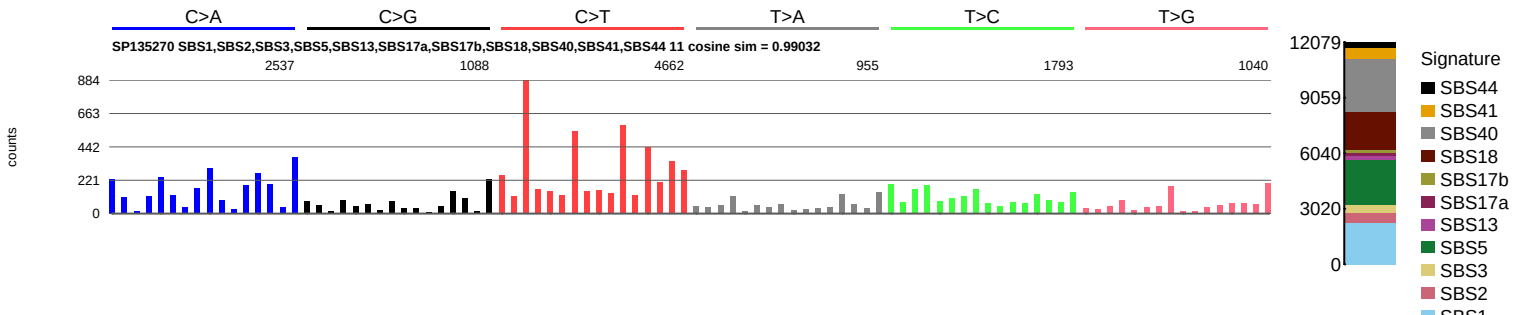

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99039
2539
1085
4670
911
1811
1064
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

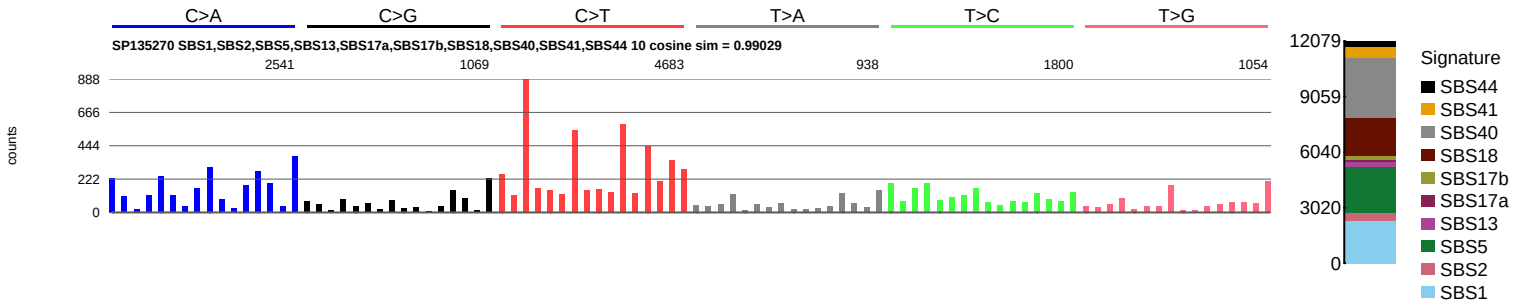

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.99038
2527
1080
4684
936
1793
1062
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

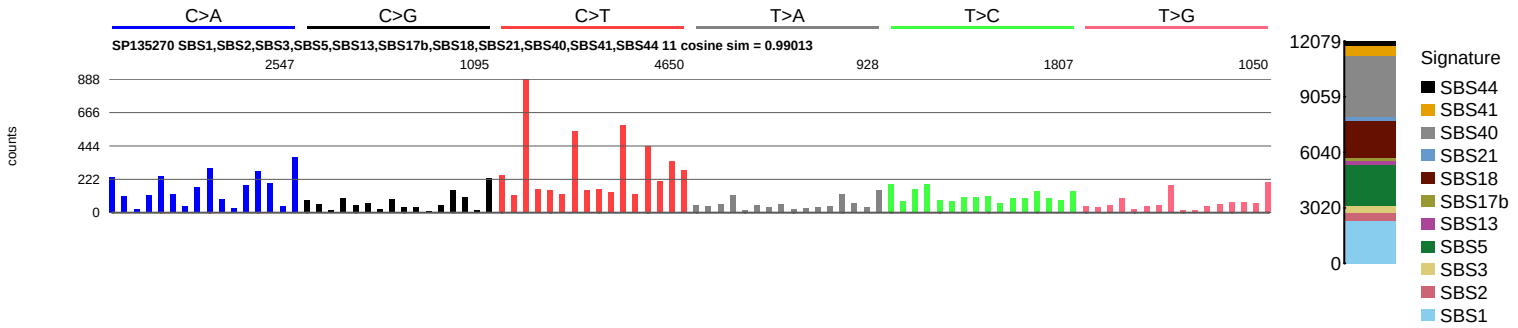

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99038
2536
1100
4659
923
1801
1059
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

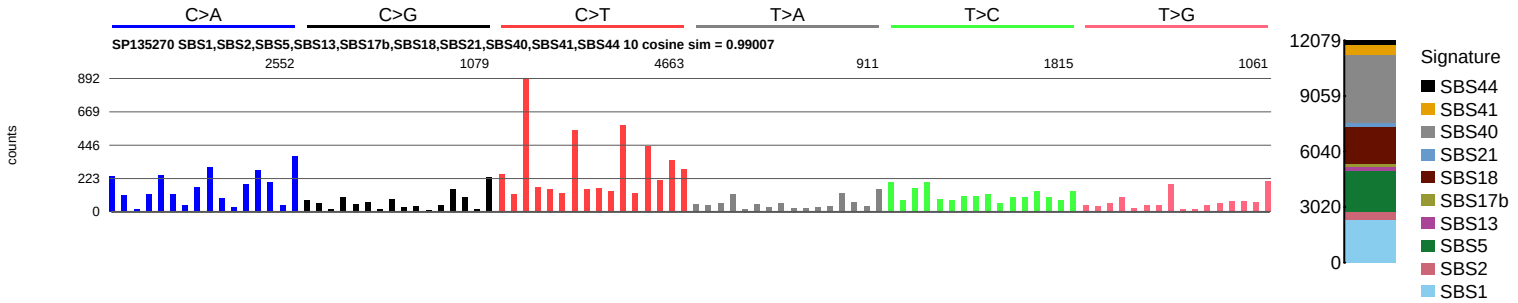

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99037
2540
1082
4678
907
1805
1063
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

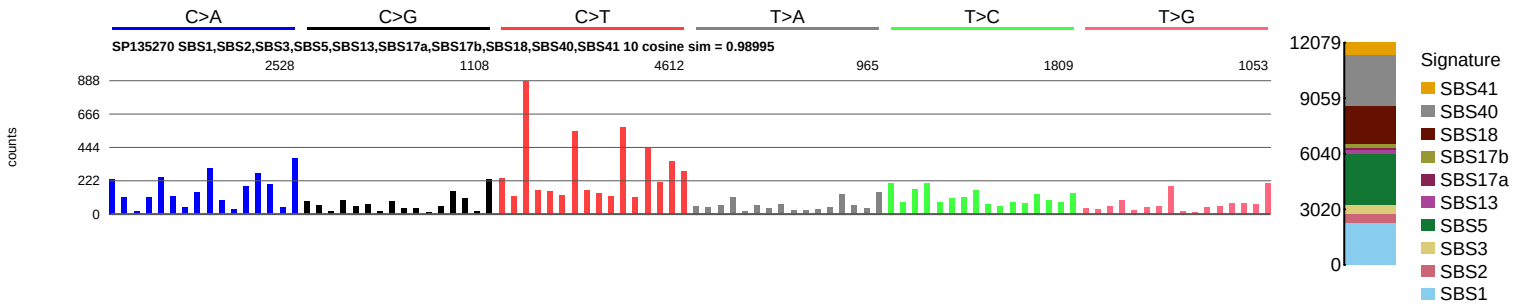

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.99034
2540
1096
4665
921
1800
1056
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

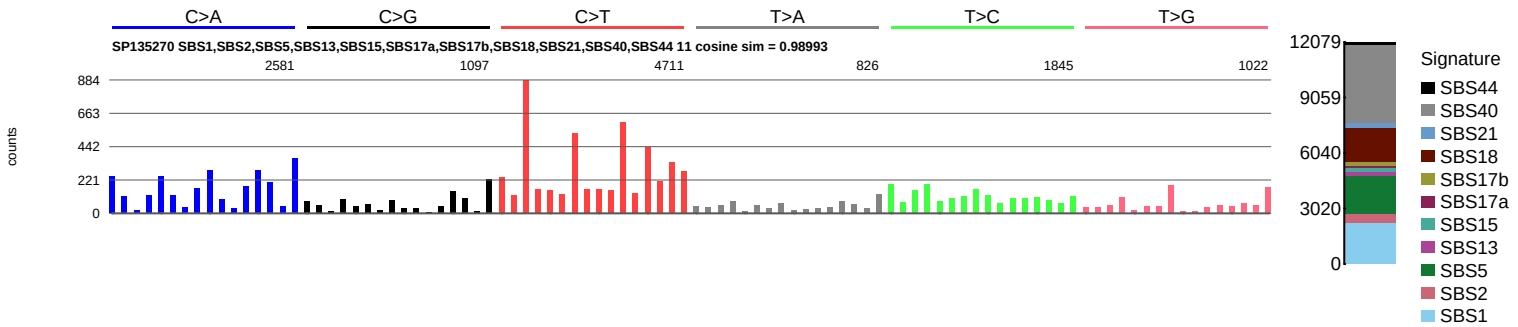

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.99032
2537
1088
4662
955
1793
1040
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

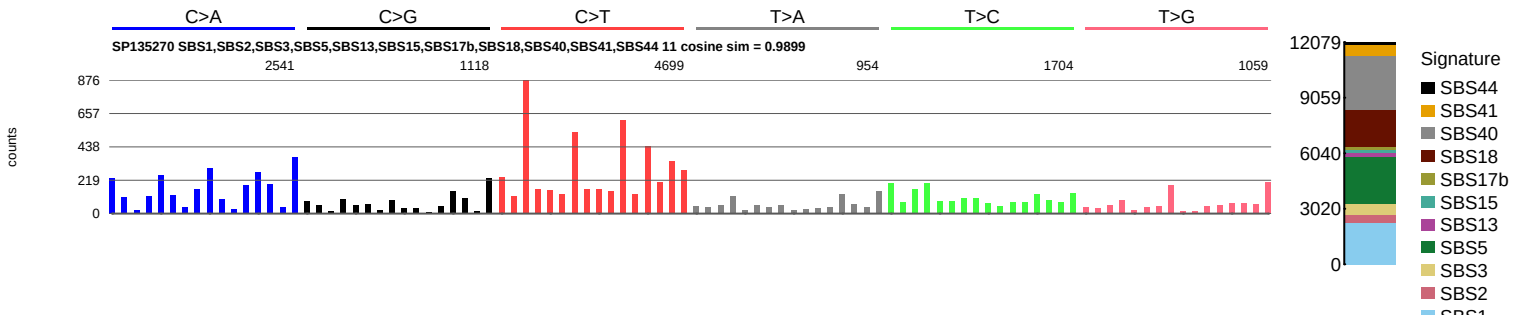

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99029
2541
1069
4683
938
1800
1054
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

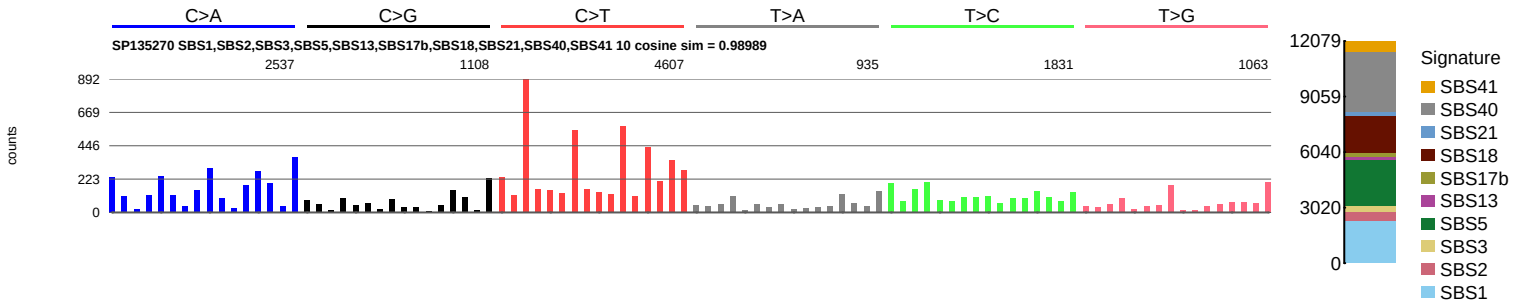

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99013
2547
1095
4650
928
1807
1050
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

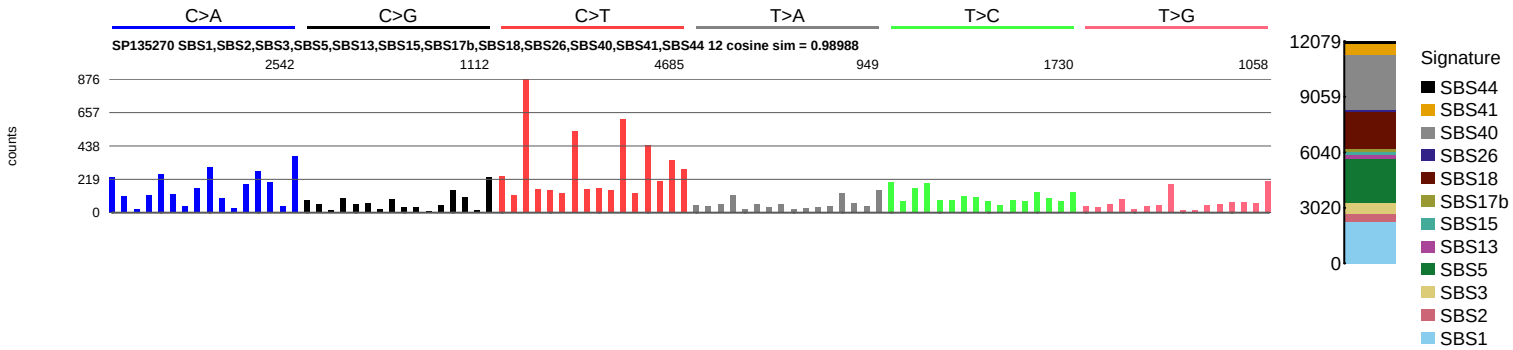

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99007
2552
1079
4663
911
1815
1061
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

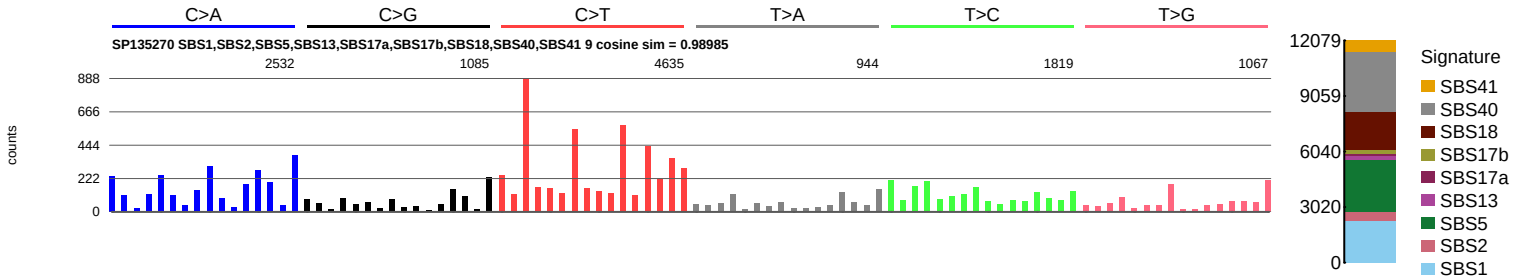

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98995
2528
1108
4612
965
1809
1053
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98993
2581
1097
4711
826
1845
1022
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.9899
2541
1118
4699
954
1704
1059
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98989
2537
1108
4607
935
1831
1063
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98988
2542
1112
4685
949
1730
1058
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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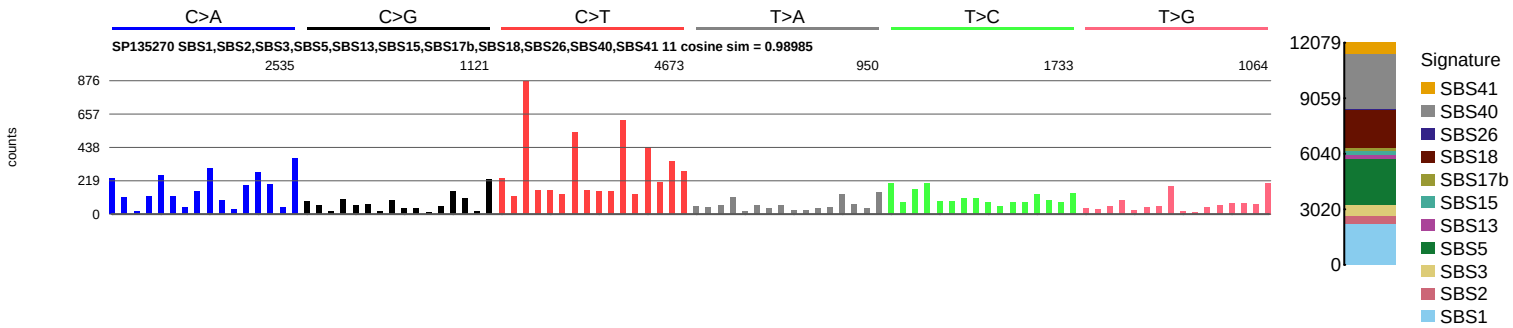
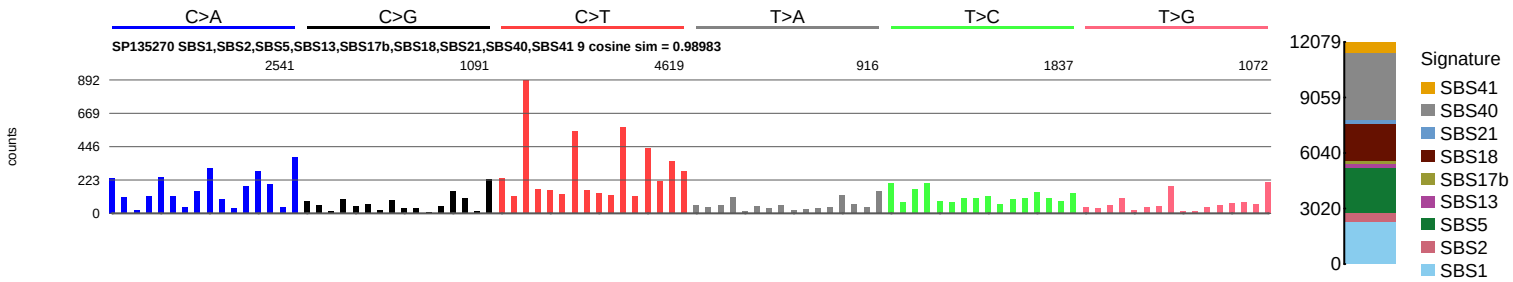
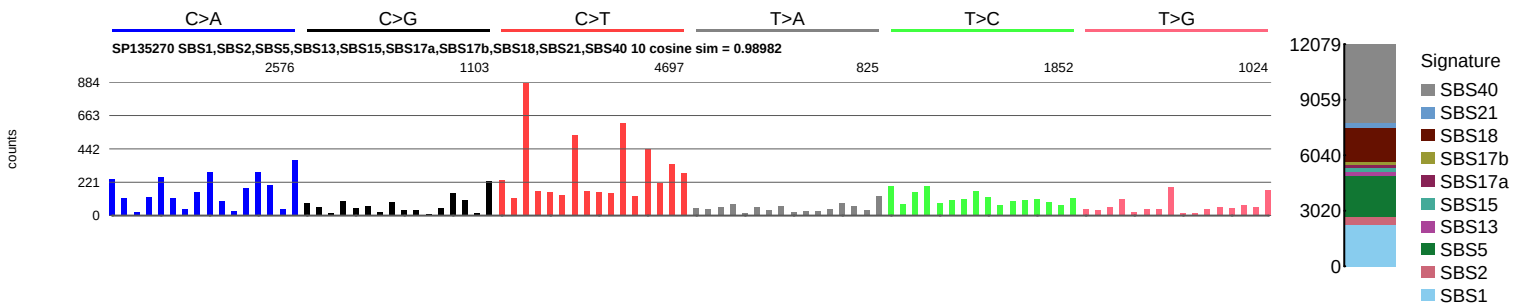
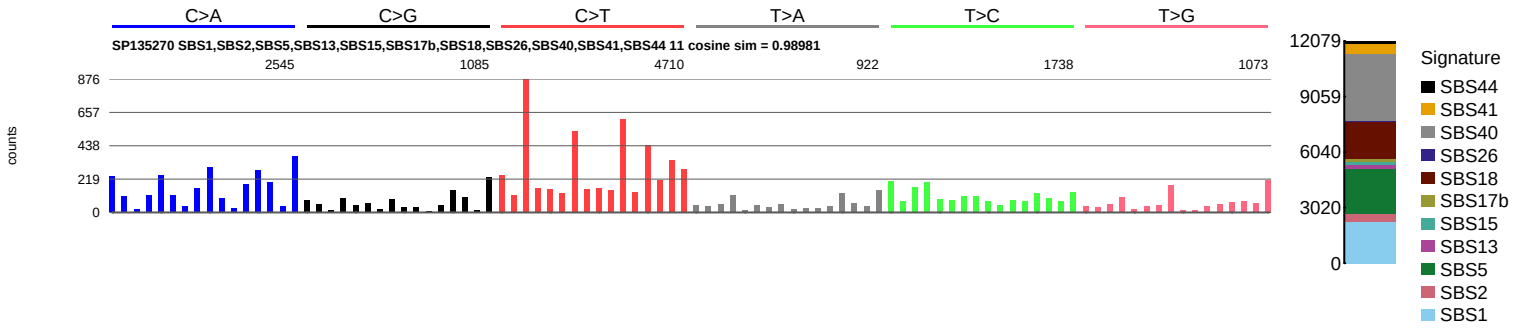
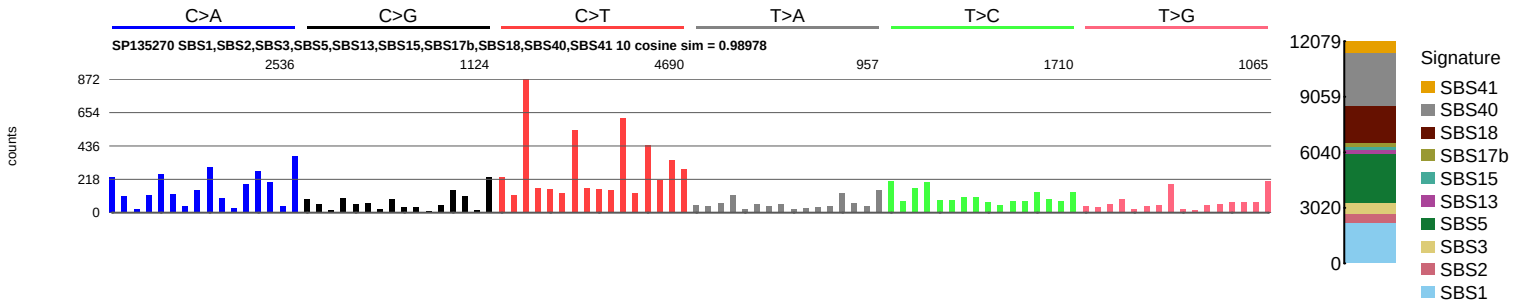
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98985
2532
1085
4635
944
1819
1067
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

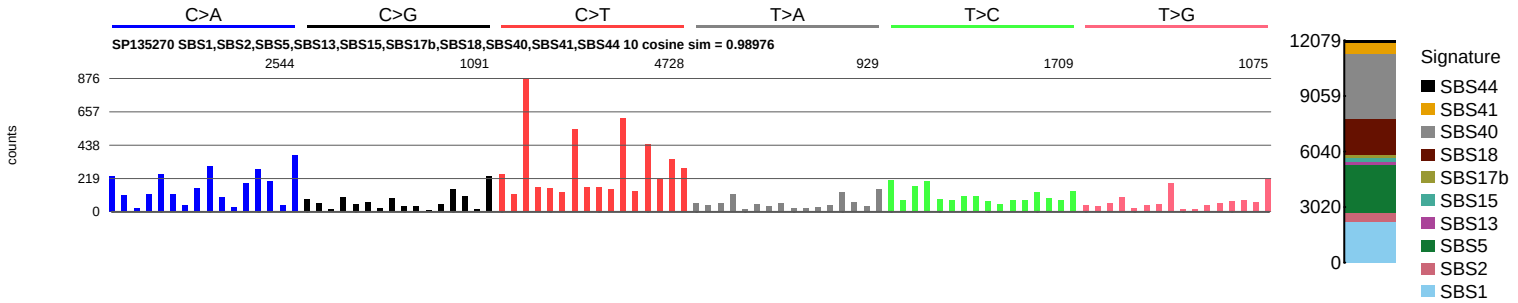

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98985
2535
1121
4673
950
1733
1064
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

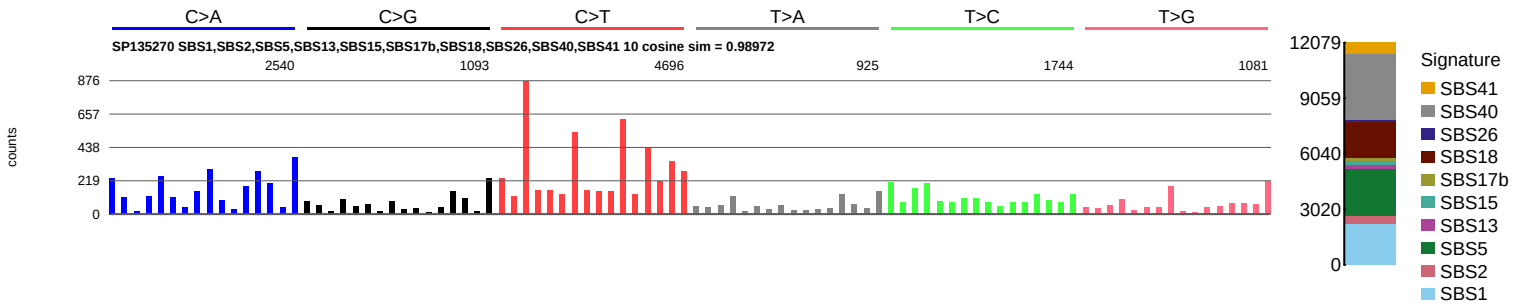

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98983
2541
1091
4619
916
1837
1072
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

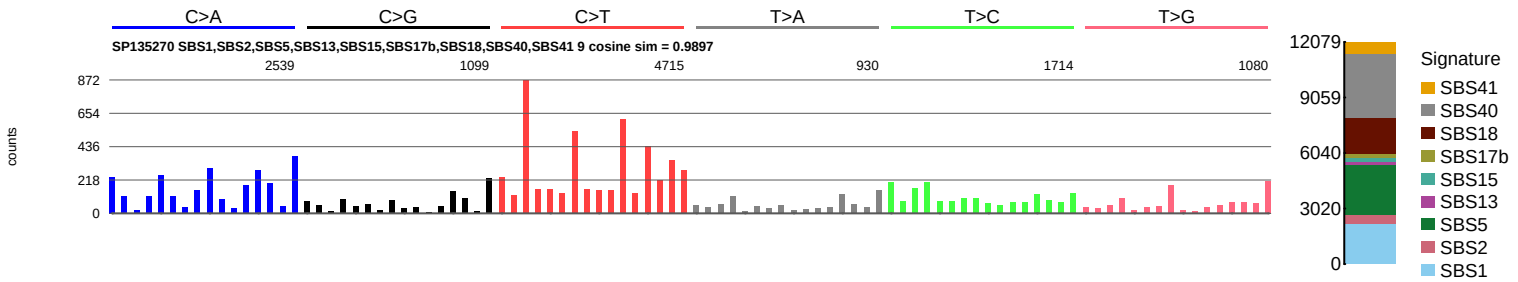

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98982
2576
1103
4697
825
1852
1024
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98981
2545
1085
4710
922
1738
1073
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98978
2536
1124
4690
957
1710
1065
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98976
2544
1091
4728
929
1709
1075
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98972
2540
1093
4696
925
1744
1081
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.9897
2539
1099
4715
930
1714
1080
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS41
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SBS18
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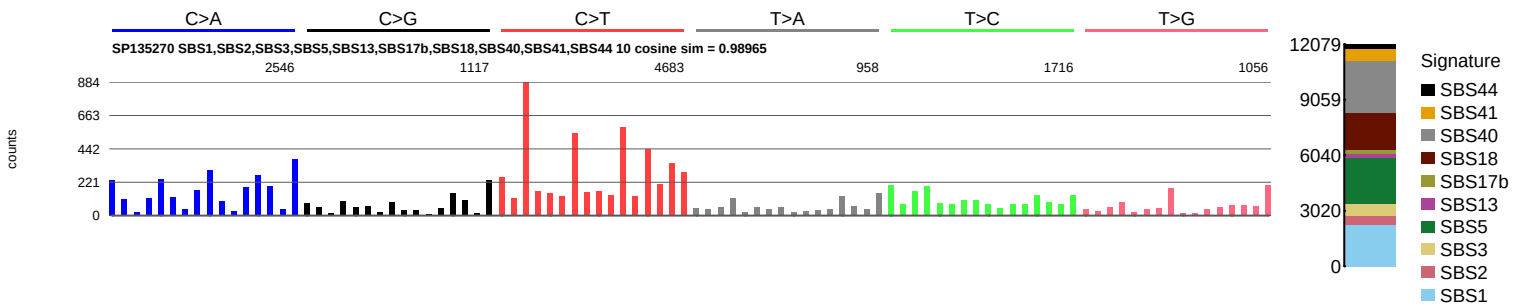
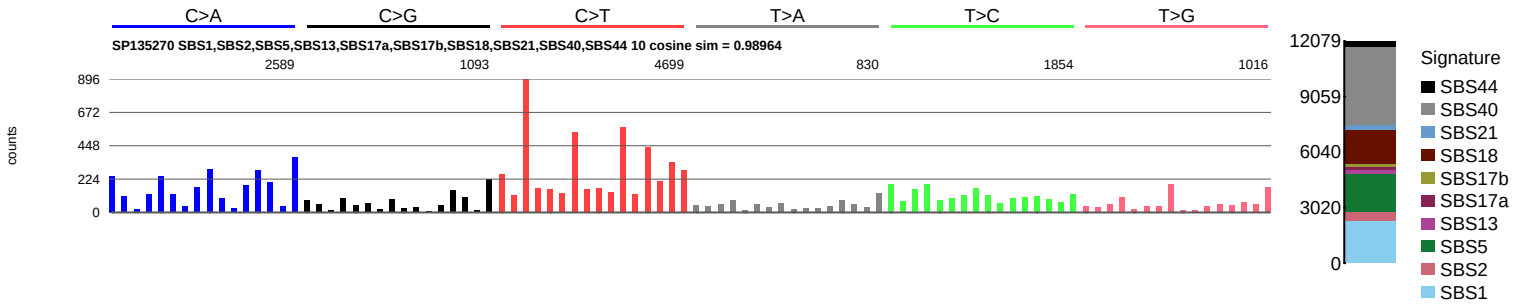
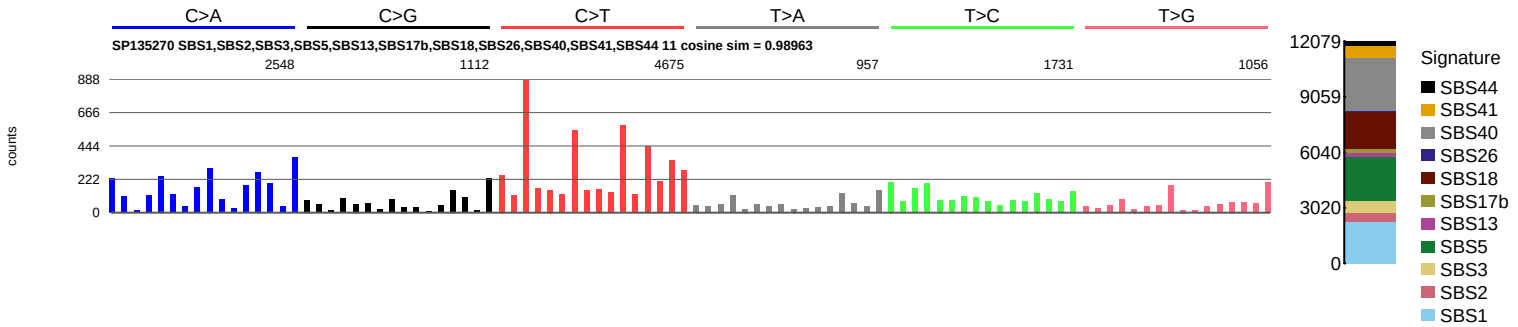
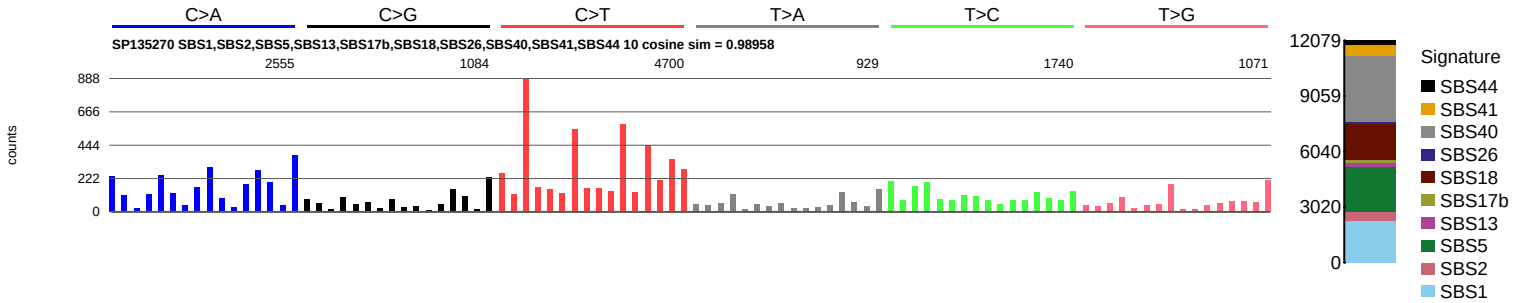
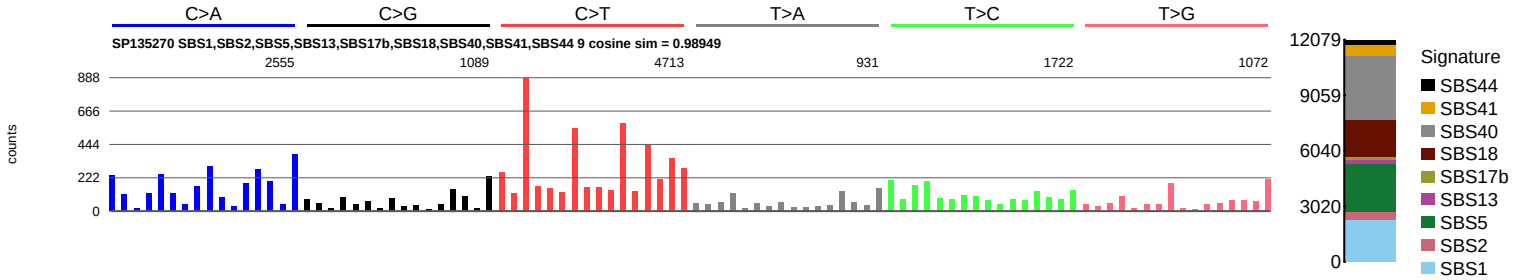
SBS2
SBS1

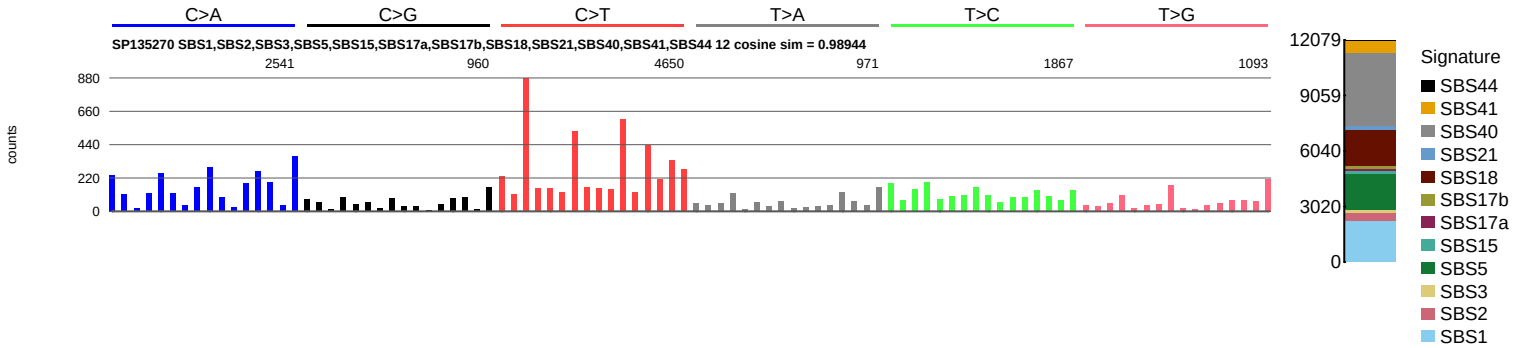

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98965
2546
1117
4683
958
1716
1056
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

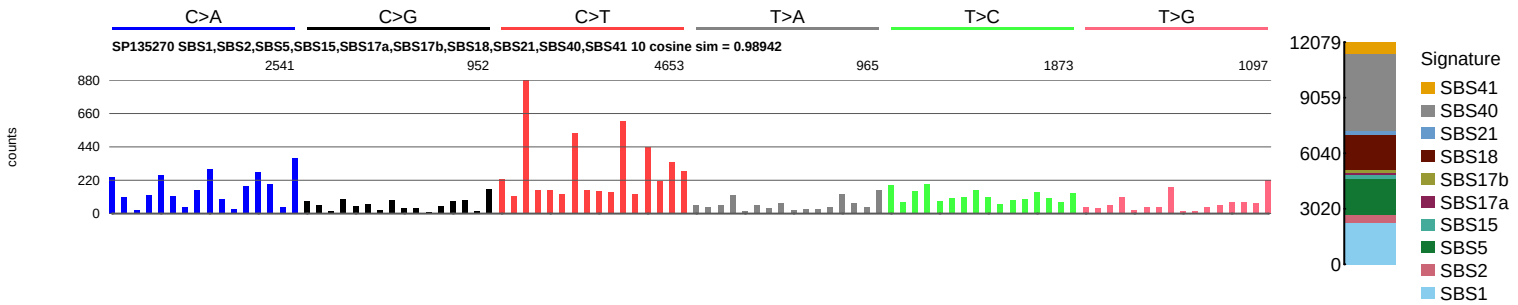

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98964
2589
1093
4699
830
1854
1016
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

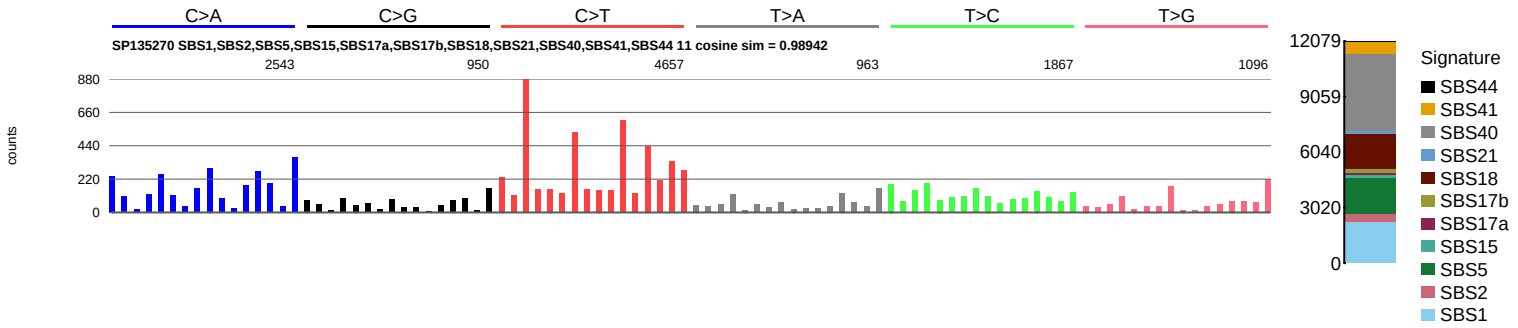

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98963
2548
1112
4675
957
1731
1056
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

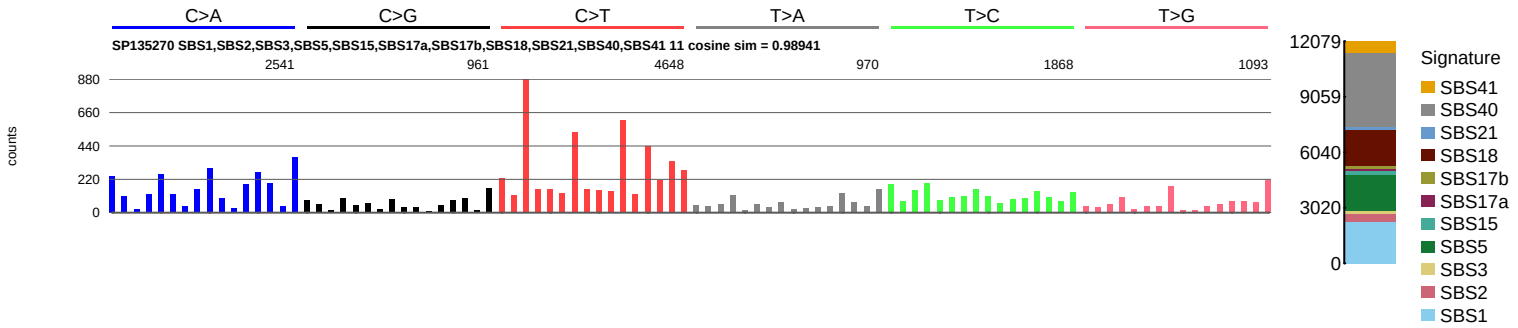

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98958
2555
1084
4700
929
1740
1071
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

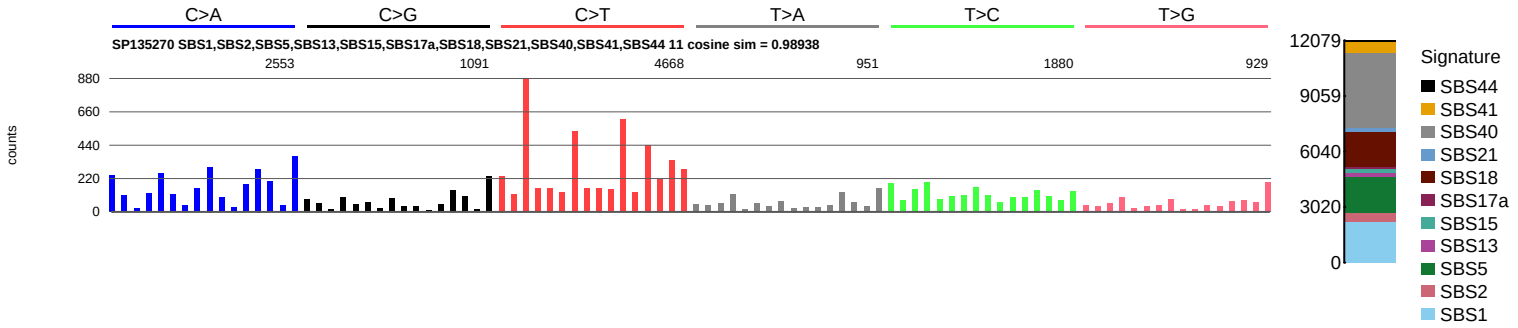

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98949
2555
1089
4713
931
1722
1072
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

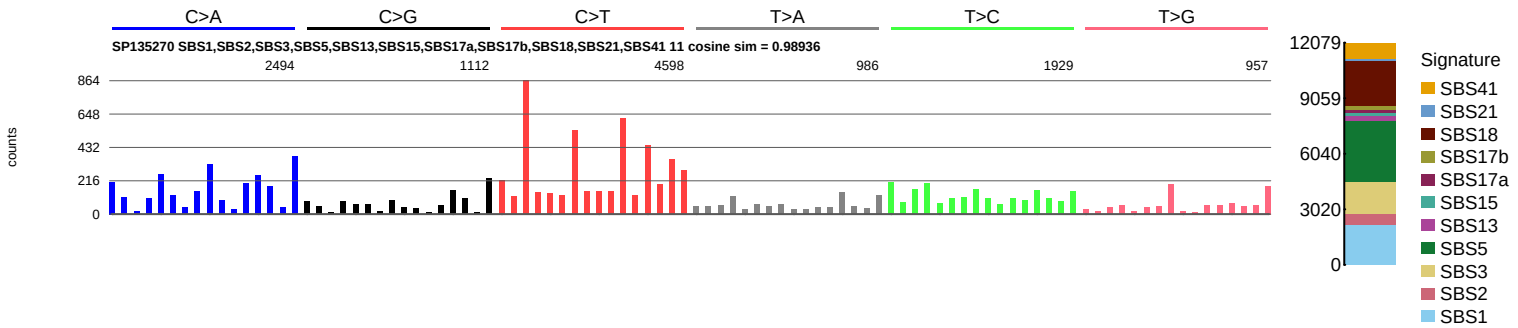

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98944
2541
960
4650
971
1867
1093
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

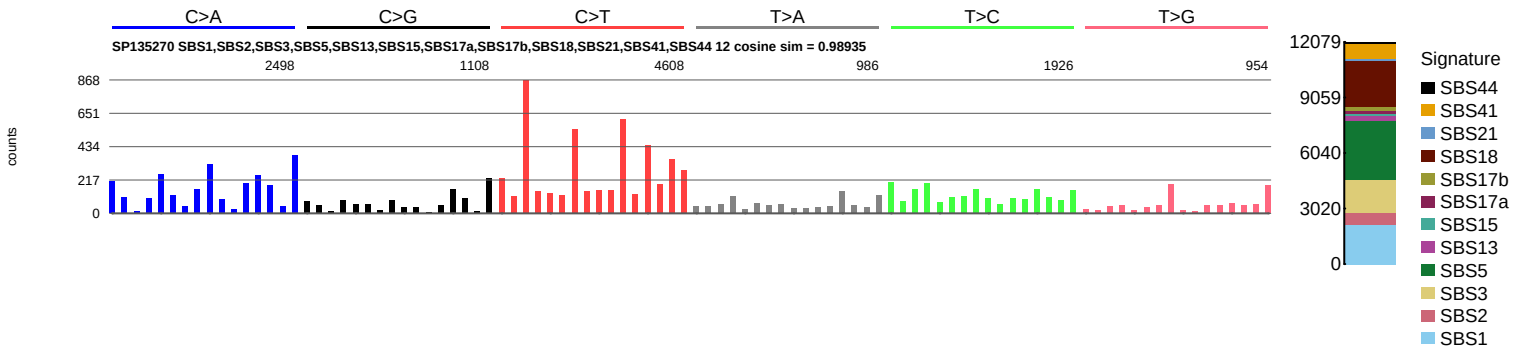

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98942
2541
952
4653
965
1873
1097
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

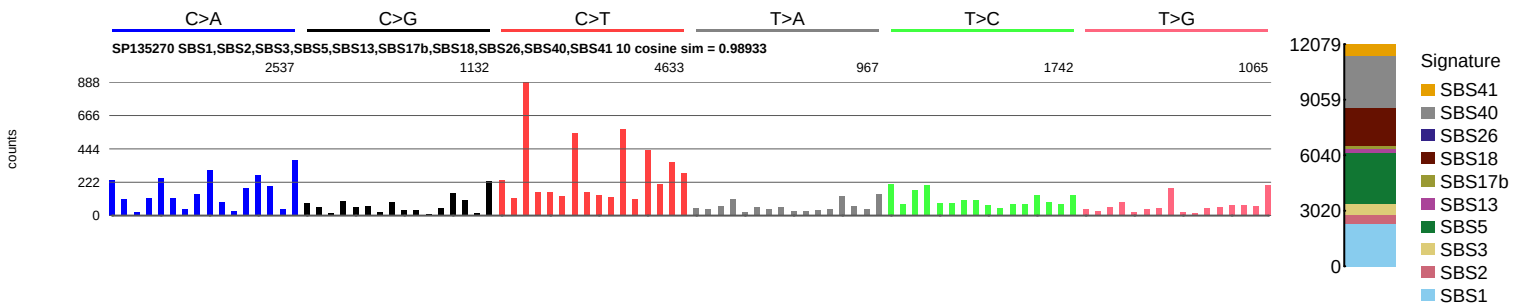

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98942
2543
950
4657
963
1867
1096
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98941
2541
961
4648
970
1868
1093
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98938
2553
1091
4668
951
1880
929
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.98936
2494
1112
4598
986
1929
957
864
648
432
216
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 12 cosine sim = 0.98935
2498
1108
4608
986
1926
954
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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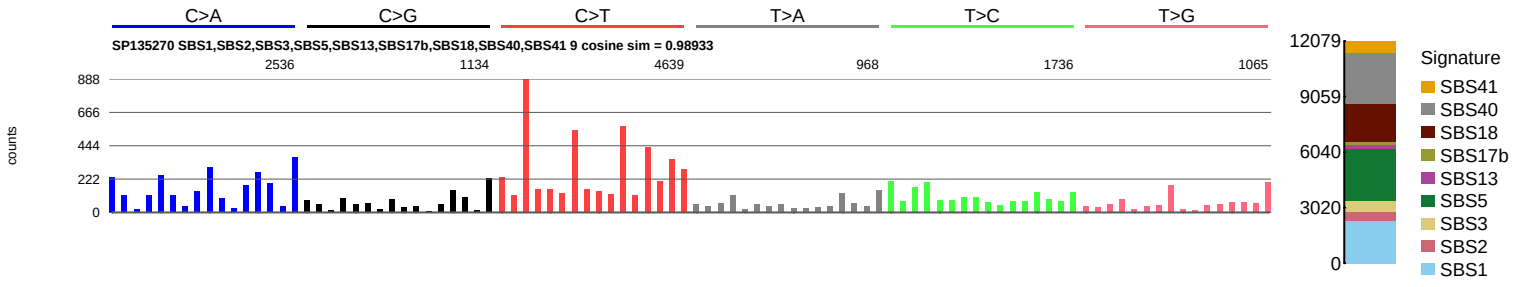
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98933
2537
1132
4633
967
1742
1065
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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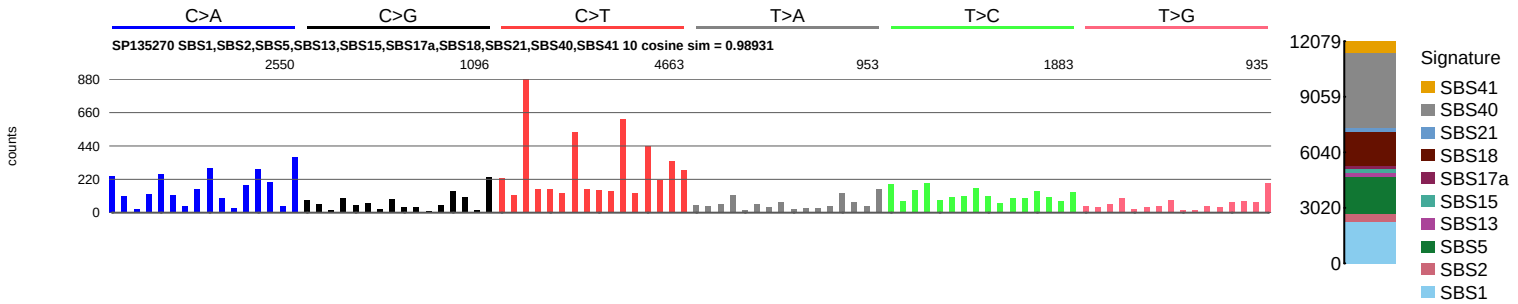
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98933
2536
1134
4639
968
1736
1065
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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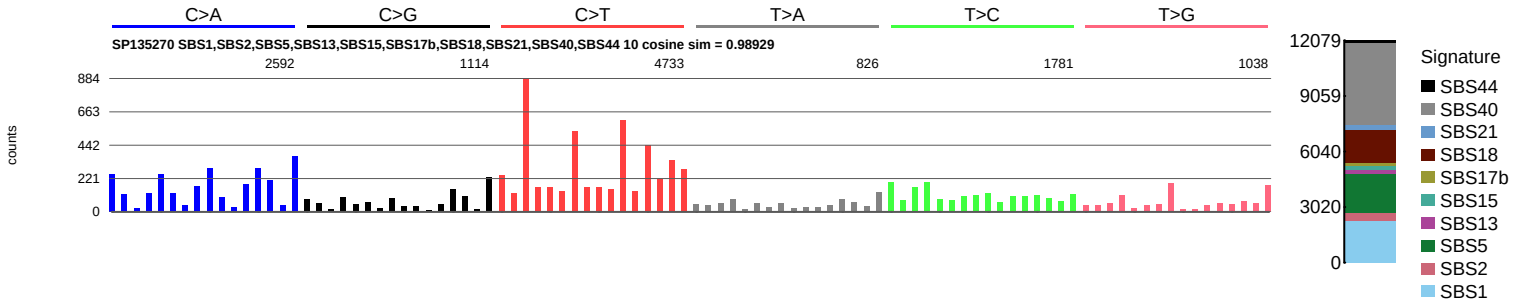
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98931
2550
1096
4663
953
1883
935
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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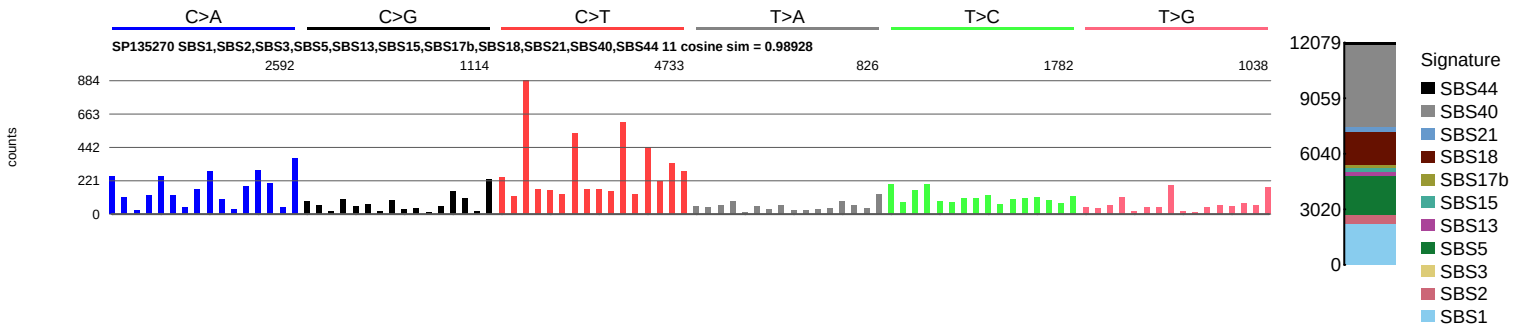
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98929
2592
1114
4733
826
1781
1038
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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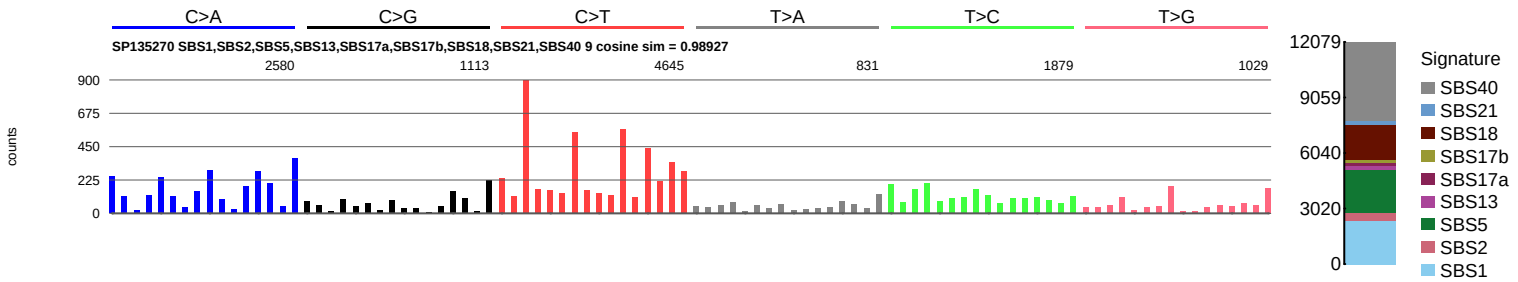
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98928
2592
1114
4733
826
1782
1038
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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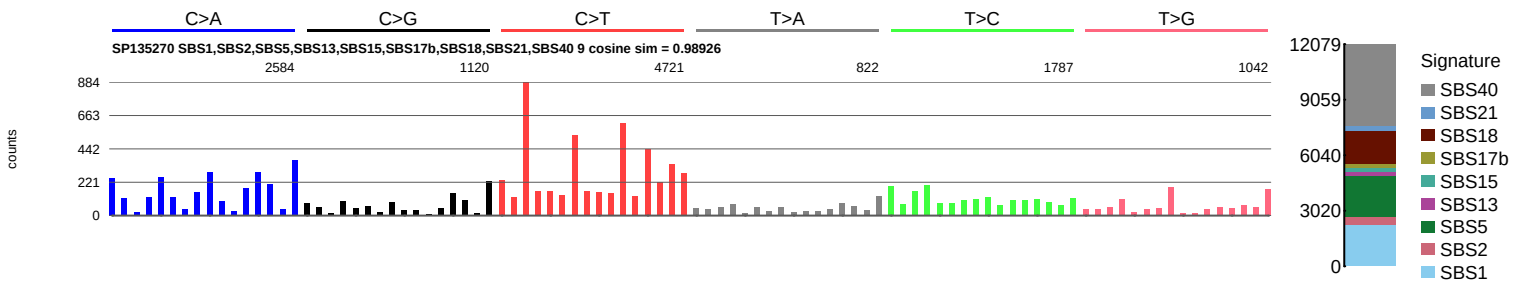
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98927
2580
1113
4645
831
1879
1029
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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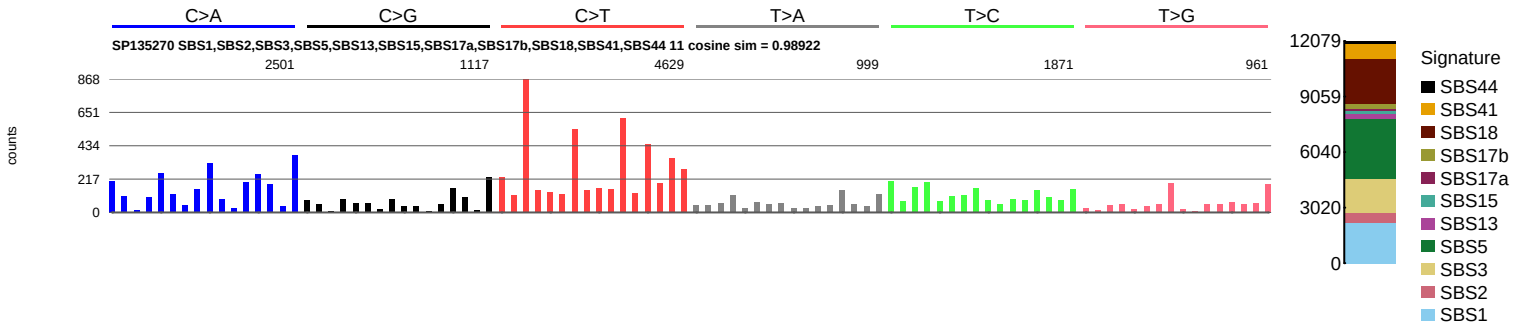
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98926
2584
1120
4721
822
1787
1042
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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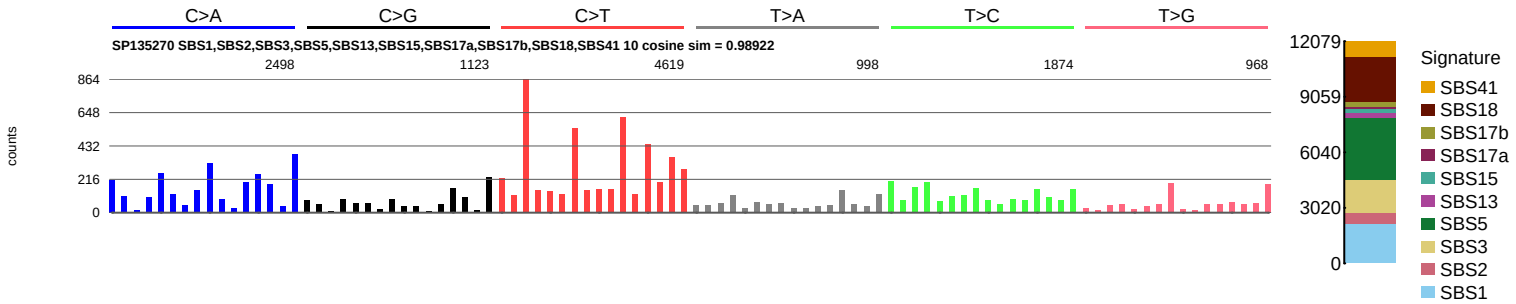
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.98922
2501
1117
4629
999
1871
961
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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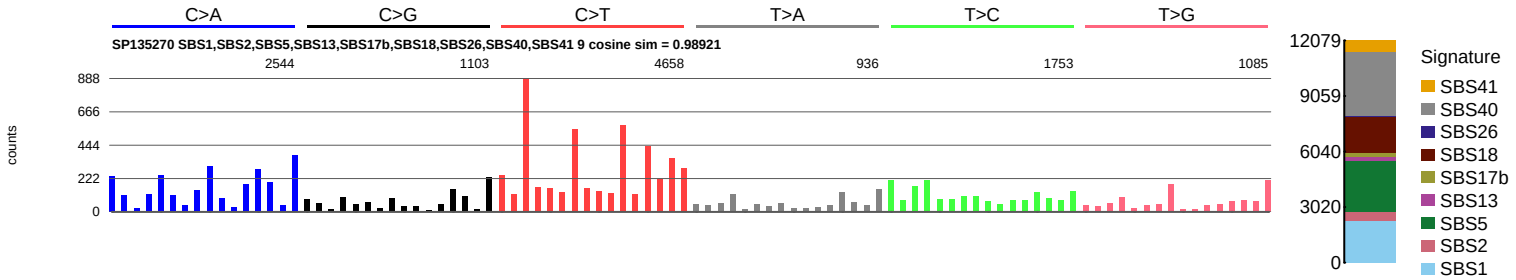
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98922
2498
1123
4619
998
1874
968
counts
864
648
432
216
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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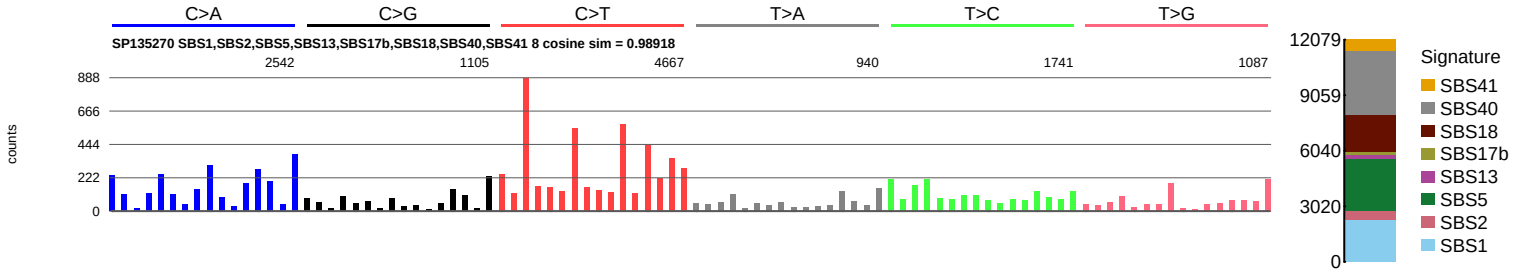
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98921
2544
1103
4658
936
1753
1085
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98918
2542
1105
4667
940
1741
1087
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
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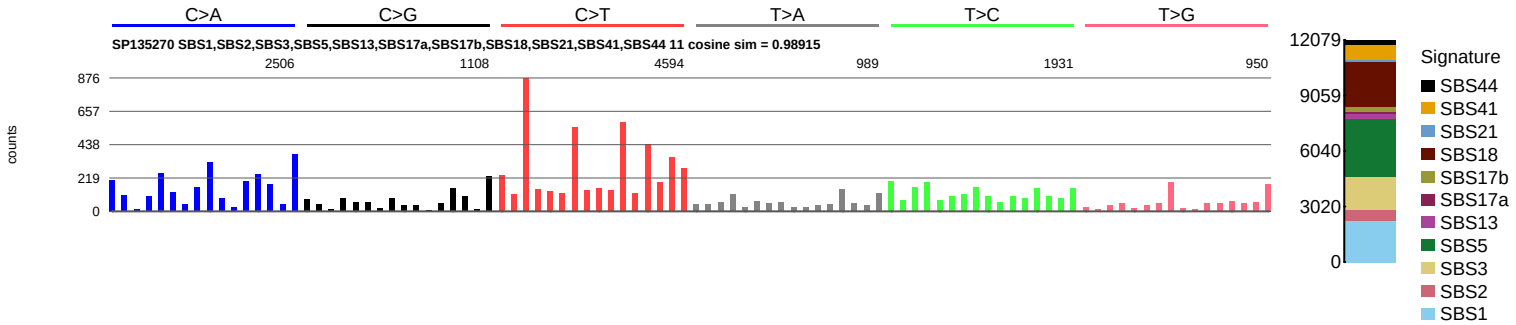
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98915
2506
1108
4594
989
1931
950
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
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SBS18
SBS17b
SBS17a
SBS13
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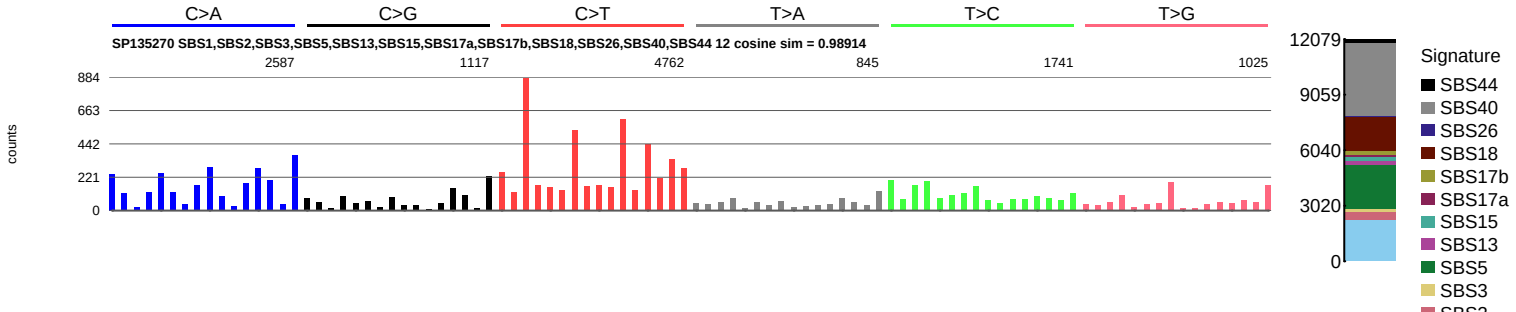
SBS3
SBS2
SBS1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C>G
C>T
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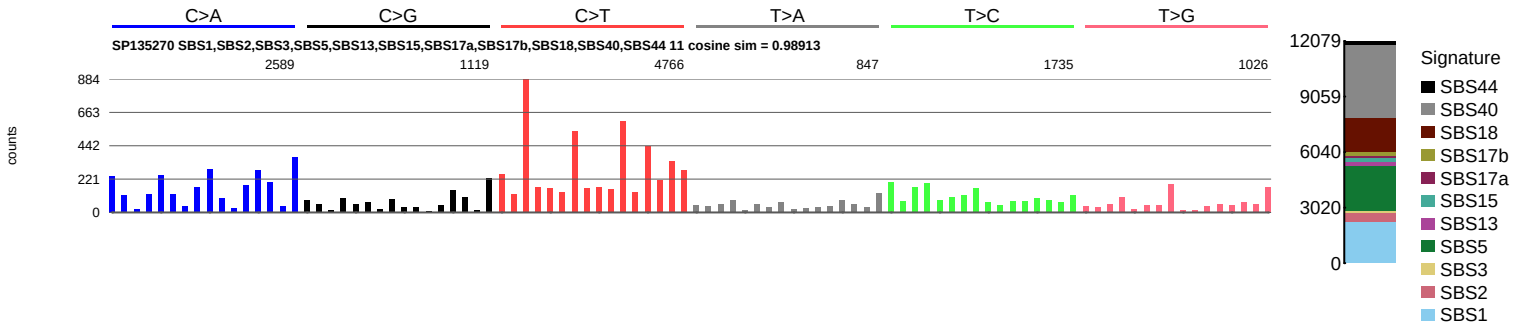
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 12 cosine sim = 0.98914
2587
1117
4762
845
1741
1025
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3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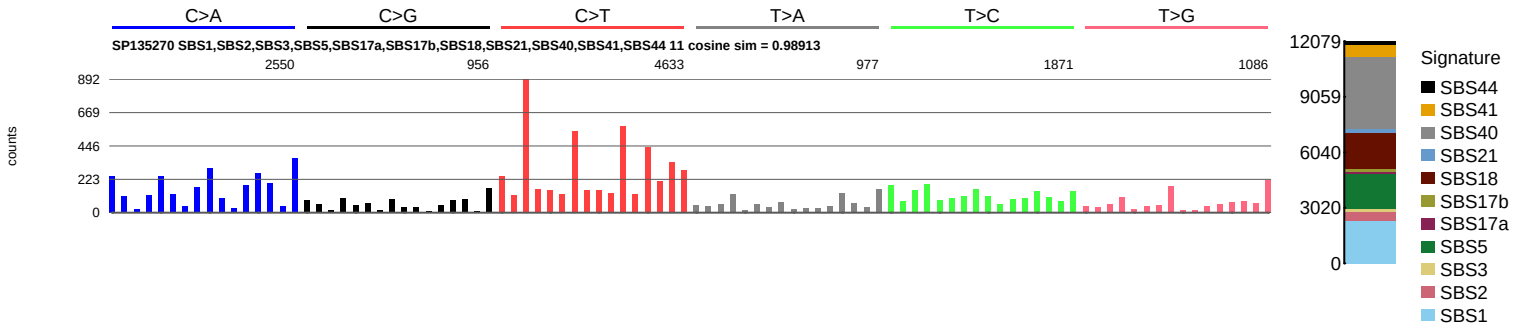
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 11 cosine sim = 0.98913
2589
1119
4766
847
1735
1026
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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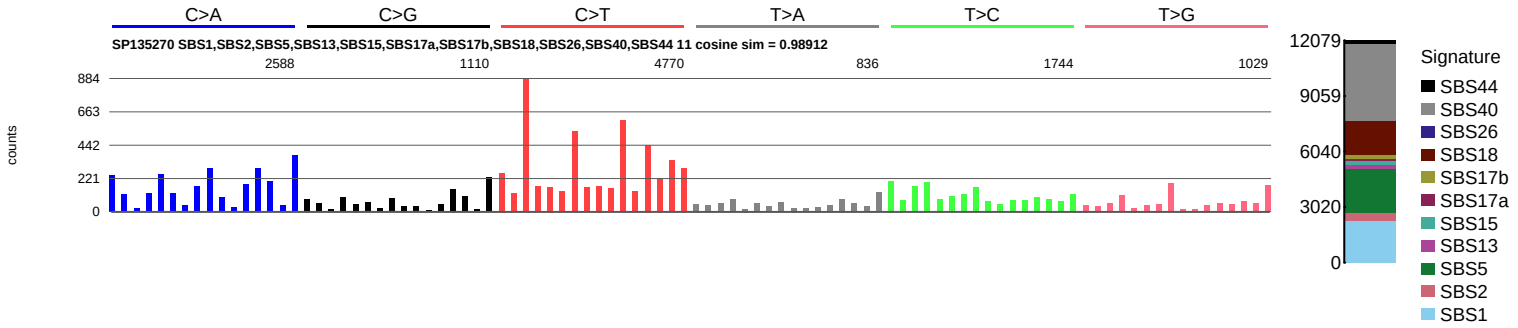
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98913
2550
956
4633
977
1871
1086
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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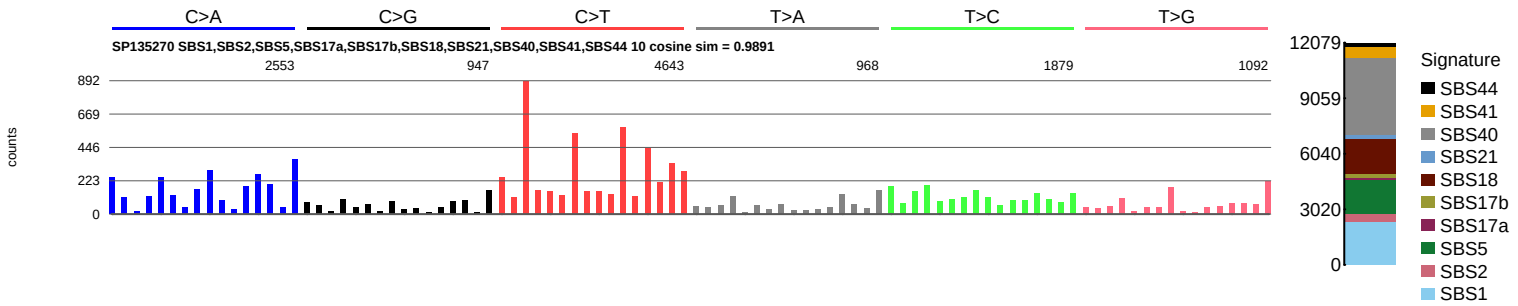
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98912
2588
1110
4770
836
1744
1029
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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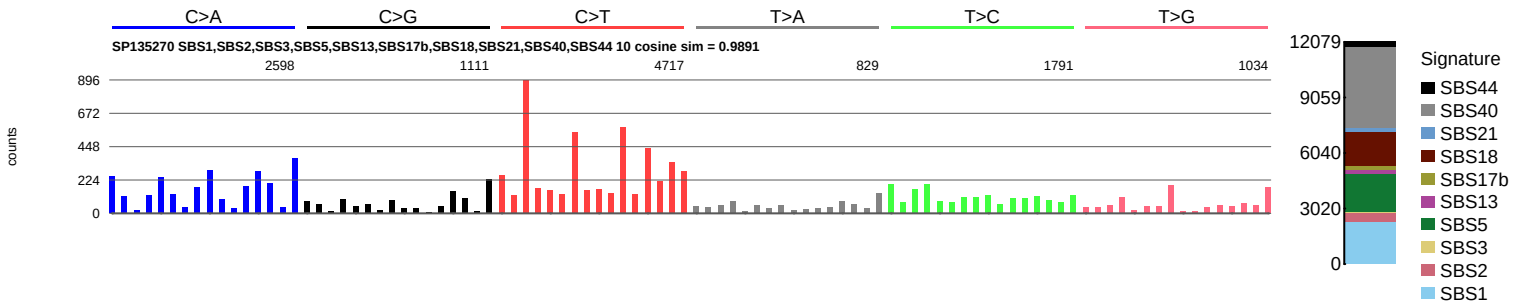
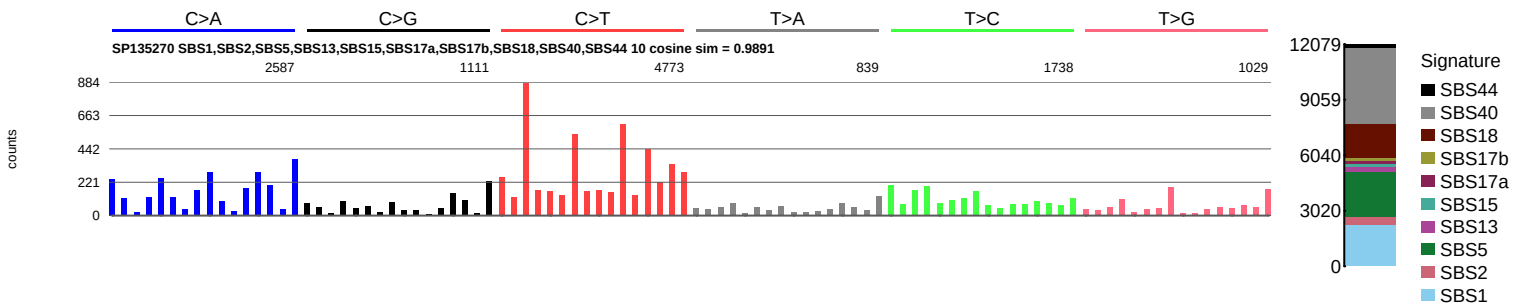
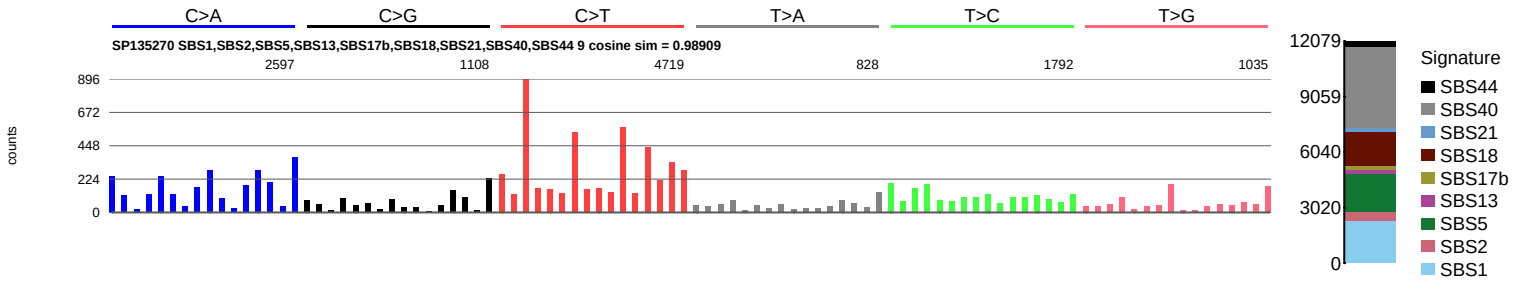
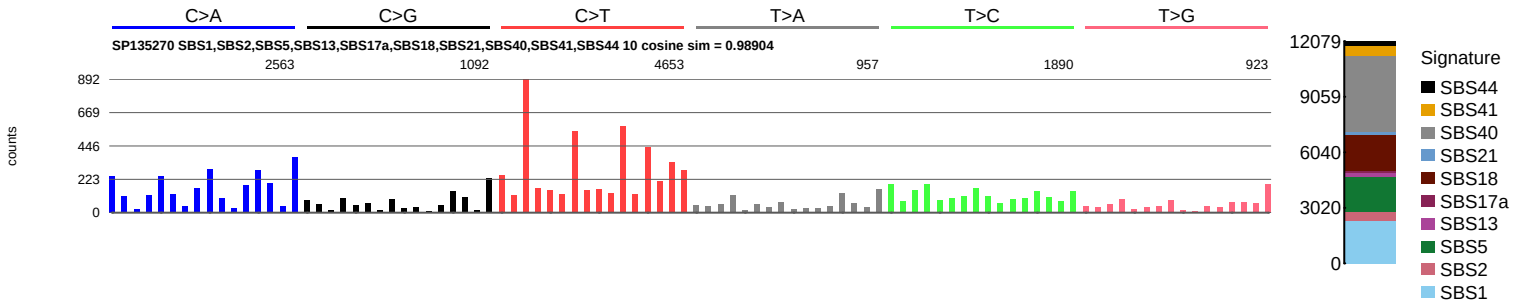
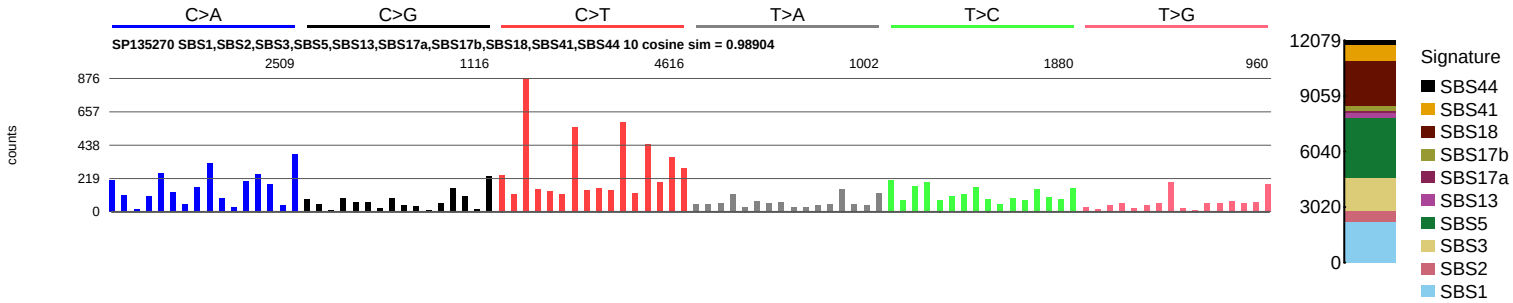
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9891
2553
947
4643
968
1879
1092
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

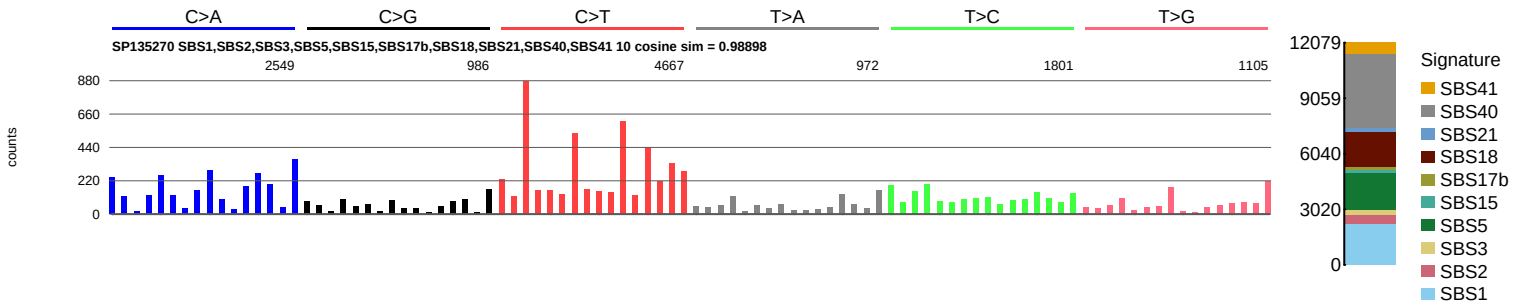

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.9891
2598
1111
4717
829
1791
1034
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

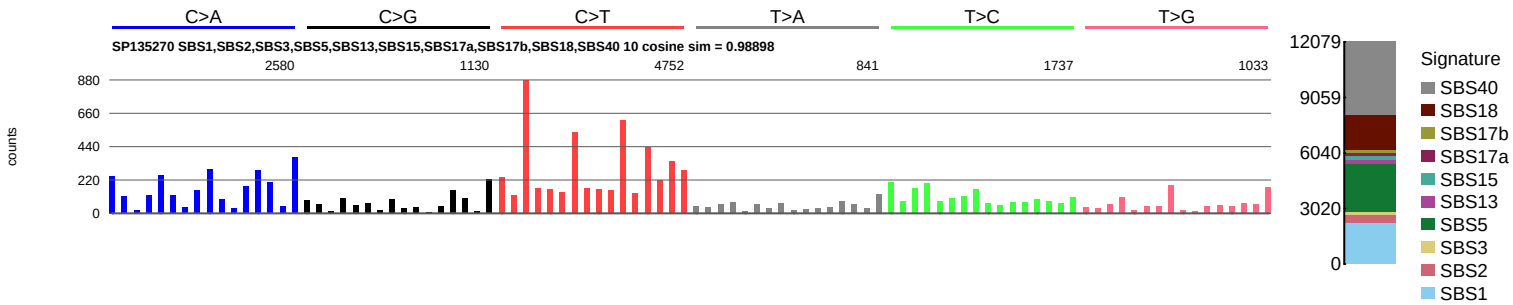

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.9891
2587
1111
4773
839
1738
1029
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98909
2597
1108
4719
828
1792
1035
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98904
2563
1092
4653
957
1890
923
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98904
2509
1116
4616
1002
1880
960
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98898
2549
986
4667
972
1801
1105
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 10 cosine sim = 0.98898
2580
1130
4752
841
1737
1033
counts
880
660
440
220
0
12079
9059
6040
3020
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Signature
SBS40
SBS18
SBS17b
SBS17a
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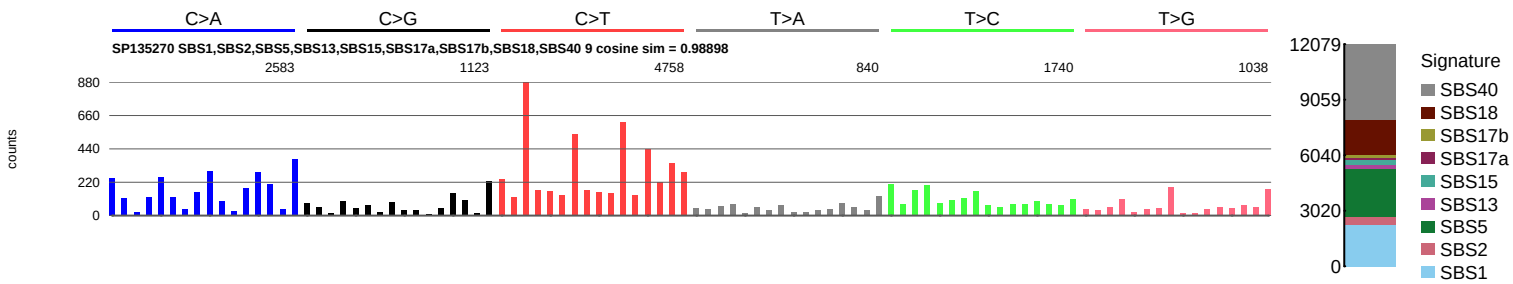
SBS1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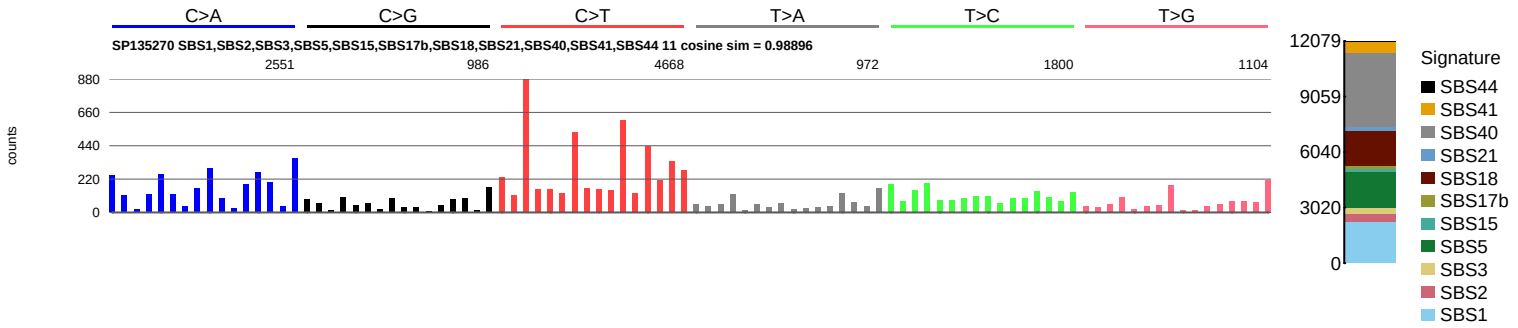
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98898
2583
1123
4758
840
1740
1038
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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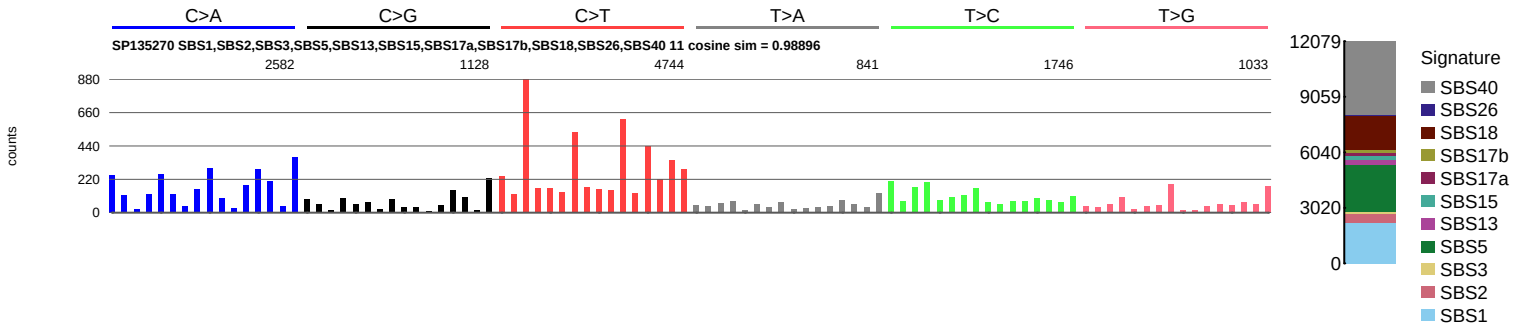
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98896
2551
986
4668
972
1800
1104
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 11 cosine sim = 0.98896
2582
1128
4744
841
1746
1033
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS40
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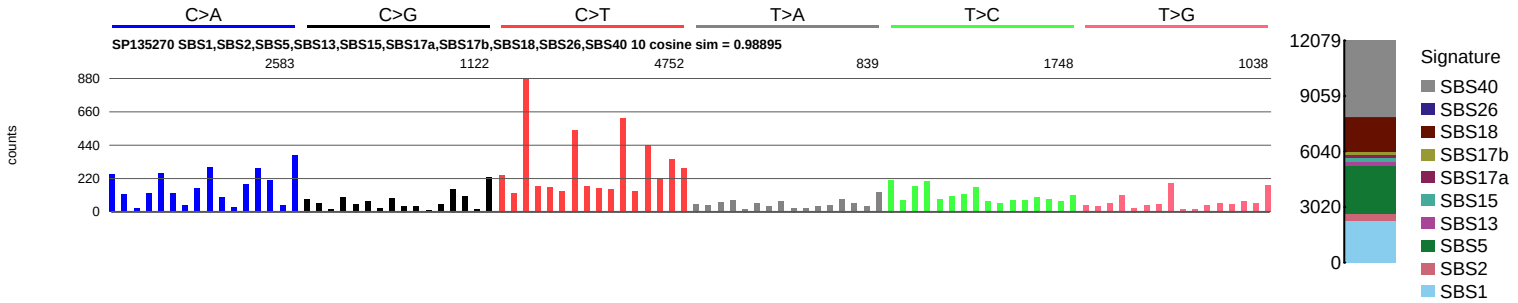
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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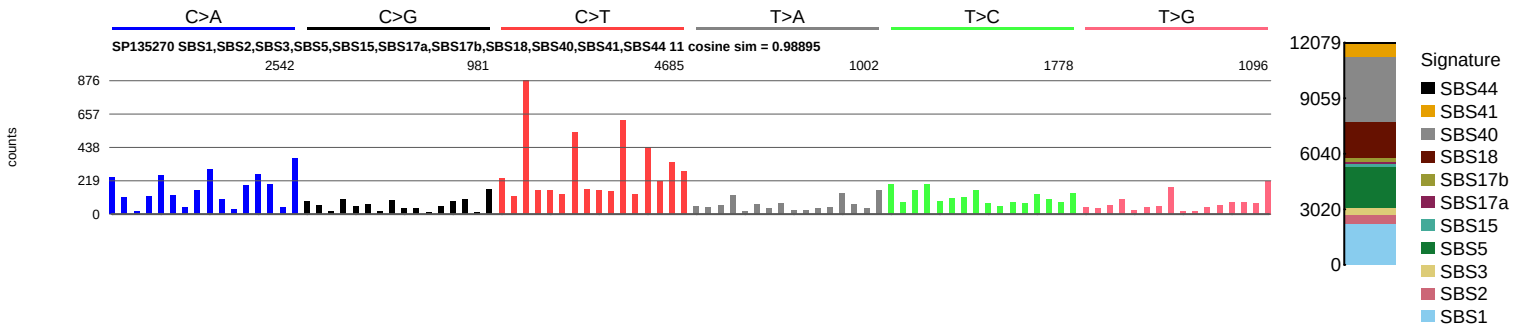
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98895
2583
1122
4752
839
1748
1038
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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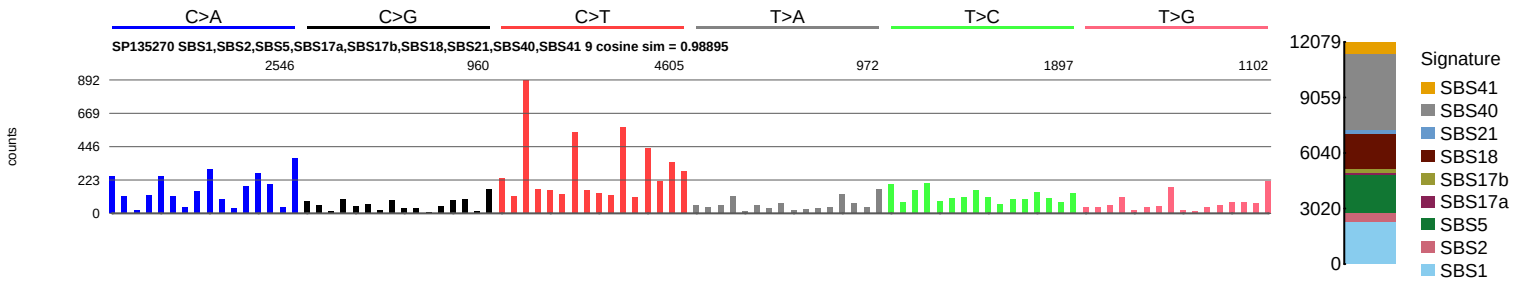
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98895
2542
981
4685
1002
1778
1096
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
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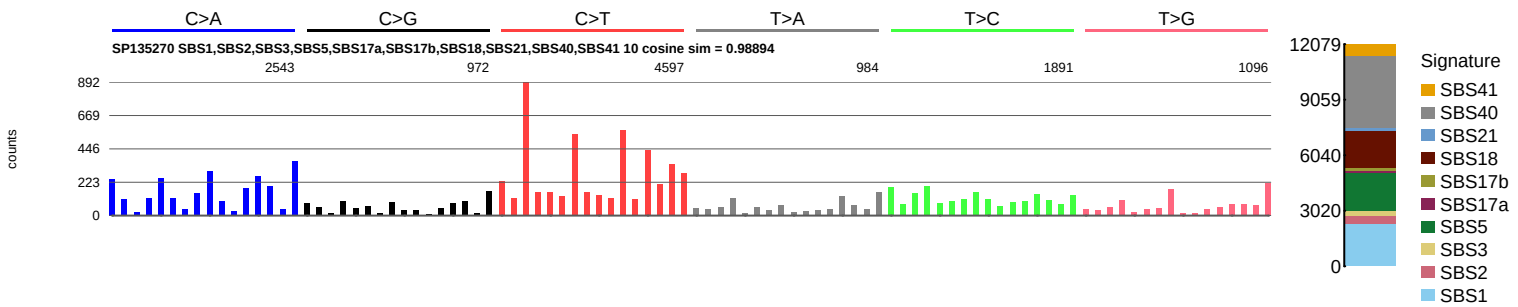
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98895
2546
960
4605
972
1897
1102
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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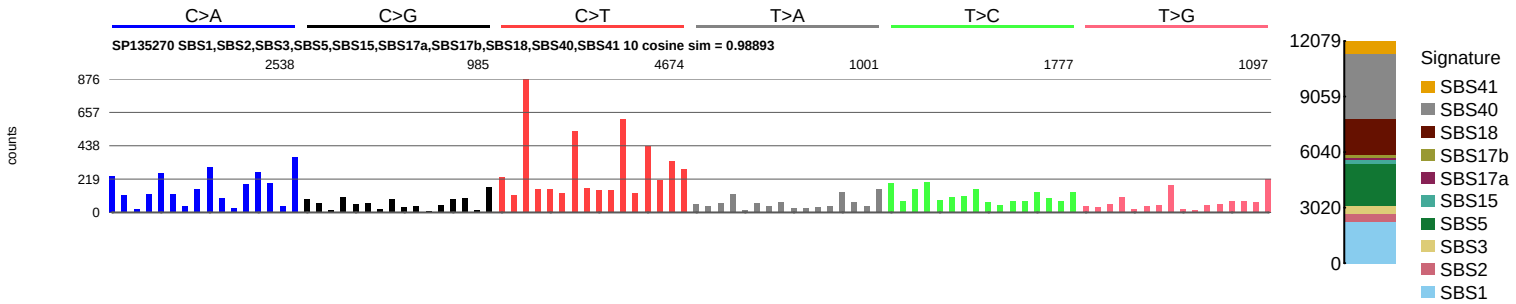
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98894
2543
972
4597
984
1891
1096
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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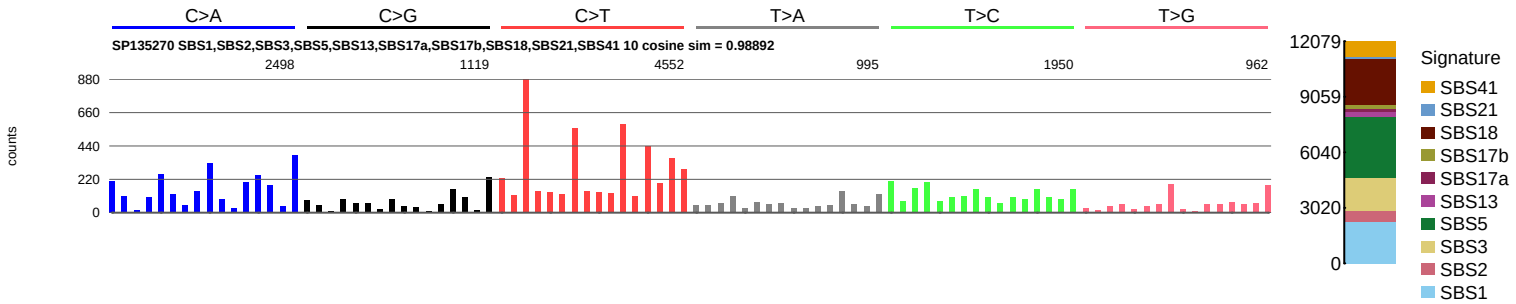
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98893
2538
985
4674
1001
1777
1097
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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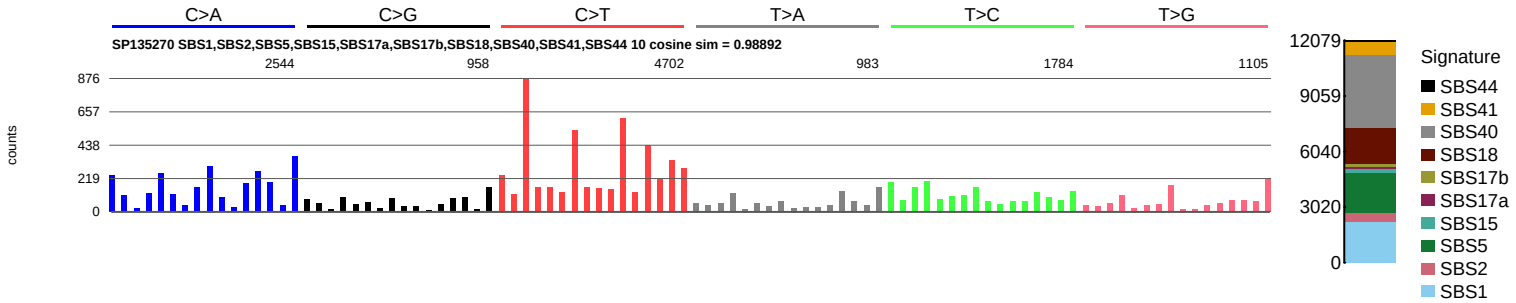
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98892
2498
1119
4552
995
1950
962
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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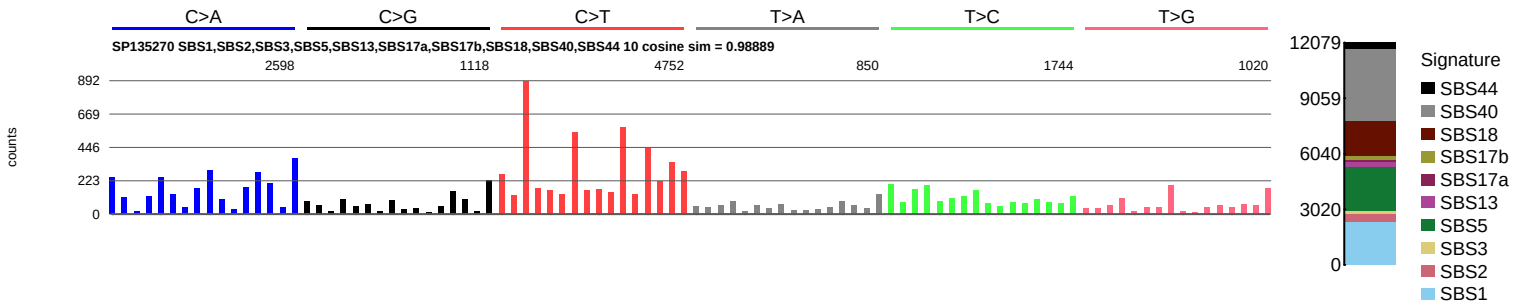
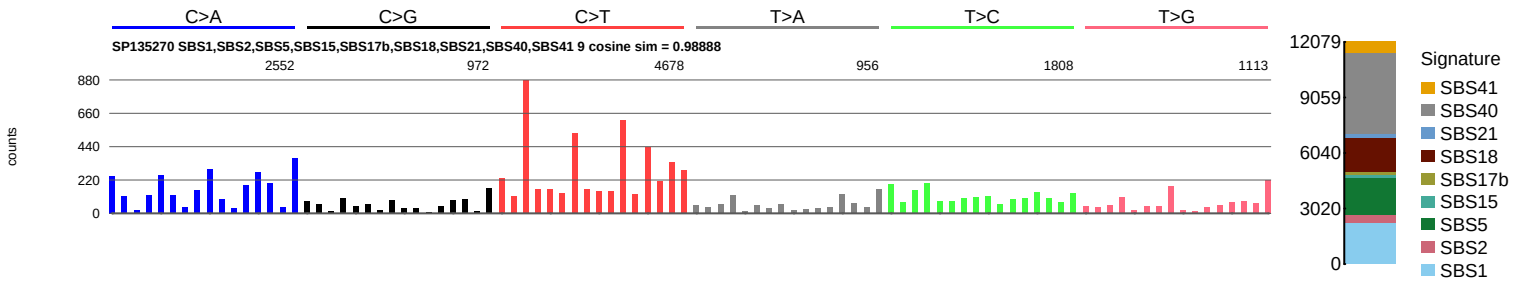
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98892
2544
958
4702
983
1784
1105
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98889
2598
1118
4752
850
1744
1020
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98888
2552
972
4678
956
1808
1113
counts
880
660
440
220
0
12079
9059
6040
3020
0
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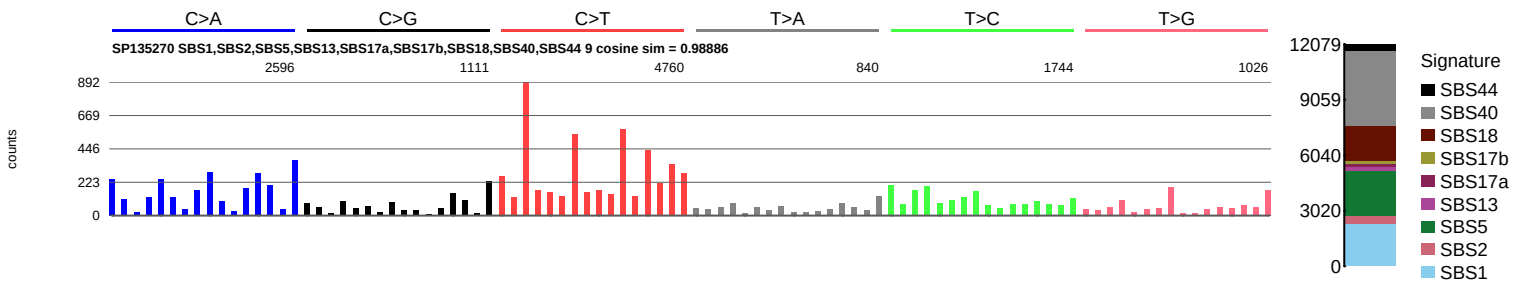
SBS2
SBS1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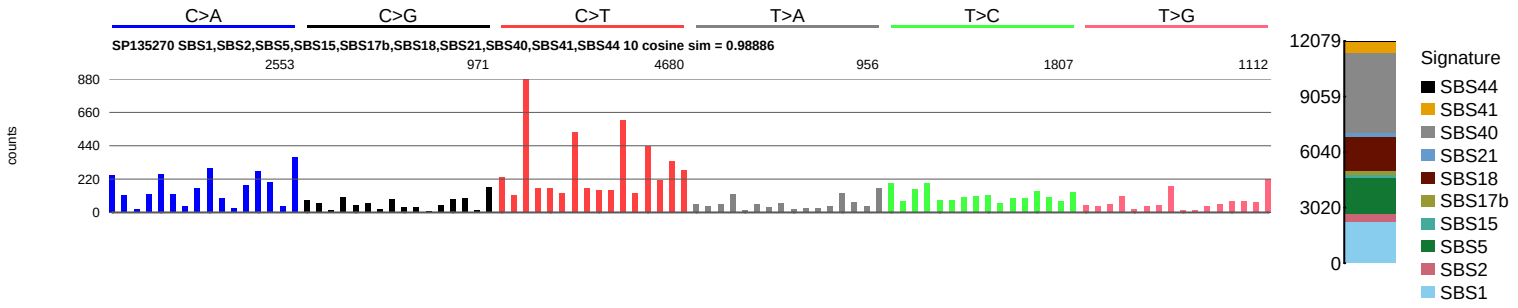
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98886
2596
1111
4760
840
1744
1026
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>T
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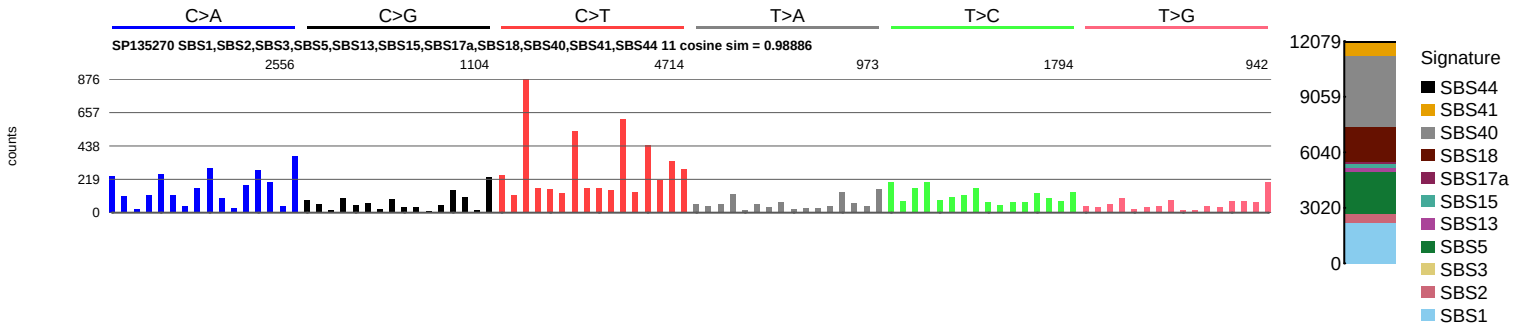
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98886
2553
971
4680
956
1807
1112
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>G
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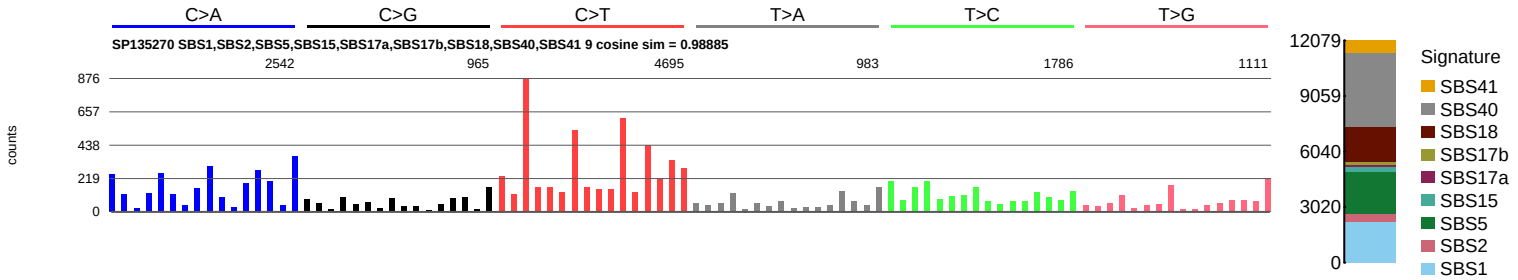
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98886
2556
1104
4714
973
1794
942
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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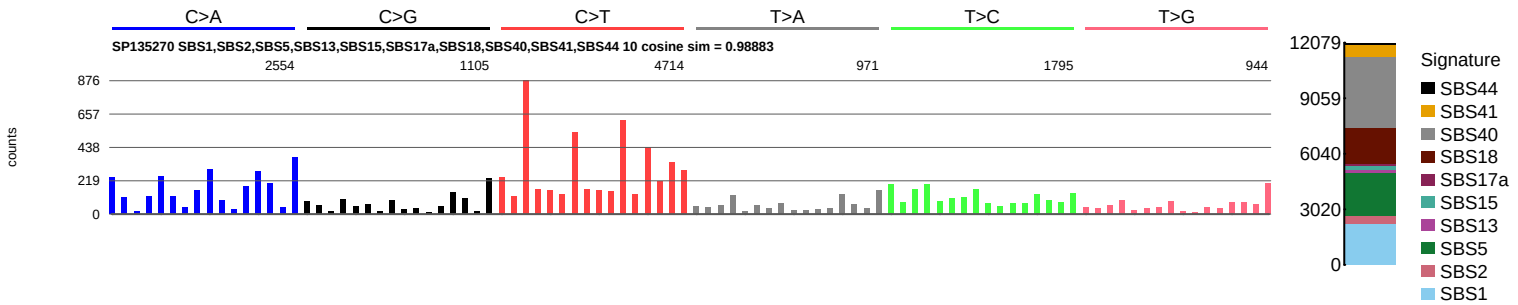
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98885
2542
965
4695
983
1786
1111
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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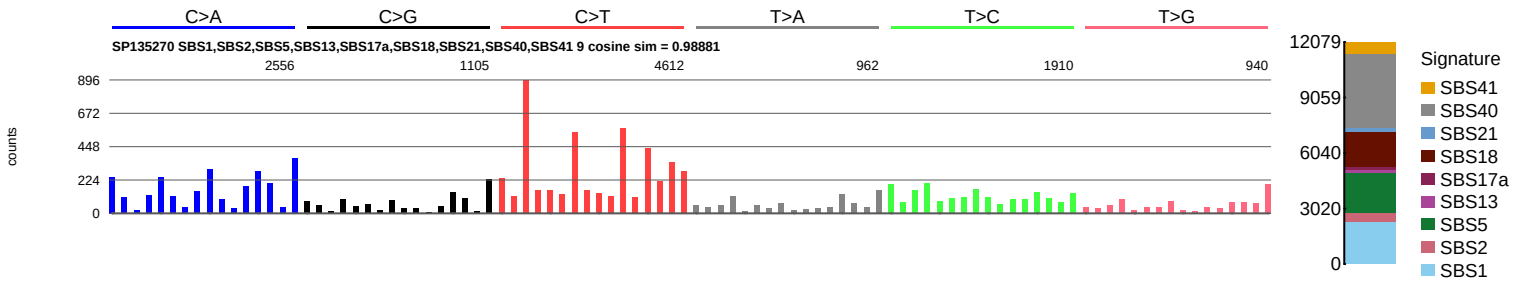
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98883
2554
1105
4714
971
1795
944
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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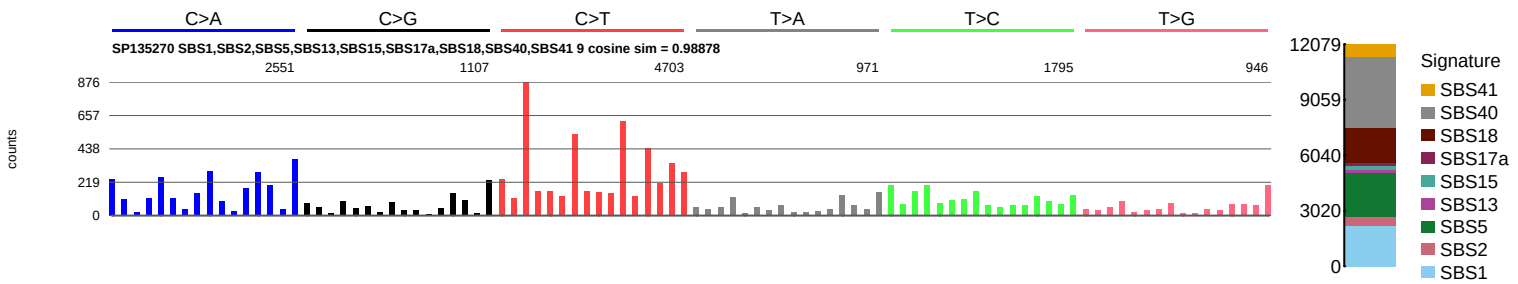
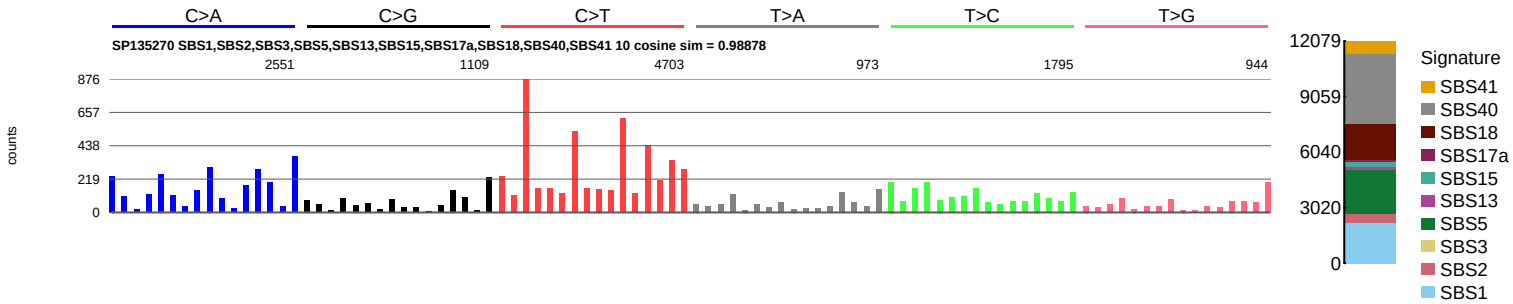
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98881
2556
1105
4612
962
1910
940
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98878
2551
1107
4703
971
1795
946
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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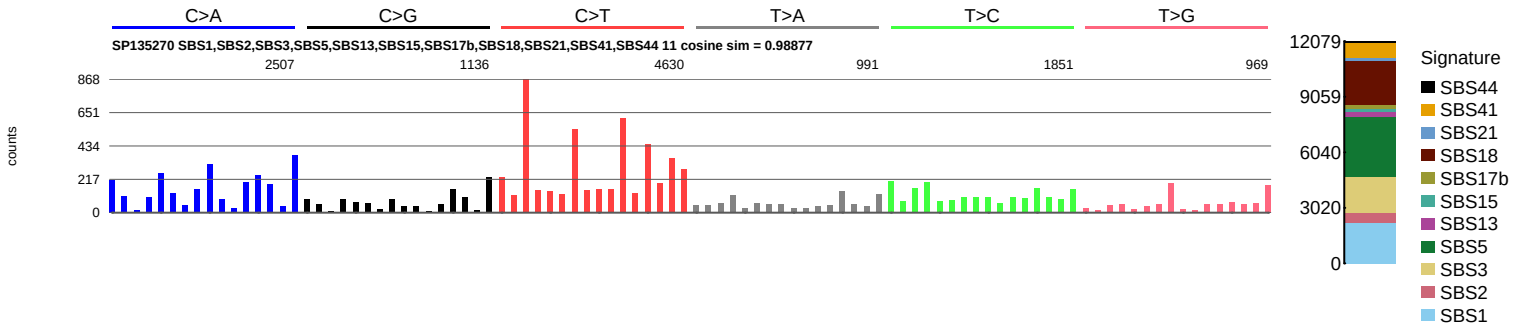
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 10 cosine sim = 0.98878
2551
1109
4703
973
1795
944
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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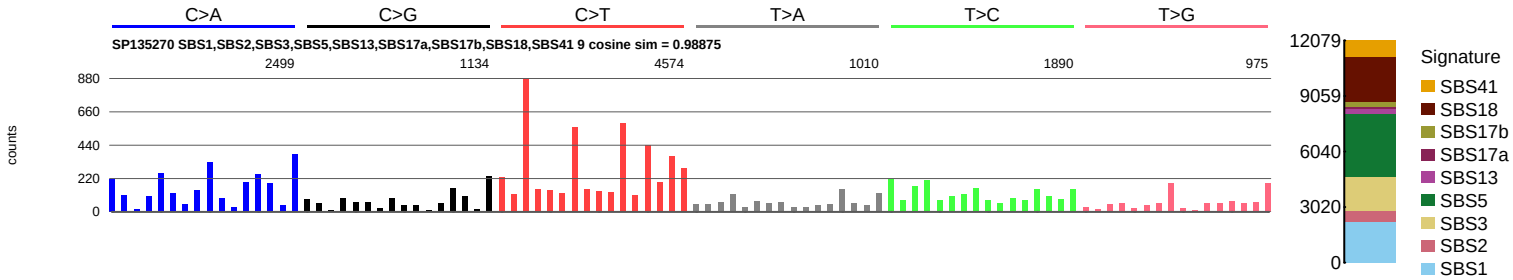
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98877
2507
1136
4630
991
1851
969
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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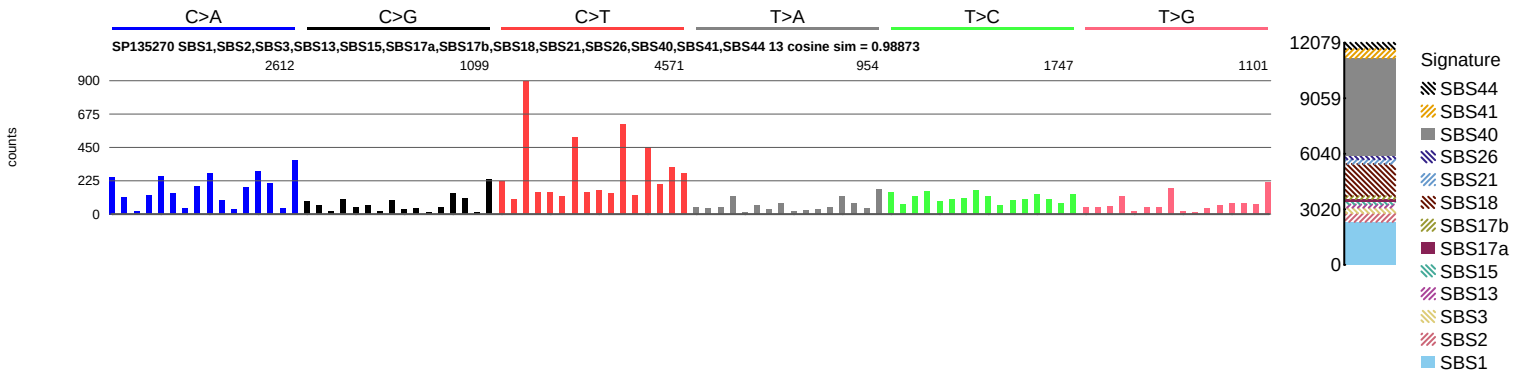
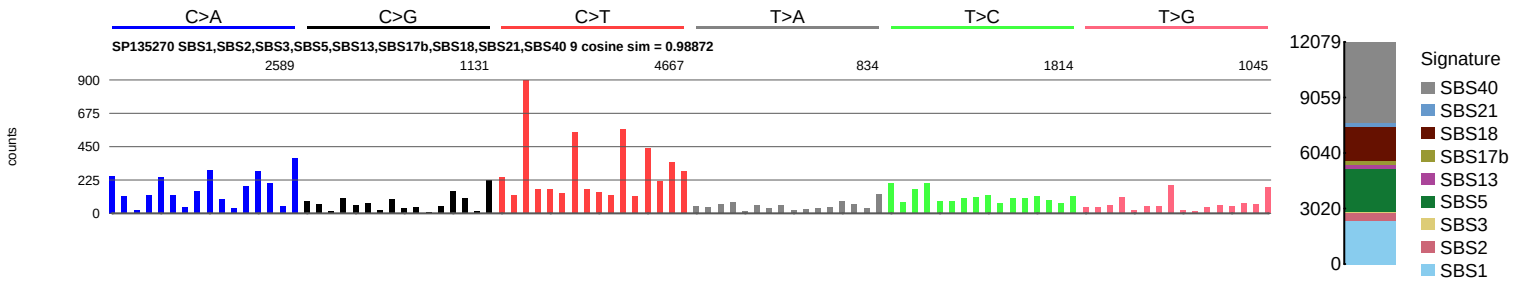
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98875
2499
1134
4574
1010
1890
975
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 13 cosine sim = 0.98873
2612
1099
4571
954
1747
1101
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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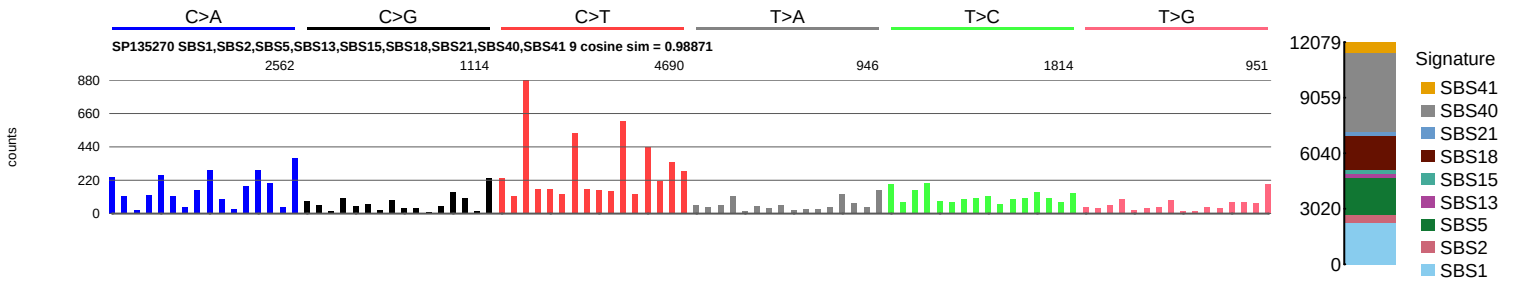
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98872
2589
1131
4667
834
1814
1045
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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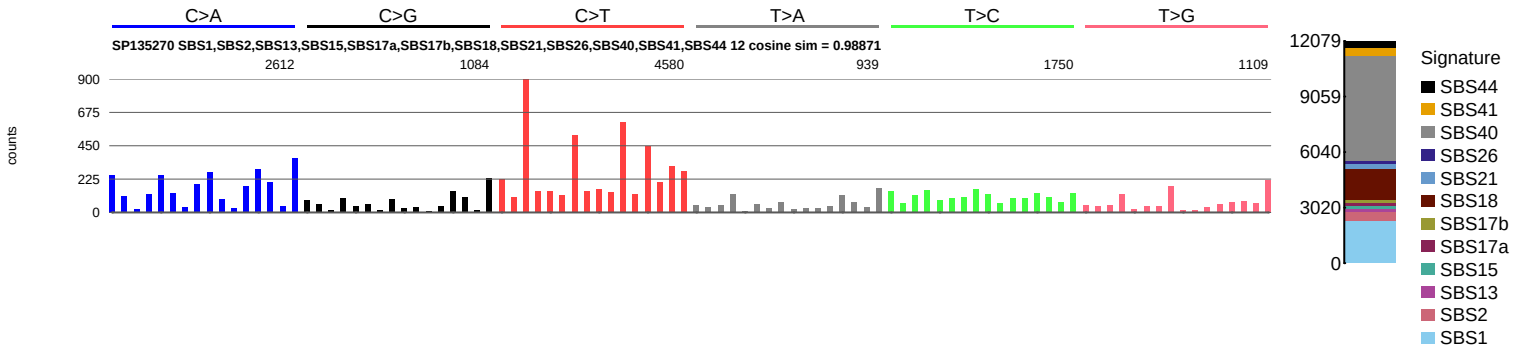
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98871
2562
1114
4690
946
1814
951
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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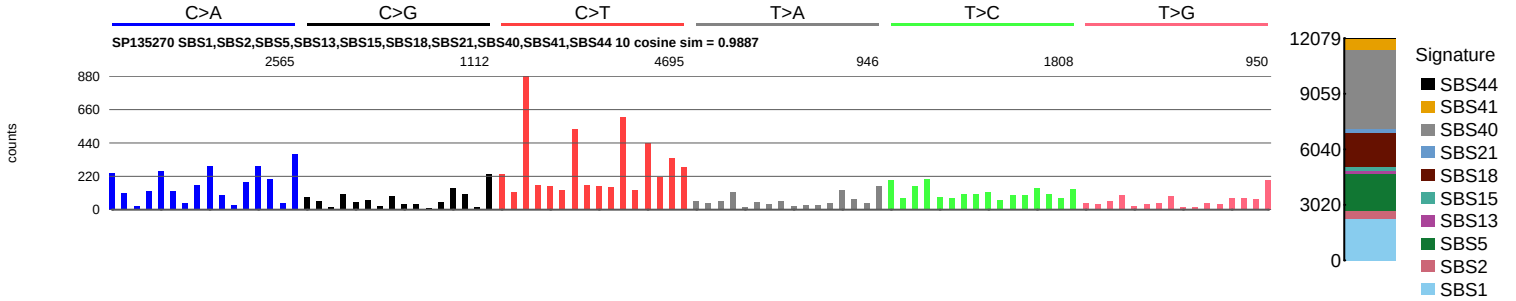
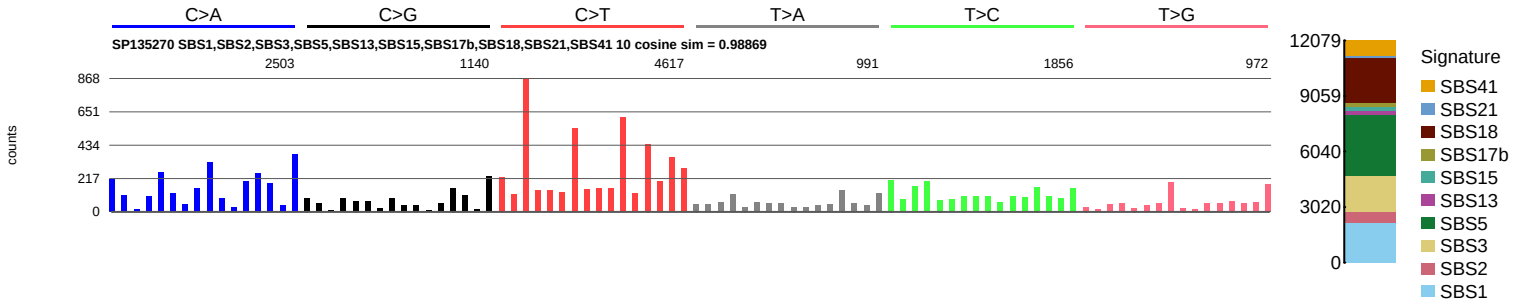
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98871
2612
1084
4580
939
1750
1109
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9887
2565
1112
4695
946
1808
950
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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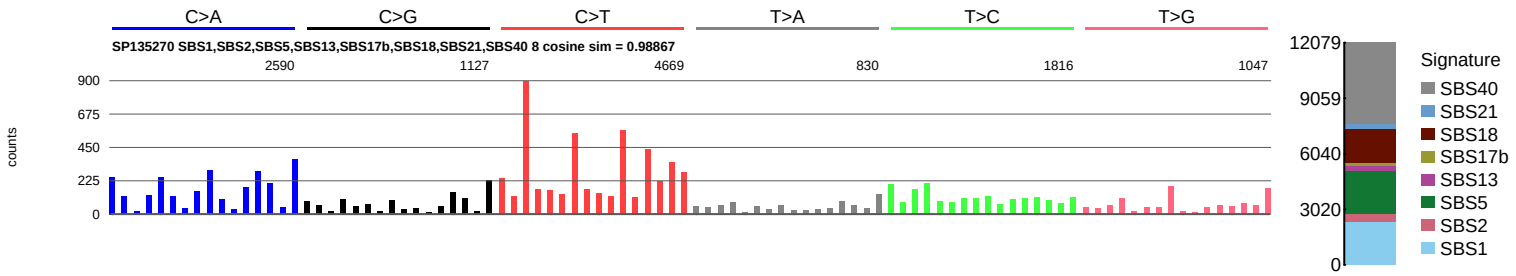
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98869
2503
1140
4617
991
1856
972
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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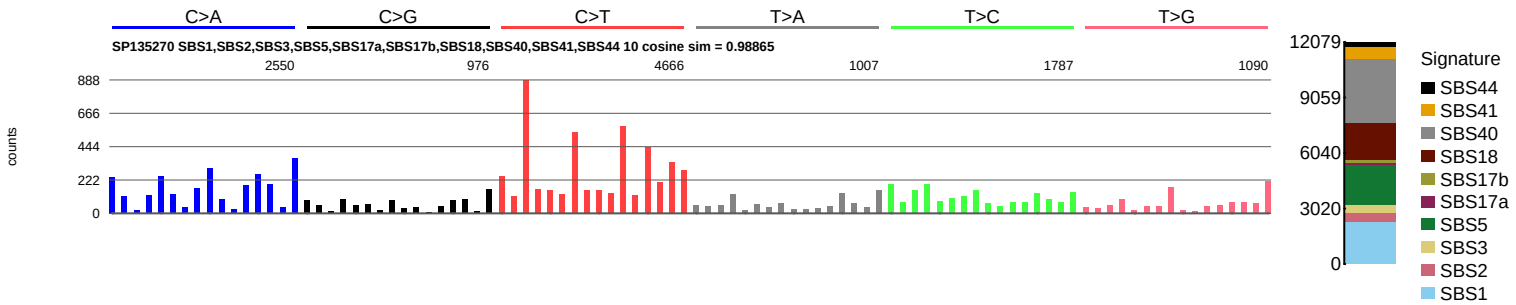
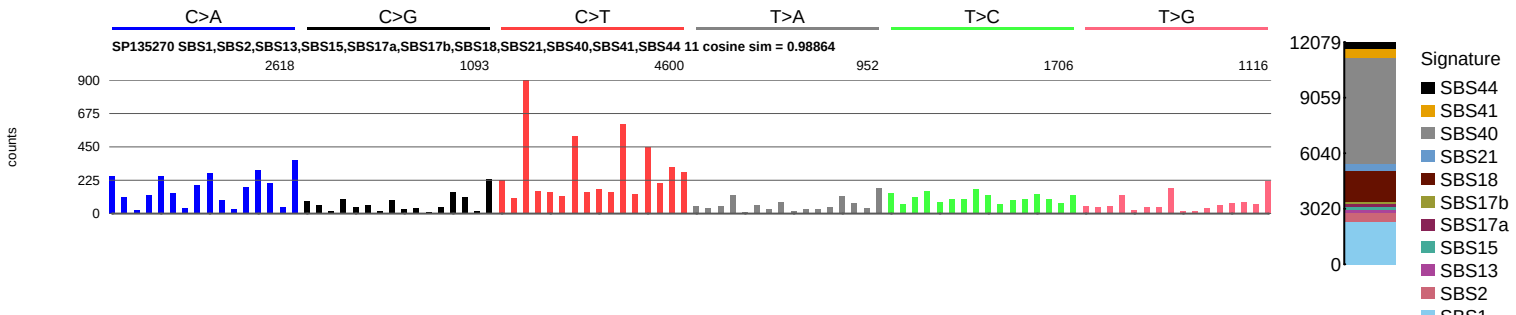
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98867
2590
1127
4669
830
1816
1047
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98865
2550
976
4666
1007
1787
1090
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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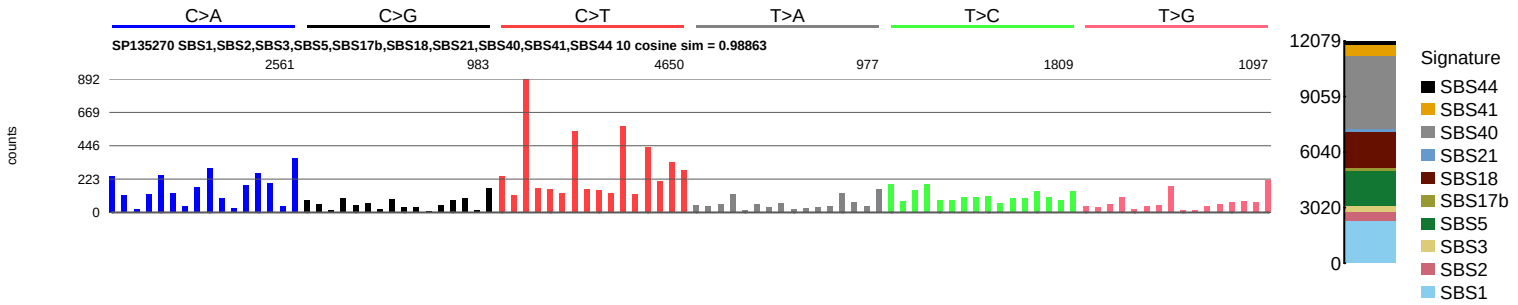
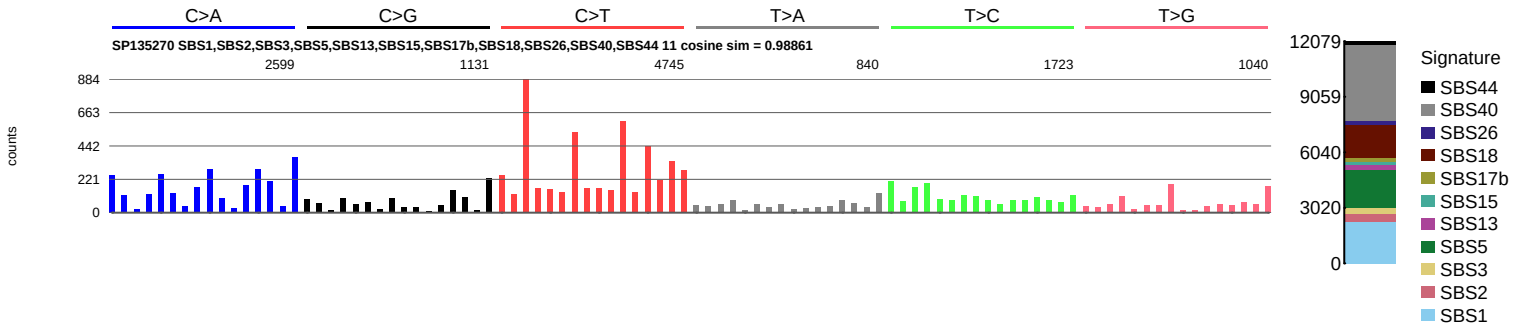
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98864
2618
1093
4600
952
1706
1116
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98863
2561
983
4650
977
1809
1097
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98861
2599
1131
4745
840
1723
1040
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
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SBS26
SBS18
SBS17b
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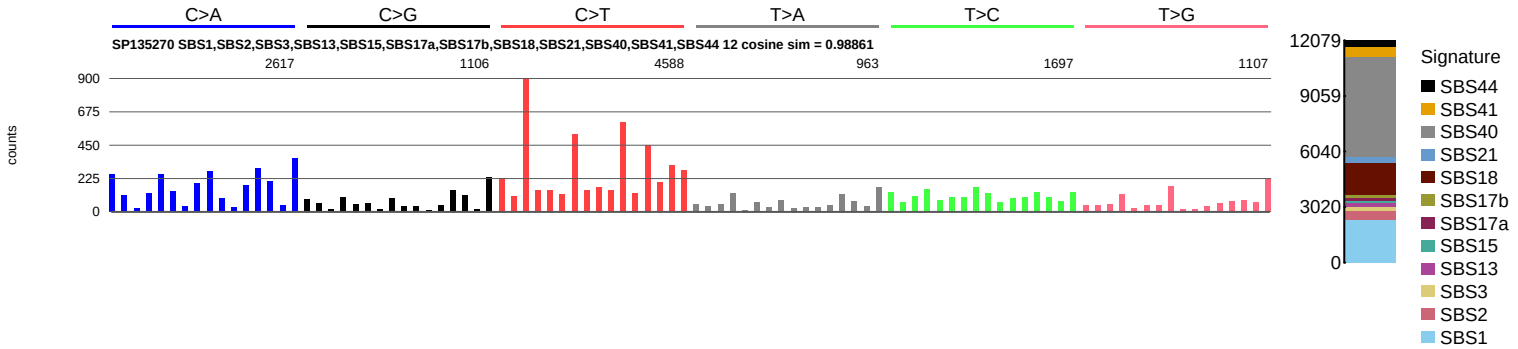
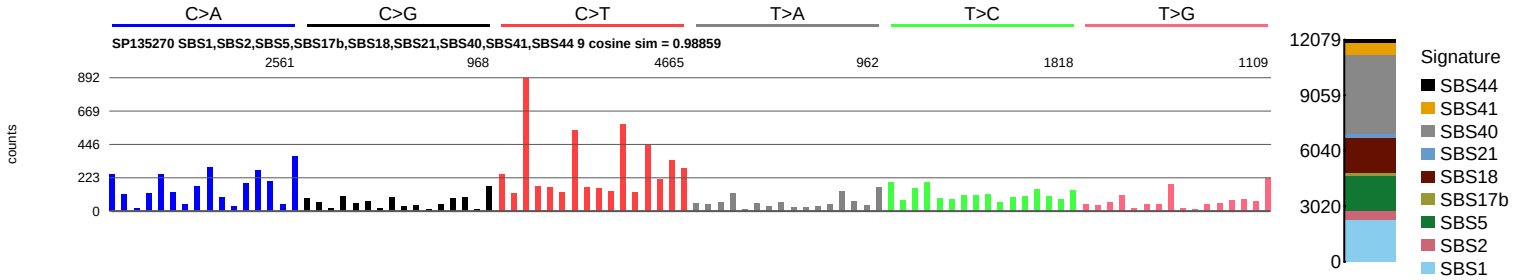
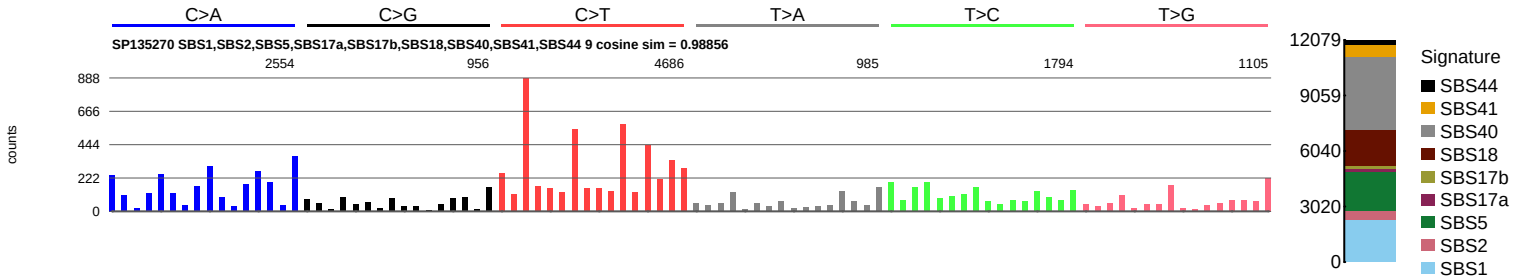
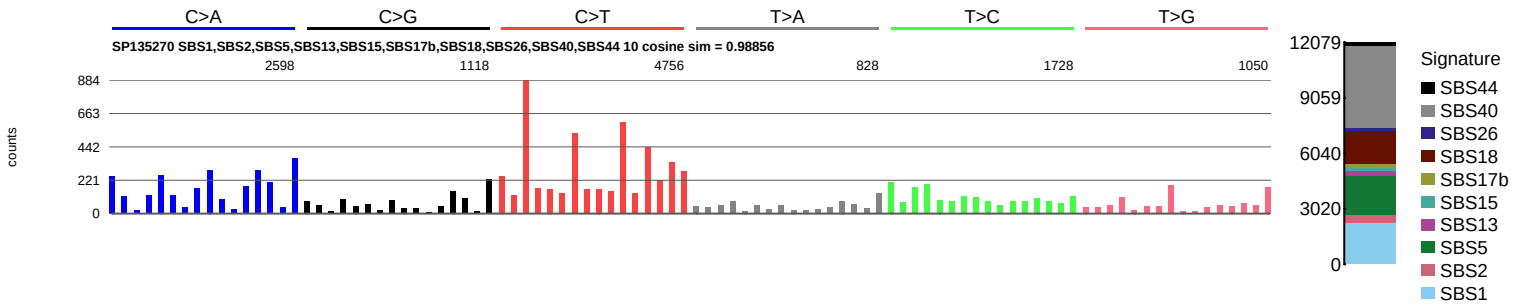
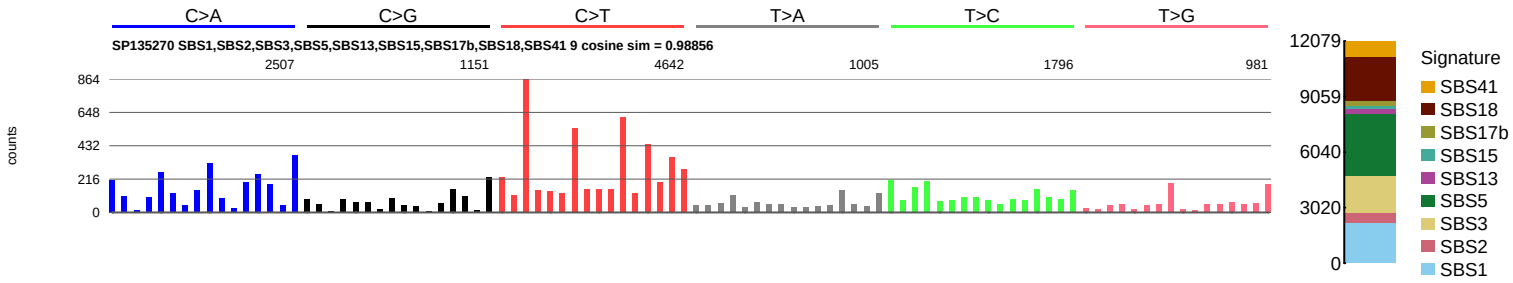
SBS3
SBS2
SBS1

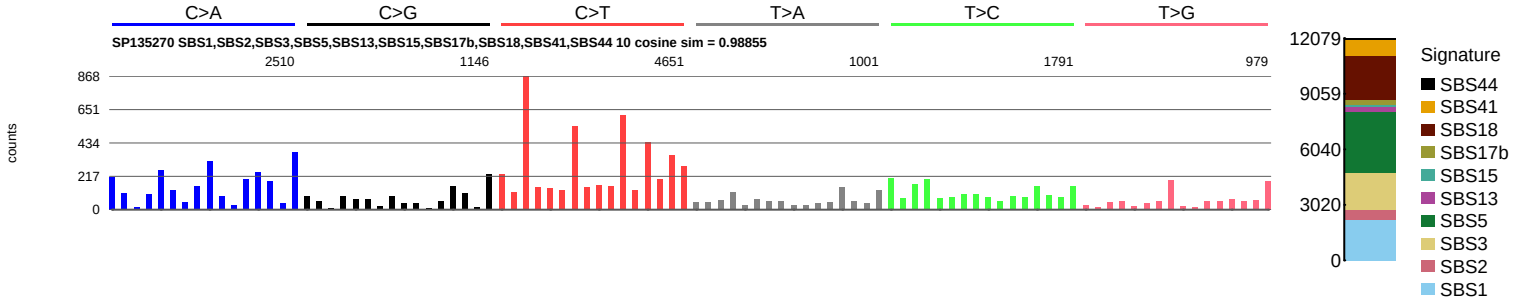

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98861
2617
1106
4588
963
1697
1107
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98859
2561
968
4665
962
1818
1109
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98856
2554
956
4686
985
1794
1105
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98856
2598
1118
4756
828
1728
1050
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 9 cosine sim = 0.98856
2507
1151
4642
1005
1796
981
counts
864
648
432
216
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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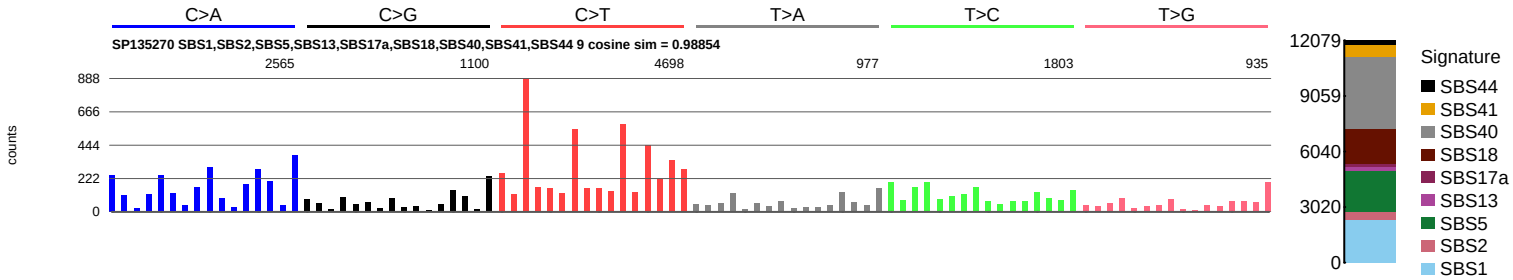
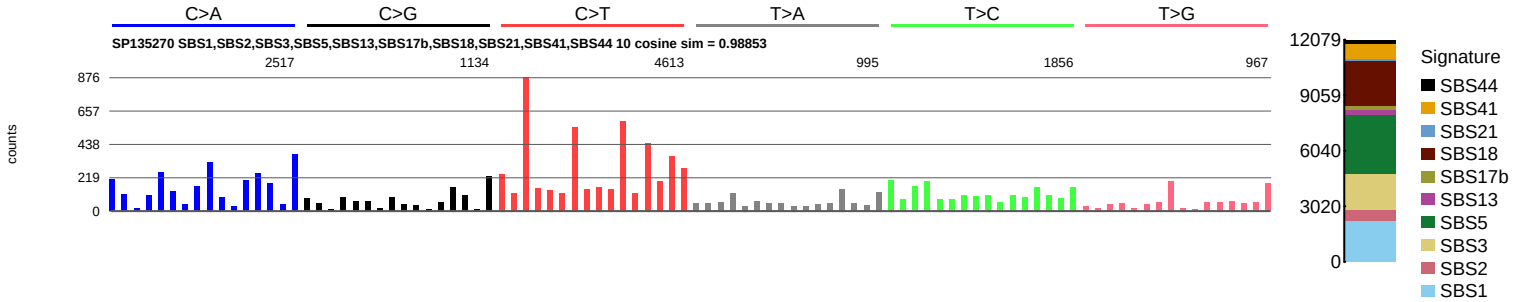
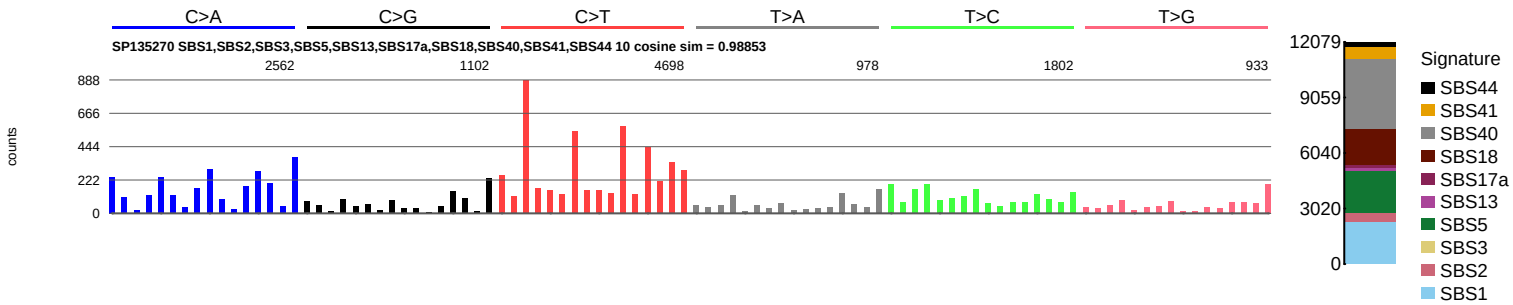
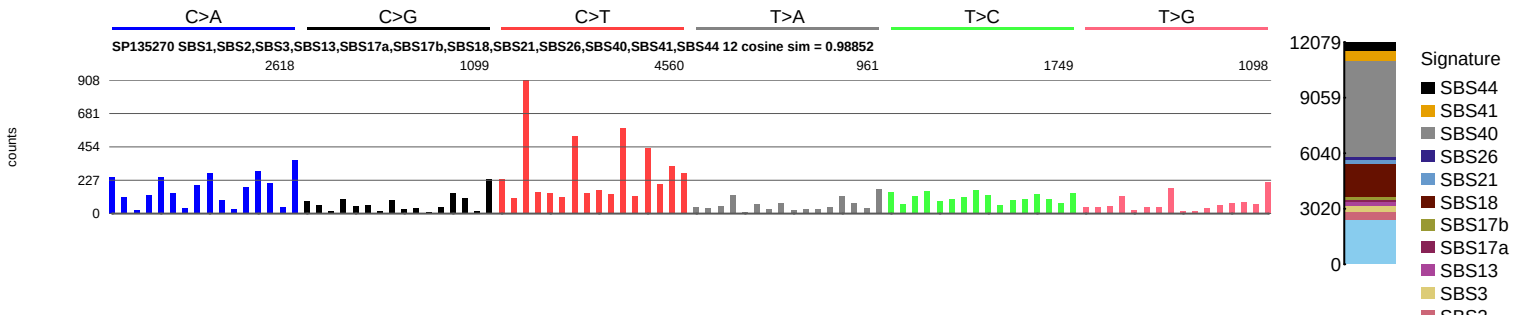
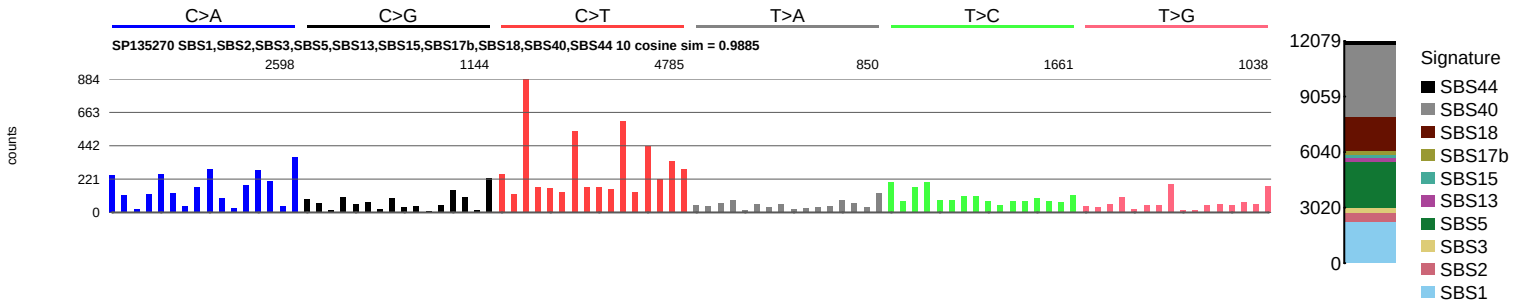
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98855
2510
1146
4651
1001
1791
979
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

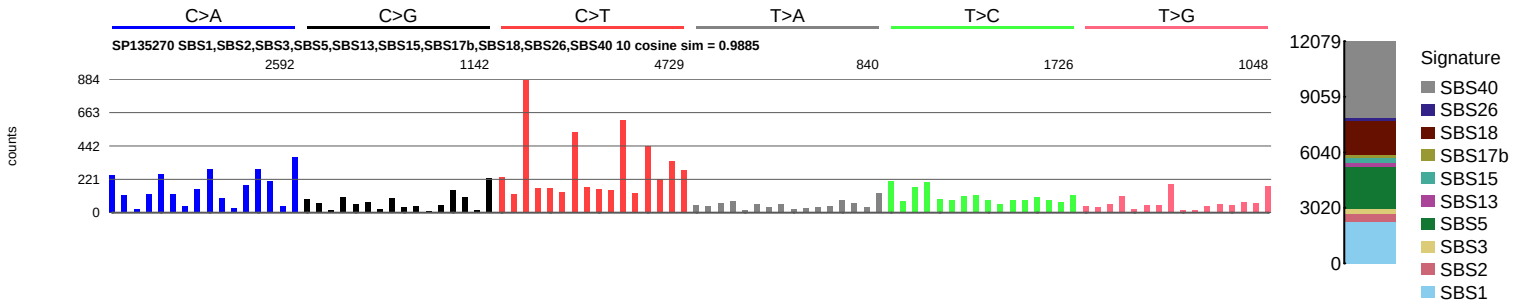

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98854
2565
1100
4698
977
1803
935
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98853
2517
1134
4613
995
1856
967
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98853
2562
1102
4698
978
1802
933
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98852
2618
1099
4560
961
1749
1098
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.9885
2598
1144
4785
850
1661
1038
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.9885
2592
1142
4729
840
1726
1048
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
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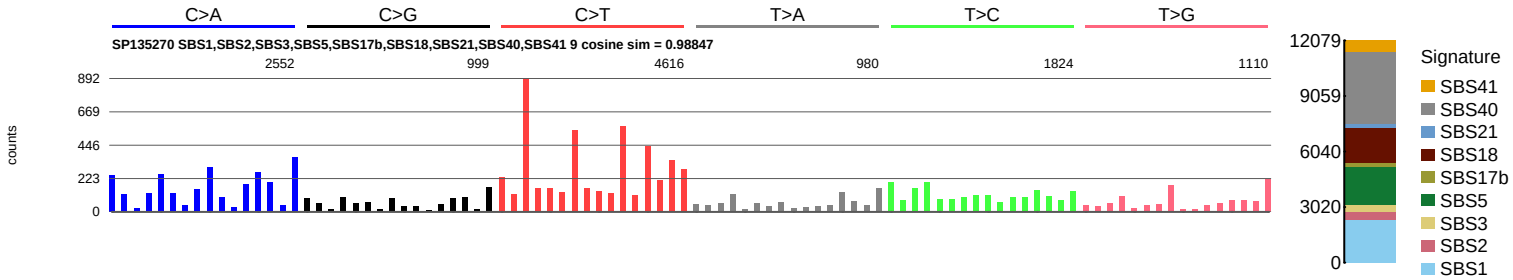
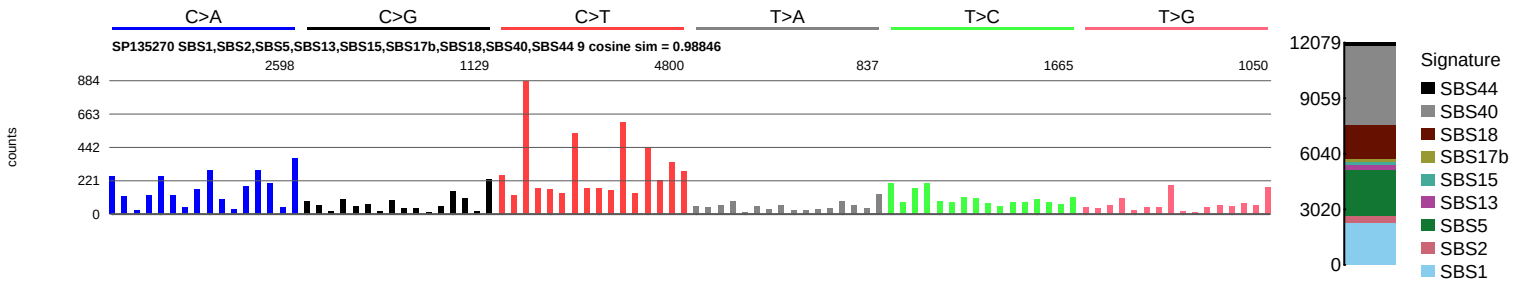
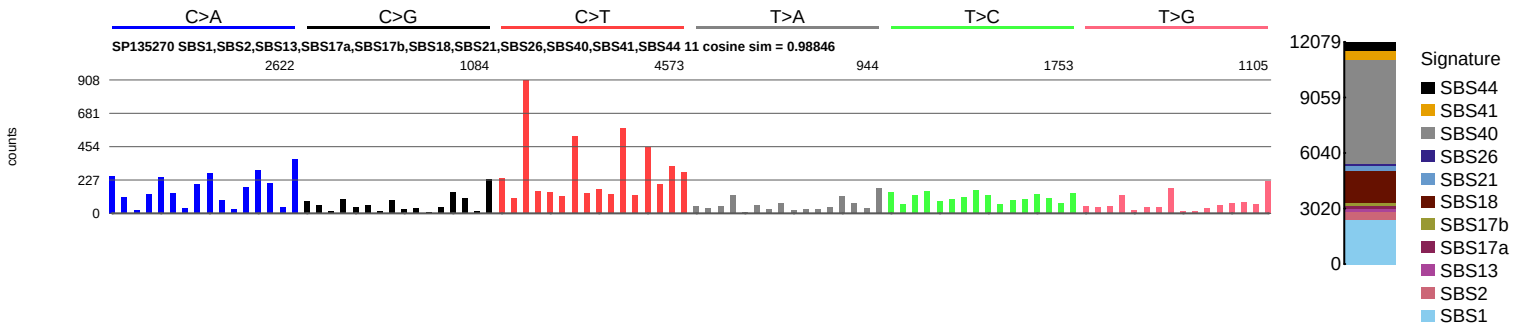
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98847
2552
999
4616
980
1824
1110
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98846
2598
1129
4800
837
1665
1050
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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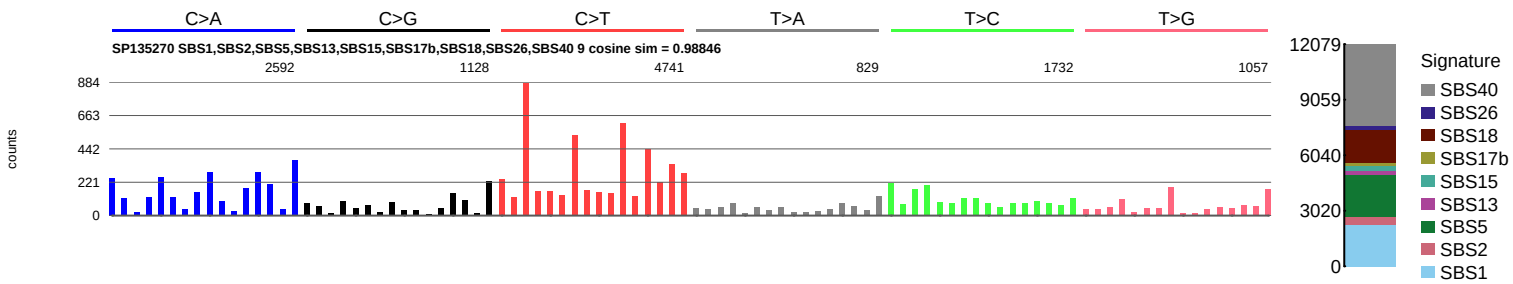
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98846
2622
1084
4573
944
1753
1105
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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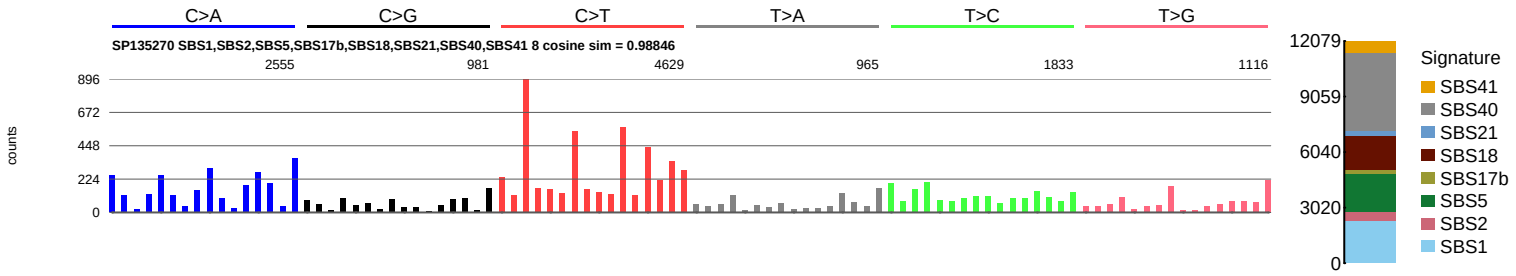
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98846
2592
1128
4741
829
1732
1057
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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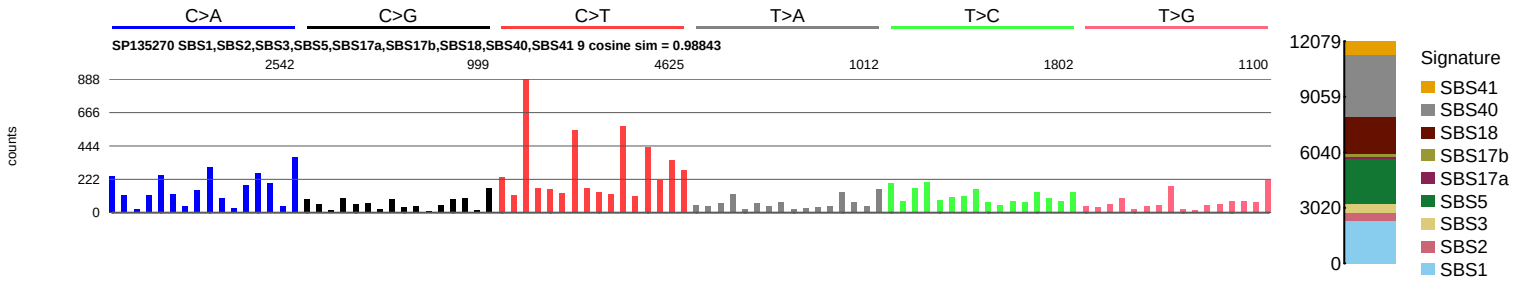
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98846
2555
981
4629
965
1833
1116
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1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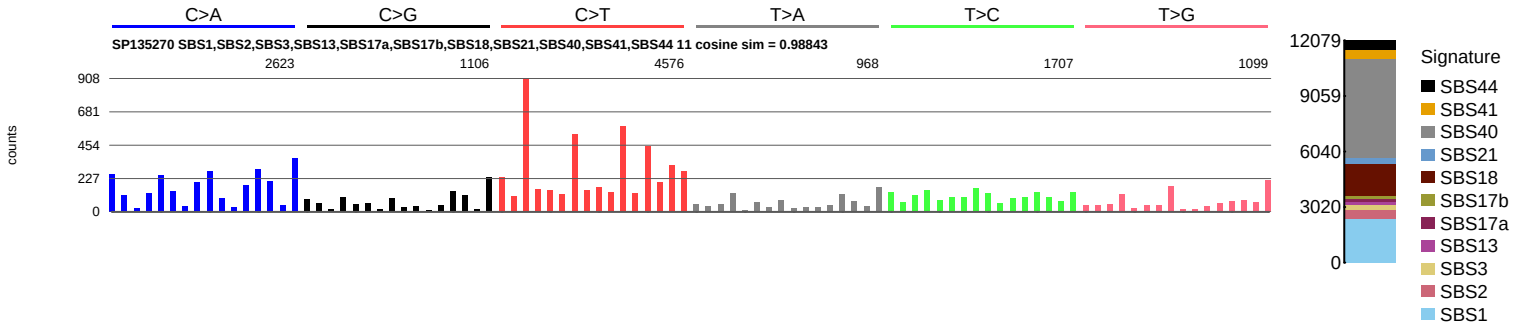
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98843
2542
999
4625
1012
1802
1100
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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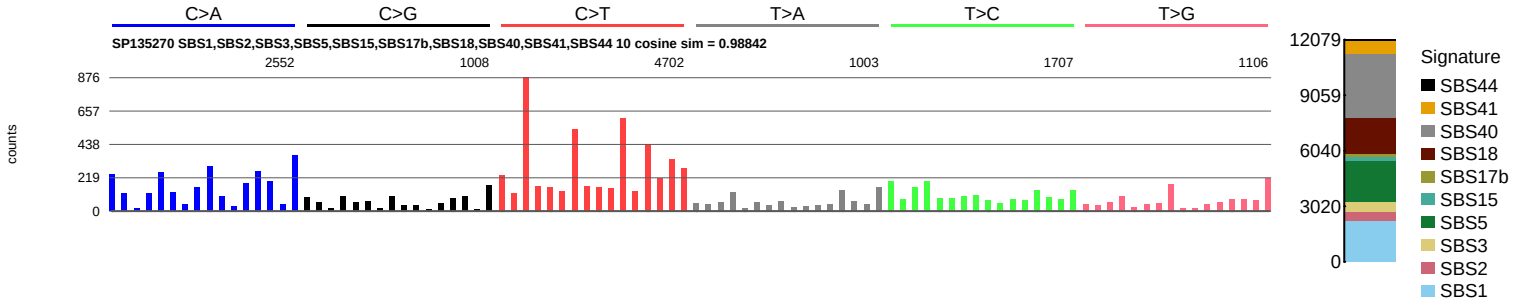
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98843
2623
1106
4576
968
1707
1099
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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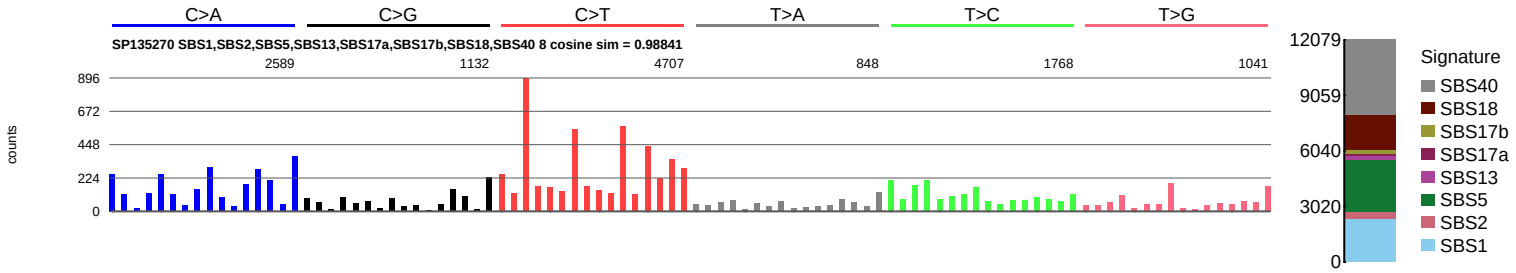
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98842
2552
1008
4702
1003
1707
1106
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98841
2589
1132
4707
848
1768
1041
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS40
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SBS17b
SBS17a
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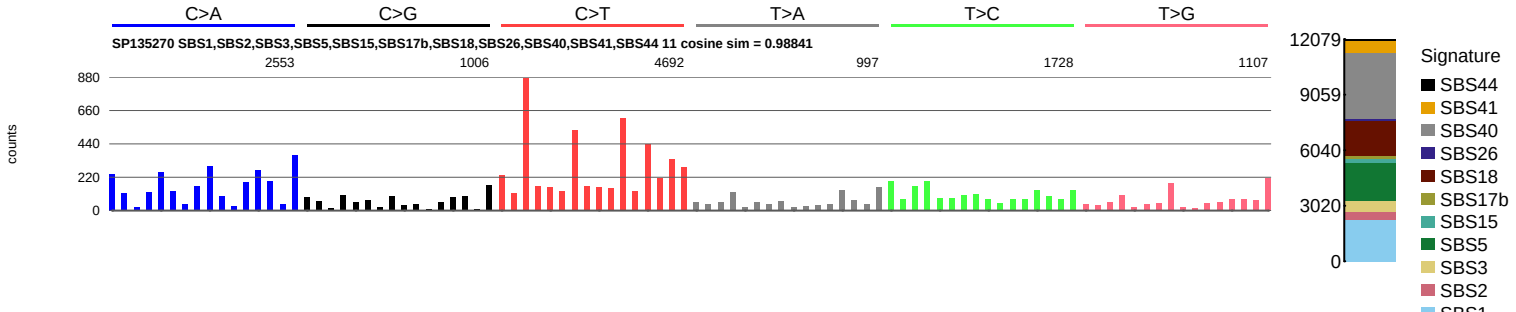
SBS5
SBS2
SBS1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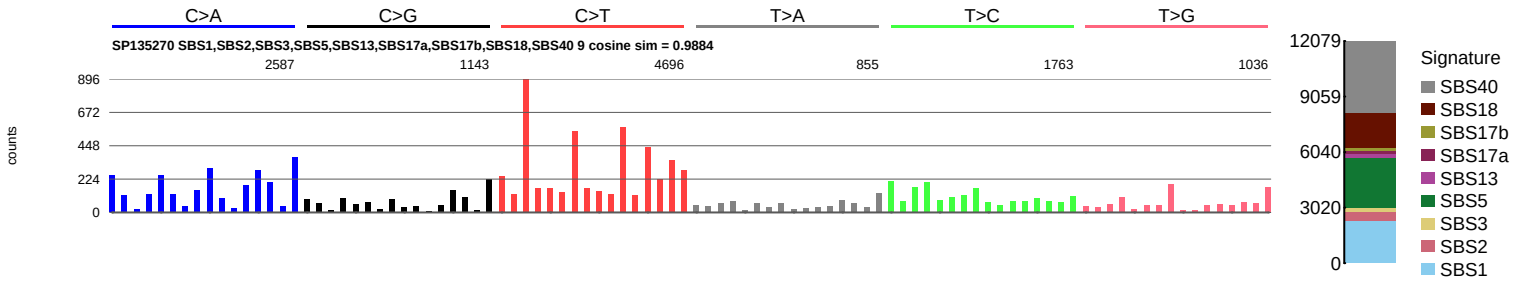
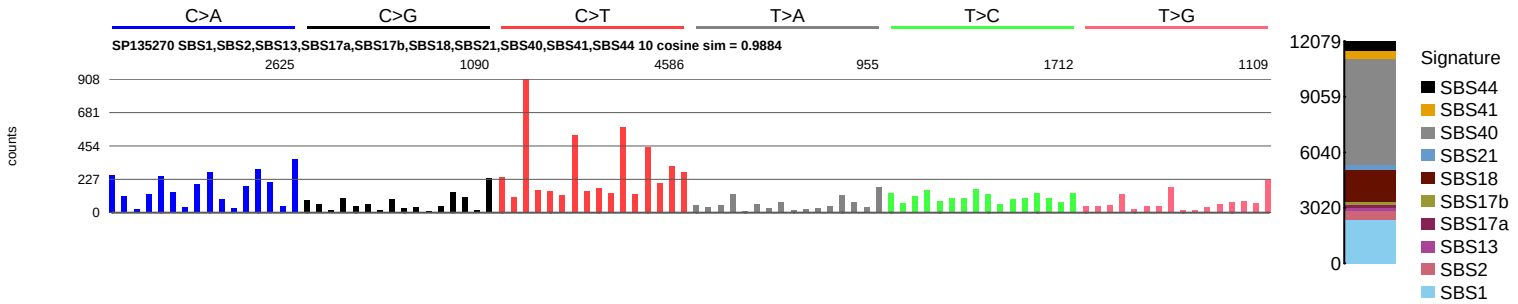
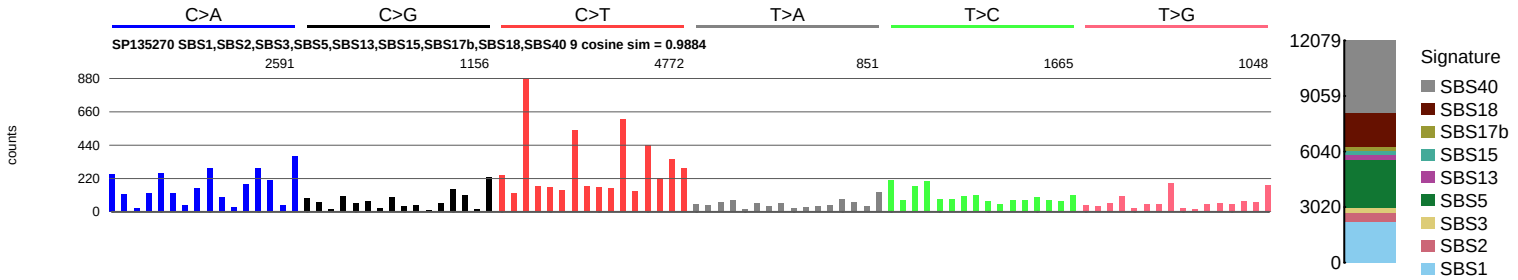
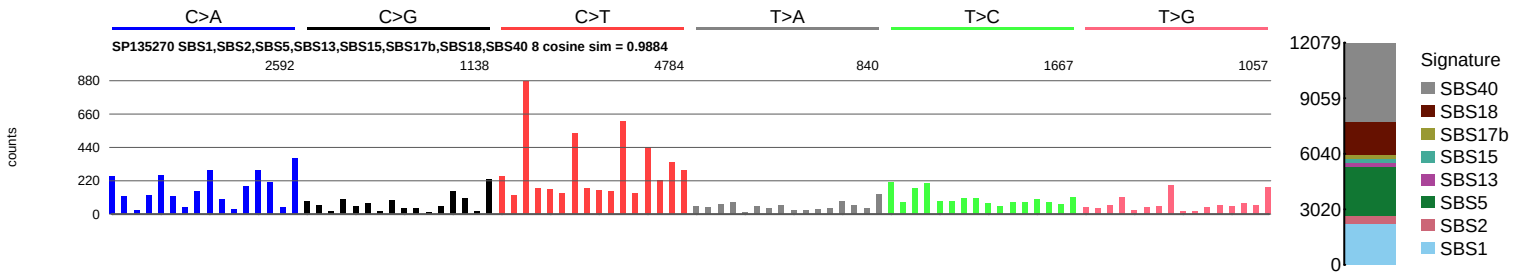
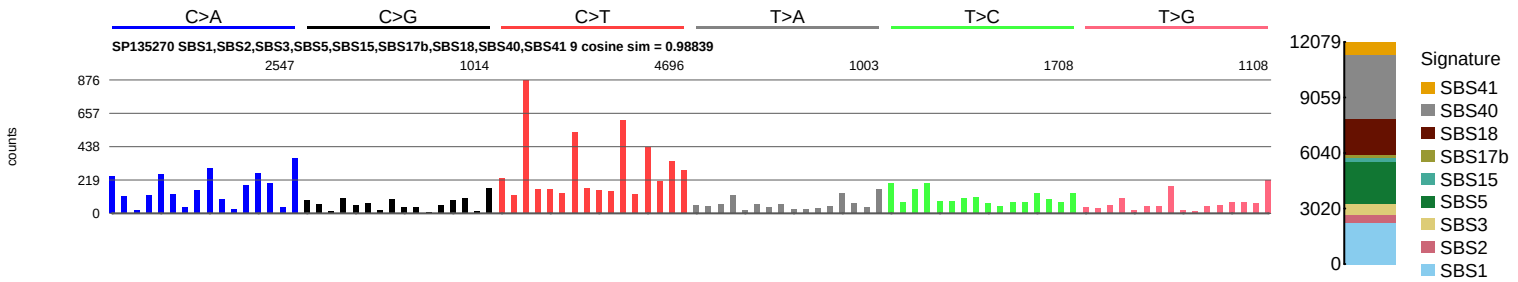
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98841
2553
1006
4692
997
1728
1107
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.9884
2587
1143
4696
855
1763
1036
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9884
2625
1090
4586
955
1712
1109
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 9 cosine sim = 0.9884
2591
1156
4772
851
1665
1048
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.9884
2592
1138
4784
840
1667
1057
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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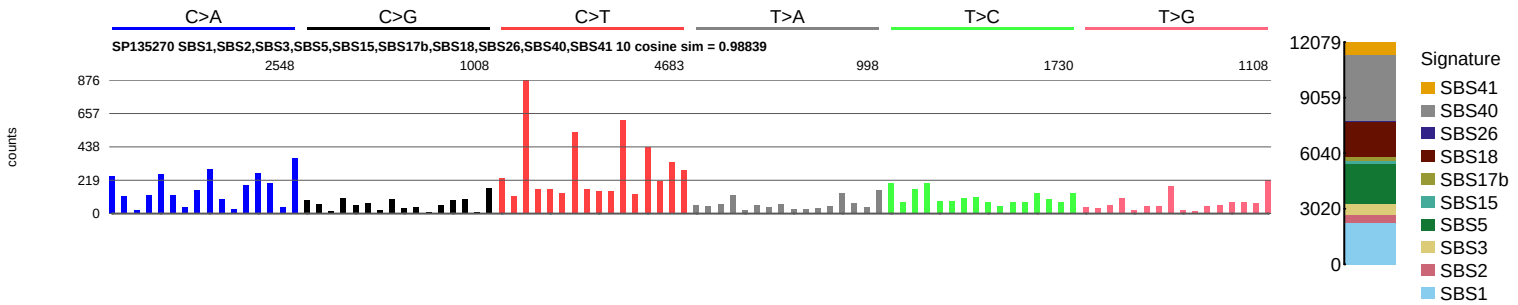
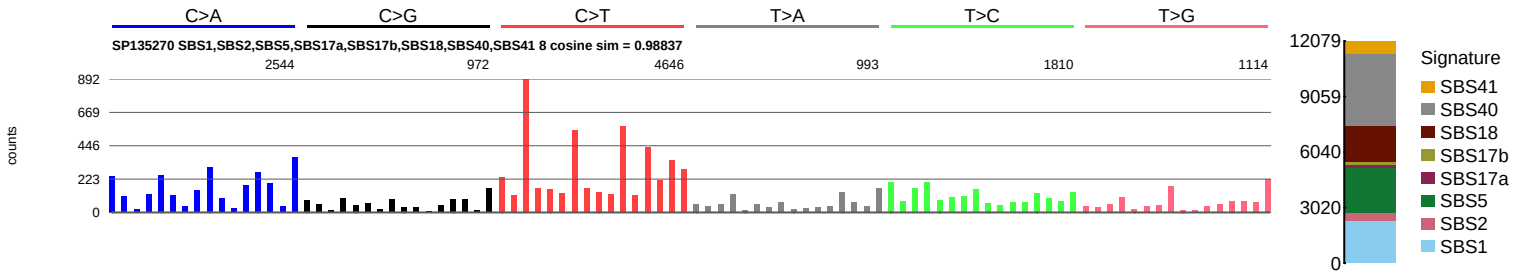
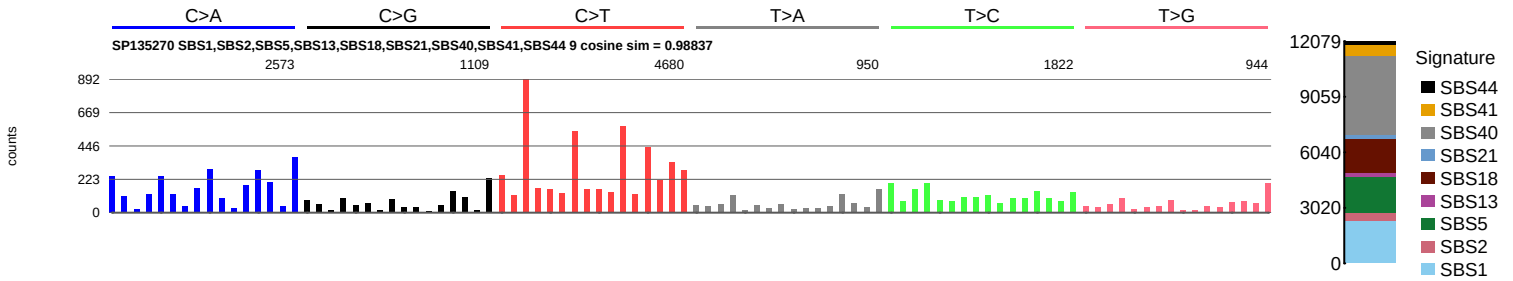
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98839
2547
1014
4696
1003
1708
1108
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98839
2548
1008
4683
998
1730
1108
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98837
2544
972
4646
993
1810
1114
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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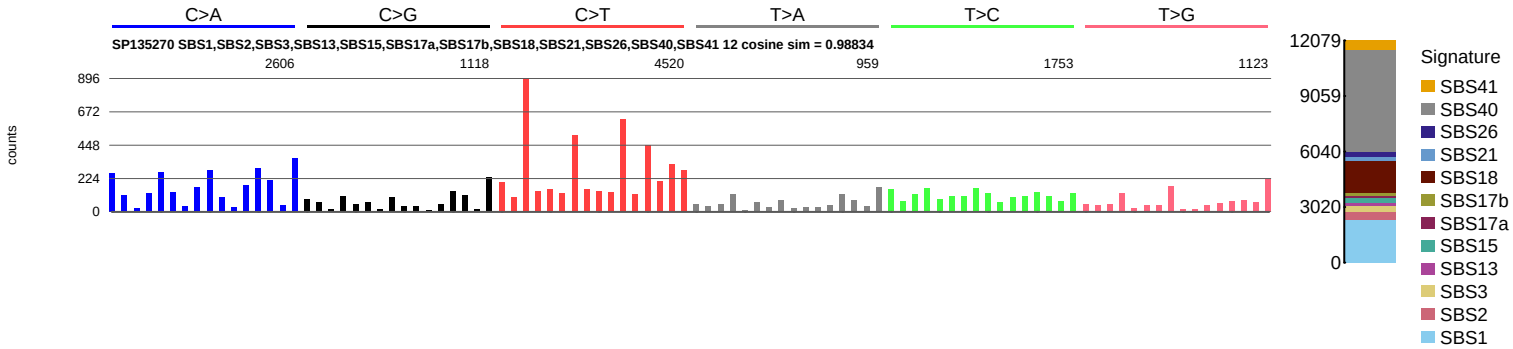
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98837
2573
1109
4680
950
1822
944
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1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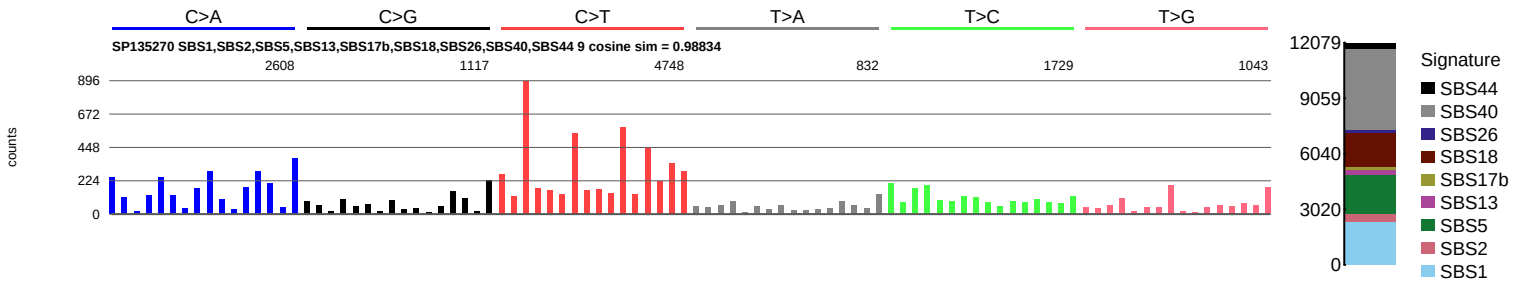
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 12 cosine sim = 0.98834
2606
1118
4520
959
1753
1123
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98834
2608
1117
4748
832
1729
1043
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
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SBS40
SBS26
SBS18
SBS17b
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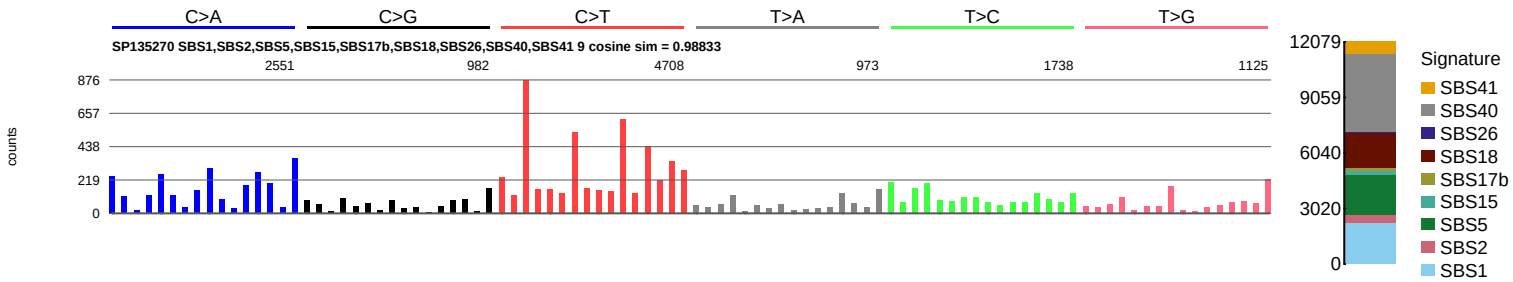
SBS5
SBS2
SBS1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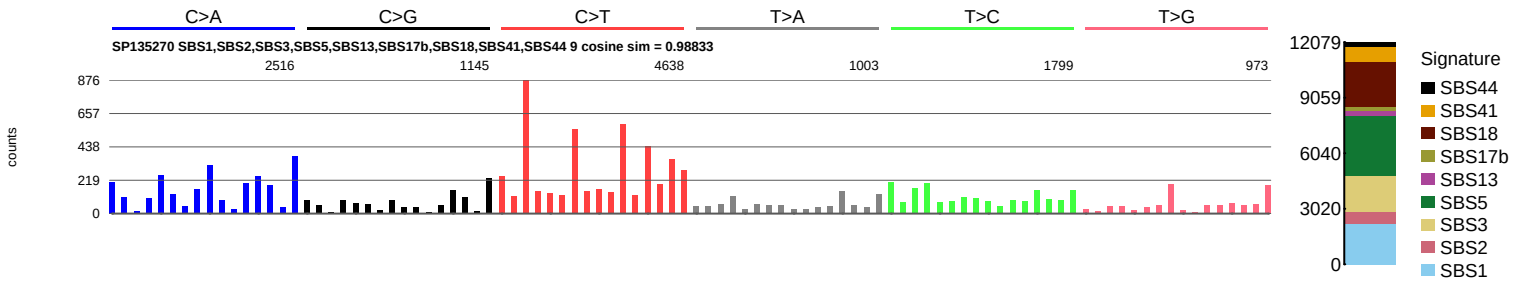
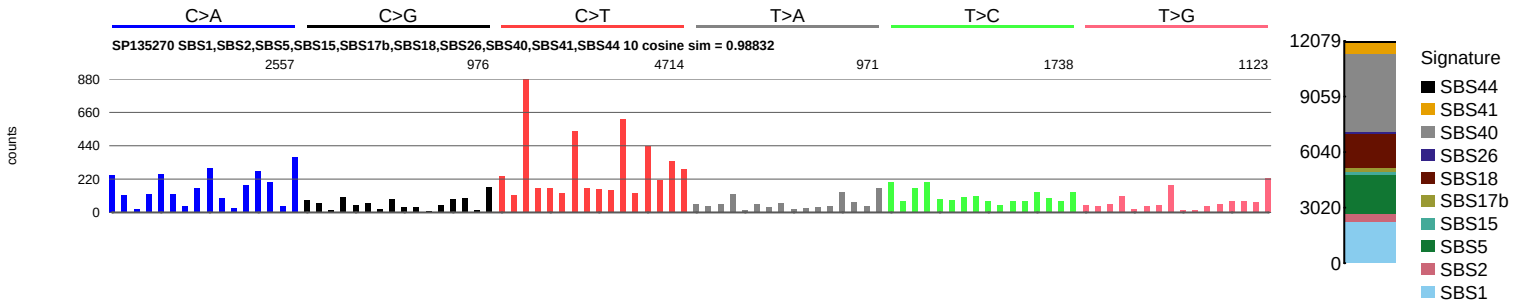
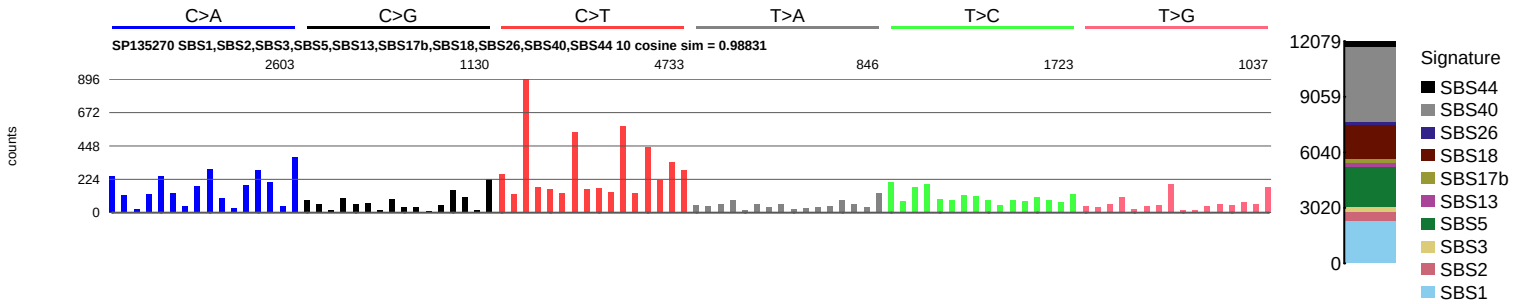
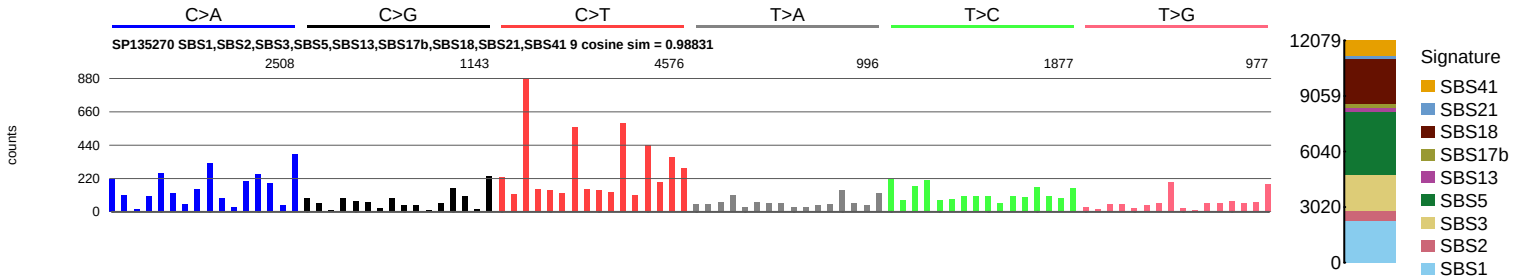
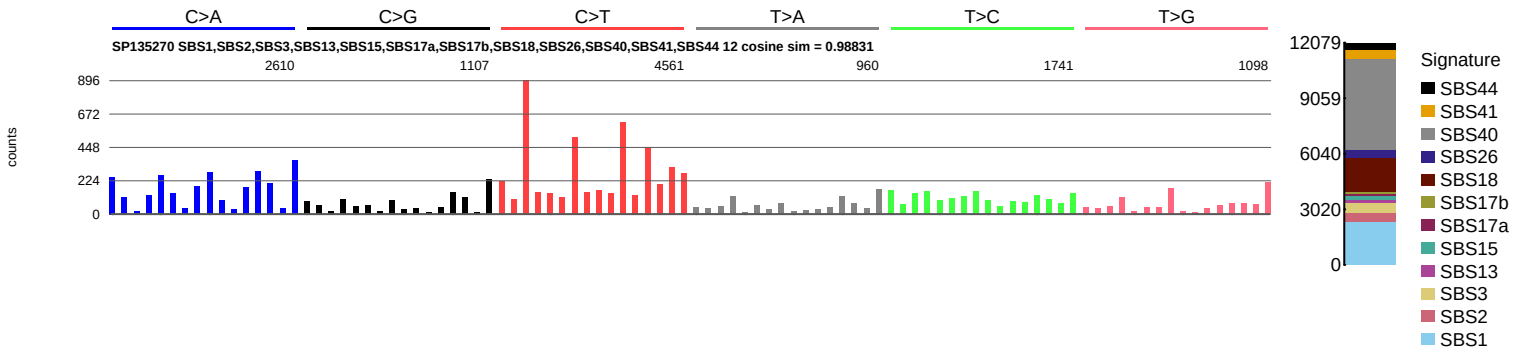
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98833
2551
982
4708
973
1738
1125
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

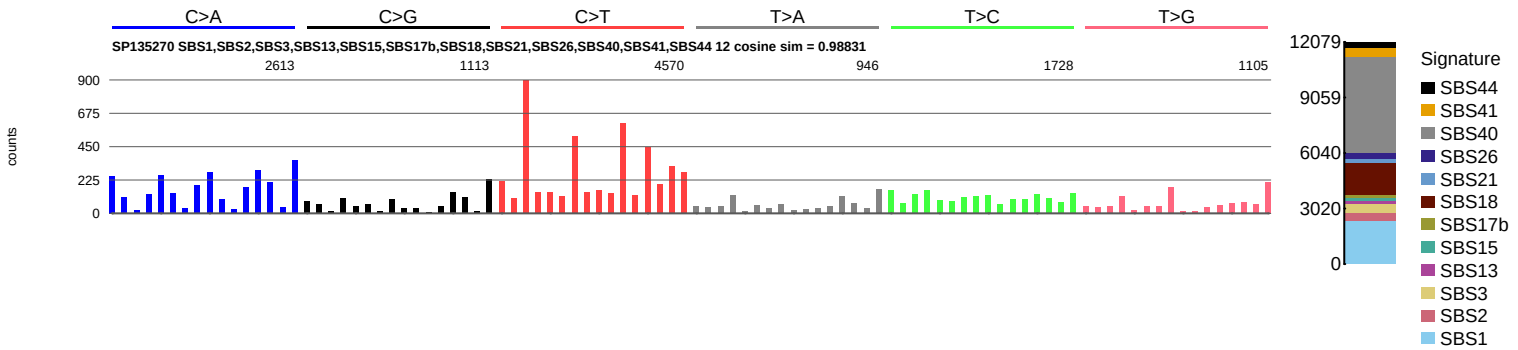

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98833
2516
1145
4638
1003
1799
973
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98832
2557
976
4714
971
1738
1123
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98831
2603
1130
4733
846
1723
1037
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98831
2508
1143
4576
996
1877
977
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98831
2610
1107
4561
960
1741
1098
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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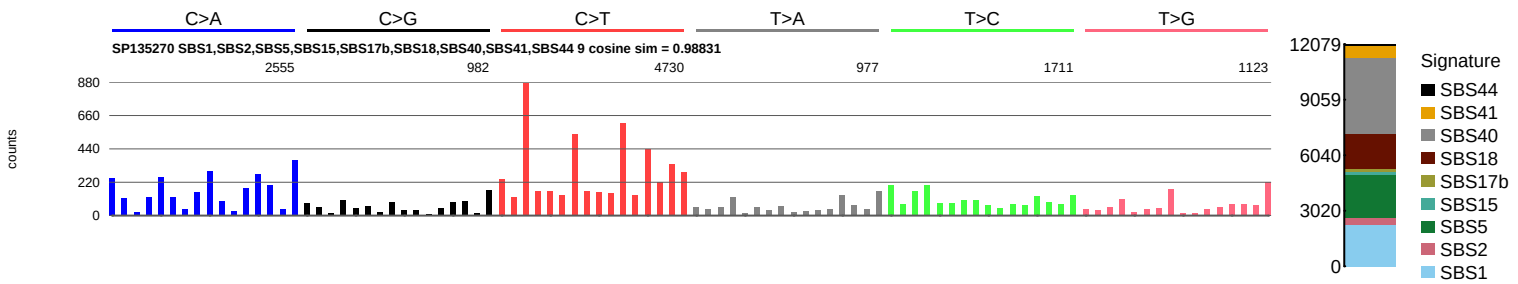
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98831
2613
1113
4570
946
1728
1105
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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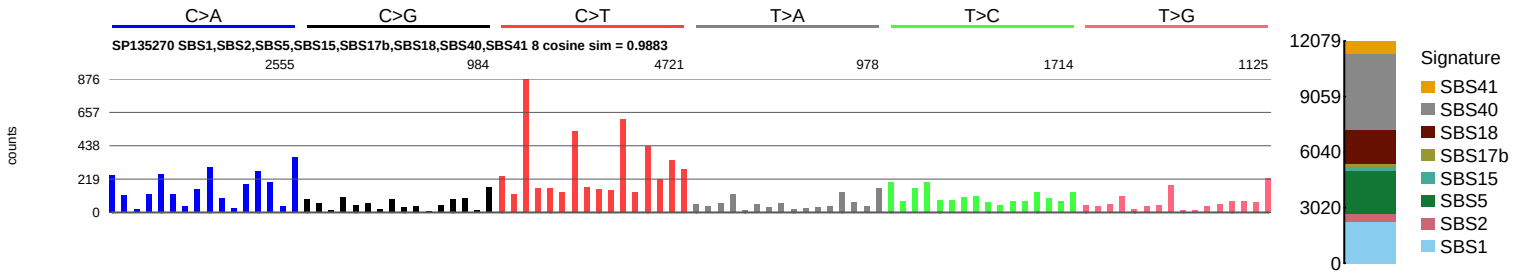
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98831
2555
982
4730
977
1711
1123
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9883
2555
984
4721
978
1714
1125
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
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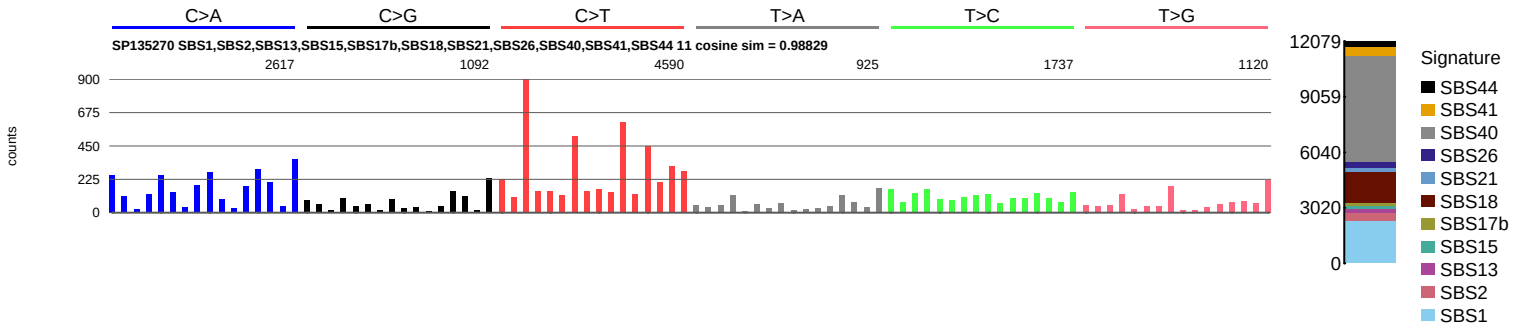
SBS2
SBS1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SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98829
2617
1092
4590
925
1737
1120
counts
900
675
450
225
0
12079
9059
6040
3020
0
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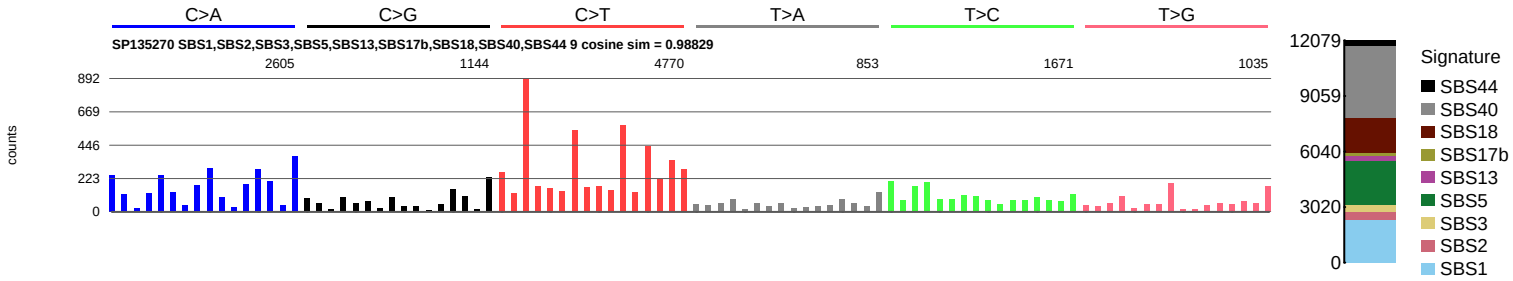
SBS2
SBS1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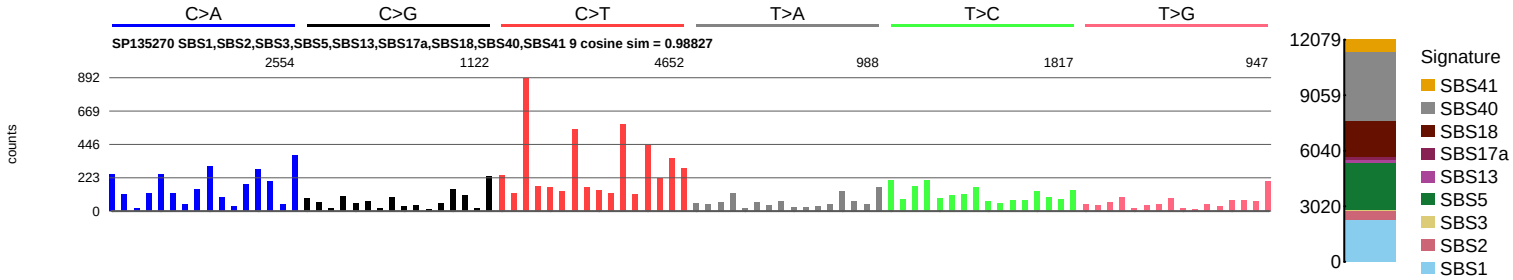
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98829
2605
1144
4770
853
1671
1035
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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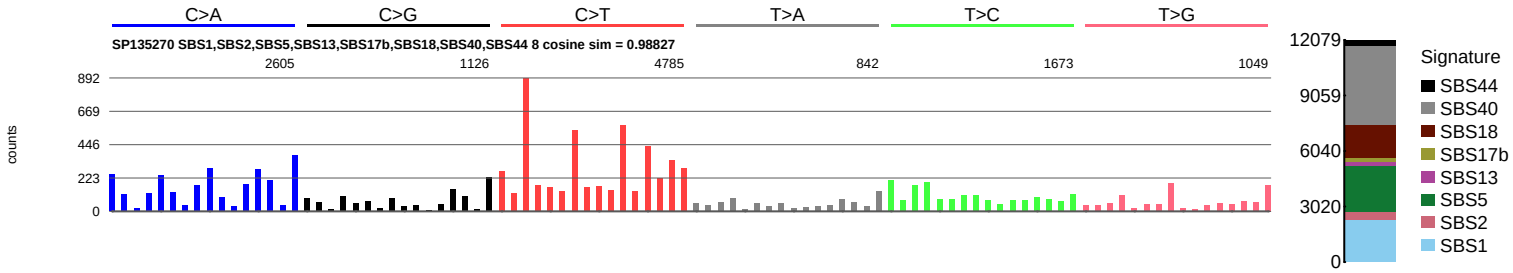
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98827
2554
1122
4652
988
1817
947
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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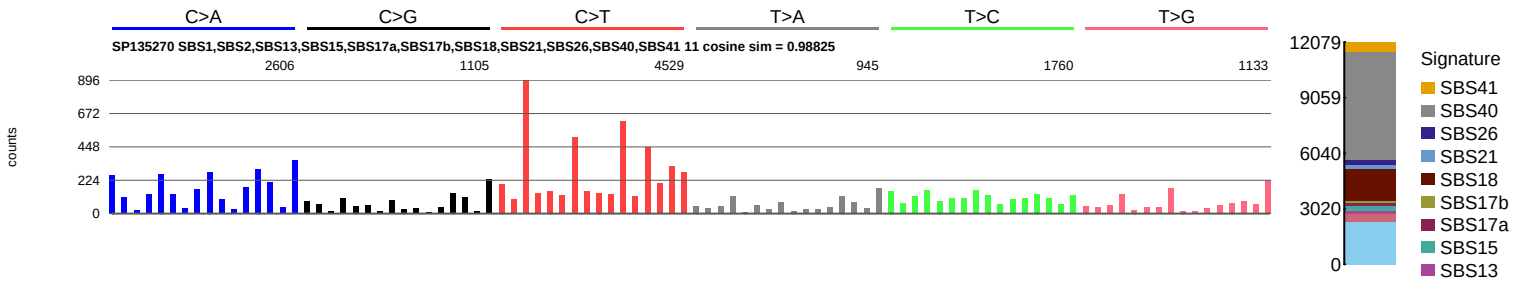
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98827
2605
1126
4785
842
1673
1049
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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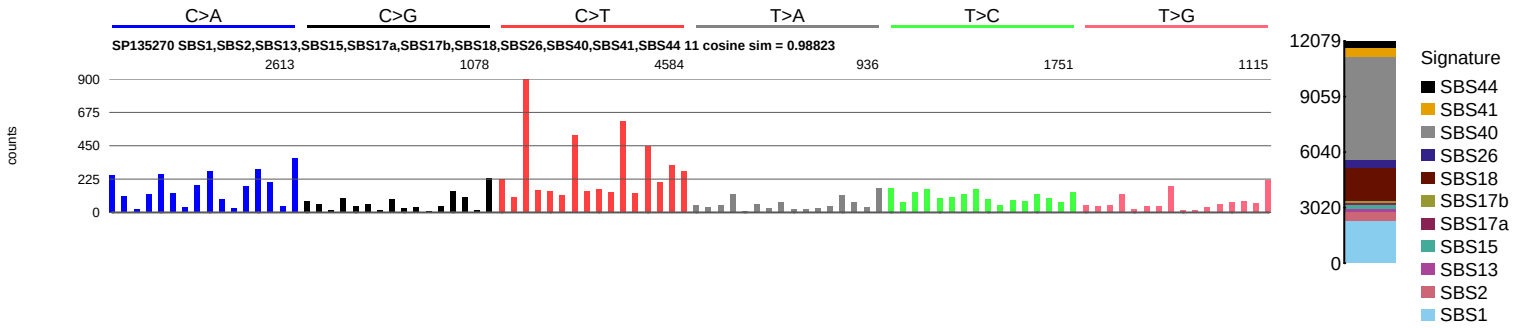
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98825
2606
1105
4529
945
1760
1133
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2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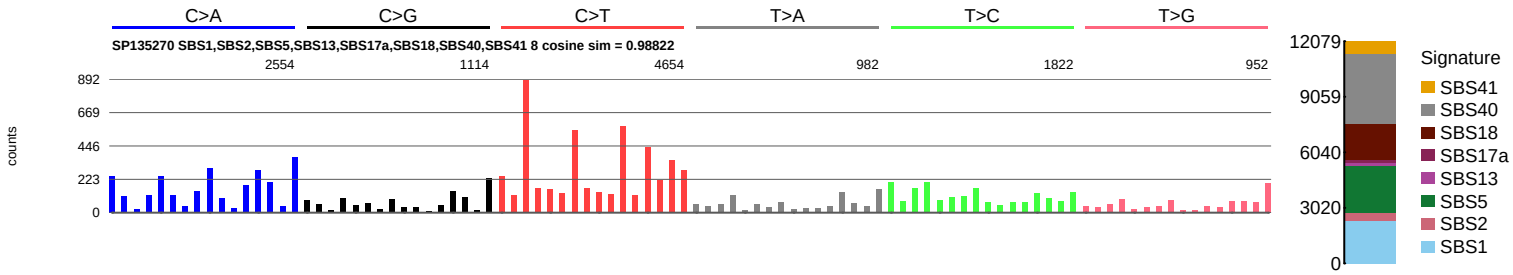
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98823
2613
1078
4584
936
1751
1115
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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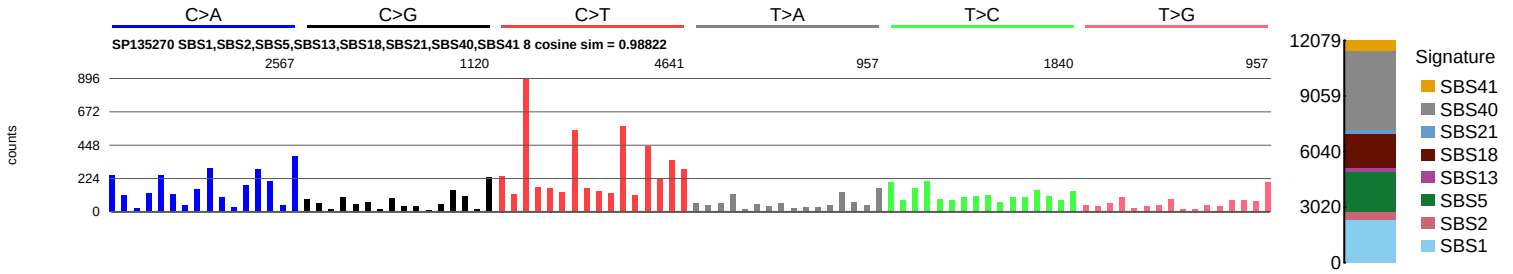
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98822
2554
1114
4654
982
1822
952
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1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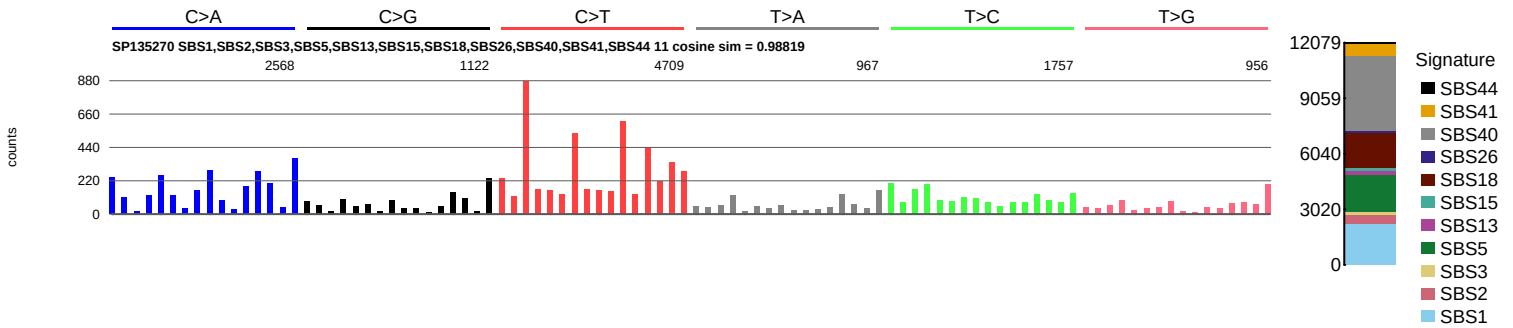
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98822
2567
1120
4641
957
1840
957
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1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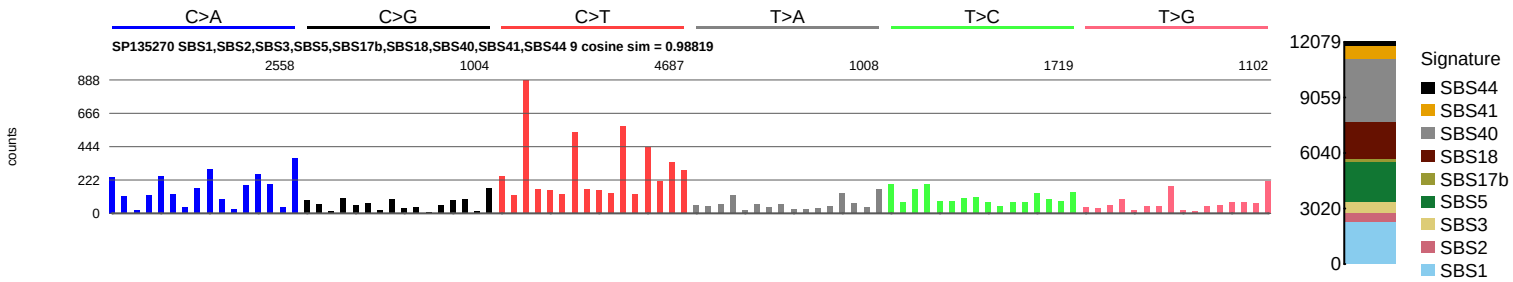
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98819
2568
1122
4709
967
1757
956
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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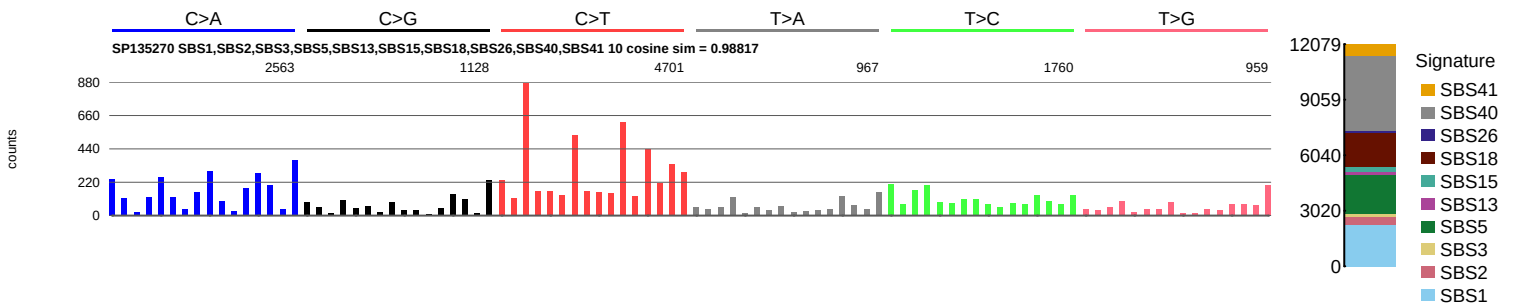
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98819
2558
1004
4687
1008
1719
1102
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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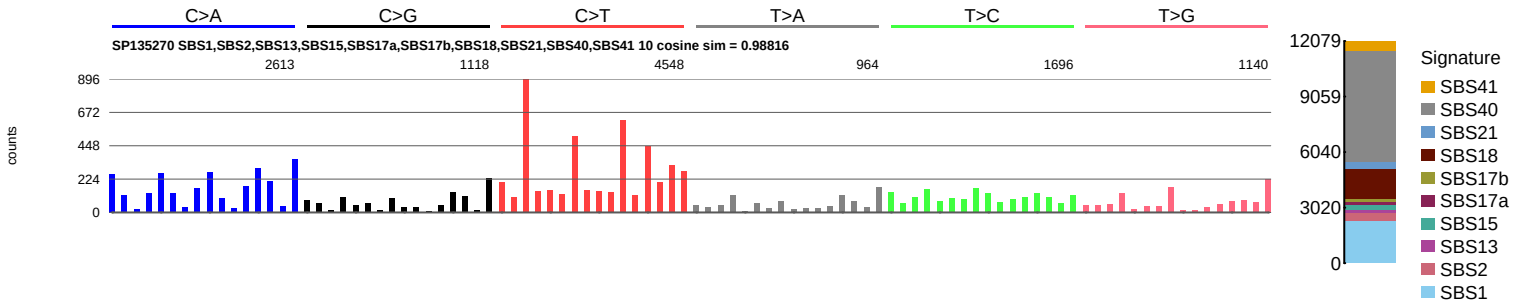
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98817
2563
1128
4701
967
1760
959
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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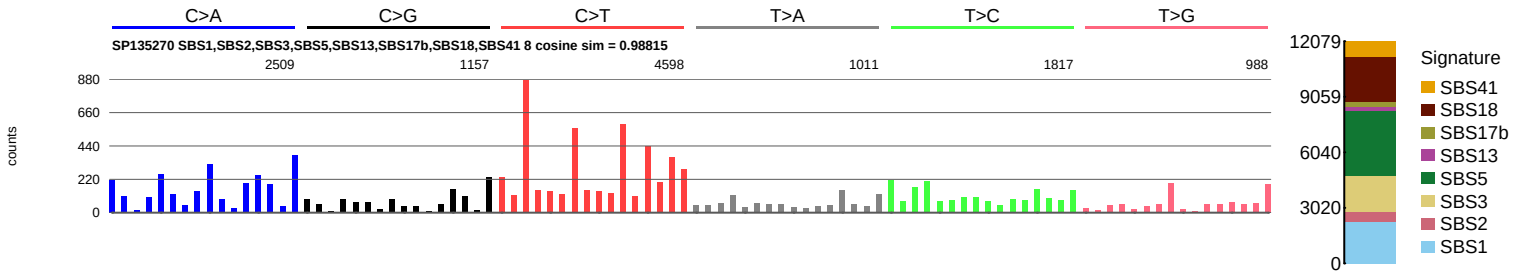
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98816
2613
1118
4548
964
1696
1140
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 8 cosine sim = 0.98815
2509
1157
4598
1011
1817
988
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
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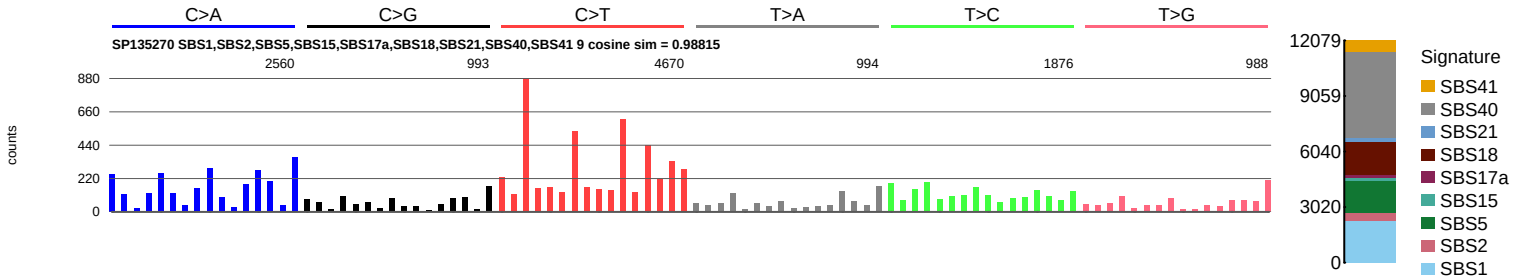
SBS2
SBS1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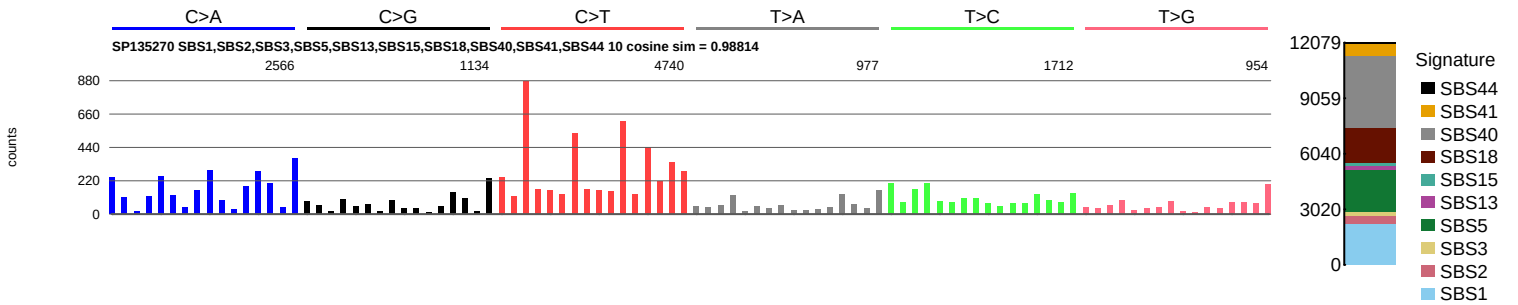
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98815
2560
993
4670
994
1876
988
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98814
2566
1134
4740
977
1712
954
counts
880
660
440
220
0
12079
9059
6040
3020
0
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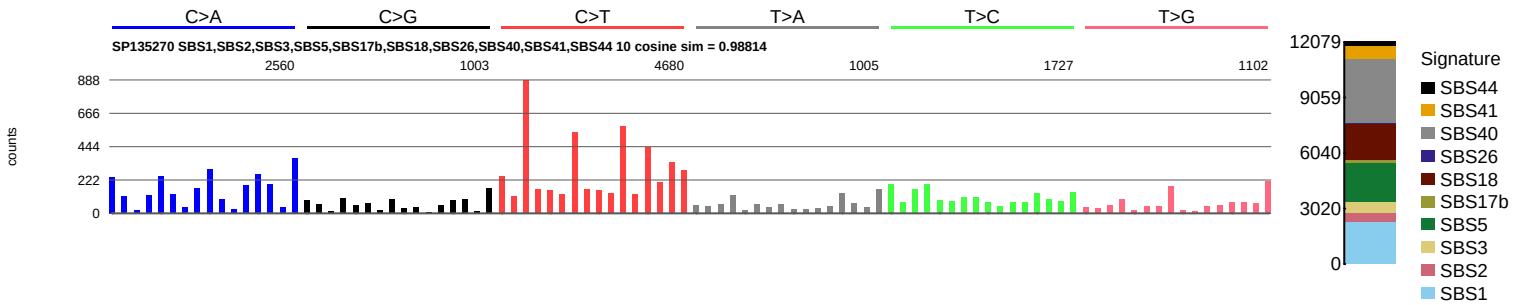
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98814
2560
1003
4680
1005
1727
1102
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
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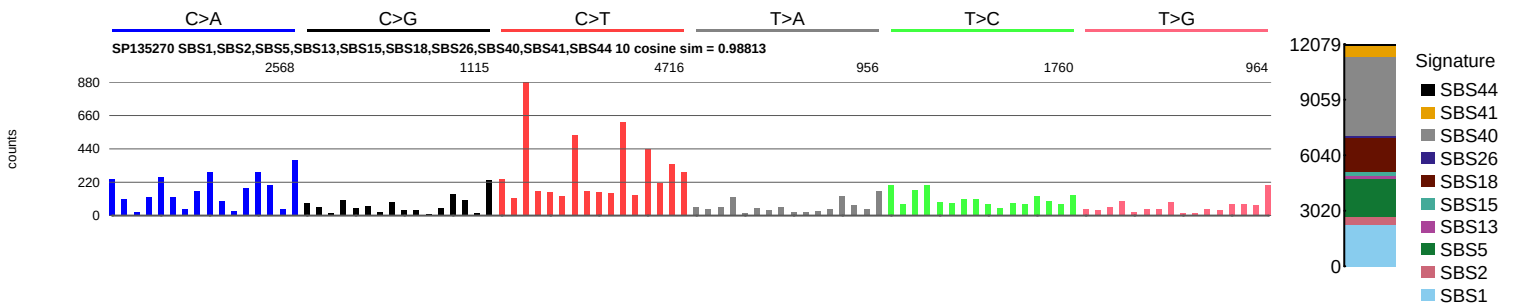
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98813
2568
1115
4716
956
1760
964
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
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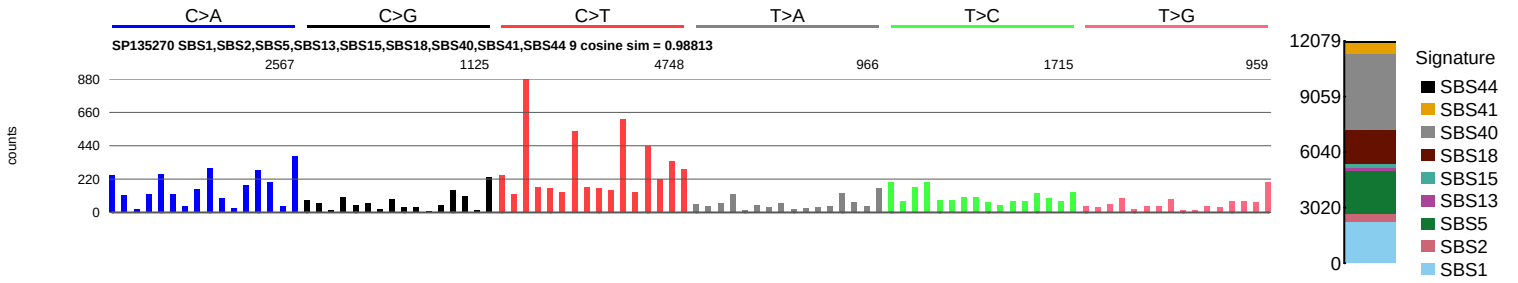
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98813
2567
1125
4748
966
1715
959
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
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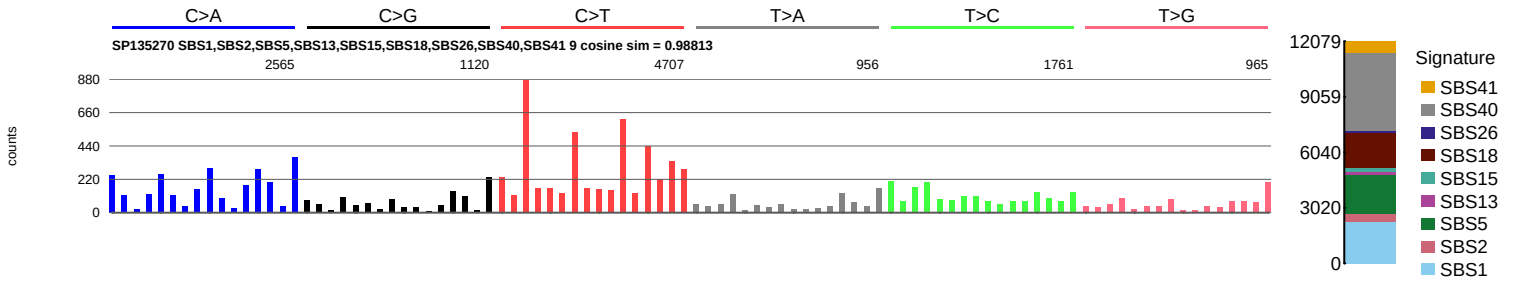
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98813
2565
1120
4707
956
1761
965
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
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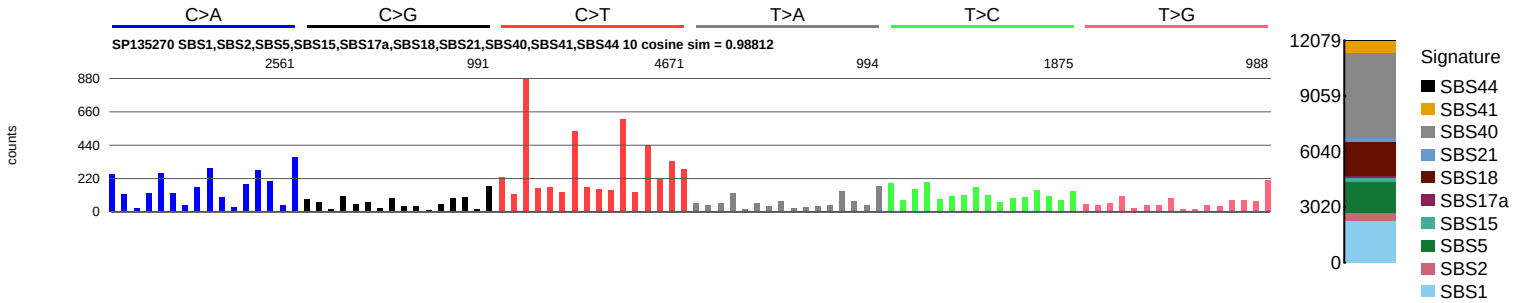
SBS2
SBS1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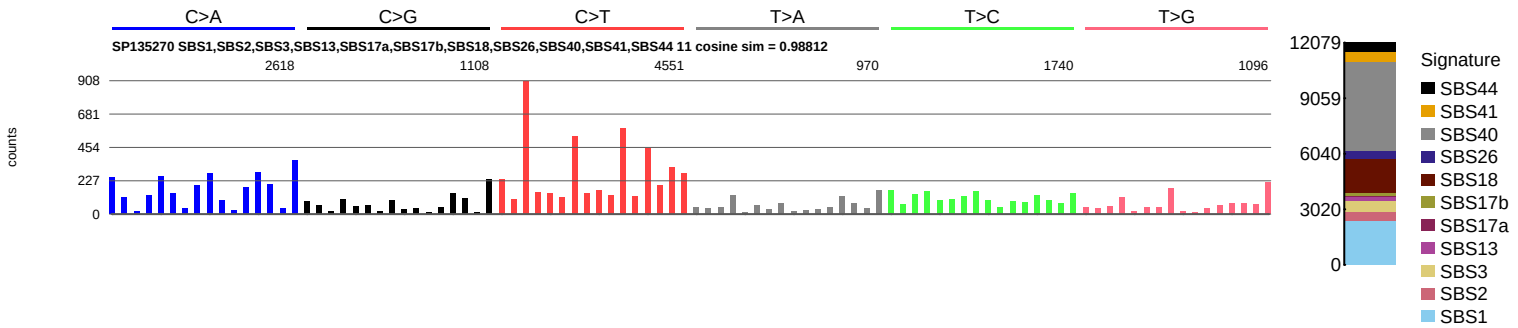
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98812
2561
991
4671
994
1875
988
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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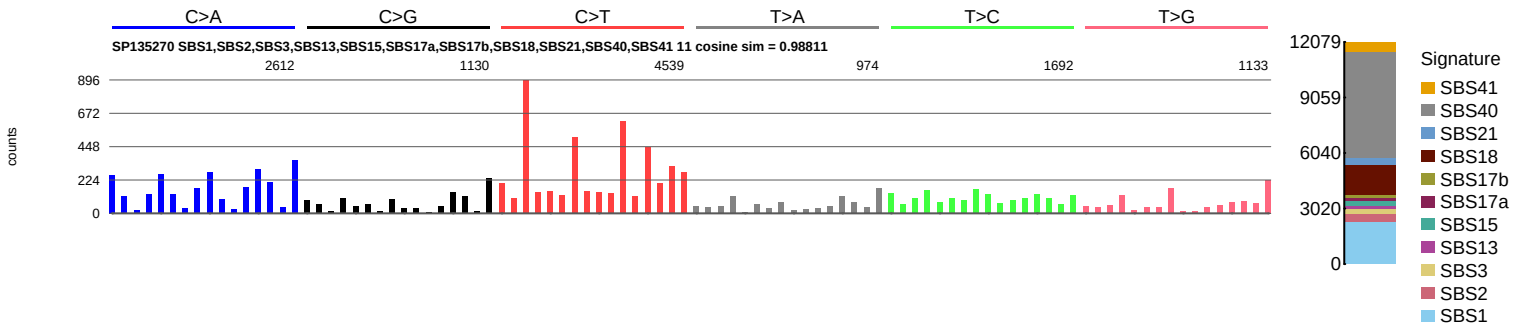
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98812
2618
1108
4551
970
1740
1096
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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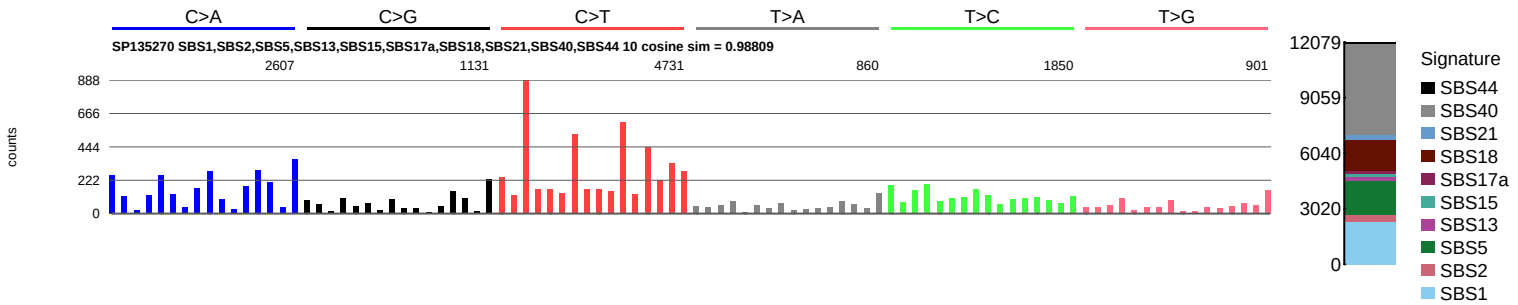
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98811
2612
1130
4539
974
1692
1133
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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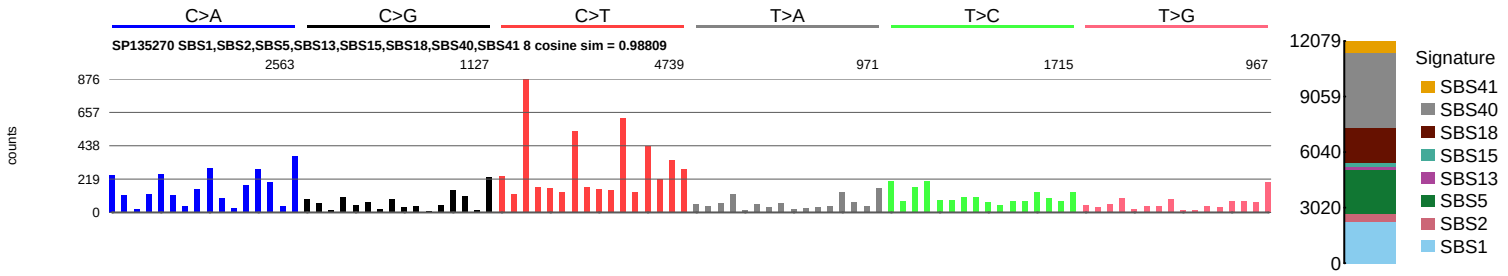
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98809
2607
1131
4731
860
1850
901
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41 8 cosine sim = 0.98809
2563
1127
4739
971
1715
967
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
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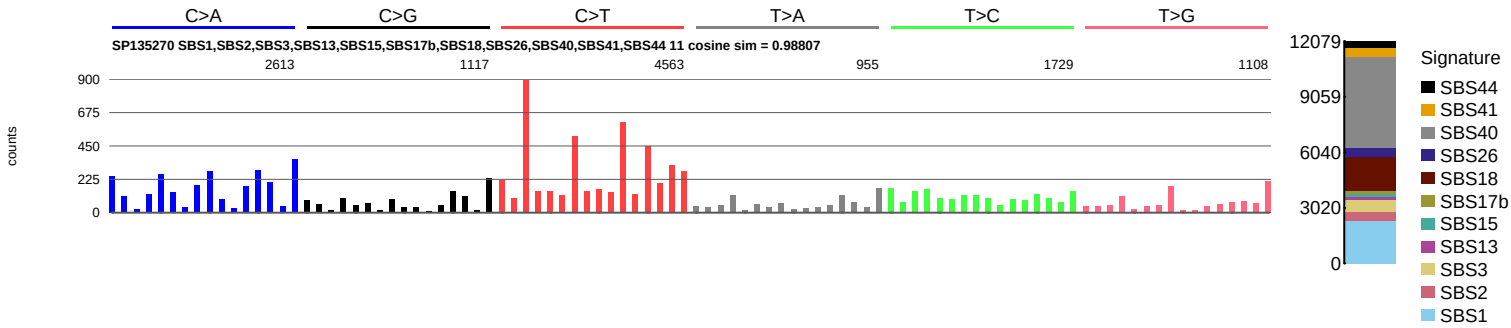
SBS2
SBS1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T>C
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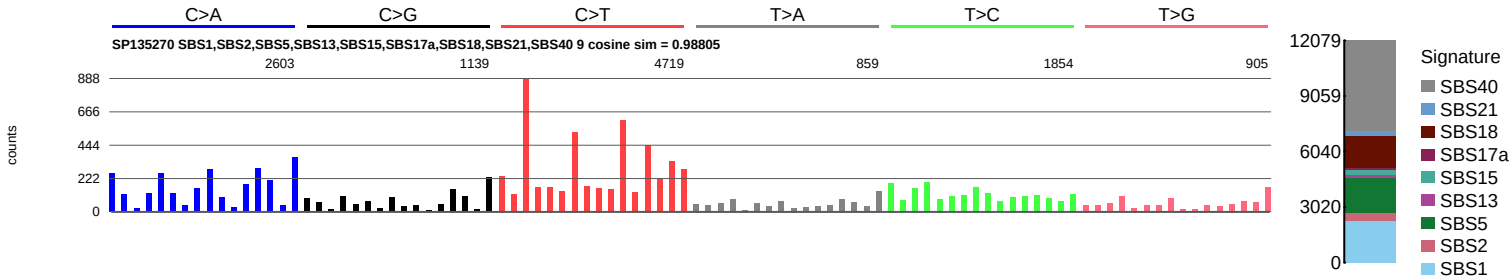
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98807
2613
1117
4563
955
1729
1108
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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C>T
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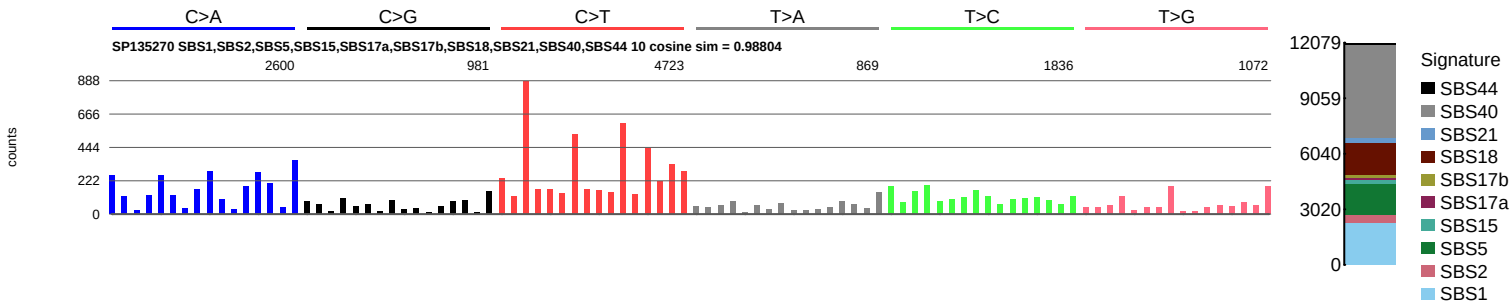
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.98805
2603
1139
4719
859
1854
905
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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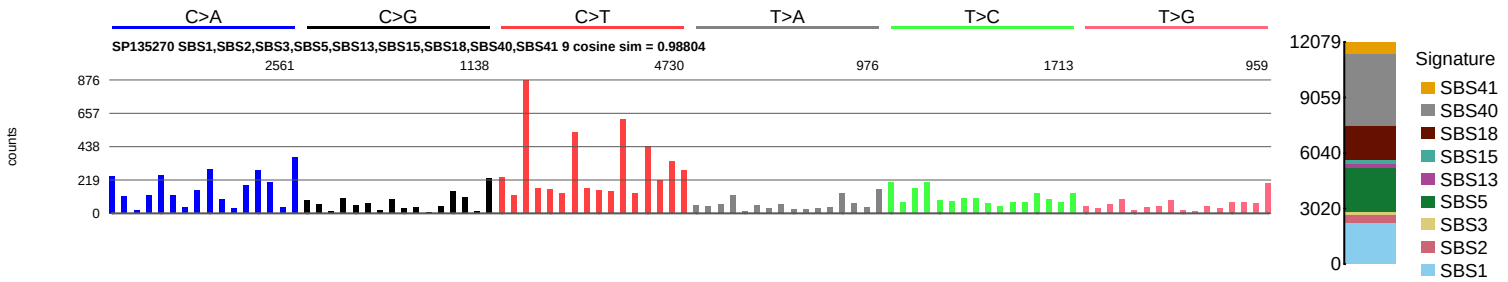
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98804
2600
981
4723
869
1836
1072
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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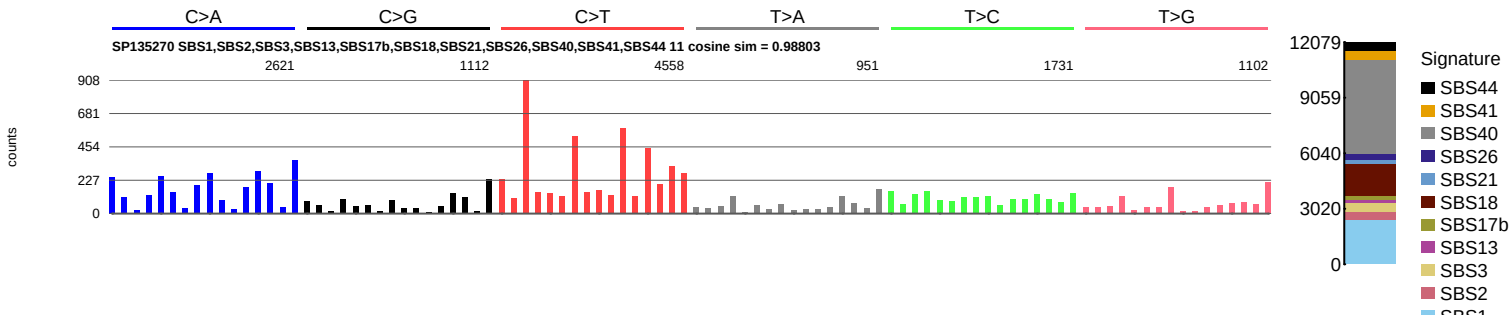
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41 9 cosine sim = 0.98804
2561
1138
4730
976
1713
959
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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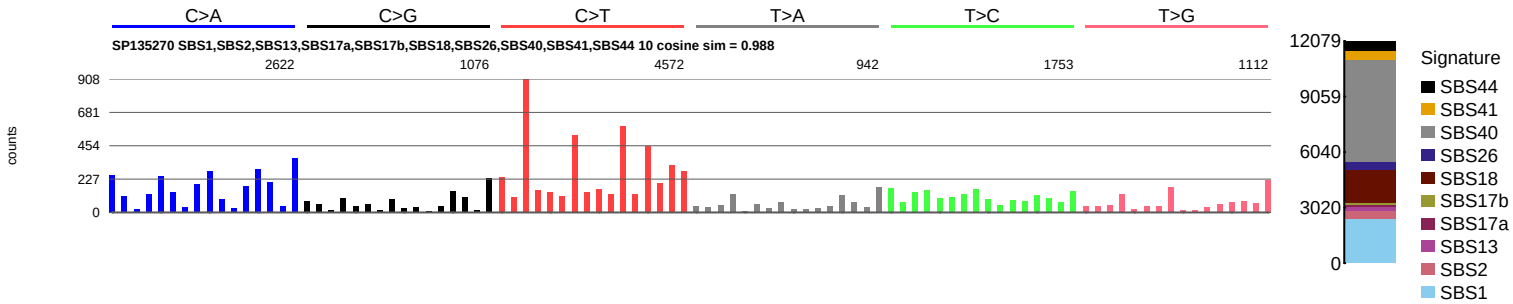
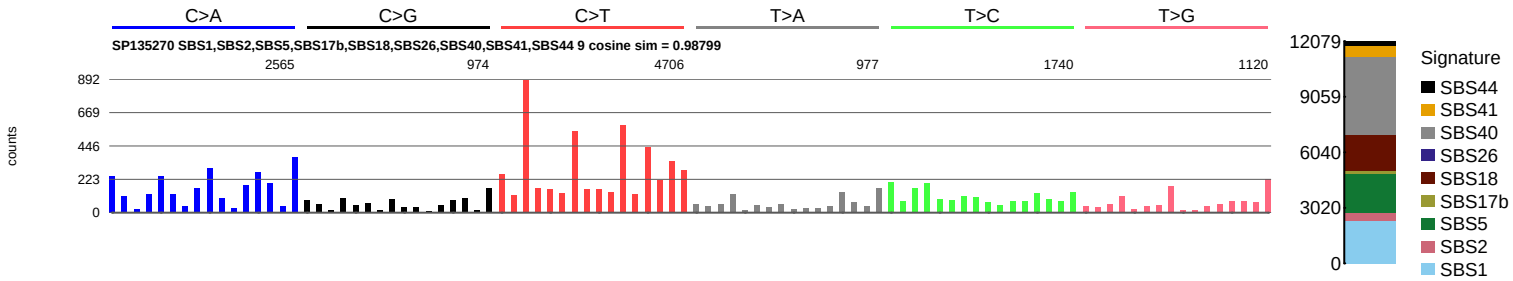
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98803
2621
1112
4558
951
1731
1102
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.988
2622
1076
4572
942
1753
1112
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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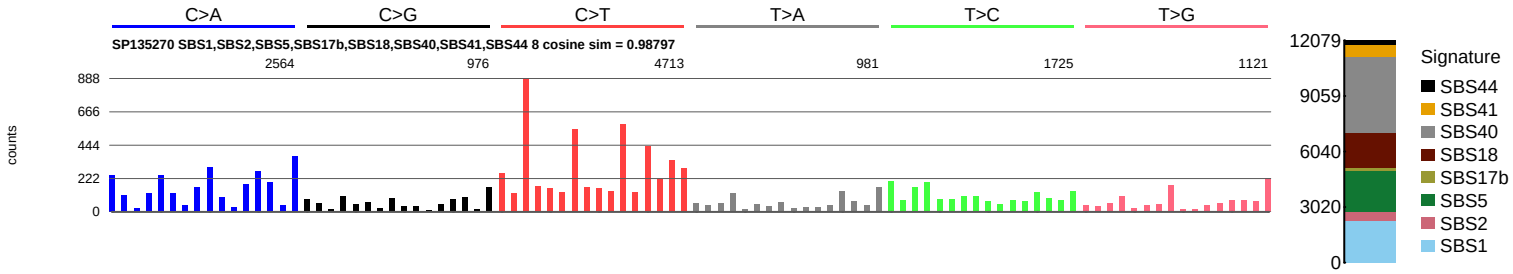
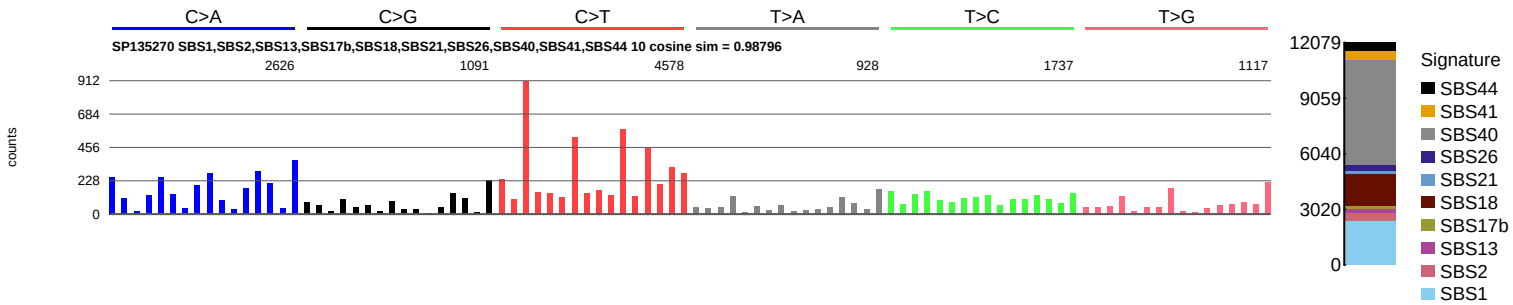
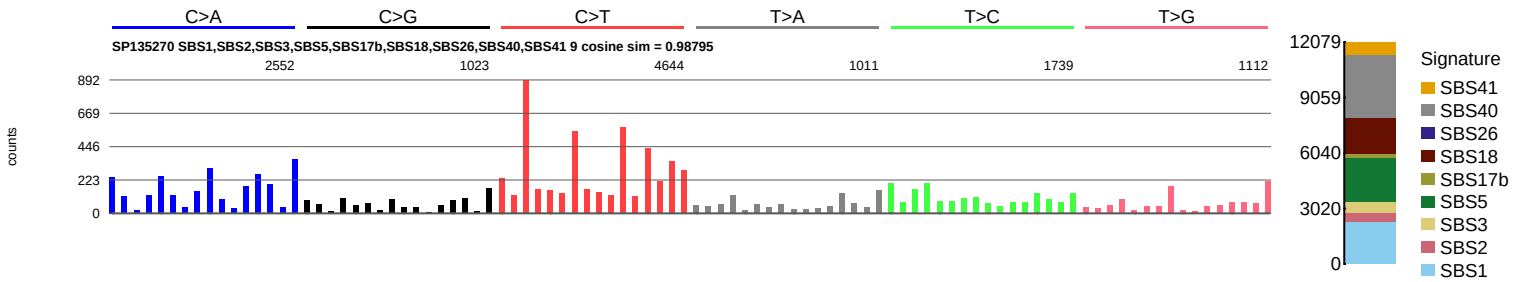
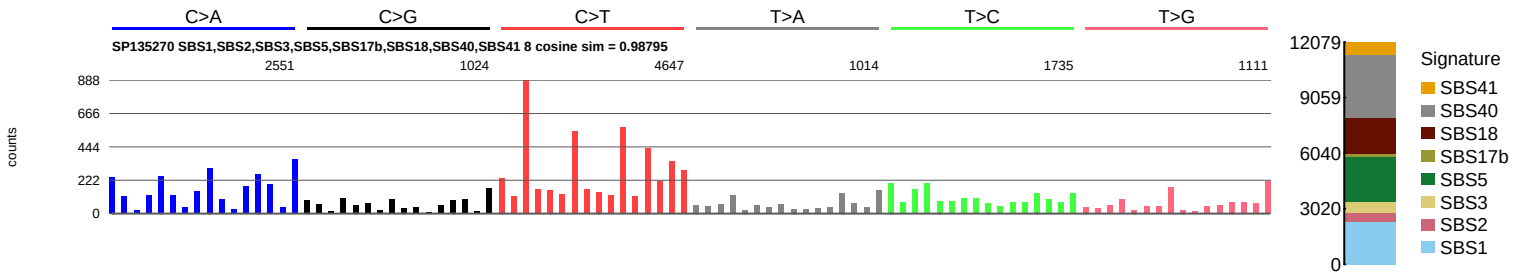
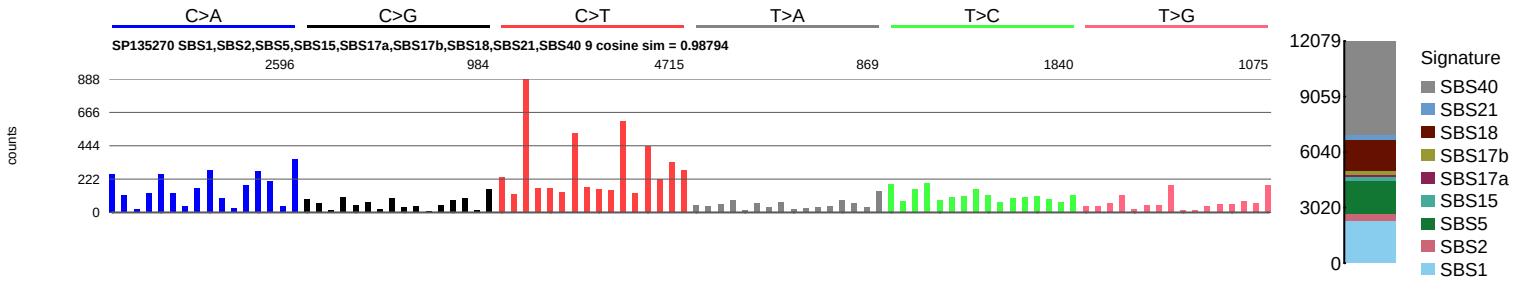
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98799
2565
974
4706
977
1740
1120
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98797
2564
976
4713
981
1725
1121
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98796
2626
1091
4578
928
1737
1117
912
684
456
228
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98795
2552
1023
4644
1011
1739
1112
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98795
2551
1024
4647
1014
1735
1111
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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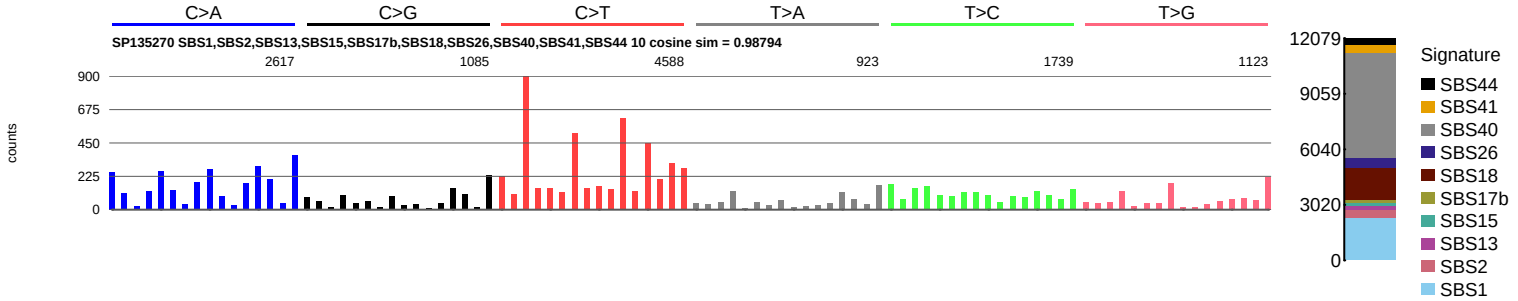
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98794
2596
984
4715
869
1840
1075
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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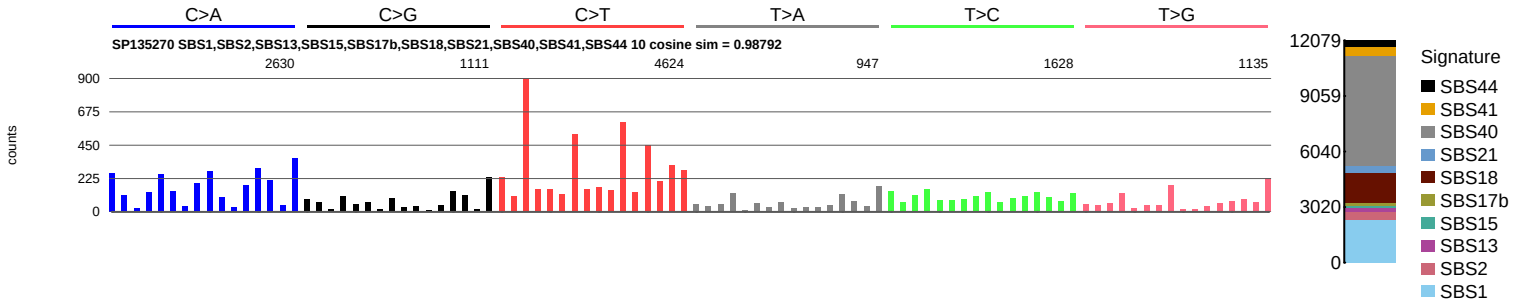
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98794
2617
1085
4588
923
1739
1123
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98792
2630
1111
4624
947
1628
1135
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
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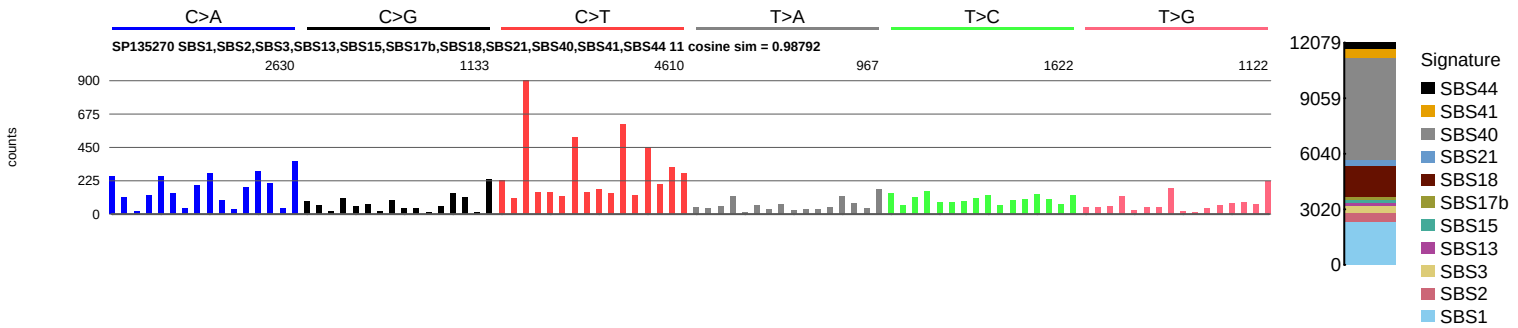
SBS2
SBS1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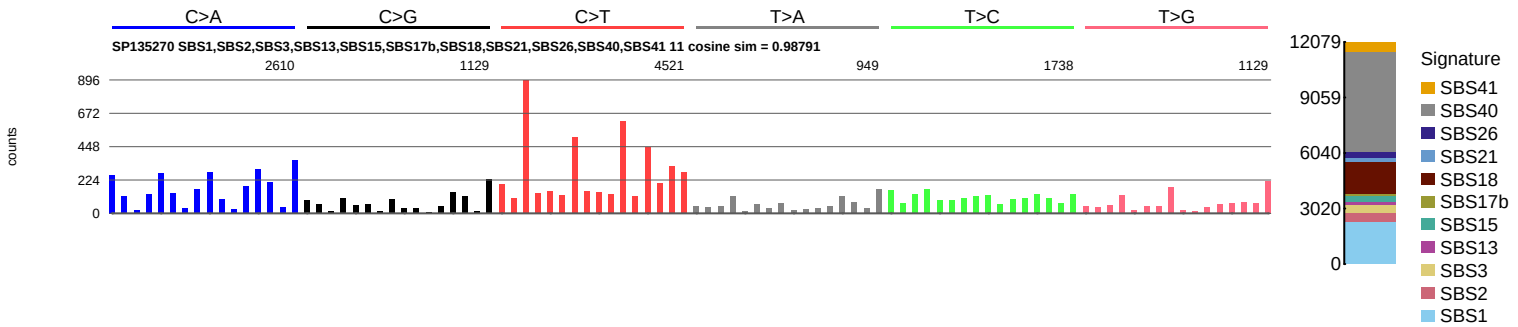
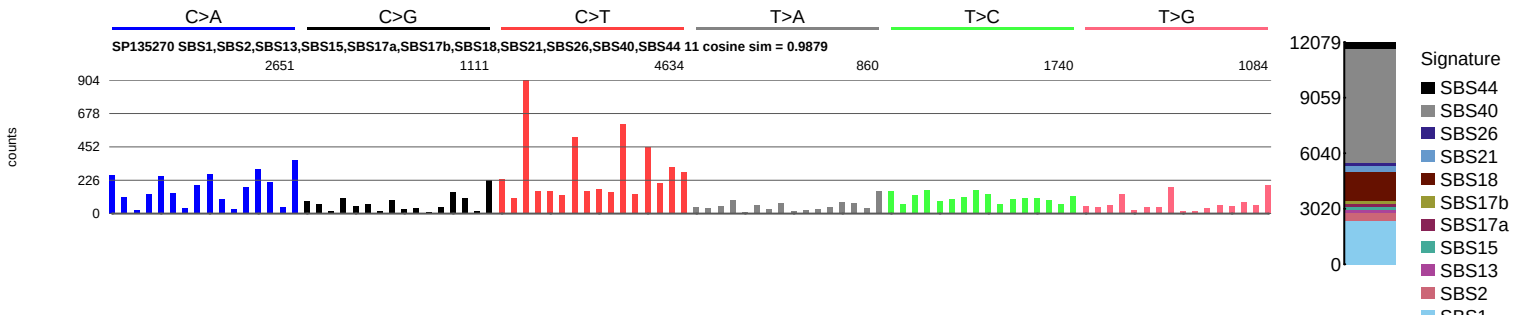
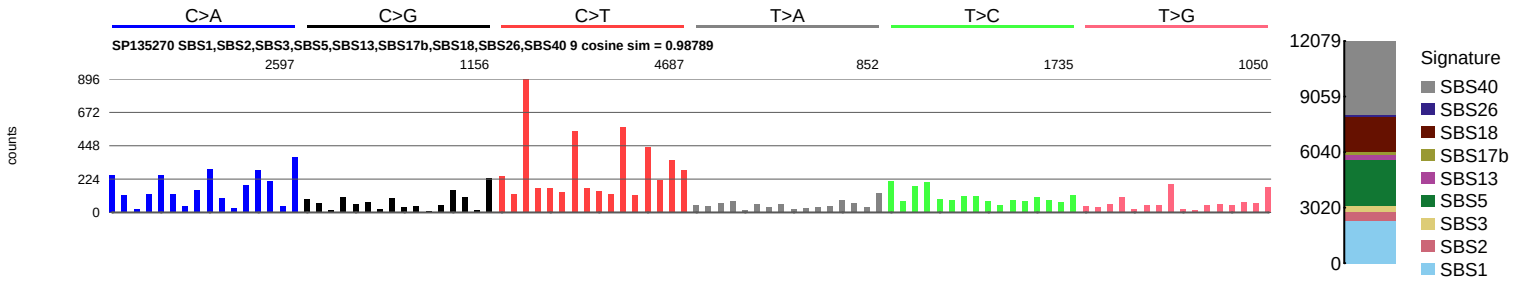
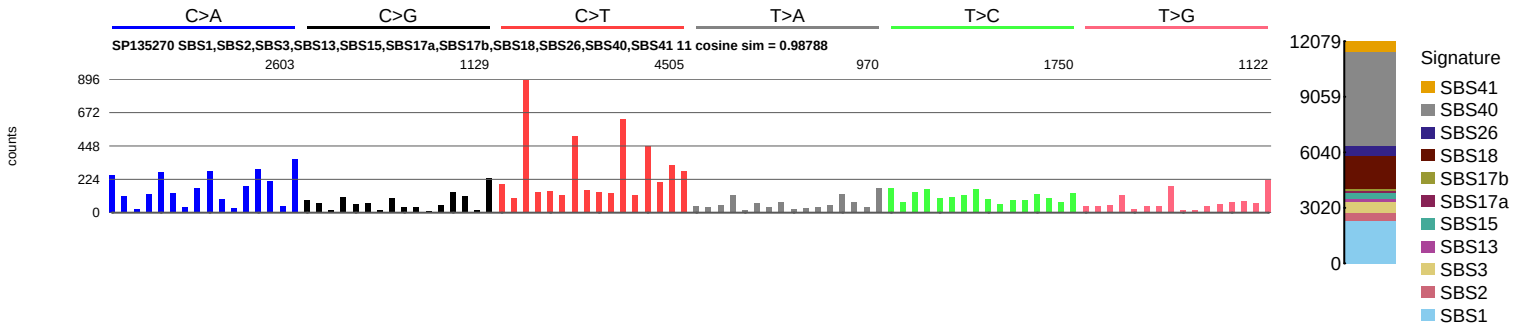
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98792
2630
1133
4610
967
1622
1122
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98791
2610
1129
4521
949
1738
1129
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.9879
2651
1111
4634
860
1740
1084
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98789
2597
1156
4687
852
1735
1050
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98788
2603
1129
4505
970
1750
1122
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
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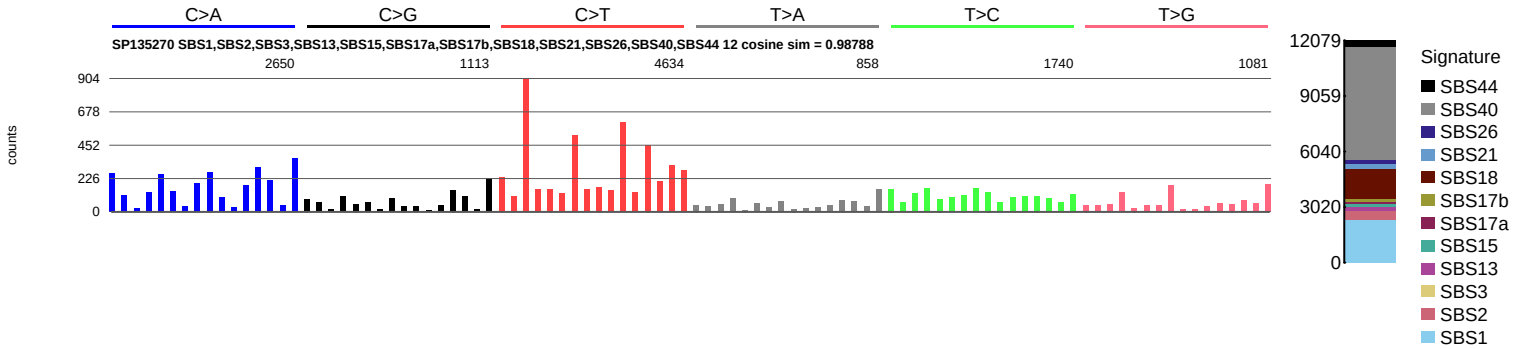
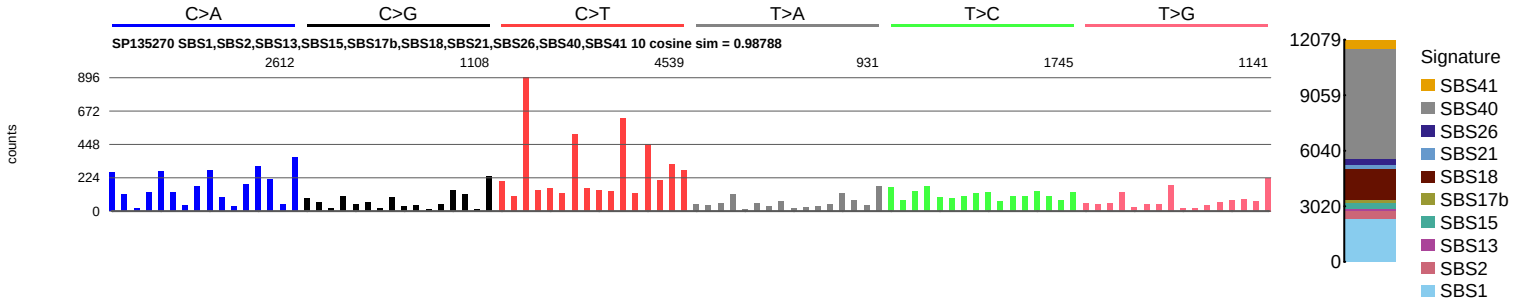
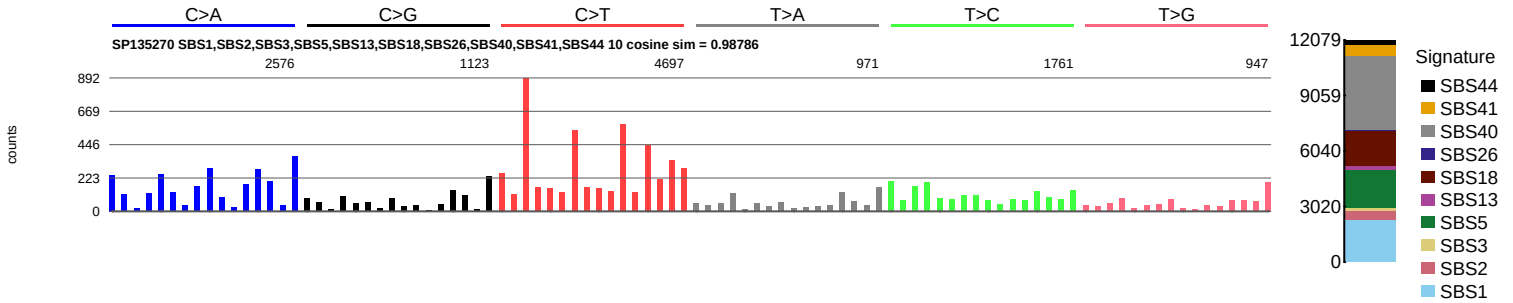
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 12 cosine sim = 0.98788
2650
1113
4634
858
1740
1081
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98788
2612
1108
4539
931
1745
1141
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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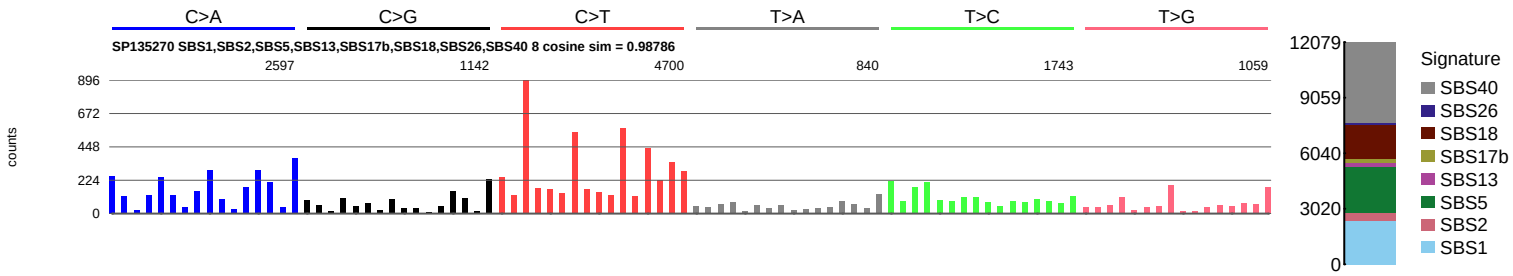
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98786
2576
1123
4697
971
1761
947
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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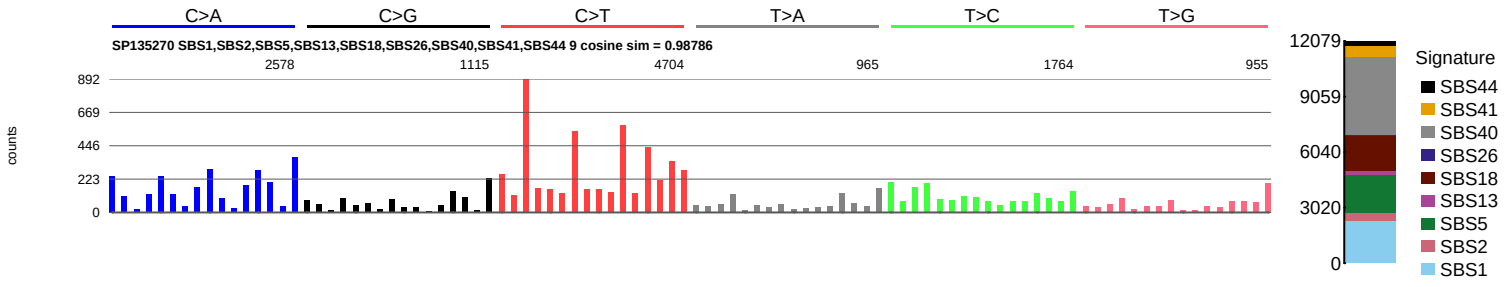
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98786
2597
1142
4700
840
1743
1059
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>T
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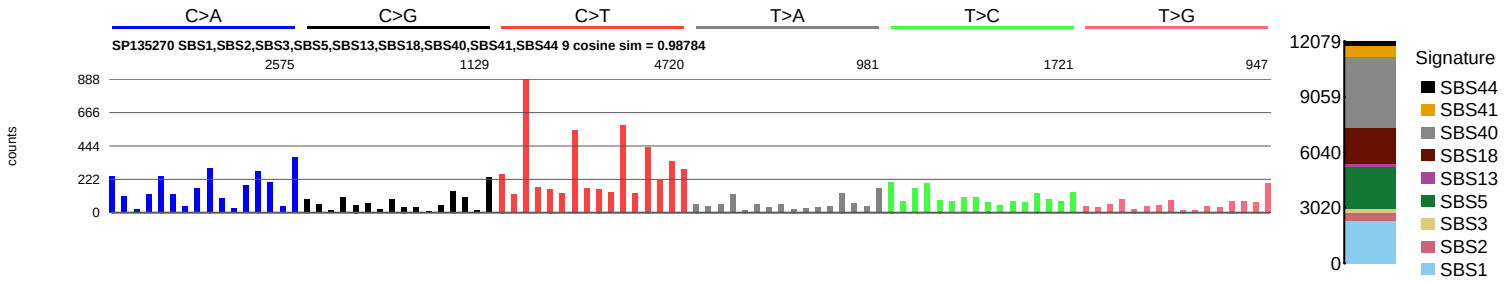
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98786
2578
1115
4704
965
1764
955
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1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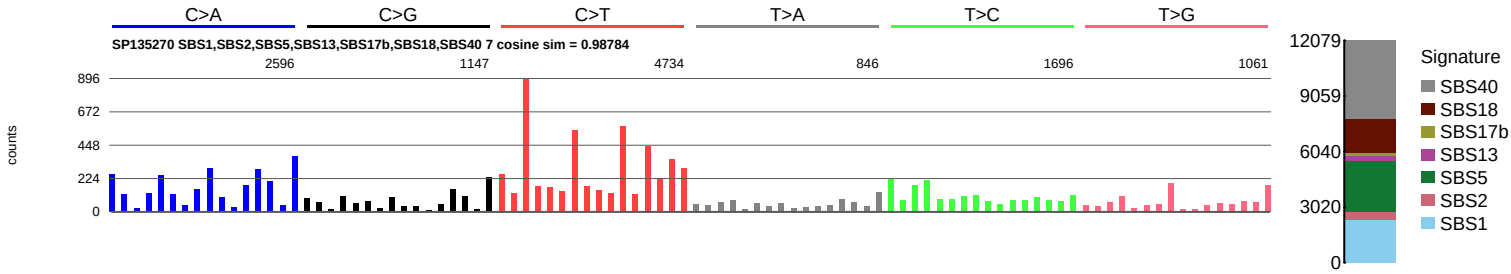
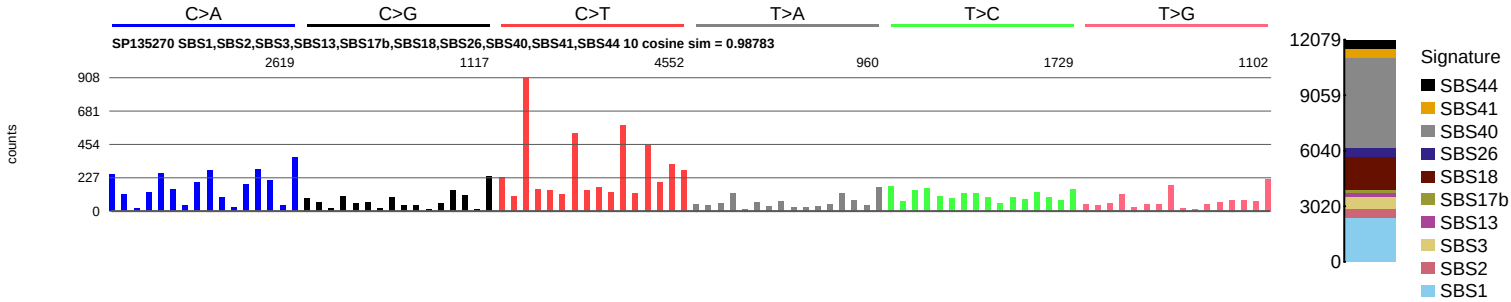
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98784
2575
1129
4720
981
1721
947
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40 7 cosine sim = 0.98784
2596
1147
4734
846
1696
1061
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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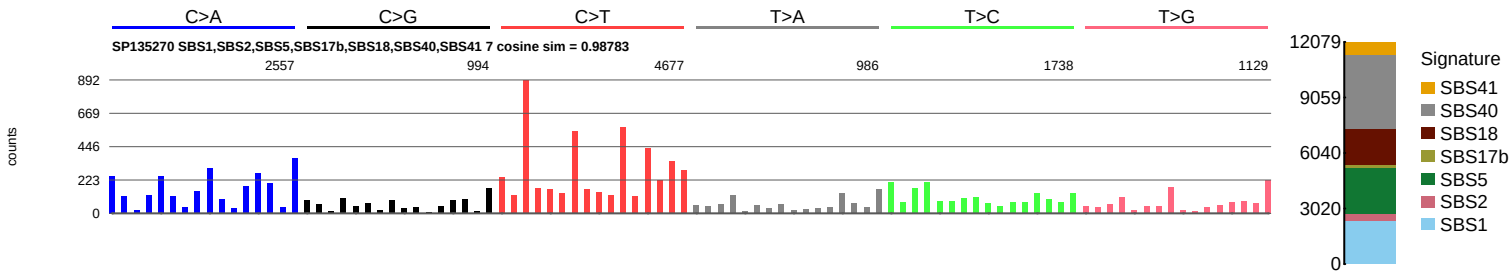
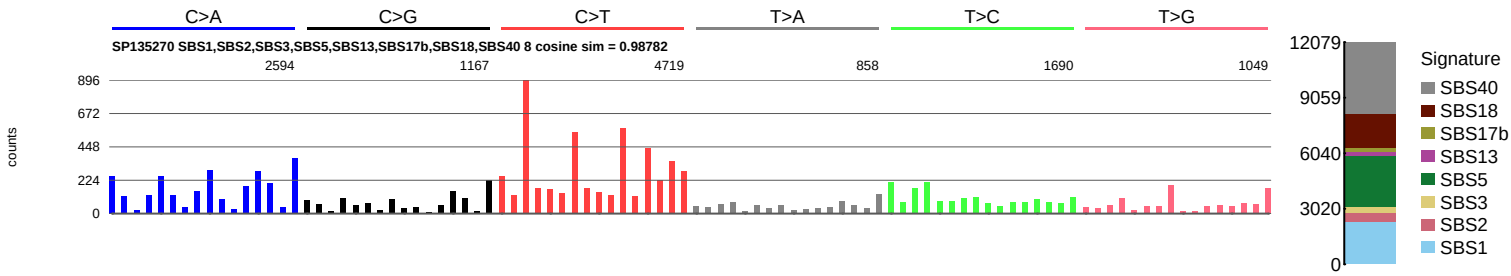
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98783
2619
1117
4552
960
1729
1102
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98783
2557
994
4677
986
1738
1129
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40 8 cosine sim = 0.98782
2594
1167
4719
858
1690
1049
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
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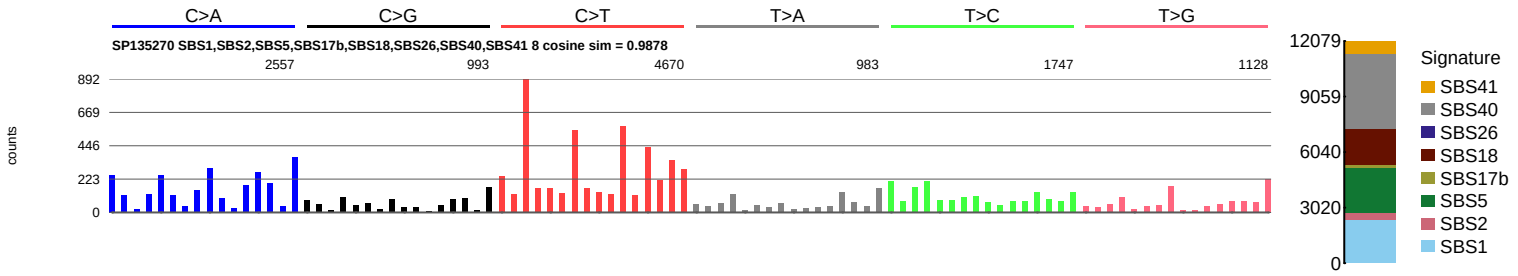
SBS2
SBS1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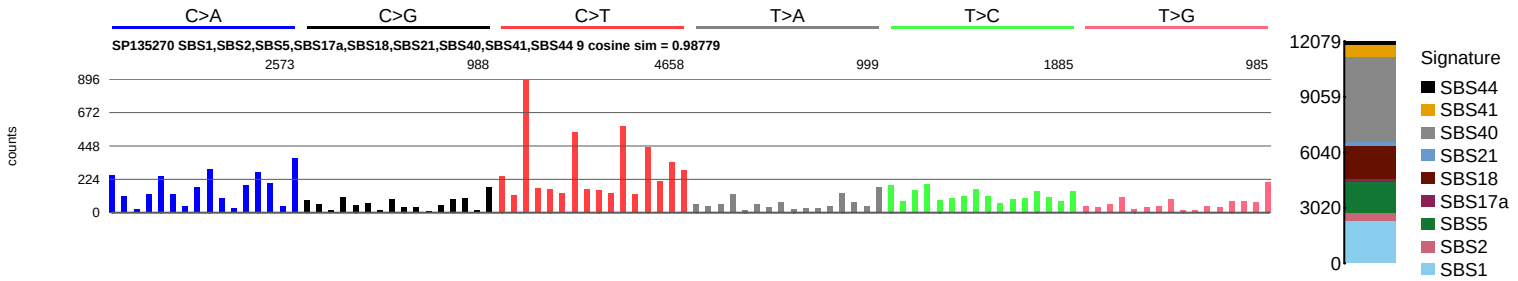
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.9878
2557
993
4670
983
1747
1128
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1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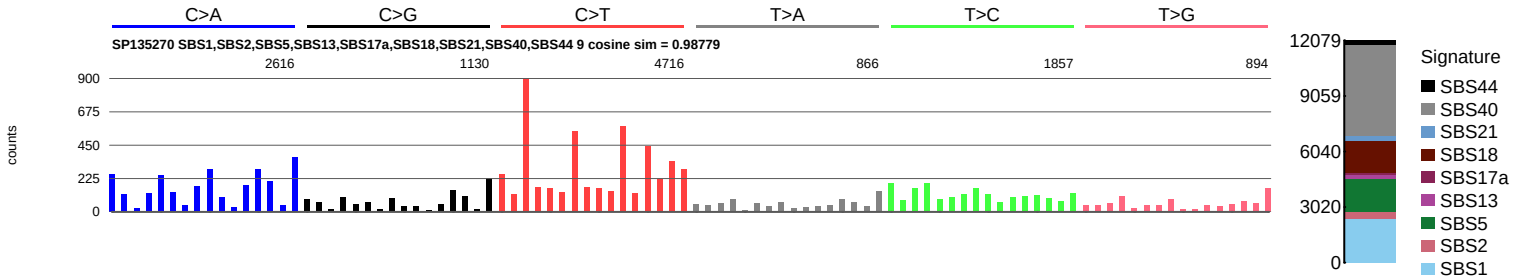
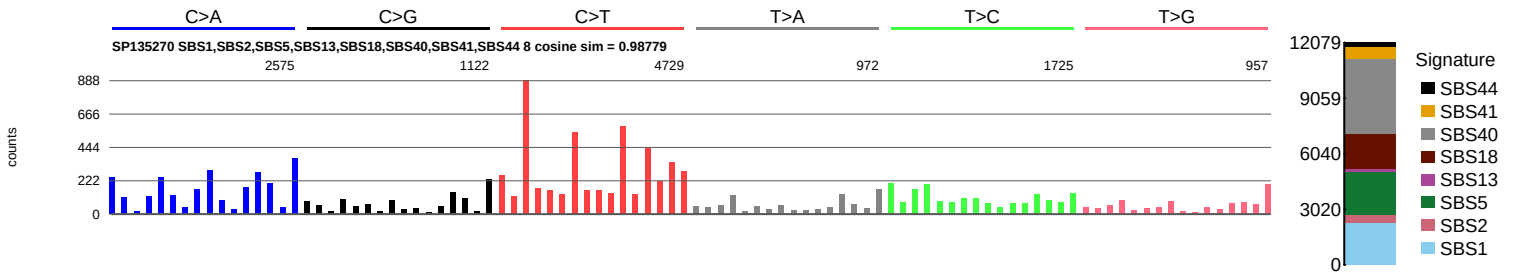
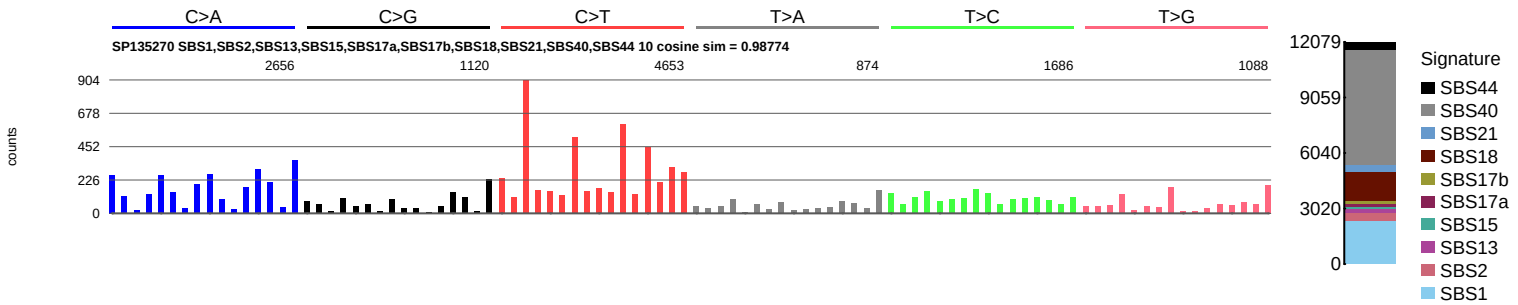
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98779
2573
988
4658
999
1885
985
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98779
2616
1130
4716
866
1857
894
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98779
2575
1122
4729
972
1725
957
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1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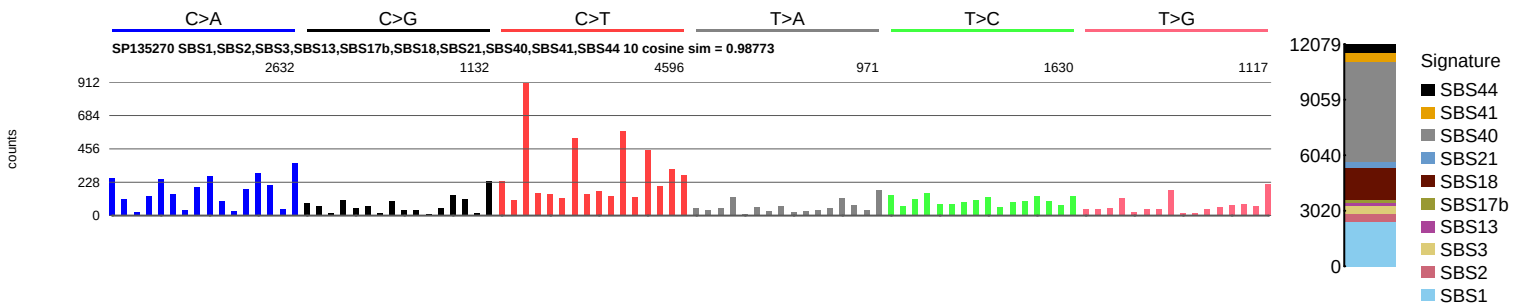
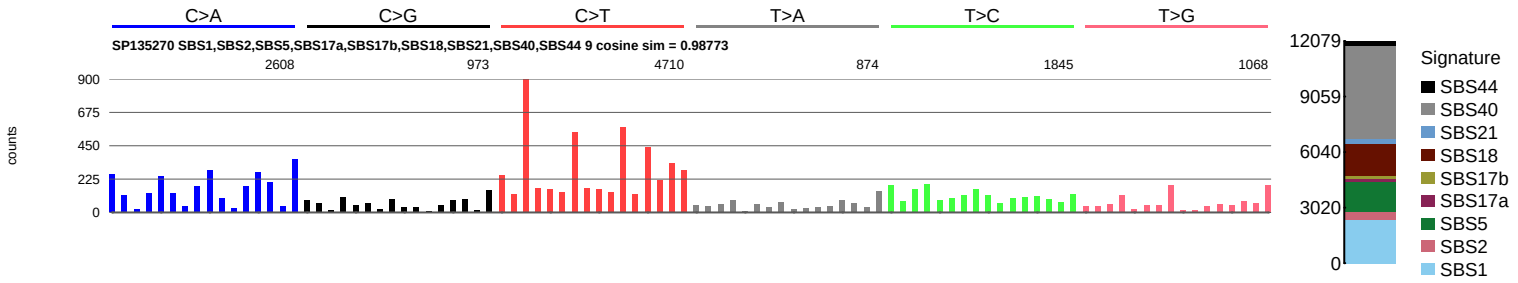
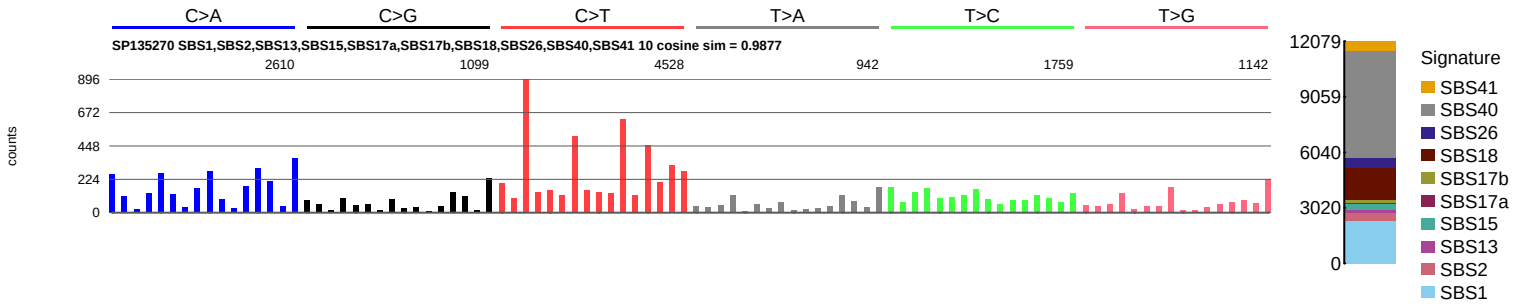
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98774
2656
1120
4653
874
1686
1088
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98773
2632
1132
4596
971
1630
1117
912
684
456
228
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98773
2608
973
4710
874
1845
1068
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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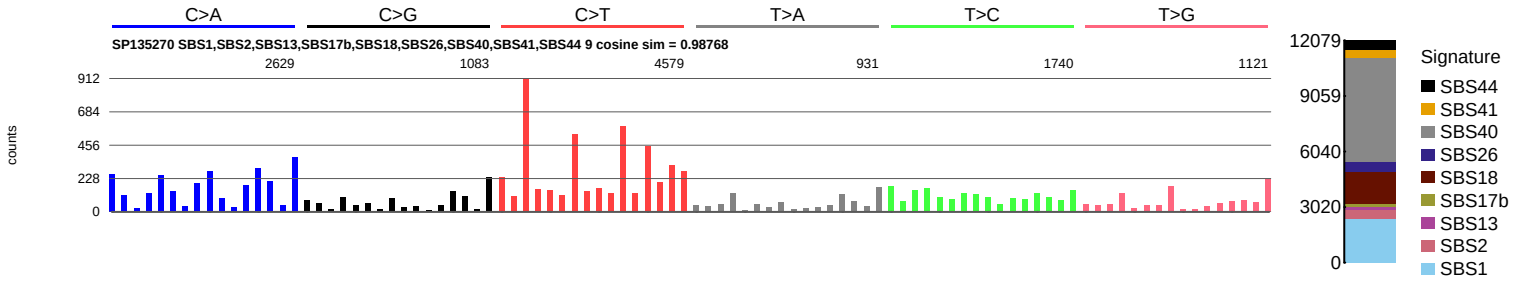
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.9877
2610
1099
4528
942
1759
1142
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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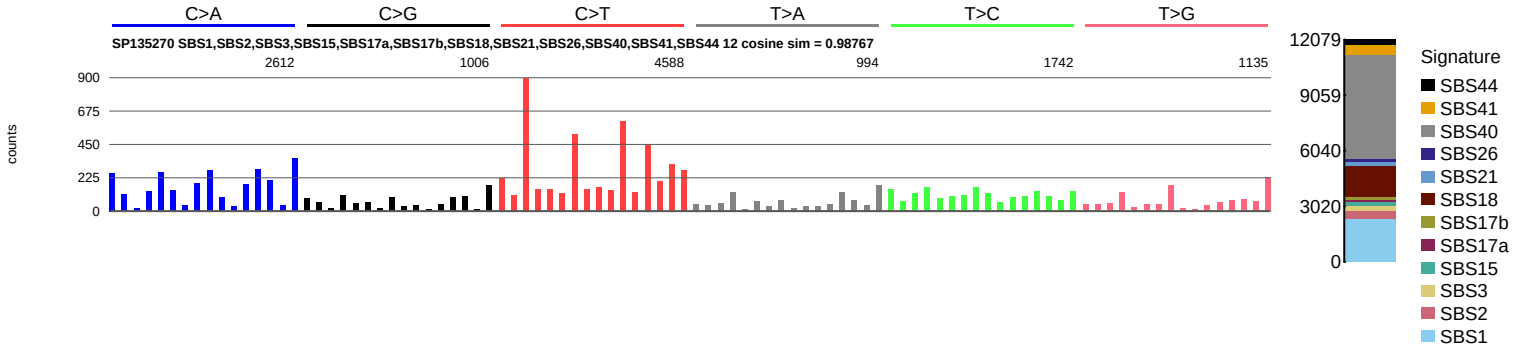
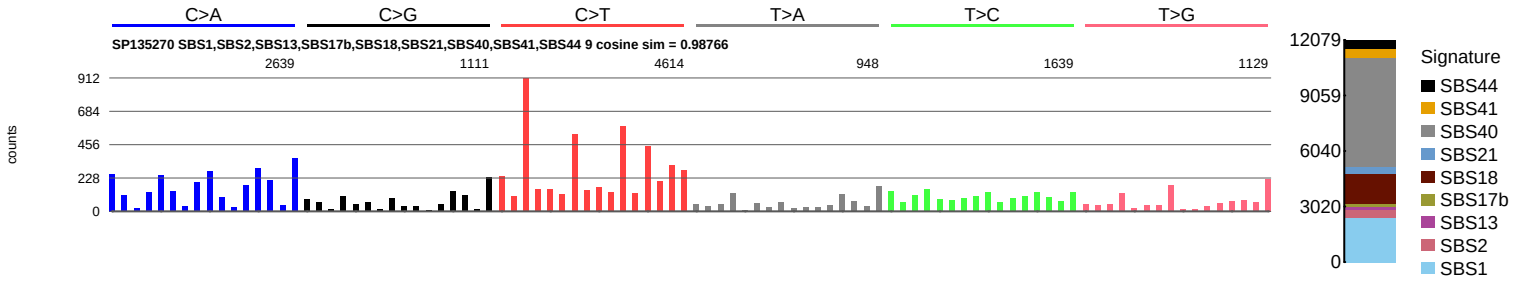
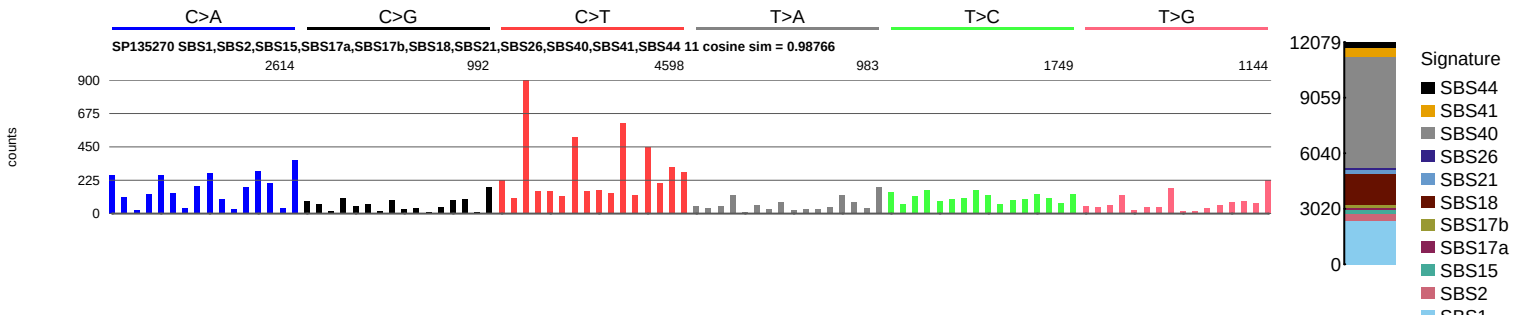
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98768
2629
1083
4579
931
1740
1121
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98767
2612
1006
4588
994
1742
1135
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98766
2639
1111
4614
948
1639
1129
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1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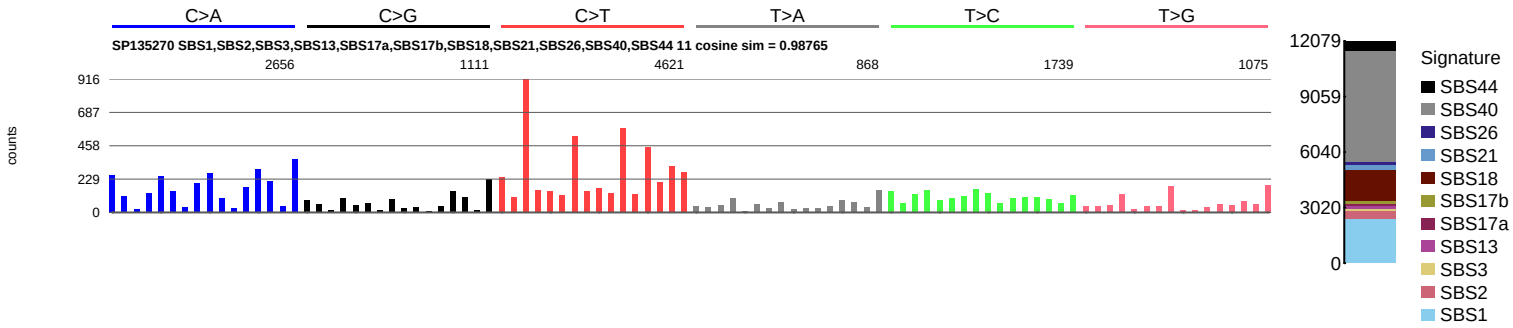
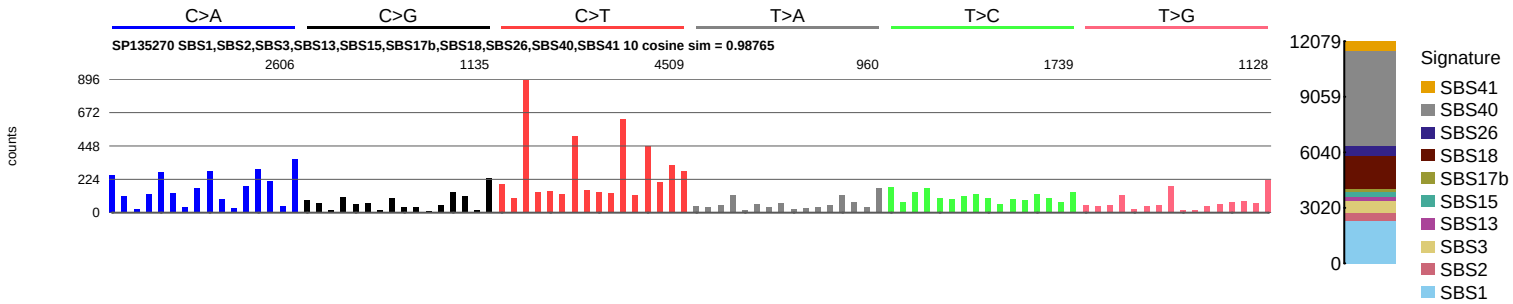
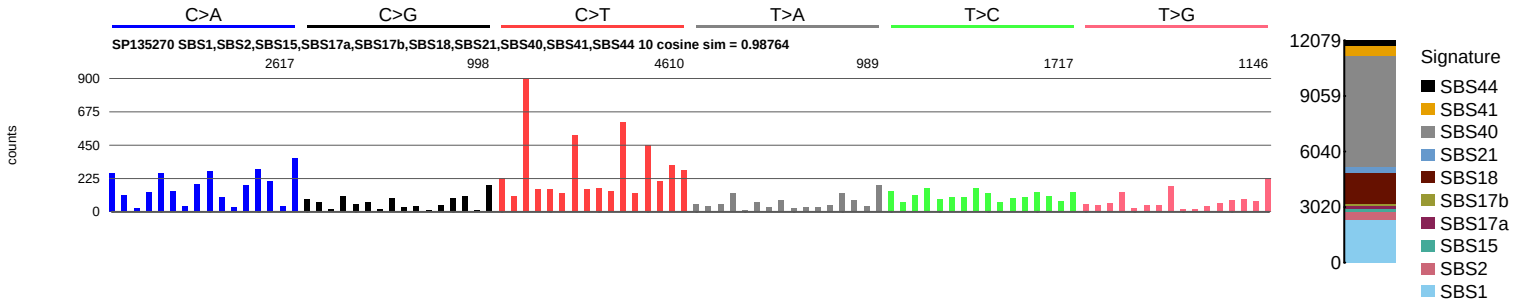
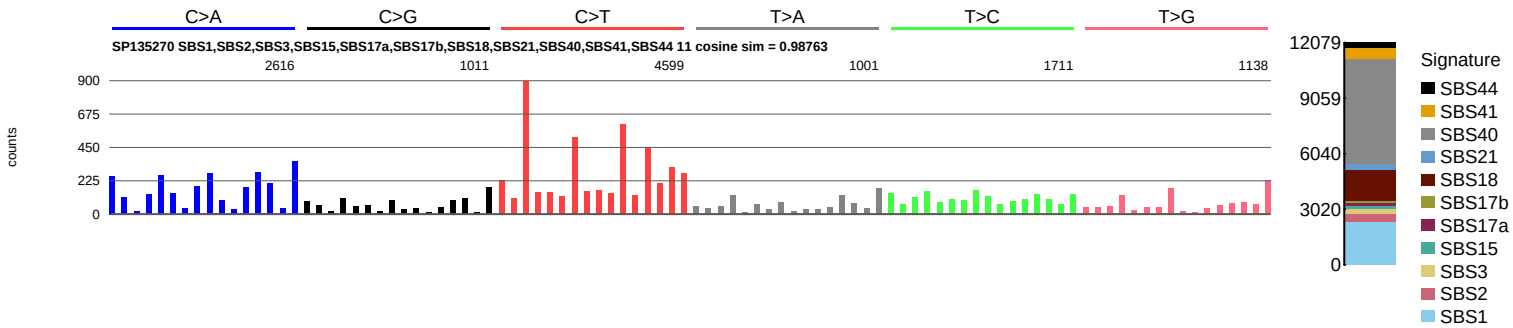
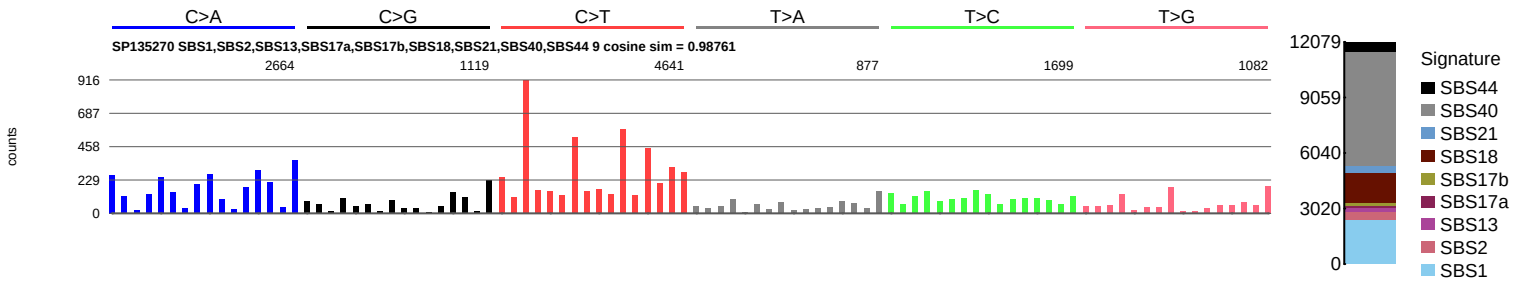
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98766
2614
992
4598
983
1749
1144
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98765
2656
1111
4621
868
1739
1075
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98765
2606
1135
4509
960
1739
1128
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98764
2617
998
4610
989
1717
1146
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98763
2616
1011
4599
1001
1711
1138
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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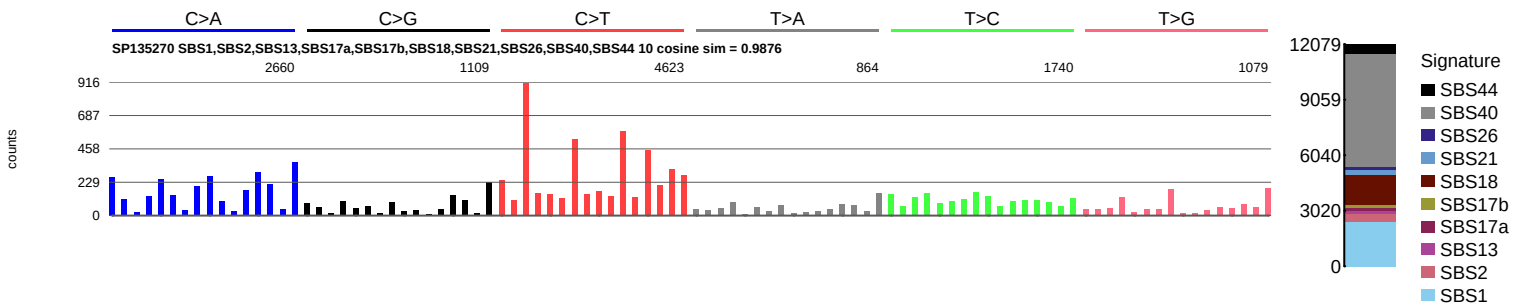
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98761
2664
1119
4641
877
1699
1082
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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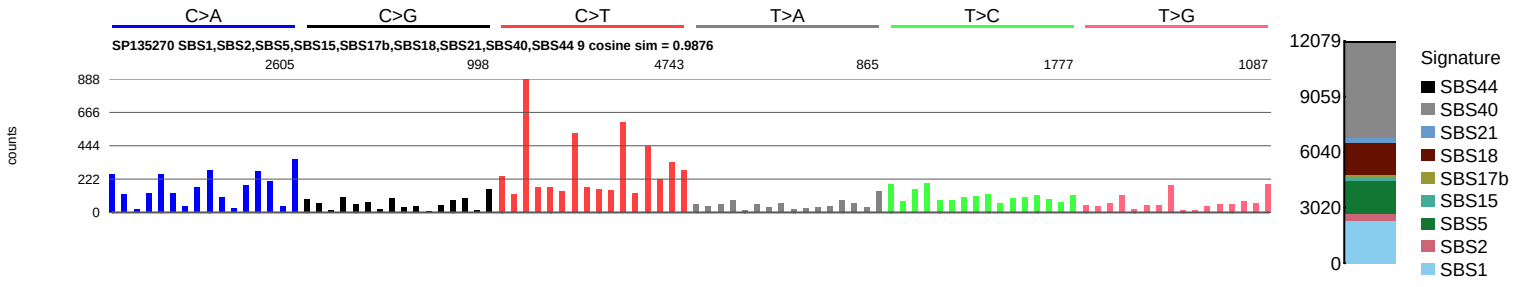
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.9876
2660
1109
4623
864
1740
1079
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.9876
2605
998
4743
865
1777
1087
888
666
444
222
0
counts
12079
9059
6040
3020
0
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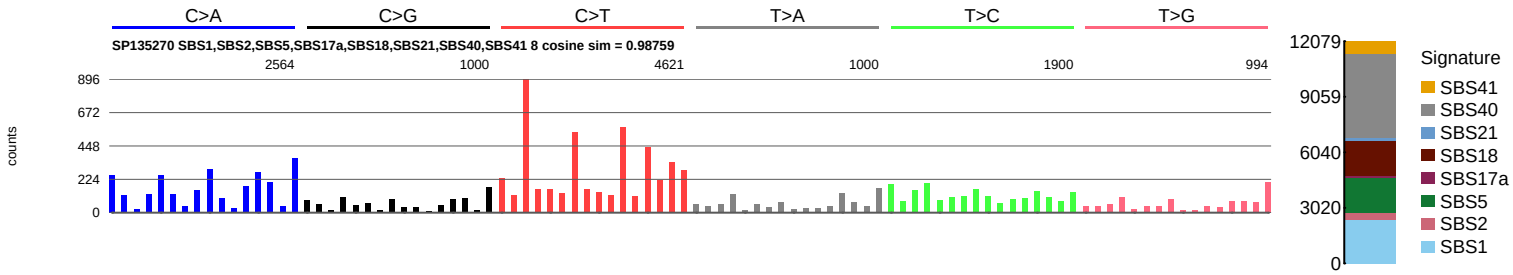
SBS2
SBS1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C>G
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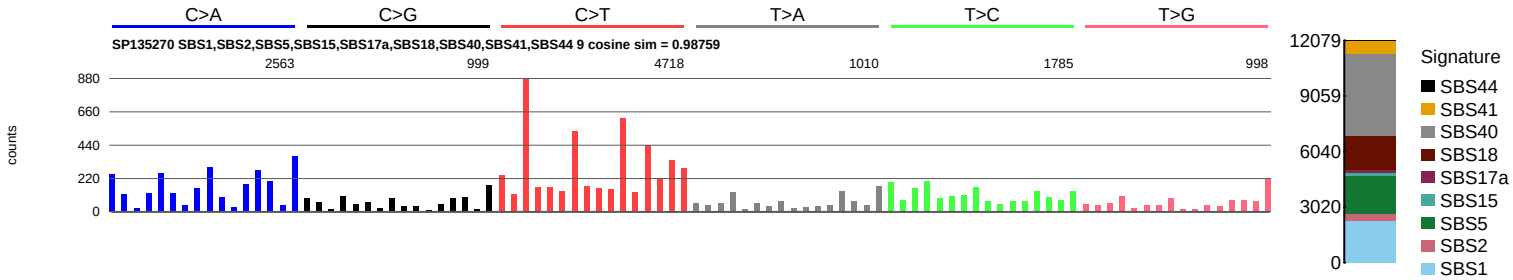
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98759
2564
1000
4621
1000
1900
994
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1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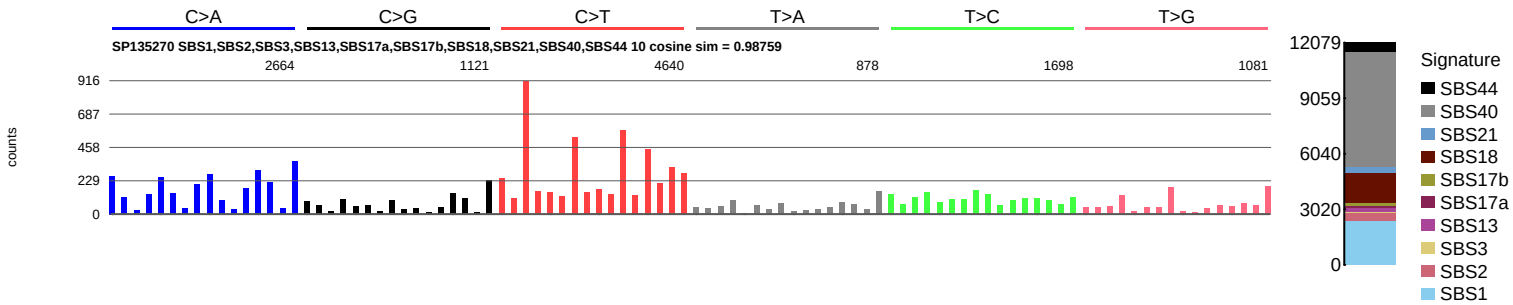
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98759
2563
999
4718
1010
1785
998
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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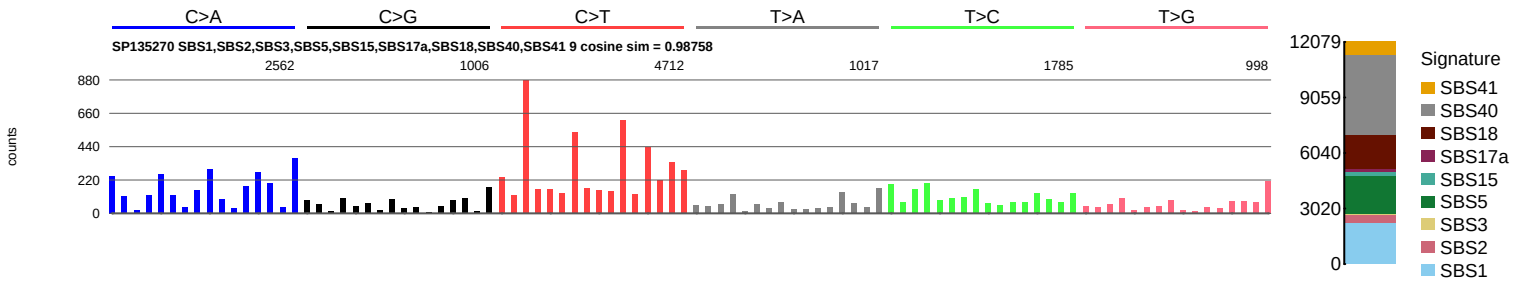
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98759
2664
1121
4640
878
1698
1081
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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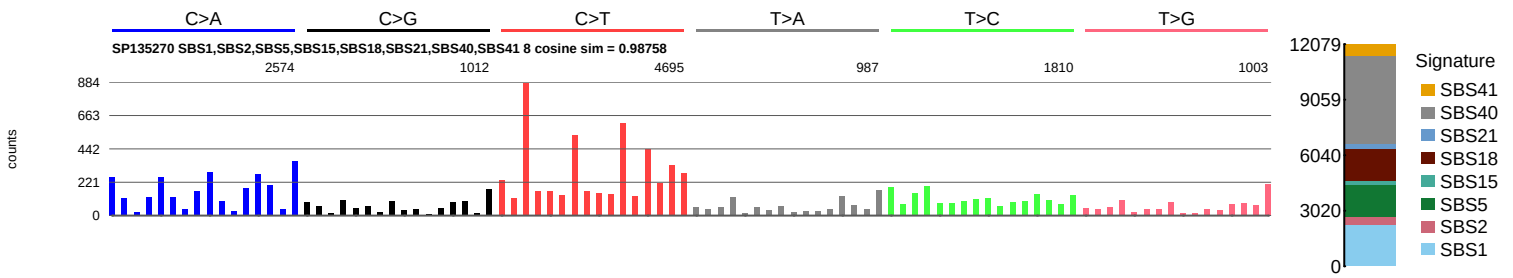
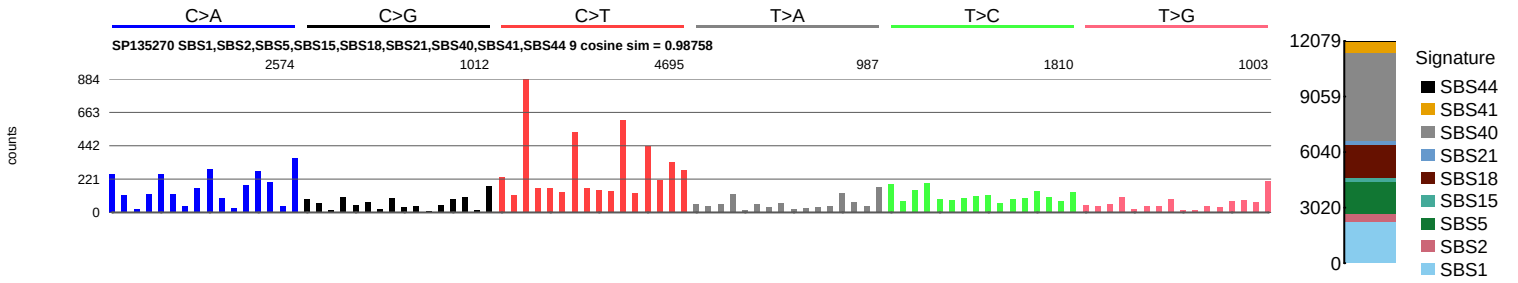
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98758
2562
1006
4712
1017
1785
998
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98758
2574
1012
4695
987
1810
1003
884
663
442
221
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1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C>G
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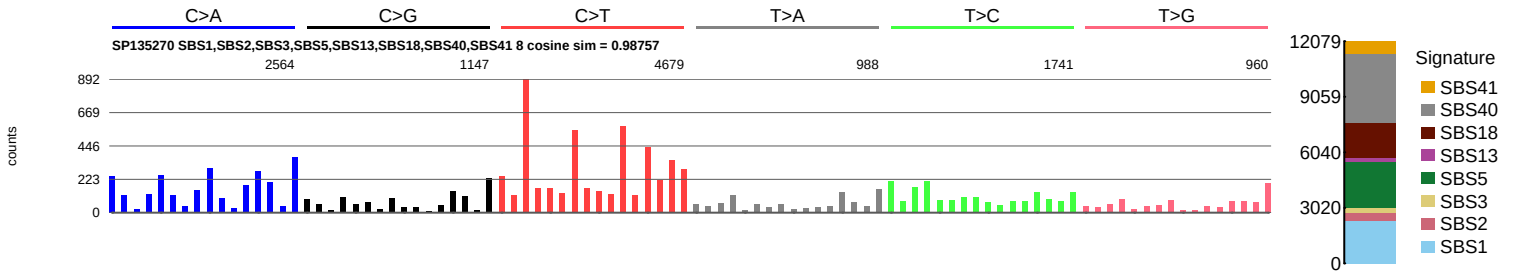
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98758
2574
1012
4695
987
1810
1003
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1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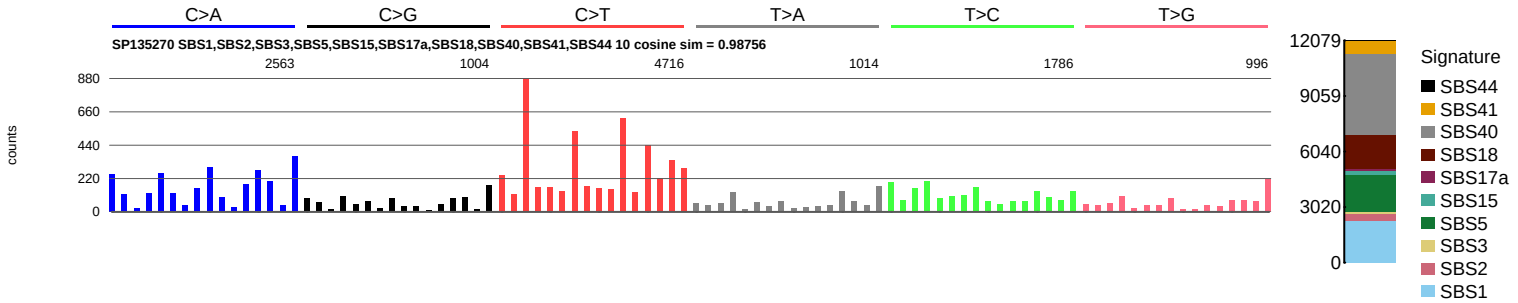
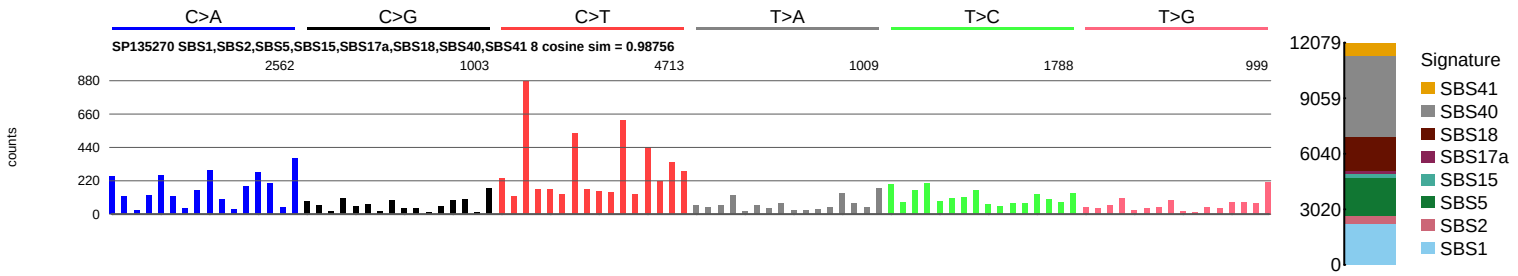
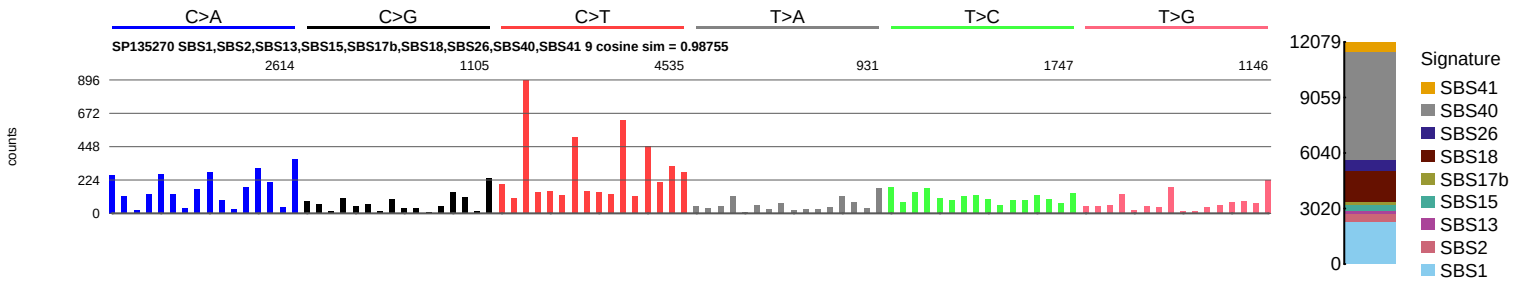
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS41 8 cosine sim = 0.98757
2564
1147
4679
988
1741
960
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98756
2563
1004
4716
1014
1786
996
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98756
2562
1003
4713
1009
1788
999
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98755
2614
1105
4535
931
1747
1146
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
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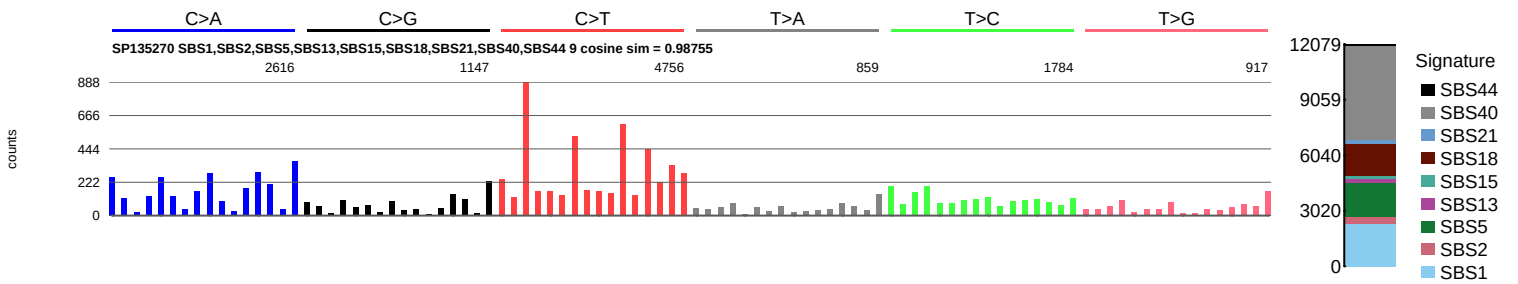
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98755
2616
1147
4756
859
1784
917
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
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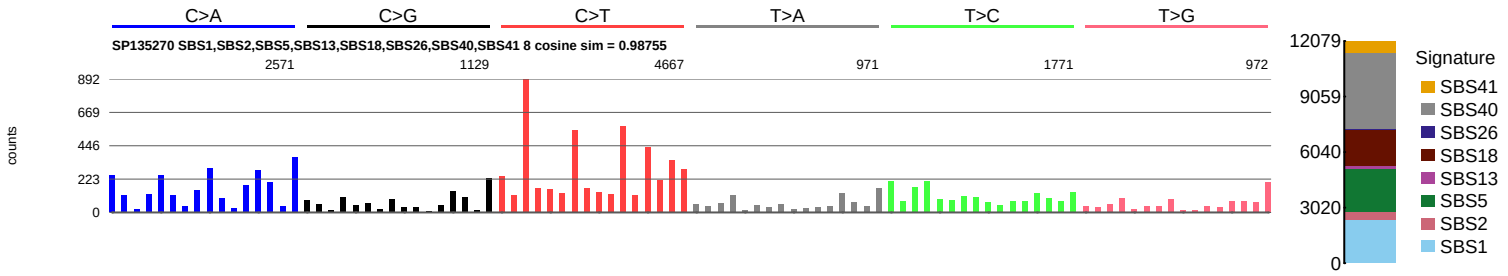
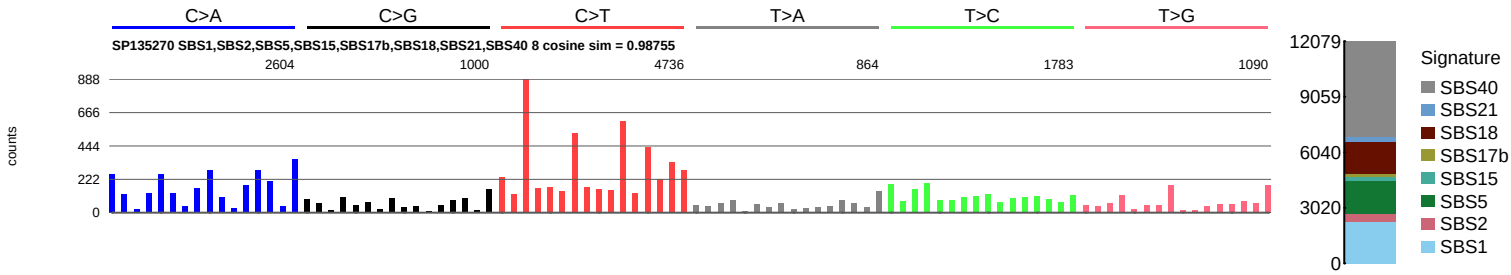
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98755
2571
1129
4667
971
1771
972
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1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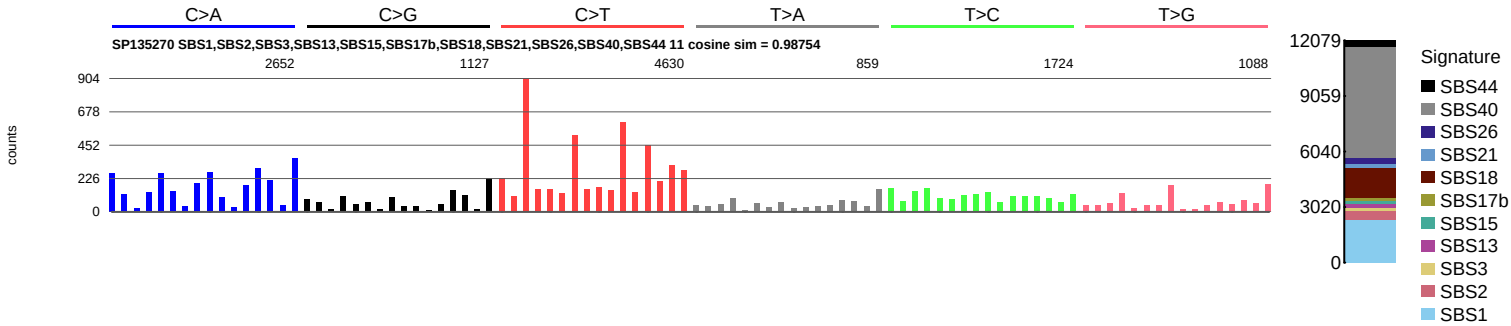
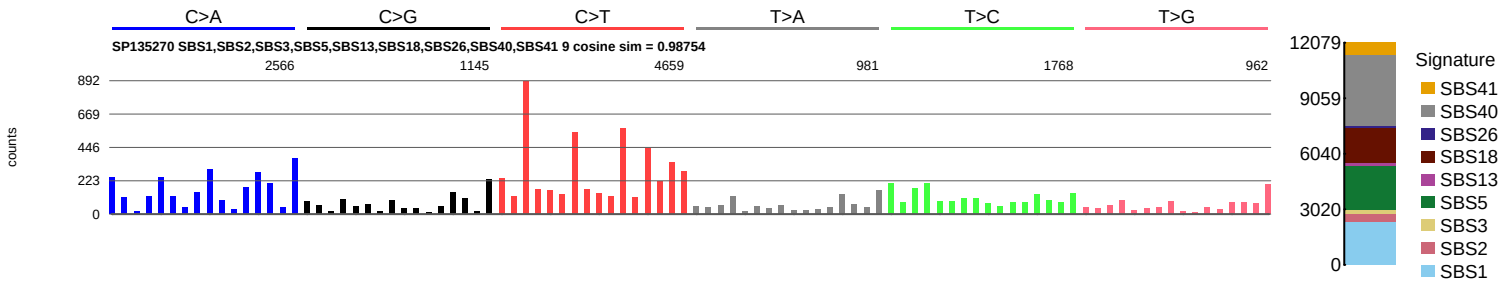
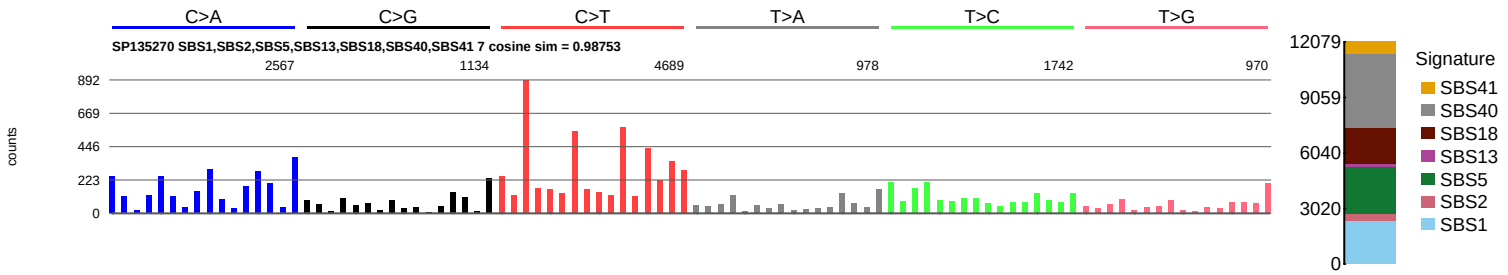
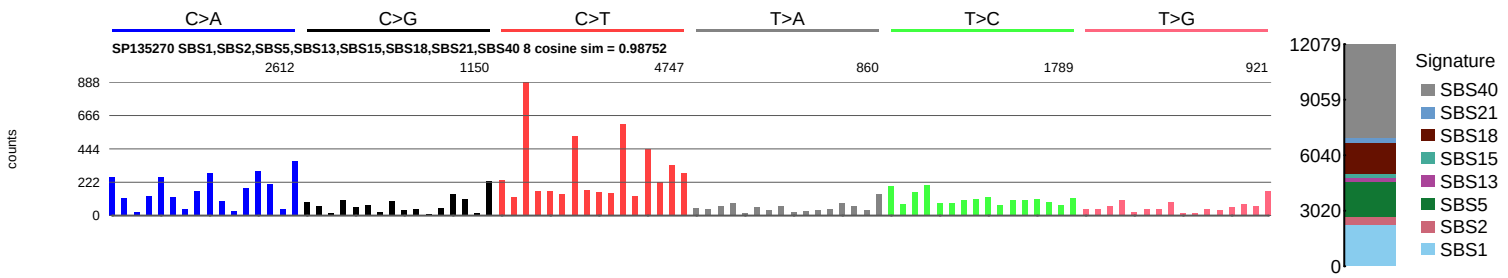
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98755
2604
1000
4736
864
1783
1090
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98754
2652
1127
4630
859
1724
1088
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98754
2566
1145
4659
981
1768
962
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS41 7 cosine sim = 0.98753
2567
1134
4689
978
1742
970
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40 8 cosine sim = 0.98752
2612
1150
4747
860
1789
921
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS15
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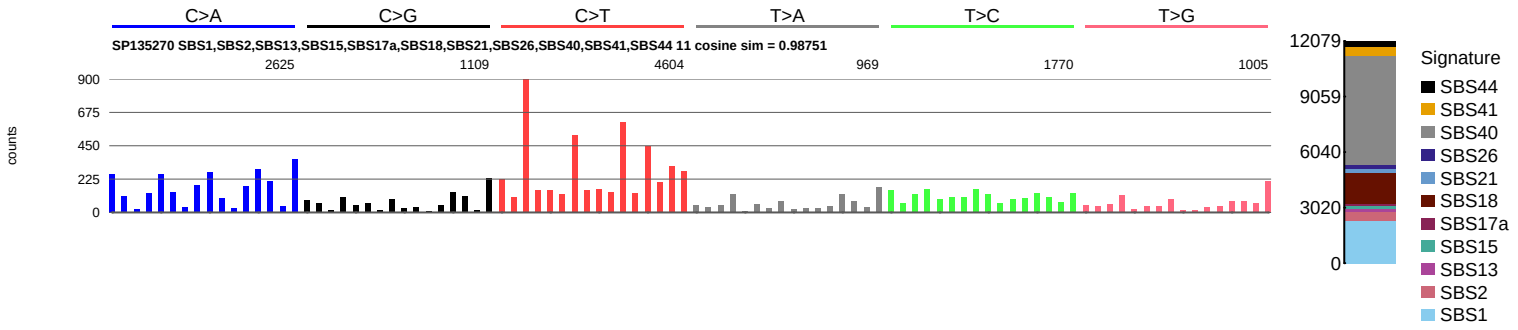
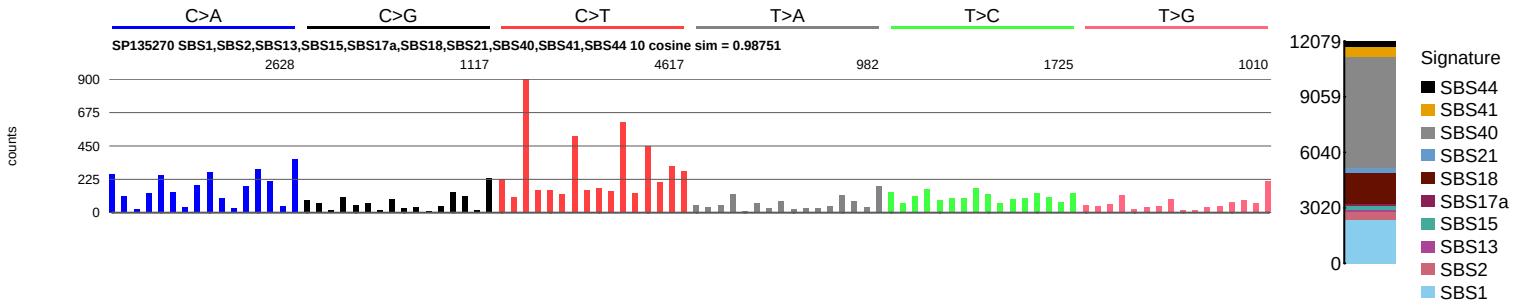
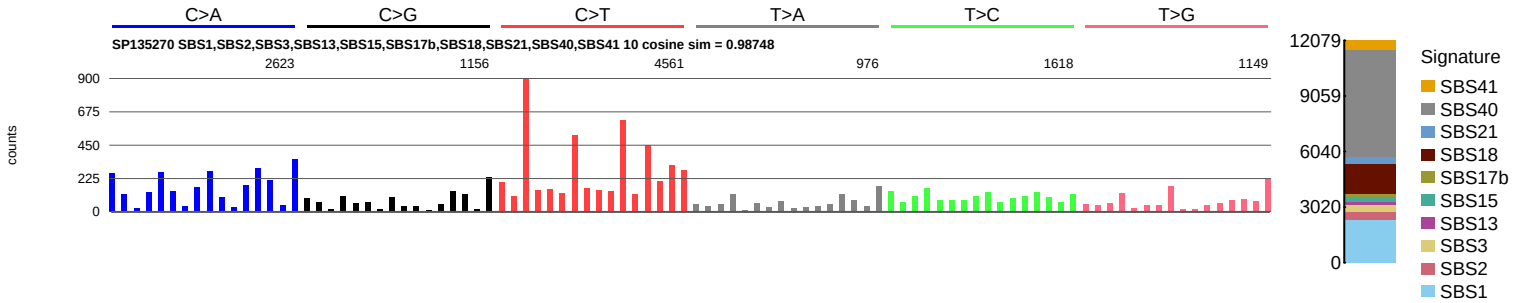
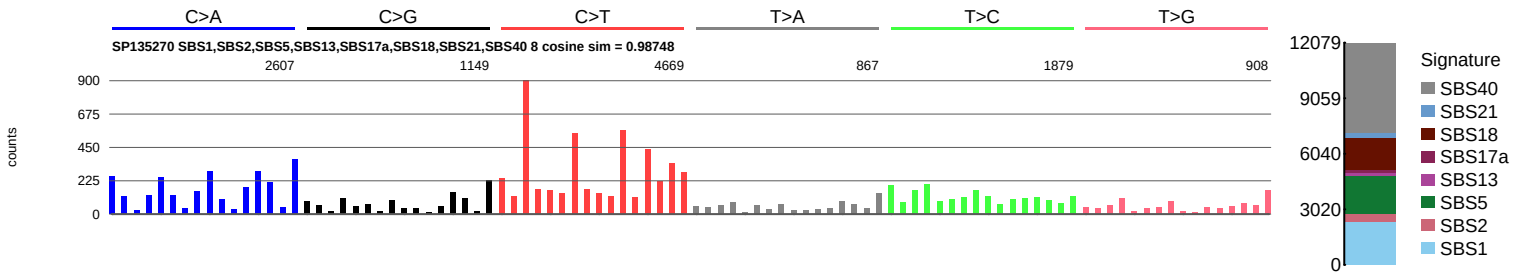
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98751
2625
1109
4604
969
1770
1005
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98751
2628
1117
4617
982
1725
1010
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98748
2623
1156
4561
976
1618
1149
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98748
2607
1149
4669
867
1879
908
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17a
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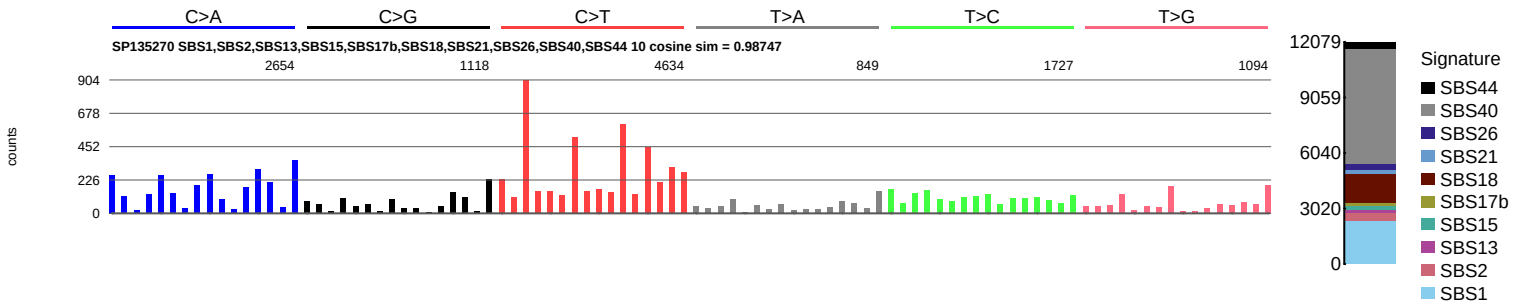
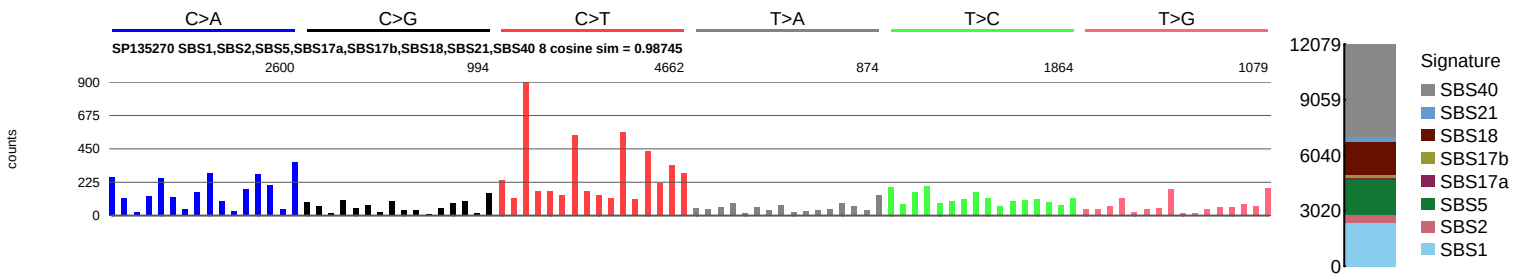
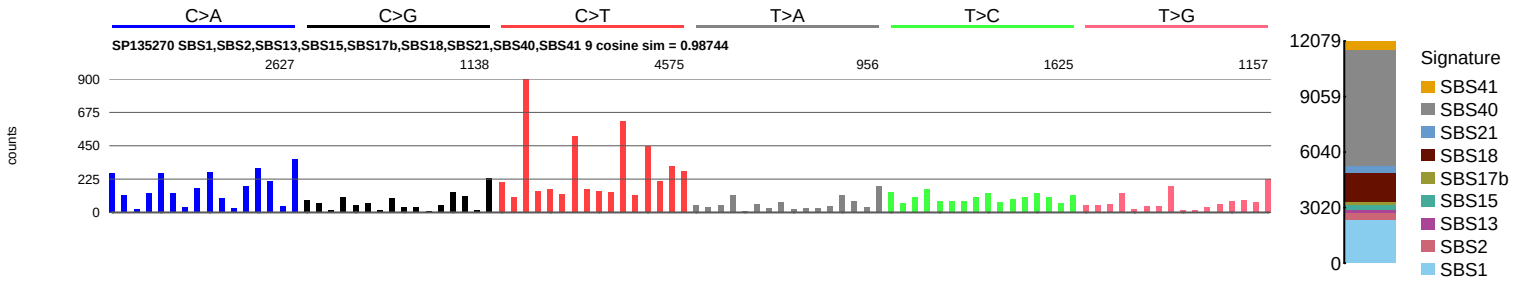
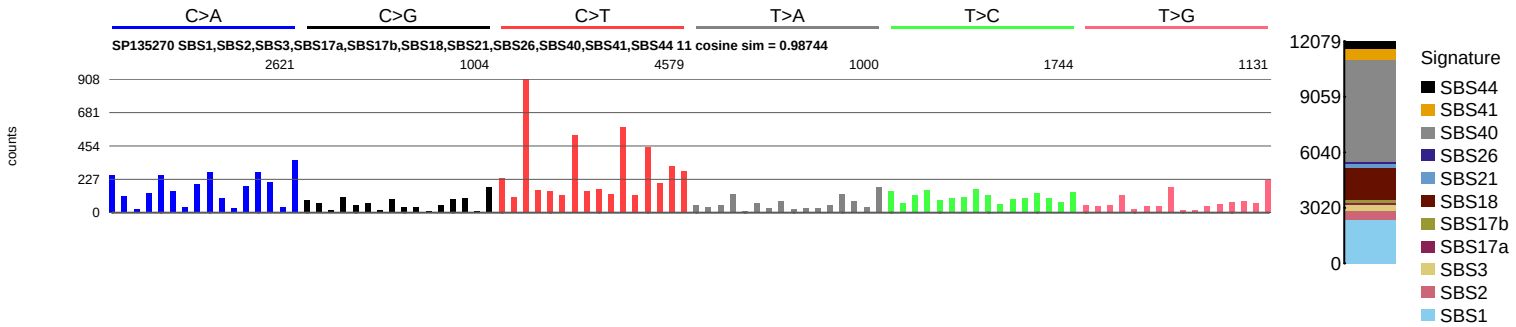
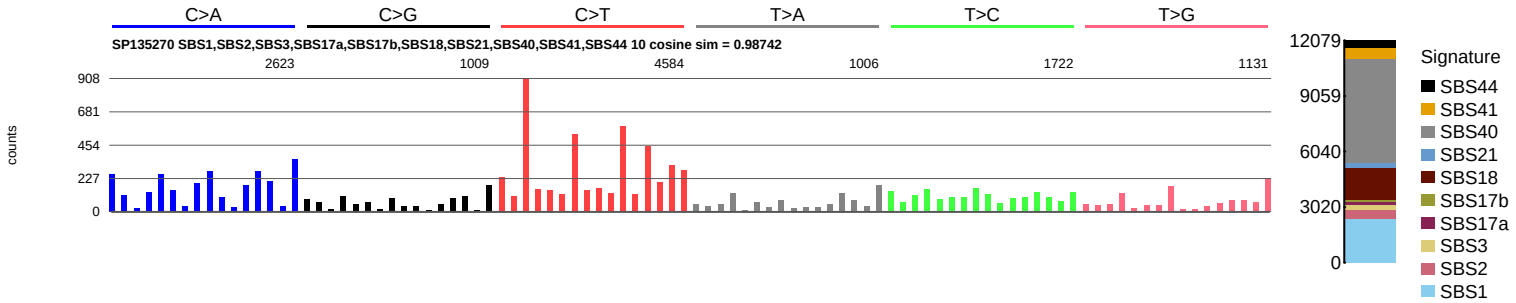
SBS5
SBS2
SBS1

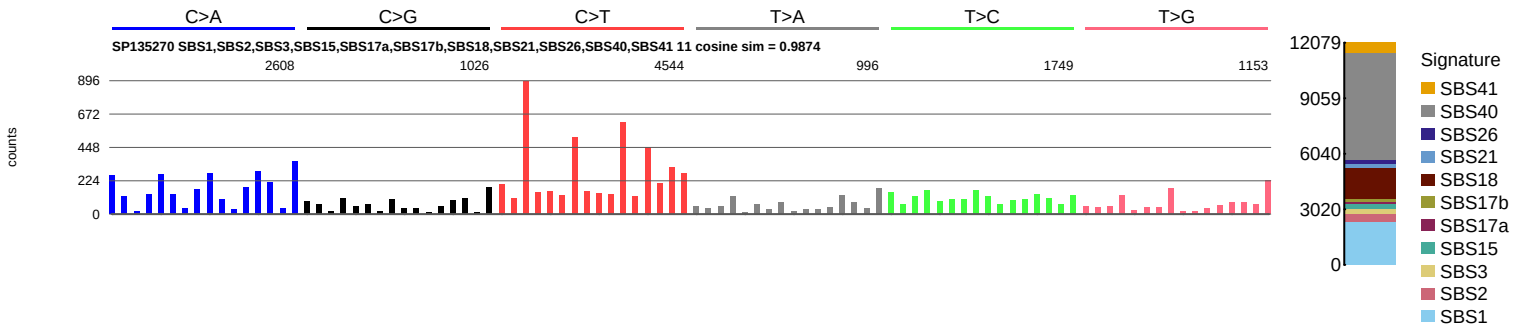

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98747
2654
1118
4634
849
1727
1094
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

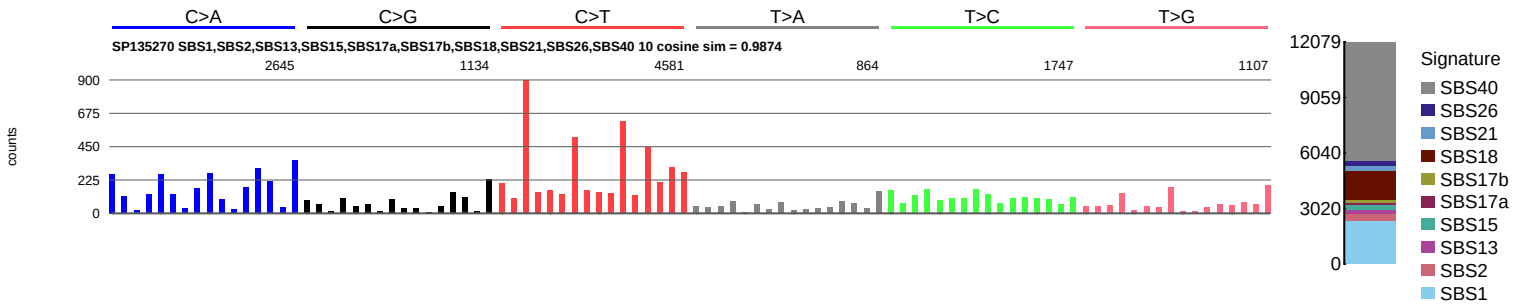

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98745
2600
994
4662
874
1864
1079
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

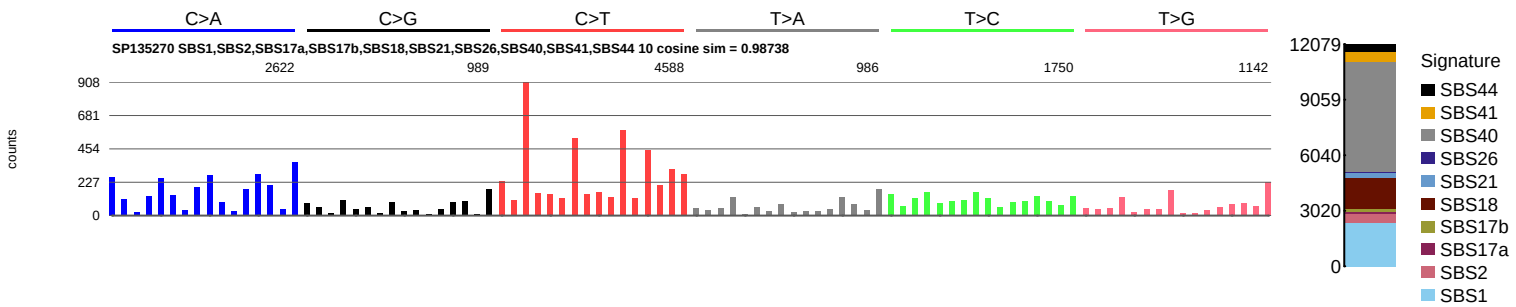

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98744
2627
1138
4575
956
1625
1157
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

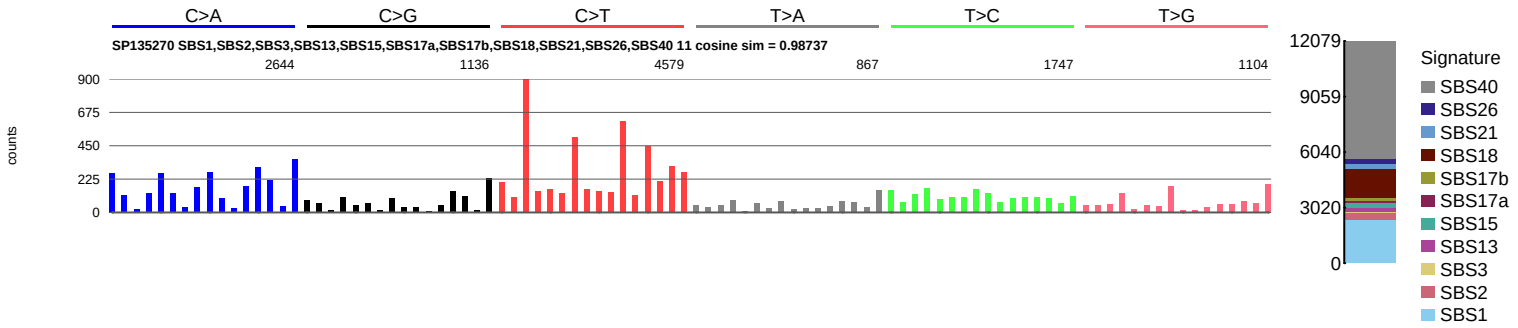

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98744
2621
1004
4579
1000
1744
1131
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

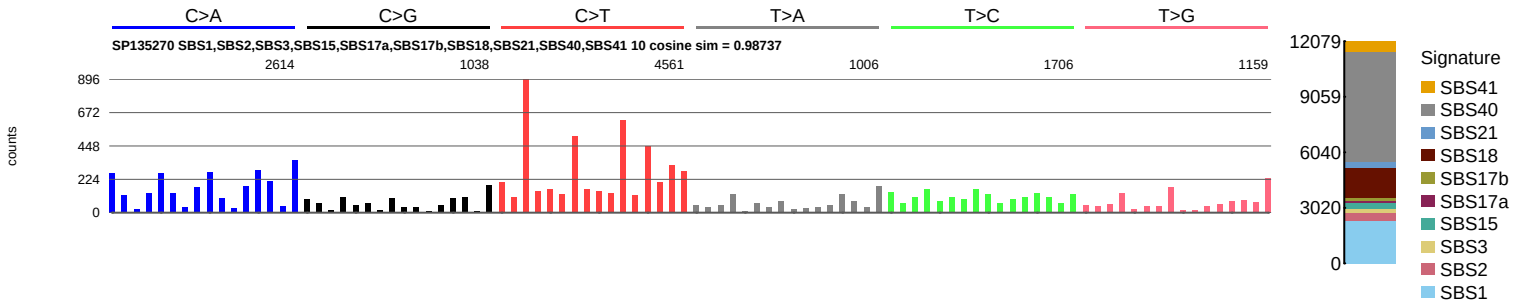

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98742
2623
1009
4584
1006
1722
1131
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

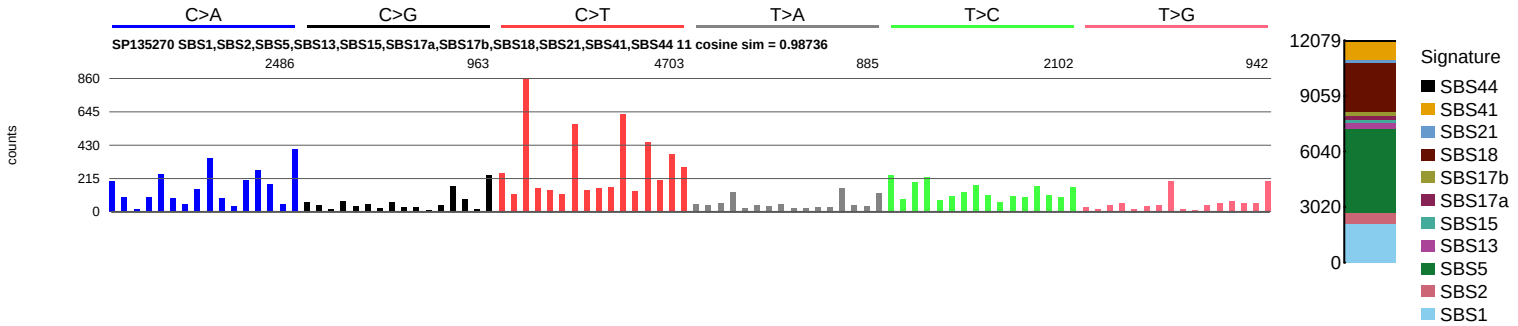

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.9874
2608
1026
4544
996
1749
1153
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

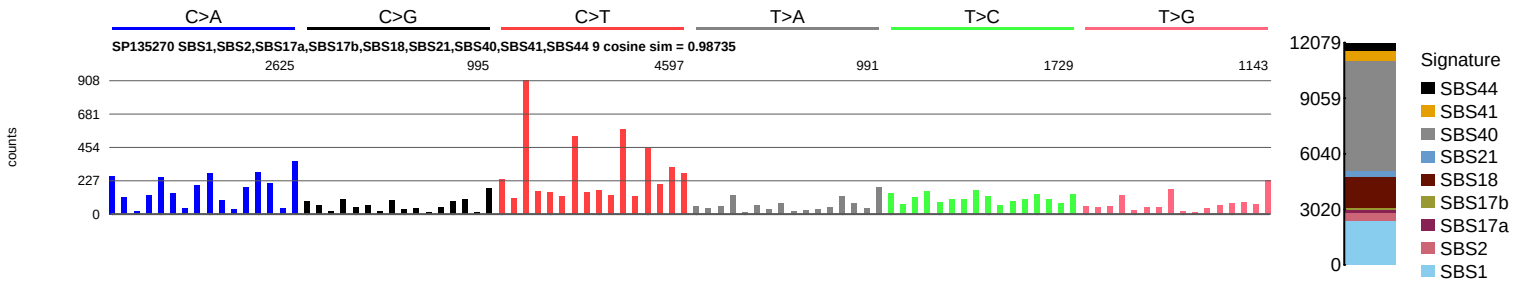

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.9874
2645
1134
4581
864
1747
1107
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

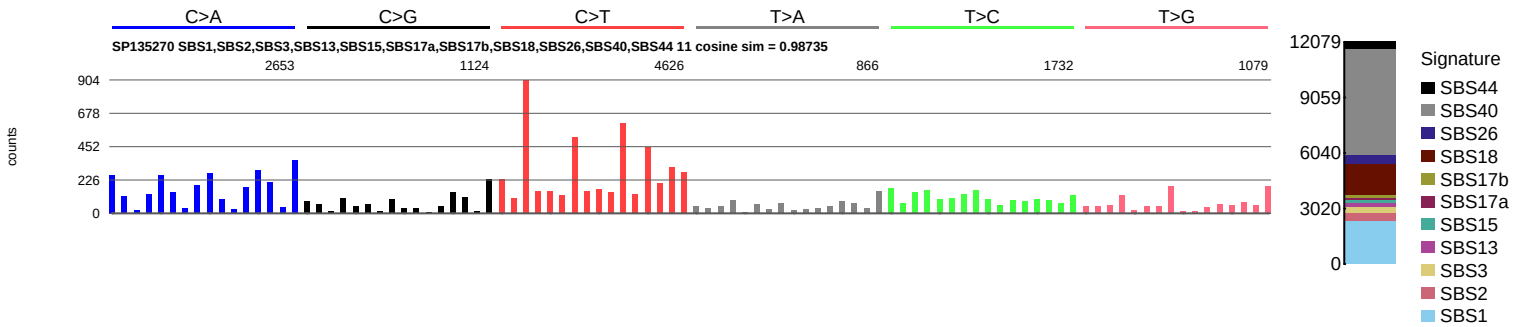

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98738
2622
989
4588
986
1750
1142
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

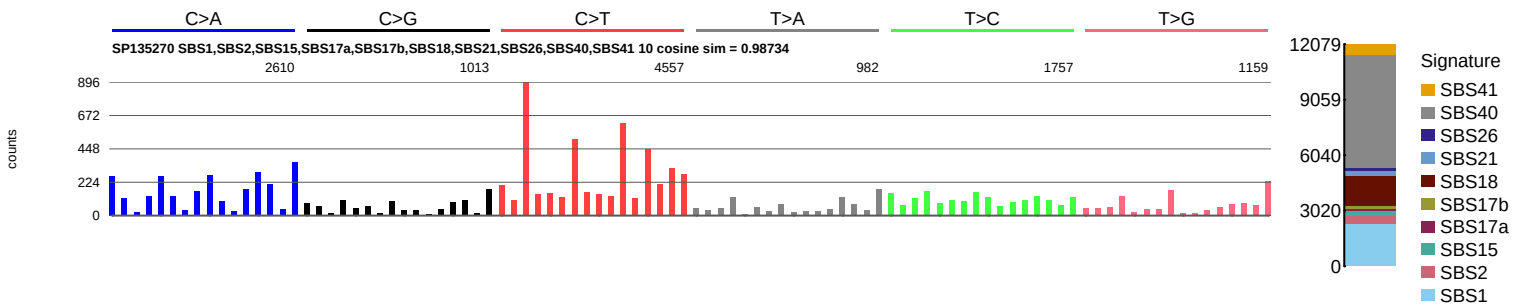

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 11 cosine sim = 0.98737
2644
1136
4579
867
1747
1104
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

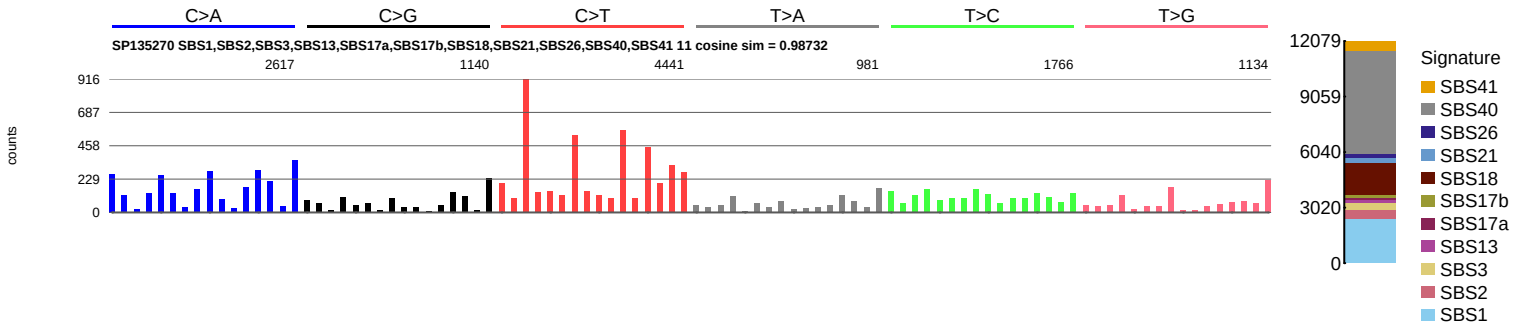

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98737
2614
1038
4561
1006
1706
1159
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

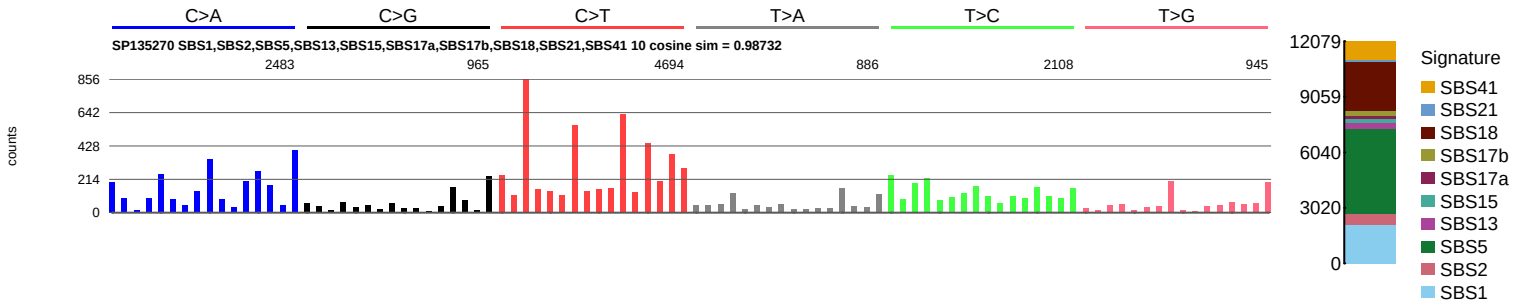

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98736
2486
963
4703
885
2102
942
counts
860
645
430
215
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

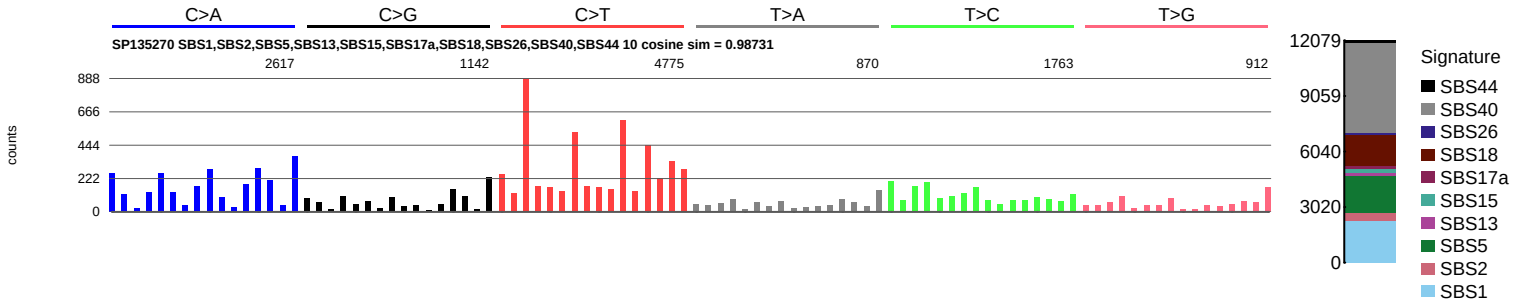

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98735
2625
995
4597
991
1729
1143
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

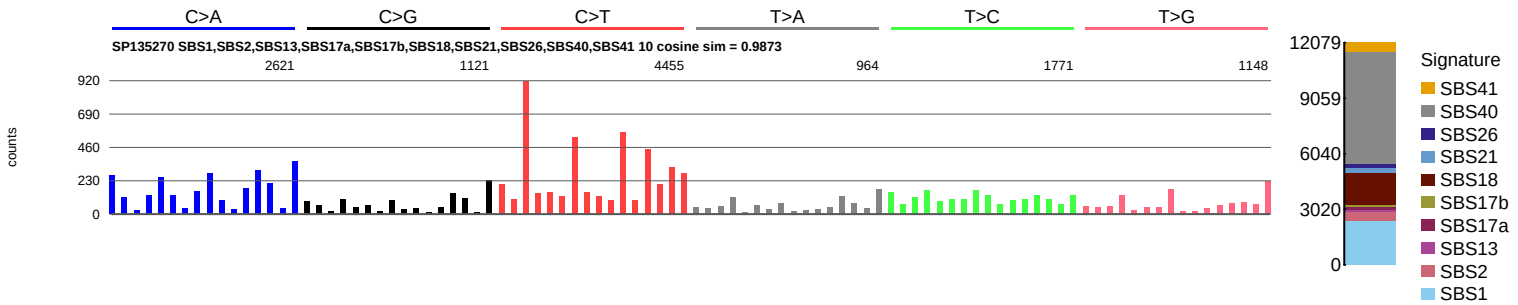

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98735
2653
1124
4626
866
1732
1079
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

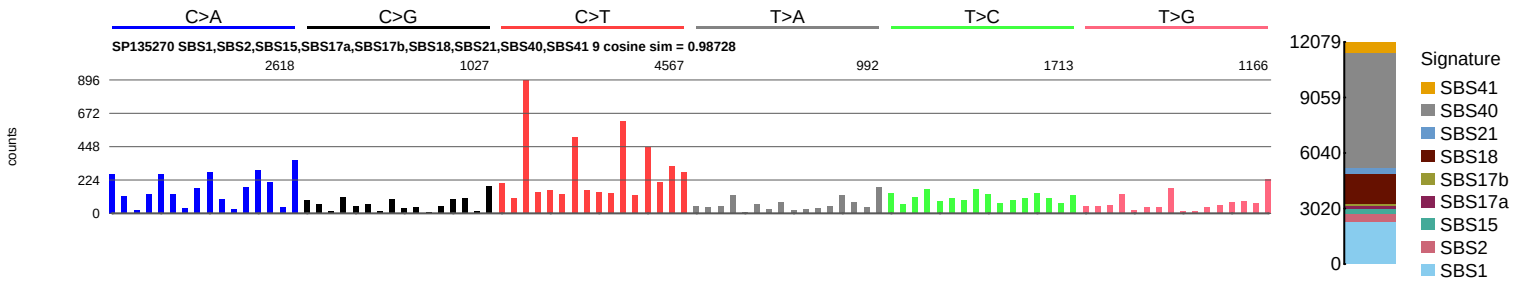

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98734
2610
1013
4557
982
1757
1159
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

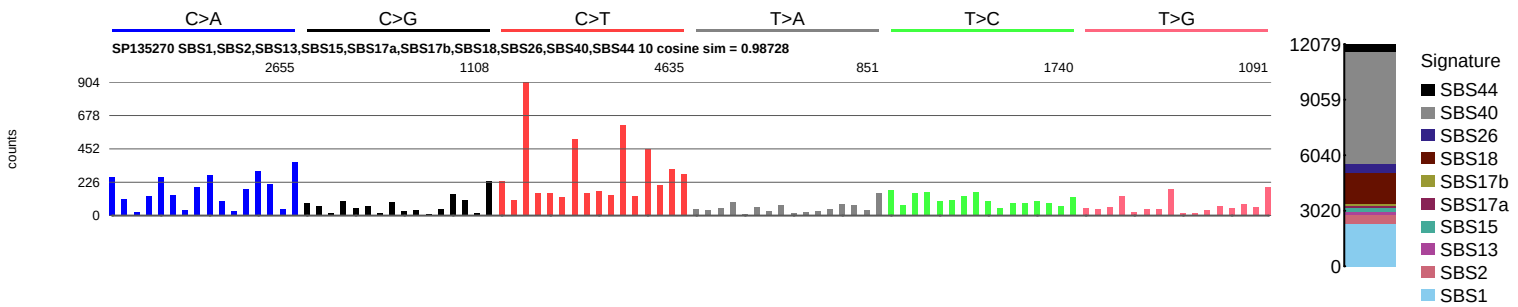

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98732
2617
1140
4441
981
1766
1134
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

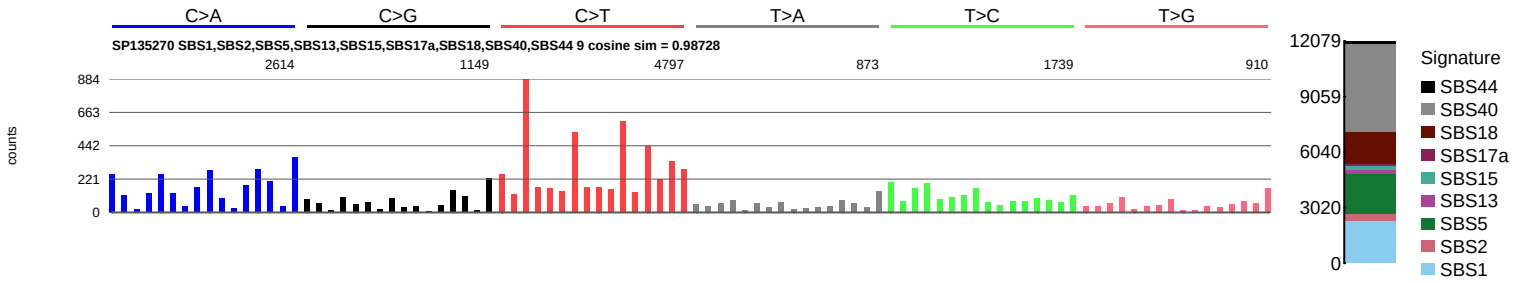

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98732
2483
965
4694
886
2108
945
counts
856
642
428
214
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

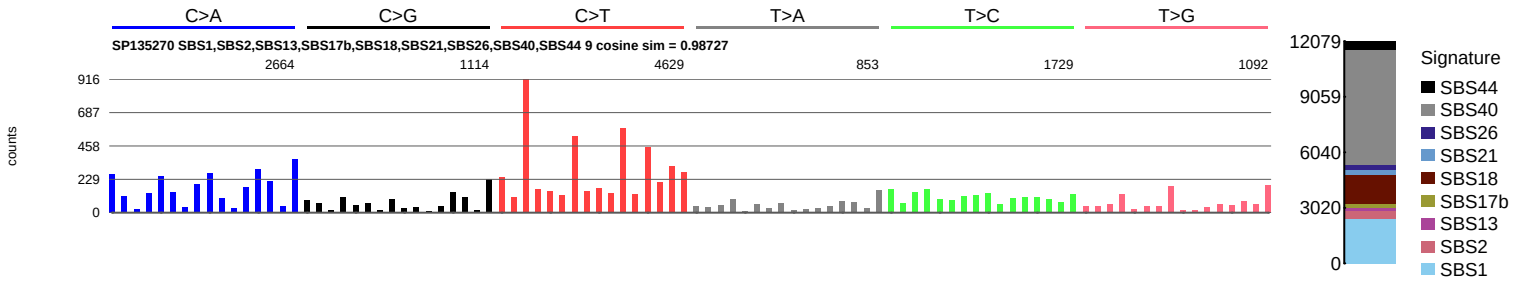

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98731
2617
1142
4775
870
1763
912
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

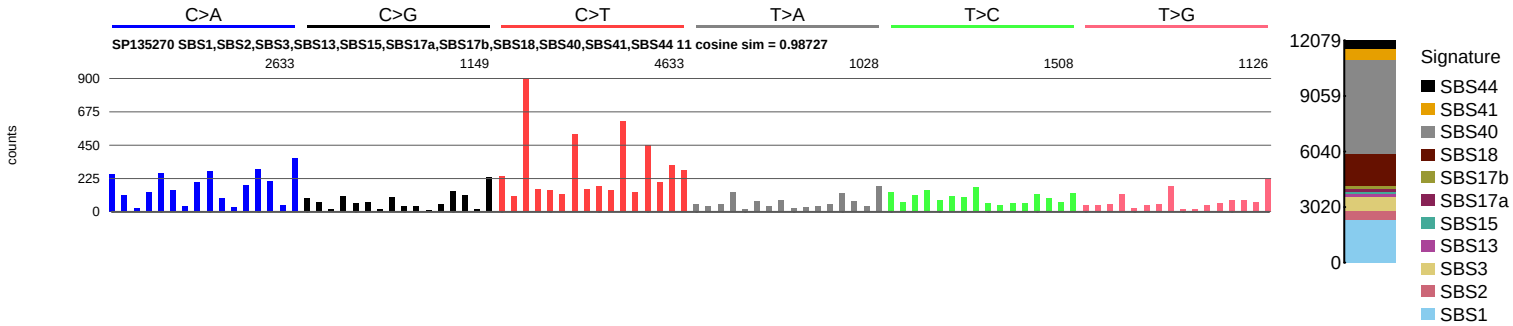

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.9873
2621
1121
4455
964
1771
1148
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

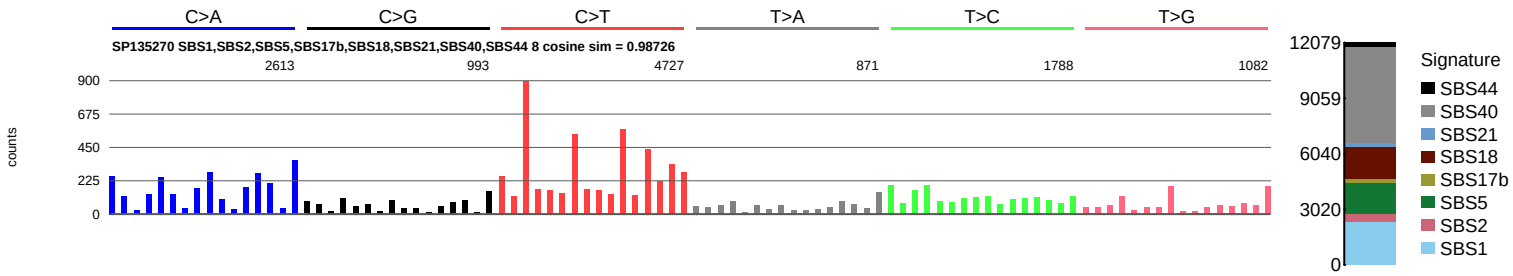

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98728
2618
1027
4567
992
1713
1166
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

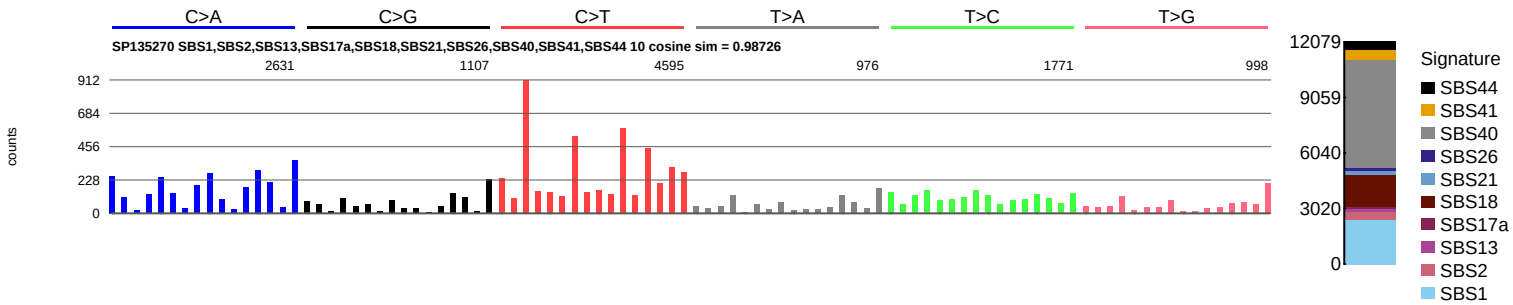

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98728
2655
1108
4635
851
1740
1091
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98728
2614
1149
4797
873
1739
910
884
663
442
221
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98727
2664
1114
4629
853
1729
1092
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98727
2633
1149
4633
1028
1508
1126
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98726
2613
993
4727
871
1788
1082
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1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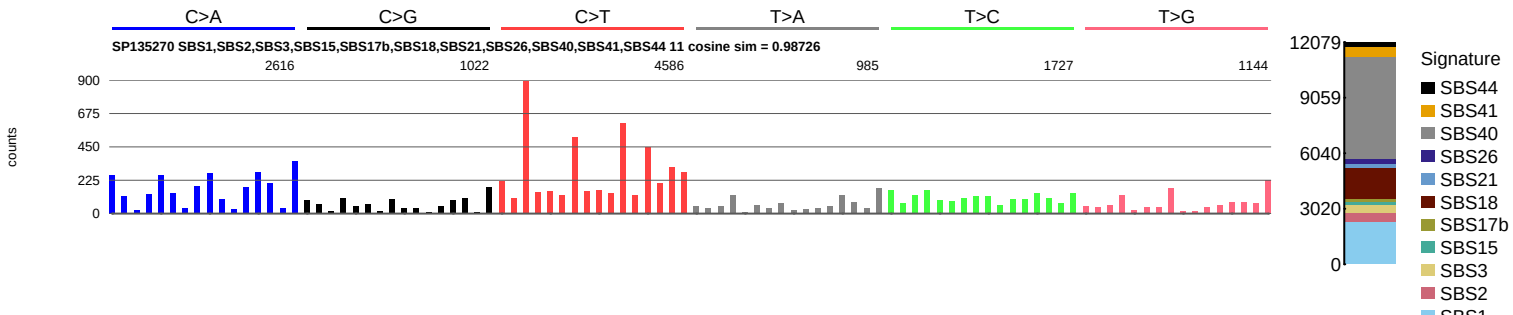
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98726
2631
1107
4595
976
1771
998
912
684
456
228
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1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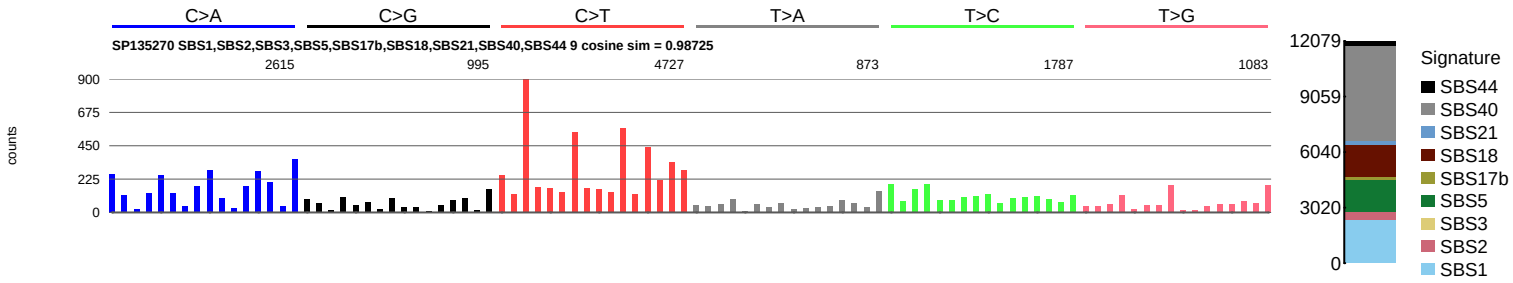
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98726
2616
1022
4586
985
1727
1144
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2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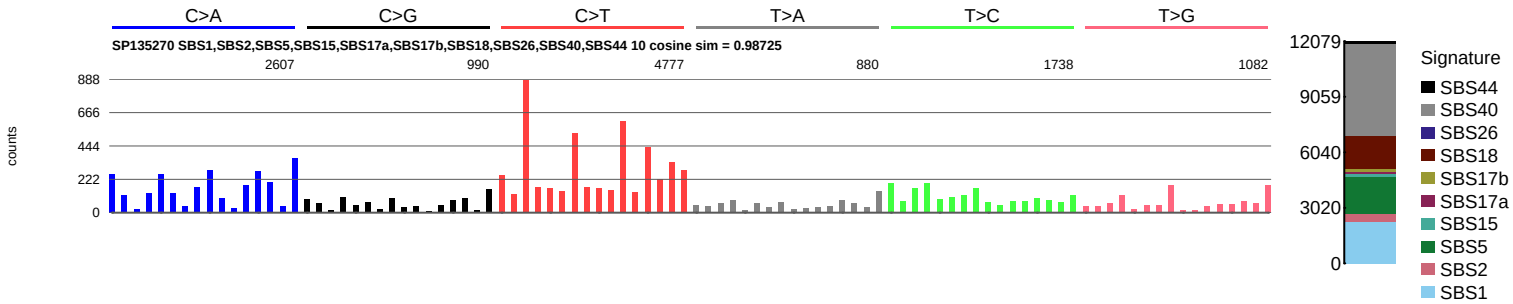
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98725
2615
995
4727
873
1787
1083
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98725
2607
990
4777
880
1738
1082
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
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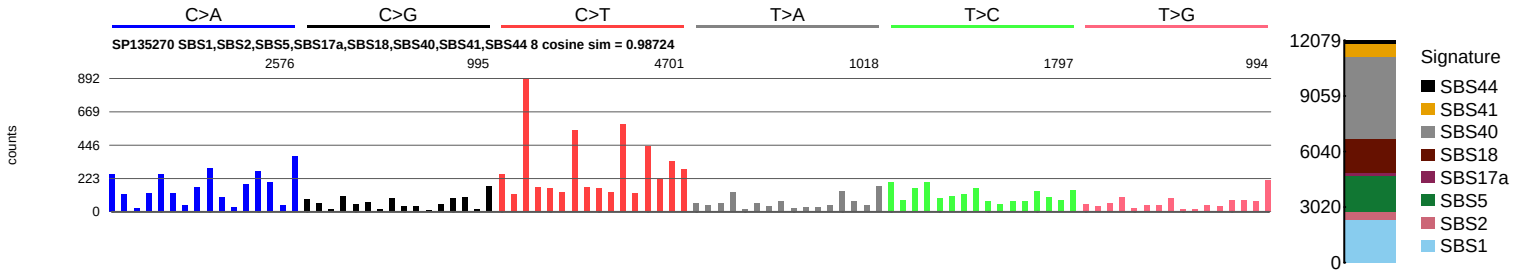
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
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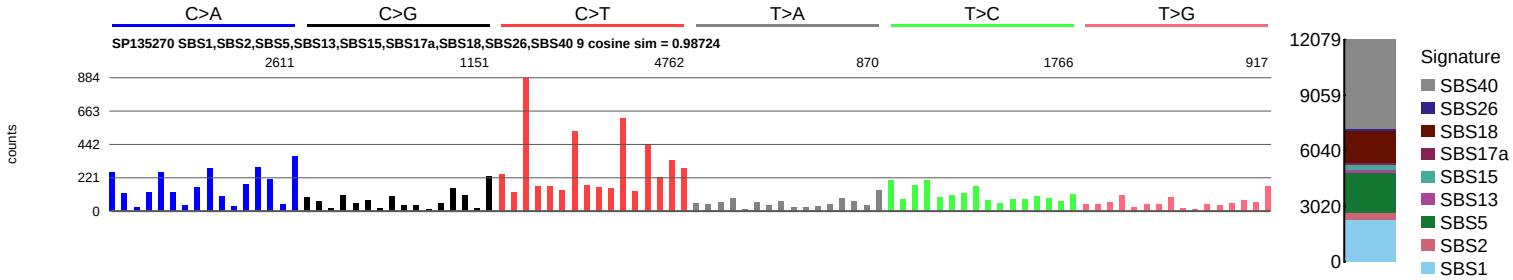
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98724
2576
995
4701
1018
1797
994
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
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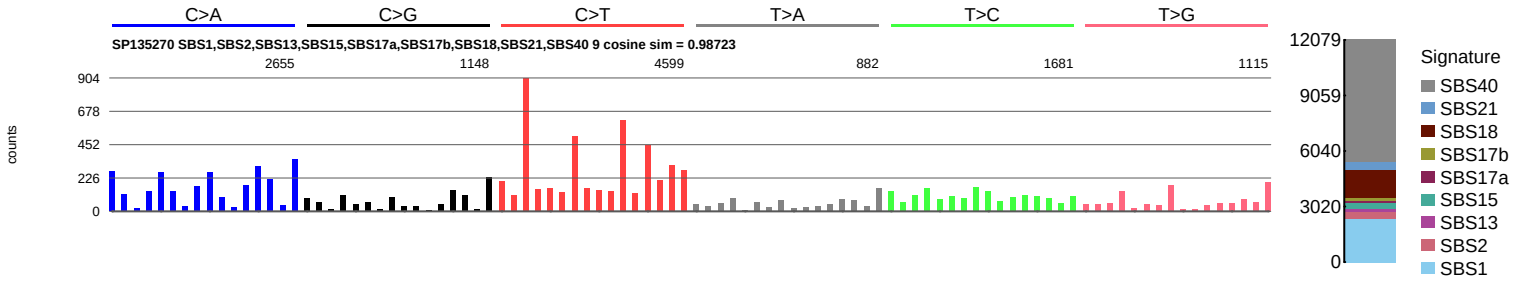
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40 9 cosine sim = 0.98724
2611
1151
4762
870
1766
917
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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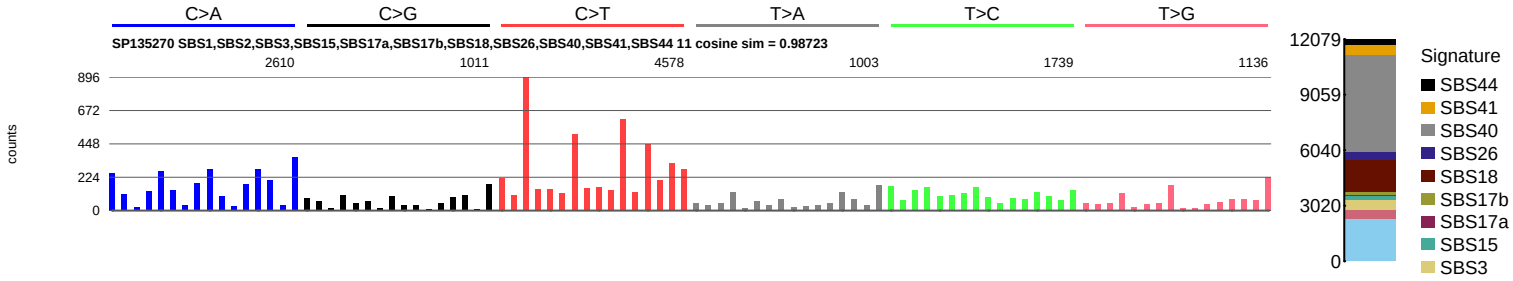
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98723
2655
1148
4599
882
1681
1115
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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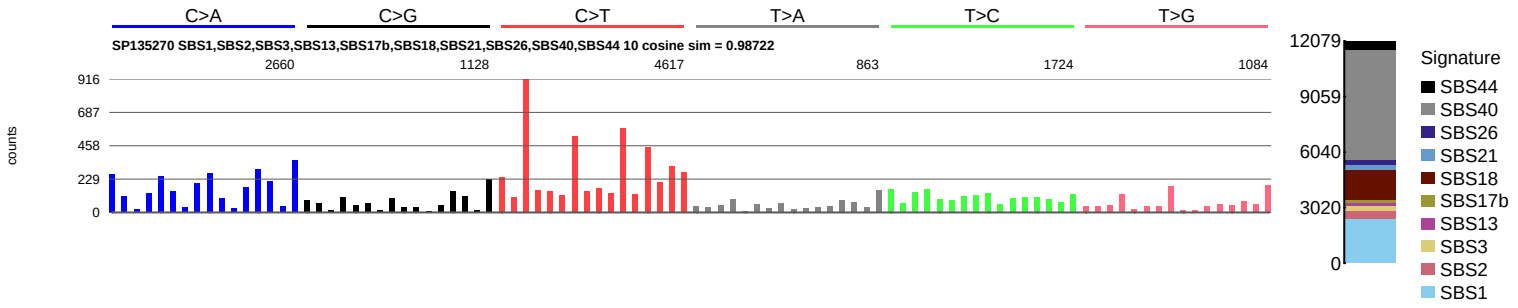
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98723
2610
1011
4578
1003
1739
1136
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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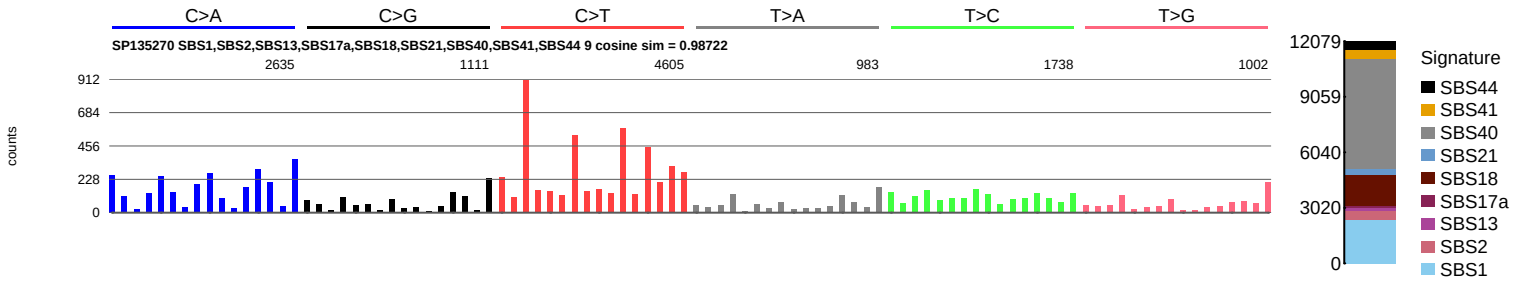
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98722
2660
1128
4617
863
1724
1084
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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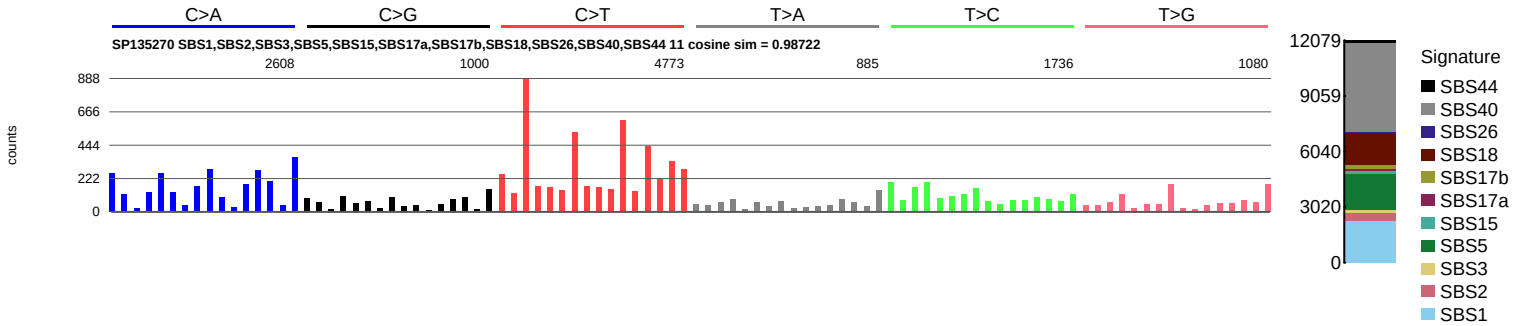
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98722
2635
1111
4605
983
1738
1002
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1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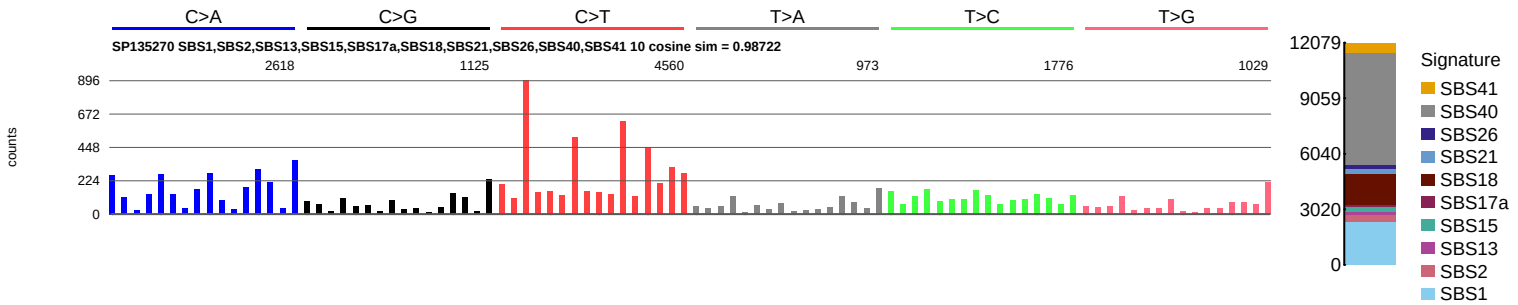
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98722
2608
1000
4773
885
1736
1080
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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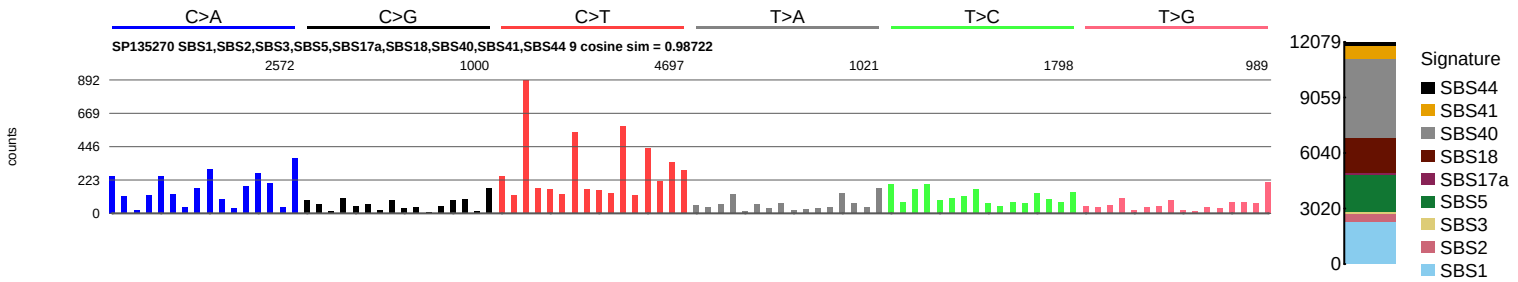
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98722
2618
1125
4560
973
1776
1029
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1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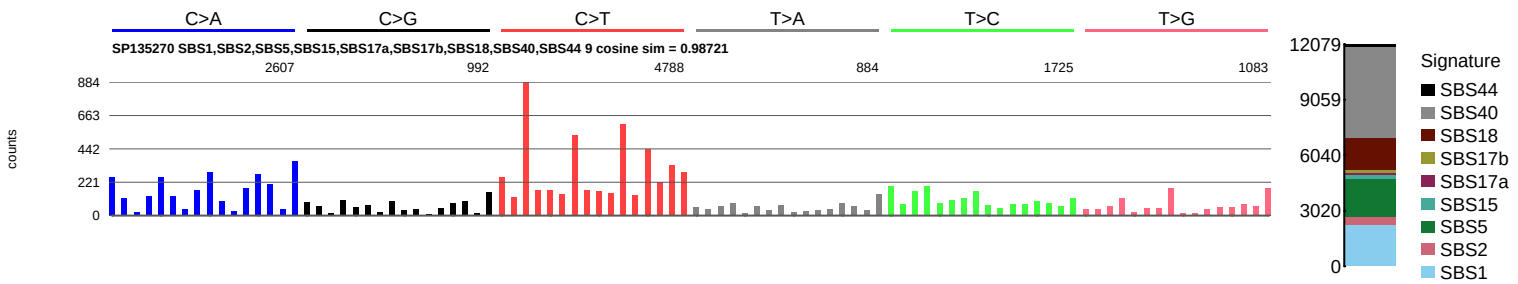
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98722
2572
1000
4697
1021
1798
989
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1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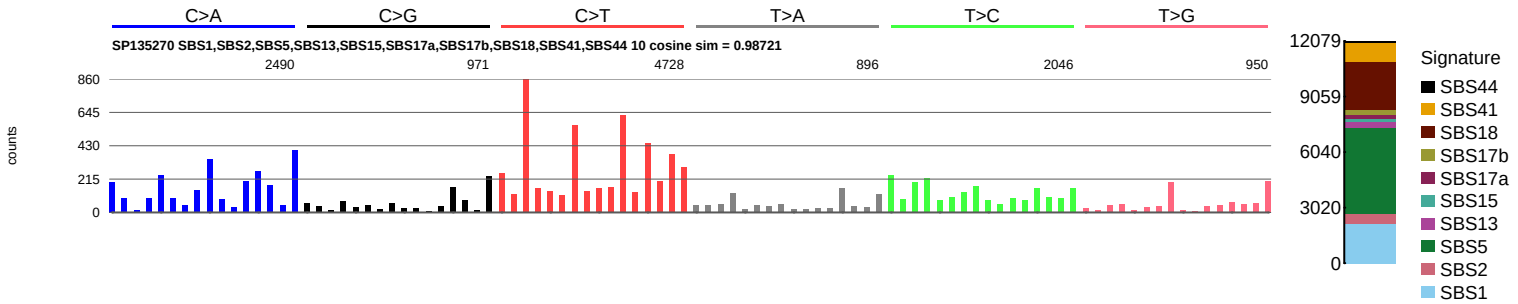
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98721
2607
992
4788
884
1725
1083
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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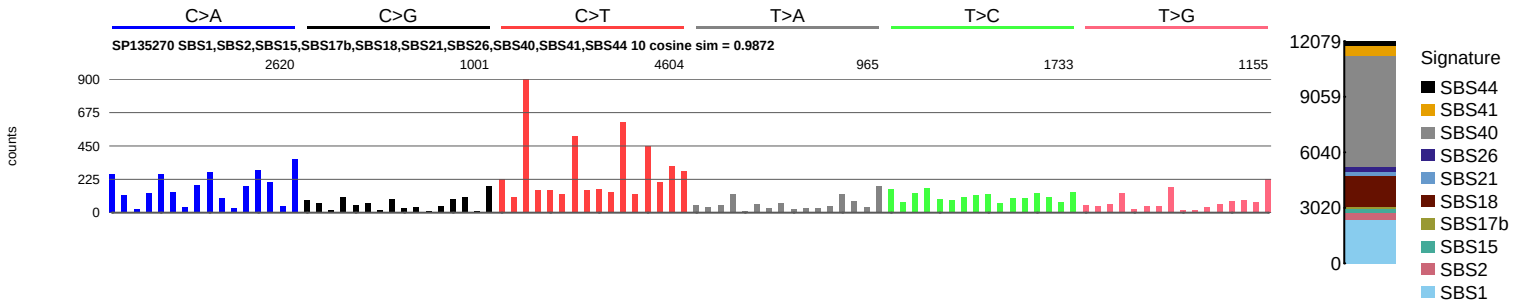
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98721
2490
971
4728
896
2046
950
counts
860
645
430
215
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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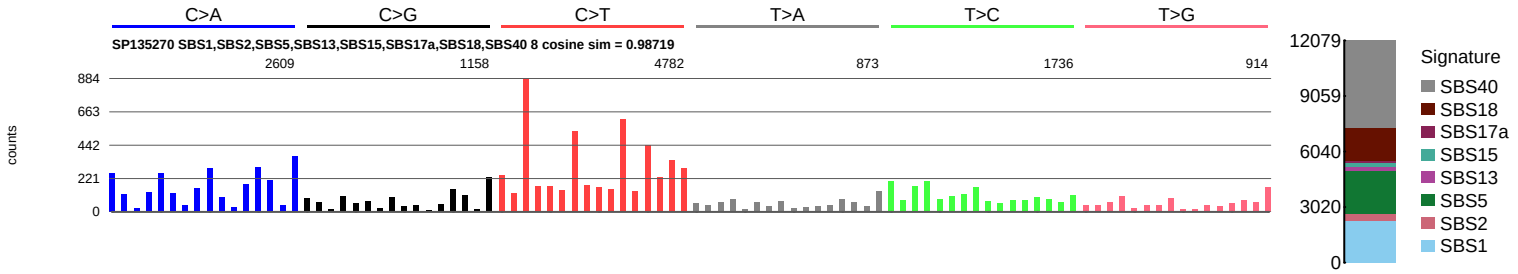
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9872
2620
1001
4604
965
1733
1155
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1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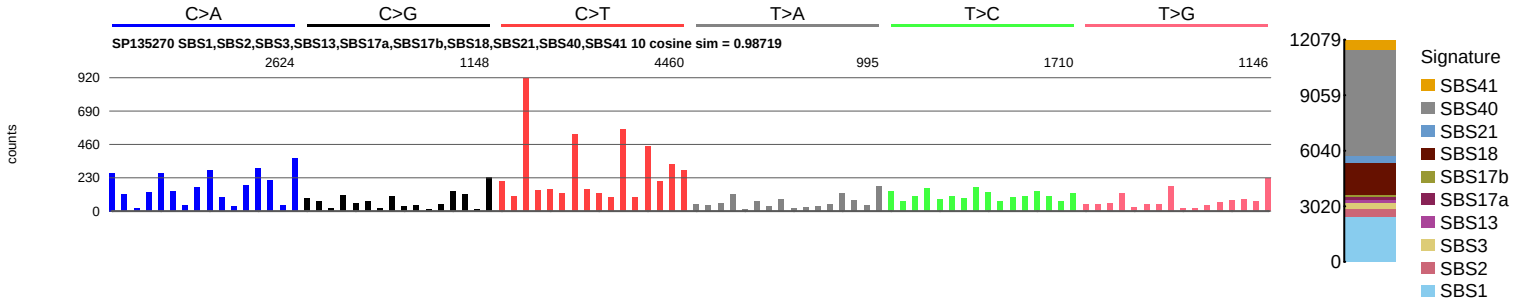
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40 8 cosine sim = 0.98719
2609
1158
4782
873
1736
914
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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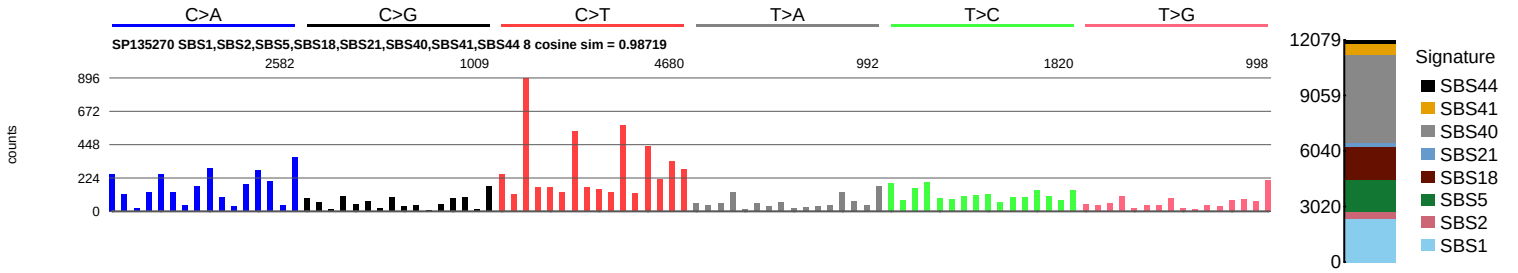
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98719
2624
1148
4460
995
1710
1146
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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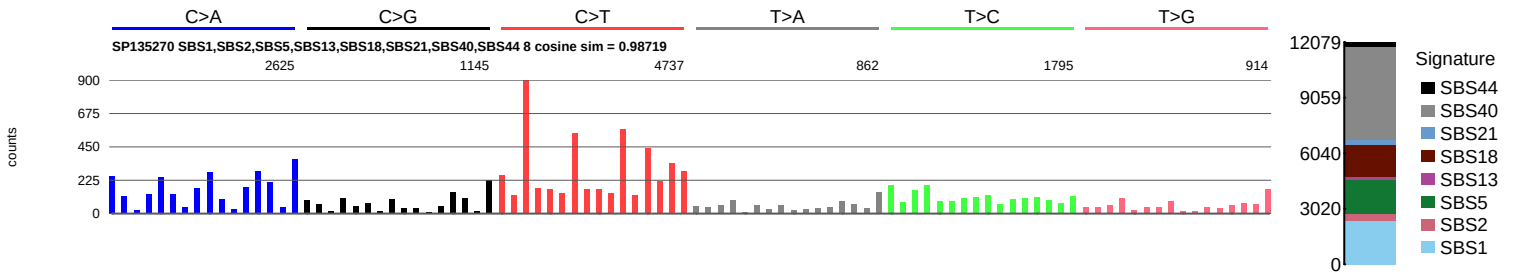
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98719
2582
1009
4680
992
1820
998
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98719
2625
1145
4737
862
1795
914
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
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SBS21
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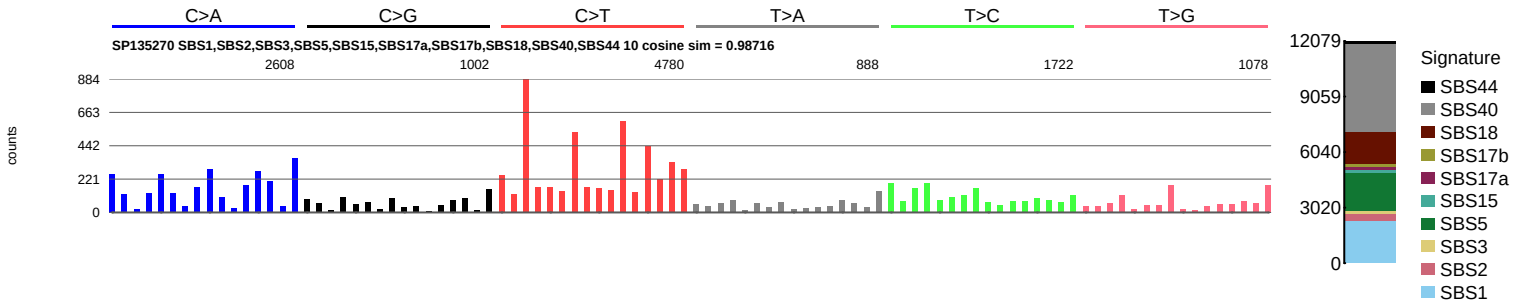
SBS5
SBS2
SBS1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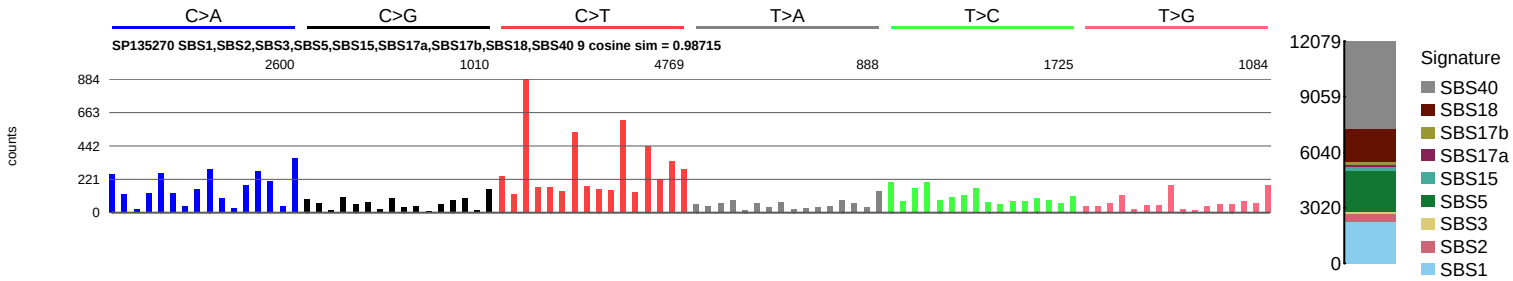
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98716
2608
1002
4780
888
1722
1078
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98715
2600
1010
4769
888
1725
1084
counts
884
663
442
221
0
12079
9059
6040
3020
0
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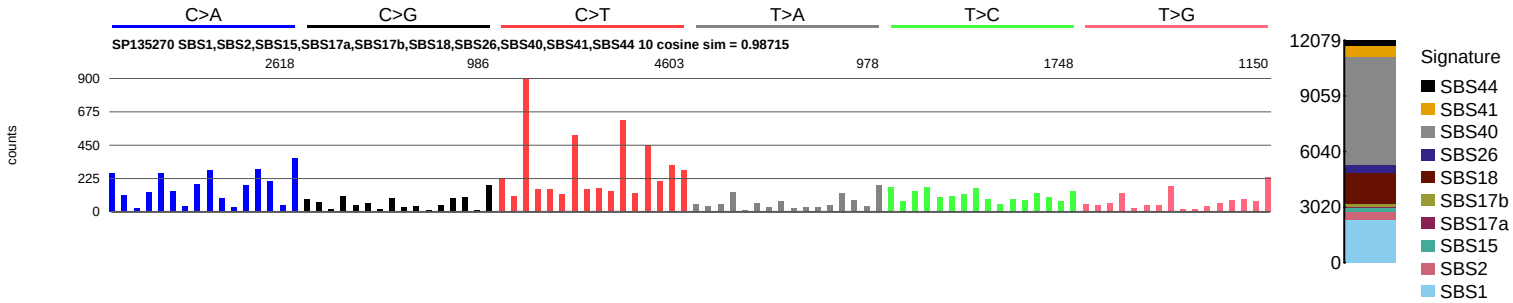
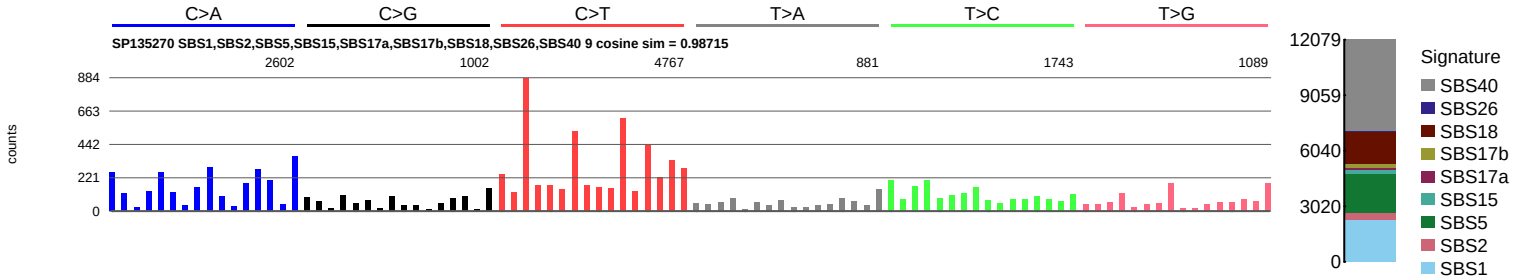
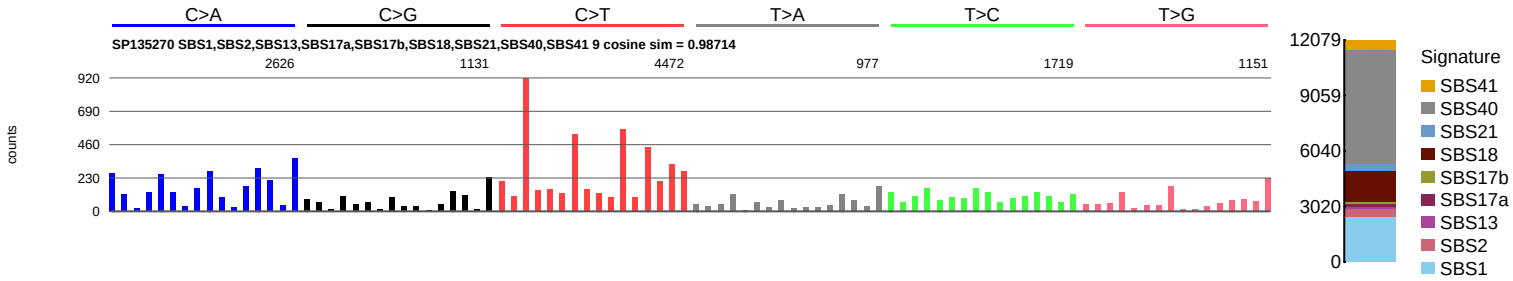
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98715
2618
986
4603
978
1748
1150
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98715
2602
1002
4767
881
1743
1089
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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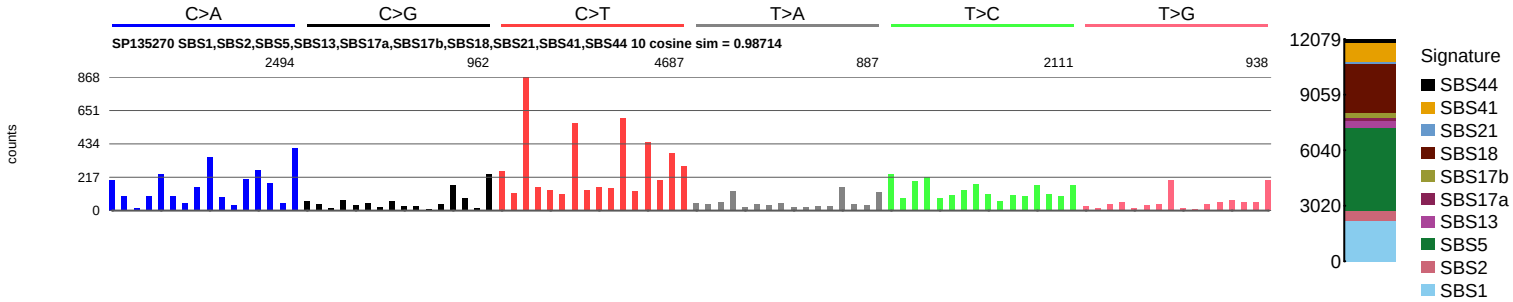
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98714
2626
1131
4472
977
1719
1151
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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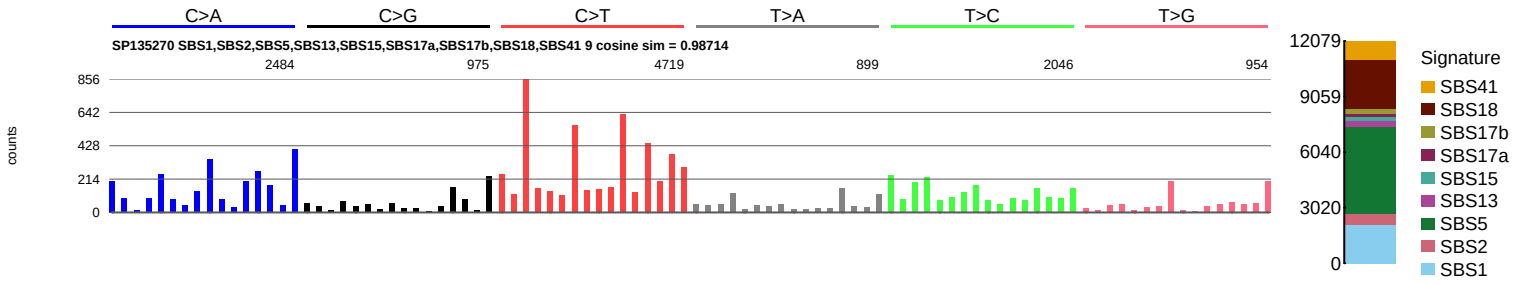
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98714
2494
962
4687
887
2111
938
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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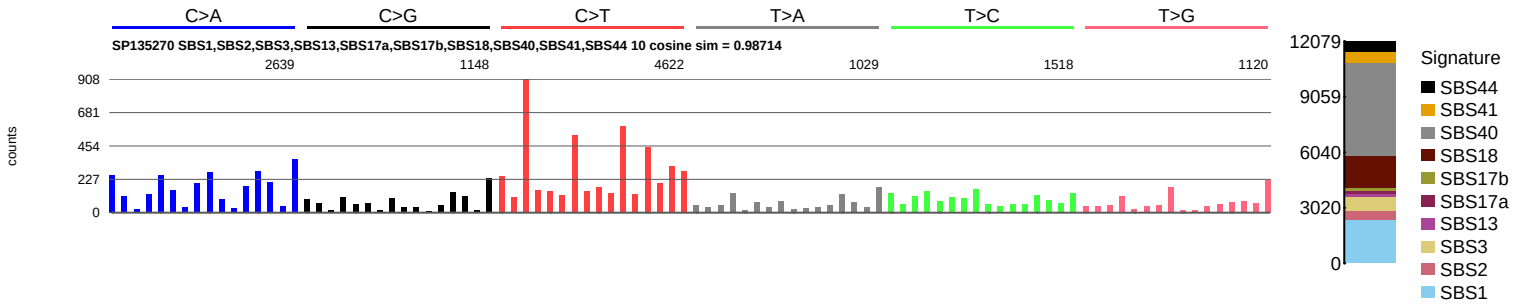
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98714
2484
975
4719
899
2046
954
856
642
428
214
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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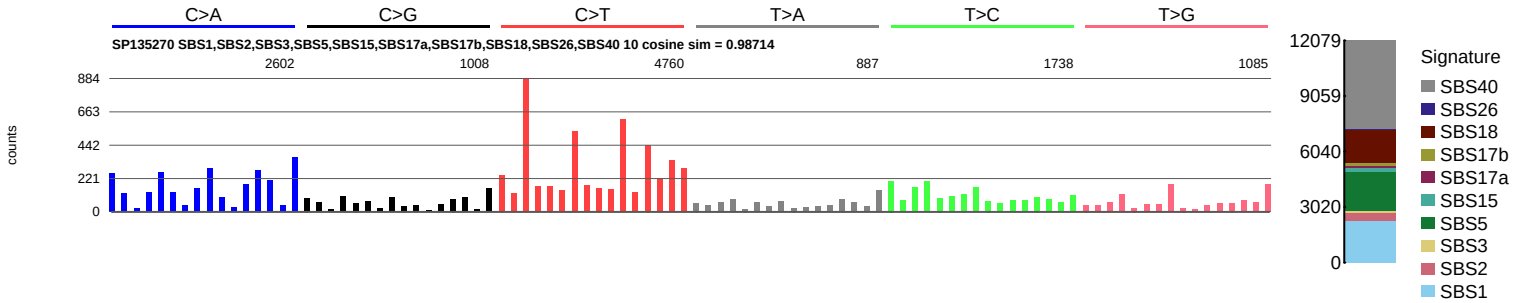
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98714
2639
1148
4622
1029
1518
1120
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
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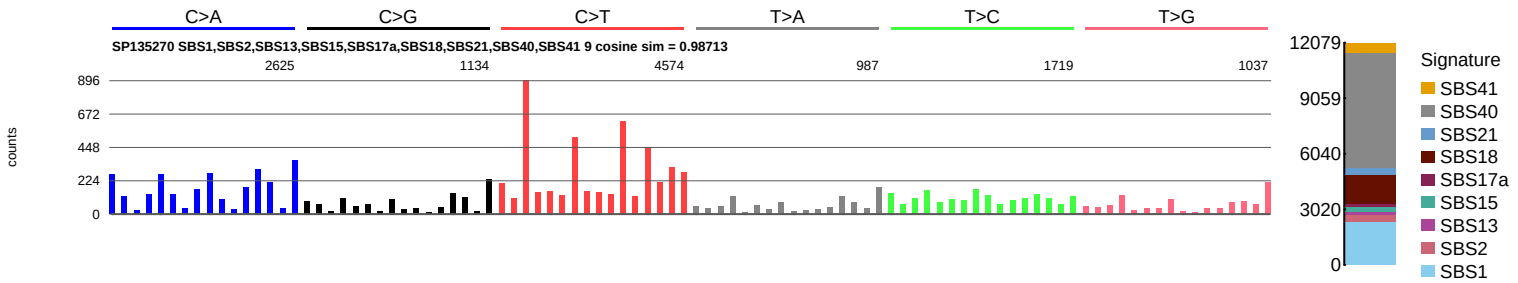
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98714
2602
1008
4760
887
1738
1085
884
663
442
221
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98713
2625
1134
4574
987
1719
1037
896
672
448
224
0
counts
12079
9059
6040
3020
0
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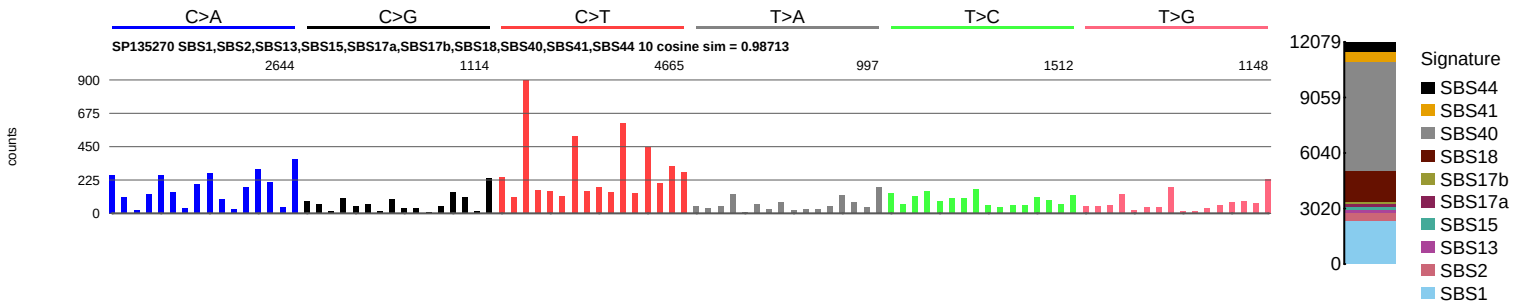
SBS2
SBS1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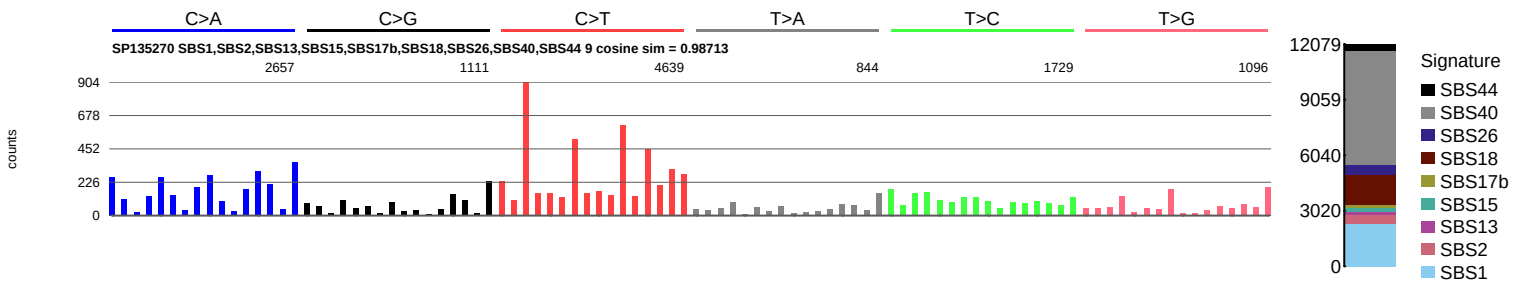
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98713
2644
1114
4665
997
1512
1148
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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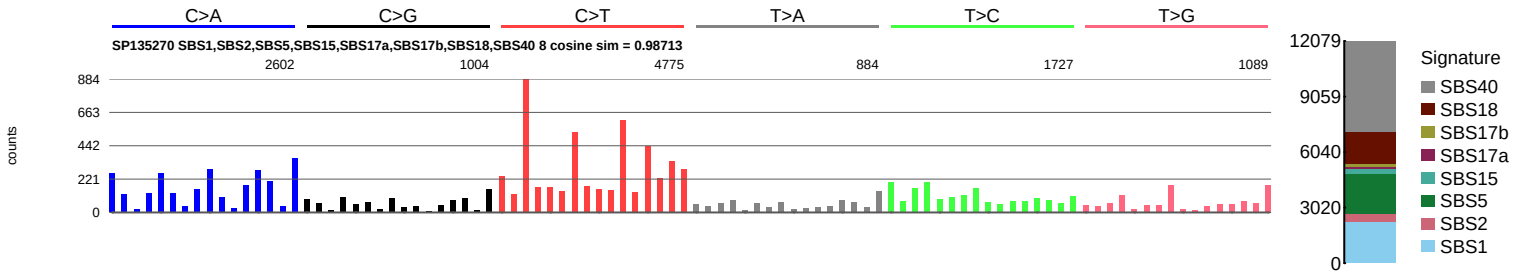
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98713
2657
1111
4639
844
1729
1096
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98713
2602
1004
4775
884
1727
1089
884
663
442
221
0
counts
12079
9059
6040
3020
0
Signature
SBS40
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SBS17b
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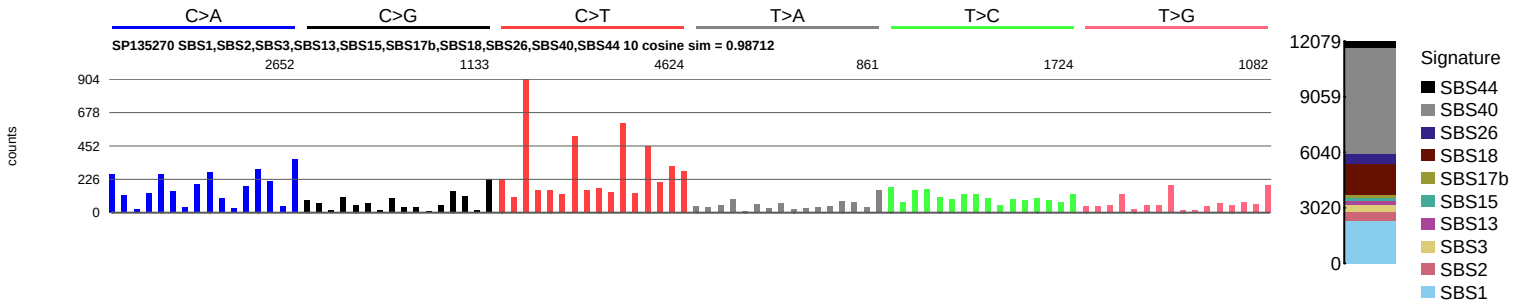
SBS5
SBS2
SBS1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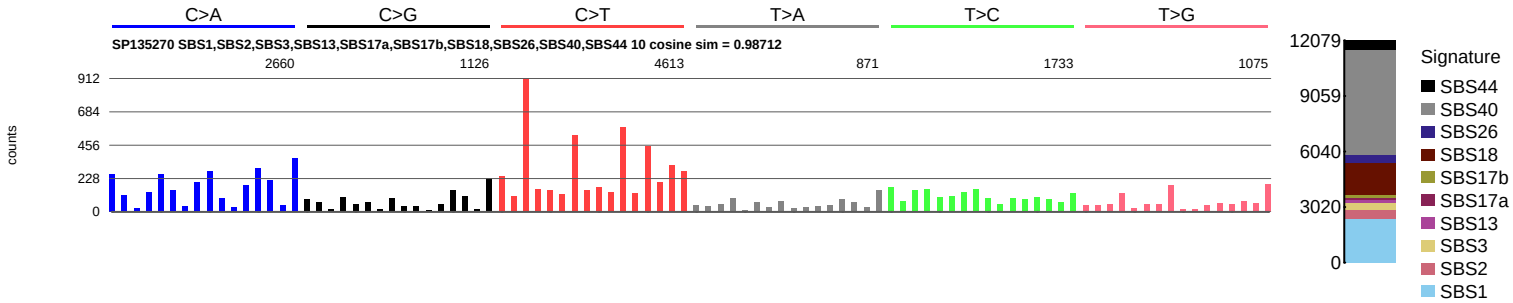
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98712
2652
1133
4624
861
1724
1082
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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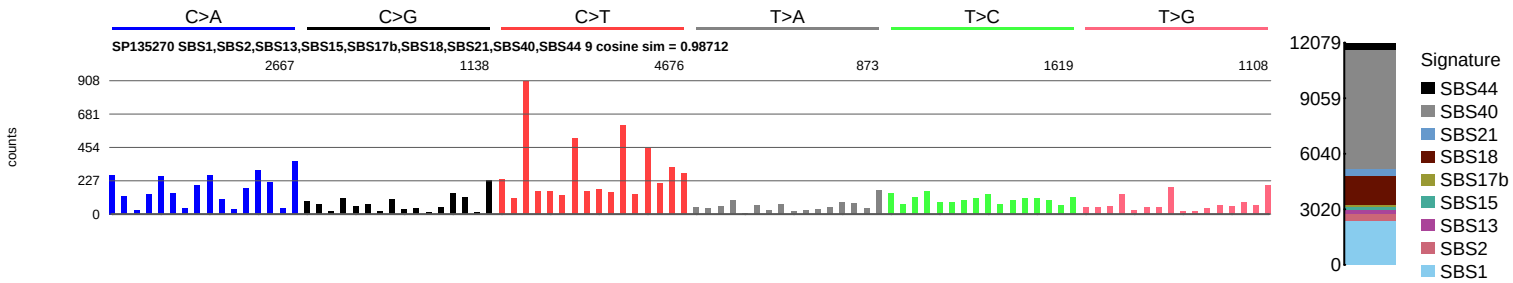
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98712
2660
1126
4613
871
1733
1075
912
684
456
228
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98712
2667
1138
4676
873
1619
1108
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
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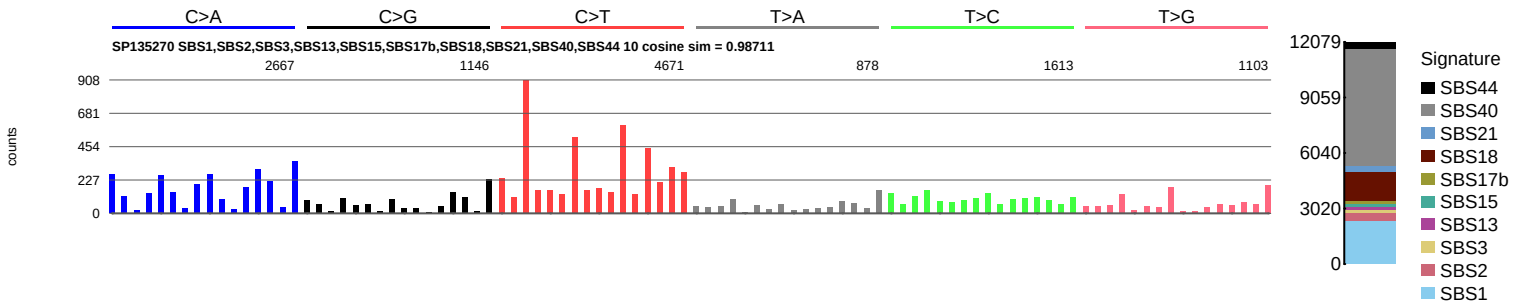
SBS2
SBS1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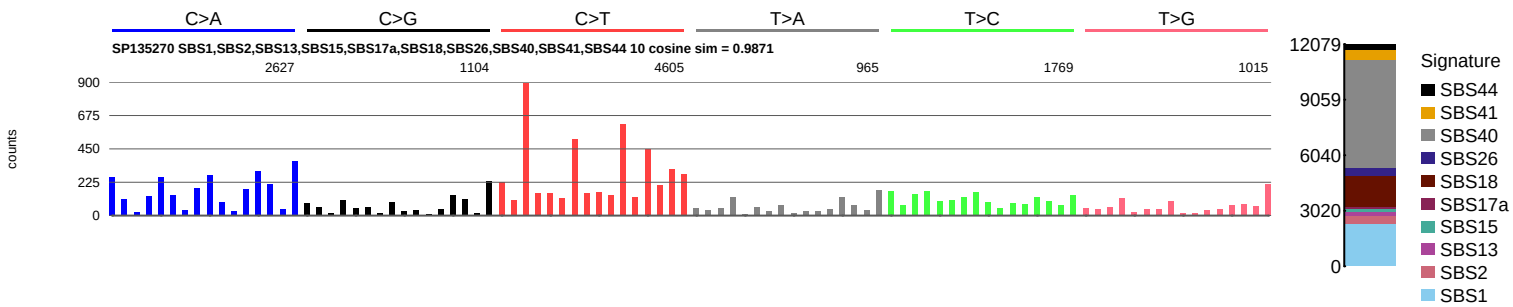
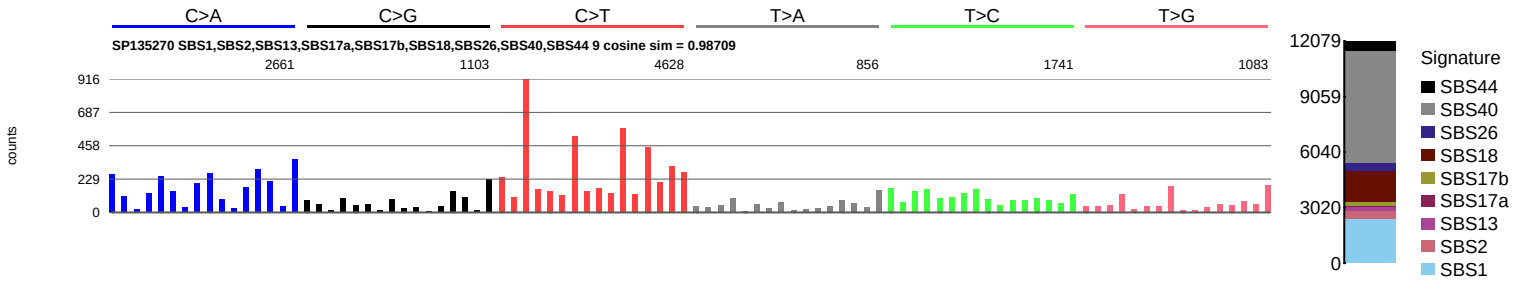
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98711
2667
1146
4671
878
1613
1103
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9871
2627
1104
4605
965
1769
1015
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1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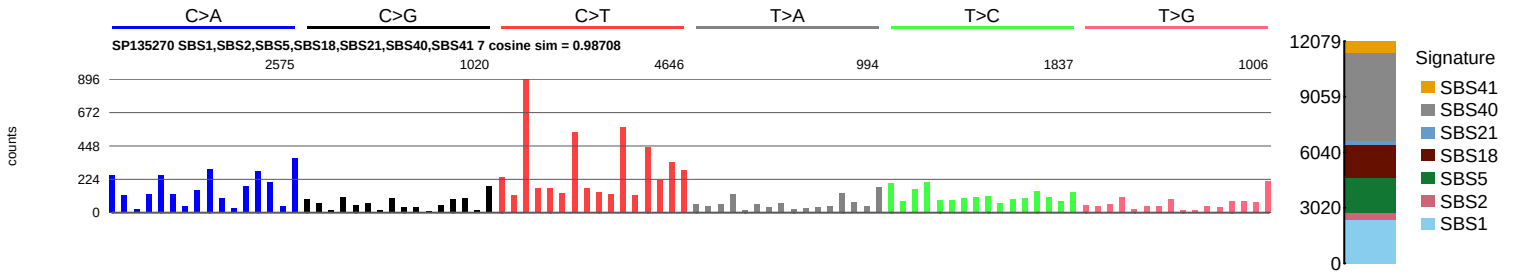
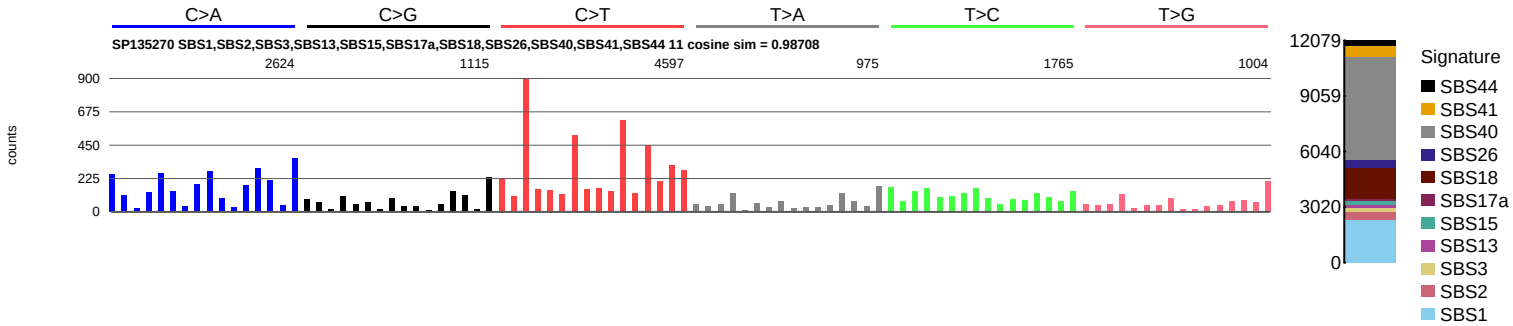
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98709
2661
1103
4628
856
1741
1083
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98708
2575
1020
4646
994
1837
1006
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1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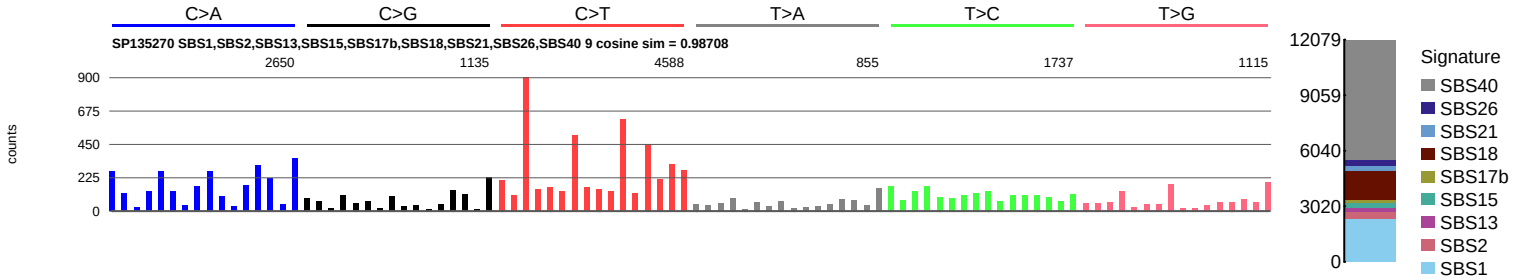
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98708
2624
1115
4597
975
1765
1004
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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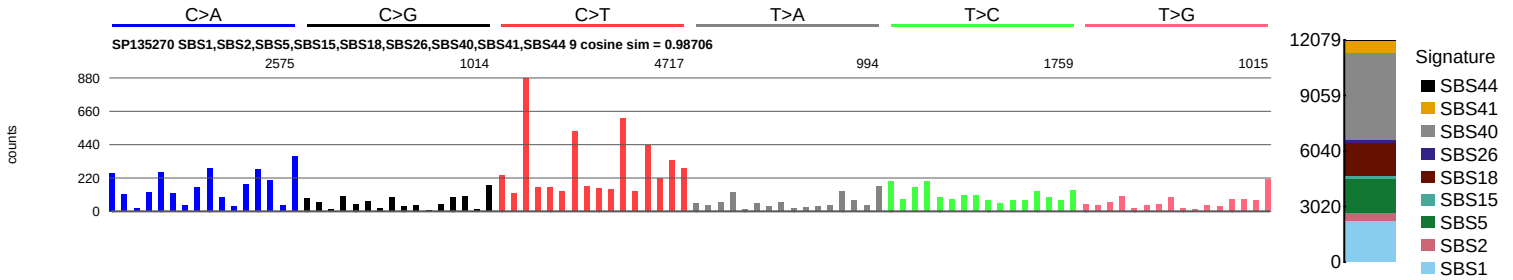
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98708
2650
1135
4588
855
1737
1115
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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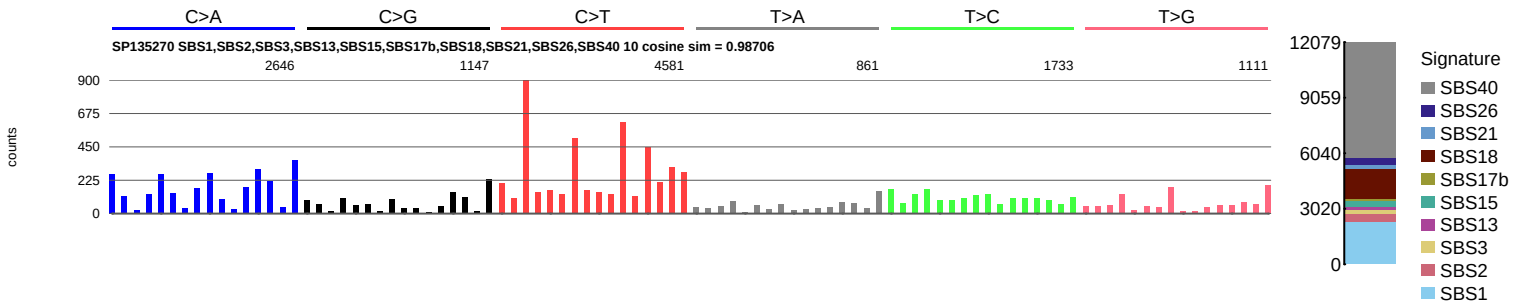
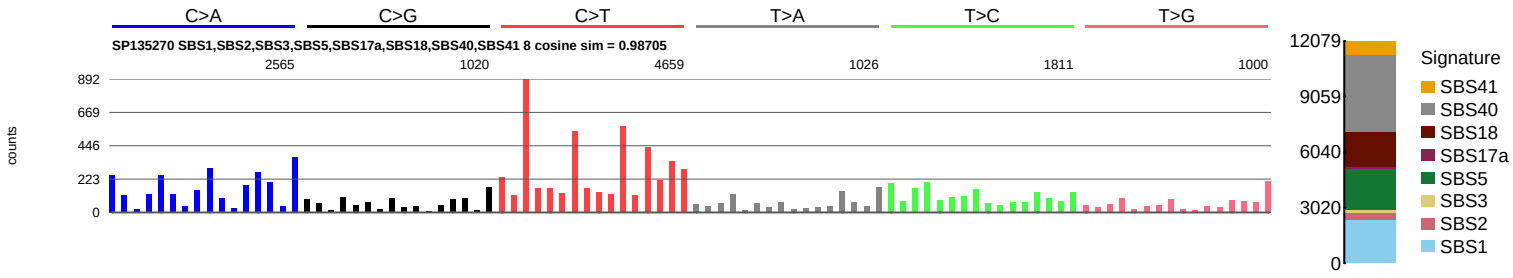
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98706
2575
1014
4717
994
1759
1015
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98706
2646
1147
4581
861
1733
1111
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98705
2565
1020
4659
1026
1811
1000
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
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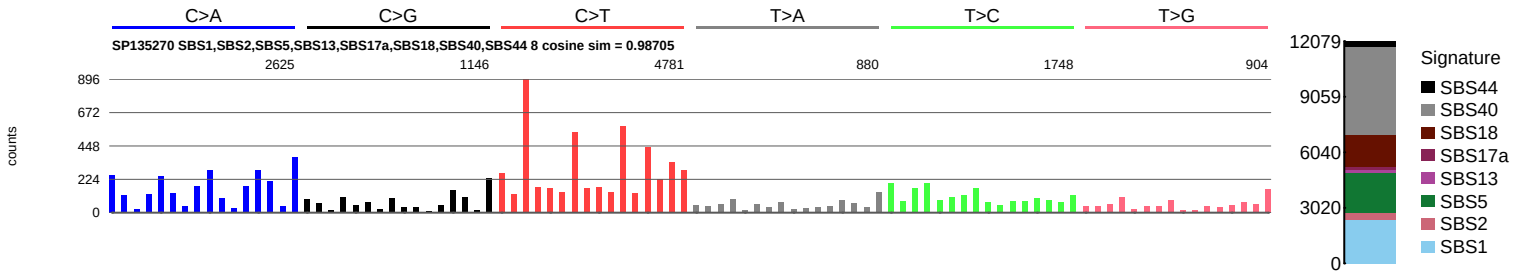
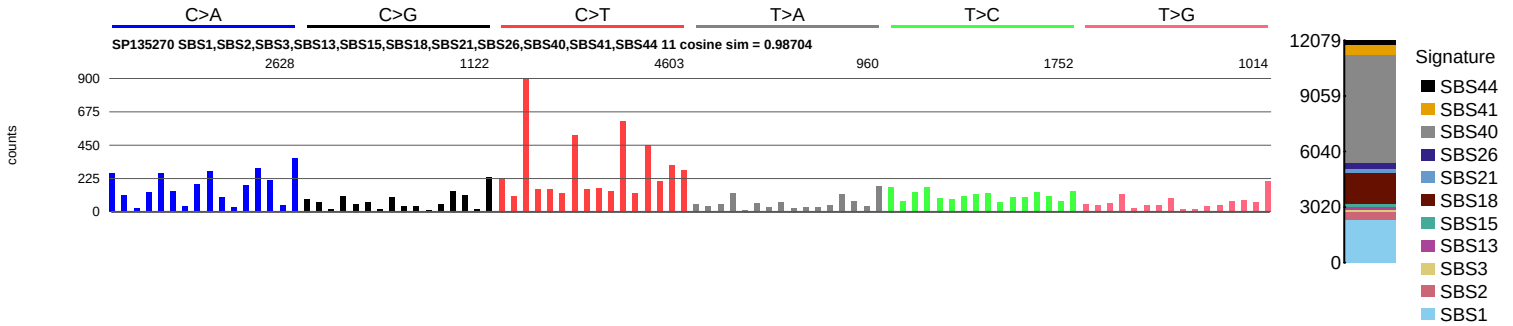
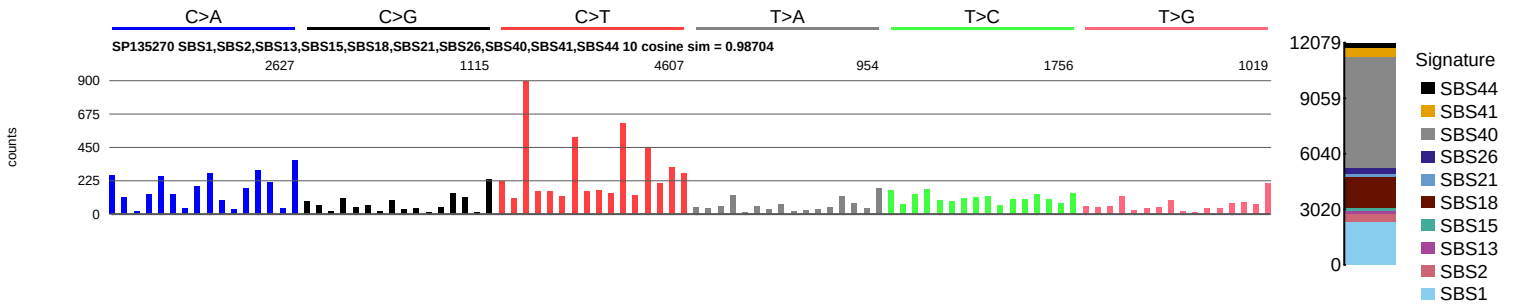
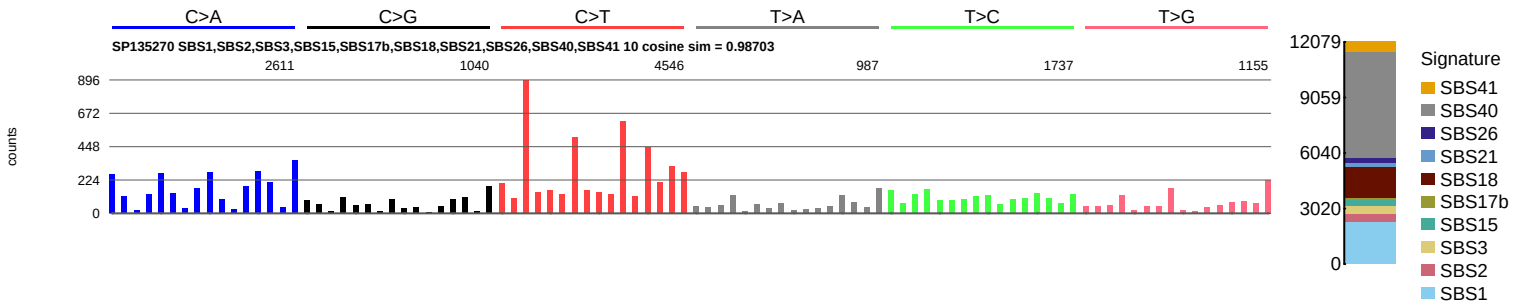
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98705
2625
1146
4781
880
1748
904
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98704
2628
1122
4603
960
1752
1014
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98704
2627
1115
4607
954
1756
1019
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1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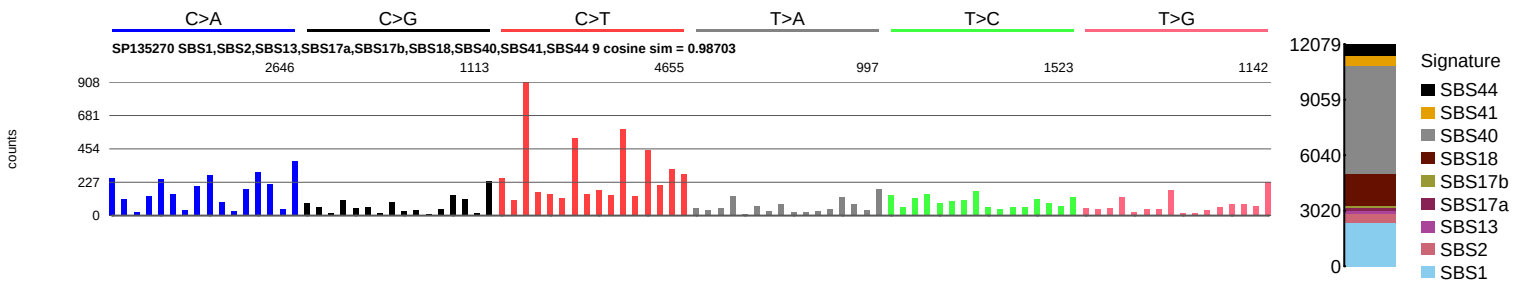
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98703
2611
1040
4546
987
1737
1155
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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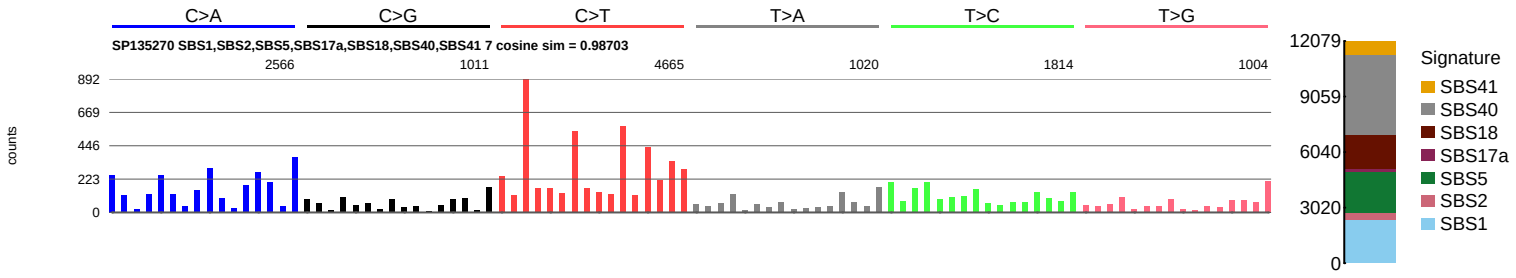
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98703
2646
1113
4655
997
1523
1142
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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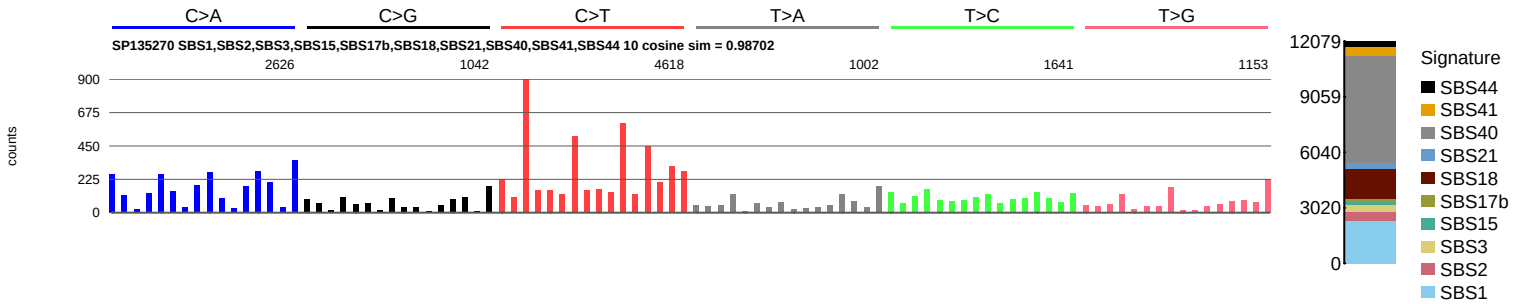
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98703
2566
1011
4665
1020
1814
1004
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1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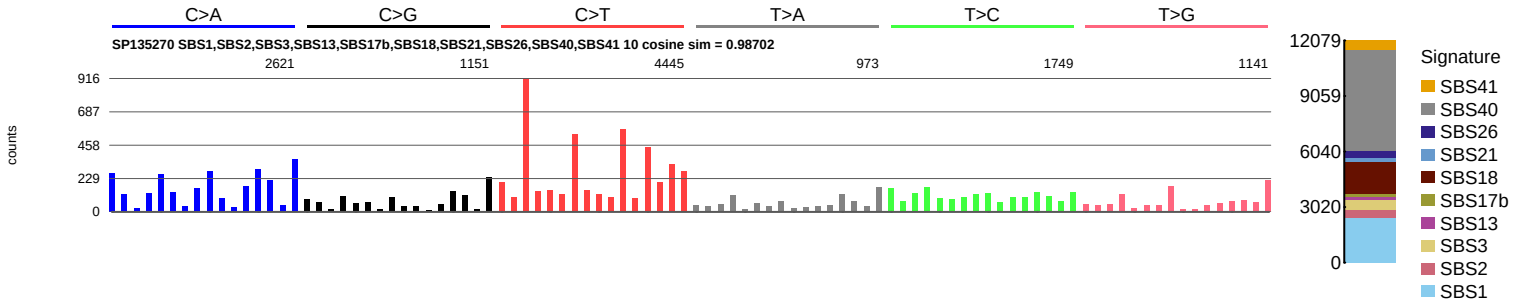
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98702
2626
1042
4618
1002
1641
1153
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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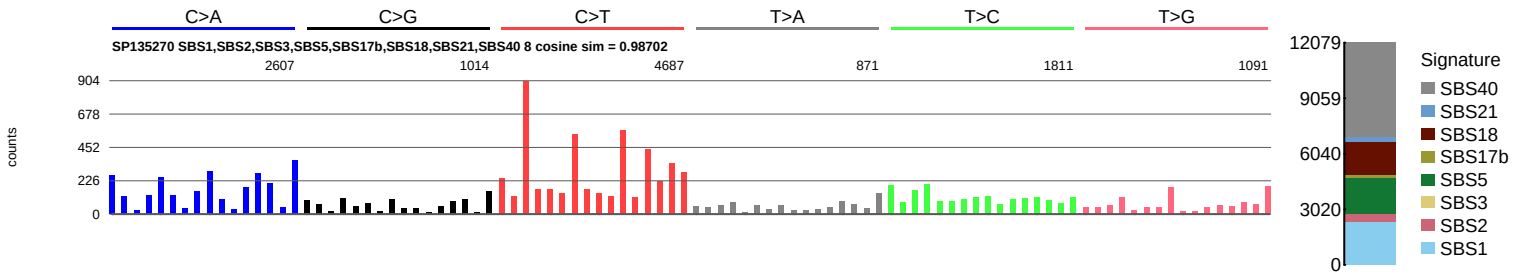
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98702
2621
1151
4445
973
1749
1141
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98702
2607
1014
4687
871
1811
1091
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
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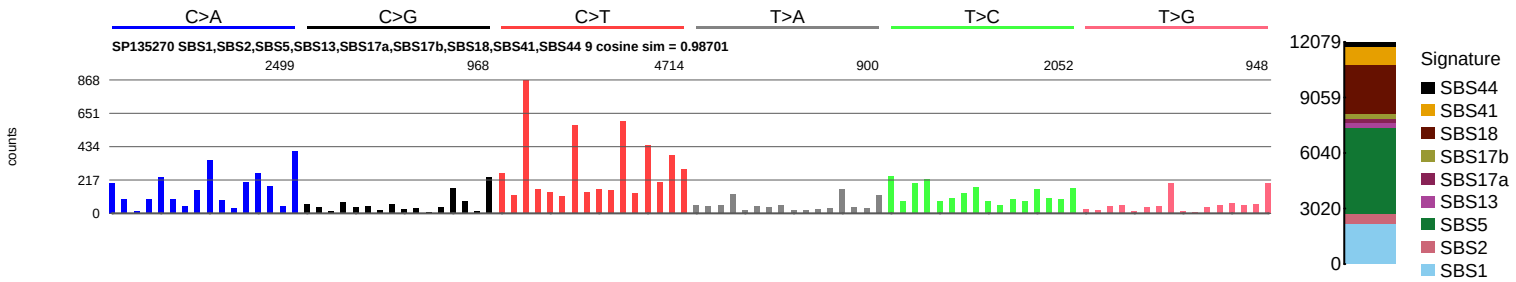
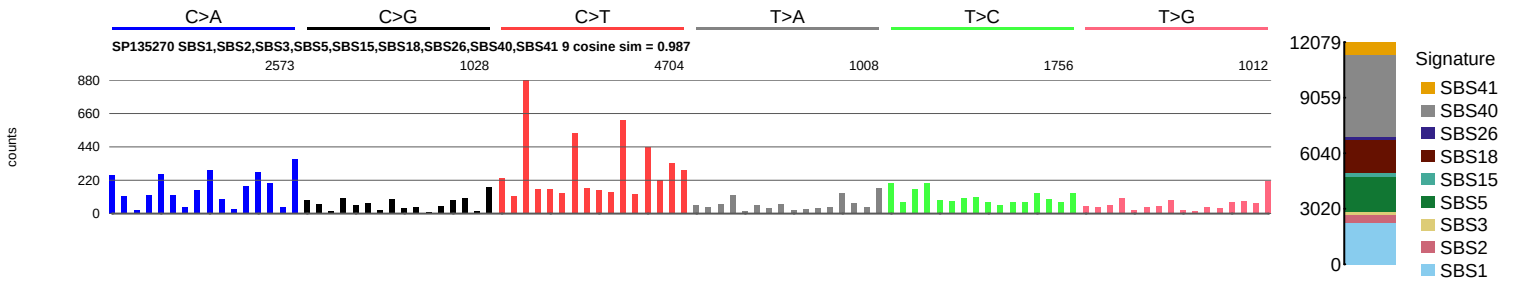
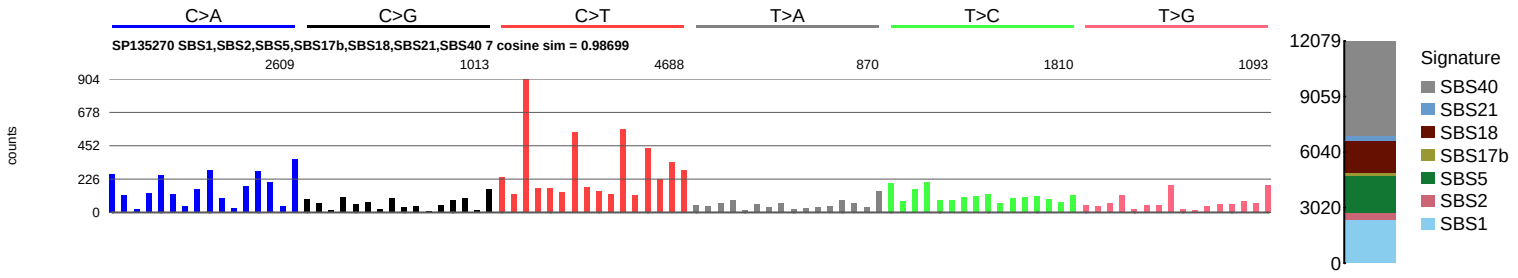
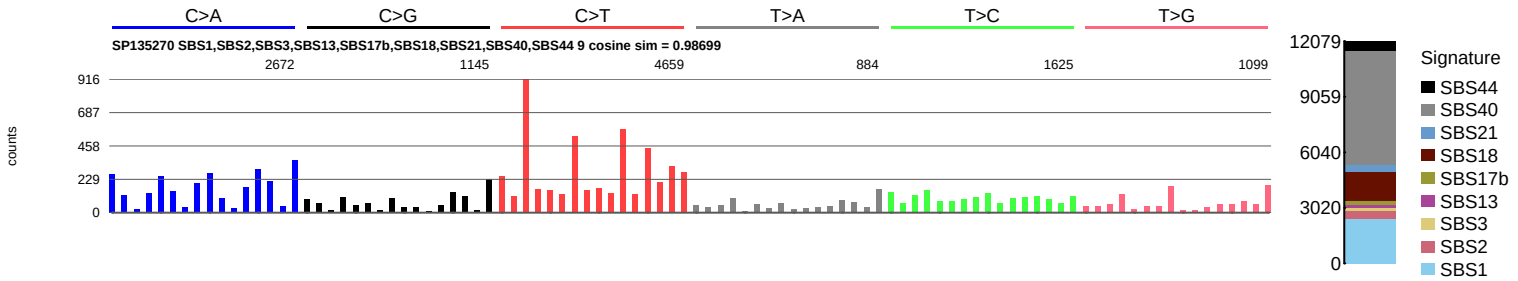
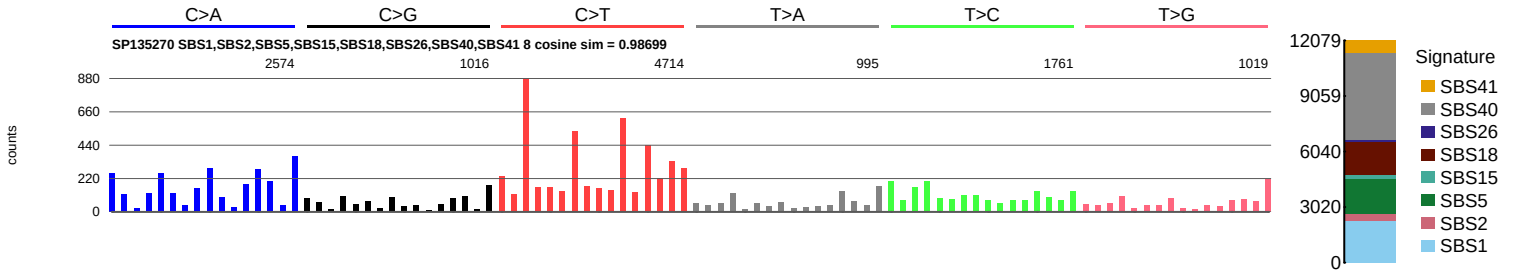
SBS3
SBS2
SBS1

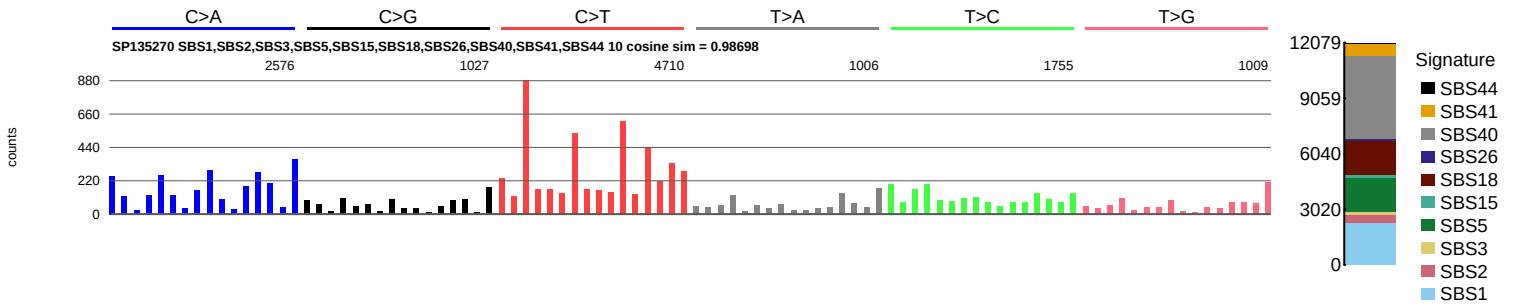

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98701
2499
968
4714
900
2052
948
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.987
2573
1028
4704
1008
1756
1012
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98699
2609
1013
4688
870
1810
1093
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98699
2672
1145
4659
884
1625
1099
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98699
2574
1016
4714
995
1761
1019
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98698
2576
1027
4710
1006
1755
1009
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
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SBS18
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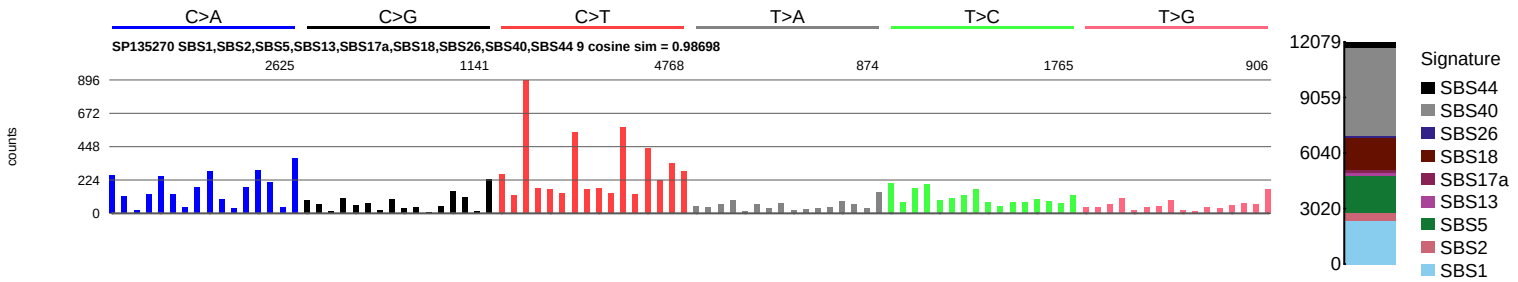
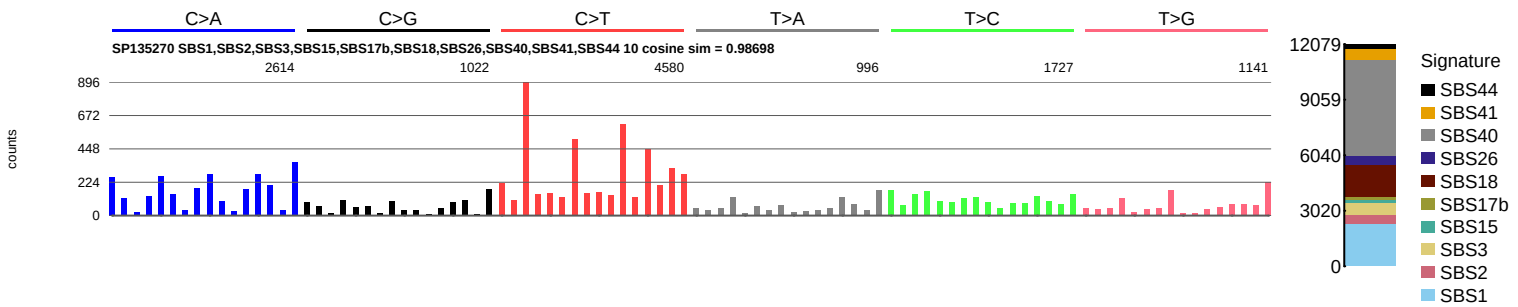
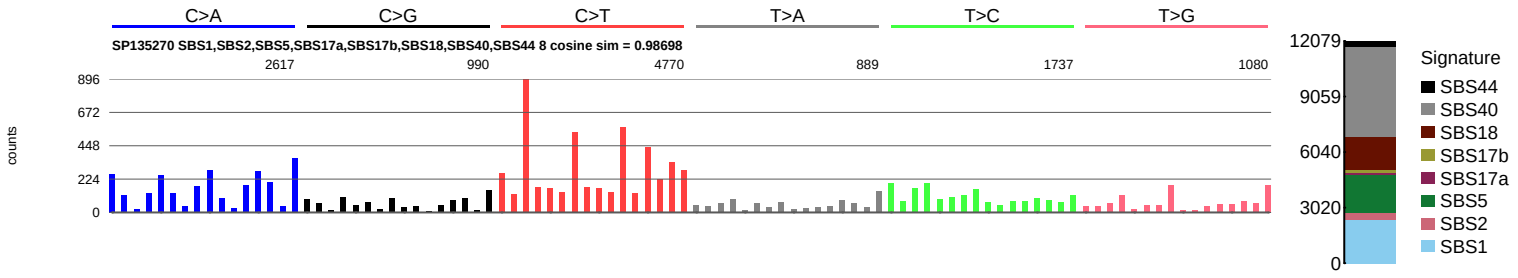
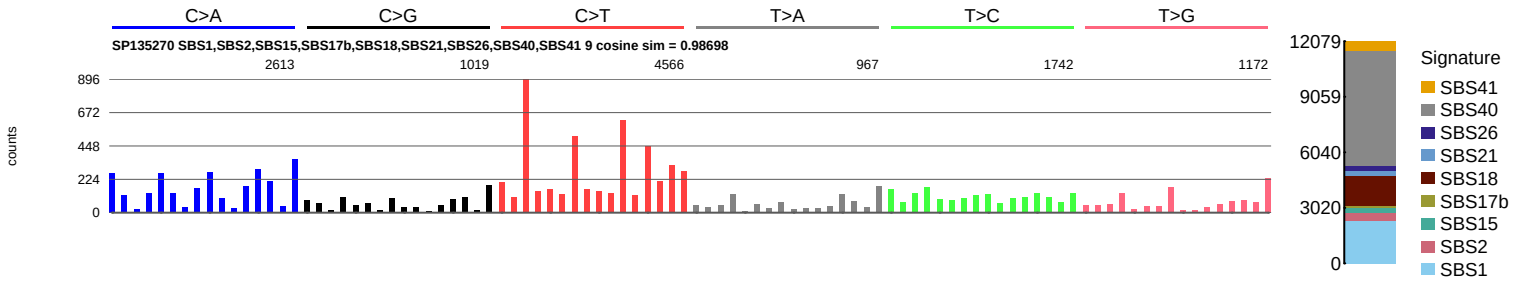
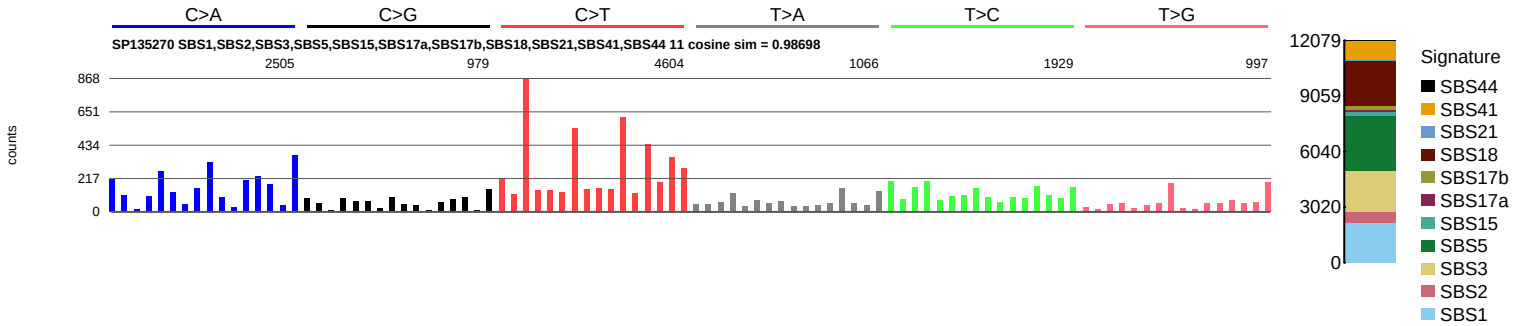
SBS3
SBS2
SBS1

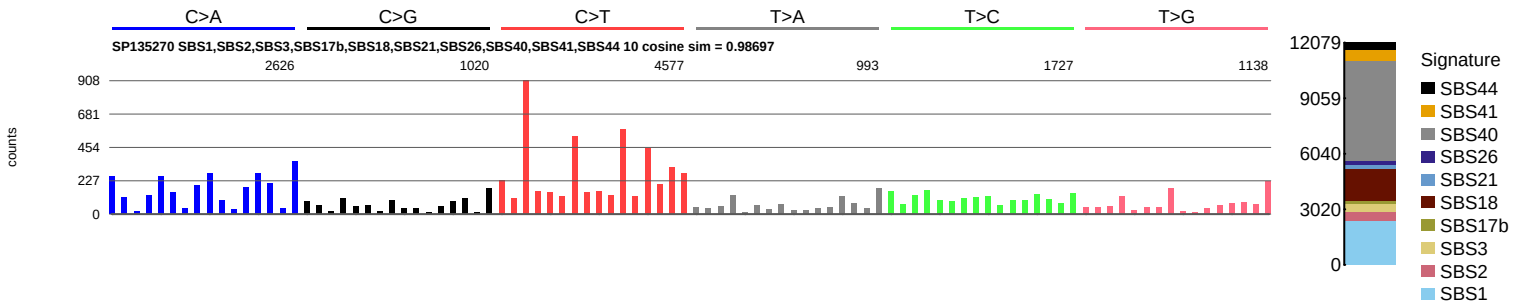

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98698
2625
1141
4768
874
1765
906
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98698
2614
1022
4580
996
1727
1141
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98698
2617
990
4770
889
1737
1080
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98698
2613
1019
4566
967
1742
1172
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98698
2505
979
4604
1066
1929
997
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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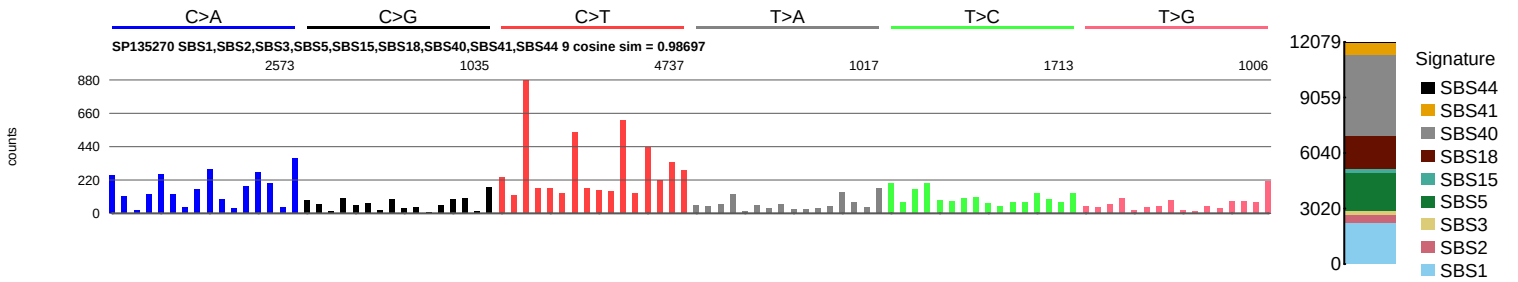
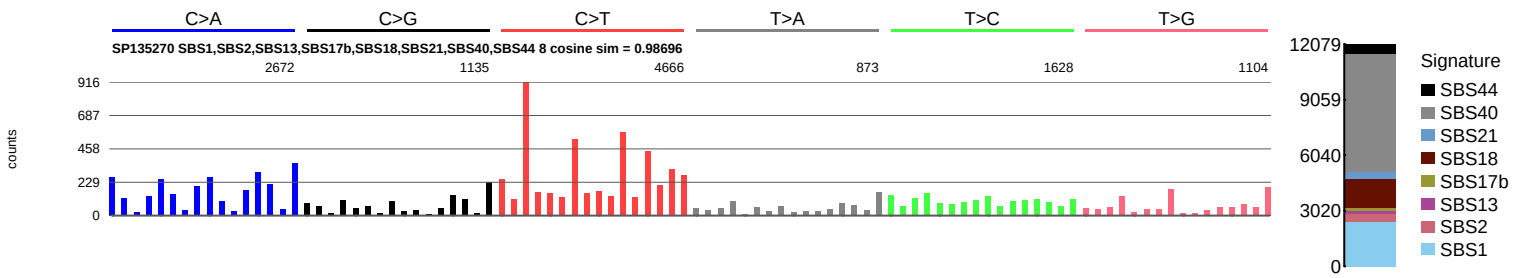
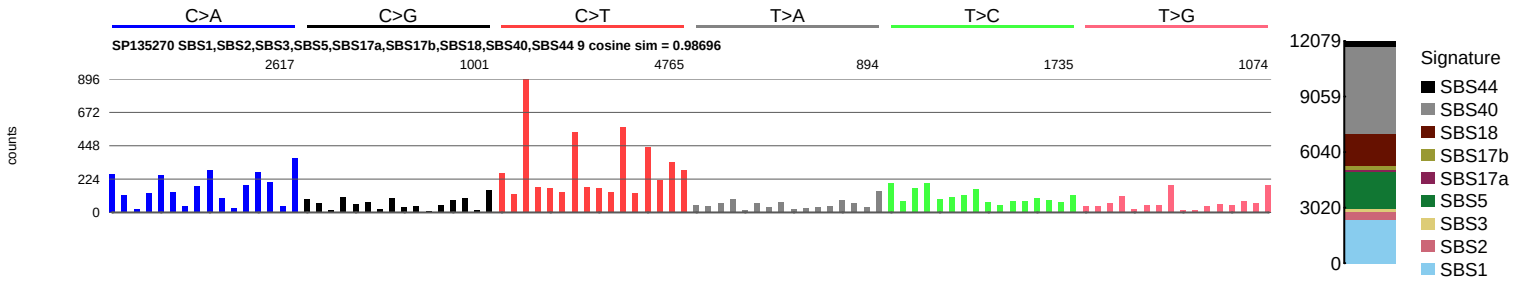
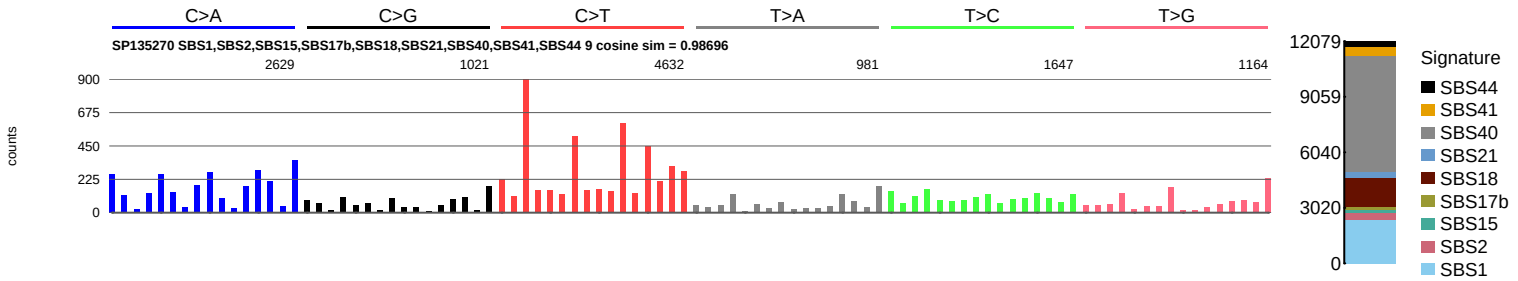
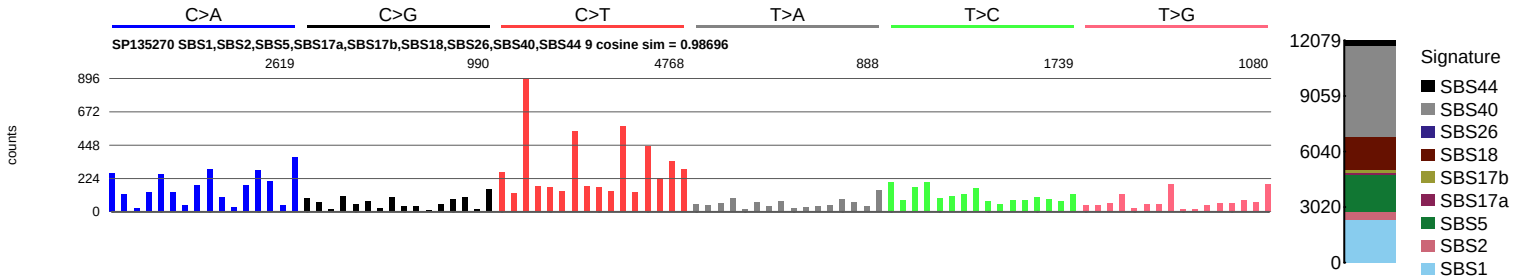
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98697
2626
1020
4577
993
1727
1138
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1

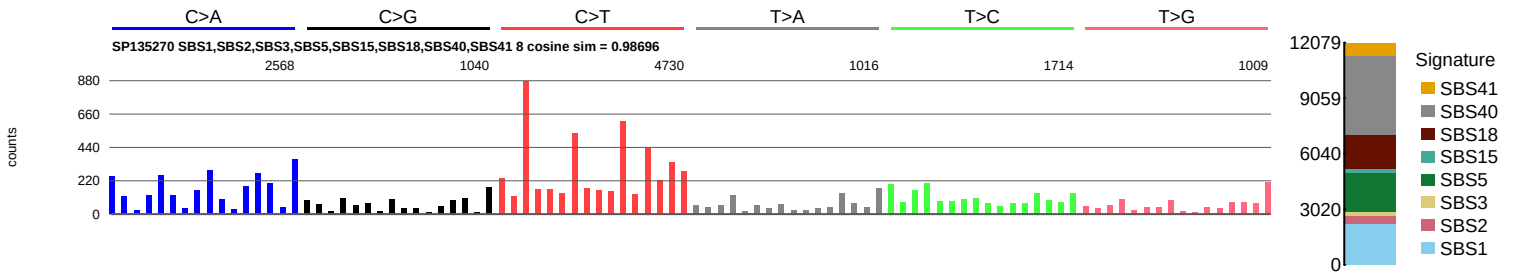

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98697
2573
1035
4737
1017
1713
1006
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98696
2672
1135
4666
873
1628
1104
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98696
2617
1001
4765
894
1735
1074
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98696
2629
1021
4632
981
1647
1164
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98696
2619
990
4768
888
1739
1080
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS40,SBS41 8 cosine sim = 0.98696
2568
1040
4730
1016
1714
1009
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
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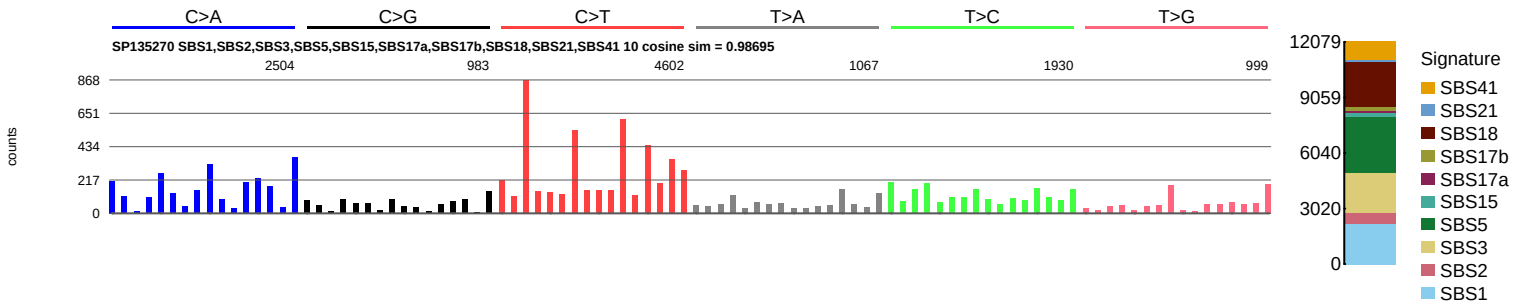
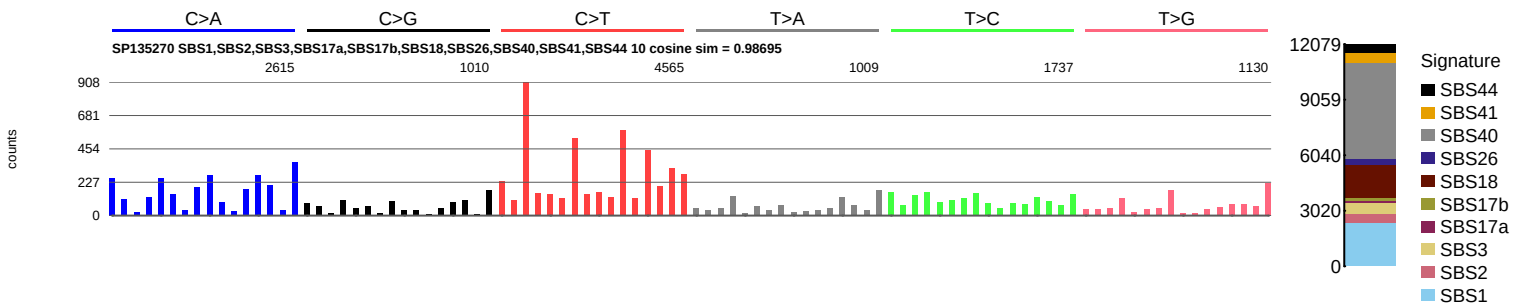
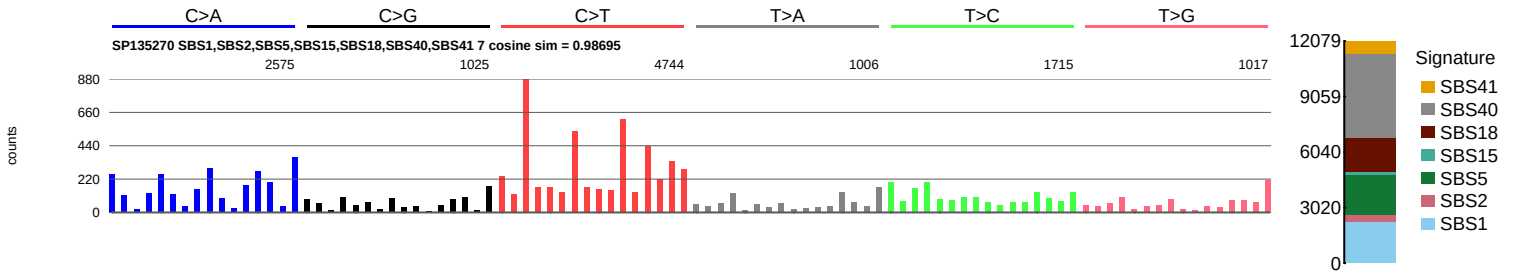
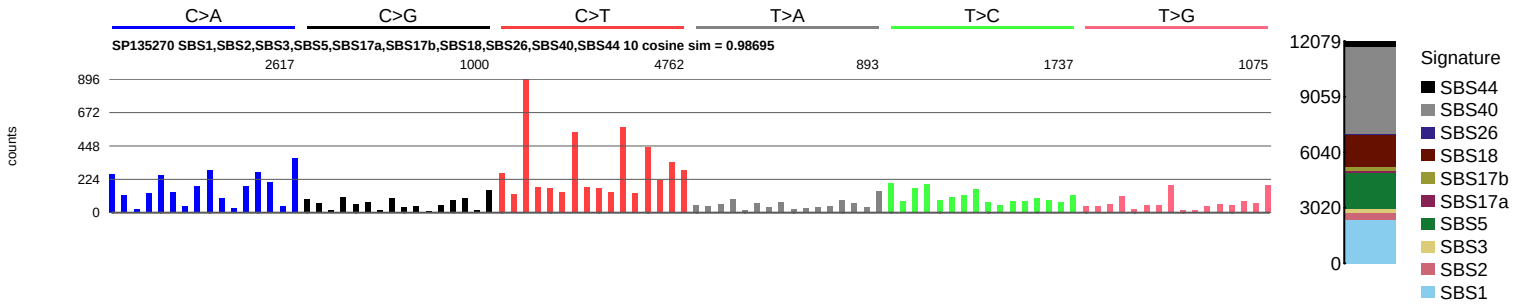
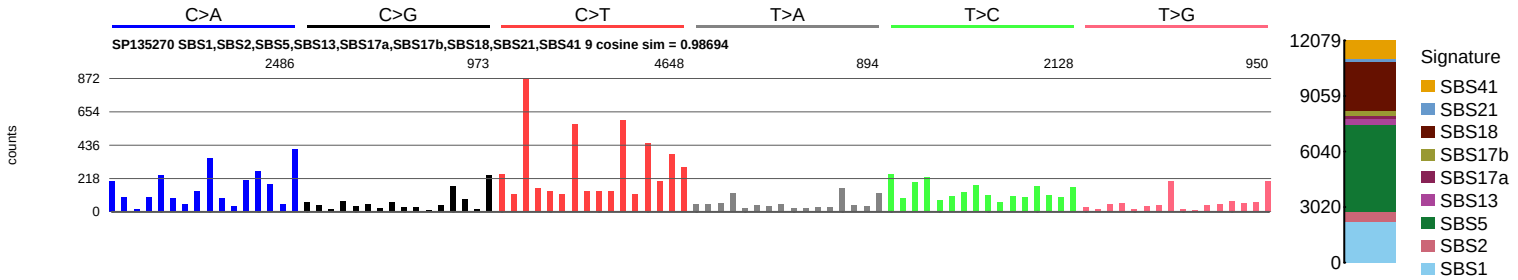
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98695
2504
983
4602
1067
1930
999
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98695
2615
1010
4565
1009
1737
1130
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.98695
2575
1025
4744
1006
1715
1017
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98695
2617
1000
4762
893
1737
1075
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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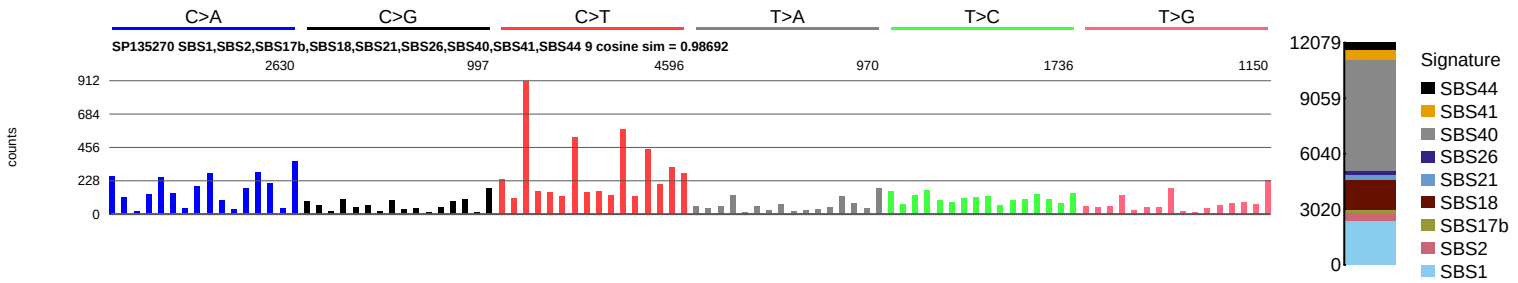
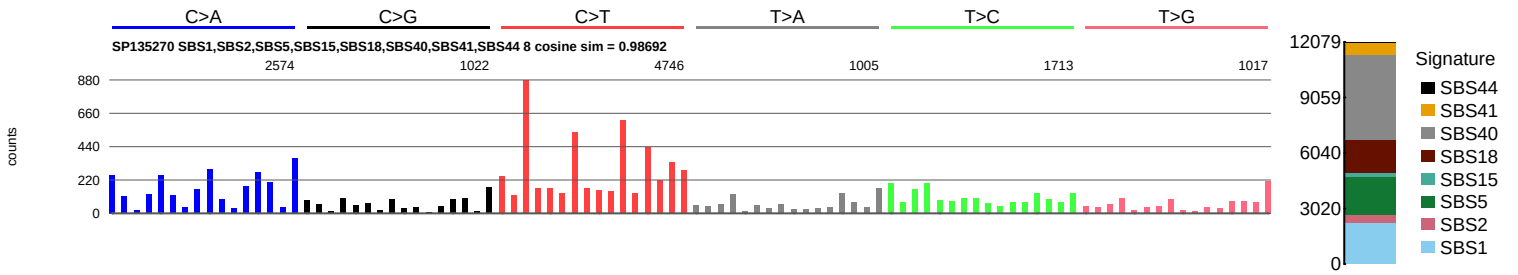
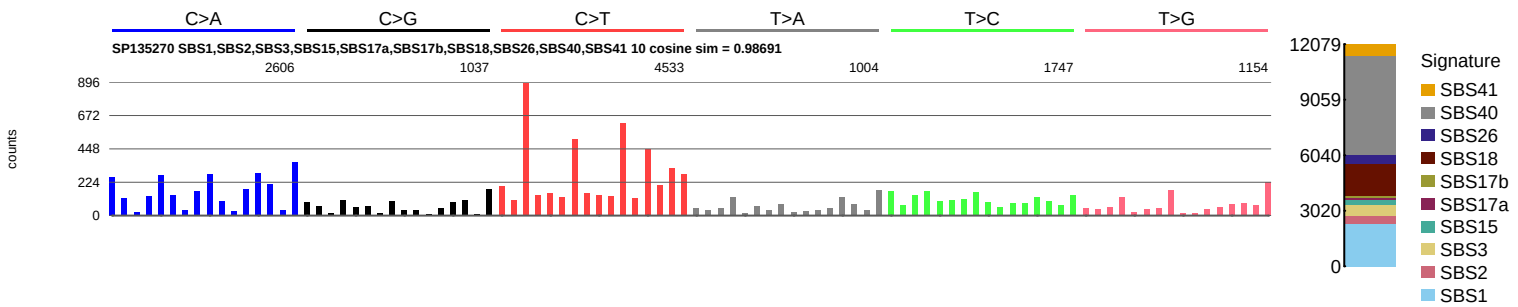
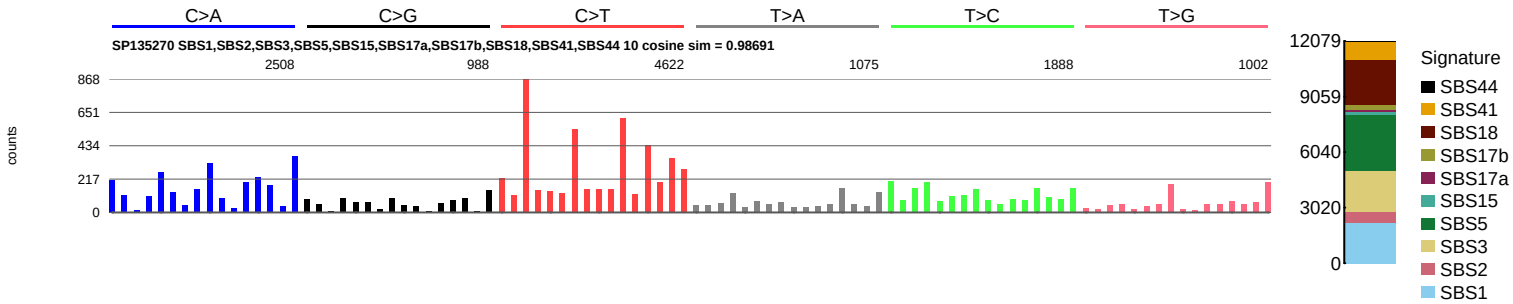
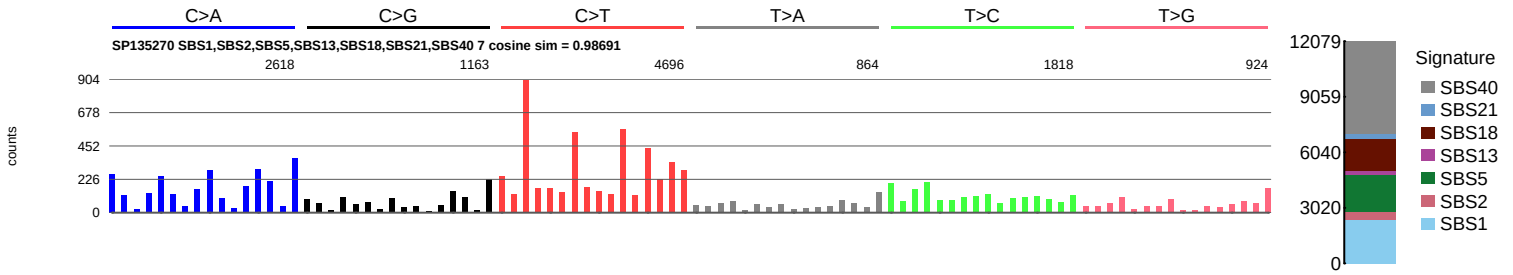
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98694
2486
973
4648
894
2128
950
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

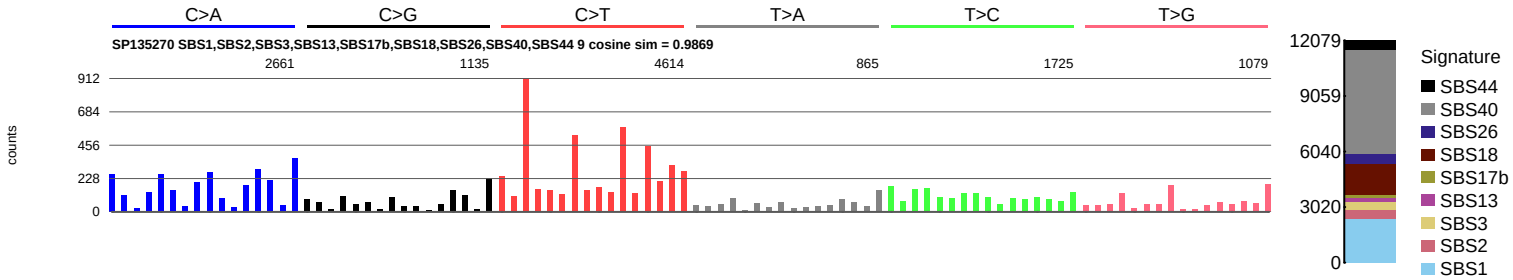

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98692
2630
997
4596
970
1736
1150
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1

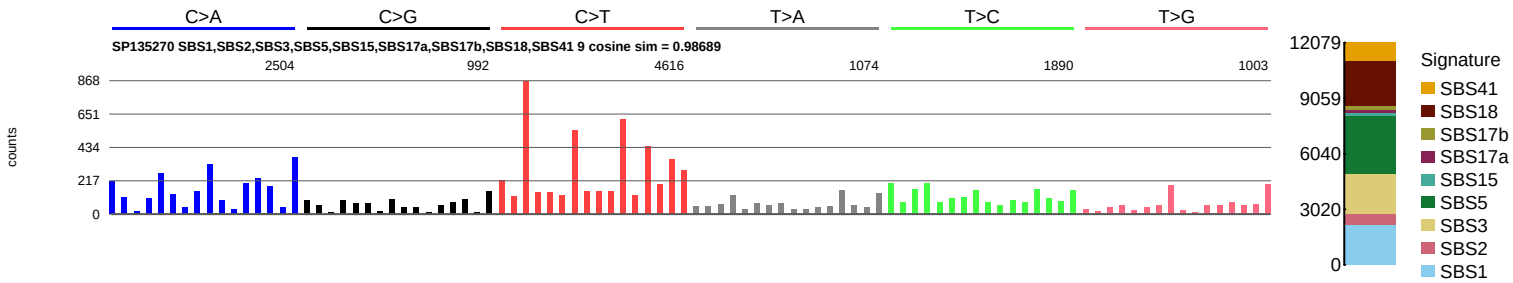

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98692
2574
1022
4746
1005
1713
1017
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS5
SBS2
SBS1

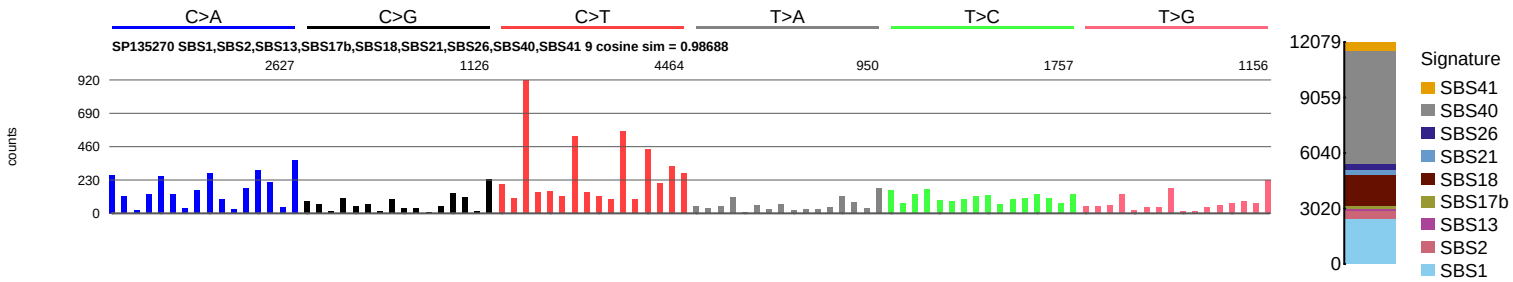

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98691
2606
1037
4533
1004
1747
1154
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

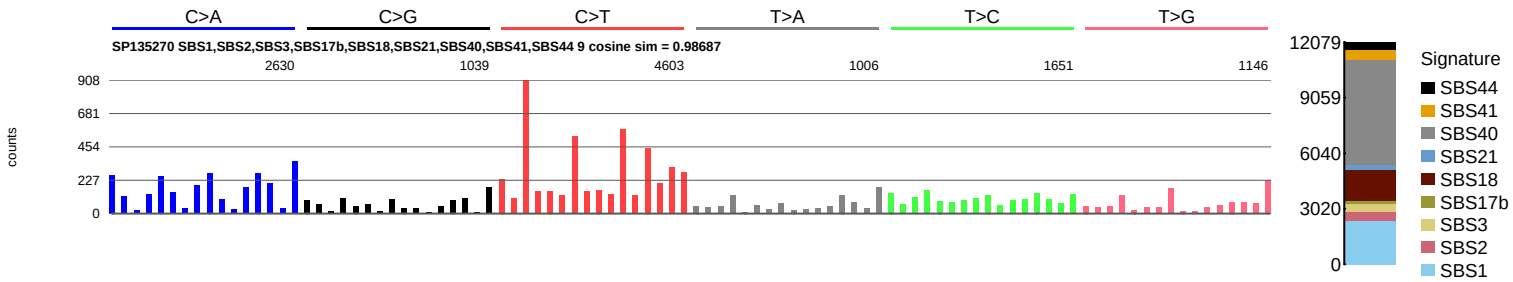

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98691
2508
988
4622
1075
1888
1002
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

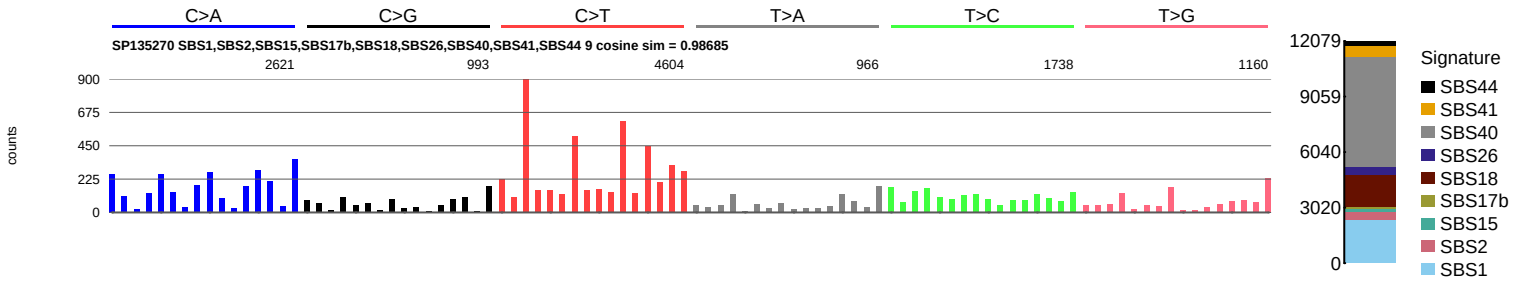

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40 7 cosine sim = 0.98691
2618
1163
4696
864
1818
924
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1

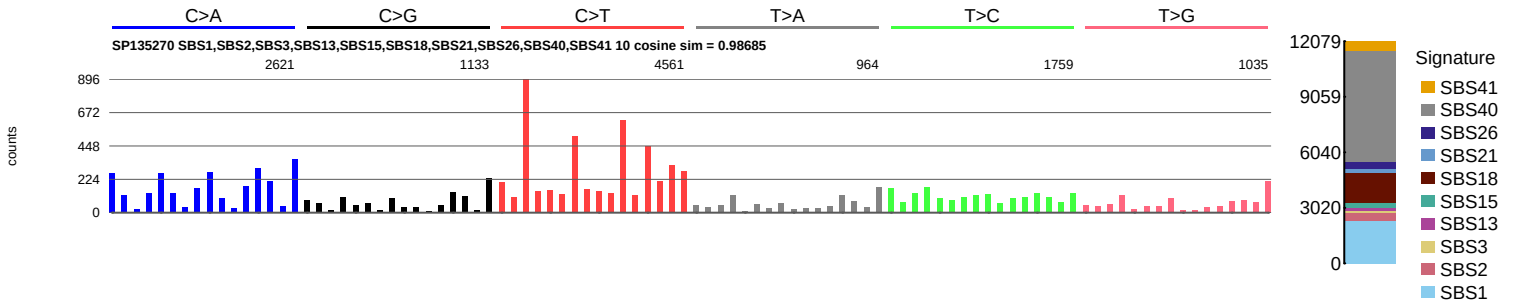

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.9869
2661
1135
4614
865
1725
1079
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

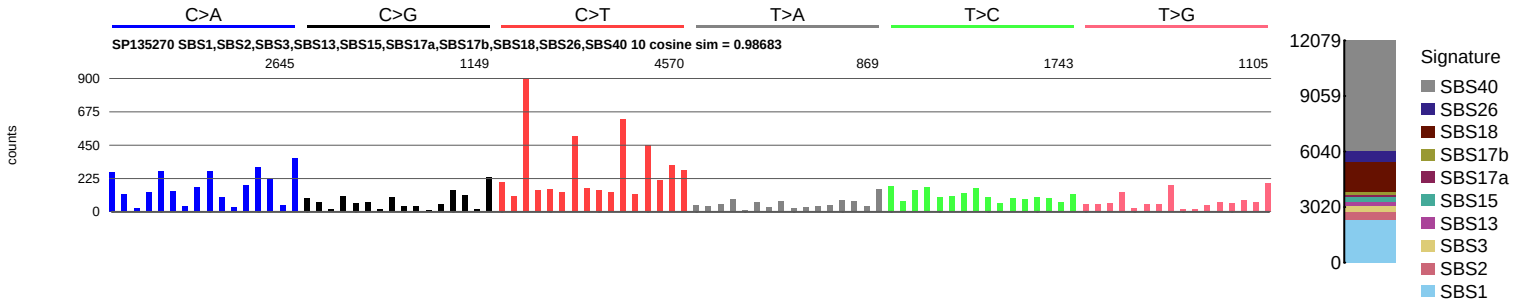

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98689
2504
992
4616
1074
1890
1003
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

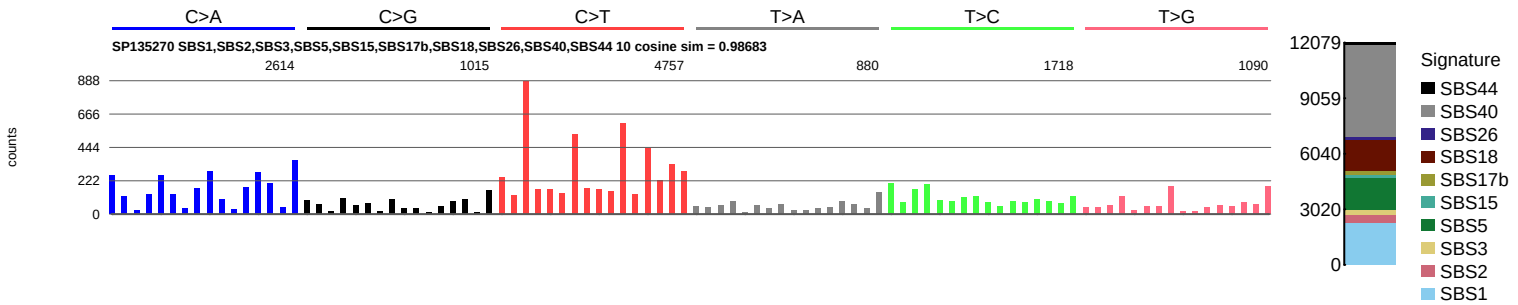

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98688
2627
1126
4464
950
1757
1156
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

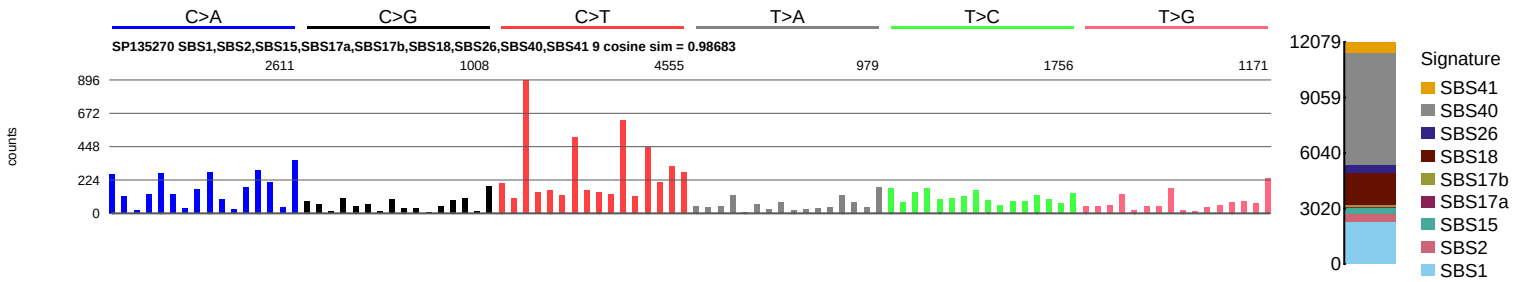

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98687
2630
1039
4603
1006
1651
1146
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1

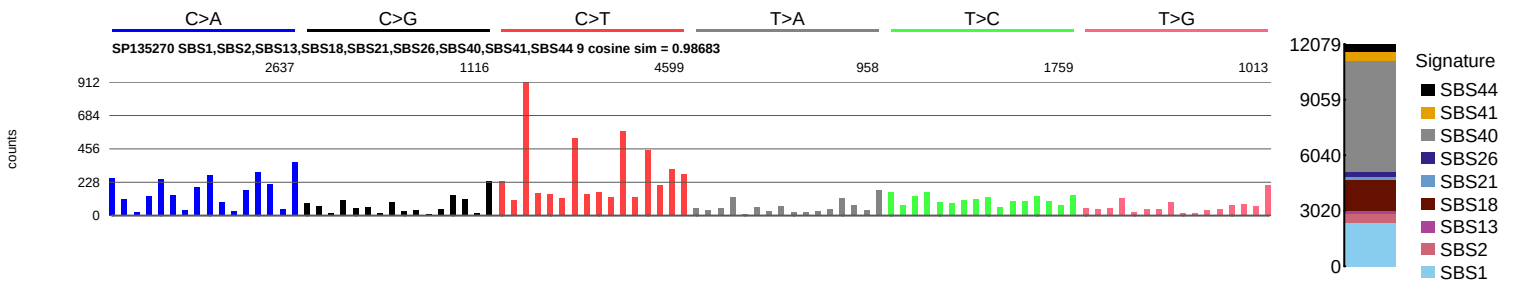

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98685
2621
993
4604
966
1738
1160
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1

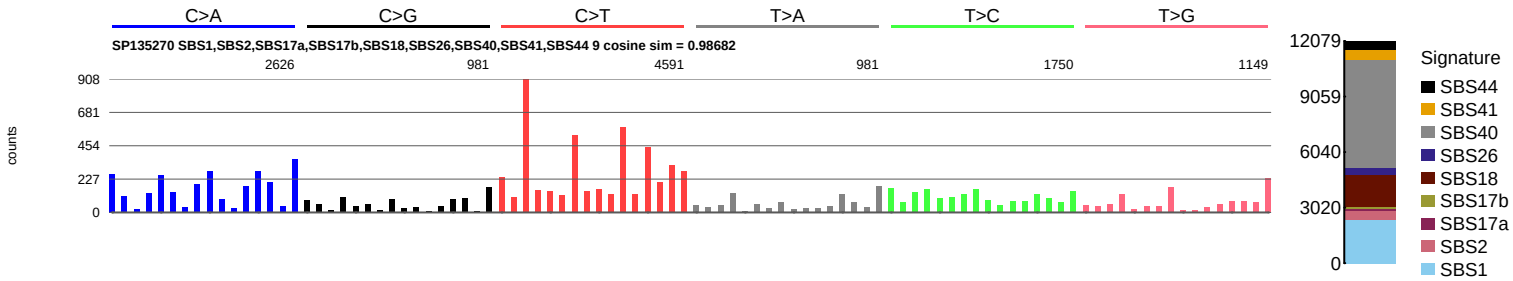

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98685
2621
1133
4561
964
1759
1035
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

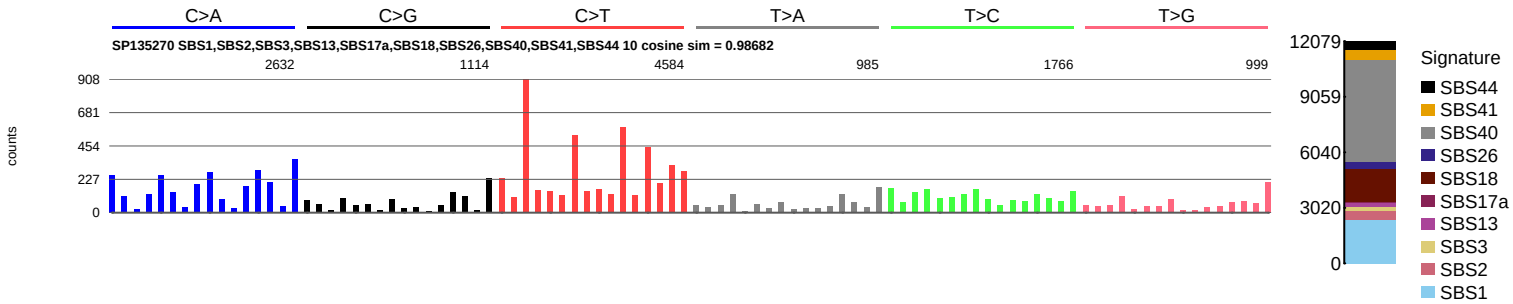

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98683
2645
1149
4570
869
1743
1105
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

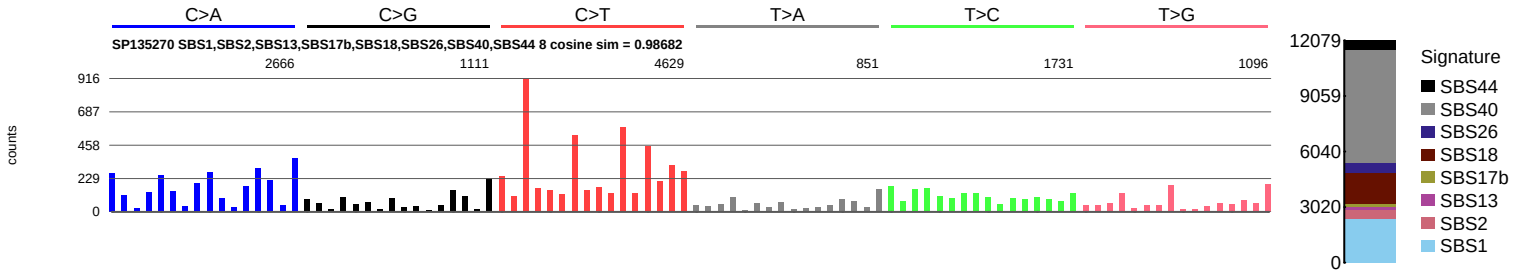

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98683
2614
1015
4757
880
1718
1090
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

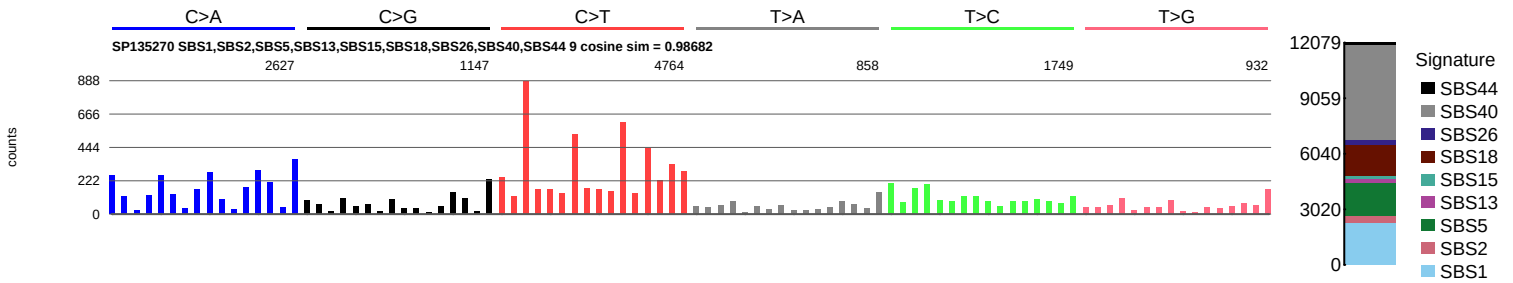

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98683
2611
1008
4555
979
1756
1171
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

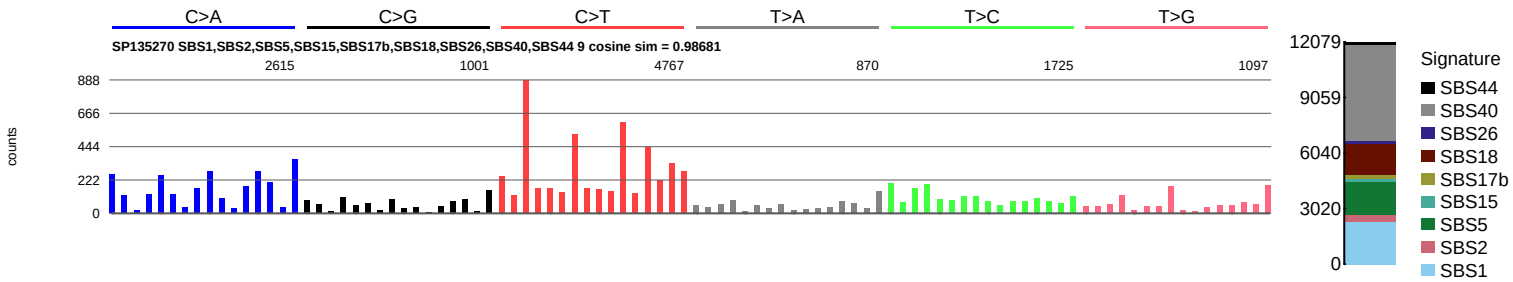

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98683
2637
1116
4599
958
1759
1013
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1

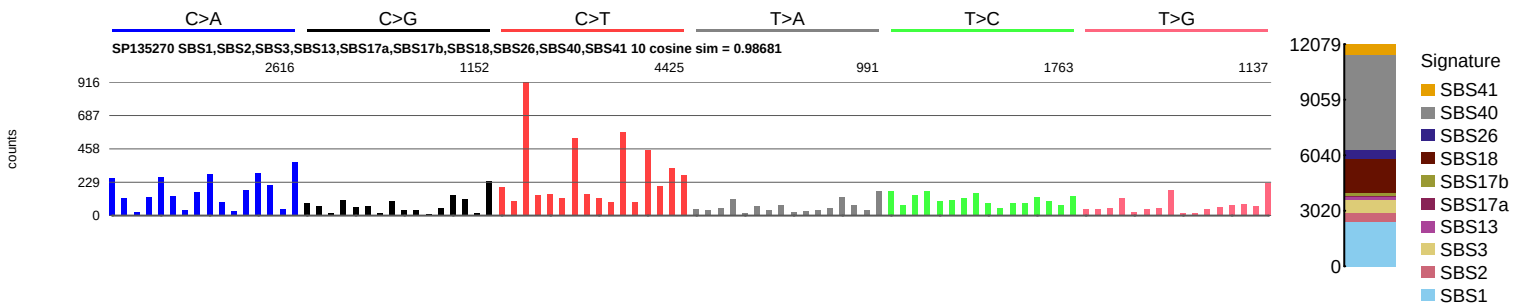

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98682
2626
981
4591
981
1750
1149
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98682
2632
1114
4584
985
1766
999
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98682
2666
1111
4629
851
1731
1096
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98682
2627
1147
4764
858
1749
932
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98681
2615
1001
4767
870
1725
1097
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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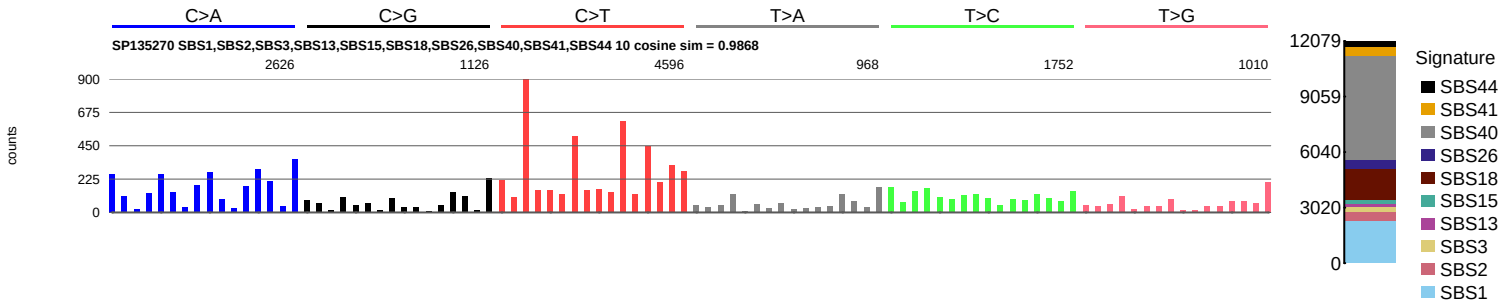
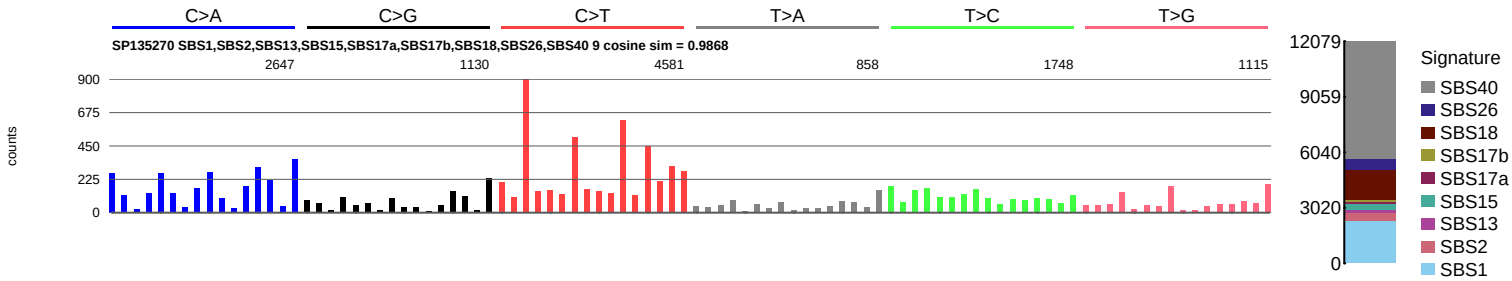
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98681
2616
1152
4425
991
1763
1137
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9868
2626
1126
4596
968
1752
1010
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1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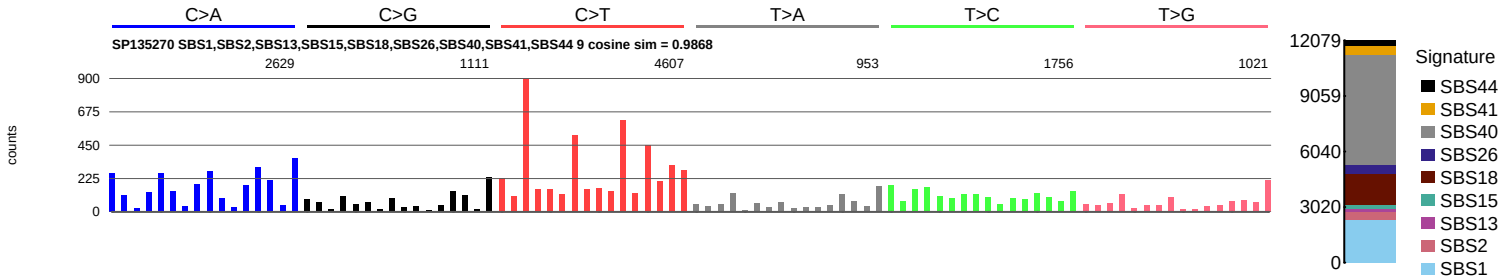
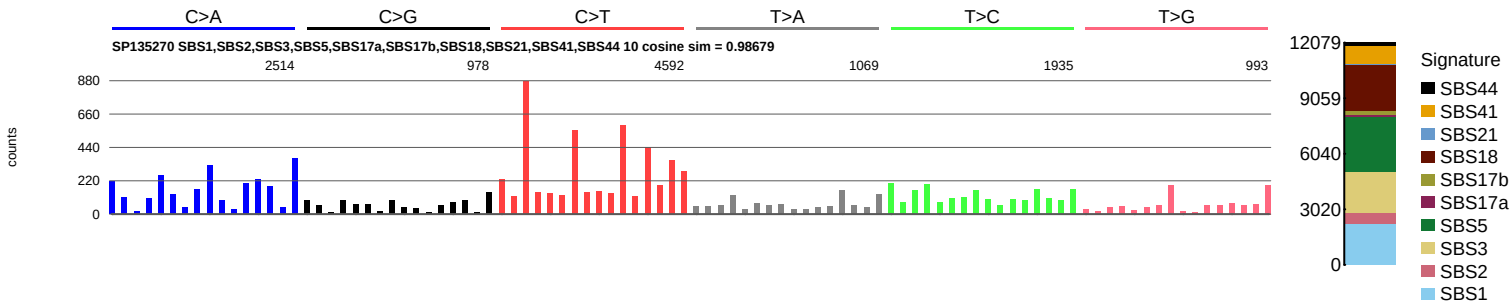
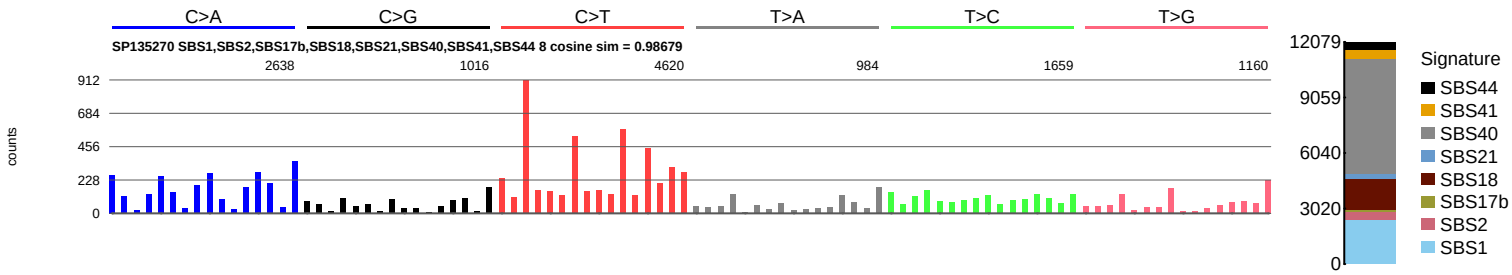
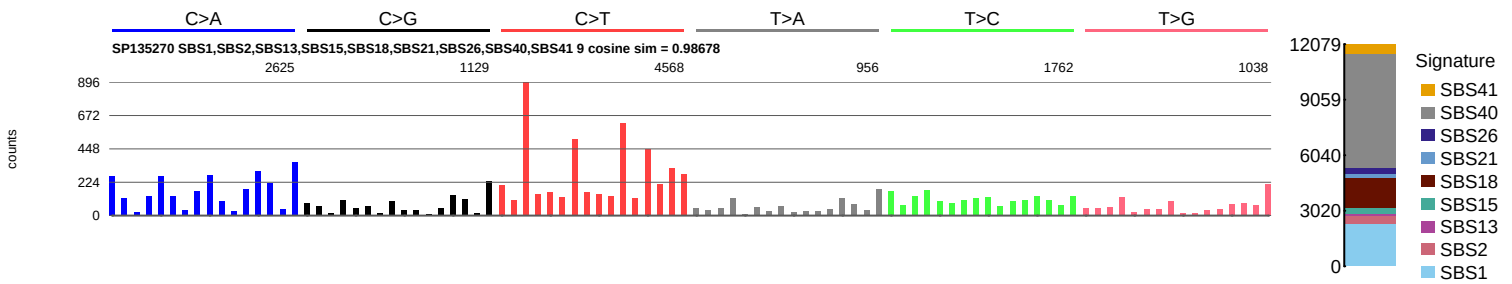
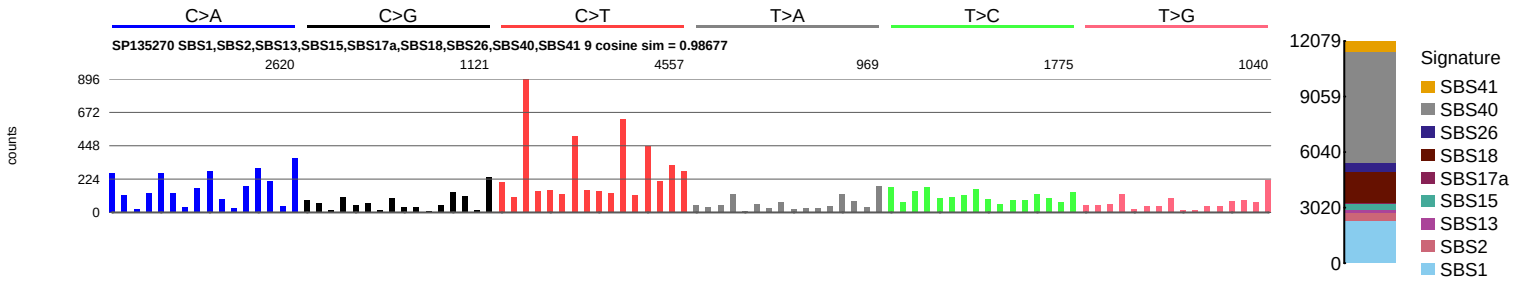
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.9868
2647
1130
4581
858
1748
1115
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

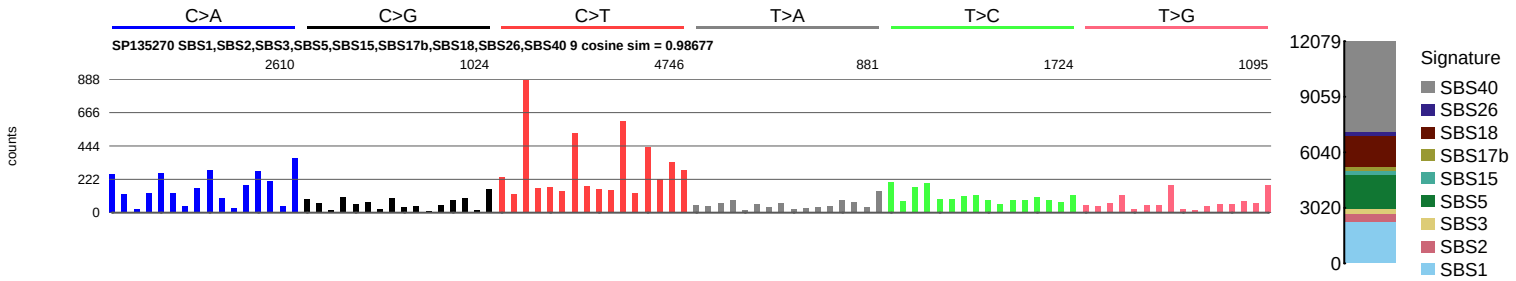

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9868
2629
1111
4607
953
1756
1021
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1

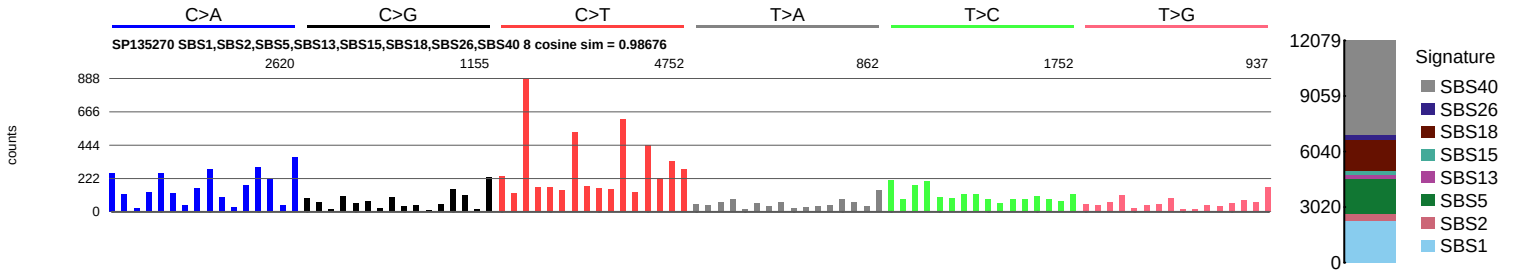

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98679
2514
978
4592
1069
1935
993
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

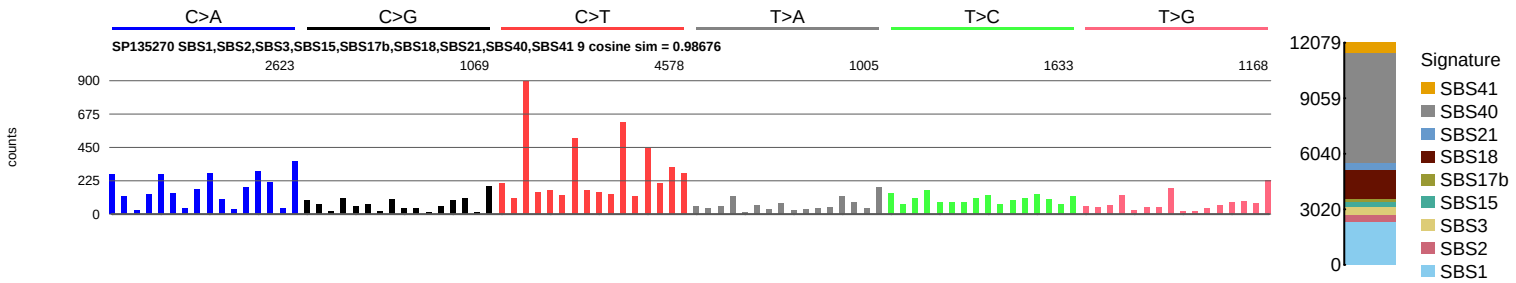

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98679
2638
1016
4620
984
1659
1160
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1

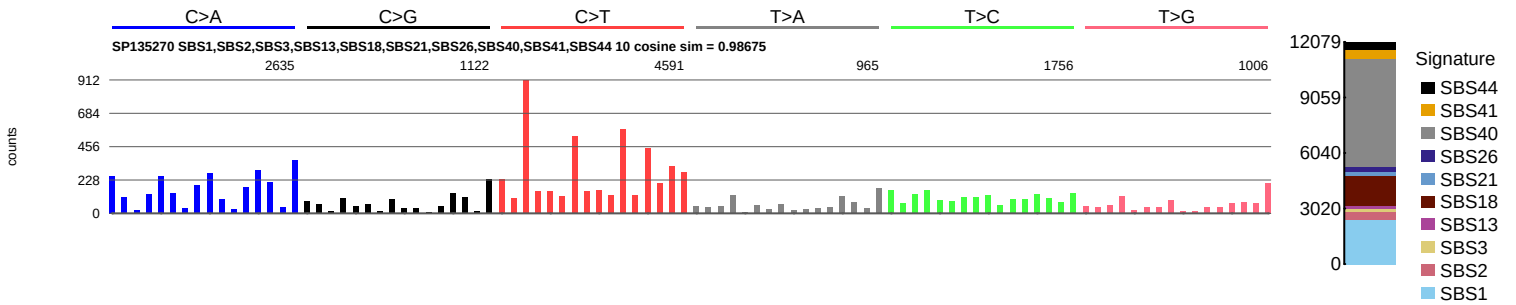

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98678
2625
1129
4568
956
1762
1038
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

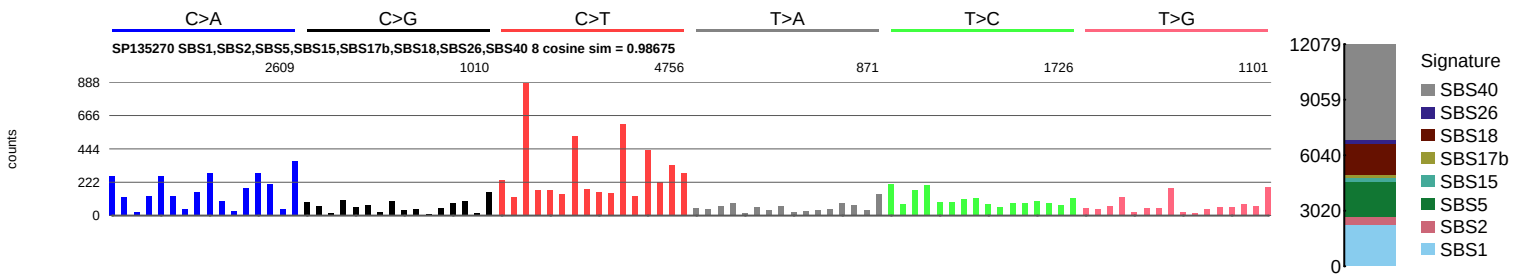

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98677
2620
1121
4557
969
1775
1040
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

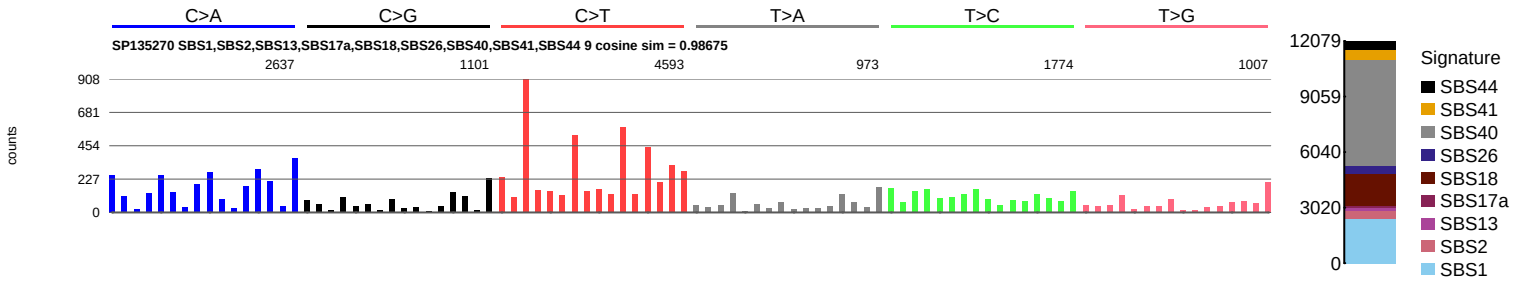

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98677
2610
1024
4746
881
1724
1095
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

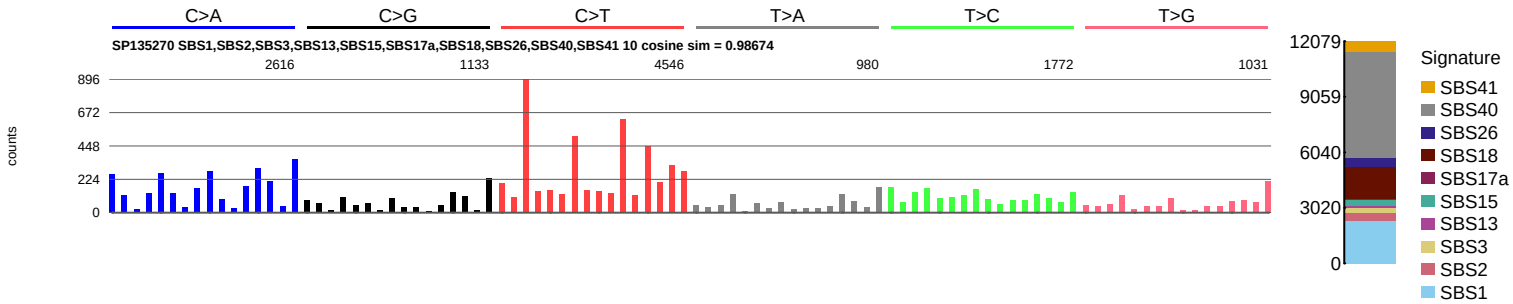

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40 8 cosine sim = 0.98676
2620
1155
4752
862
1752
937
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1

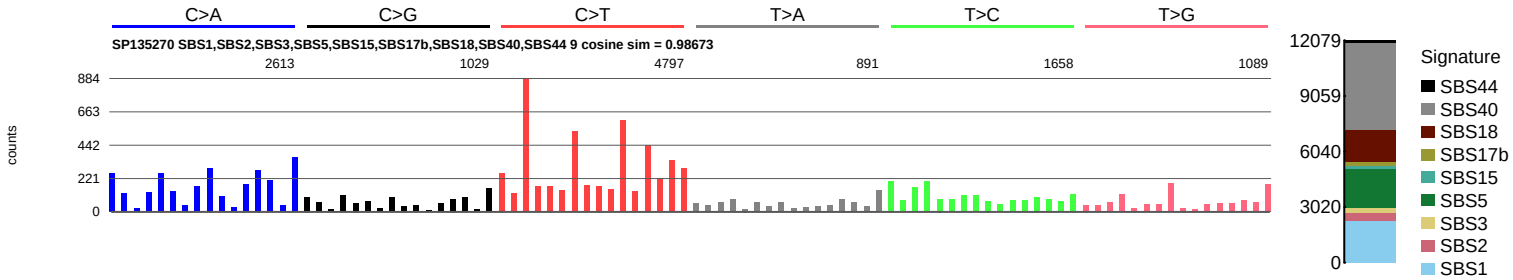

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98676
2623
1069
4578
1005
1633
1168
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

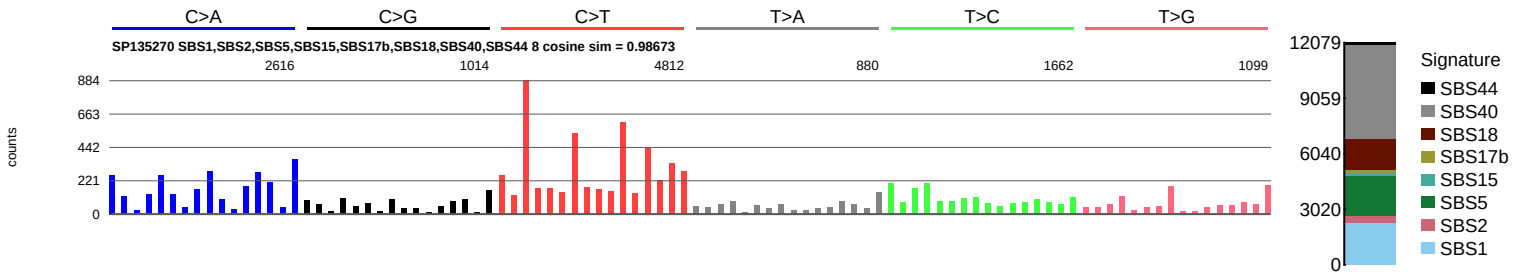

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98675
2635
1122
4591
965
1756
1006
912
684
456
228
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1

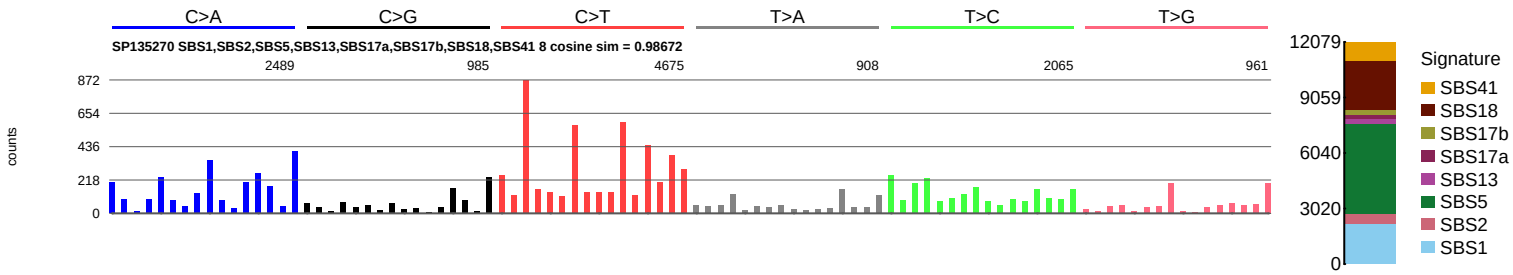

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98675
2609
1010
4756
871
1726
1101
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

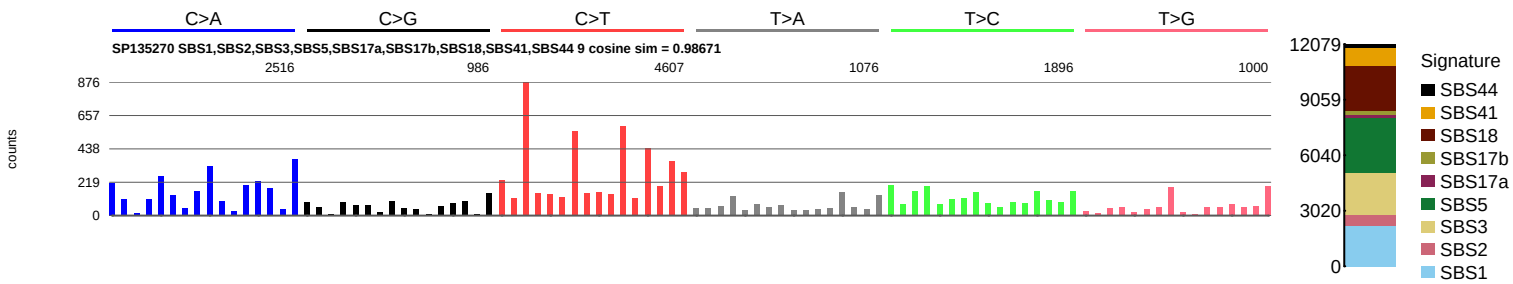

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98675
2637
1101
4593
973
1774
1007
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1

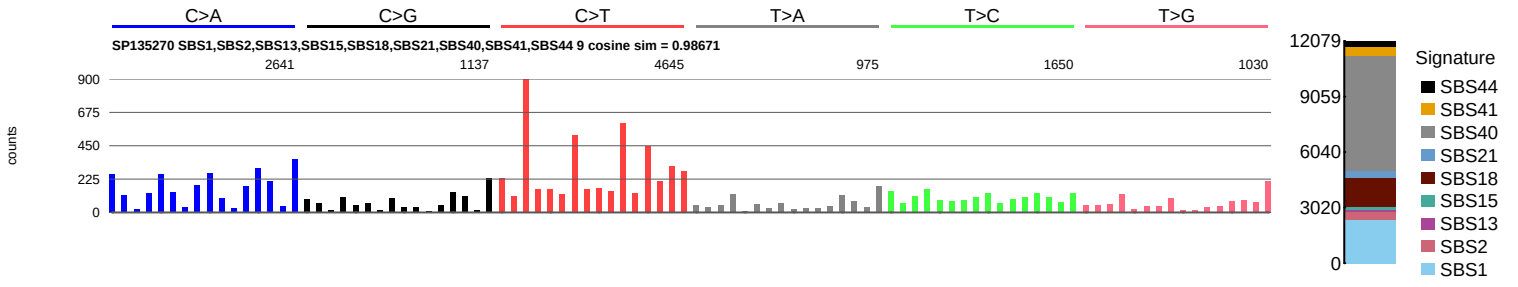

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98674
2616
1133
4546
980
1772
1031
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

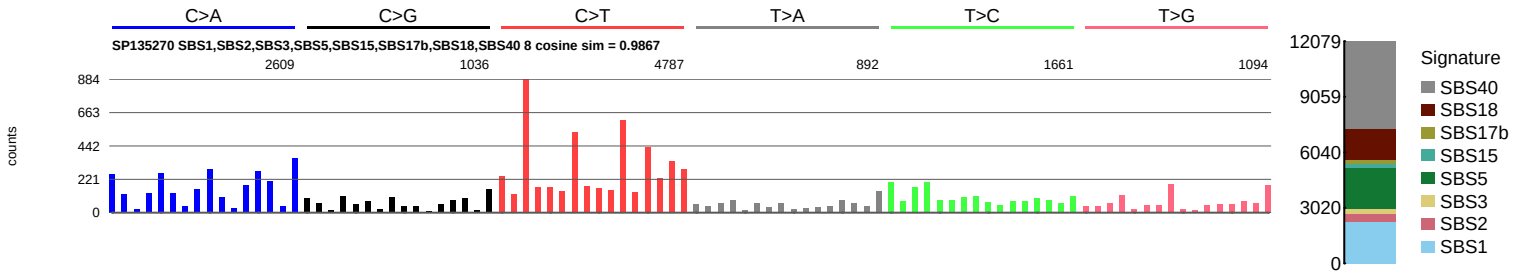

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98673
2613
1029
4797
891
1658
1089
884
663
442
221
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

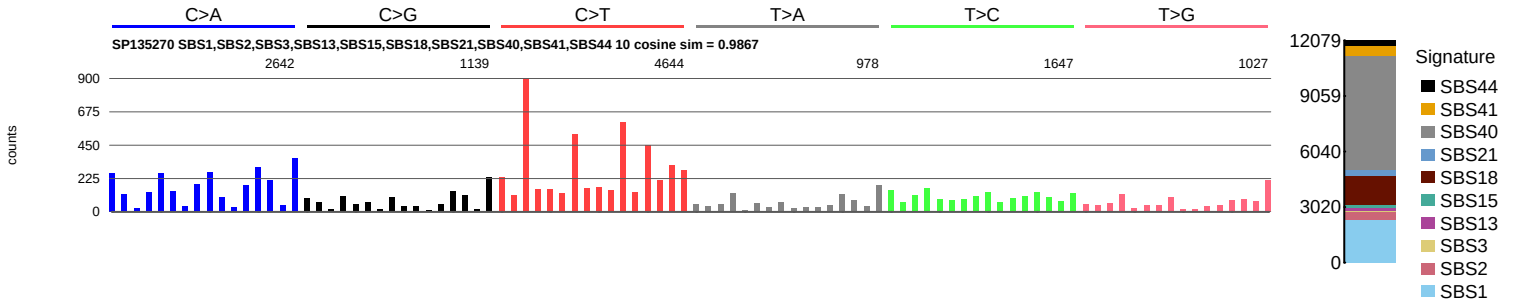

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98673
2616
1014
4812
880
1662
1099
884
663
442
221
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

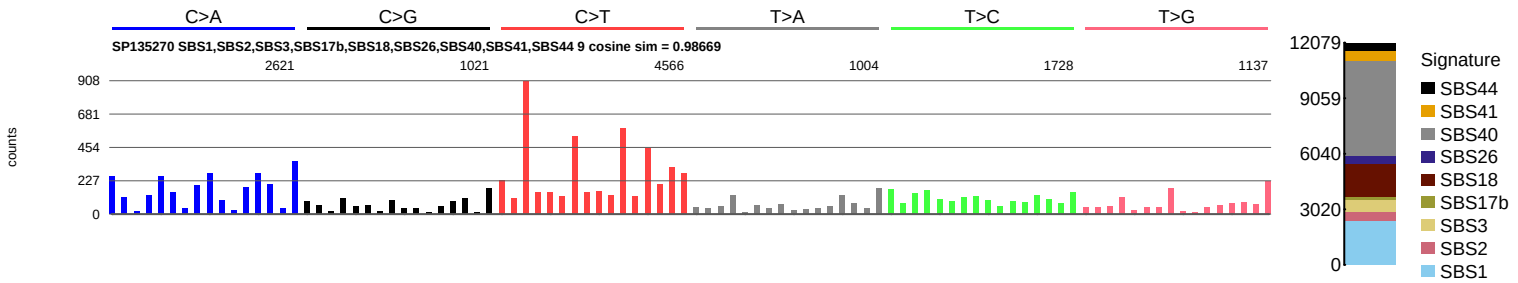

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98672
2489
985
4675
908
2065
961
872
654
436
218
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

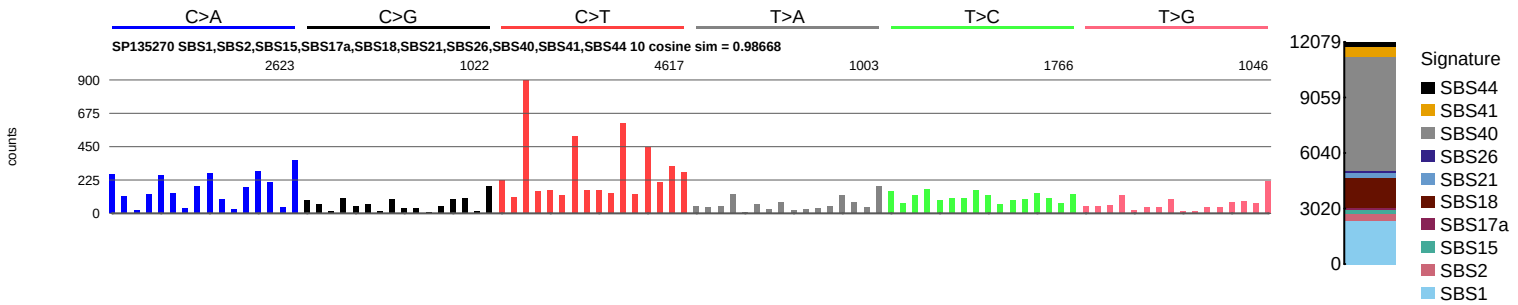

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98671
2516
986
4607
1076
1896
1000
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

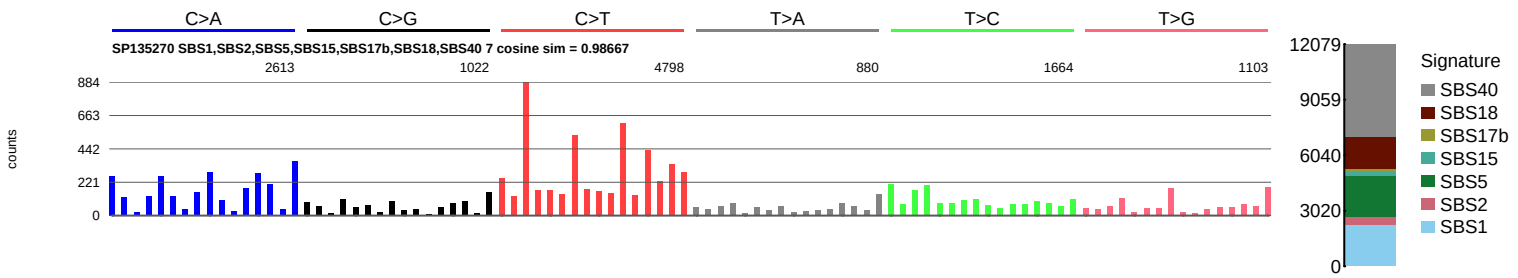

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98671
2641
1137
4645
975
1650
1030
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.9867
2609
1036
4787
892
1661
1094
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9867
2642
1139
4644
978
1647
1027
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98669
2621
1021
4566
1004
1728
1137
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98668
2623
1022
4617
1003
1766
1046
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1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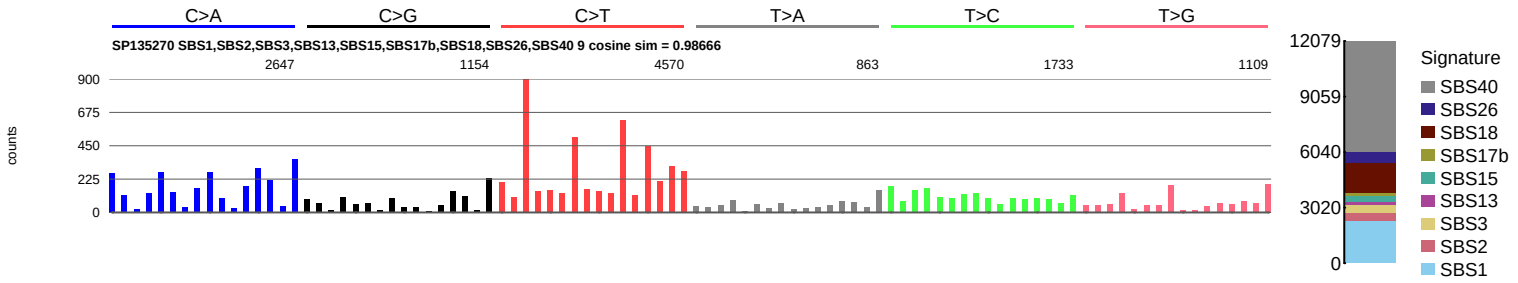
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98667
2613
1022
4798
880
1664
1103
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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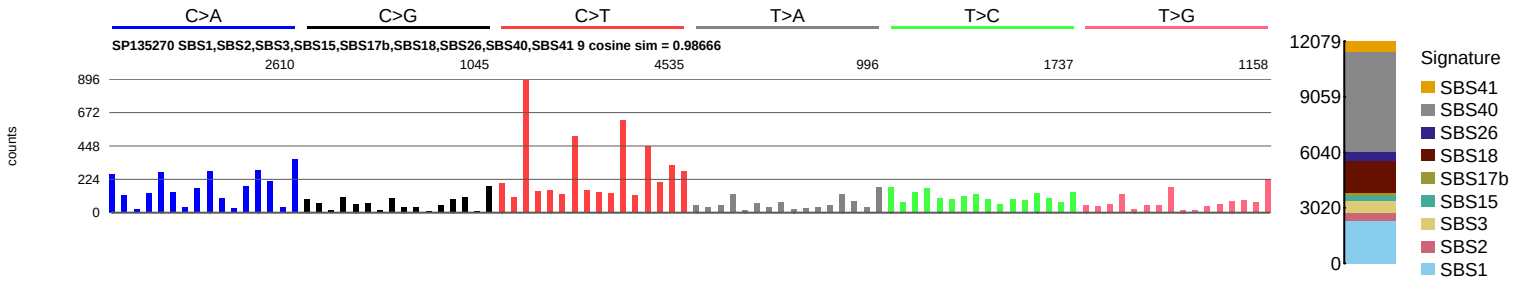
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98666
2647
1154
4570
863
1733
1109
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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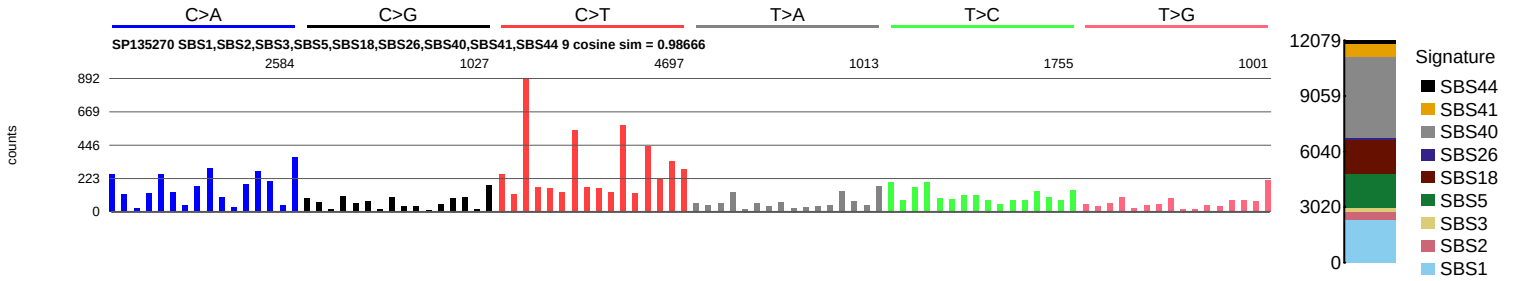
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98666
2610
1045
4535
996
1737
1158
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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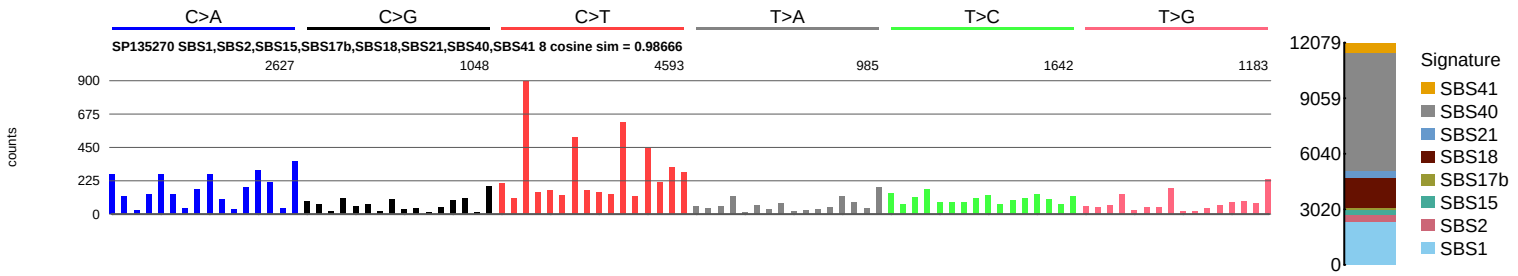
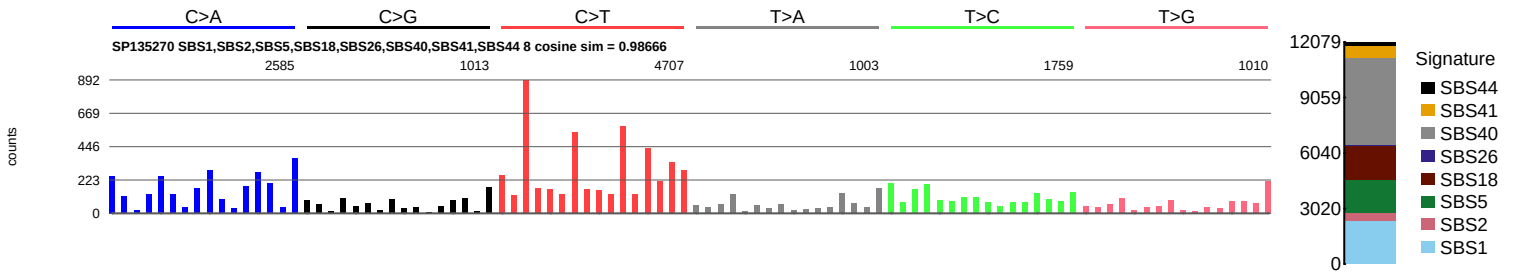
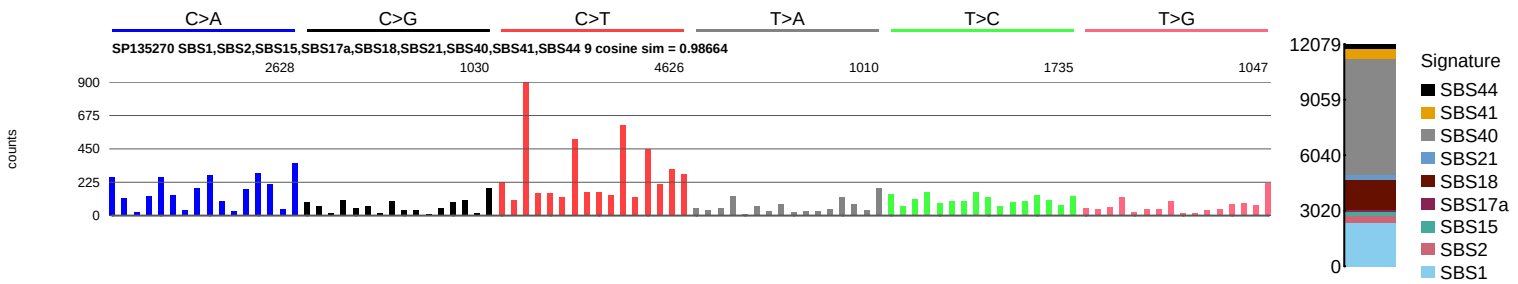
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98666
2584
1027
4697
1013
1755
1001
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98666
2627
1048
4593
985
1642
1183
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98666
2585
1013
4707
1003
1759
1010
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1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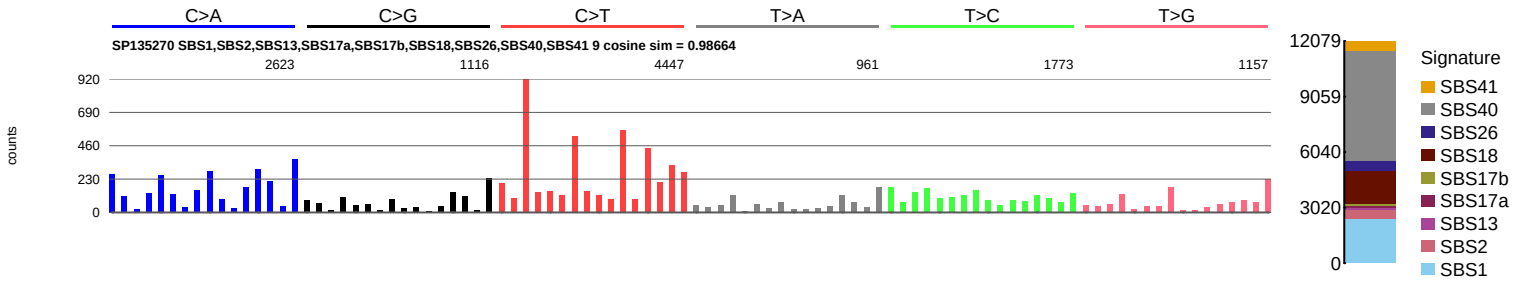
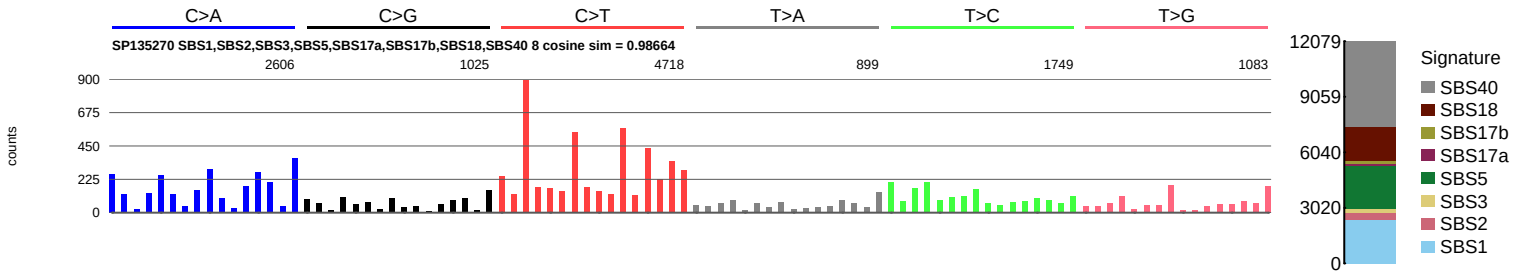
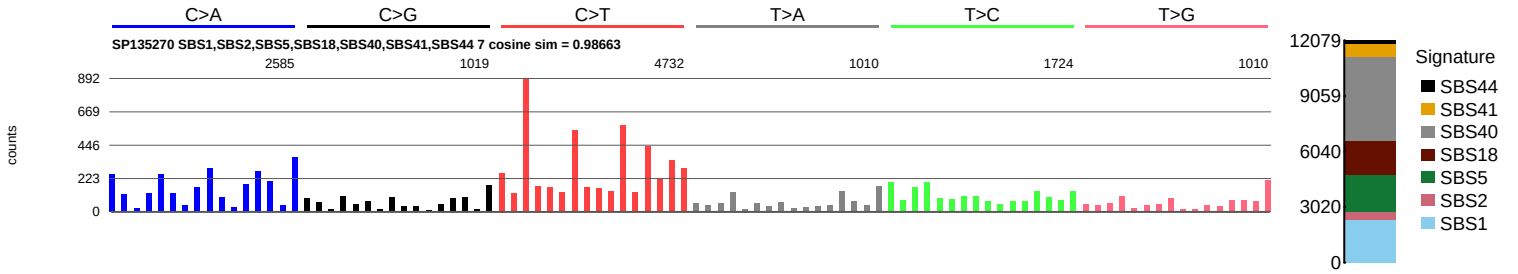
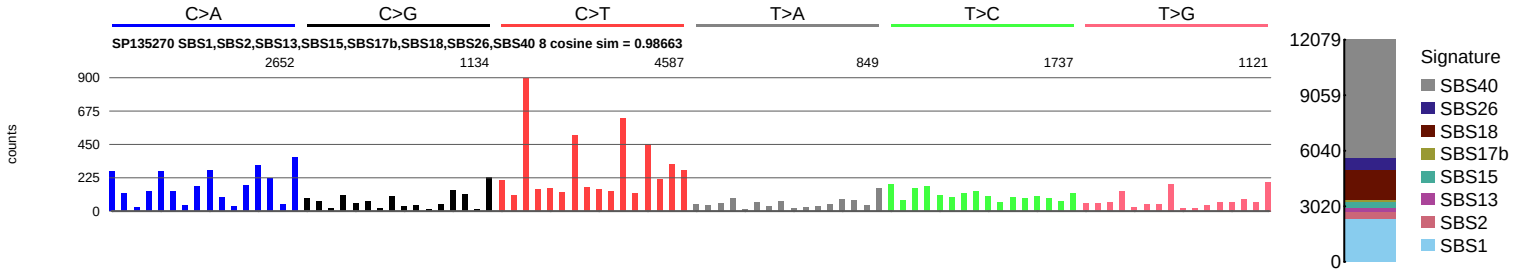
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98664
2628
1030
4626
1010
1735
1047
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98664
2623
1116
4447
961
1773
1157
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98664
2606
1025
4718
899
1749
1083
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98663
2585
1019
4732
1010
1724
1010
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98663
2652
1134
4587
849
1737
1121
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
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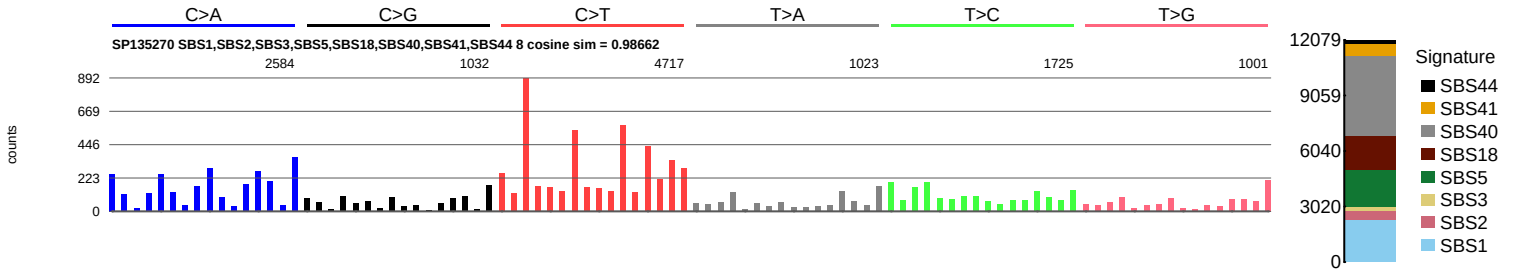
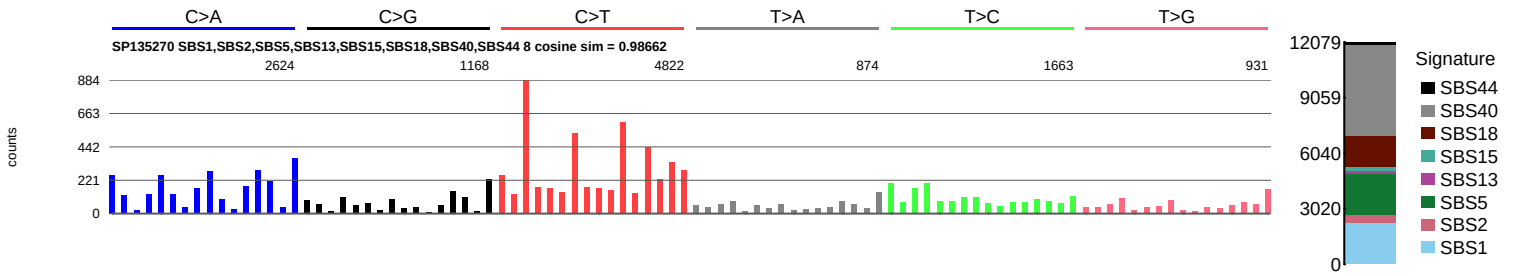
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98662
2584
1032
4717
1023
1725
1001
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS44 8 cosine sim = 0.98662
2624
1168
4822
874
1663
931
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS15
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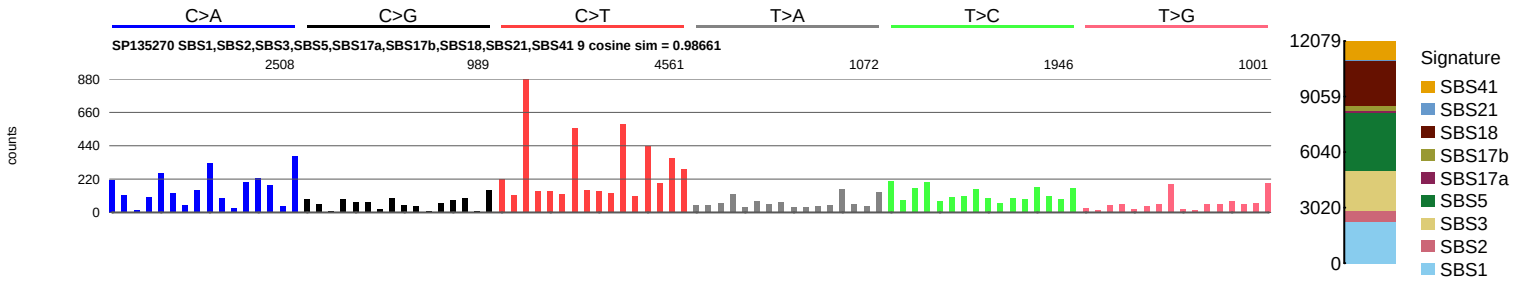
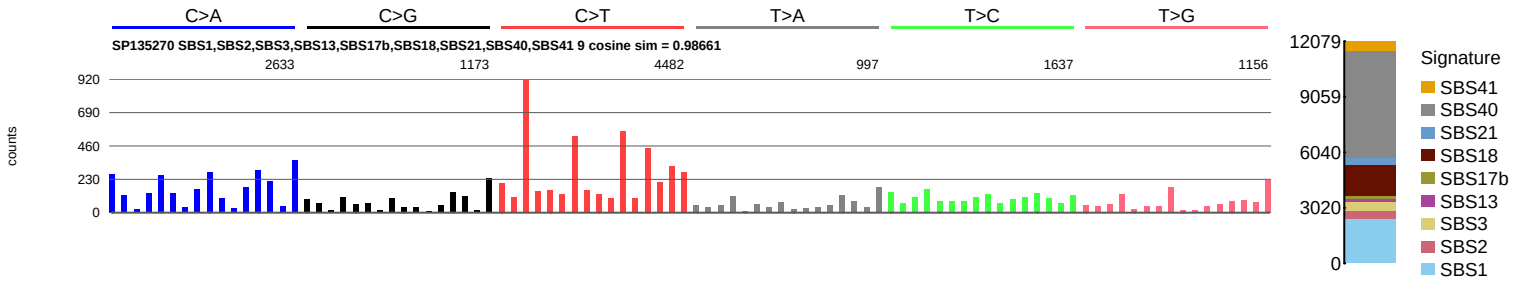
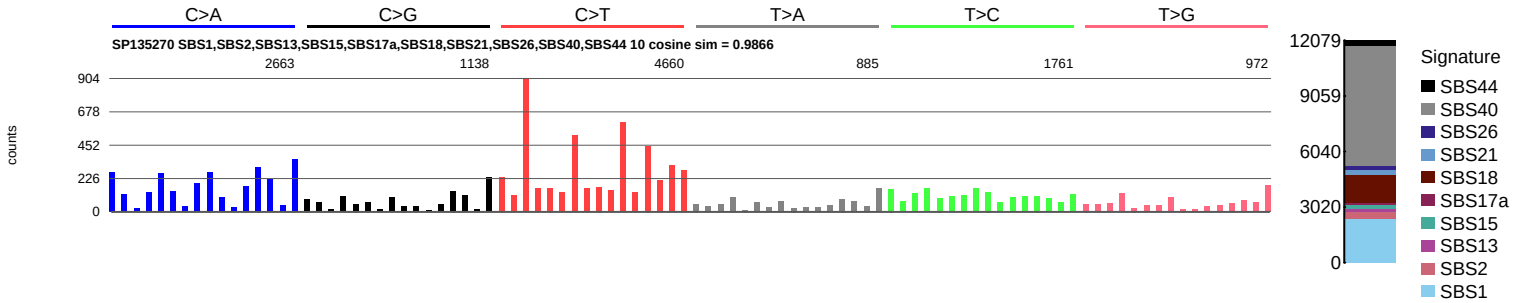
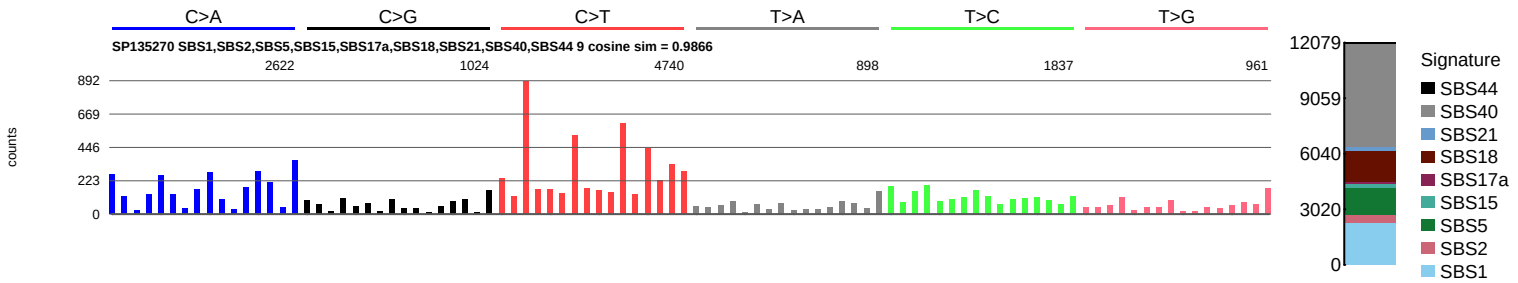
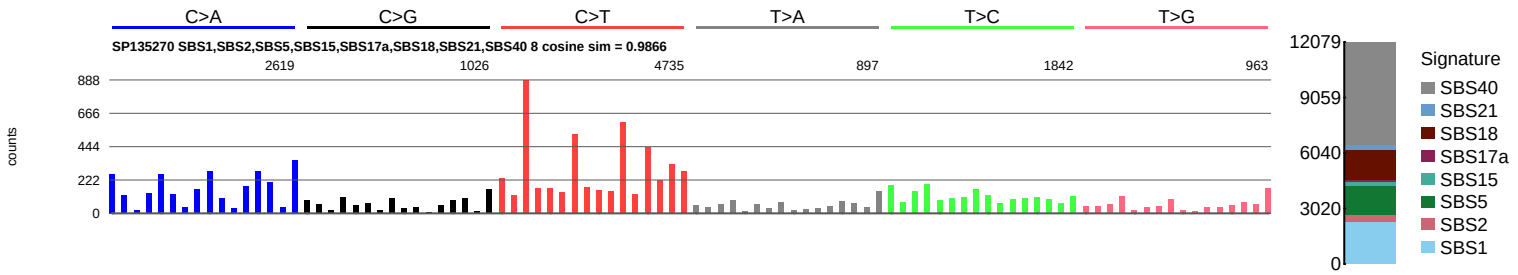
SBS5
SBS2
SBS1

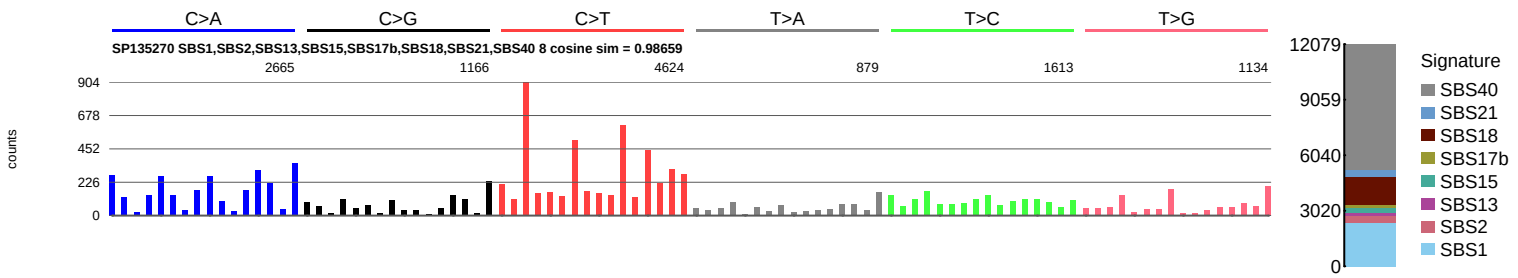

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98661
2508
989
4561
1072
1946
1001
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

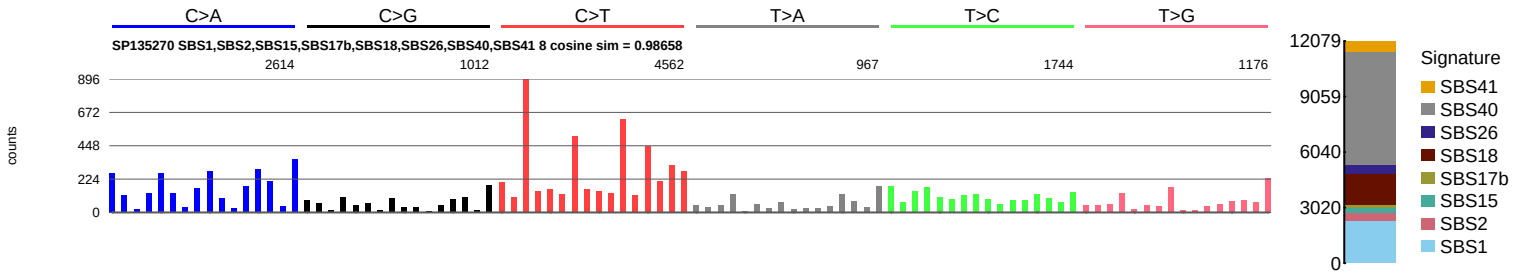

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98661
2633
1173
4482
997
1637
1156
920
690
460
230
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

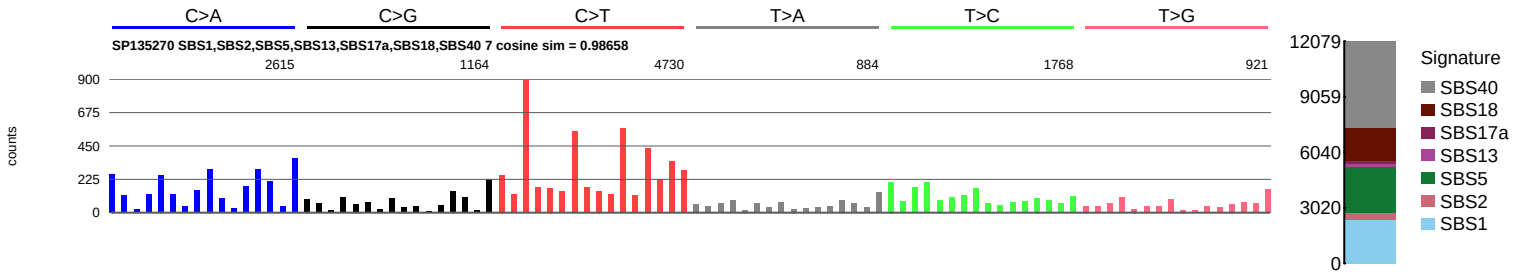

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.9866
2663
1138
4660
885
1761
972
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

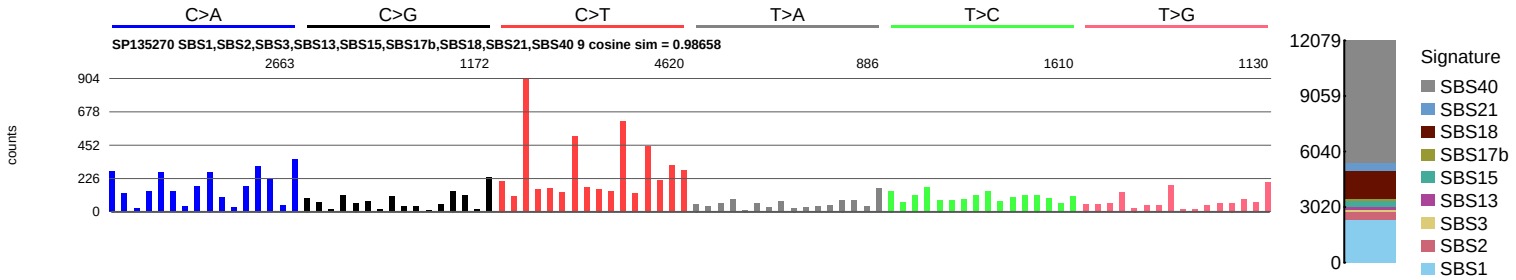

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.9866
2622
1024
4740
898
1837
961
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1

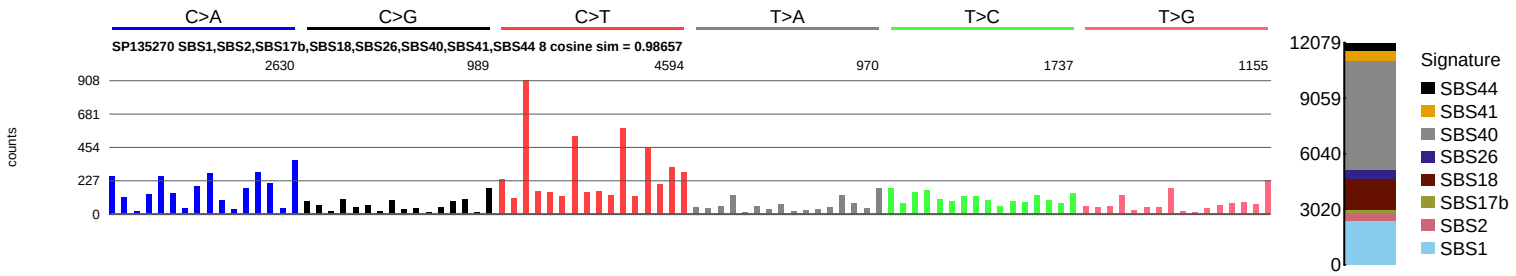

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.9866
2619
1026
4735
897
1842
963
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1

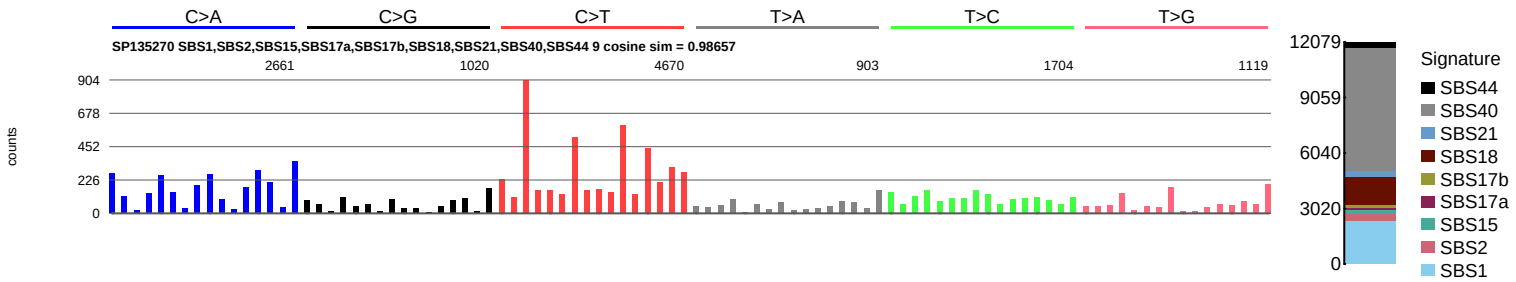

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98659
2665
1166
4624
879
1613
1134
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

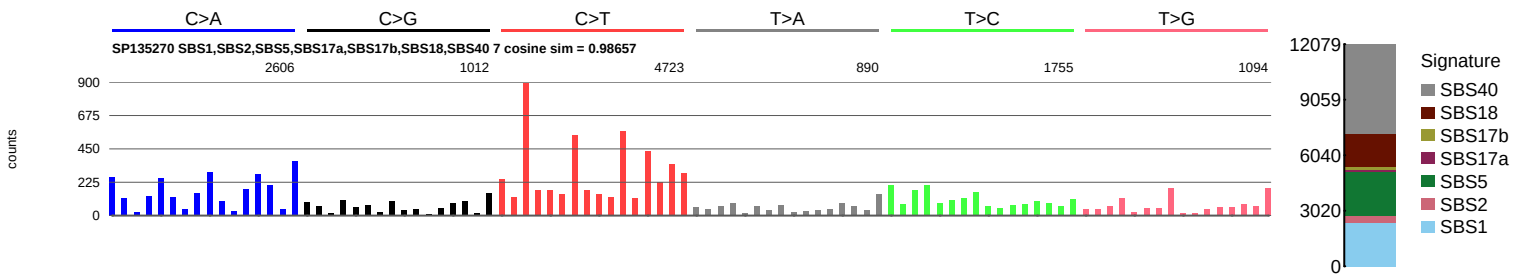

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98658
2614
1012
4562
967
1744
1176
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1

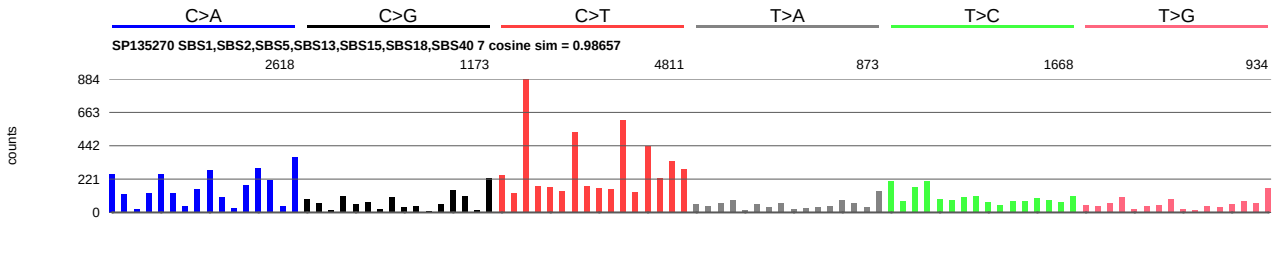

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40 7 cosine sim = 0.98658
2615
1164
4730
884
1768
921
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98658
2663
1172
4620
886
1610
1130
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98657
2630
989
4594
970
1737
1155
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98657
2661
1020
4670
903
1704
1119
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98657
2606
1012
4723
890
1755
1094
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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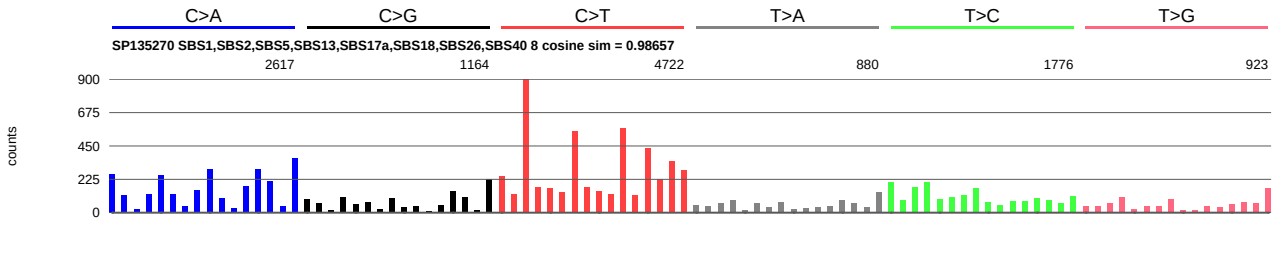
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40 7 cosine sim = 0.98657
2618
1173
4811
873
1668
934
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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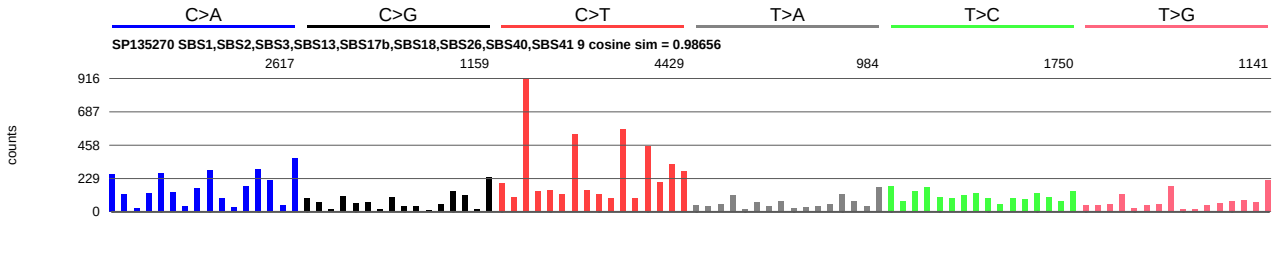
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98657
2617
1164
4722
880
1776
923
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1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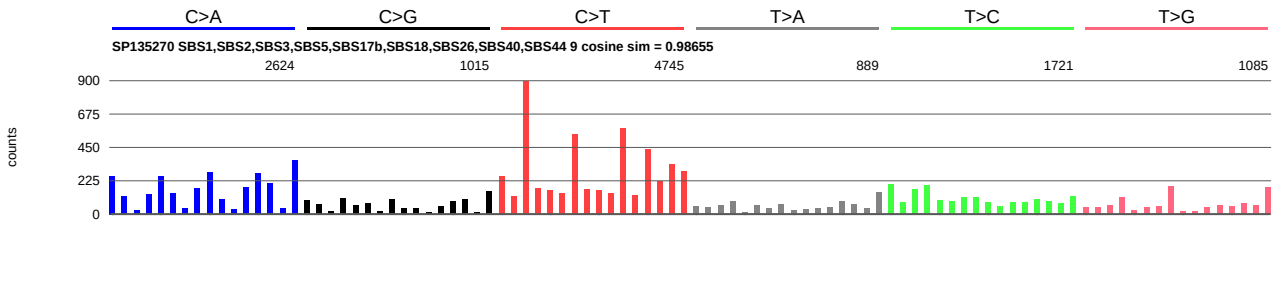
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98656
2617
1159
4429
984
1750
1141
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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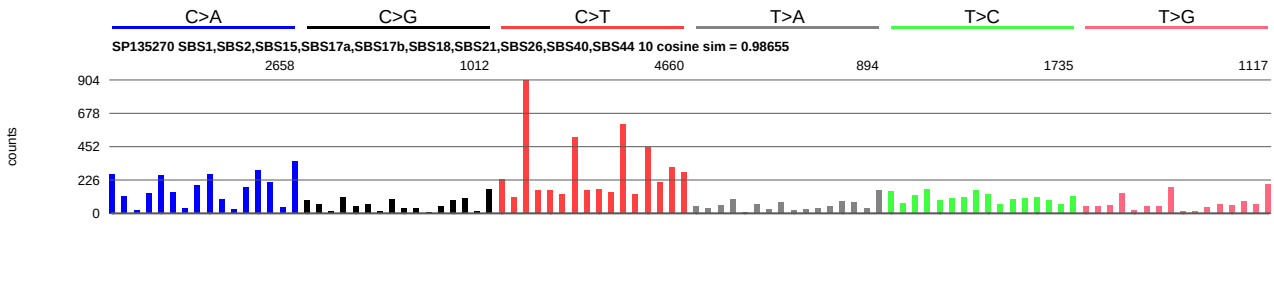
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98655
2624
1015
4745
889
1721
1085
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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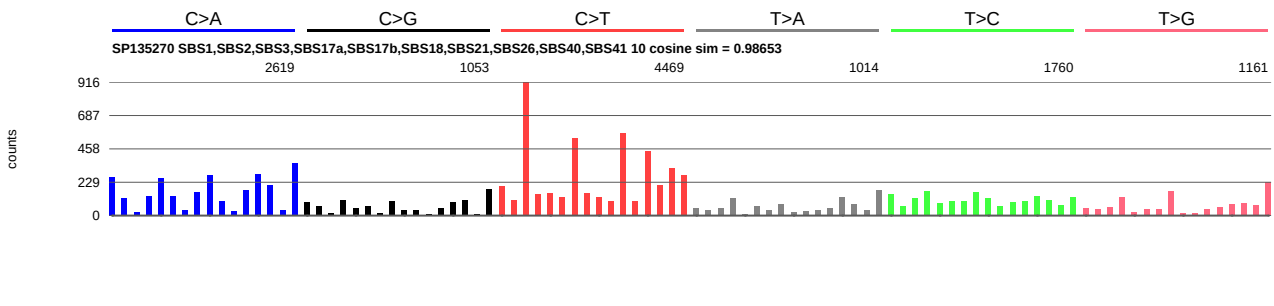
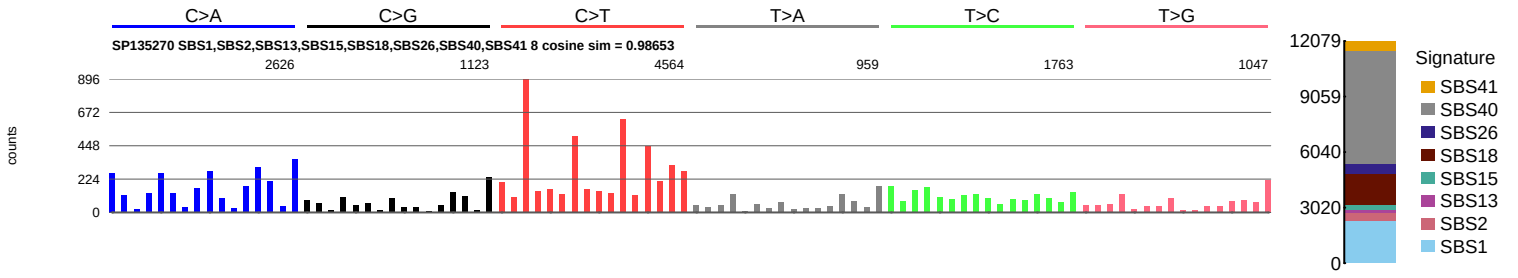
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98655
2658
1012
4660
894
1735
1117
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98653
2619
1053
4469
1014
1760
1161
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98653
2626
1123
4564
959
1763
1047
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
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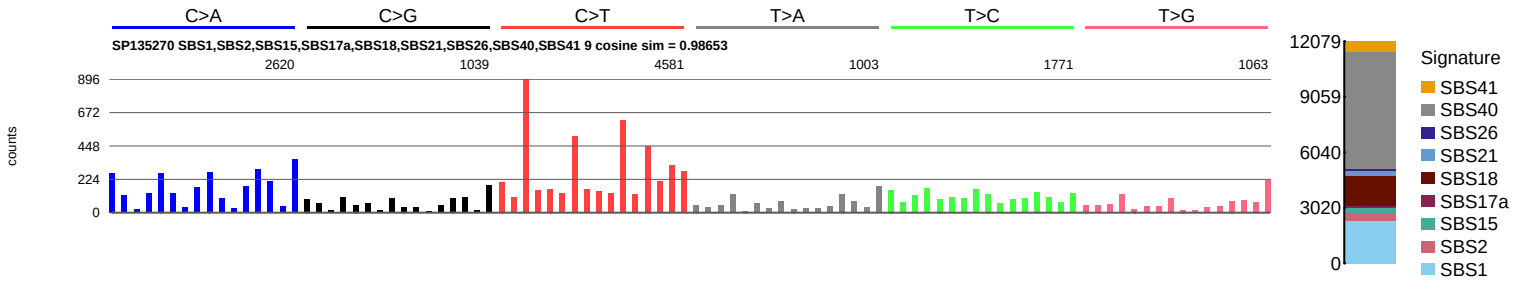
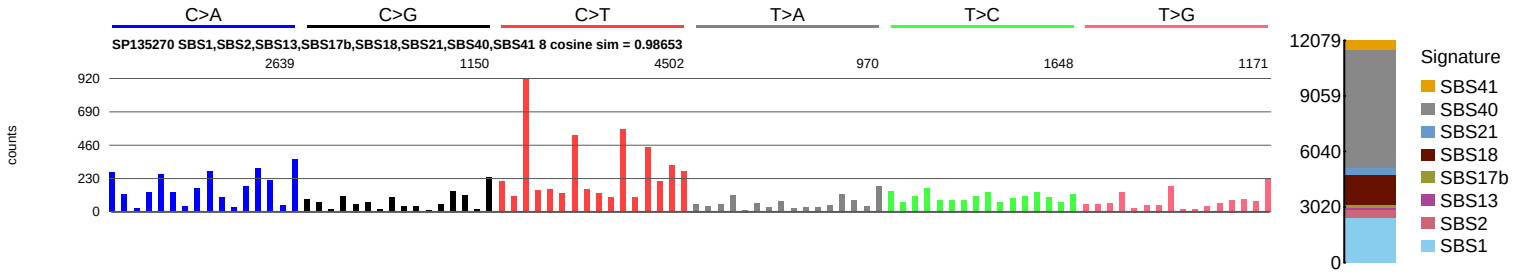
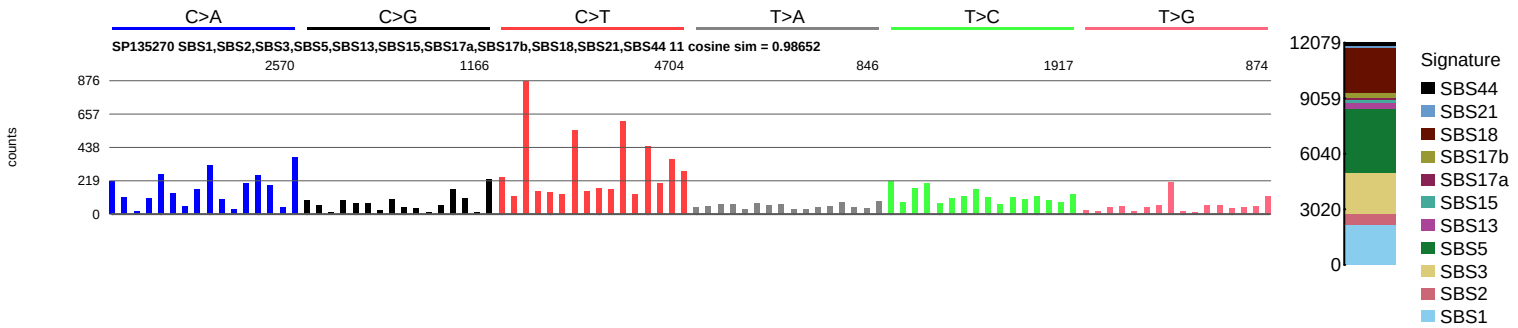
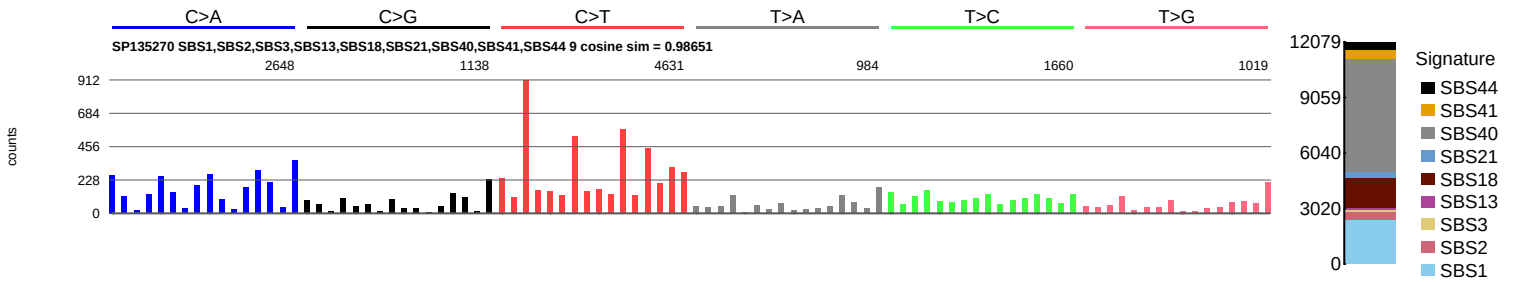
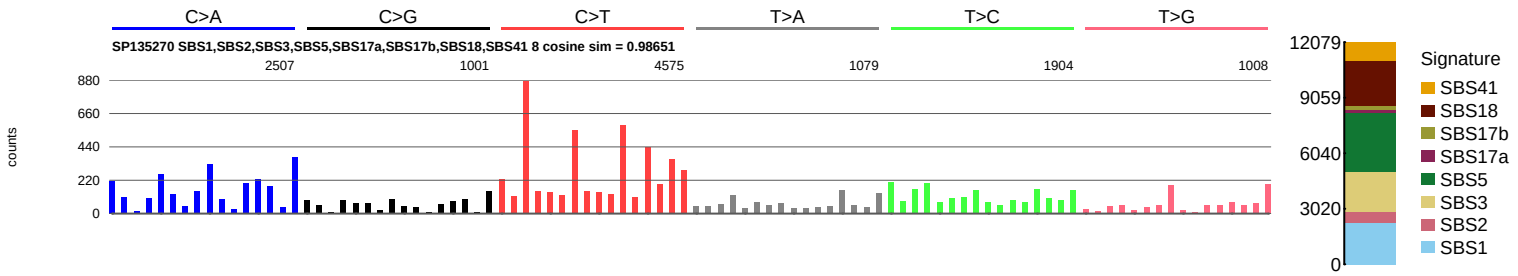
SBS13
SBS2
SBS1

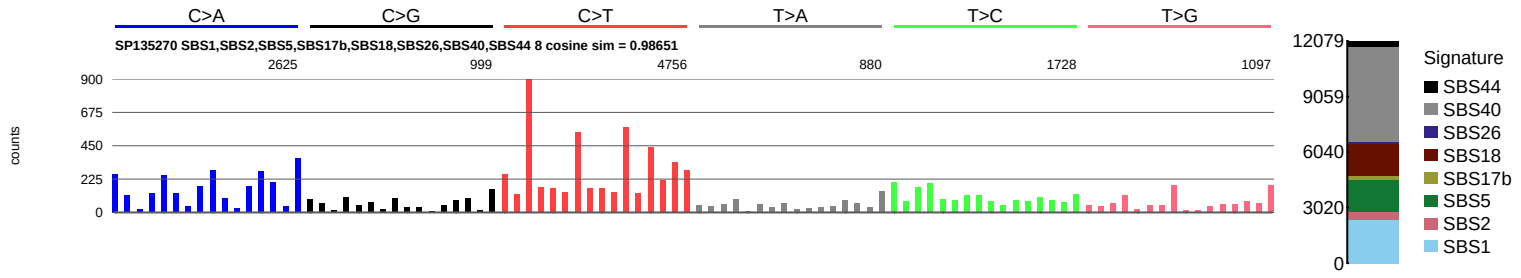

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98653
2620
1039
4581
1003
1771
1063
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1

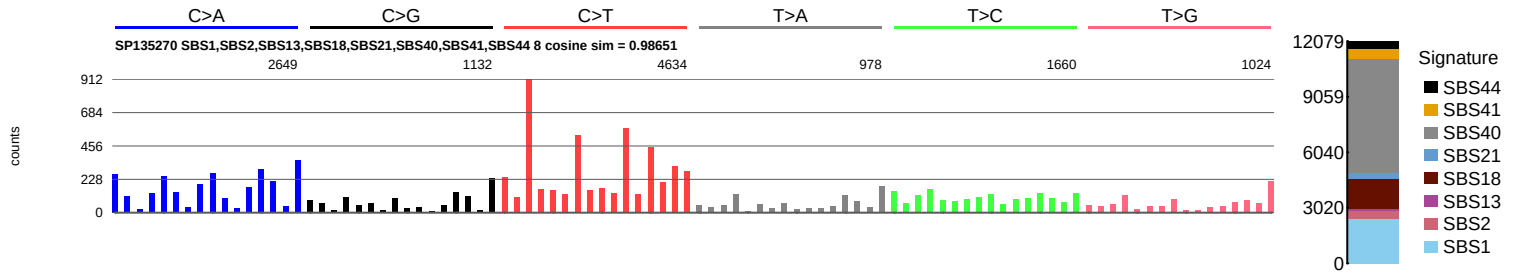

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98653
2639
1150
4502
970
1648
1171
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

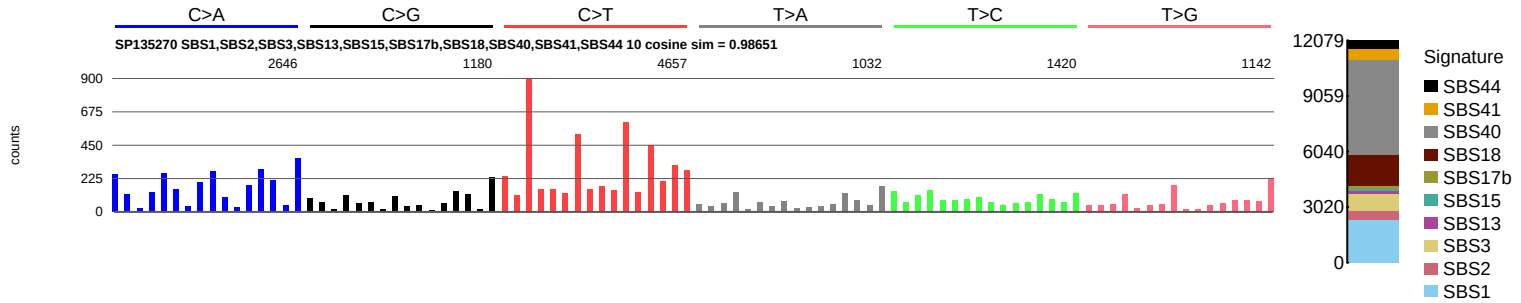

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98652
2570
1166
4704
846
1917
874
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

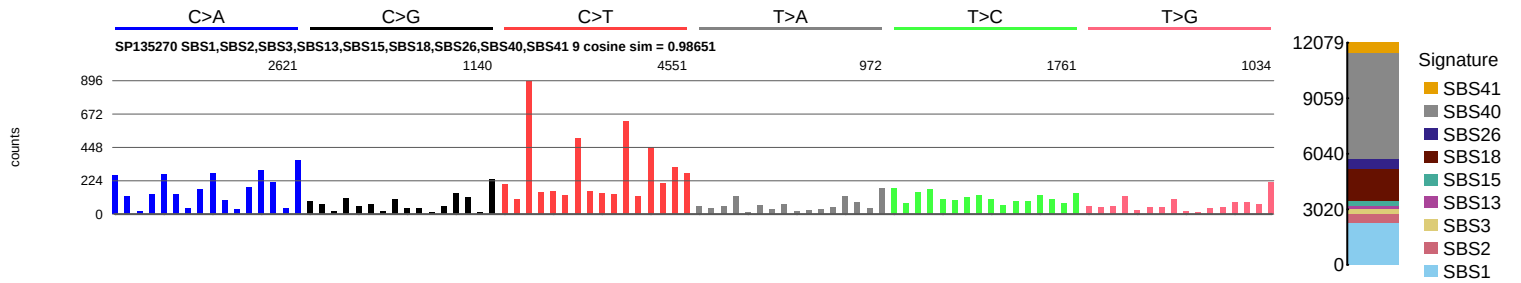

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98651
2648
1138
4631
984
1660
1019
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1

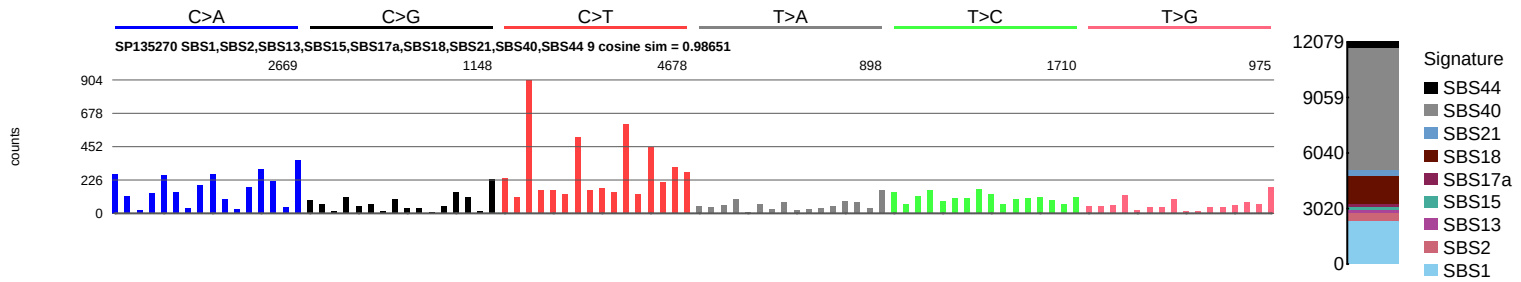

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98651
2507
1001
4575
1079
1904
1008
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

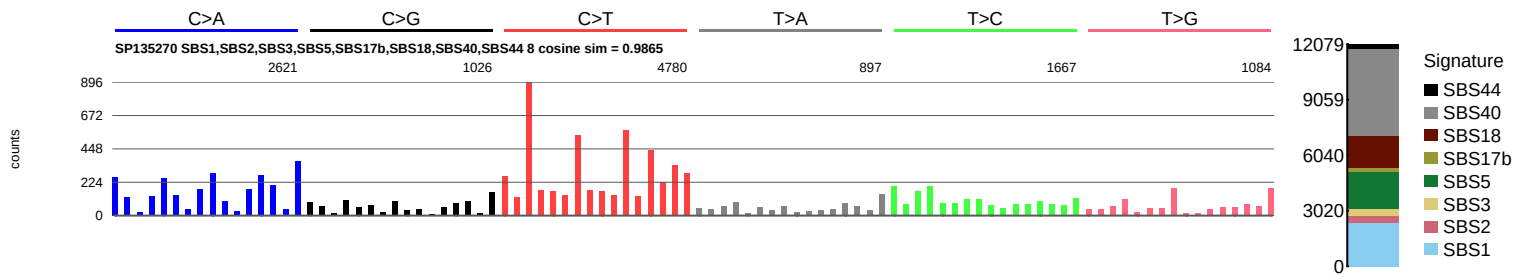

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98651
2625
999
4756
880
1728
1097
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1

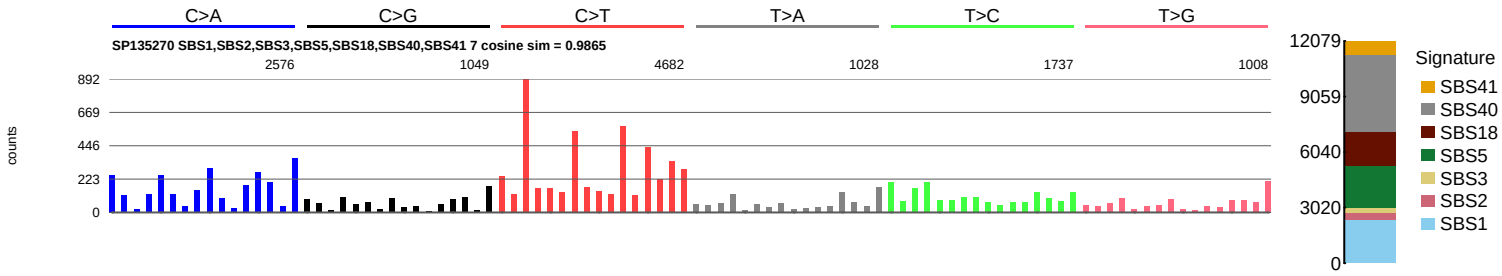

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98651
2649
1132
4634
978
1660
1024
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1

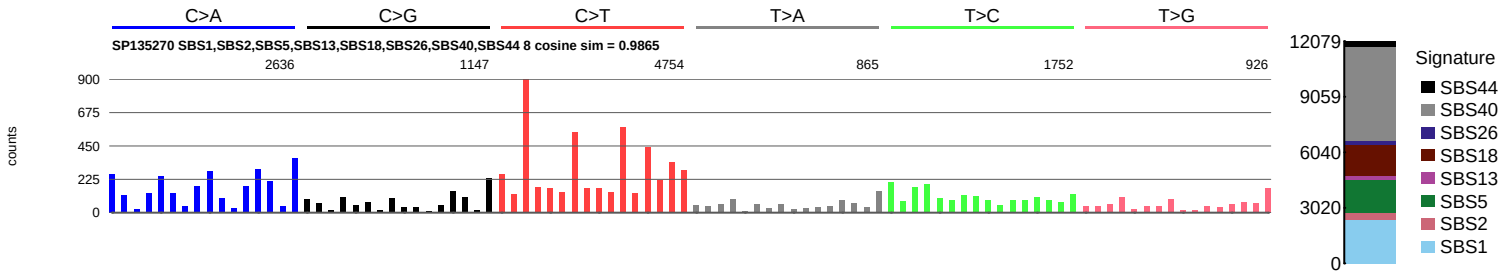

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98651
2646
1180
4657
1032
1420
1142
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

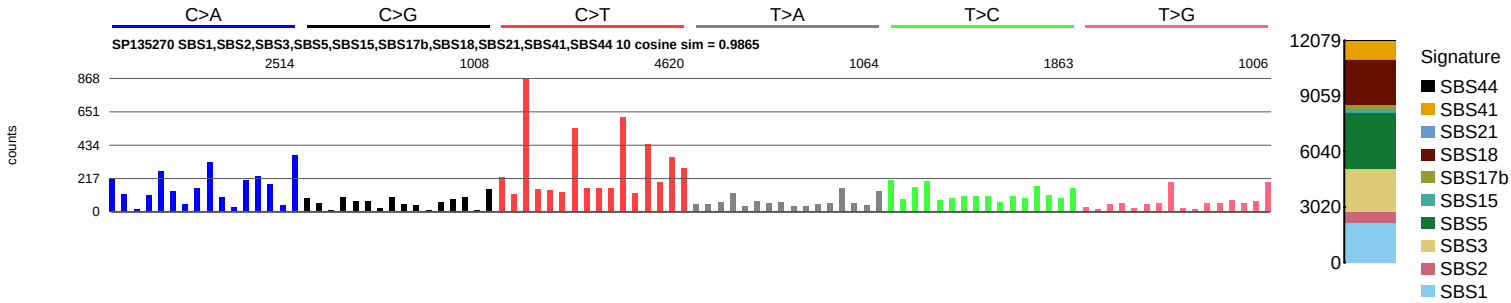

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98651
2621
1140
4551
972
1761
1034
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

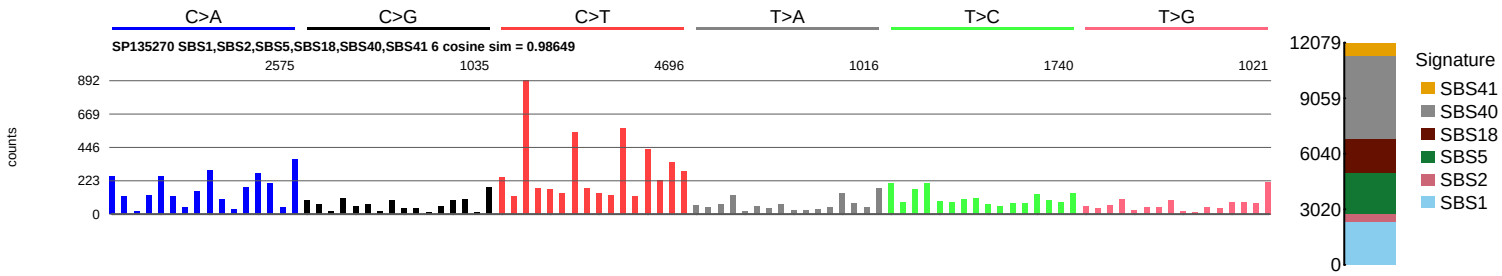

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98651
2669
1148
4678
898
1710
975
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

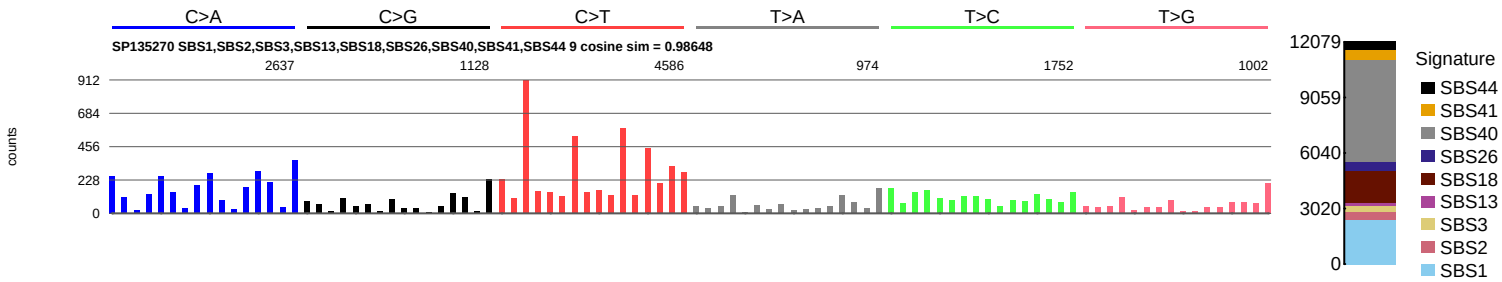

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.9865
2621
1026
4780
897
1667
1084
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

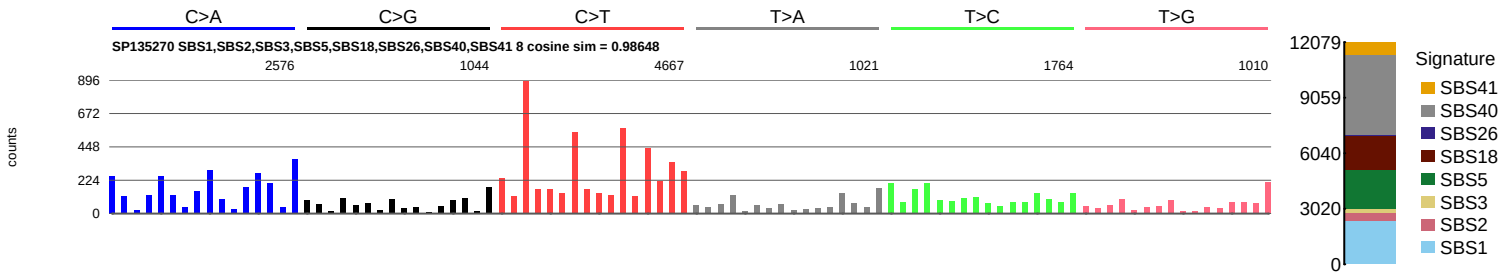

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS18,SBS40,SBS41 7 cosine sim = 0.9865
2576
1049
4682
1028
1737
1008
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS5
SBS3
SBS2
SBS1

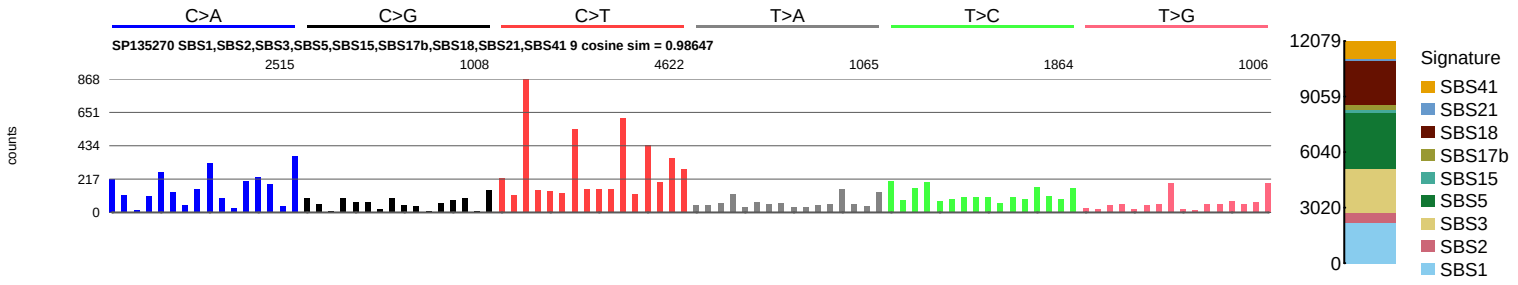

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.9865
2636
1147
4754
865
1752
926
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1

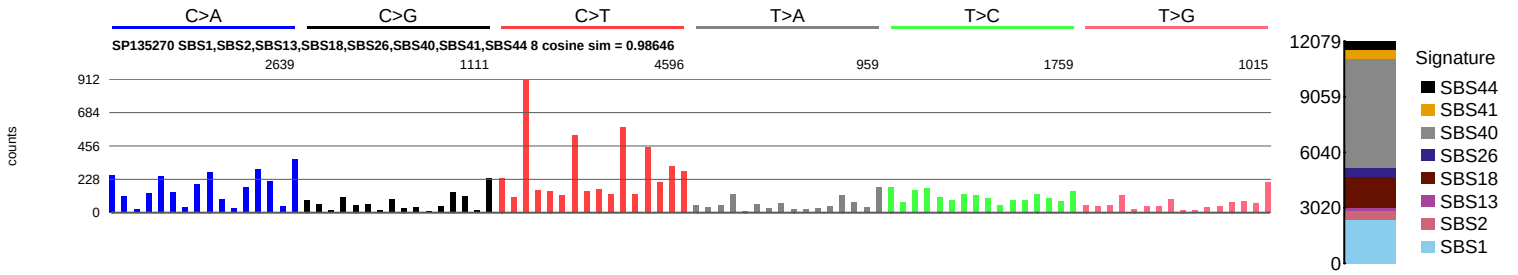

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9865
2514
1008
4620
1064
1863
1006
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

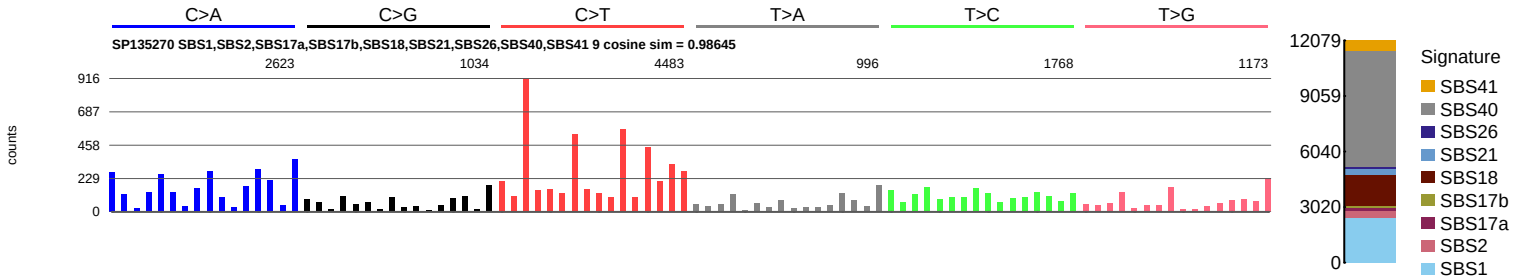

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS40,SBS41 6 cosine sim = 0.98649
2575
1035
4696
1016
1740
1021
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS5
SBS2
SBS1

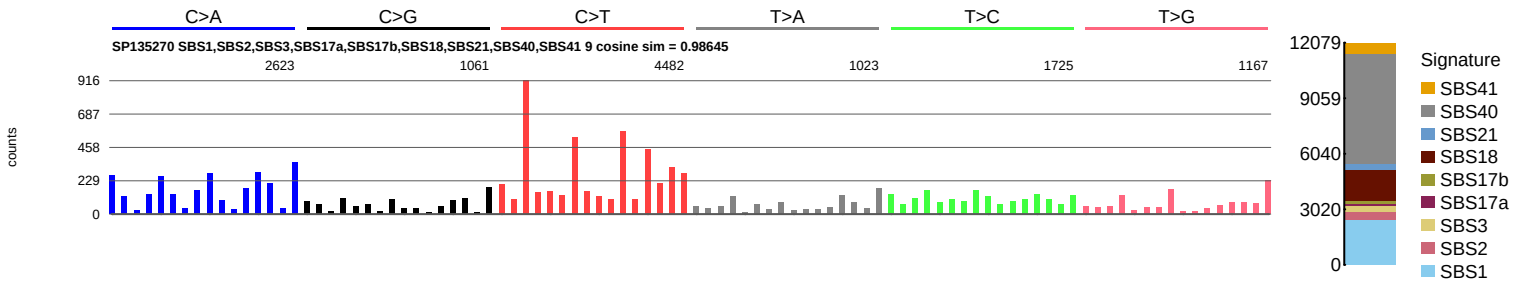

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98648
2637
1128
4586
974
1752
1002
912
684
456
228
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1

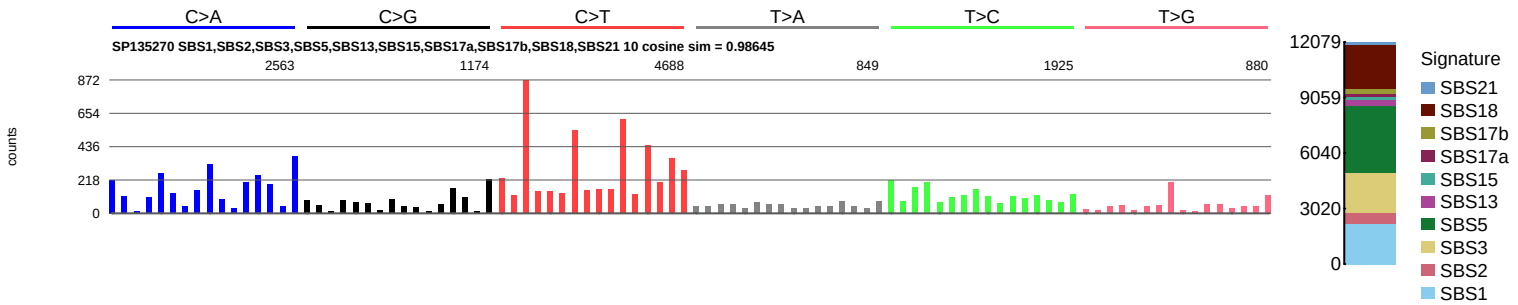

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98648
2576
1044
4667
1021
1764
1010
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS5
SBS3
SBS2
SBS1

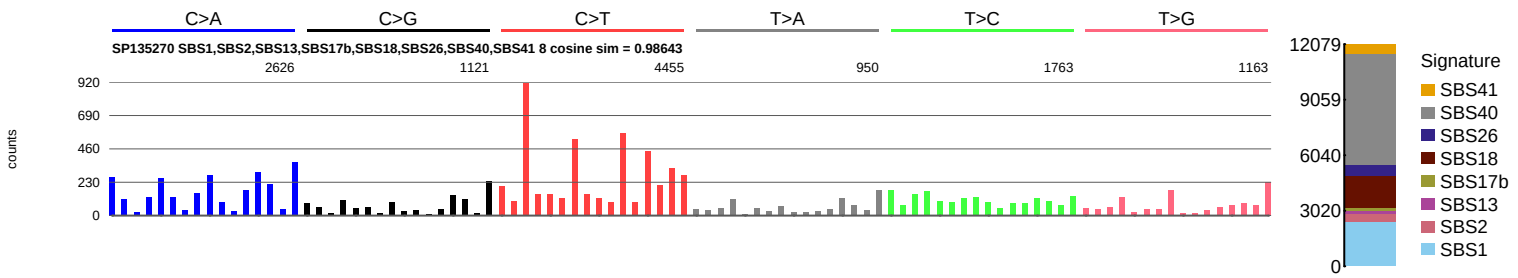

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98647
2515
1008
4622
1065
1864
1006
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

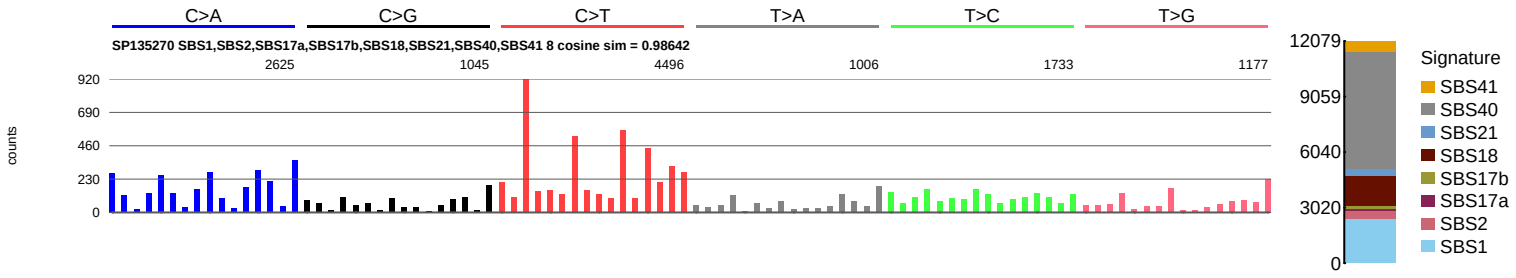

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98646
2639
1111
4596
959
1759
1015
912
684
456
228
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1

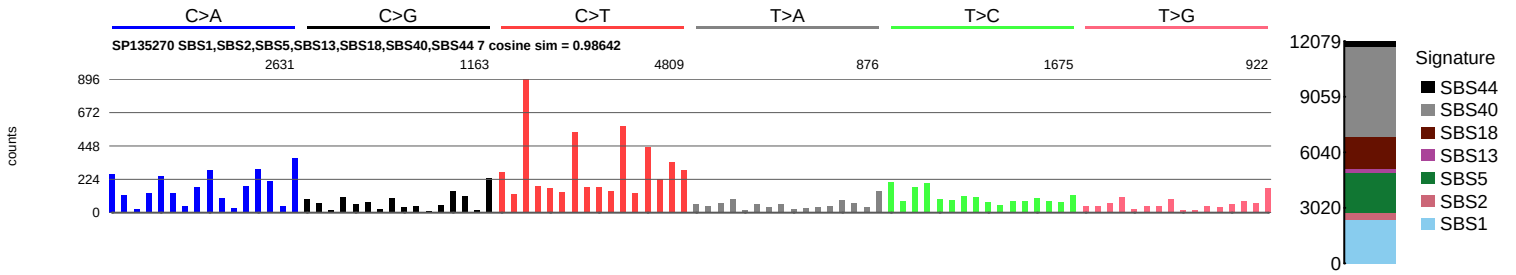

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98645
2623
1034
4483
996
1768
1173
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

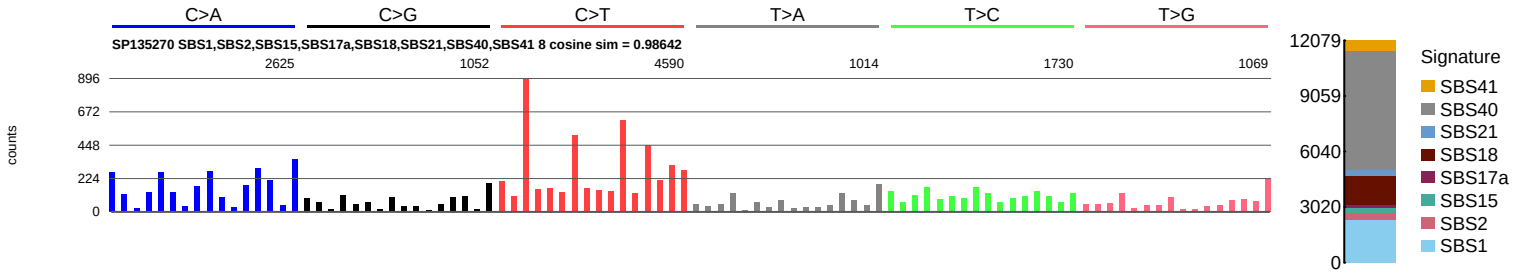

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98645
2623
1061
4482
1023
1725
1167
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

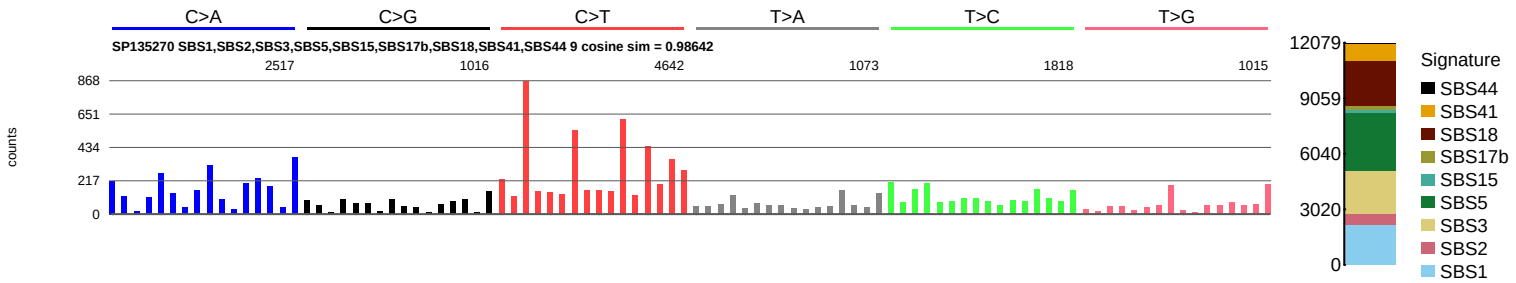

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98645
2563
1174
4688
849
1925
880
872
654
436
218
0
counts
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

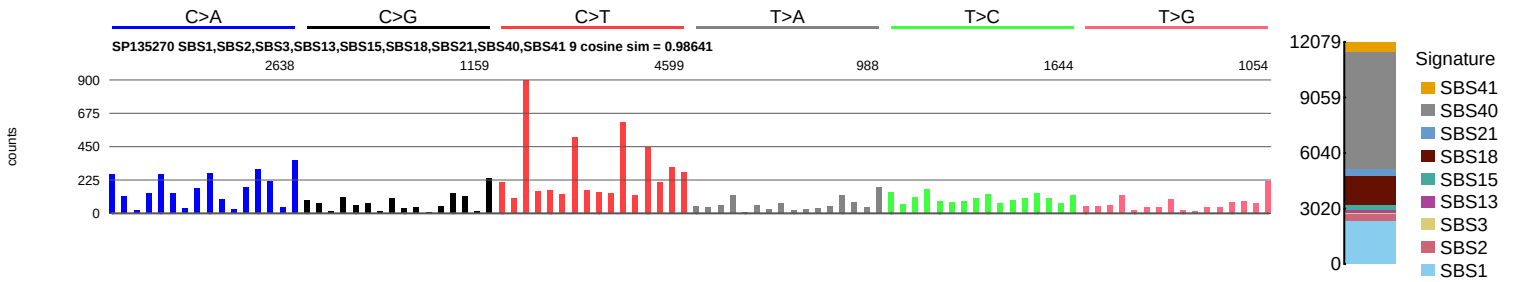

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98643
2626
1121
4455
950
1763
1163
920
690
460
230
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1

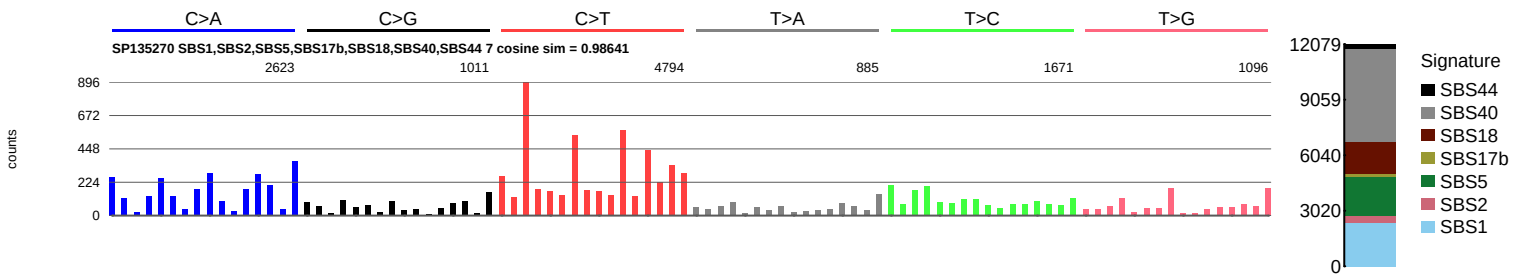

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98642
2625
1045
4496
1006
1733
1177
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

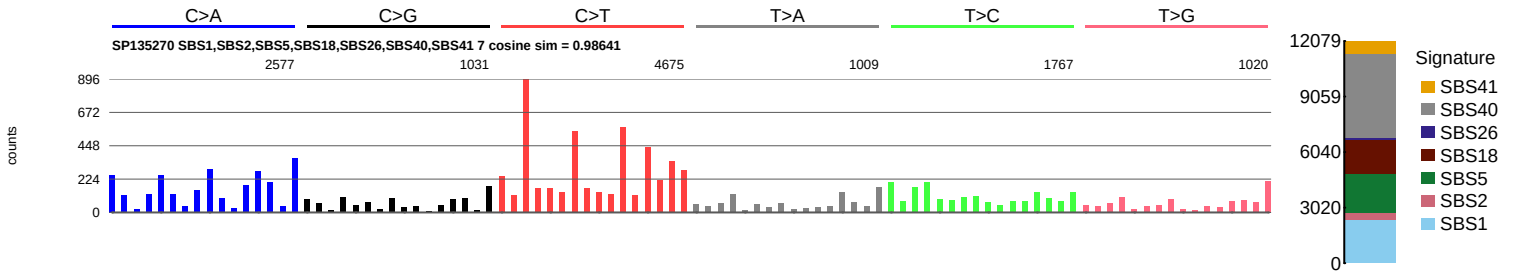

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS44 7 cosine sim = 0.98642
2631
1163
4809
876
1675
922
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1

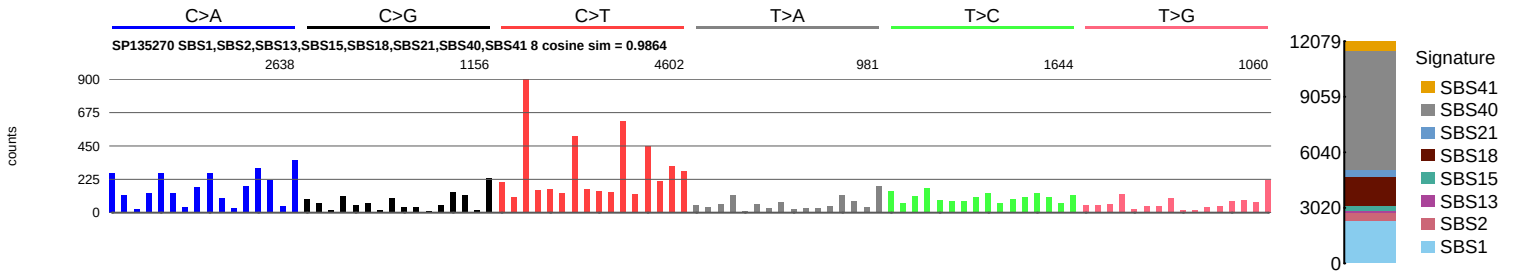

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98642
2625
1052
4590
1014
1730
1069
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1

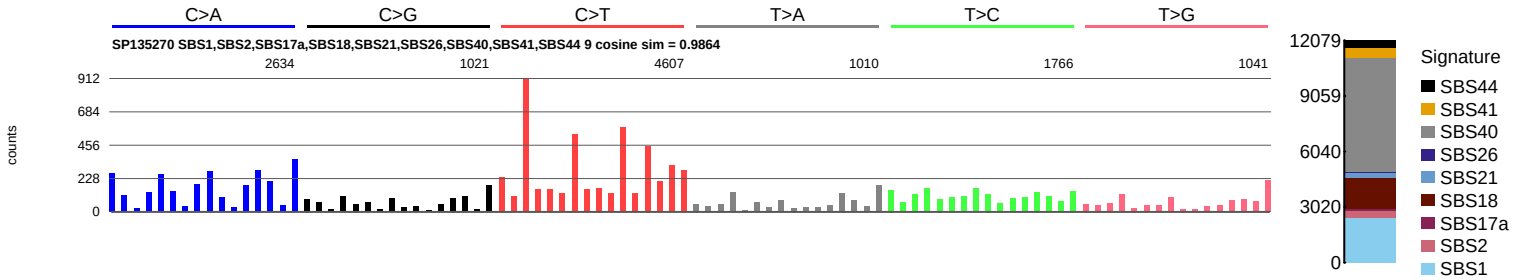

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98642
2517
1016
4642
1073
1818
1015
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

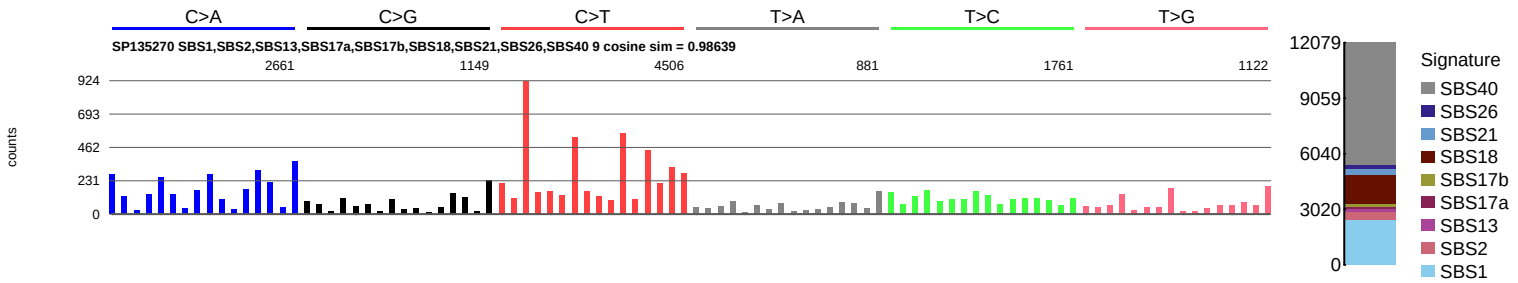

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98641
2638
1159
4599
988
1644
1054
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

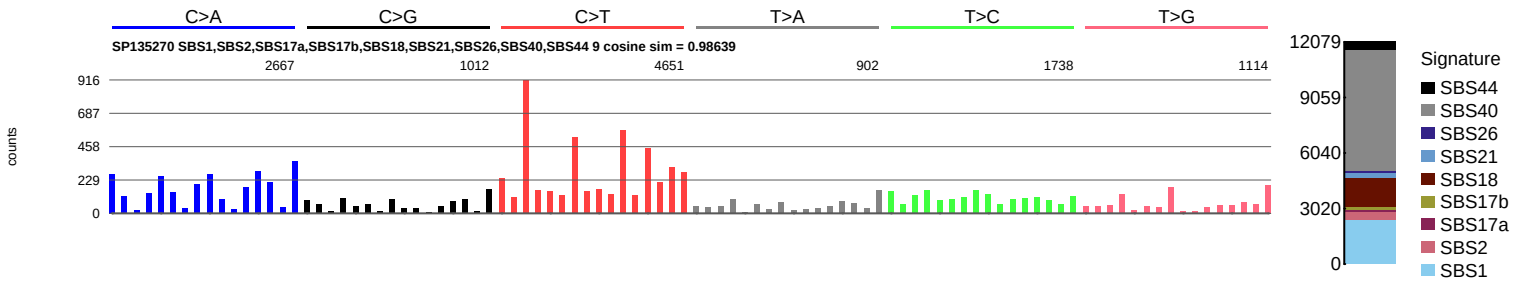

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98641
2623
1011
4794
885
1671
1096
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1

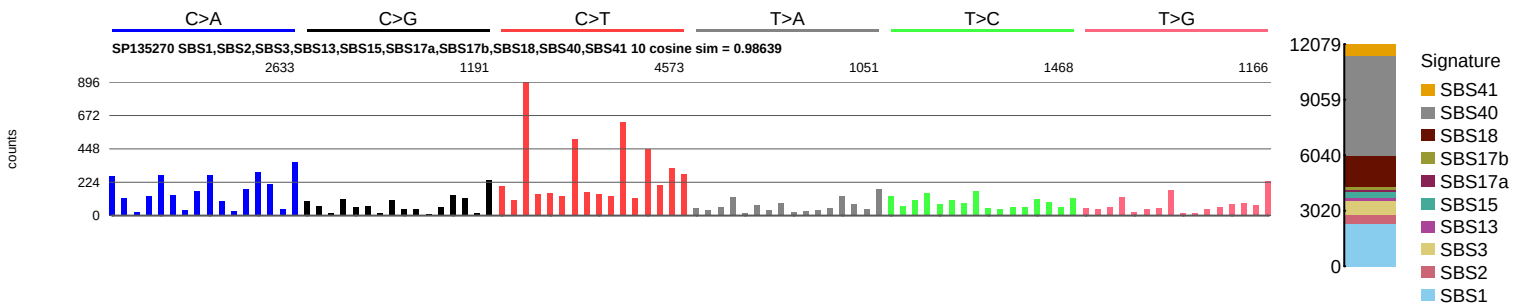

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98641
2577
1031
4675
1009
1767
1020
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1

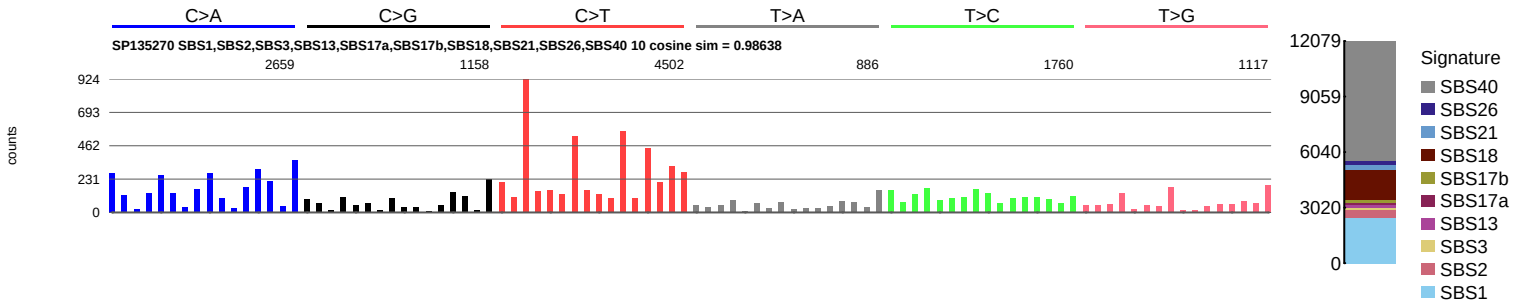

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9864
2638
1156
4602
981
1644
1060
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

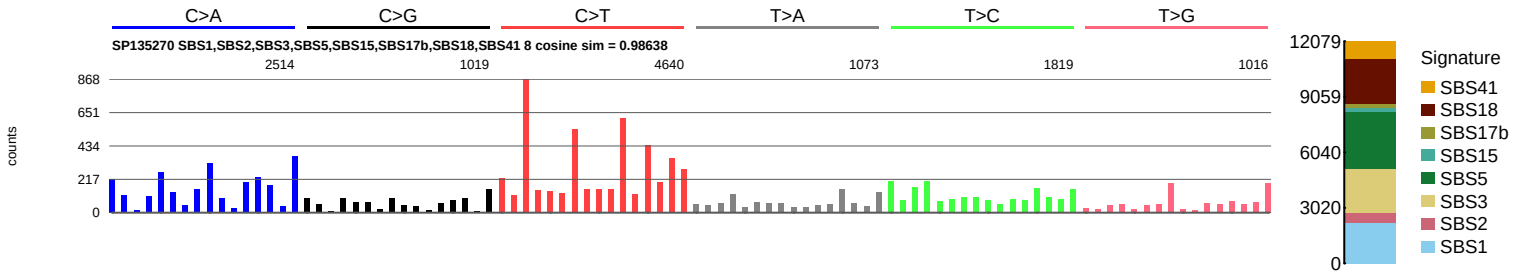

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9864
2634
1021
4607
1010
1766
1041
912
684
456
228
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS2
SBS1

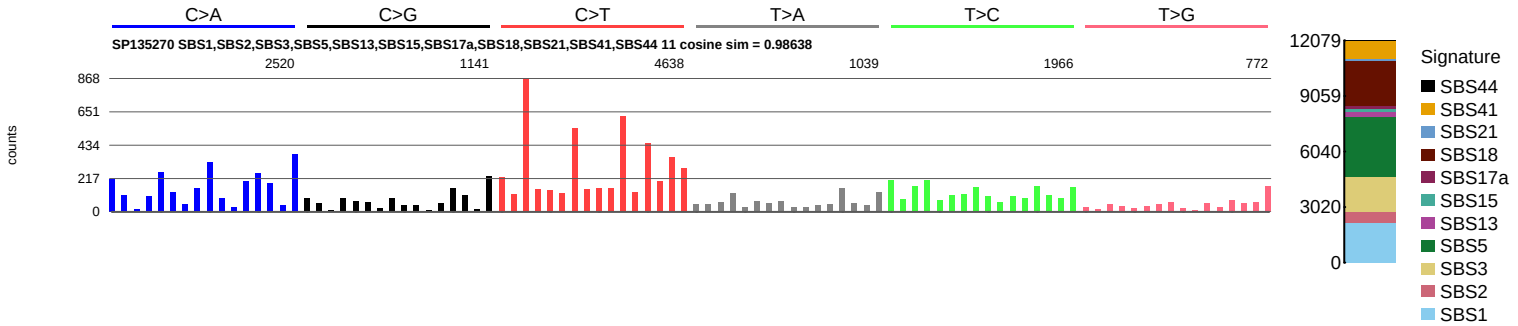

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98639
2661
1149
4506
881
1761
1122
924
693
462
231
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

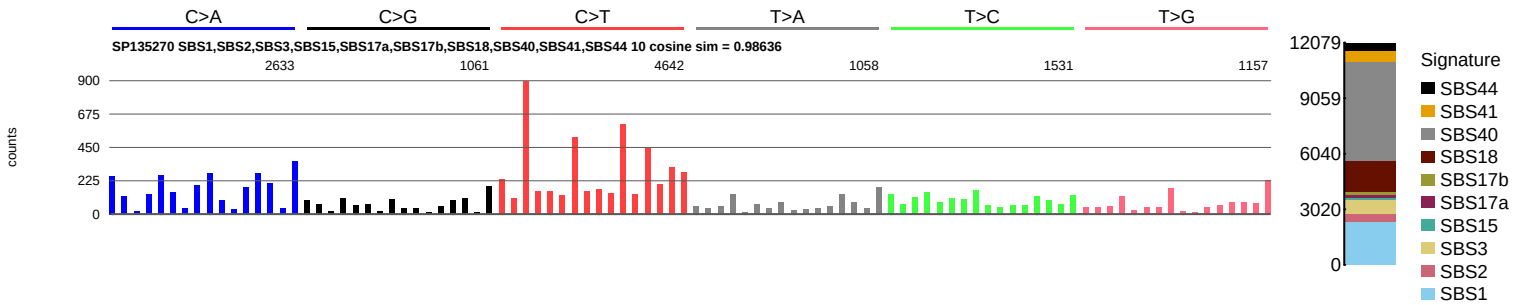

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98639
2667
1012
4651
902
1738
1114
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

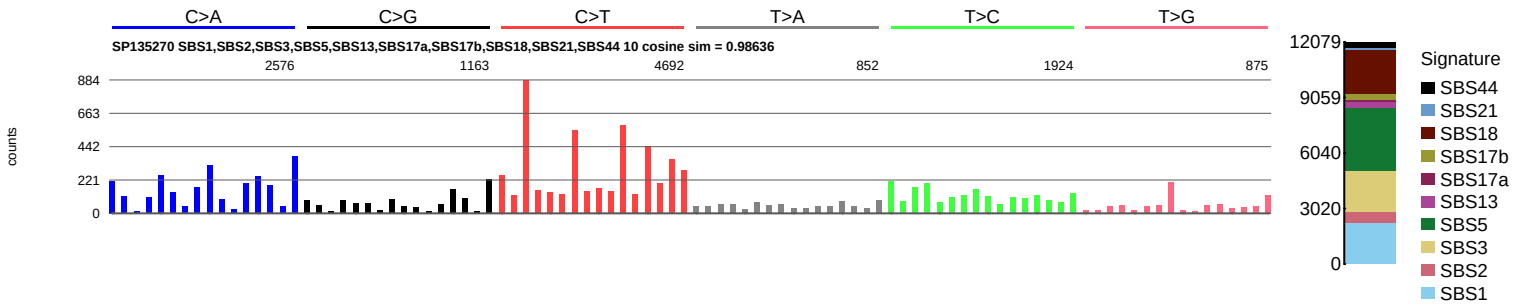

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98639
2633
1191
4573
1051
1468
1166
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

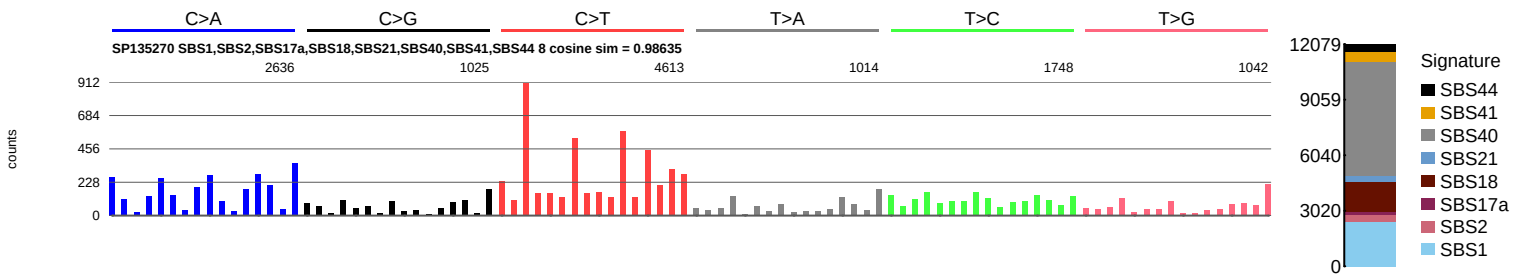

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98638
2659
1158
4502
886
1760
1117
924
693
462
231
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

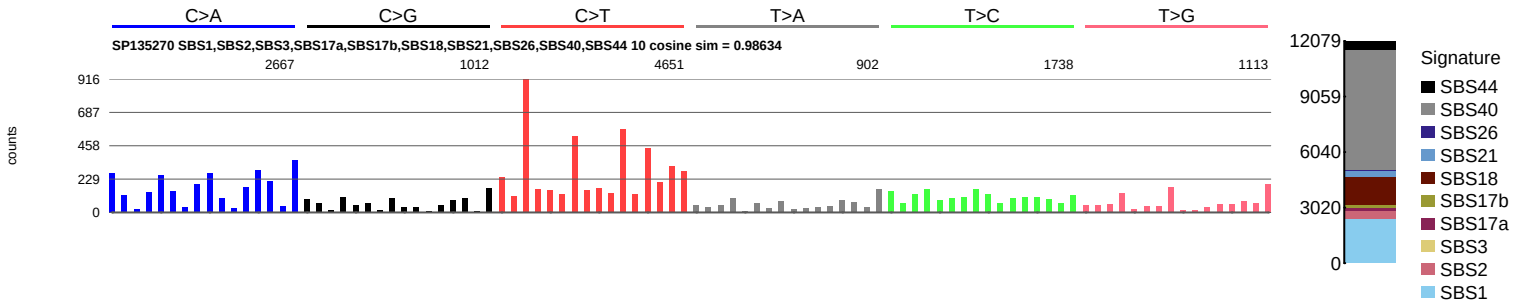

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98638
2514
1019
4640
1073
1819
1016
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

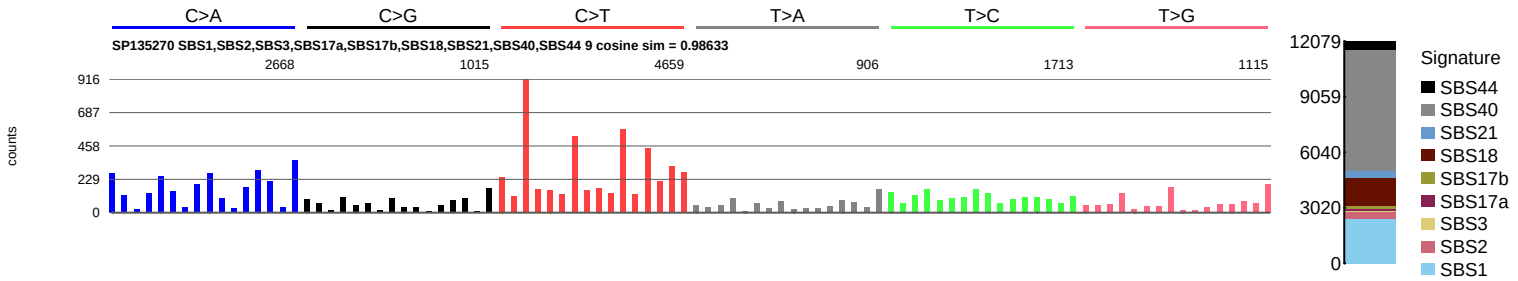

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98638
2520
1141
4638
1039
1966
772
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98636
2633
1061
4642
1058
1531
1157
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98636
2576
1163
4692
852
1924
875
884
663
442
221
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98635
2636
1025
4613
1014
1748
1042
912
684
456
228
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98634
2667
1012
4651
902
1738
1113
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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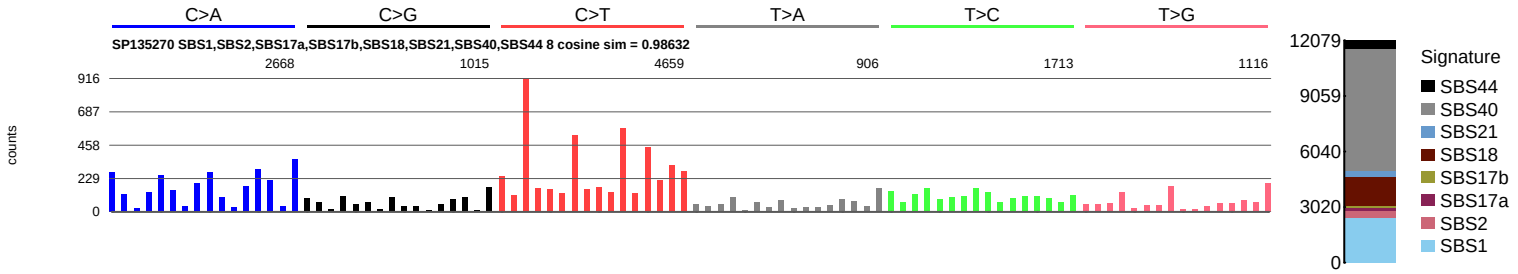
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98633
2668
1015
4659
906
1713
1115
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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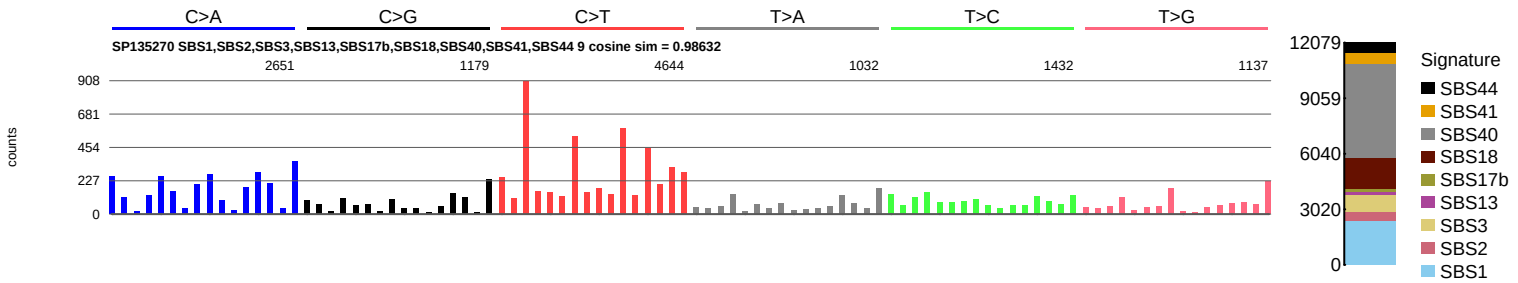
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98632
2668
1015
4659
906
1713
1116
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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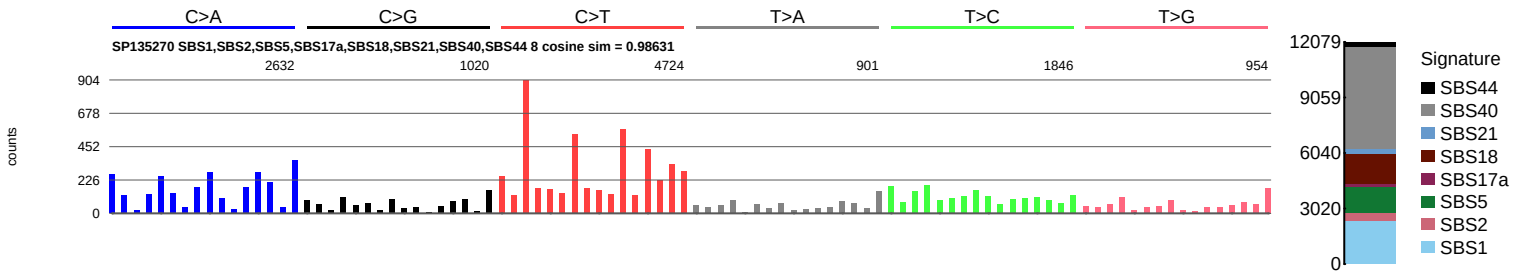
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98632
2651
1179
4644
1032
1432
1137
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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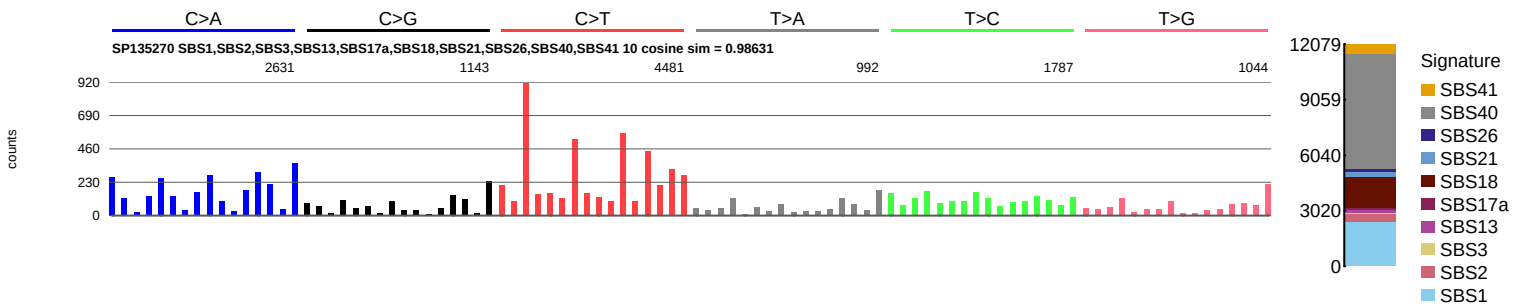
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98631
2632
1020
4724
901
1846
954
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1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C>T
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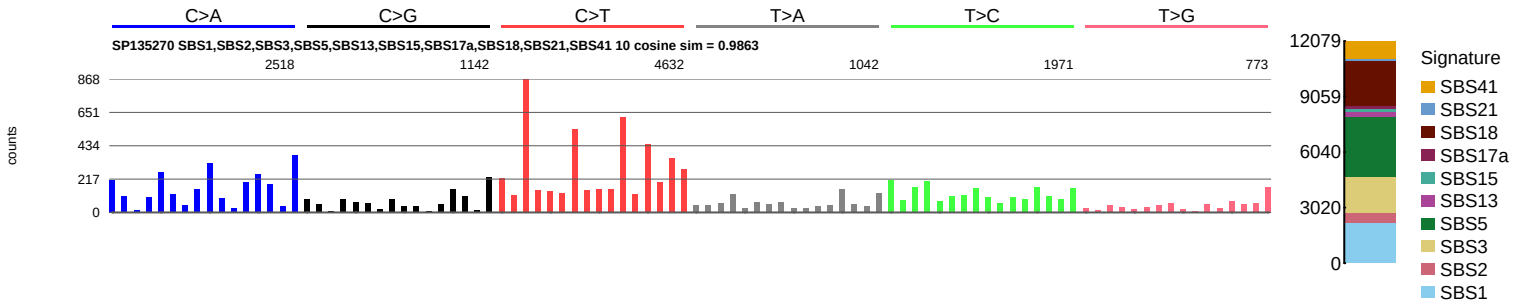
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98631
2631
1143
4481
992
1787
1044
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1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C>T
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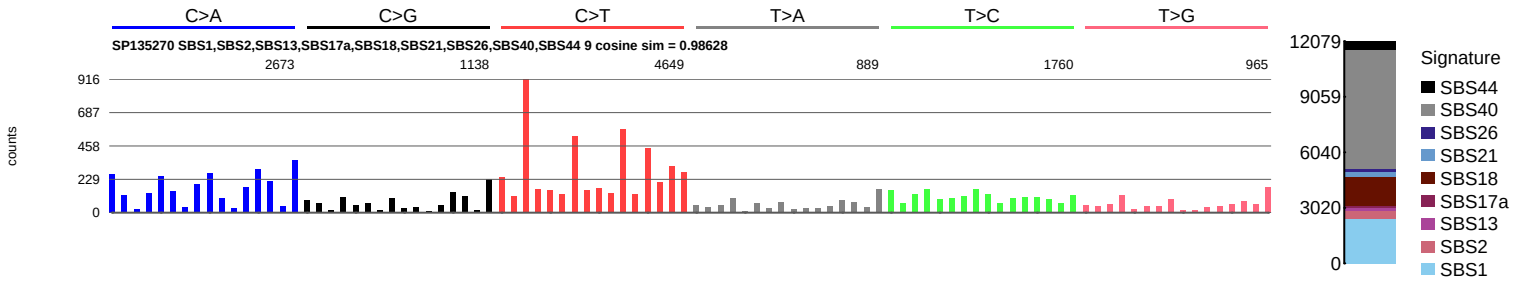
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41 10 cosine sim = 0.9863
2518
1142
4632
1042
1971
773
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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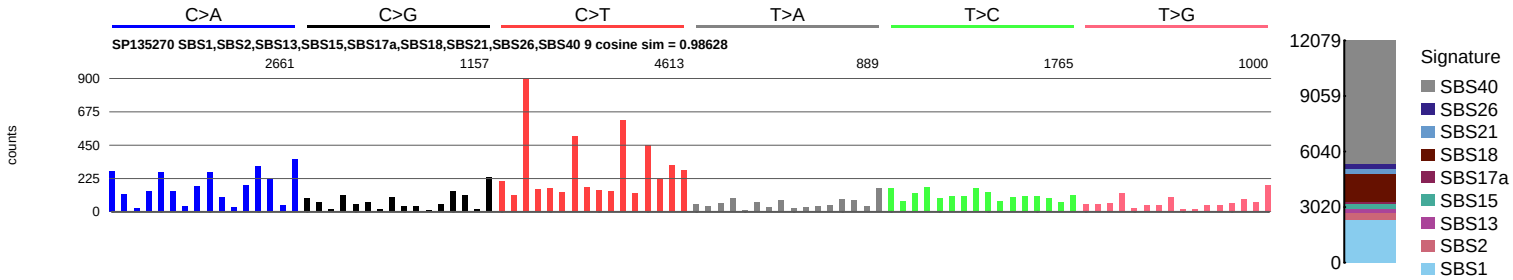
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98628
2673
1138
4649
889
1760
965
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1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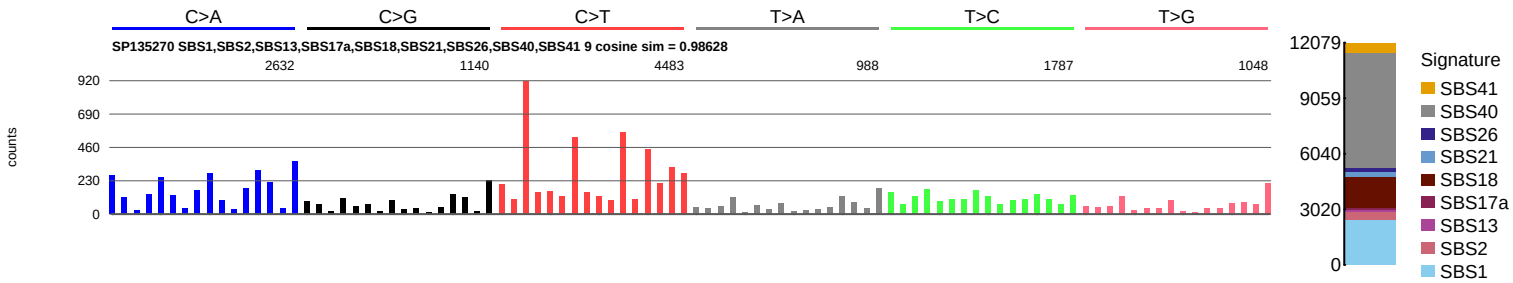
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98628
2661
1157
4613
889
1765
1000
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1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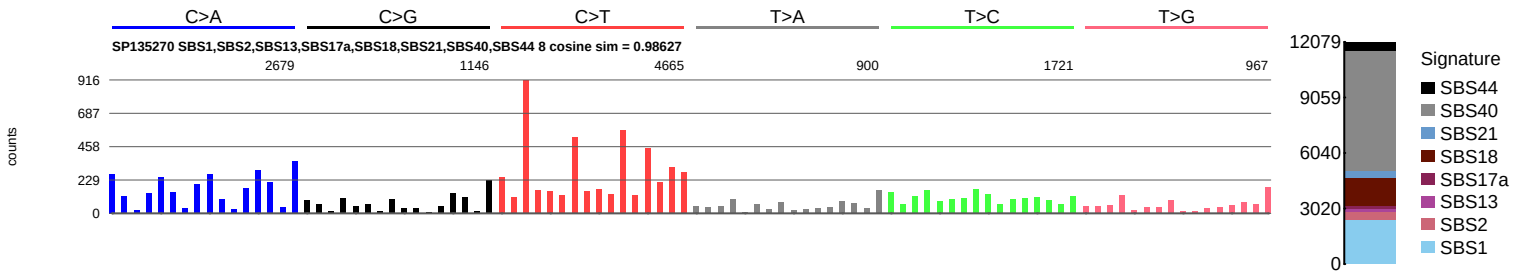
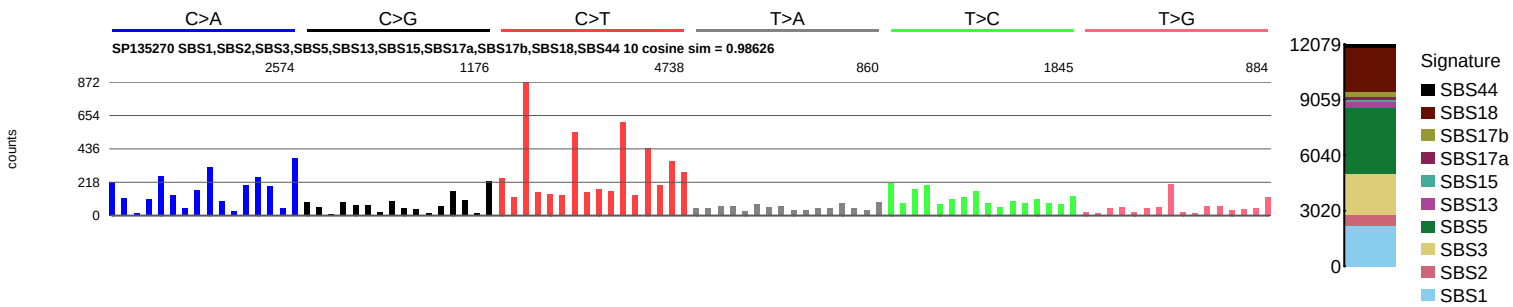
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98628
2632
1140
4483
988
1787
1048
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98627
2679
1146
4665
900
1721
967
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1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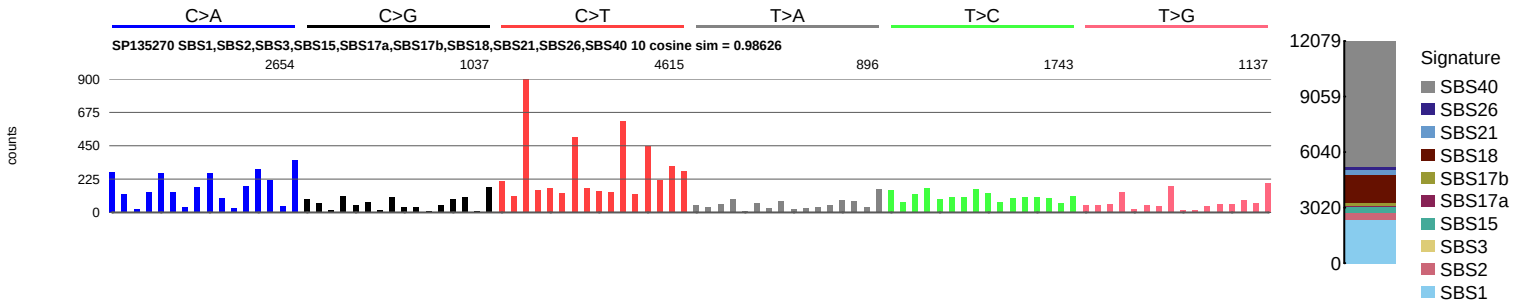
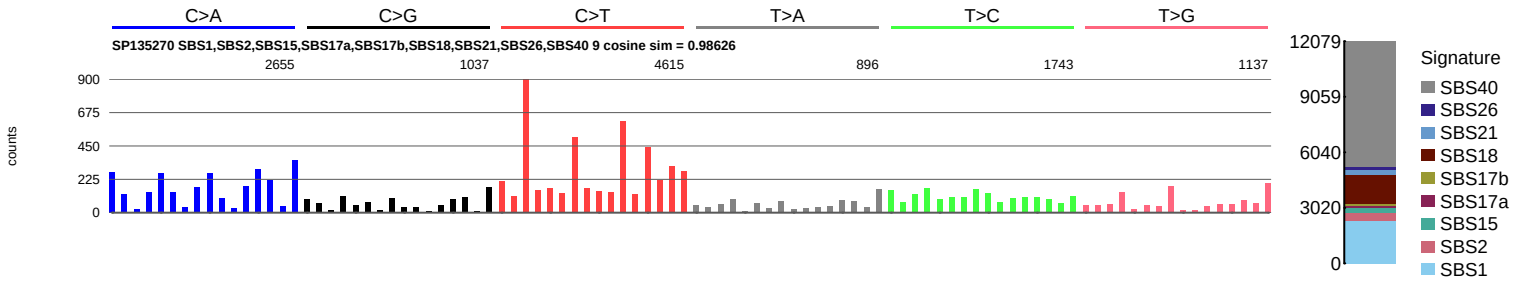
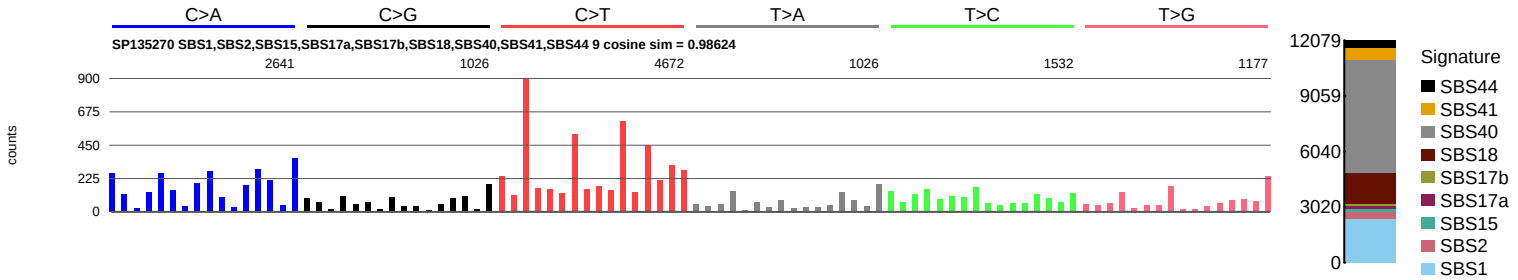
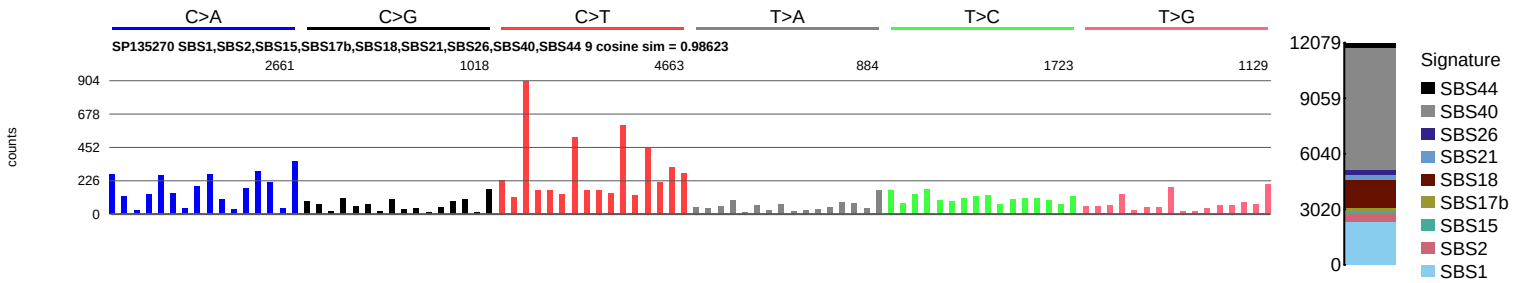
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98626
2574
1176
4738
860
1845
884
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98626
2654
1037
4615
896
1743
1137
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98626
2655
1037
4615
896
1743
1137
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98624
2641
1026
4672
1026
1532
1177
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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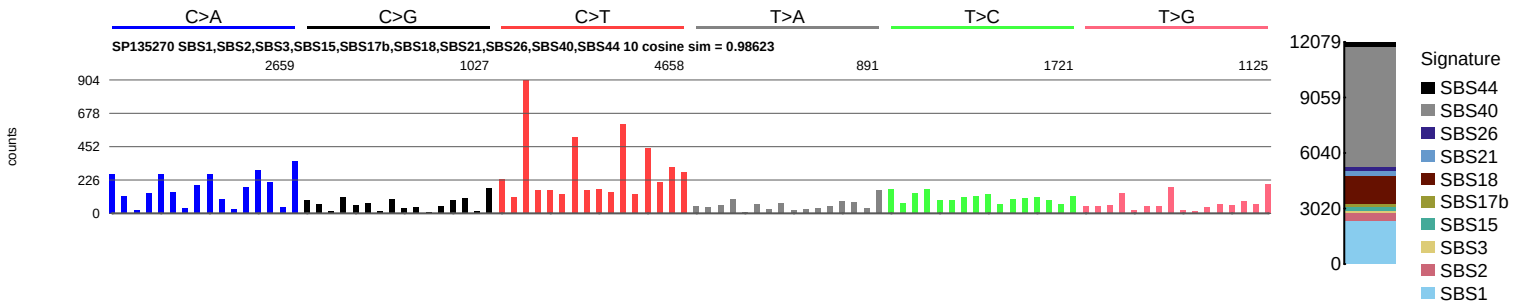
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98623
2661
1018
4663
884
1723
1129
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1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C>T
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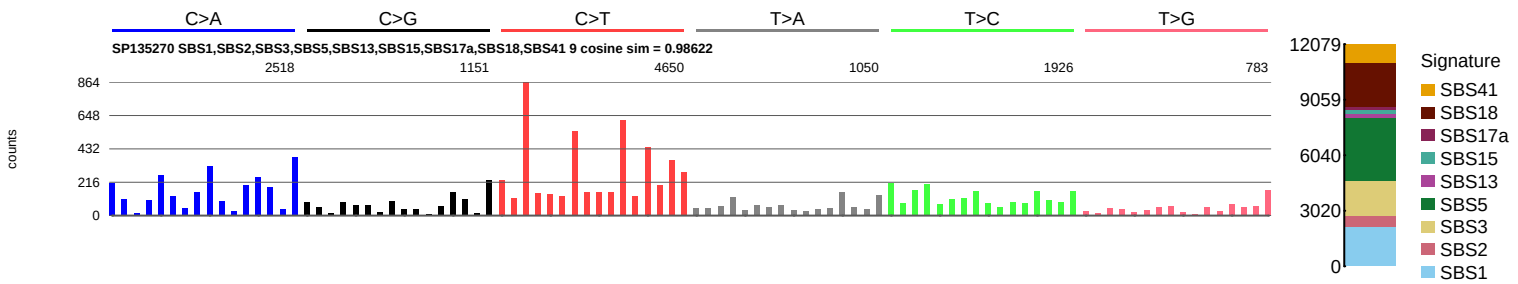
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98623
2659
1027
4658
891
1721
1125
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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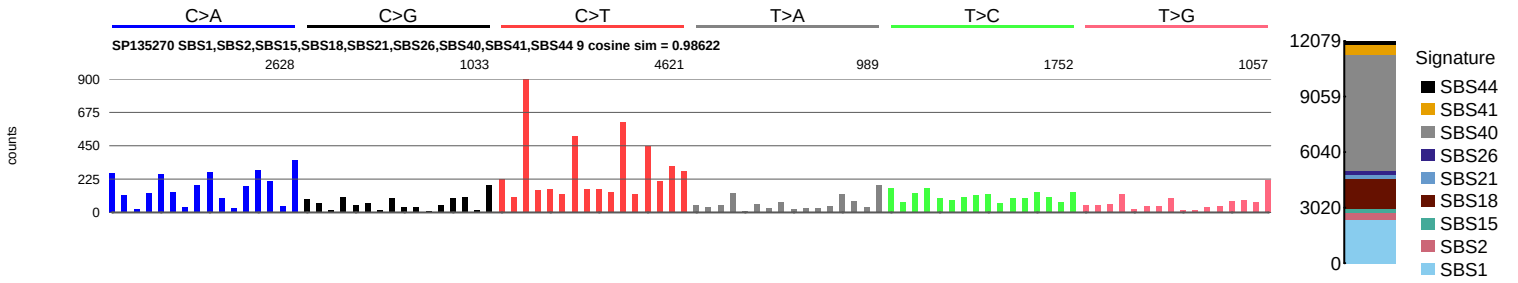
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 9 cosine sim = 0.98622
2518
1151
4650
1050
1926
783
counts
864
648
432
216
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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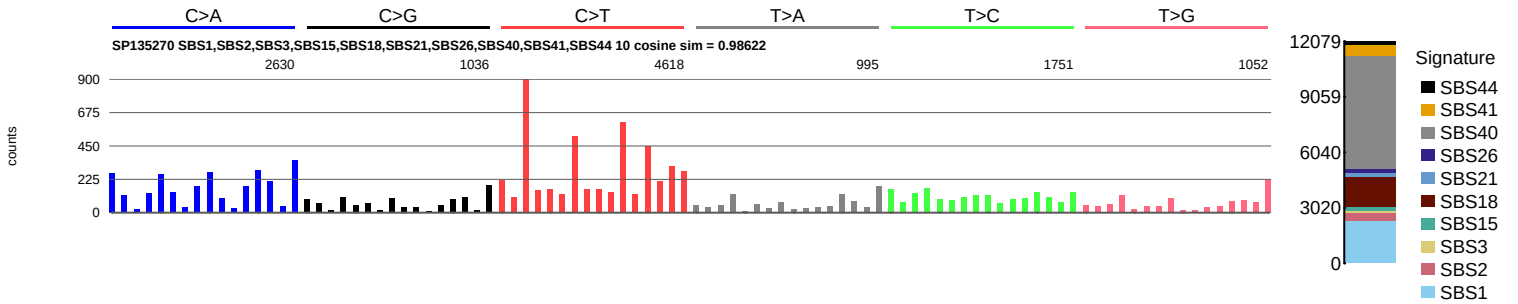
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98622
2628
1033
4621
989
1752
1057
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98622
2630
1036
4618
995
1751
1052
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
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SBS26
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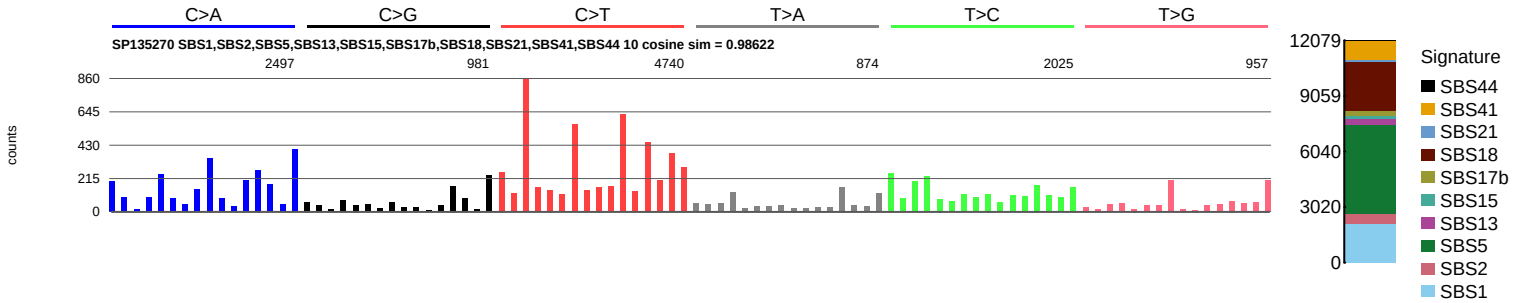
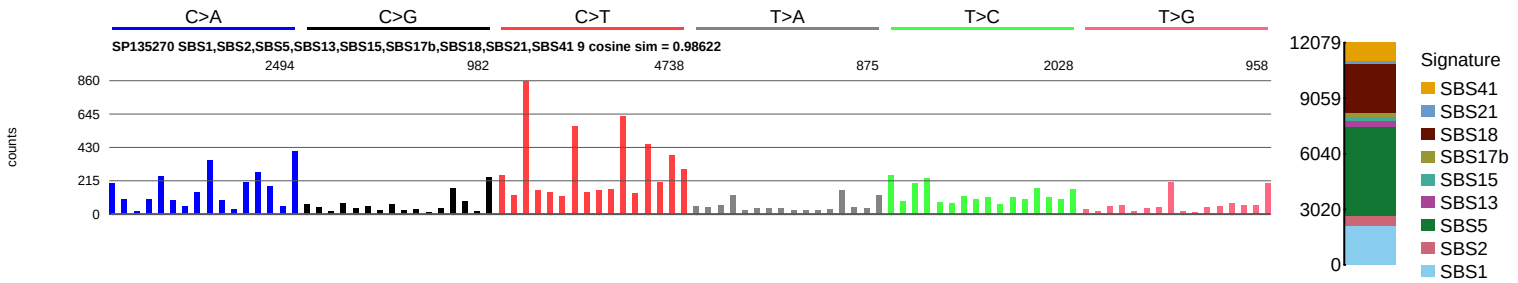
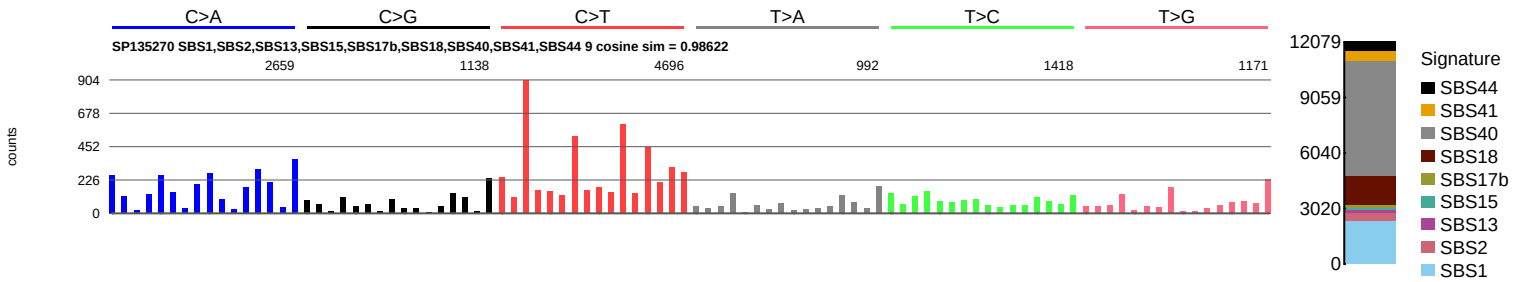
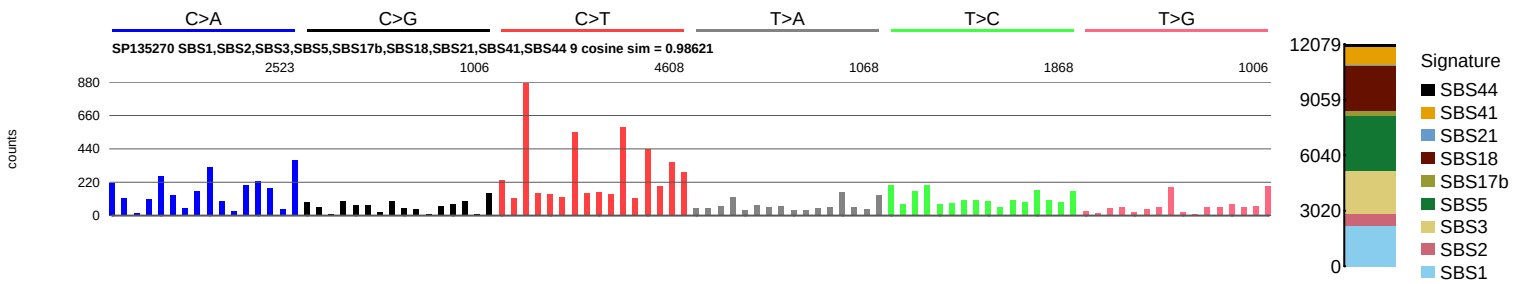
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98622
2497
981
4740
874
2025
957
counts
860
645
430
215
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98622
2494
982
4738
875
2028
958
counts
860
645
430
215
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98622
2659
1138
4696
992
1418
1171
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98621
2523
1006
4608
1068
1868
1006
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
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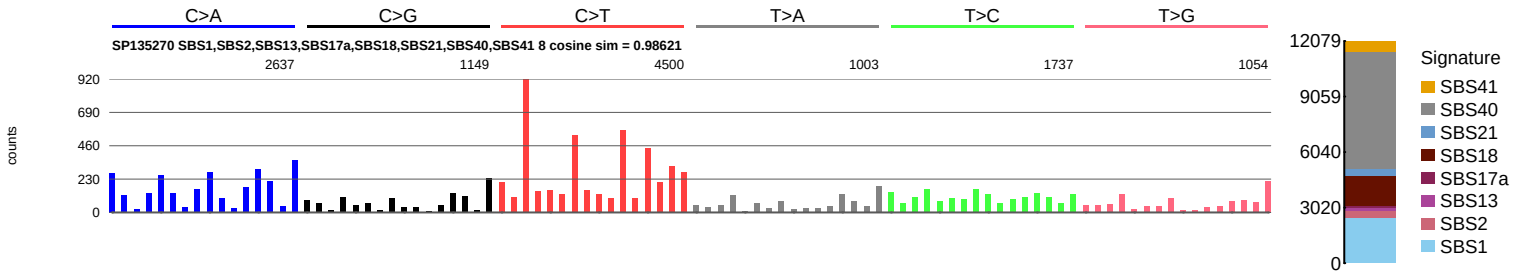
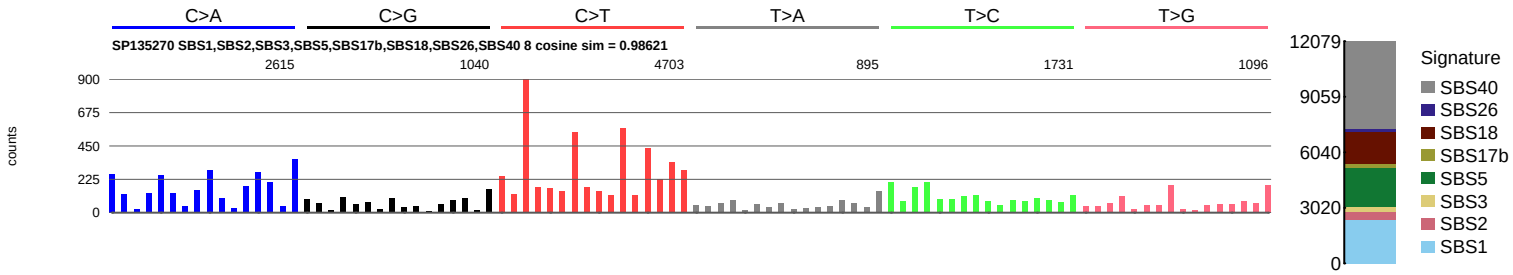
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98621
2637
1149
4500
1003
1737
1054
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
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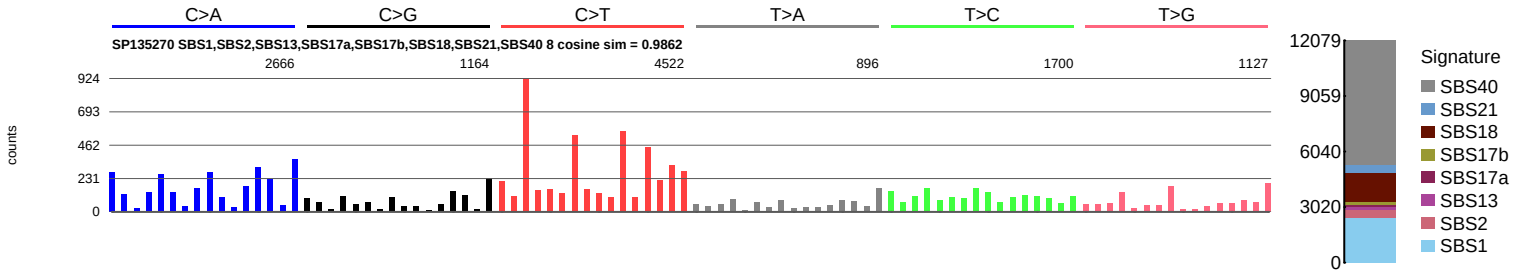
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98621
2615
1040
4703
895
1731
1096
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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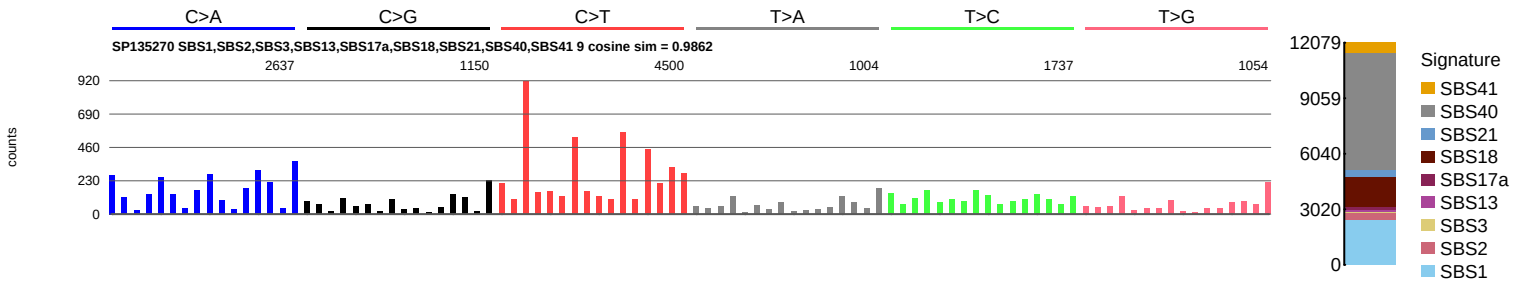
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.9862
2666
1164
4522
896
1700
1127
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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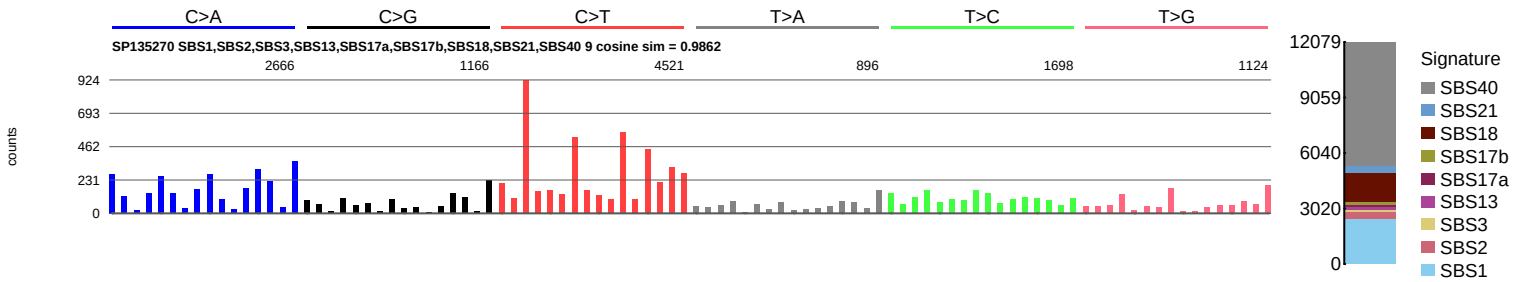
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.9862
2637
1150
4500
1004
1737
1054
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1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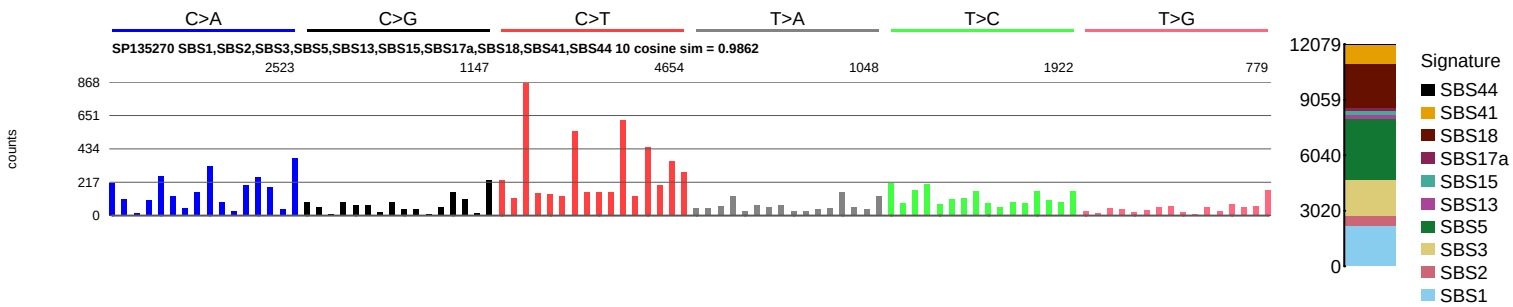
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.9862
2666
1166
4521
896
1698
1124
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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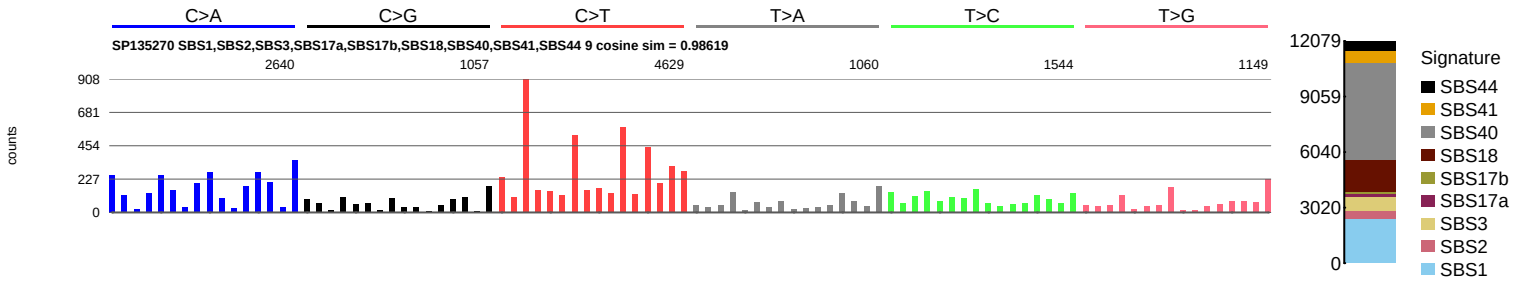
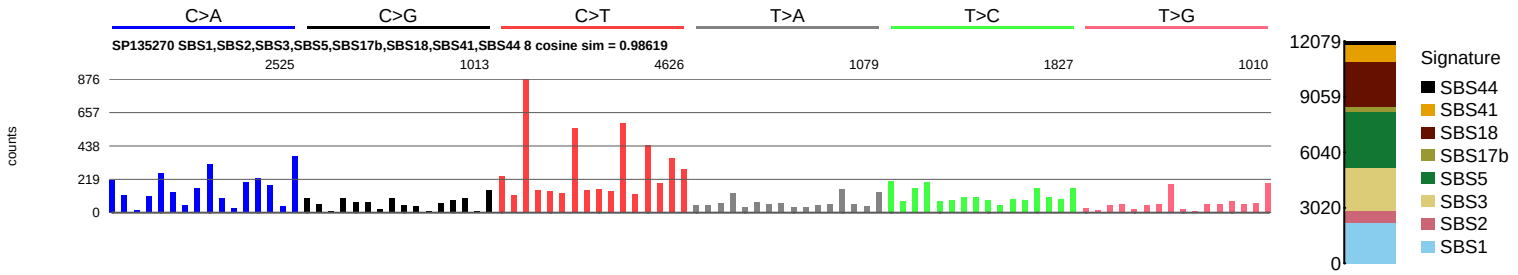
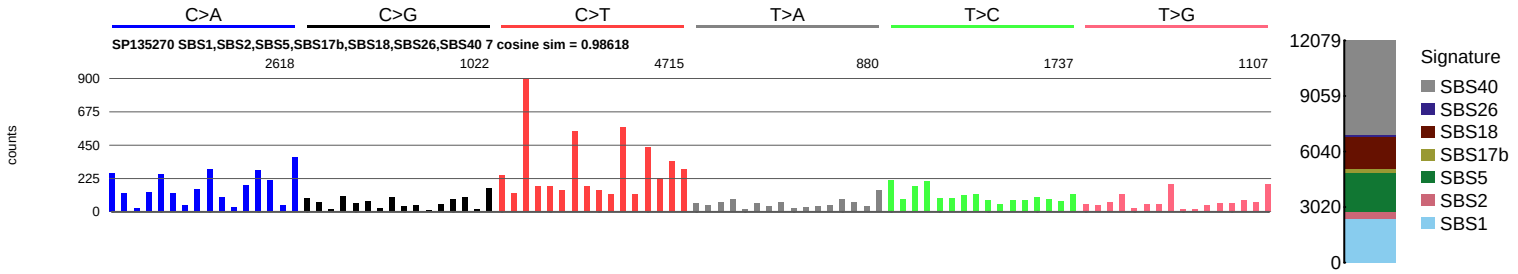
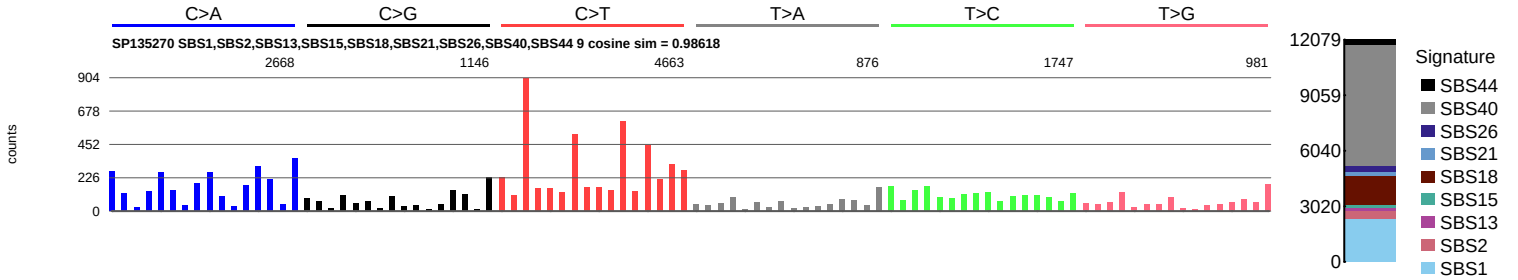
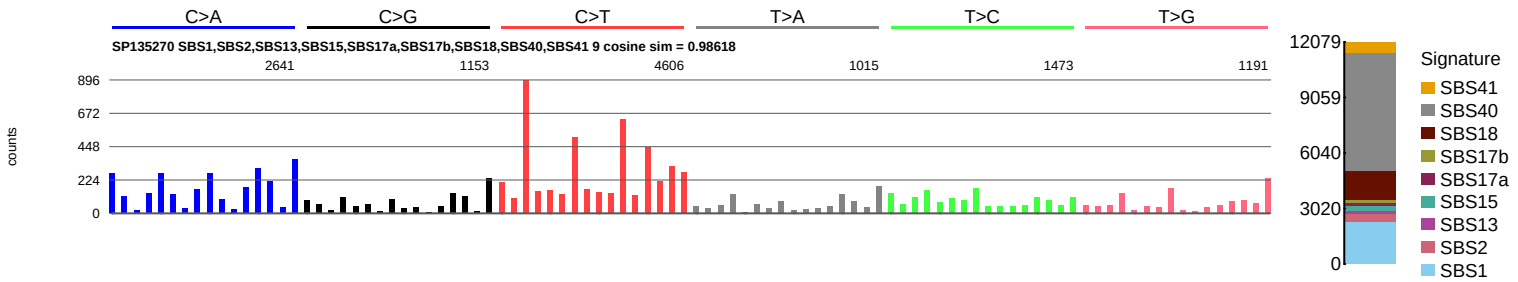
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41,SBS44 10 cosine sim = 0.9862
2523
1147
4654
1048
1922
779
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

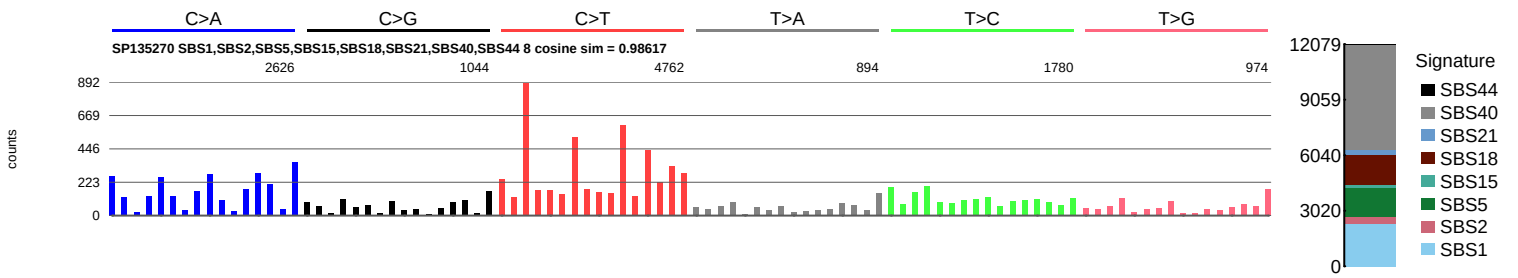

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98619
2640
1057
4629
1060
1544
1149
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

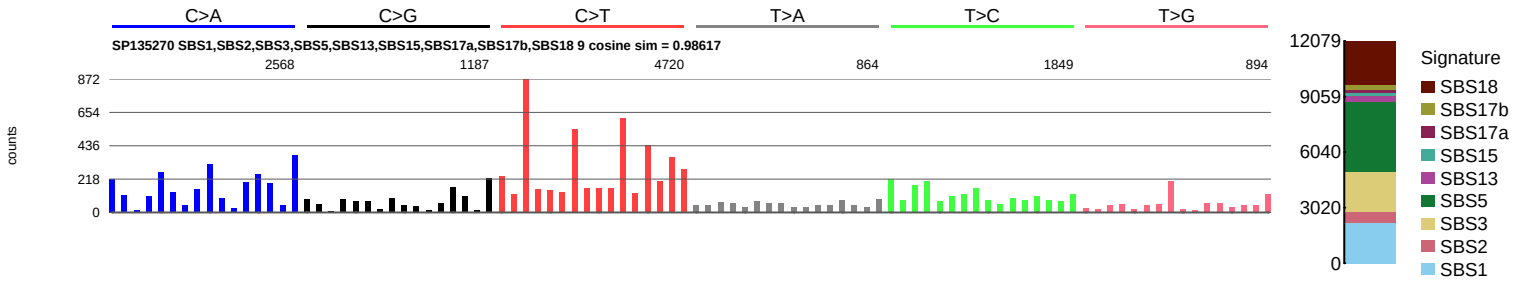

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98619
2525
1013
4626
1079
1827
1010
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

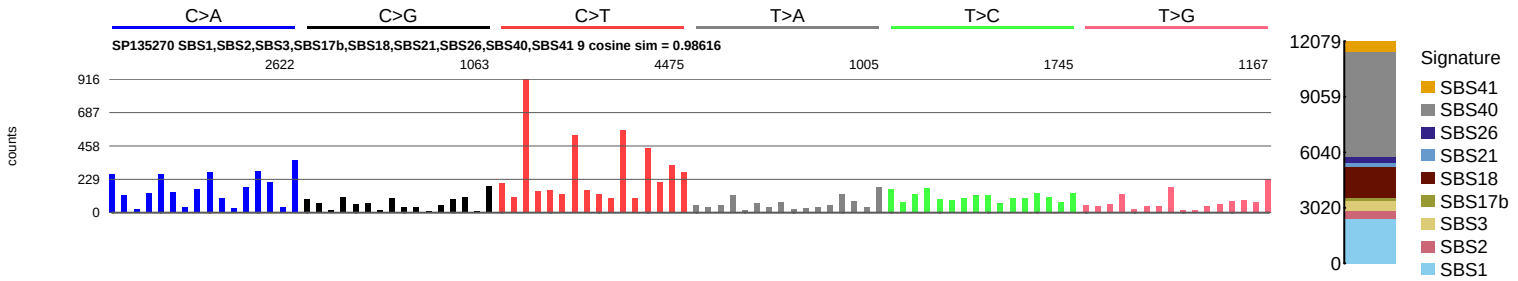

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98618
2618
1022
4715
880
1737
1107
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1

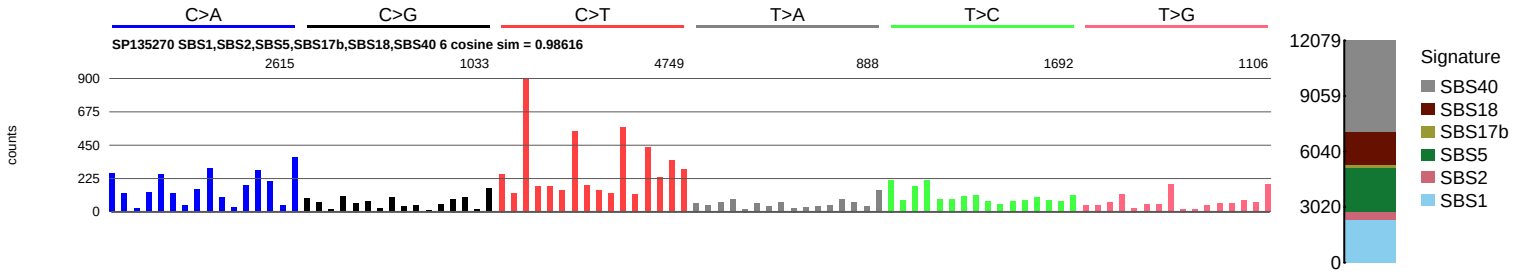

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98618
2668
1146
4663
876
1747
981
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

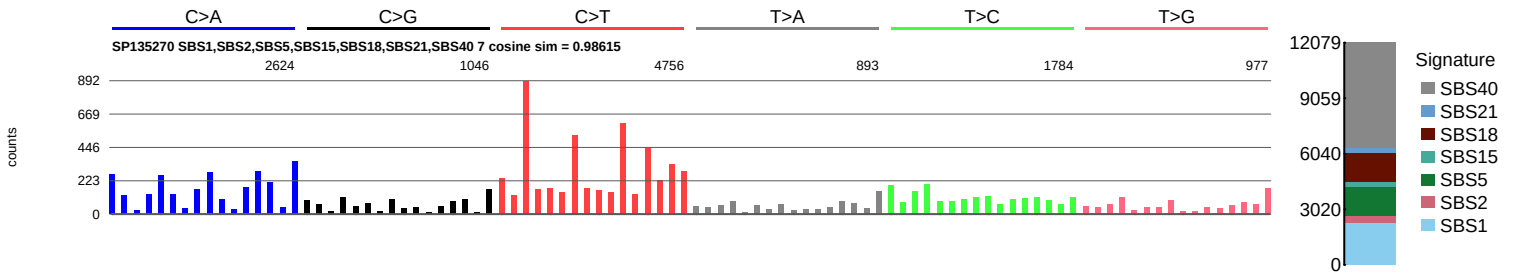

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98618
2641
1153
4606
1015
1473
1191
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

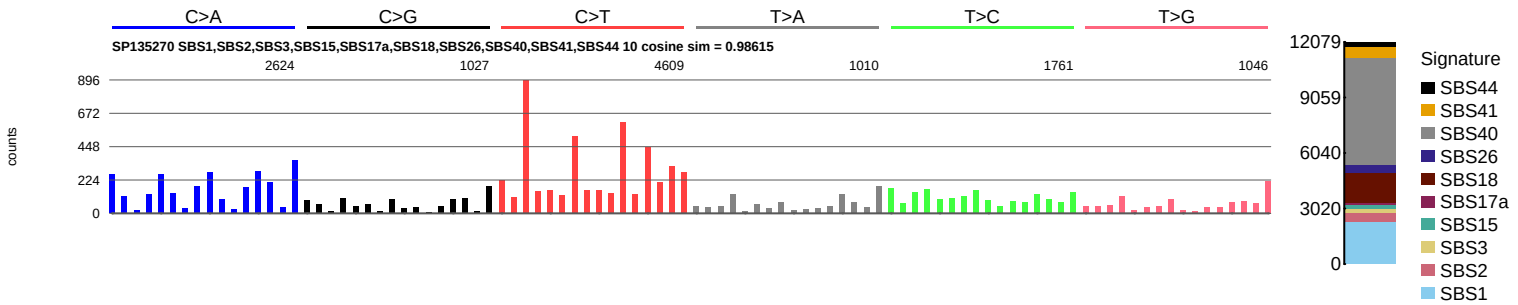

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98617
2626
1044
4762
894
1780
974
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1

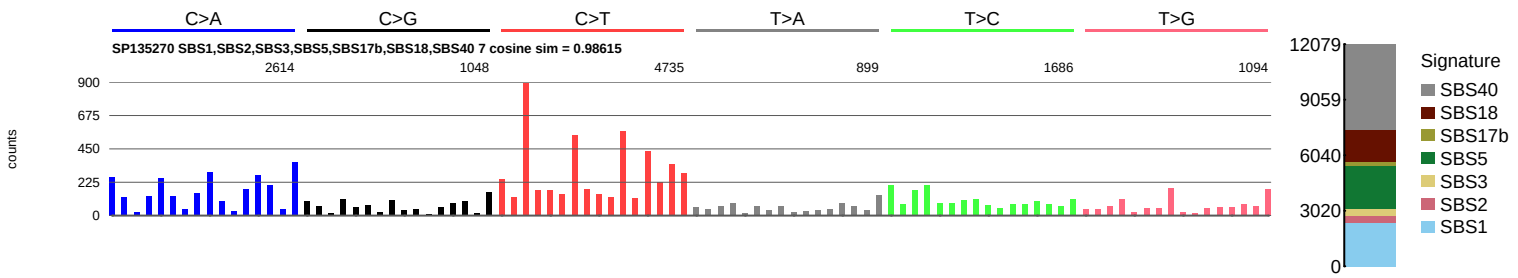

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98617
2568
1187
4720
864
1849
894
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

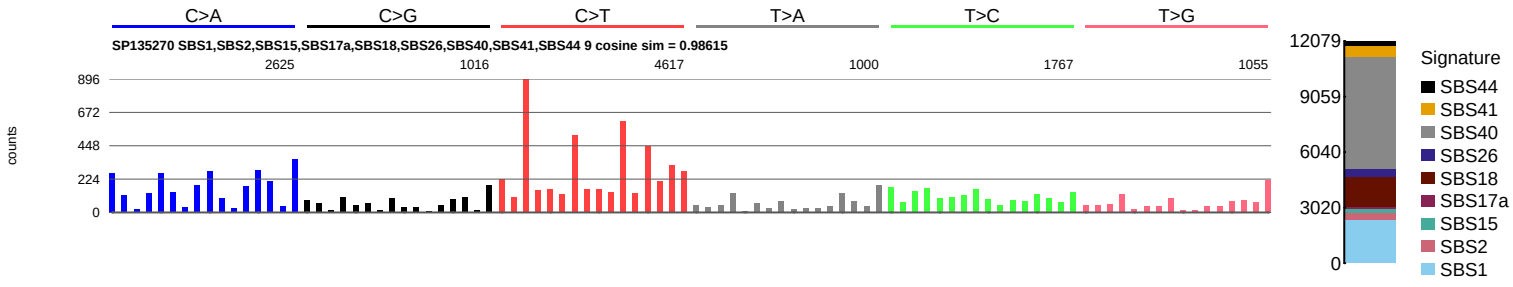

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98616
2622
1063
4475
1005
1745
1167
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1

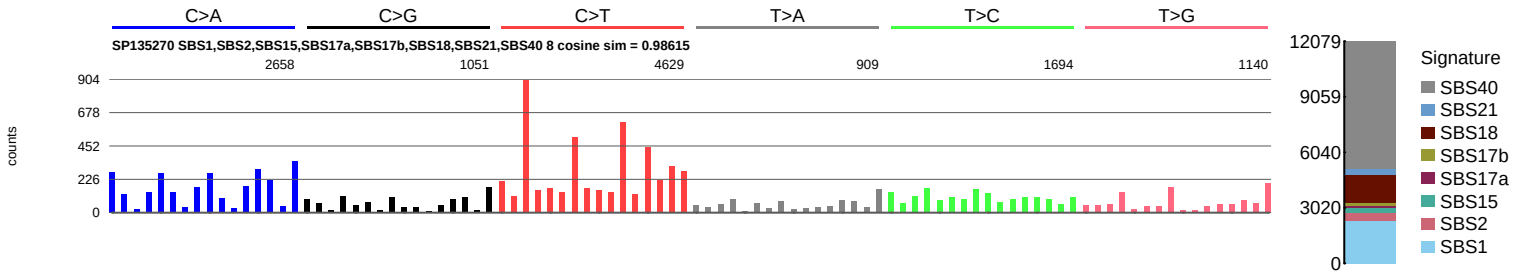

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40 6 cosine sim = 0.98616
2615
1033
4749
888
1692
1106
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40 7 cosine sim = 0.98615
2624
1046
4756
893
1784
977
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98615
2624
1027
4609
1010
1761
1046
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40 7 cosine sim = 0.98615
2614
1048
4735
899
1686
1094
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98615
2625
1016
4617
1000
1767
1055
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98615
2658
1051
4629
909
1694
1140
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
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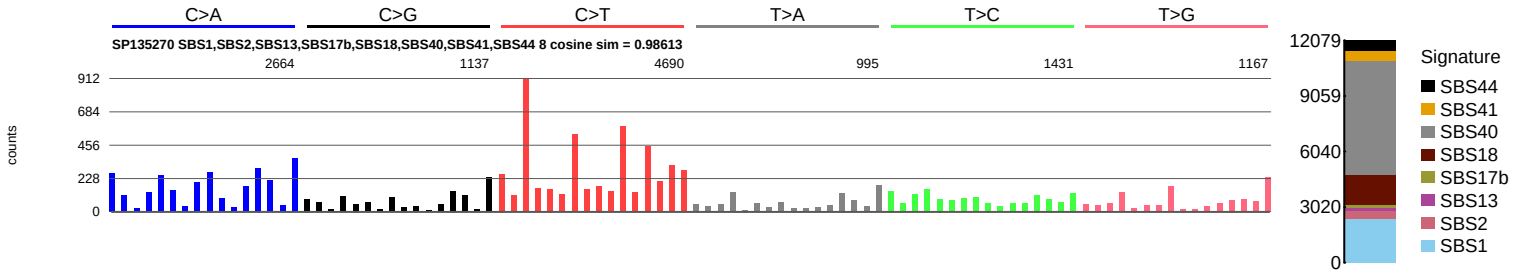
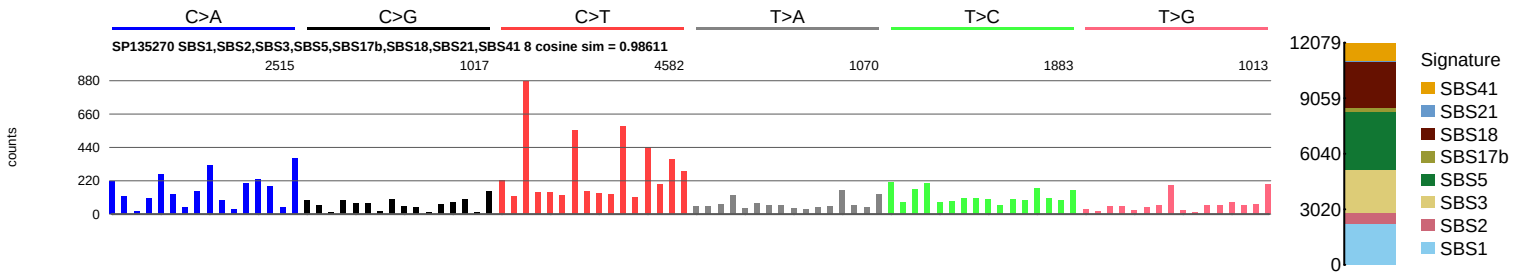
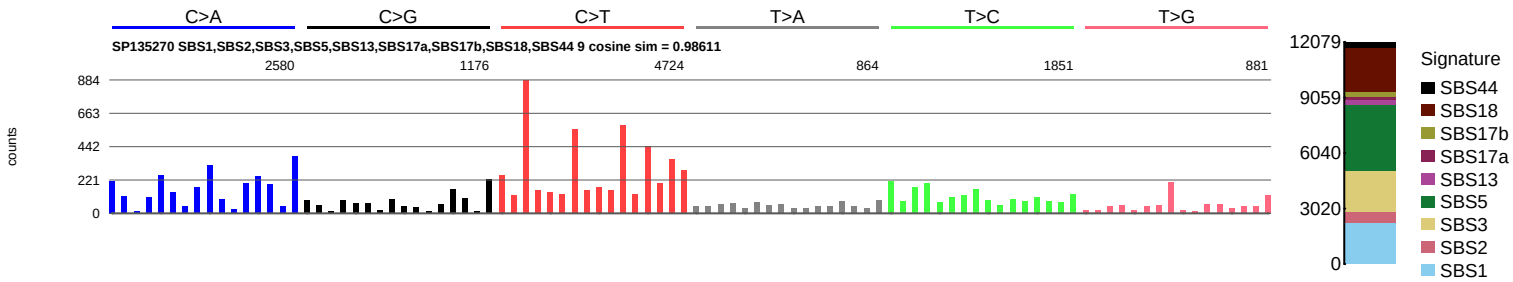
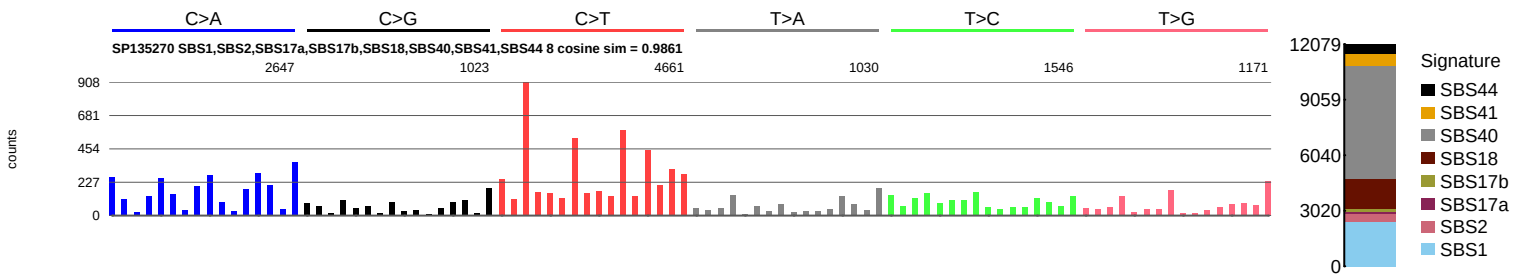
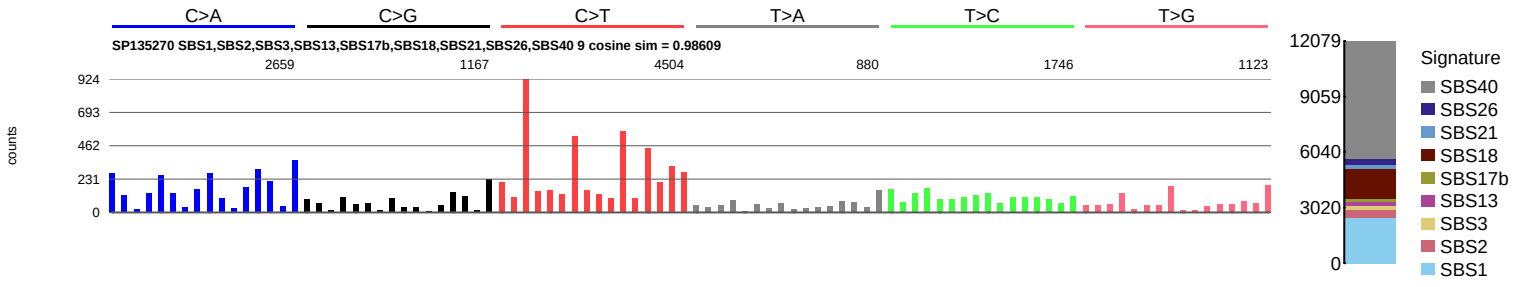
SBS15
SBS2
SBS1

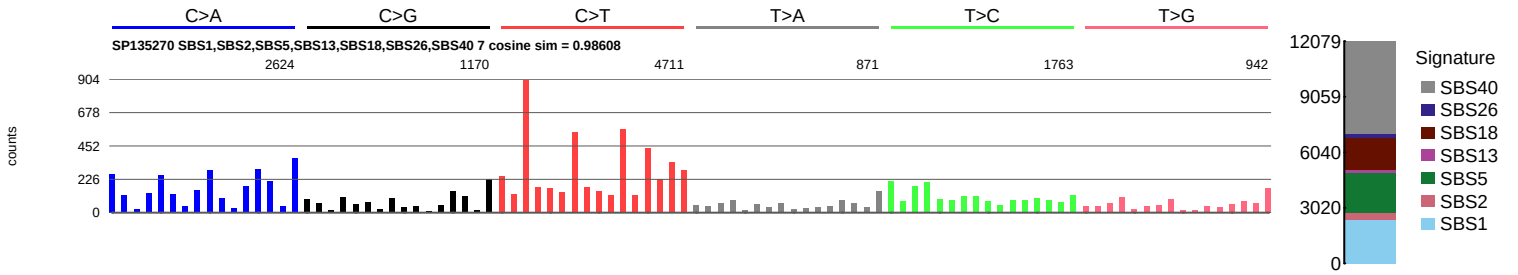

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98613
2664
1137
4690
995
1431
1167
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1

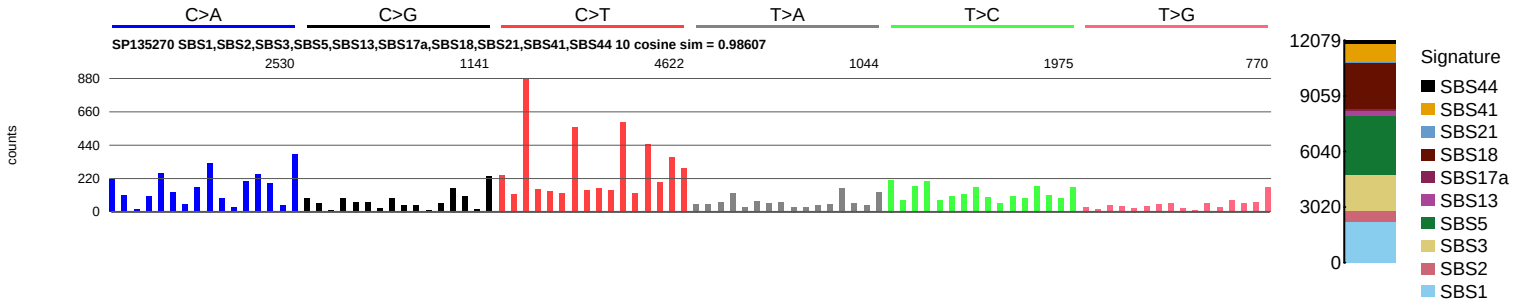

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98611
2515
1017
4582
1070
1883
1013
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

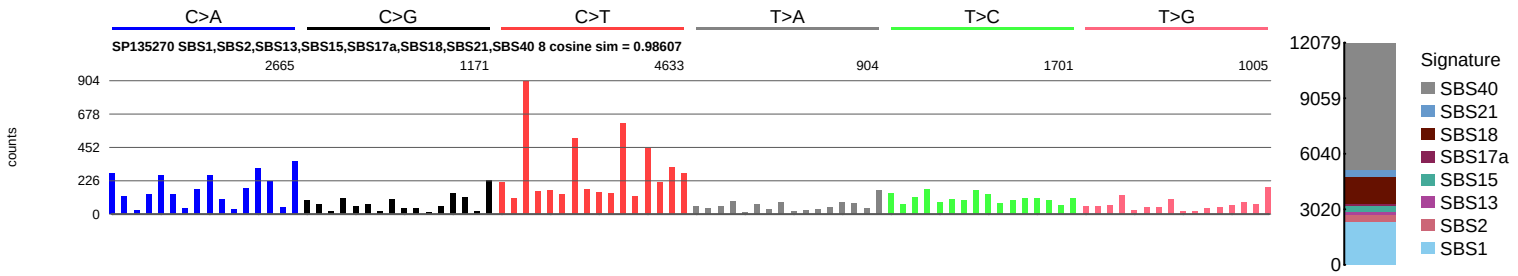

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98611
2580
1176
4724
864
1851
881
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

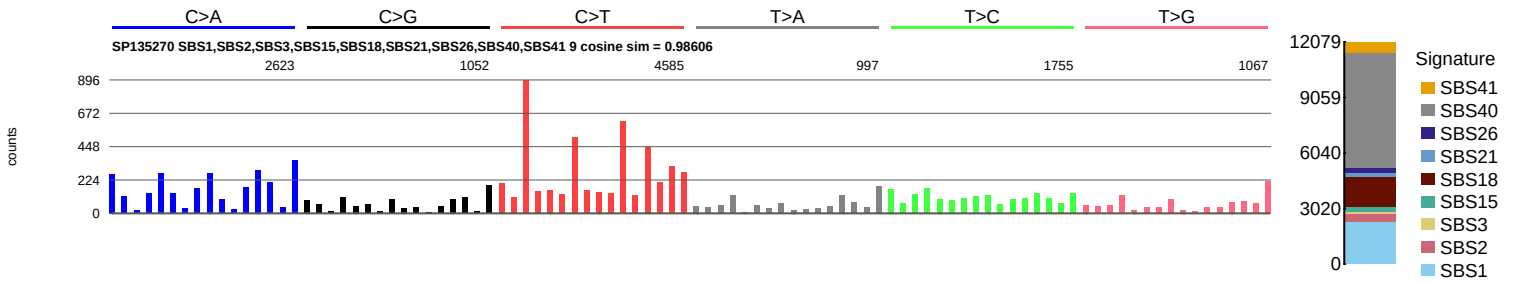

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9861
2647
1023
4661
1030
1546
1171
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

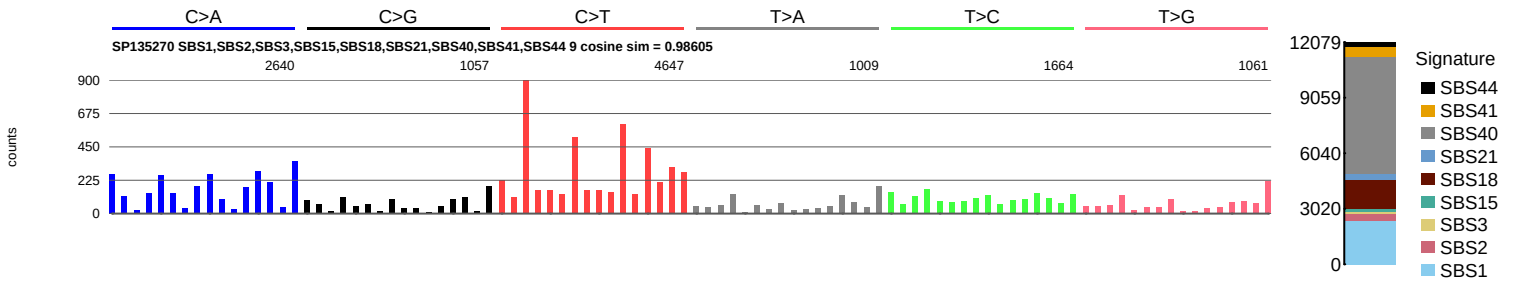

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98609
2659
1167
4504
880
1746
1123
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

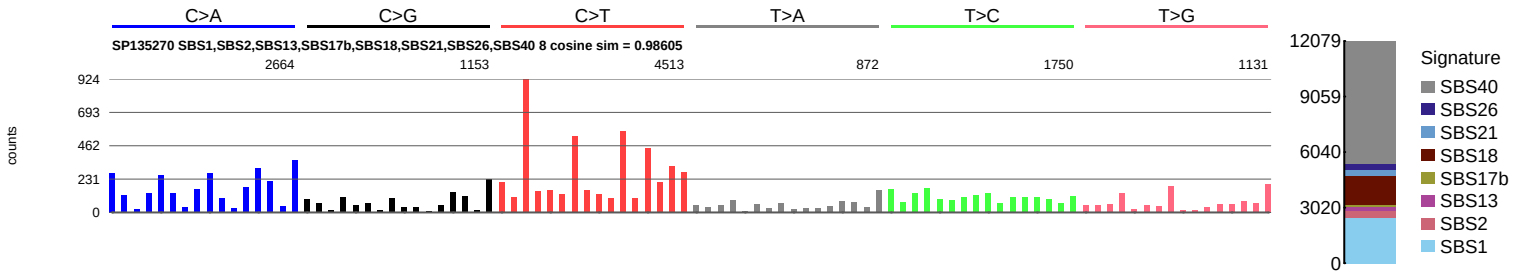

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40 7 cosine sim = 0.98608
2624
1170
4711
871
1763
942
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1

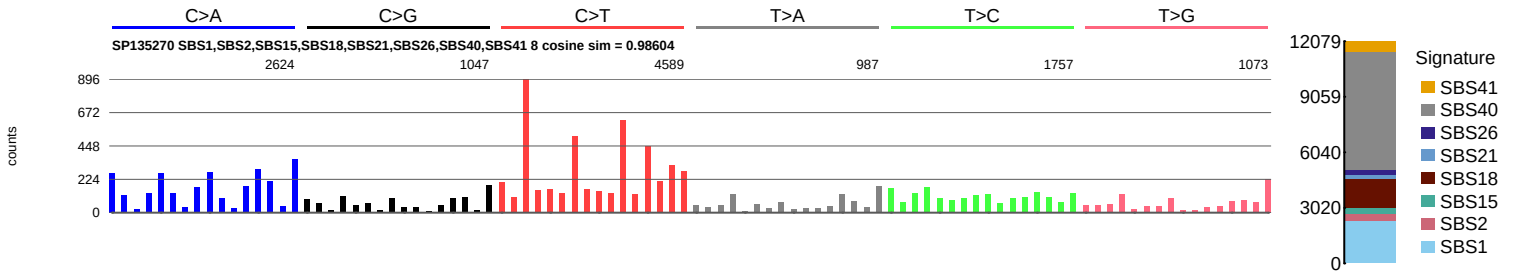

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98607
2530
1141
4622
1044
1975
770
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

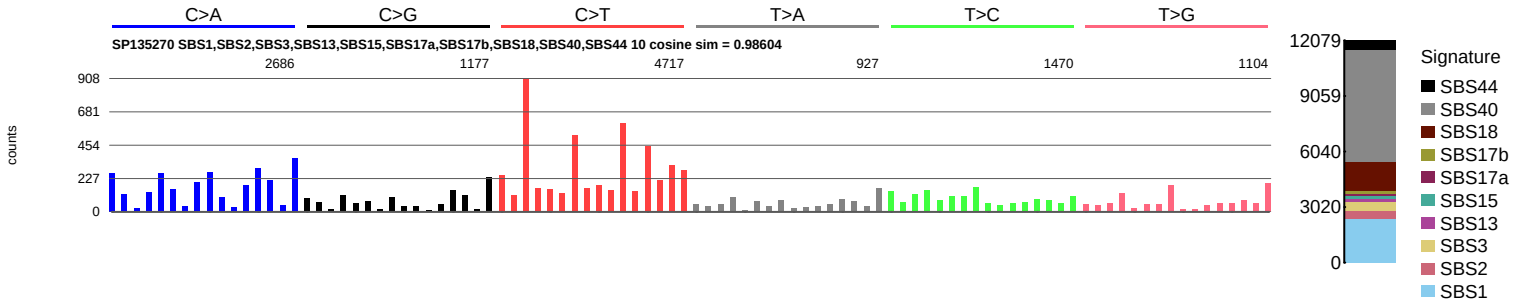

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98607
2665
1171
4633
904
1701
1005
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

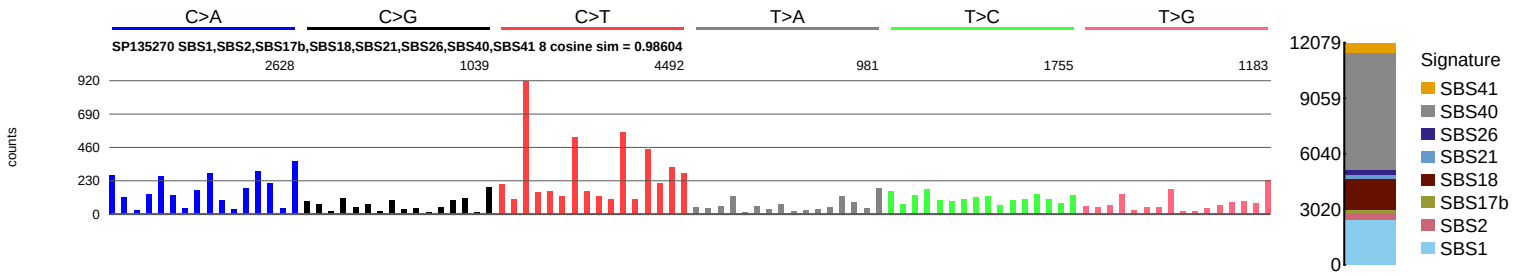

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98606
2623
1052
4585
997
1755
1067
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS3
SBS2
SBS1

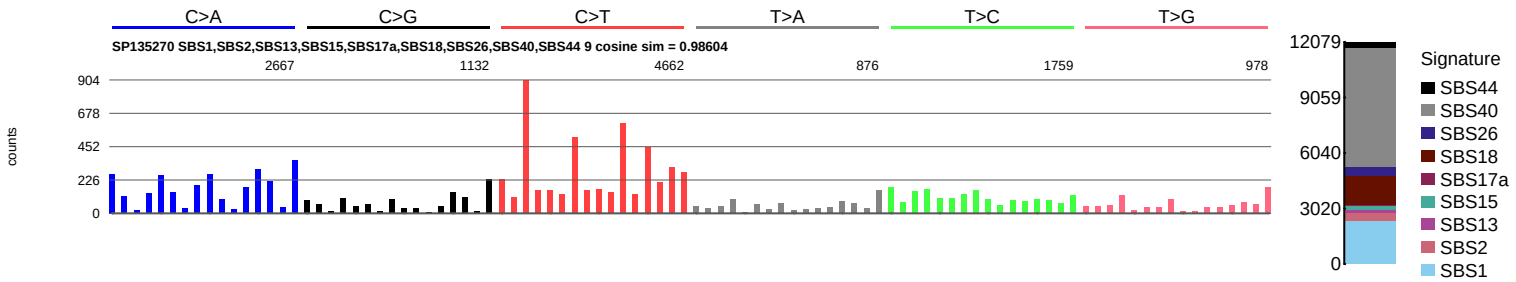

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98605
2640
1057
4647
1009
1664
1061
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS3
SBS2
SBS1

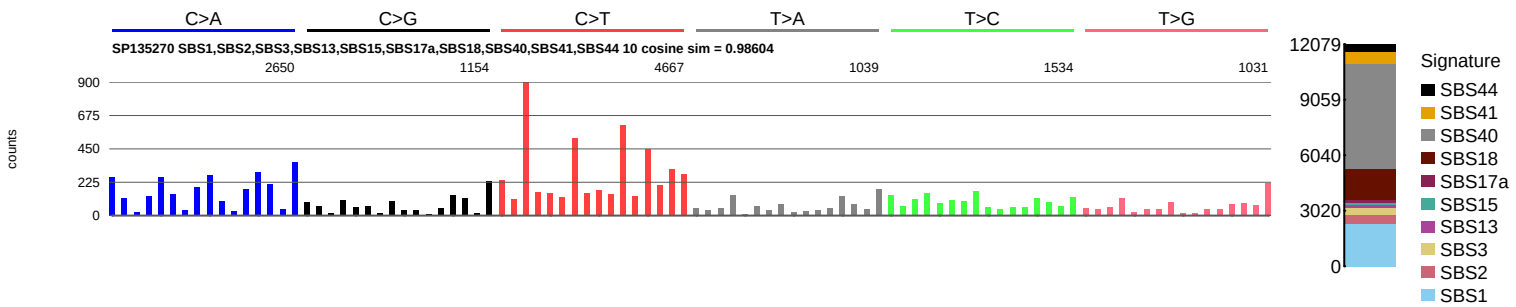

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98605
2664
1153
4513
872
1750
1131
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

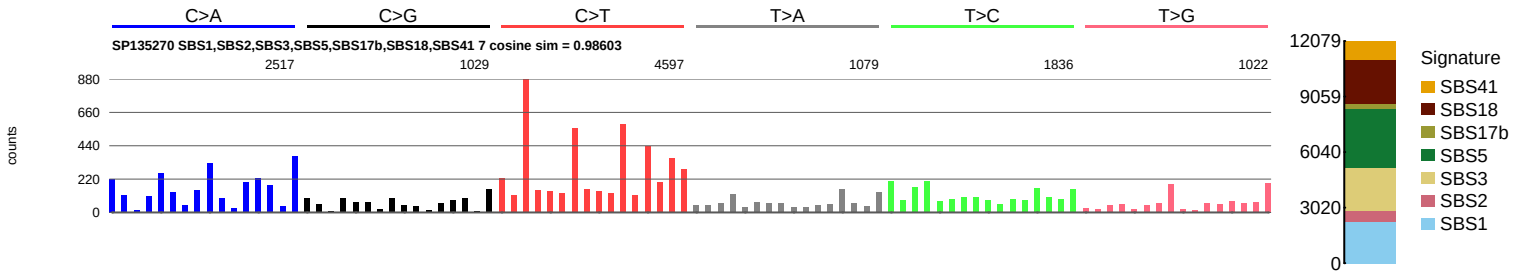

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98604
2624
1047
4589
987
1757
1073
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS2
SBS1

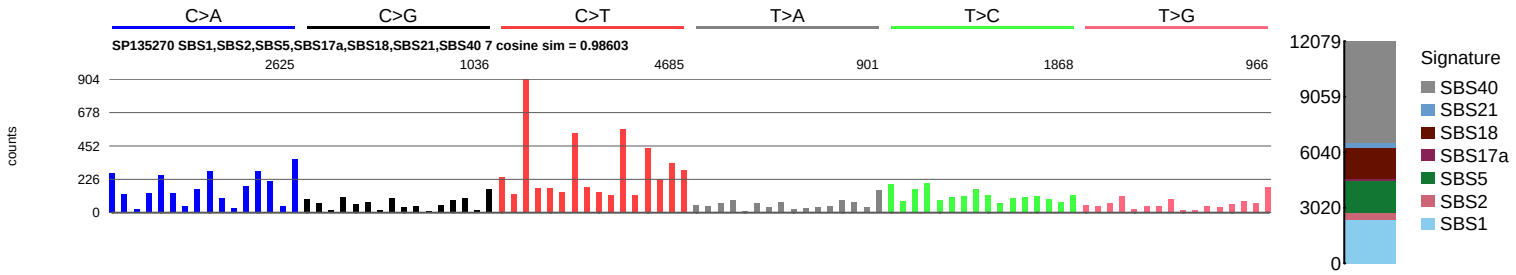

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98604
2686
1177
4717
927
1470
1104
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

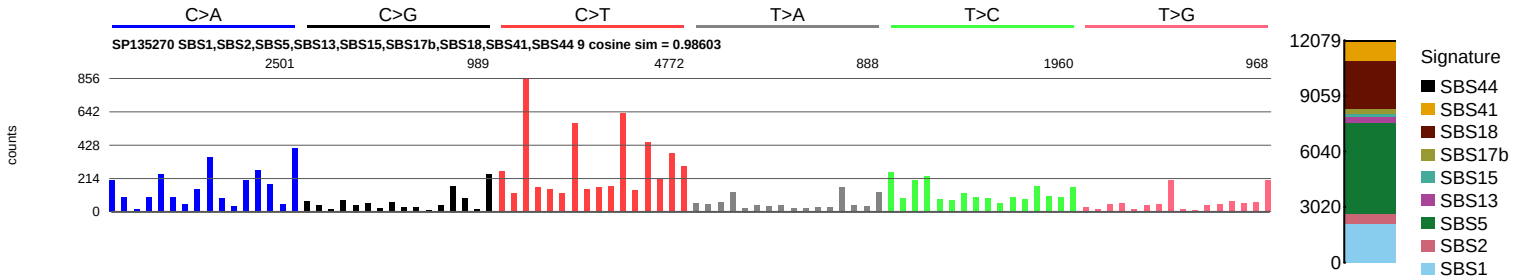

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98604
2628
1039
4492
981
1755
1183
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1

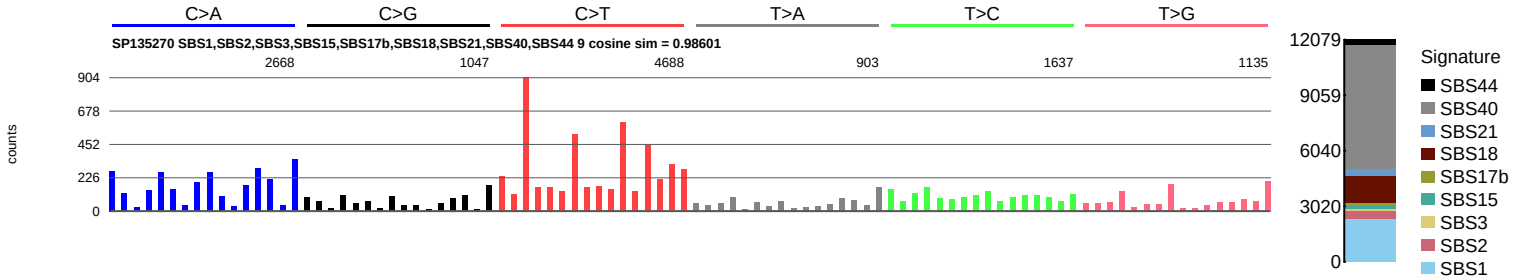

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98604
2667
1132
4662
876
1759
978
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

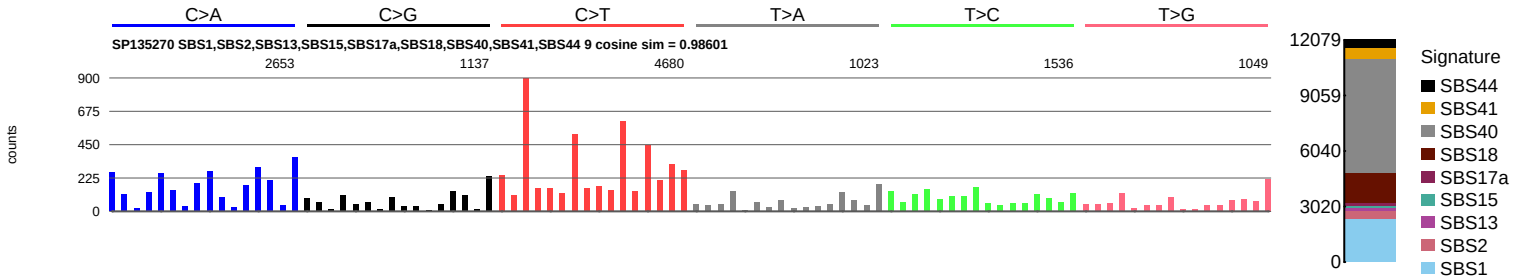

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98604
2650
1154
4667
1039
1534
1031
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

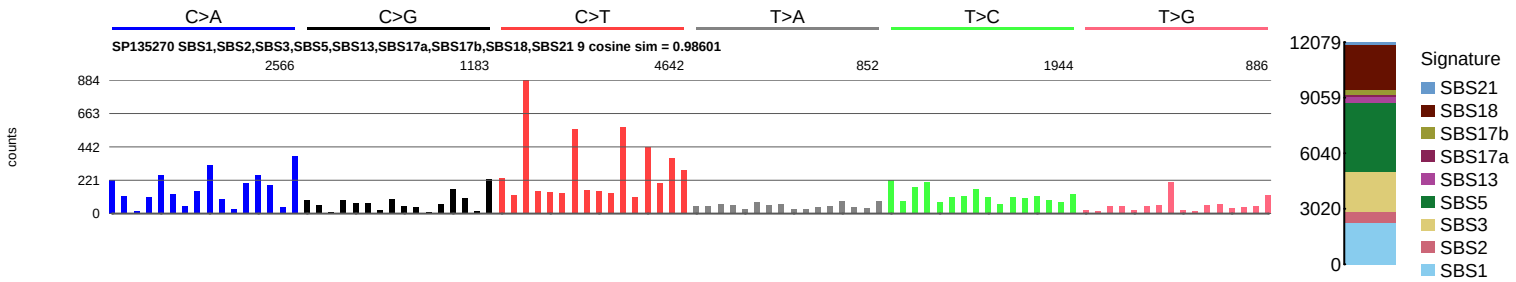

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41 7 cosine sim = 0.98603
2517
1029
4597
1079
1836
1022
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

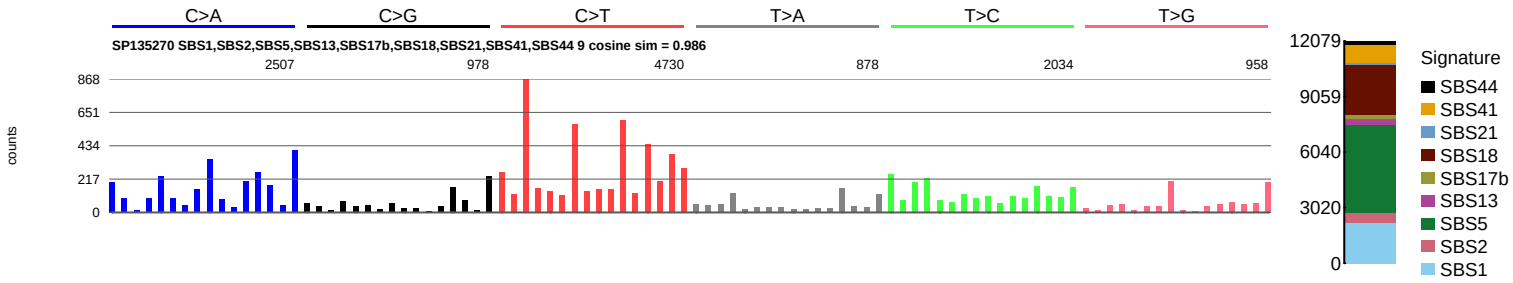

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98603
2625
1036
4685
901
1868
966
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1

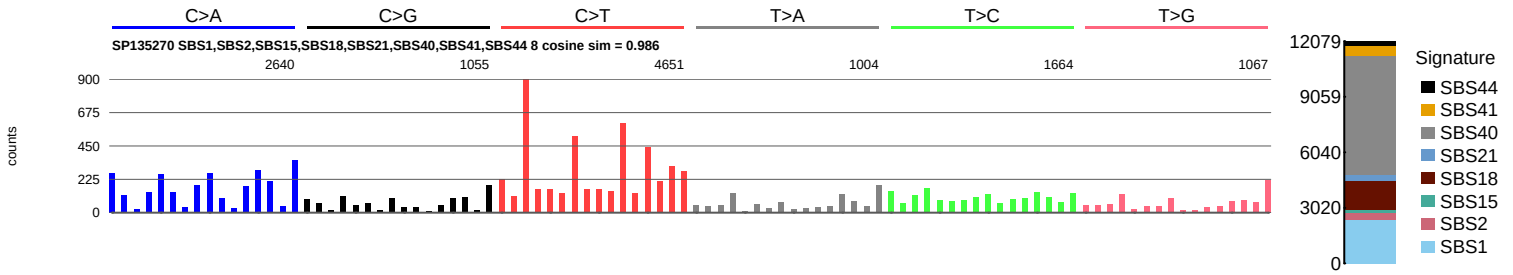

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98603
2501
989
4772
888
1960
968
856
642
428
214
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

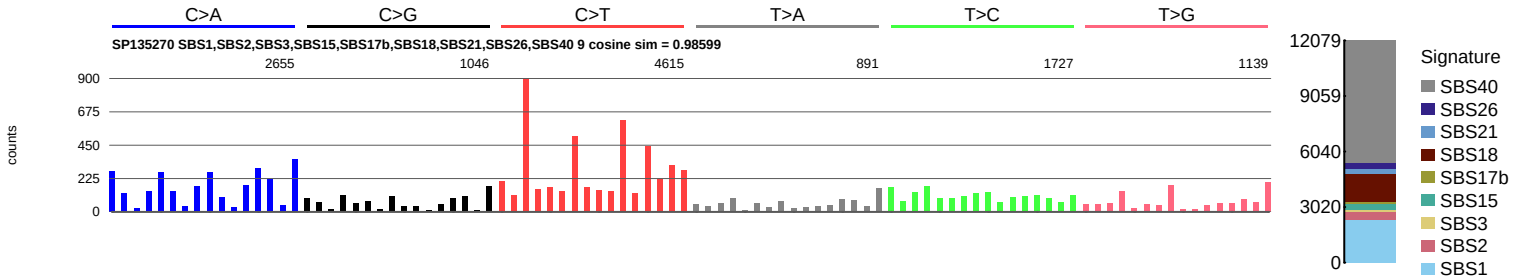

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98601
2668
1047
4688
903
1637
1135
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

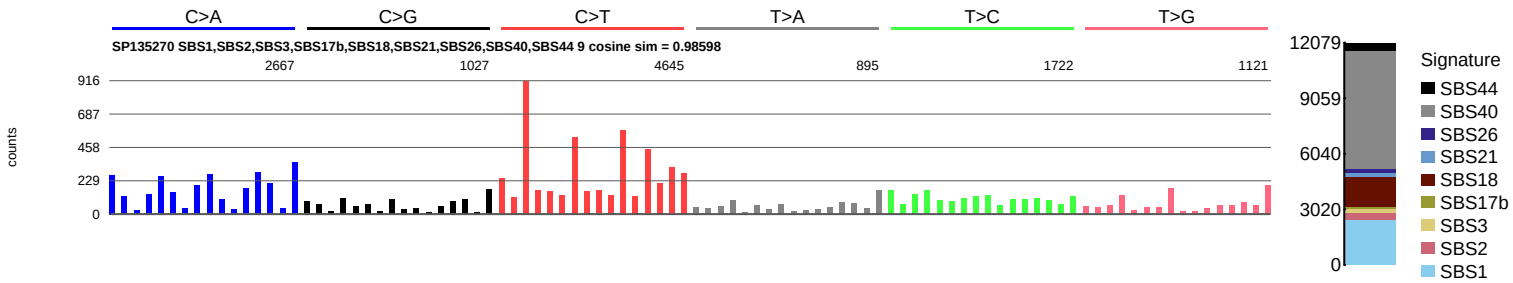

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98601
2653
1137
4680
1023
1536
1049
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

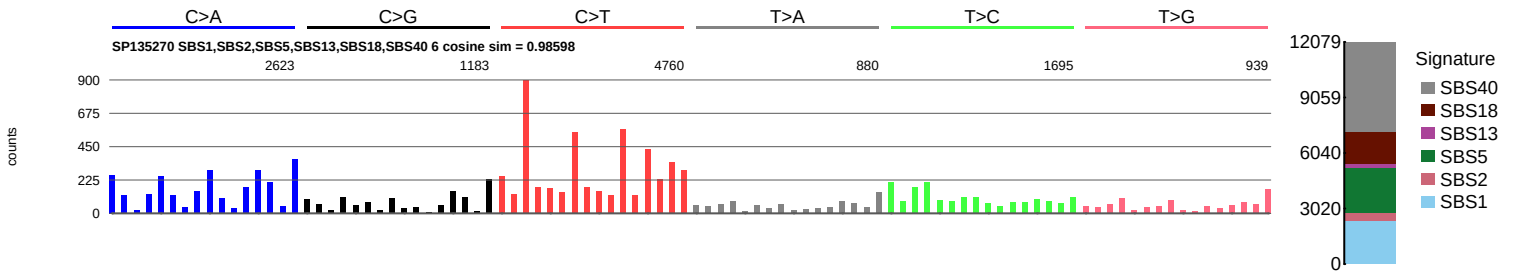

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98601
2566
1183
4642
852
1944
886
884
663
442
221
0
counts
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

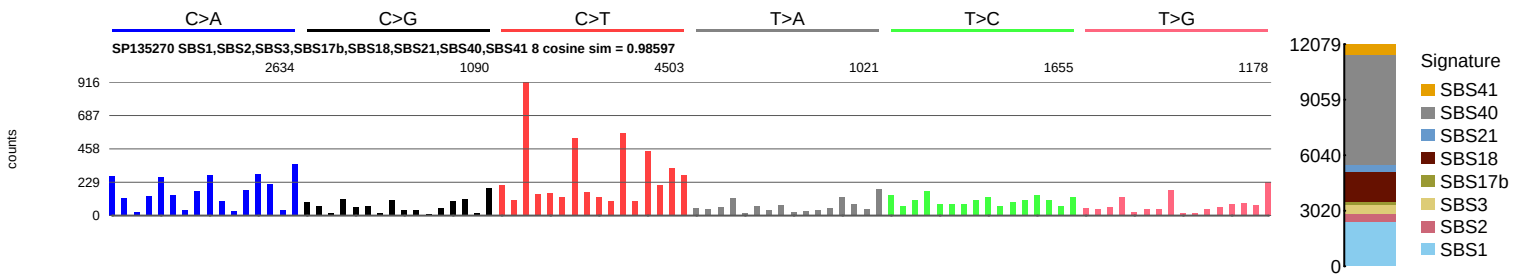

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.986
2507
978
4730
878
2034
958
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

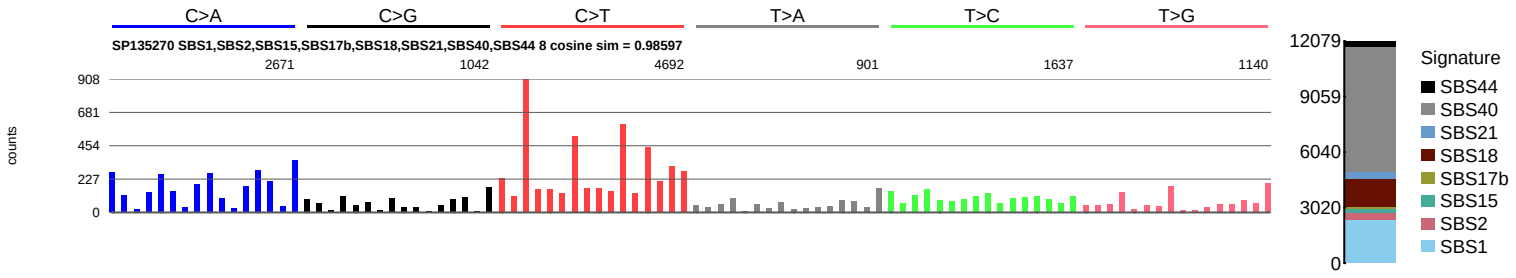

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.986
2640
1055
4651
1004
1664
1067
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS2
SBS1

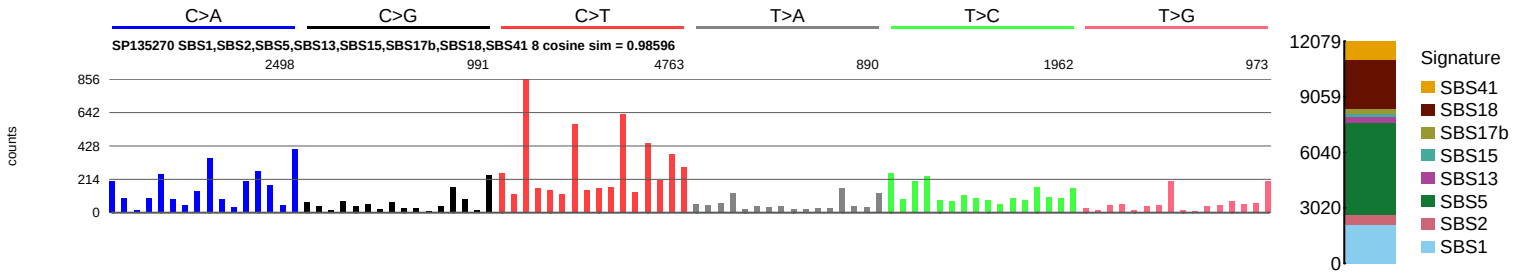

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98599
2655
1046
4615
891
1727
1139
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

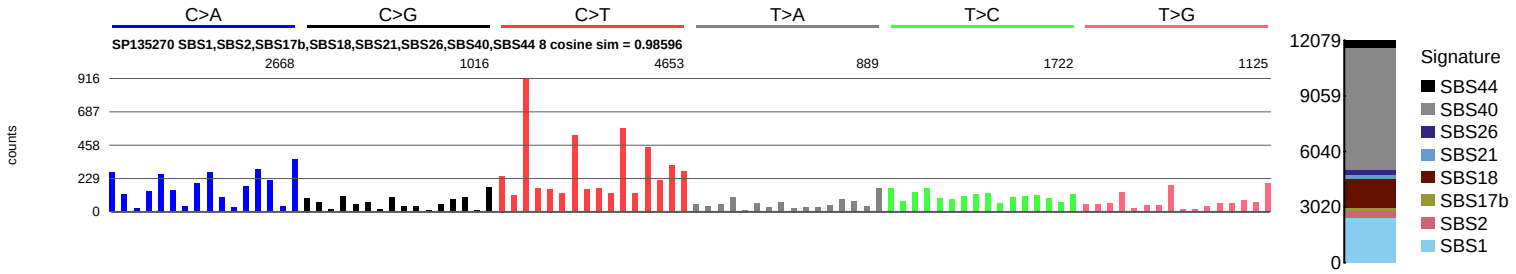

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98598
2667
1027
4645
895
1722
1121
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1

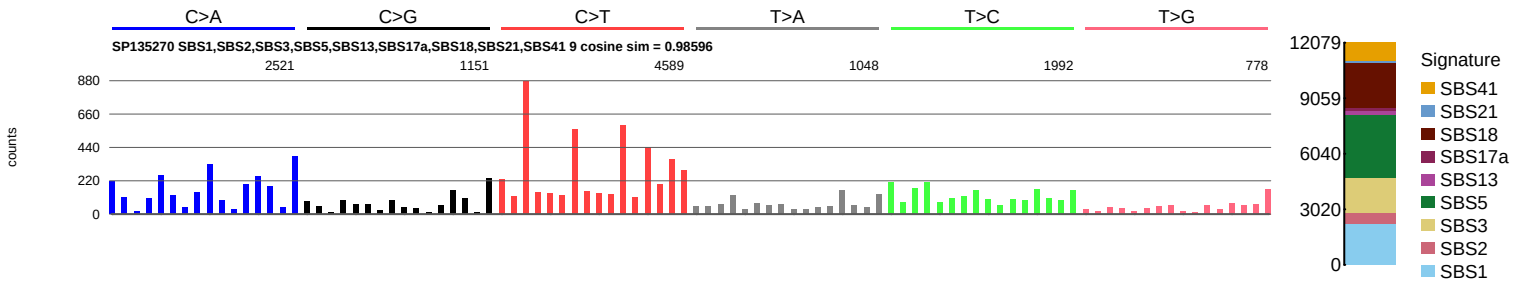

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40 6 cosine sim = 0.98598
2623
1183
4760
880
1695
939
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98597
2634
1090
4503
1021
1655
1178
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98597
2671
1042
4692
901
1637
1140
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98596
2498
991
4763
890
1962
973
856
642
428
214
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98596
2668
1016
4653
889
1722
1125
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1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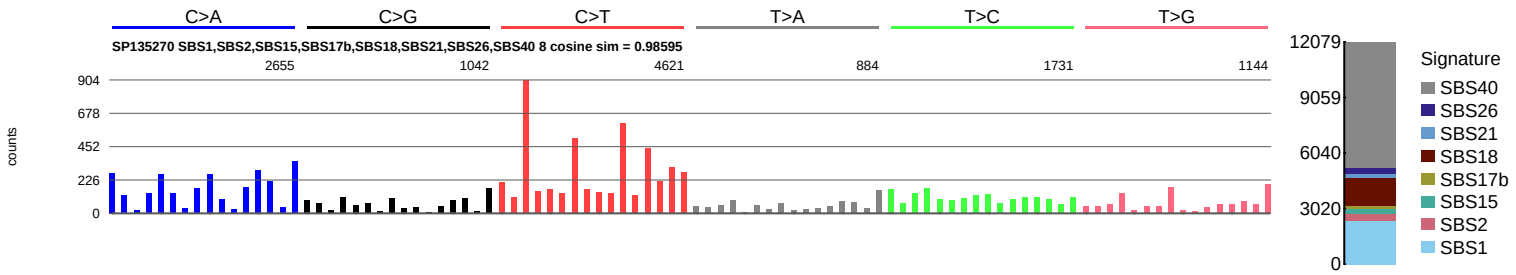
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS41 9 cosine sim = 0.98596
2521
1151
4589
1048
1992
778
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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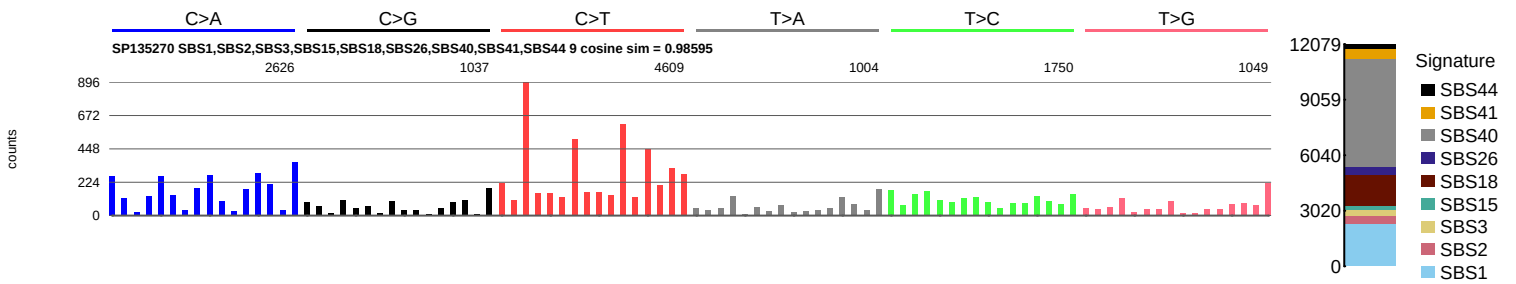
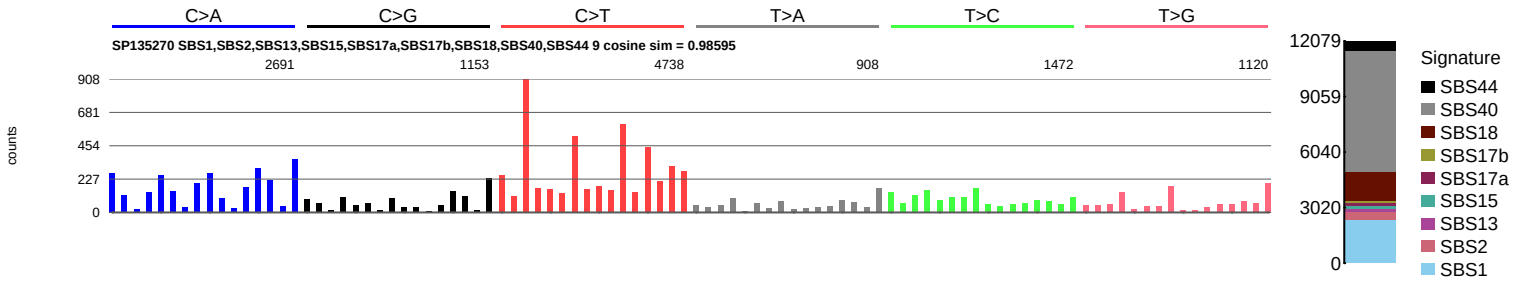
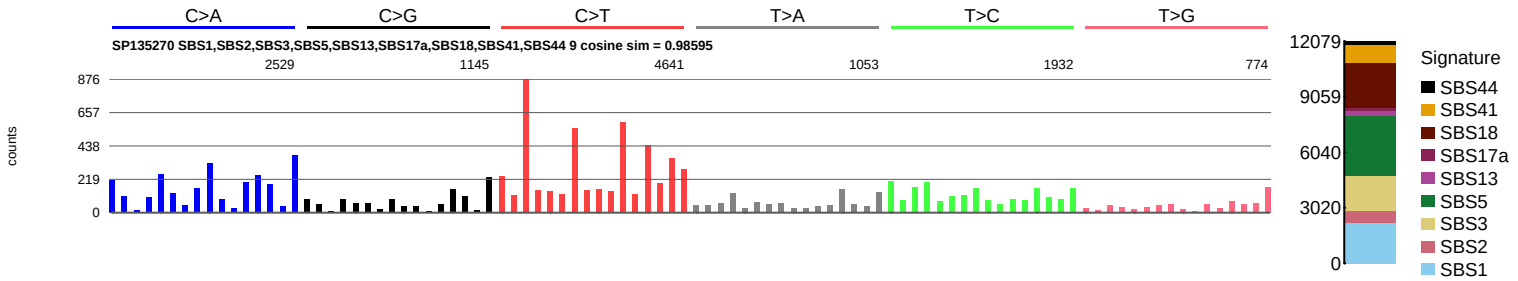
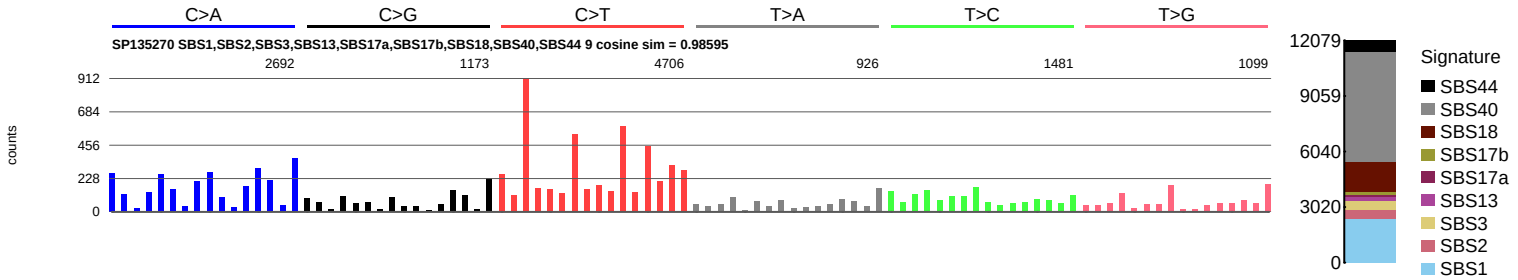
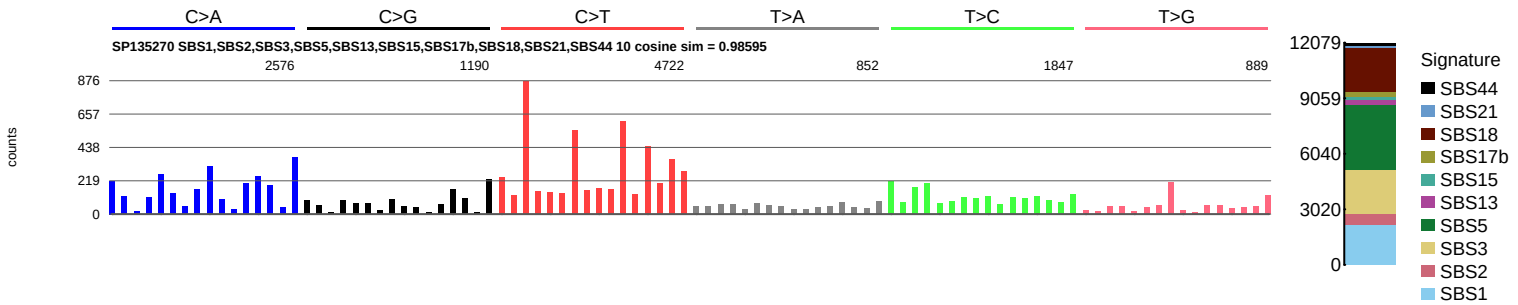
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98595
2655
1042
4621
884
1731
1144
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1

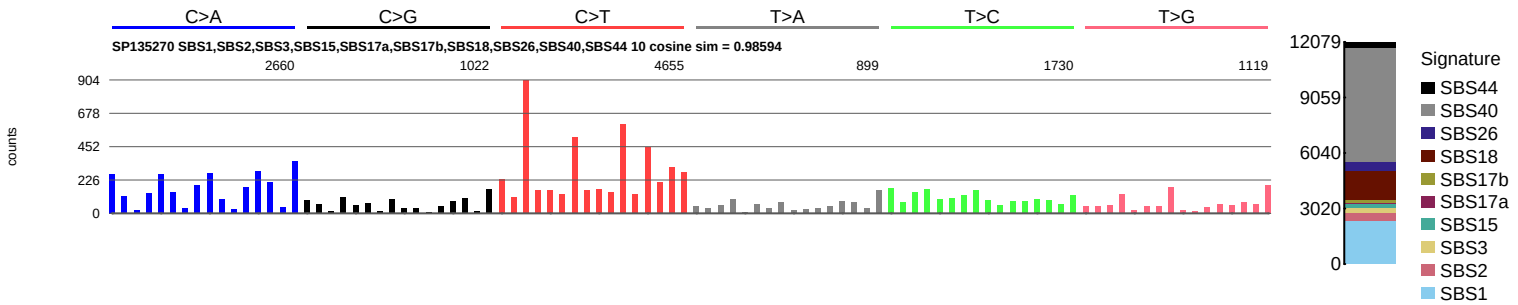

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98595
2626
1037
4609
1004
1750
1049
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS3
SBS2
SBS1

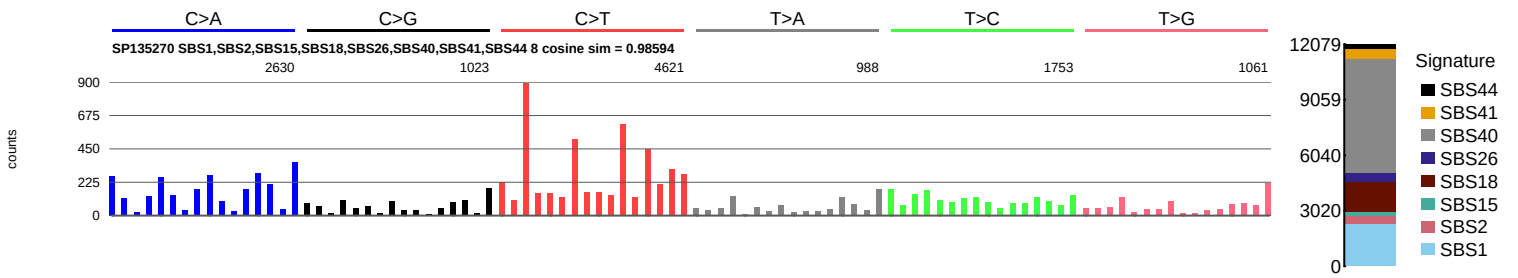

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98595
2691
1153
4738
908
1472
1120
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

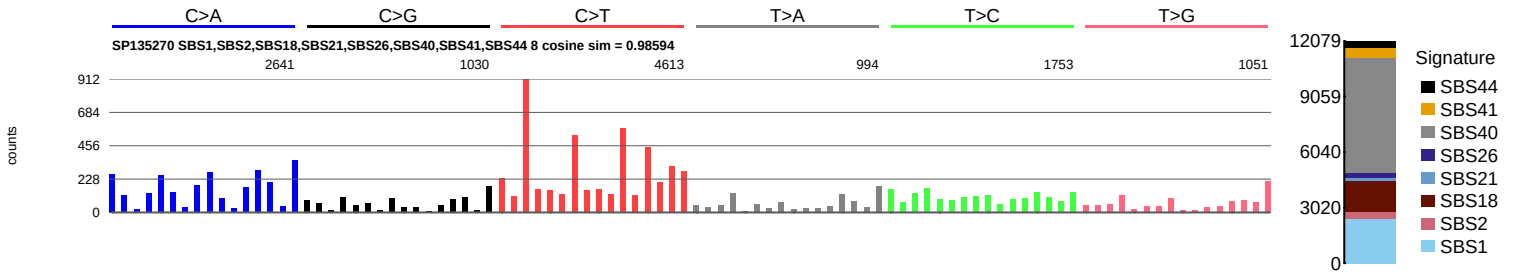

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.98595
2529
1145
4641
1053
1932
774
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

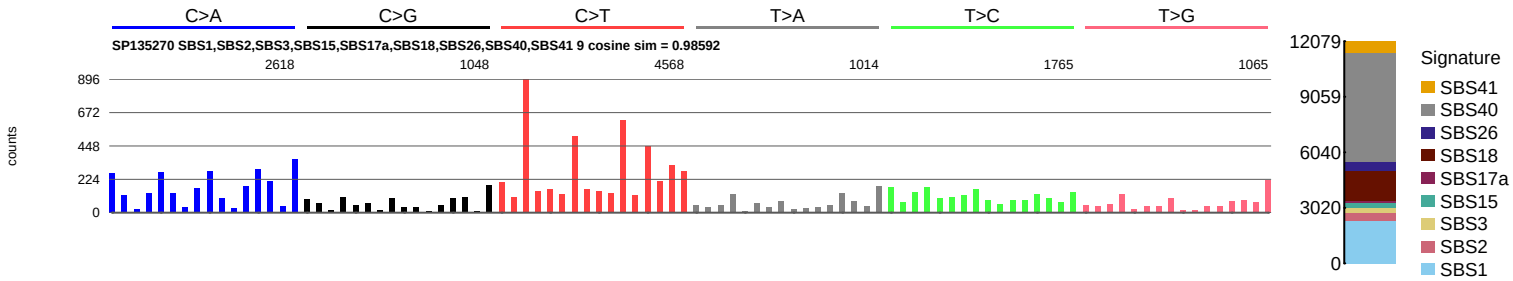

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98595
2692
1173
4706
926
1481
1099
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

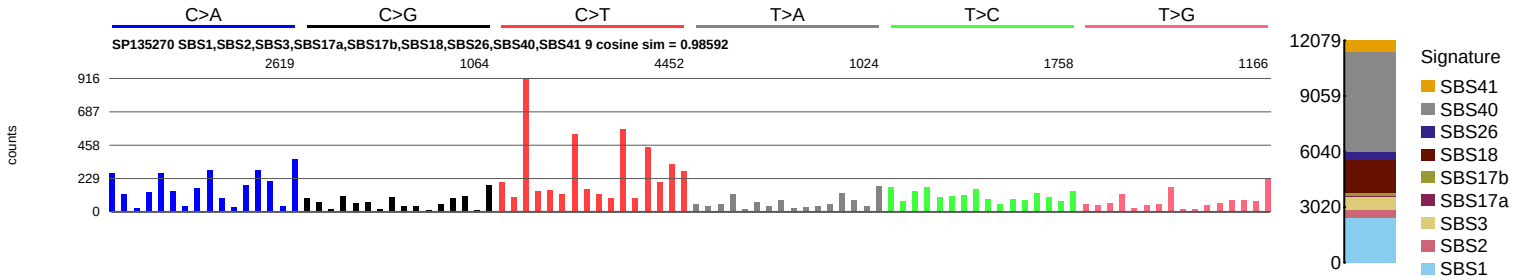

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98595
2576
1190
4722
852
1847
889
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

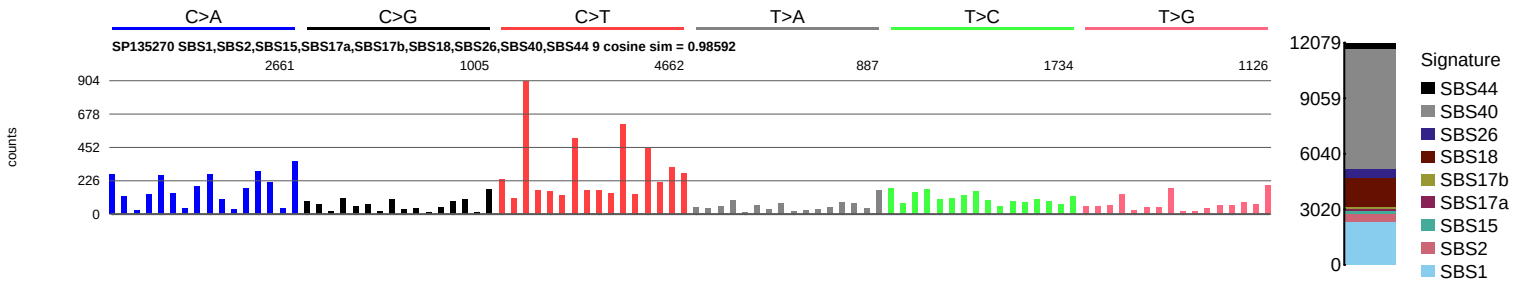

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98594
2660
1022
4655
899
1730
1119
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

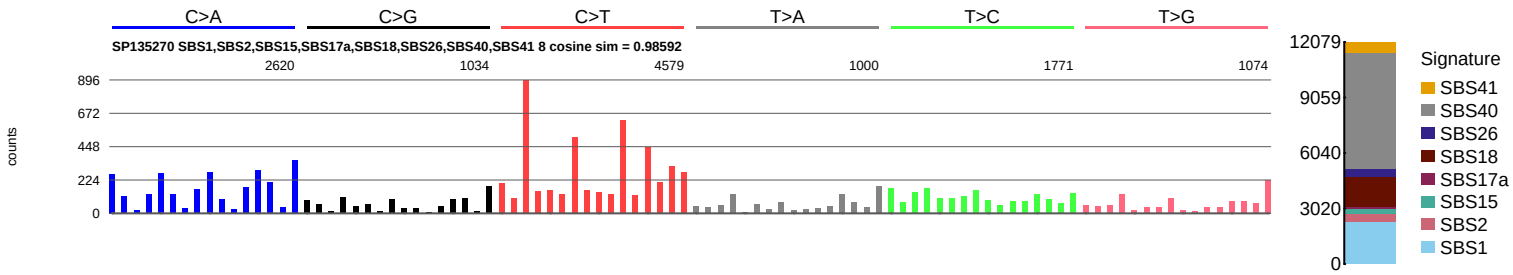

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98594
2630
1023
4621
988
1753
1061
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1

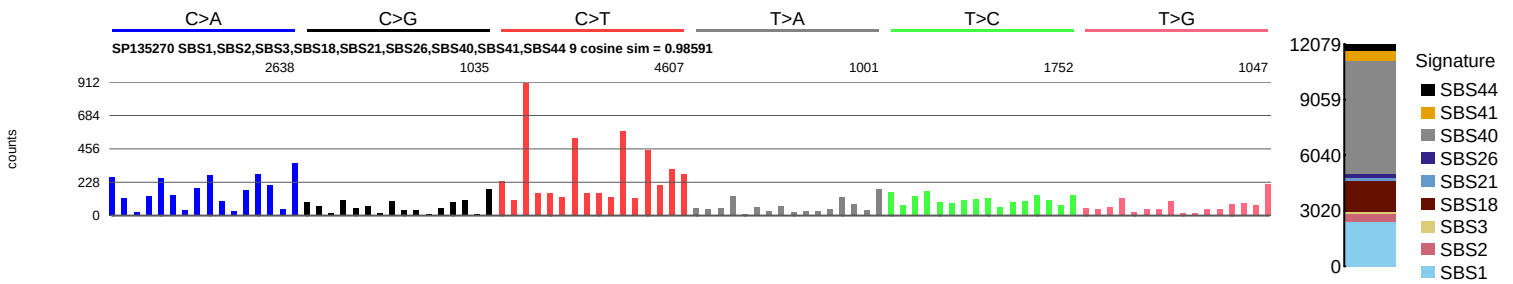

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98594
2641
1030
4613
994
1753
1051
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1

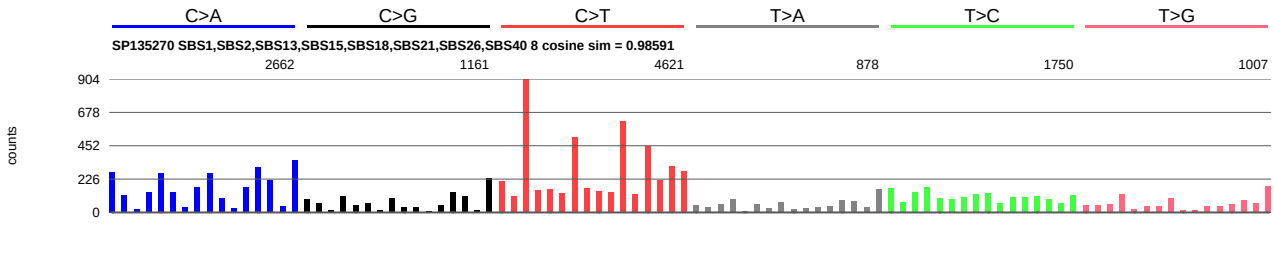

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98592
2618
1048
4568
1014
1765
1065
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1

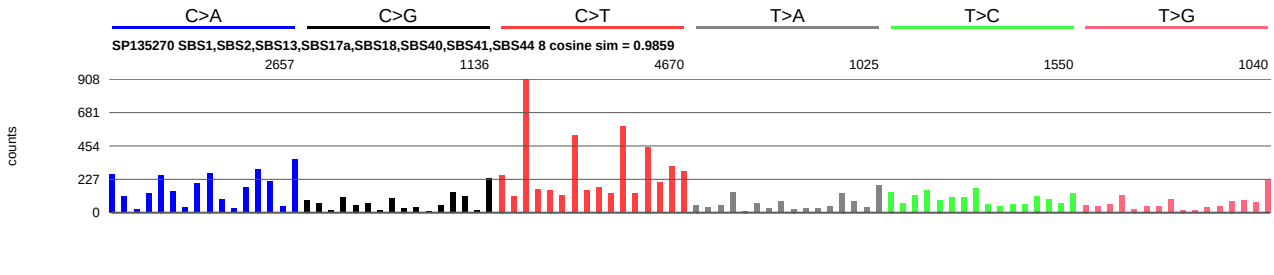

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98592
2619
1064
4452
1024
1758
1166
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

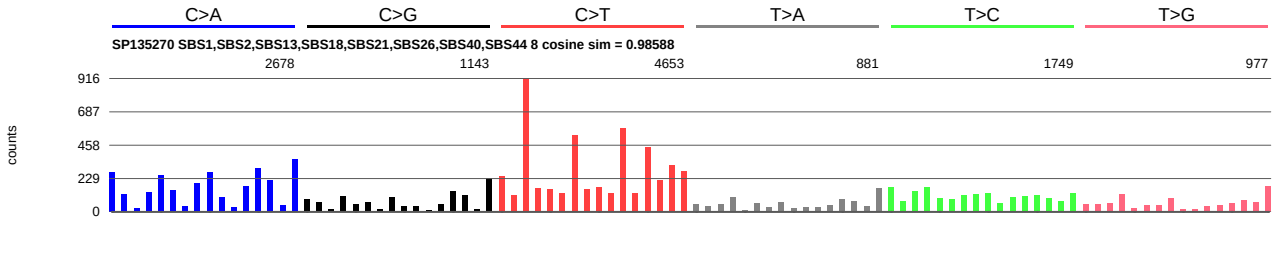

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98592
2661
1005
4662
887
1734
1126
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

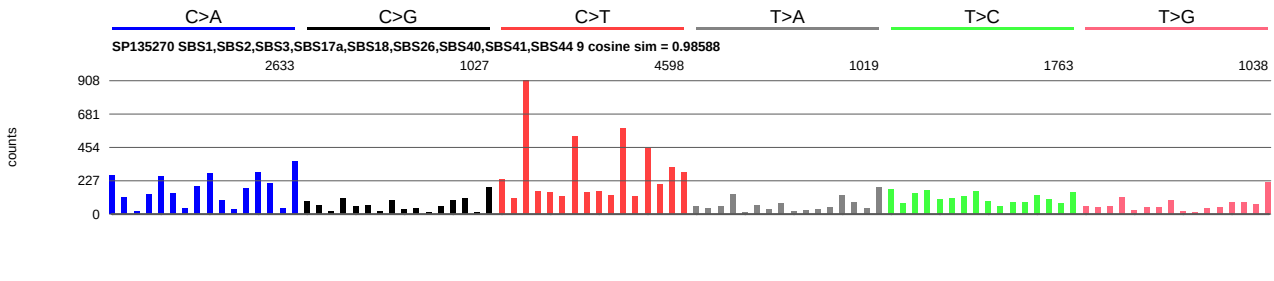

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98592
2620
1034
4579
1000
1771
1074
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1

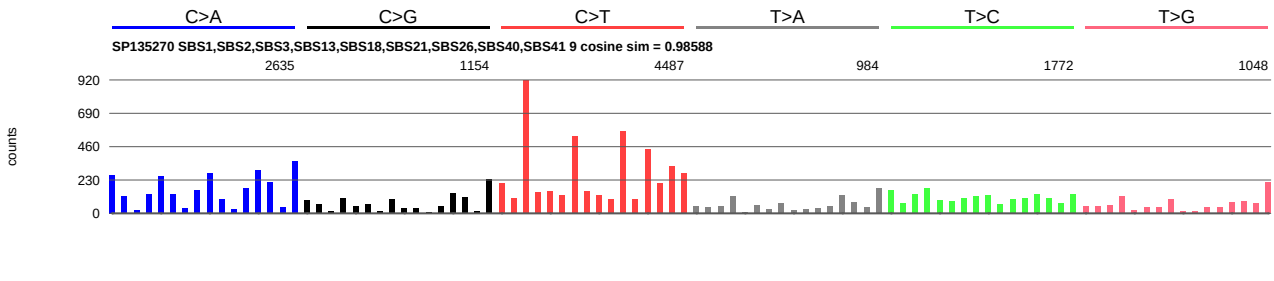

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98591
2638
1035
4607
1001
1752
1047
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS3
SBS2
SBS1

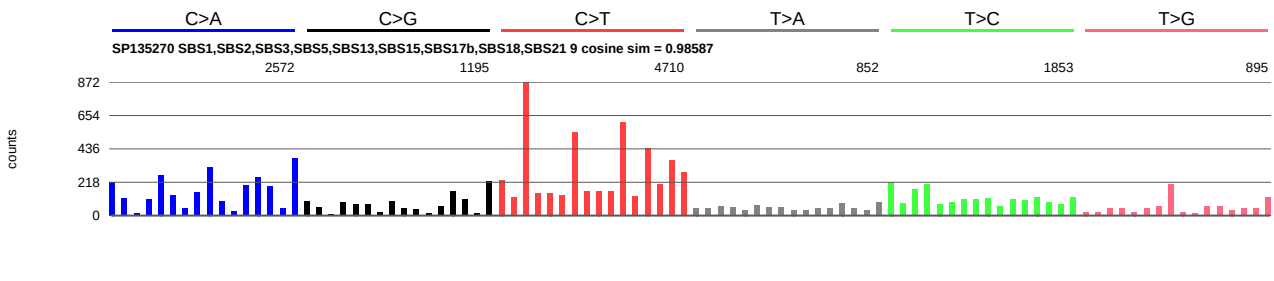

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98591
2662
1161
4621
878
1750
1007
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1

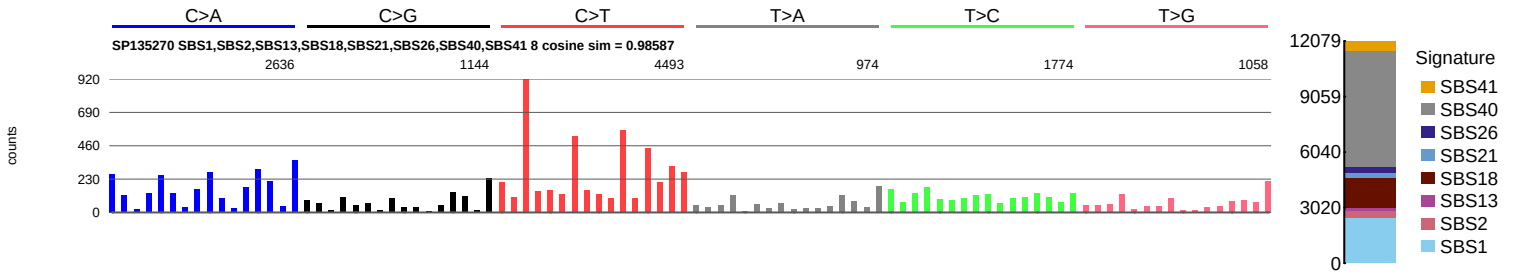

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9859
2657
1136
4670
1025
1550
1040
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98588
2678
1143
4653
881
1749
977
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98588
2633
1027
4598
1019
1763
1038
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98588
2635
1154
4487
984
1772
1048
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 9 cosine sim = 0.98587
2572
1195
4710
852
1853
895
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98587
2636
1144
4493
974
1774
1058
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
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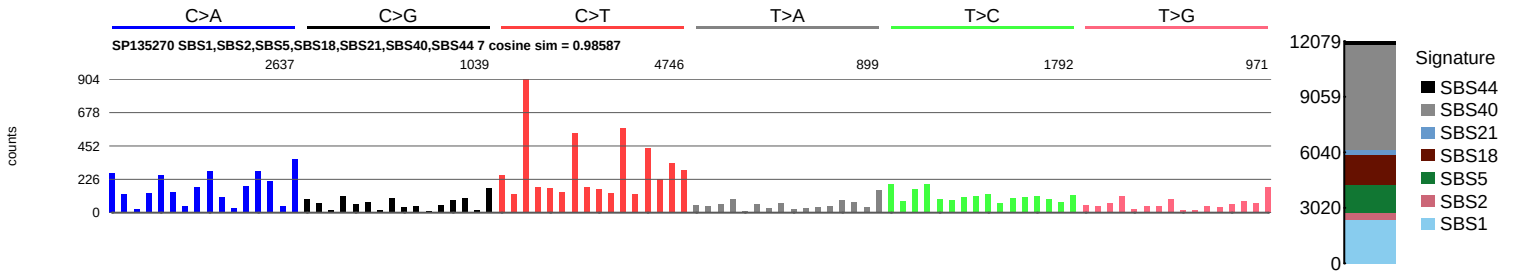
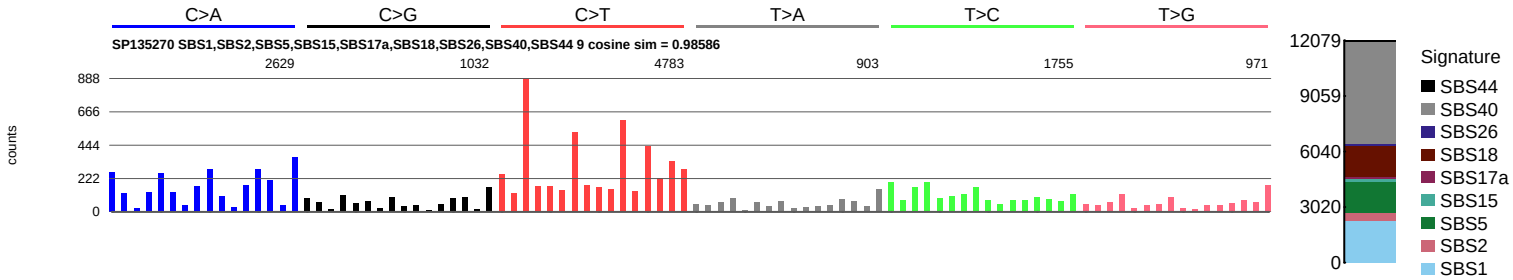
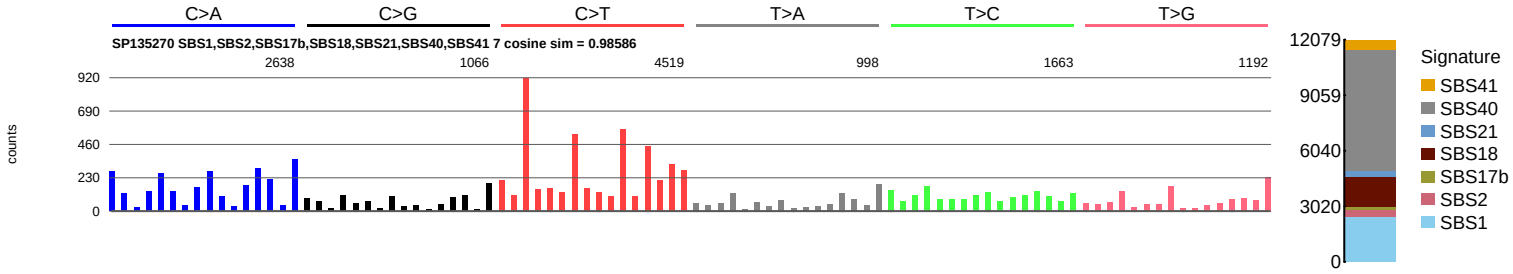
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98587
2637
1039
4746
899
1792
971
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98586
2629
1032
4783
903
1755
971
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
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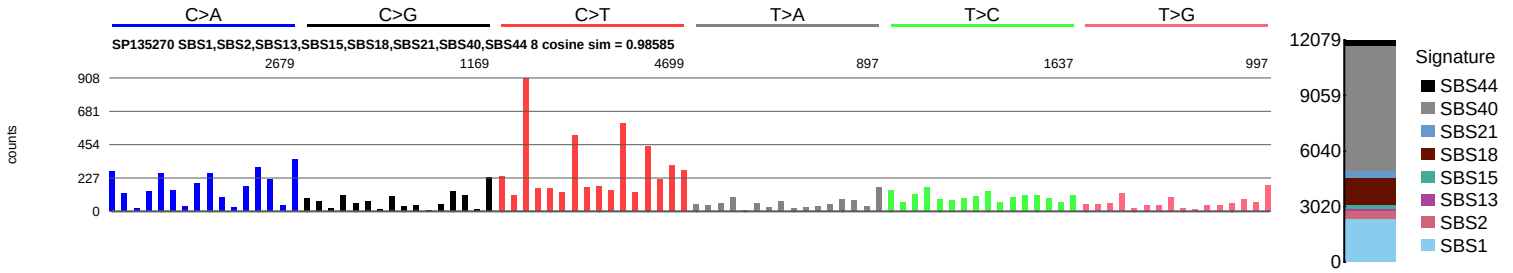
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98586
2638
1066
4519
998
1663
1192
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1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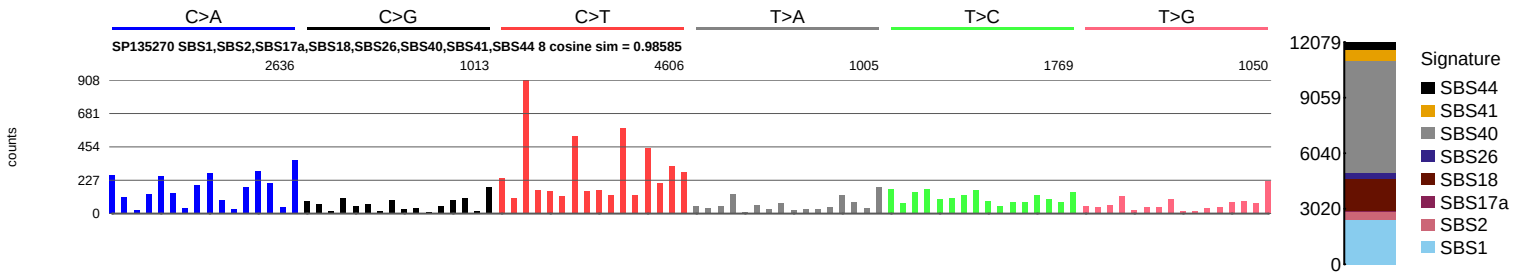
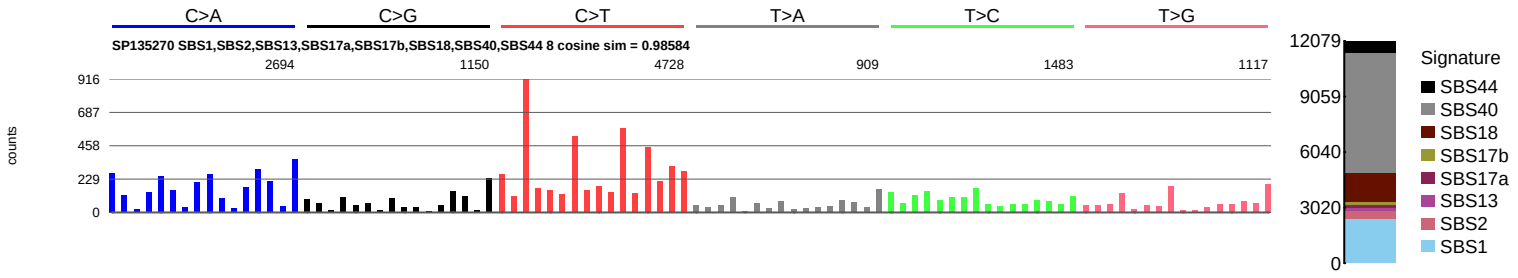
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98585
2679
1169
4699
897
1637
997
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98585
2636
1013
4606
1005
1769
1050
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1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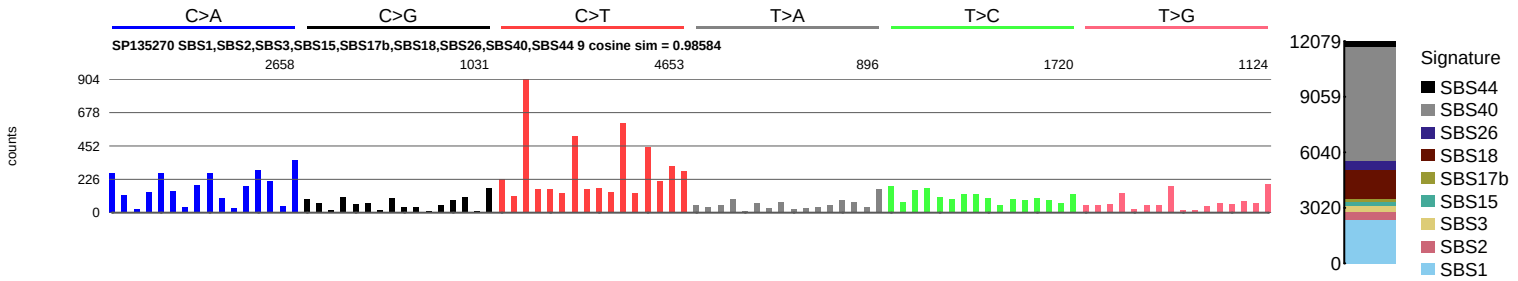
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98584
2694
1150
4728
909
1483
1117
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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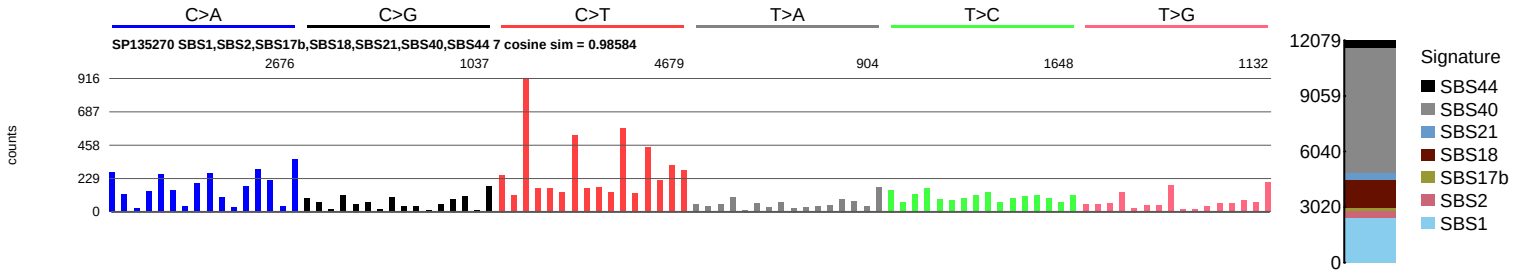
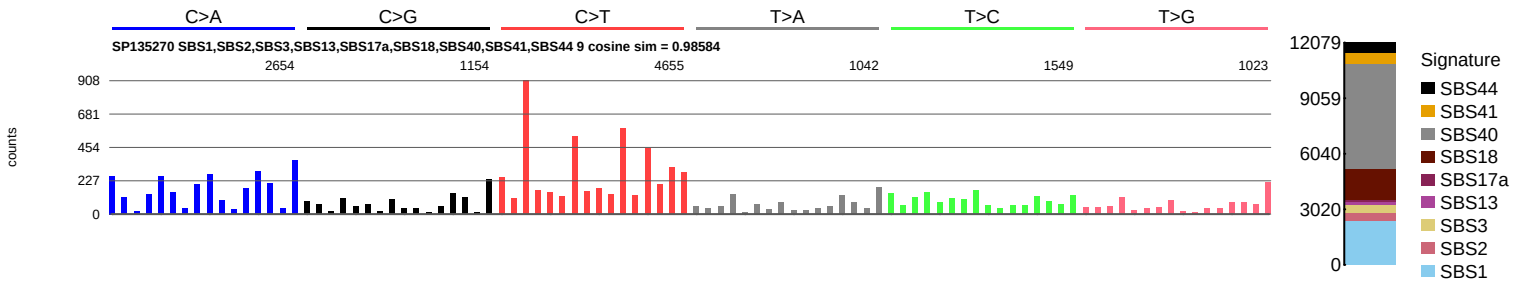
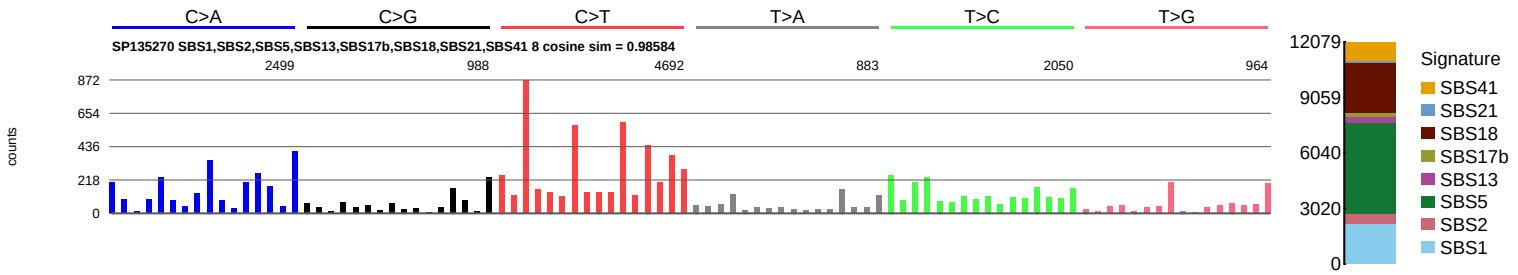
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98584
2658
1031
4653
896
1720
1124
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98584
2676
1037
4679
904
1648
1132
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98584
2654
1154
4655
1042
1549
1023
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1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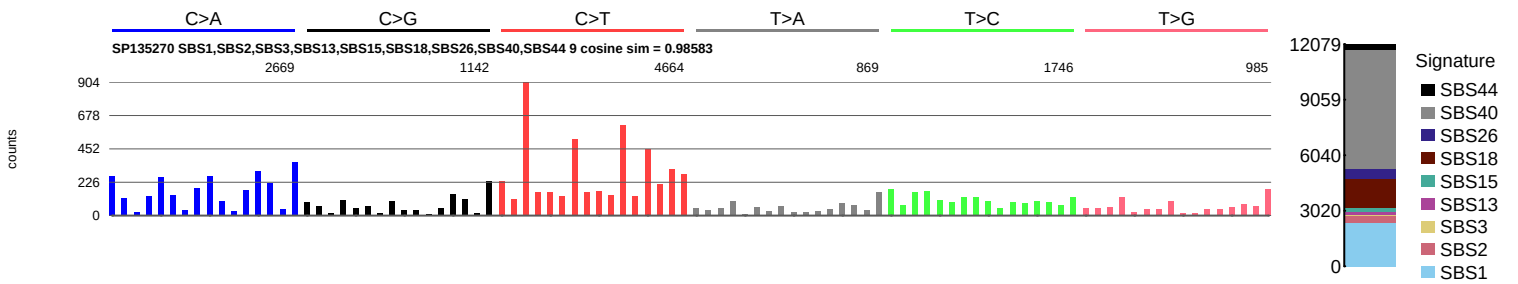
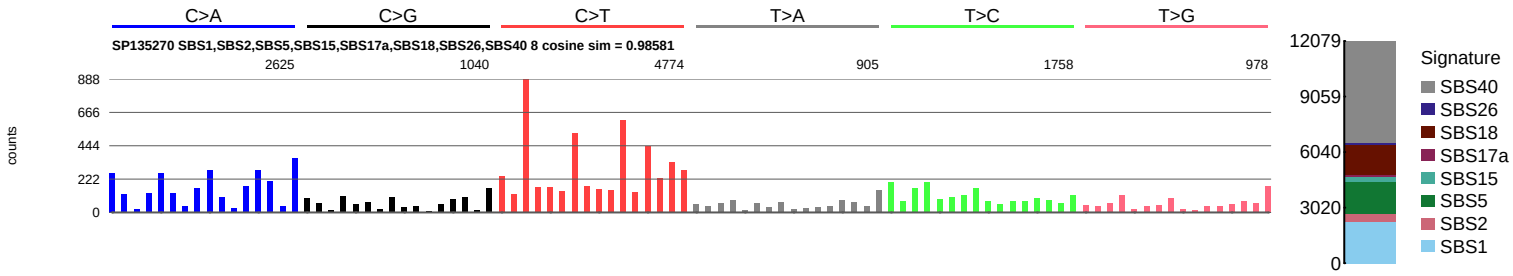
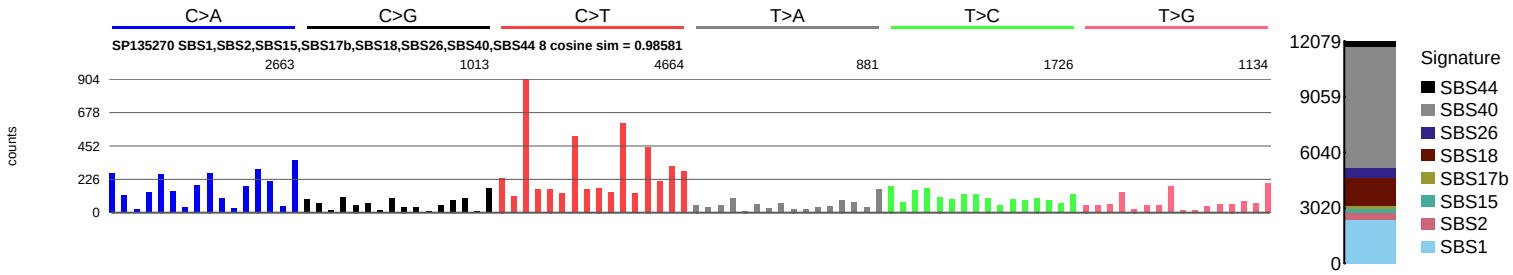
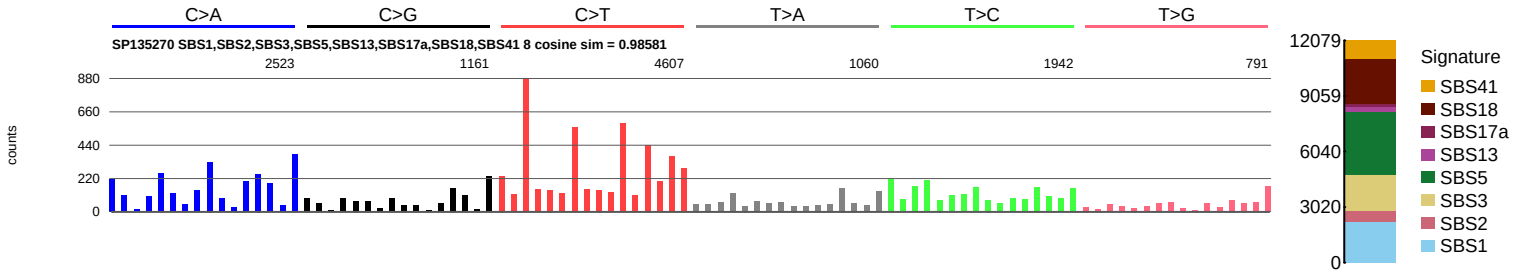
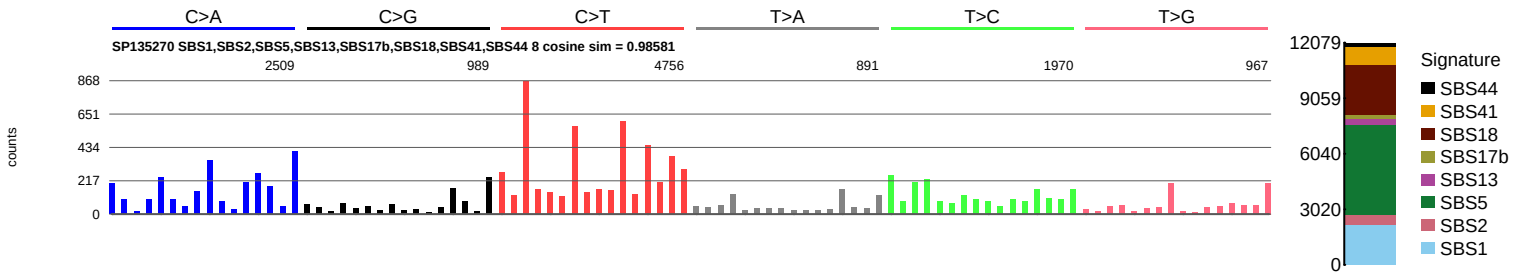
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98584
2499
988
4692
883
2050
964
872
654
436
218
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

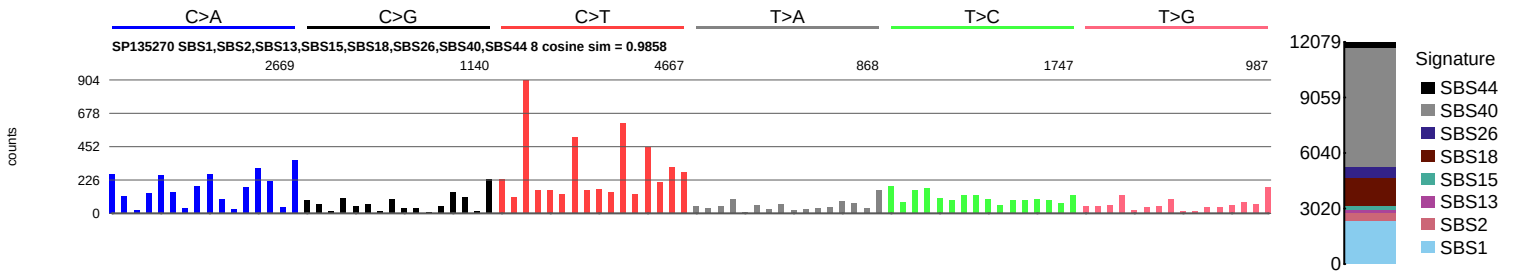

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98583
2669
1142
4664
869
1746
985
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

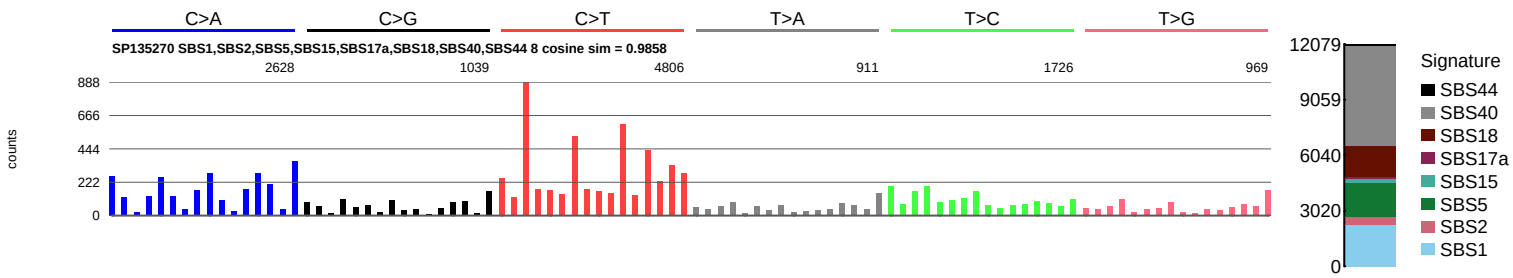

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98581
2625
1040
4774
905
1758
978
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1

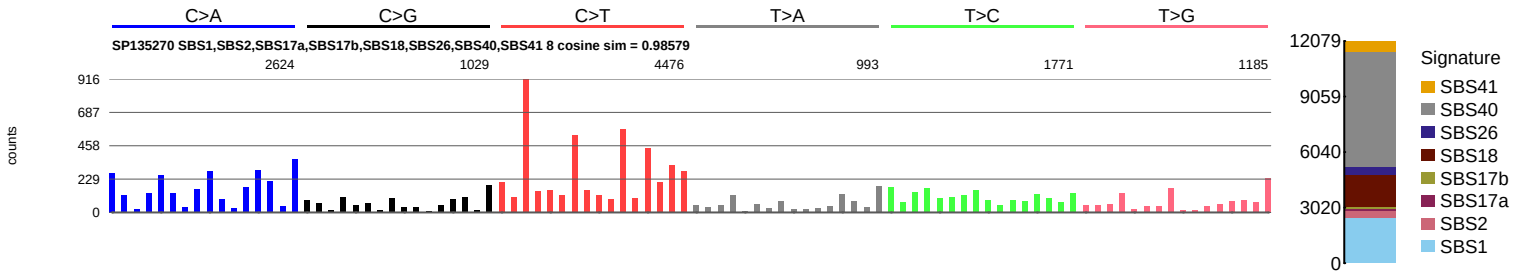

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98581
2663
1013
4664
881
1726
1134
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1

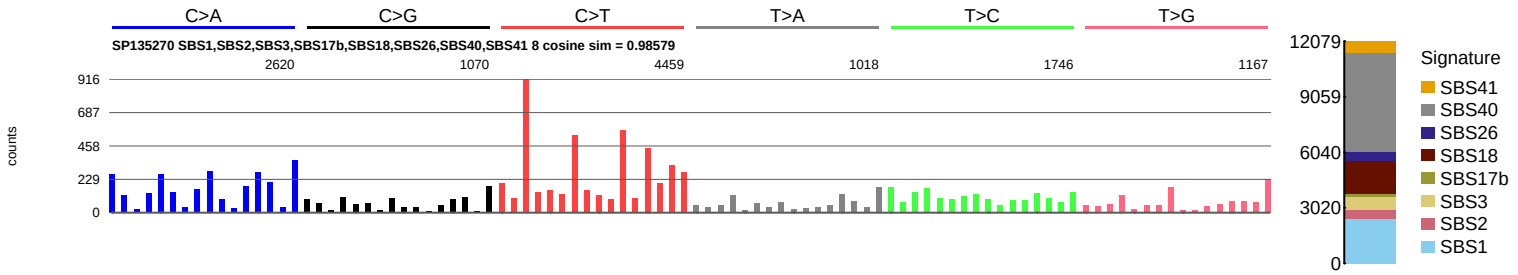

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41 8 cosine sim = 0.98581
2523
1161
4607
1060
1942
791
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

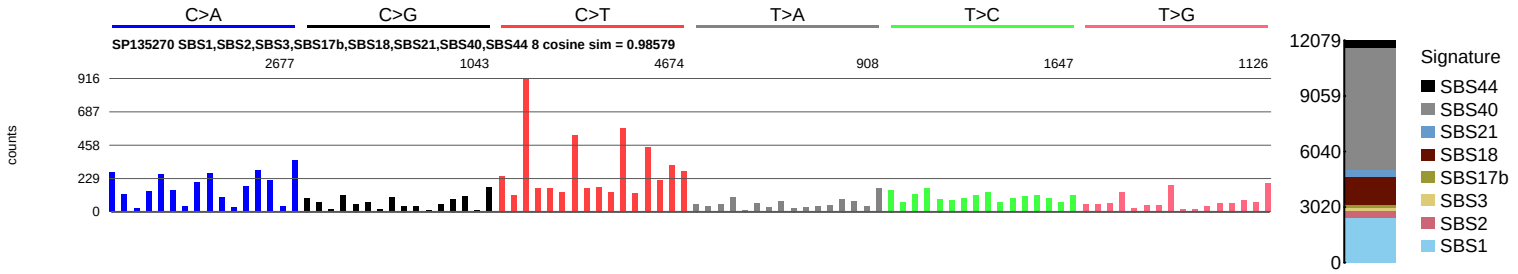

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98581
2509
989
4756
891
1970
967
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.9858
2669
1140
4667
868
1747
987
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.9858
2628
1039
4806
911
1726
969
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98579
2624
1029
4476
993
1771
1185
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98579
2620
1070
4459
1018
1746
1167
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98579
2677
1043
4674
908
1647
1126
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
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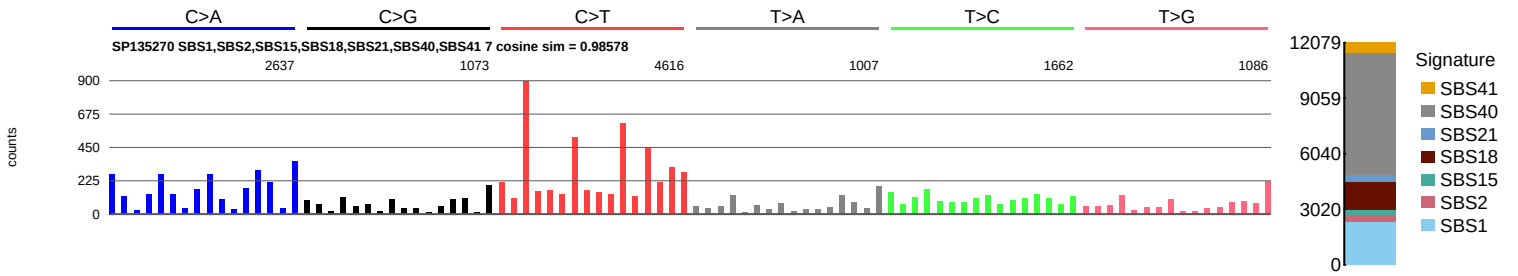
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98578
2637
1073
4616
1007
1662
1086
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
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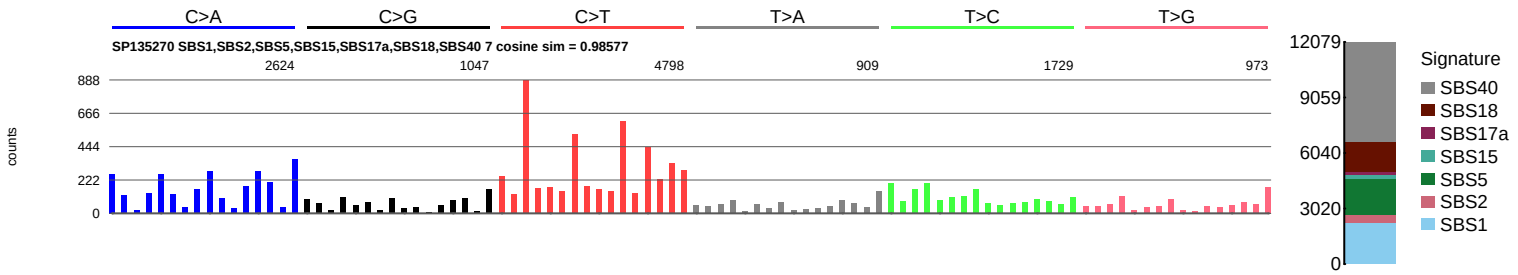
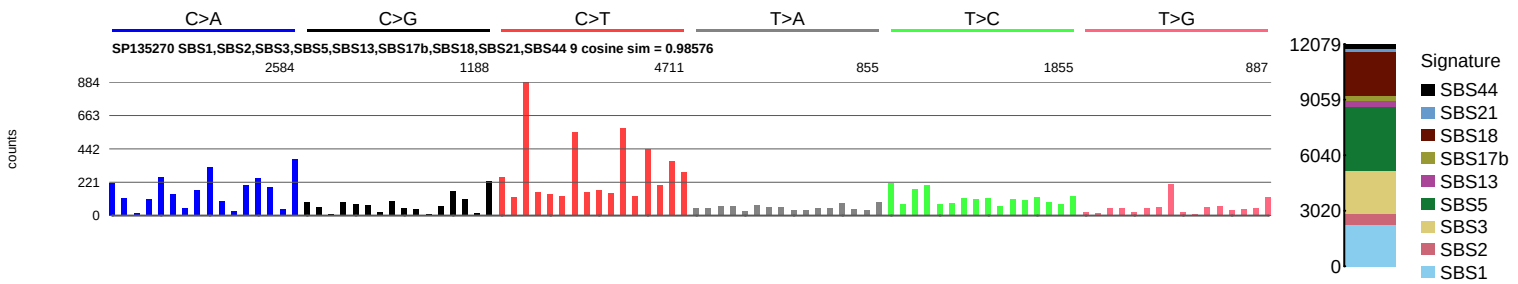
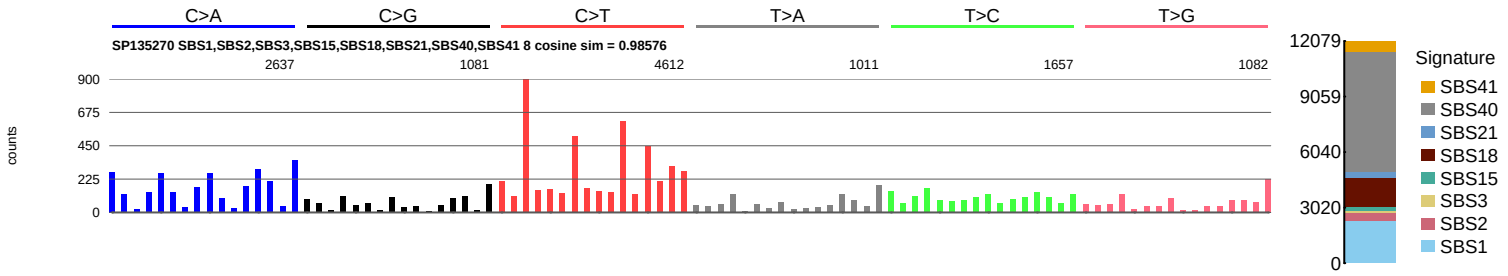
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40 7 cosine sim = 0.98577
2624
1047
4798
909
1729
973
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98576
2584
1188
4711
855
1855
887
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98576
2637
1081
4612
1011
1657
1082
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
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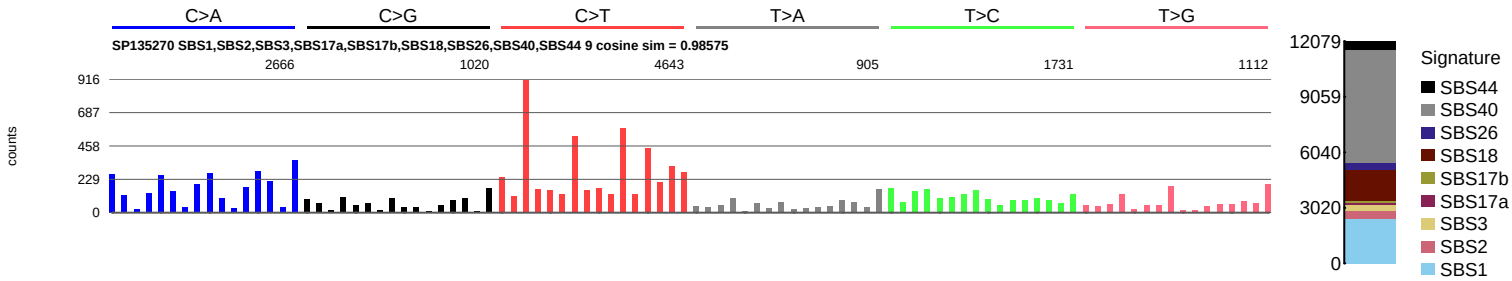
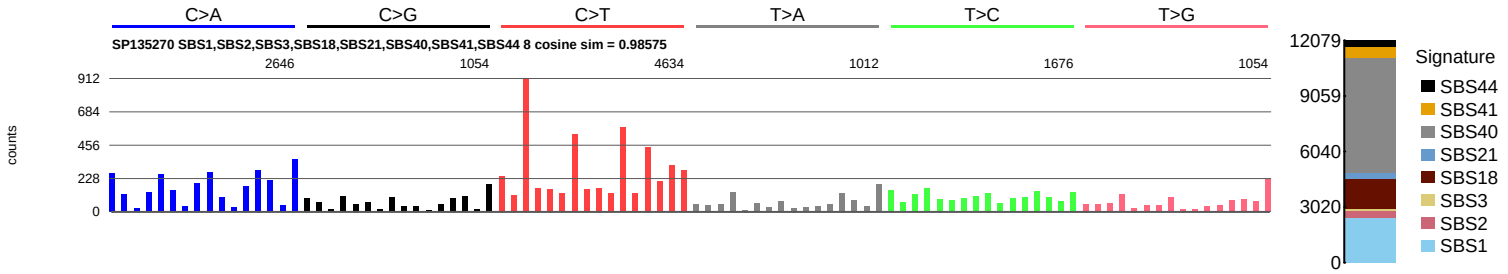
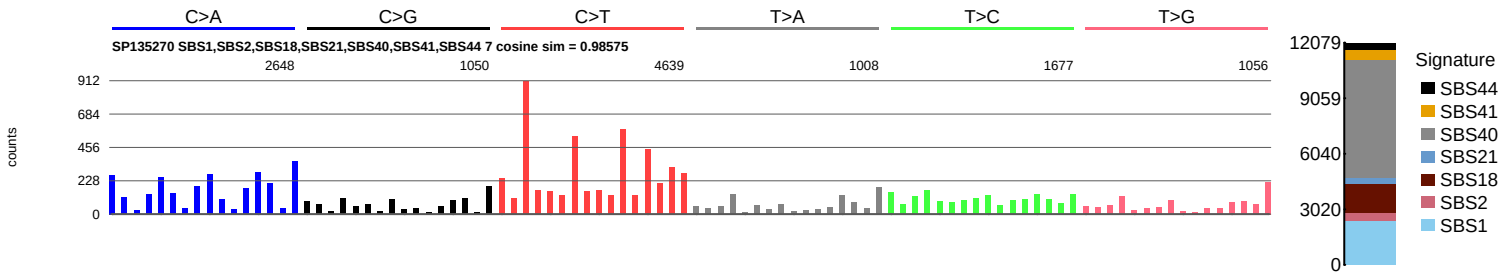
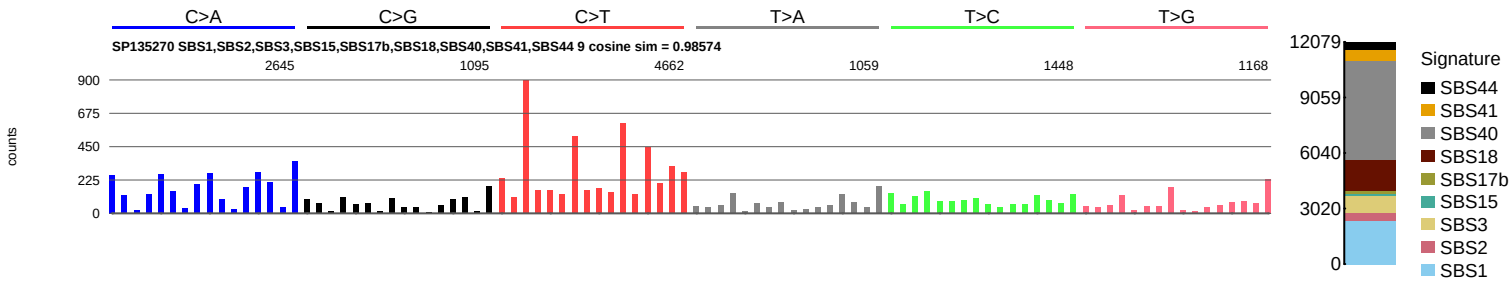
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98575
2666
1020
4643
905
1731
1112
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98575
2646
1054
4634
1012
1676
1054
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98575
2648
1050
4639
1008
1677
1056
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS2
SBS1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C>T
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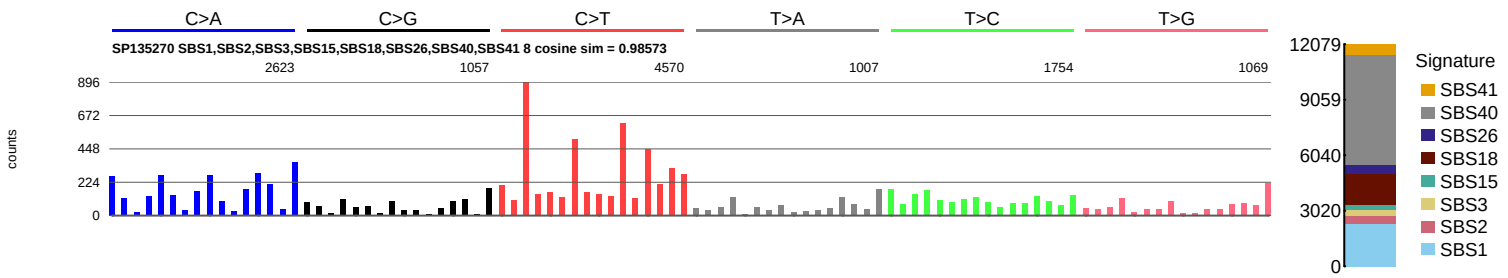
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98574
2645
1095
4662
1059
1448
1168
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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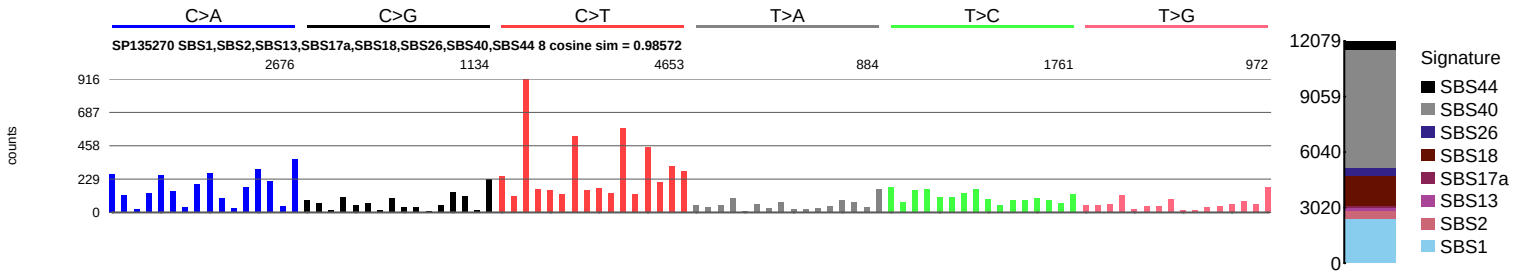
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98573
2623
1057
4570
1007
1754
1069
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS3
SBS2
SBS1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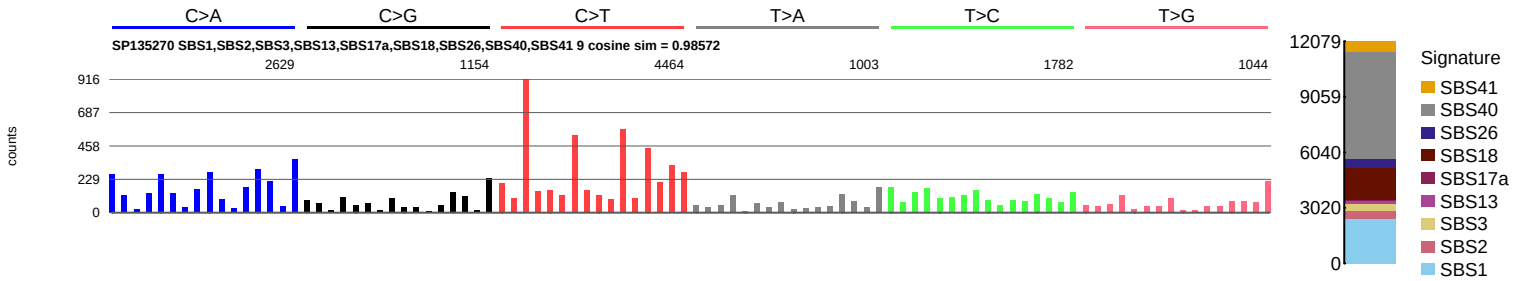
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98572
2676
1134
4653
884
1761
972
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1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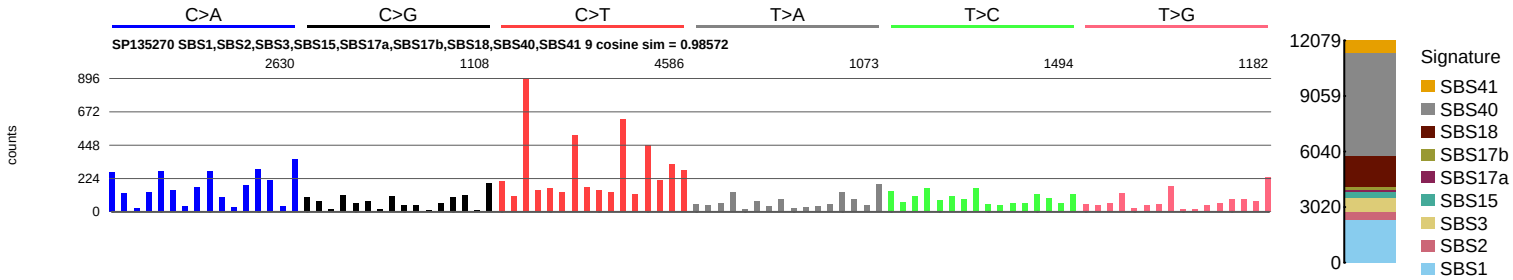
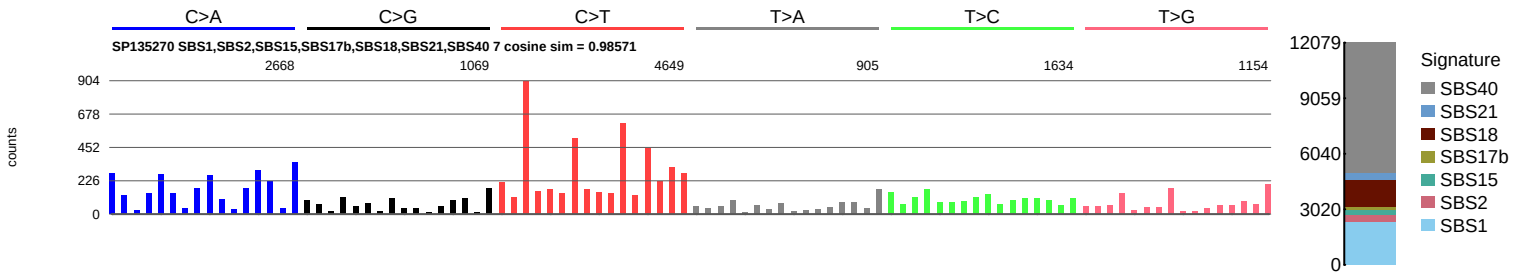
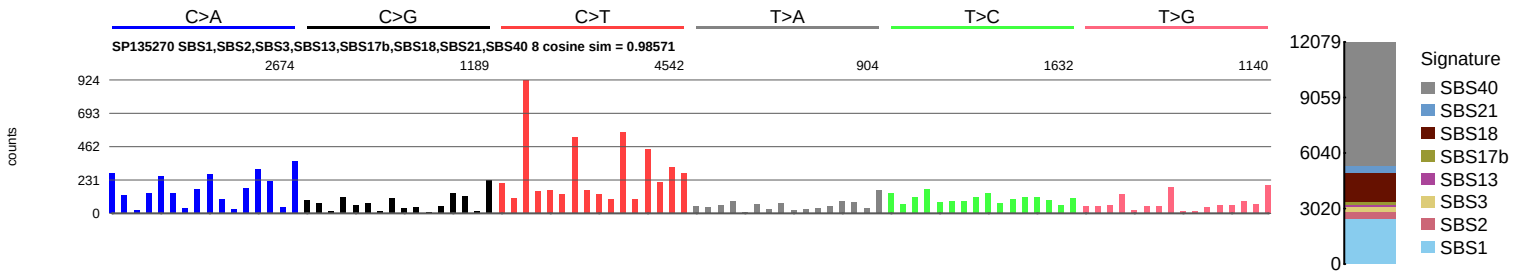
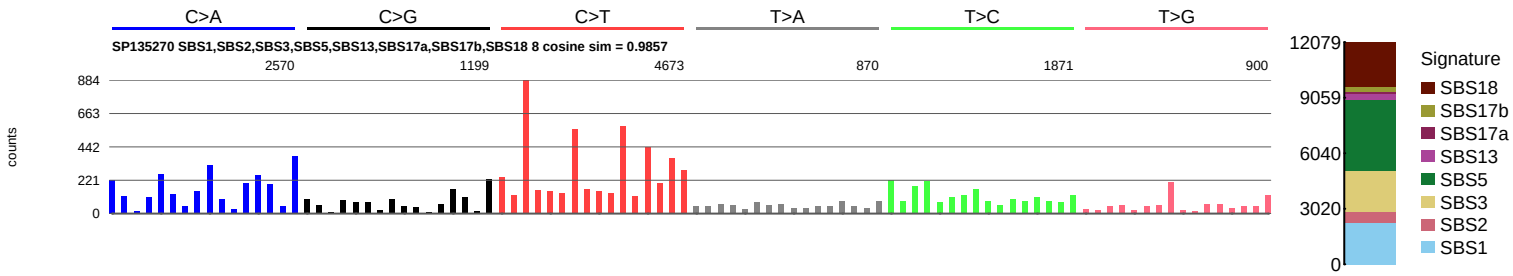
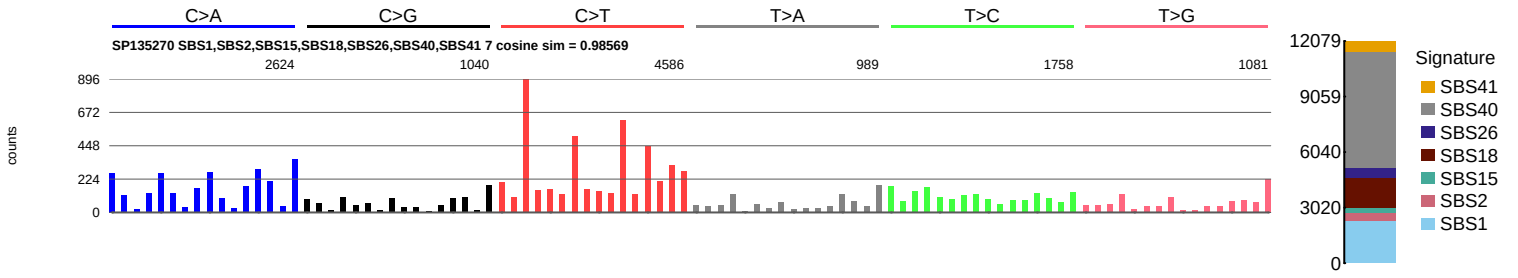
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98572
2629
1154
4464
1003
1782
1044
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

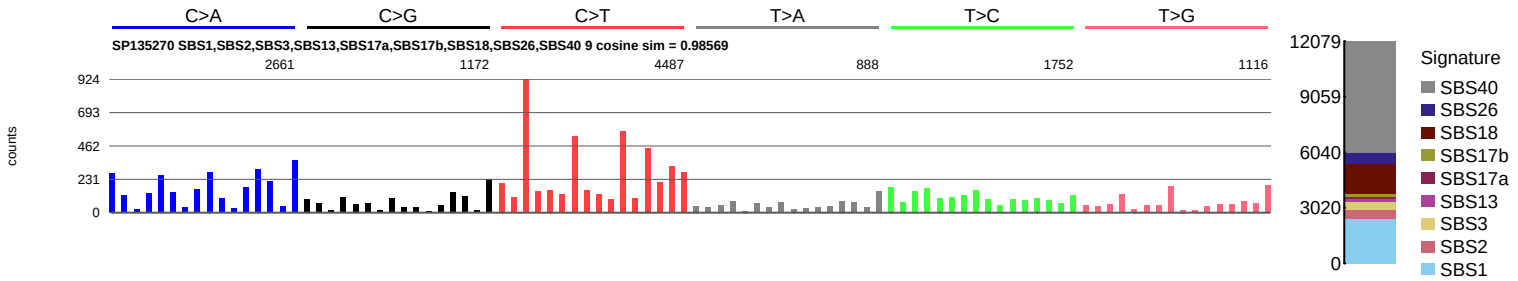

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98572
2630
1108
4586
1073
1494
1182
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

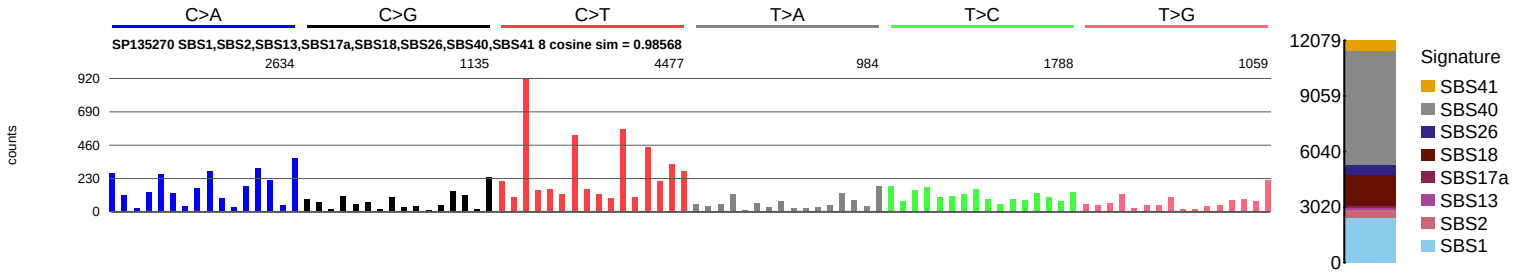

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98571
2668
1069
4649
905
1634
1154
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1

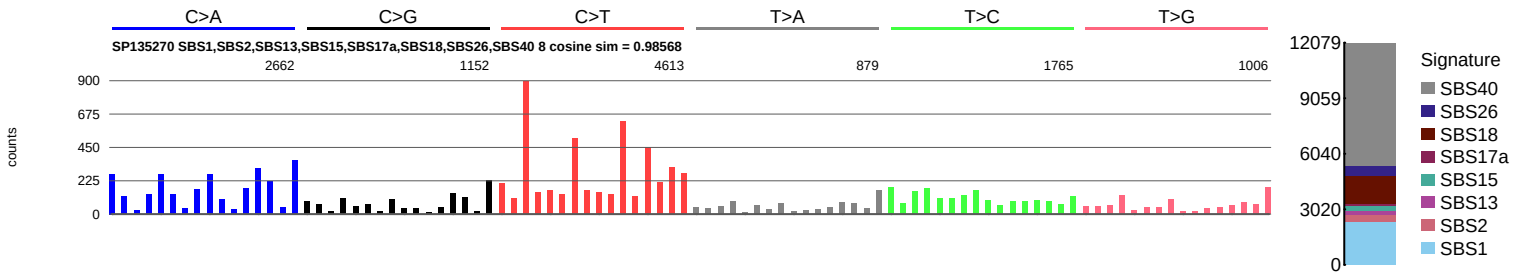

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98571
2674
1189
4542
904
1632
1140
924
693
462
231
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

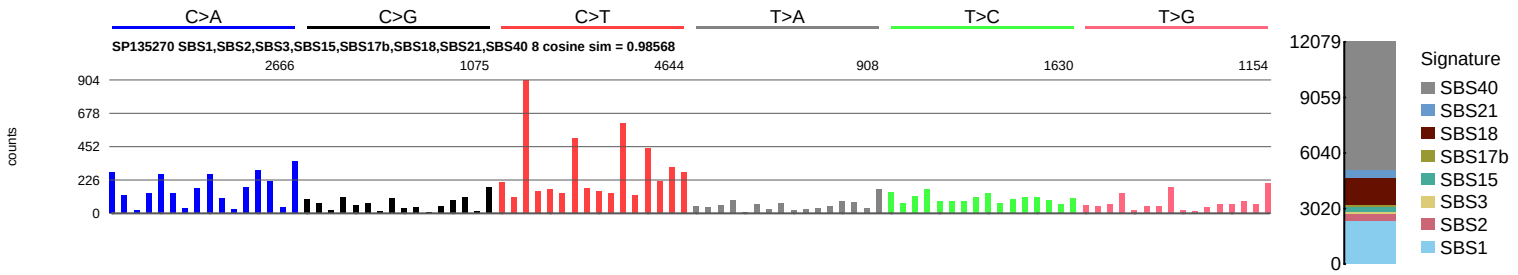

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.9857
2570
1199
4673
870
1871
900
884
663
442
221
0
counts
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98569
2624
1040
4586
989
1758
1081
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98569
2661
1172
4487
888
1752
1116
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98568
2634
1135
4477
984
1788
1059
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98568
2662
1152
4613
879
1765
1006
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98568
2666
1075
4644
908
1630
1154
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
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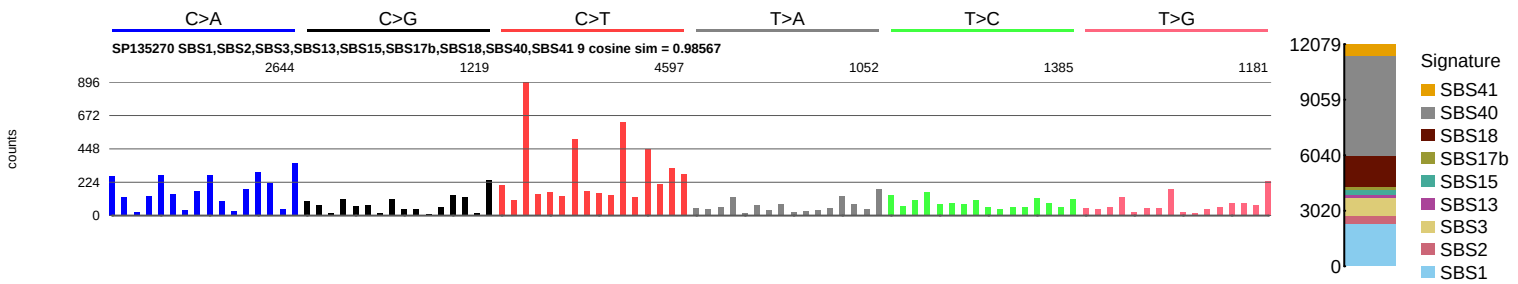
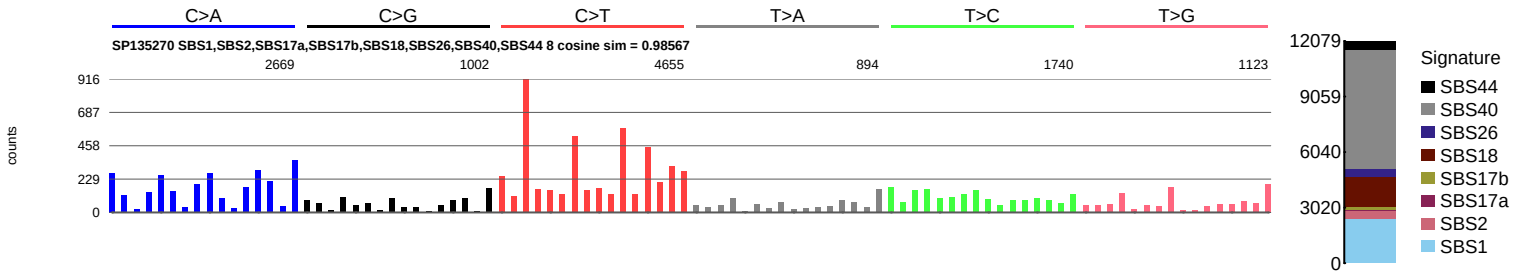
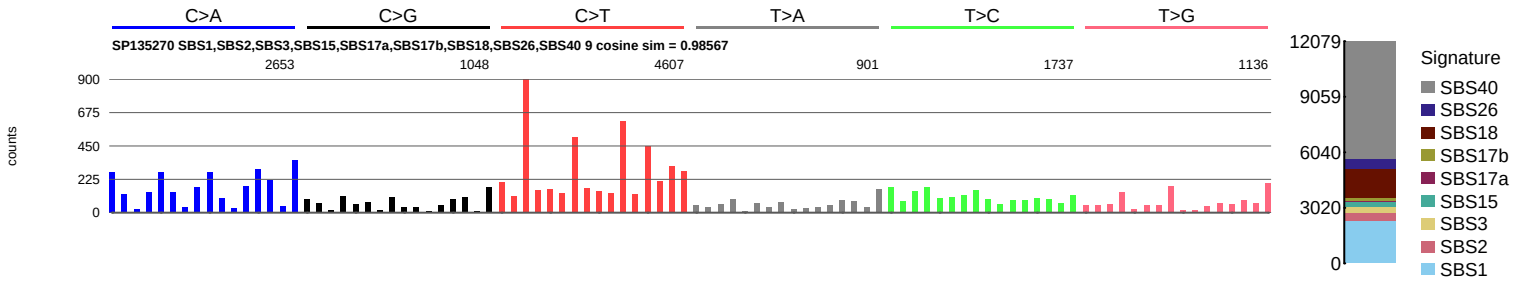
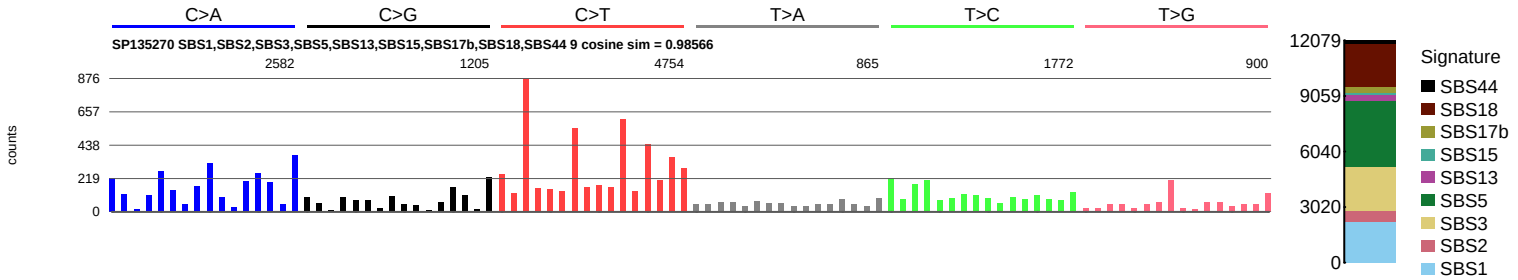
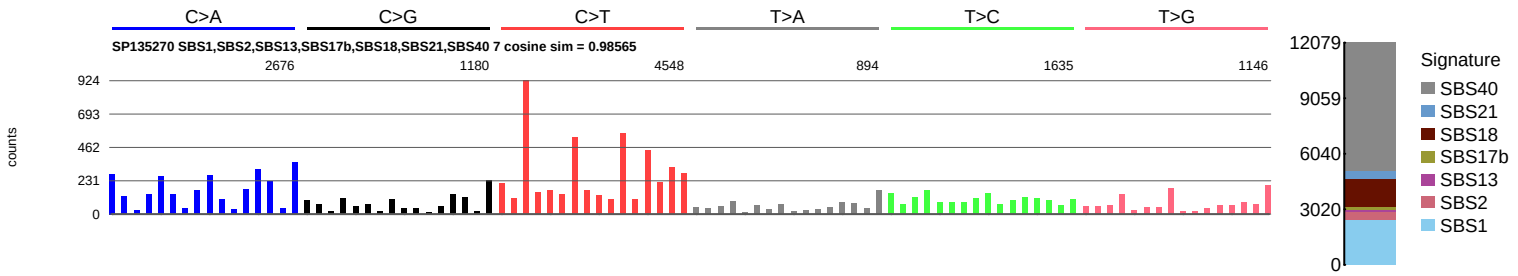
SBS3
SBS2
SBS1

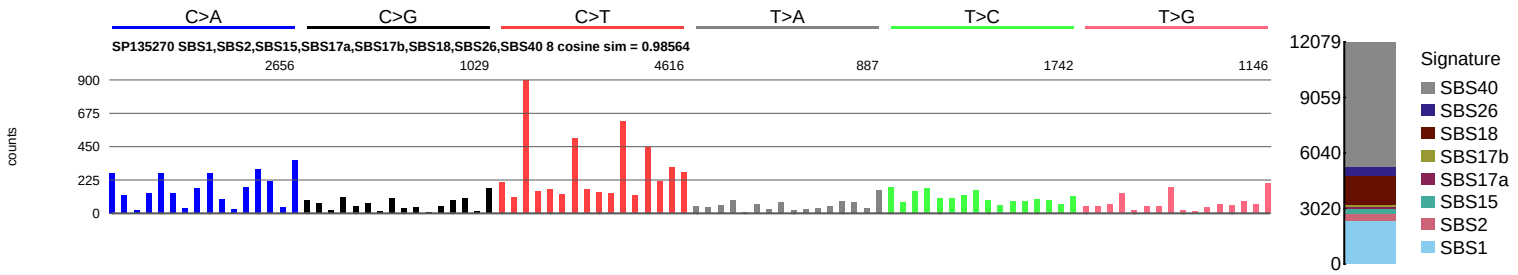

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98567
2644
1219
4597
1052
1385
1181
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

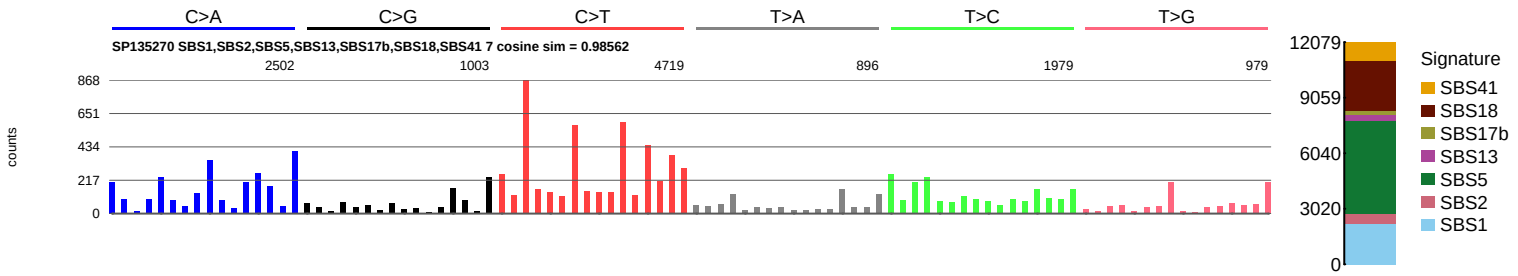

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98567
2669
1002
4655
894
1740
1123
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

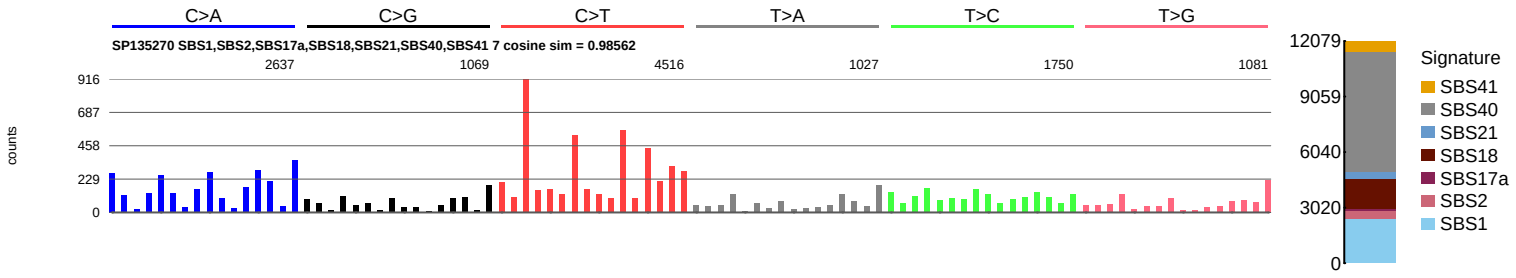

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98567
2653
1048
4607
901
1737
1136
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

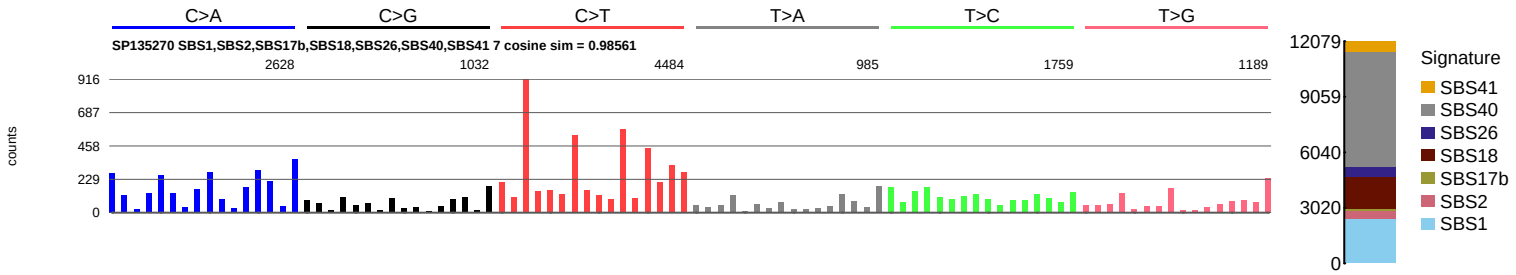

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 9 cosine sim = 0.98566
2582
1205
4754
865
1772
900
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

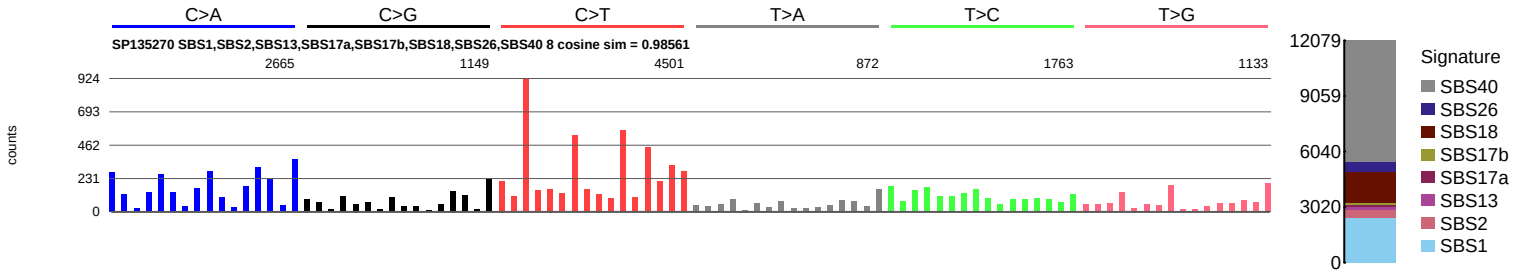

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98565
2676
1180
4548
894
1635
1146
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

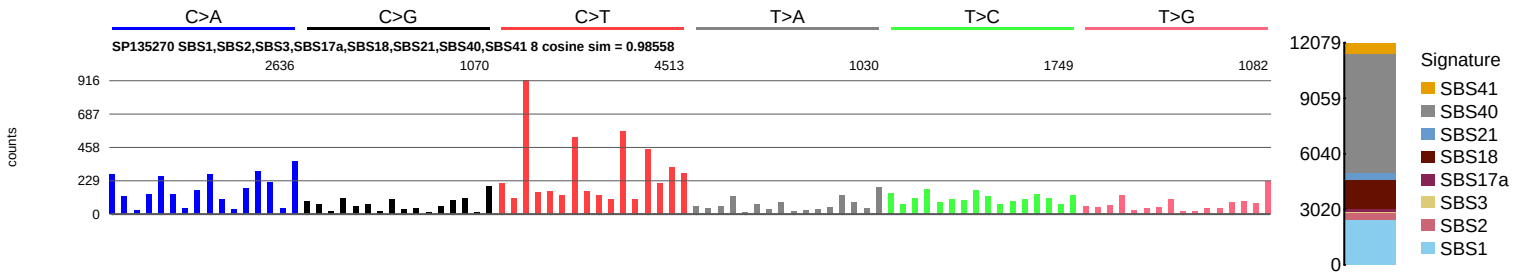

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98564
2656
1029
4616
887
1742
1146
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

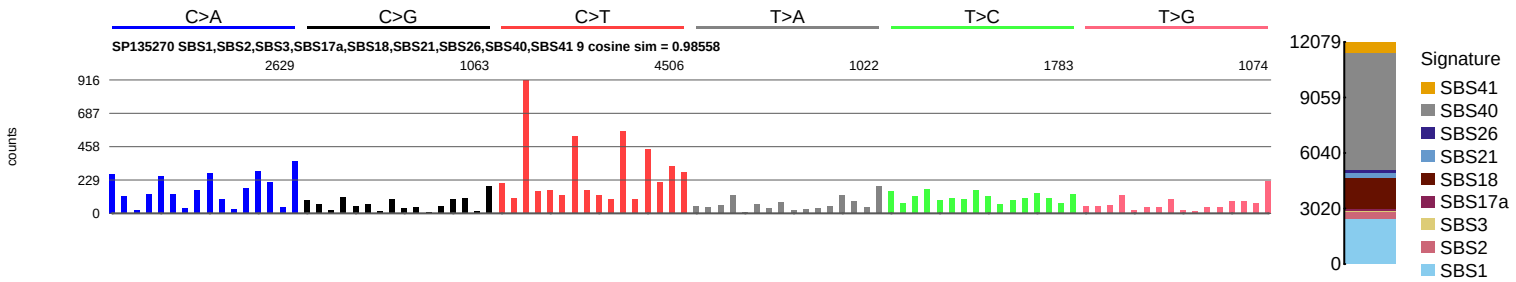

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41 7 cosine sim = 0.98562
2502
1003
4719
896
1979
979
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

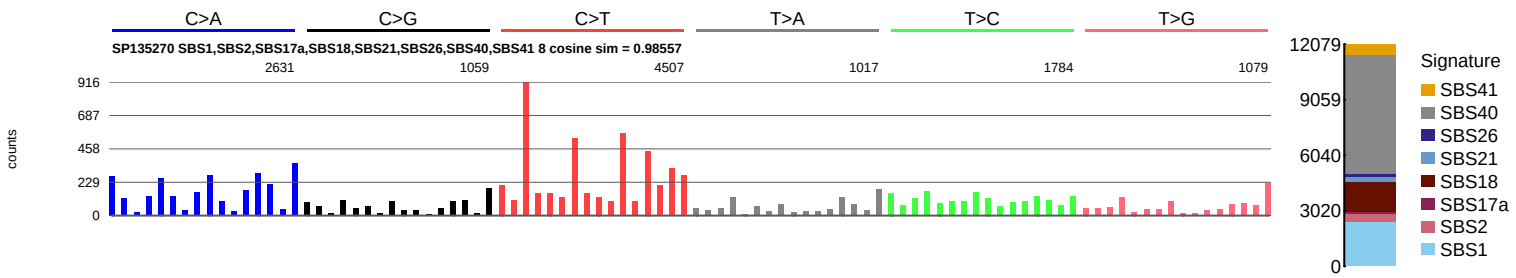

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98562
2637
1069
4516
1027
1750
1081
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98561
2628
1032
4484
985
1759
1189
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98561
2665
1149
4501
872
1763
1133
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98558
2636
1070
4513
1030
1749
1082
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98558
2629
1063
4506
1022
1783
1074
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS3
SBS2
SBS1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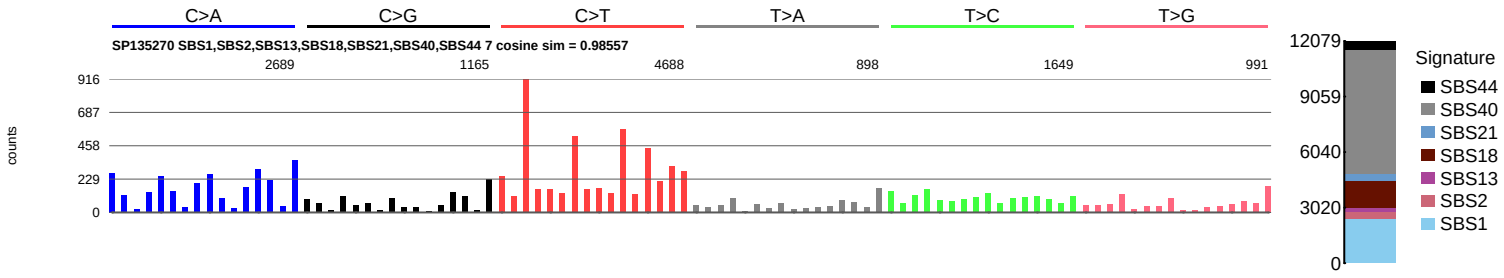
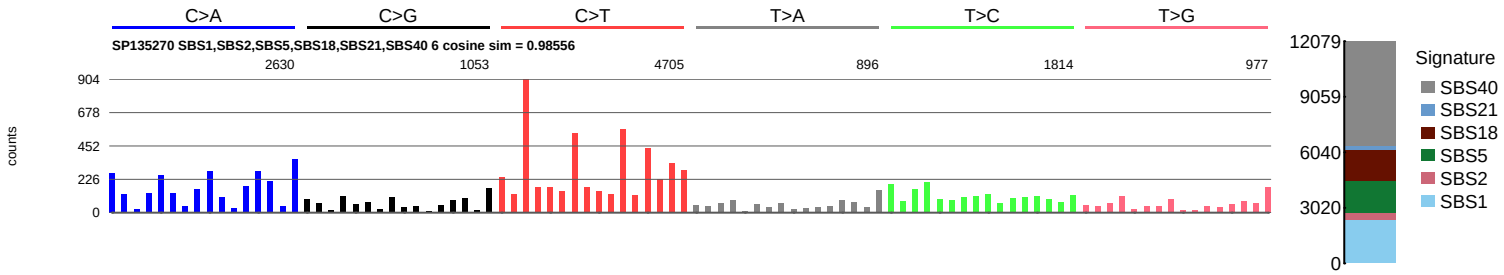
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98557
2631
1059
4507
1017
1784
1079
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98557
2689
1165
4688
898
1649
991
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1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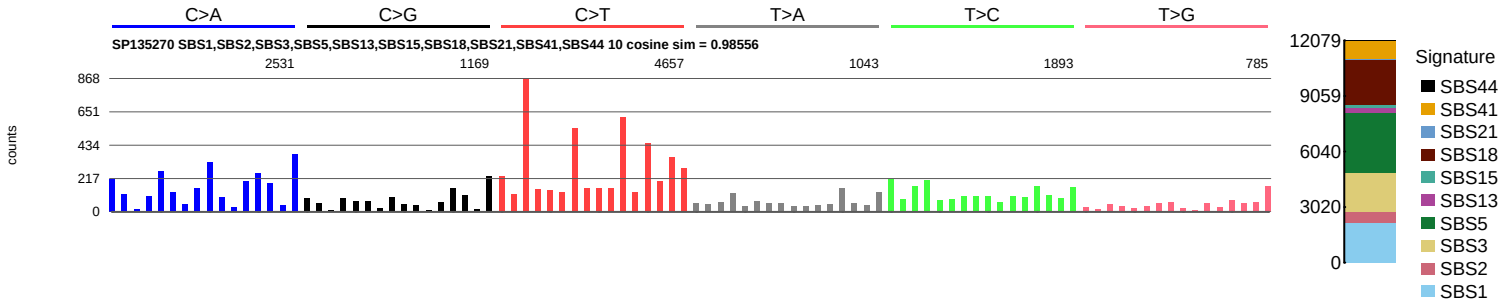
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40 6 cosine sim = 0.98556
2630
1053
4705
896
1814
977
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1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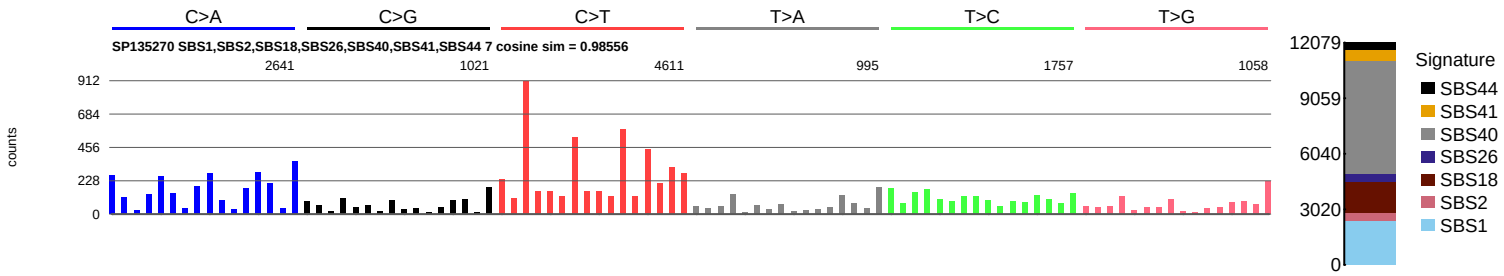
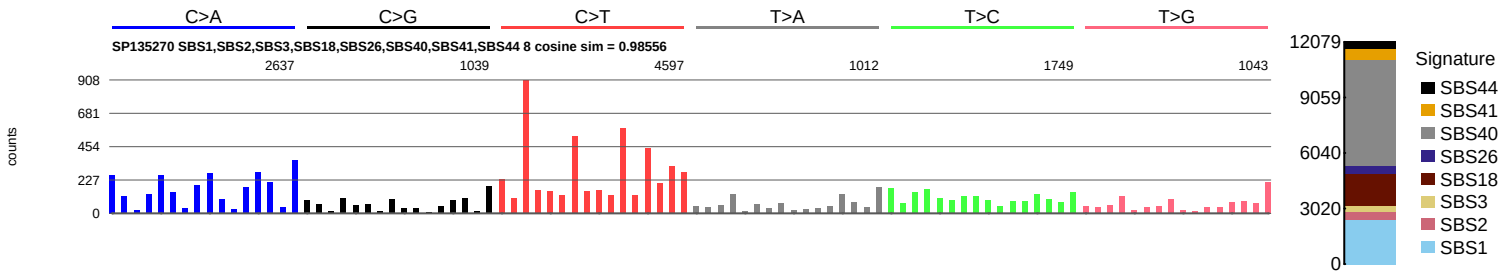
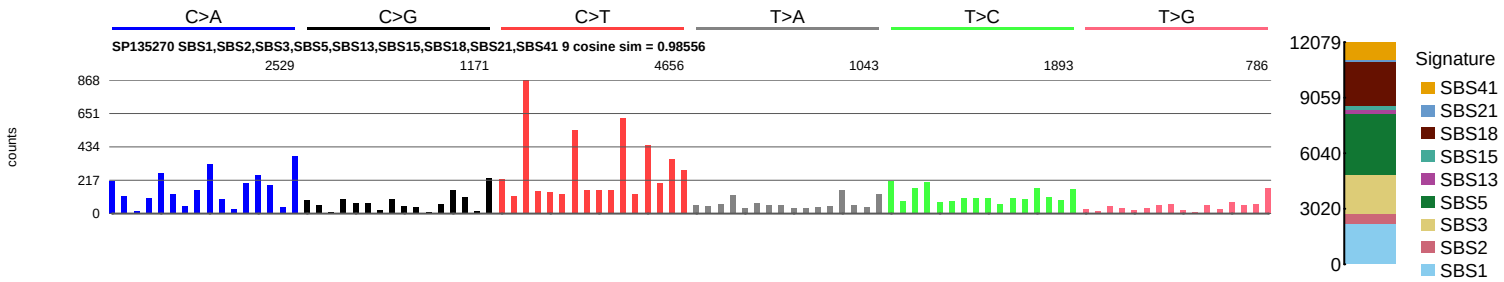
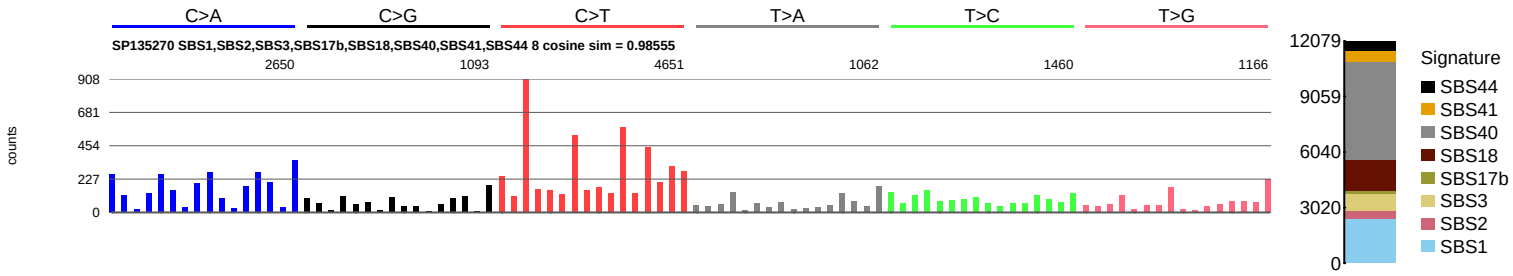
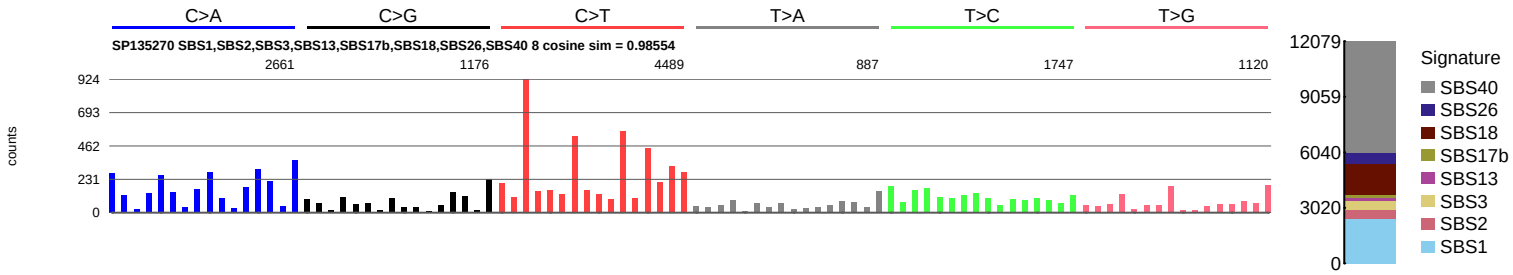
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98556
2531
1169
4657
1043
1893
785
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

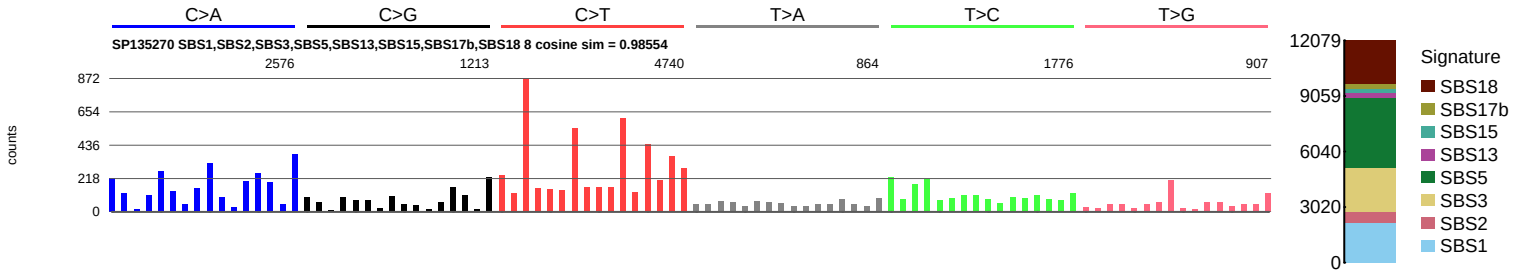

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98556
2641
1021
4611
995
1757
1058
912
684
456
228
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS2
SBS1

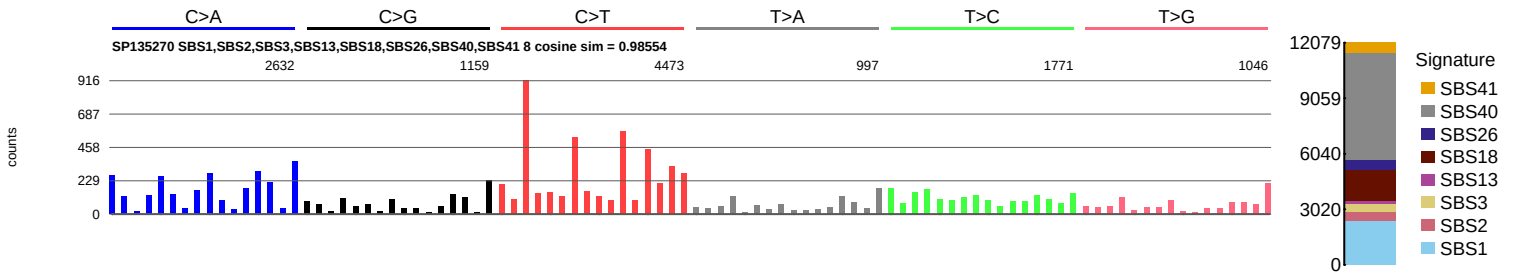

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98556
2637
1039
4597
1012
1749
1043
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS41 9 cosine sim = 0.98556
2529
1171
4656
1043
1893
786
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98555
2650
1093
4651
1062
1460
1166
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98554
2661
1176
4489
887
1747
1120
924
693
462
231
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 8 cosine sim = 0.98554
2576
1213
4740
864
1776
907
872
654
436
218
0
counts
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98554
2632
1159
4473
997
1771
1046
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
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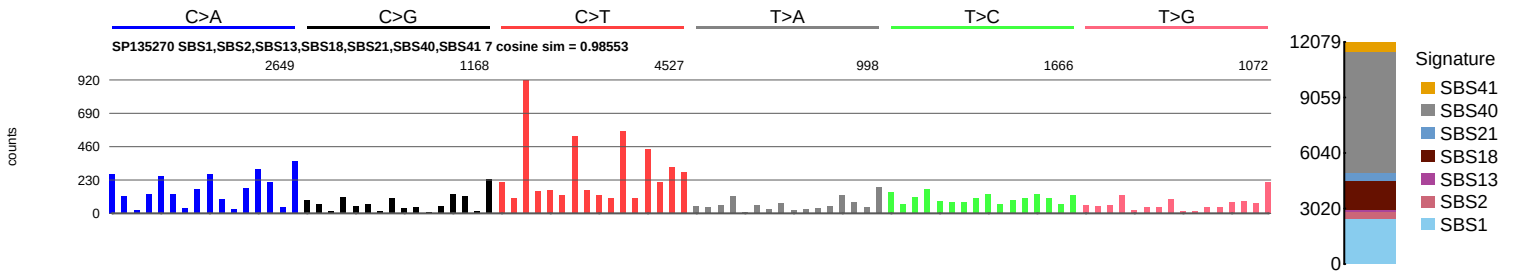
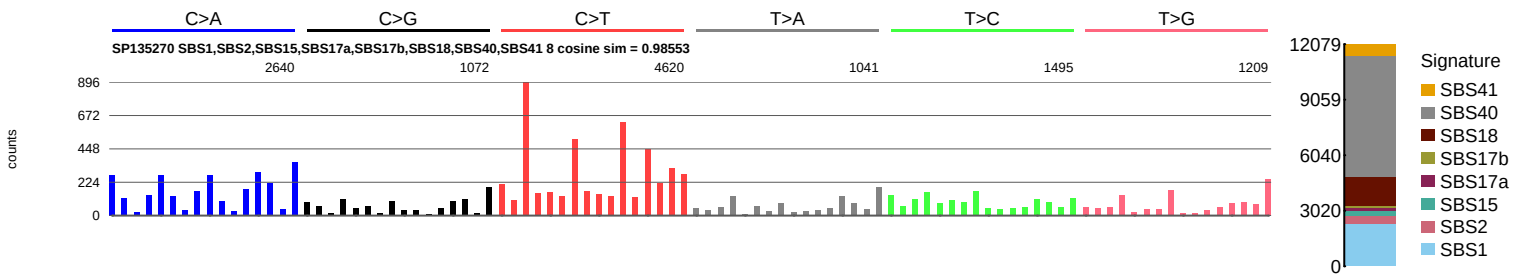
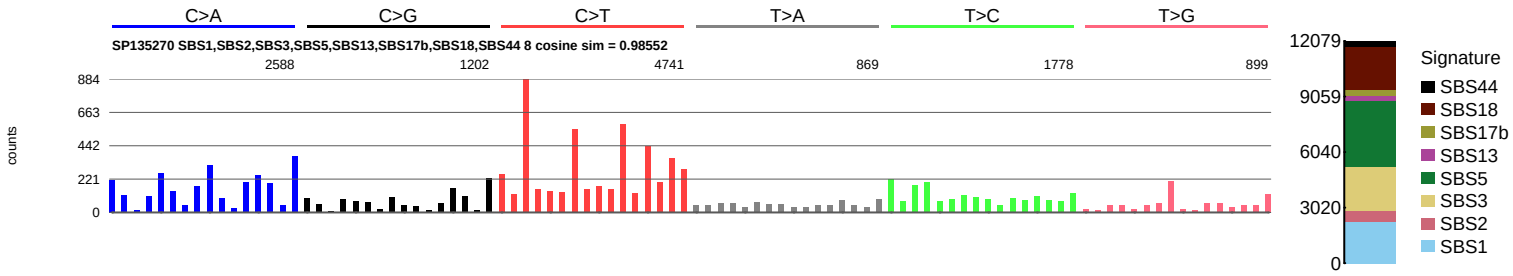
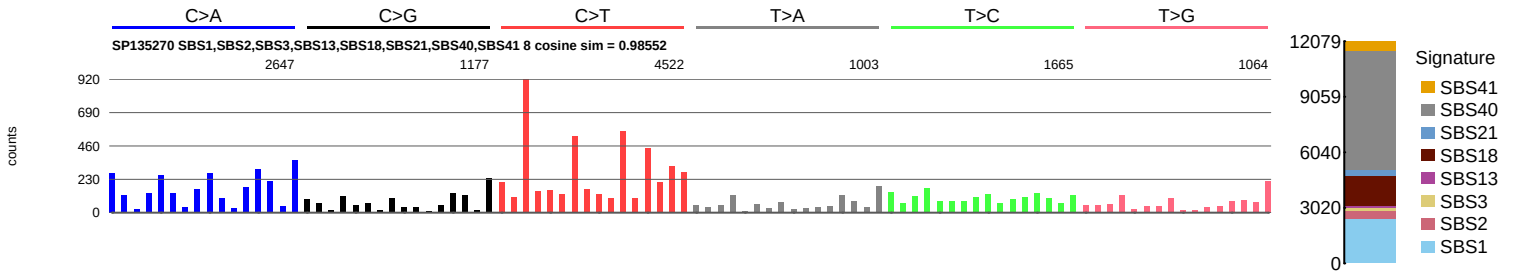
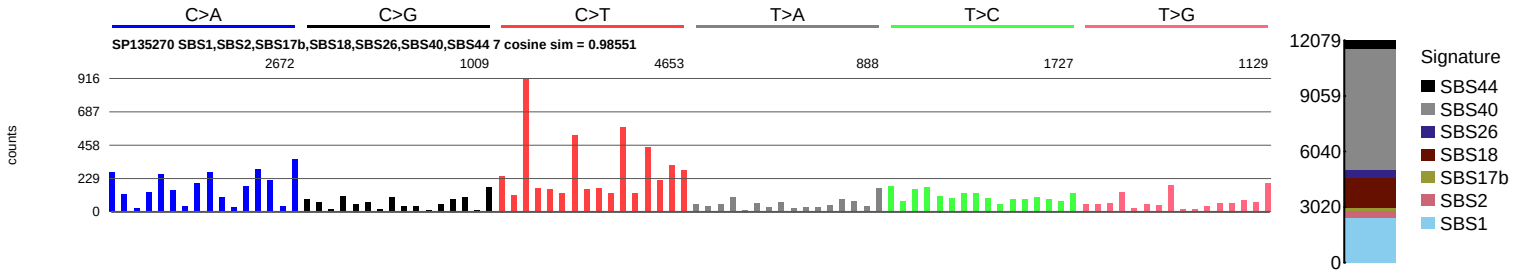
SBS3
SBS2
SBS1

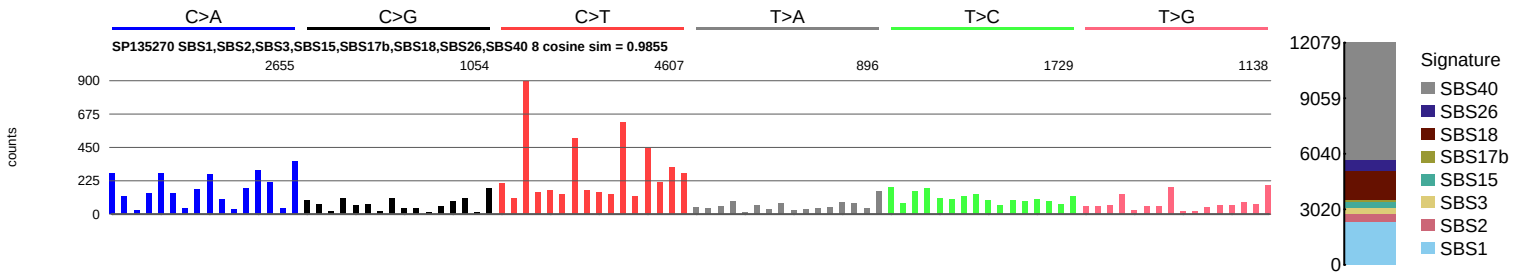

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98553
2649
1168
4527
998
1666
1072
920
690
460
230
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1

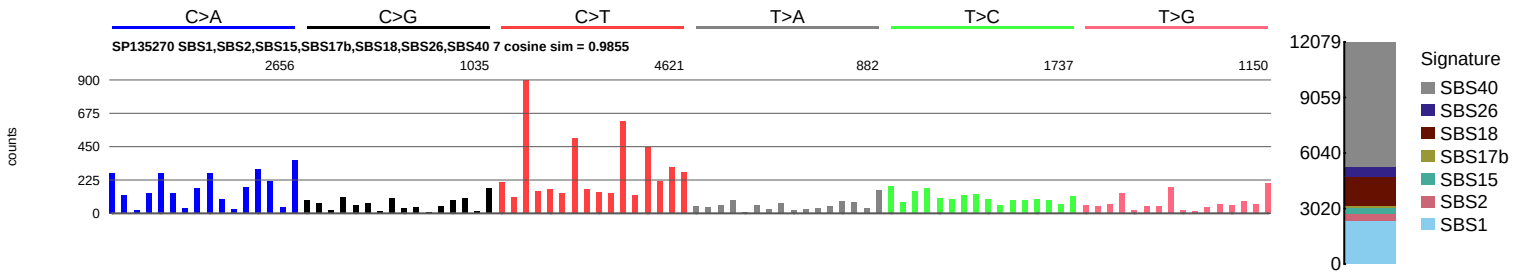

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98553
2640
1072
4620
1041
1495
1209
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

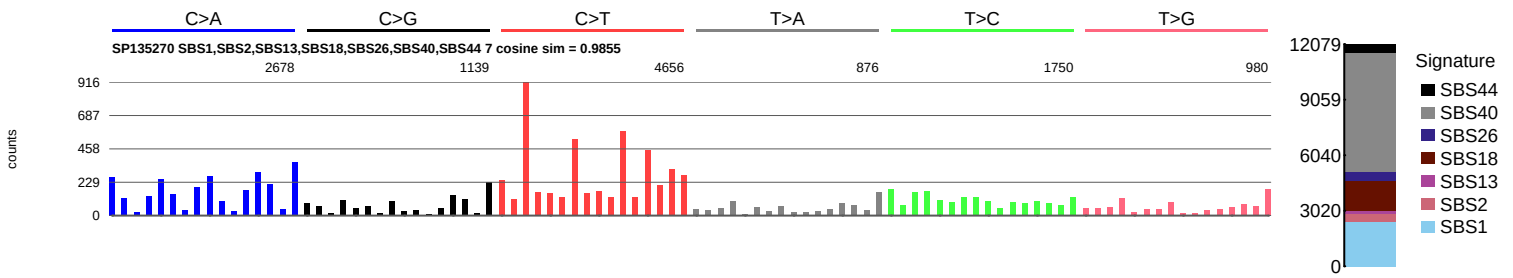

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS44 8 cosine sim = 0.98552
2588
1202
4741
869
1778
899
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

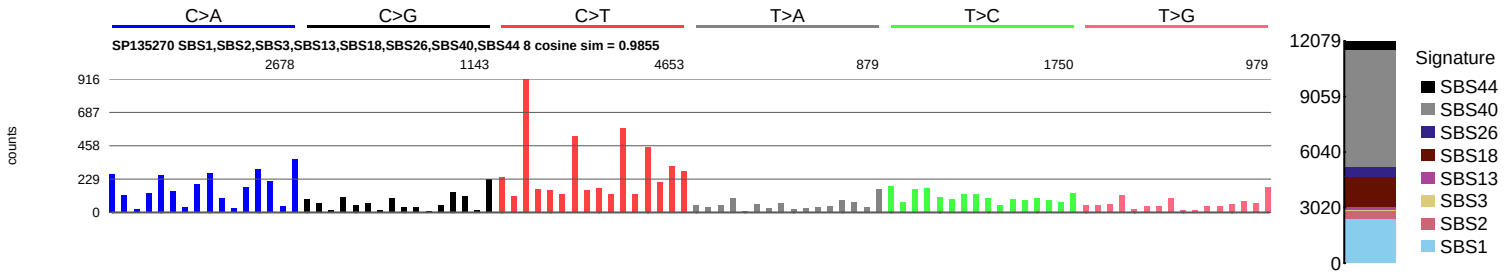

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98552
2647
1177
4522
1003
1665
1064
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS3
SBS2
SBS1

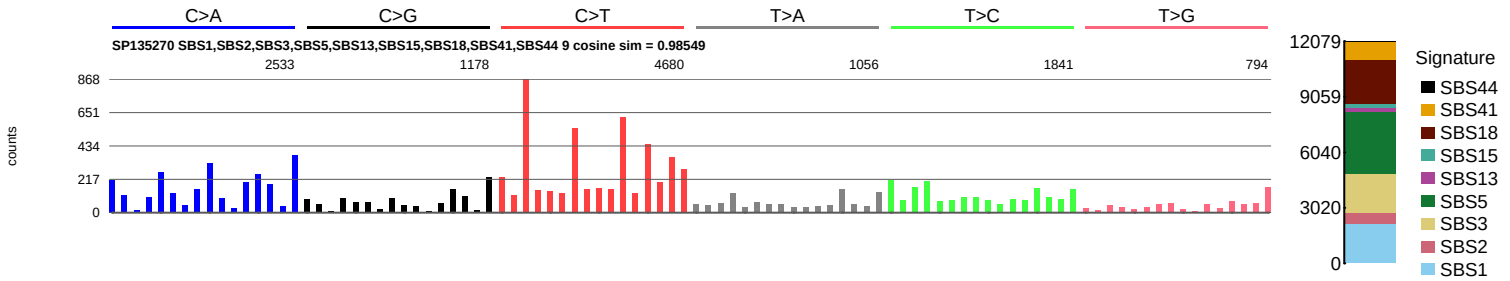

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98551
2672
1009
4653
888
1727
1129
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1

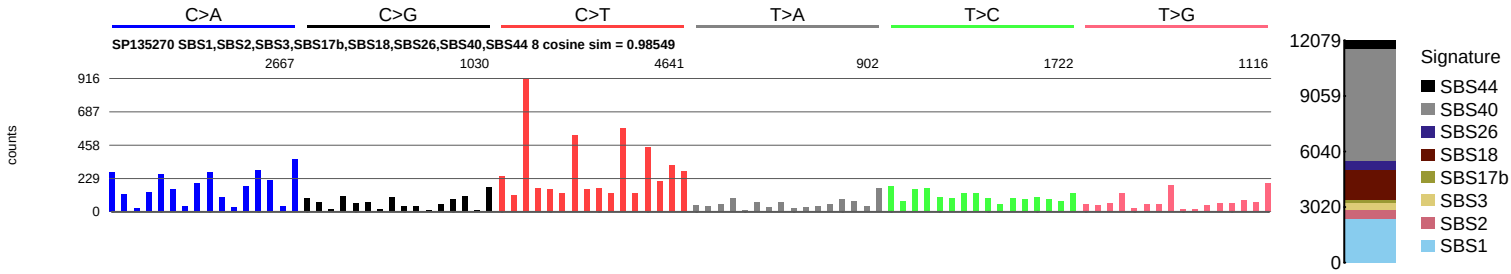

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.9855
2655
1054
4607
896
1729
1138
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

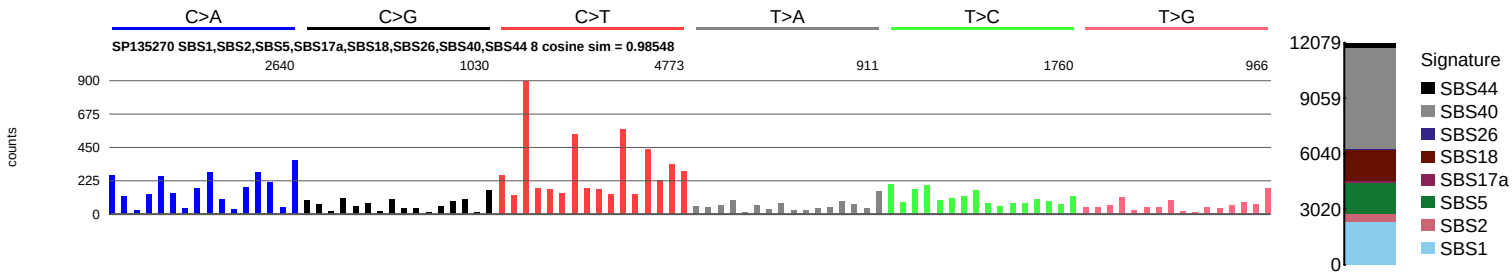

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.9855
2656
1035
4621
882
1737
1150
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1

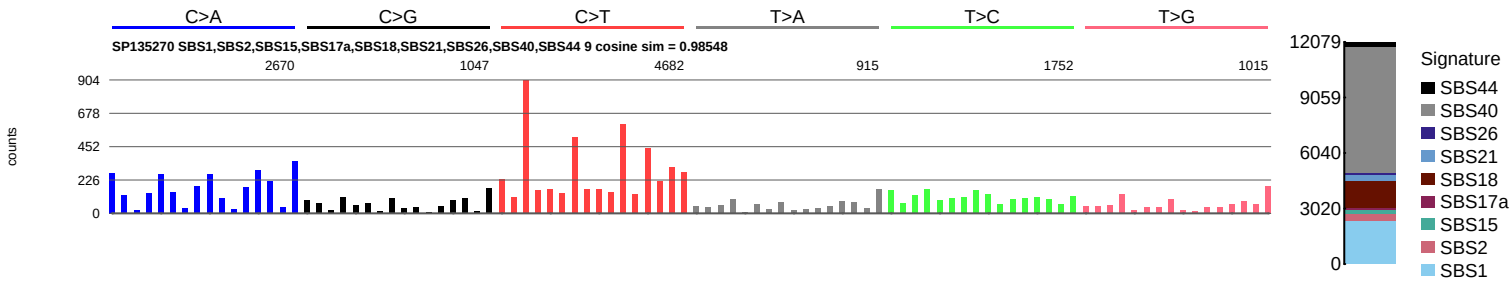

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.9855
2678
1139
4656
876
1750
980
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1

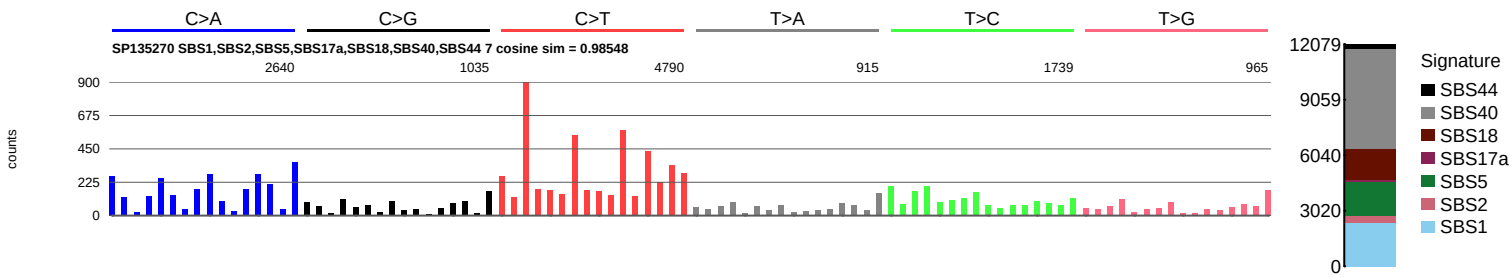

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.9855
2678
1143
4653
879
1750
979
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1

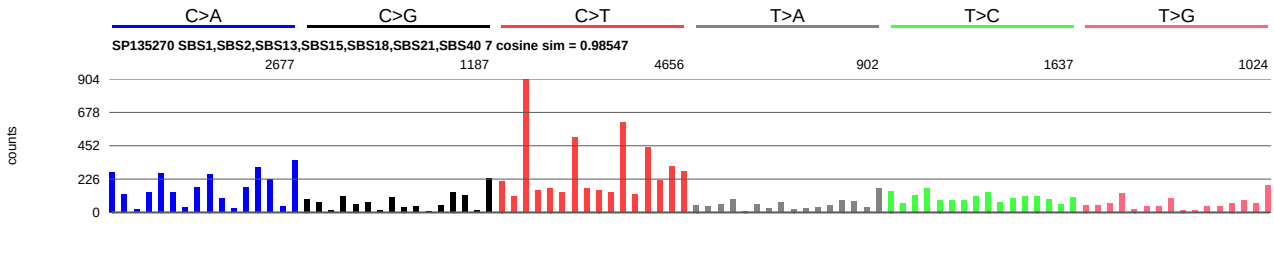

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS41,SBS44 9 cosine sim = 0.98549
2533
1178
4680
1056
1841
794
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98549
2667
1030
4641
902
1722
1116
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98548
2640
1030
4773
911
1760
966
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98548
2670
1047
4682
915
1752
1015
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.98548
2640
1035
4790
915
1739
965
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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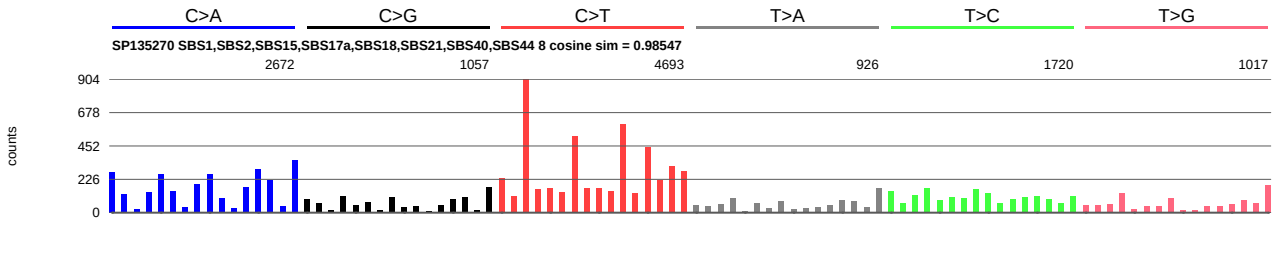
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS40 7 cosine sim = 0.98547
2677
1187
4656
902
1637
1024
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1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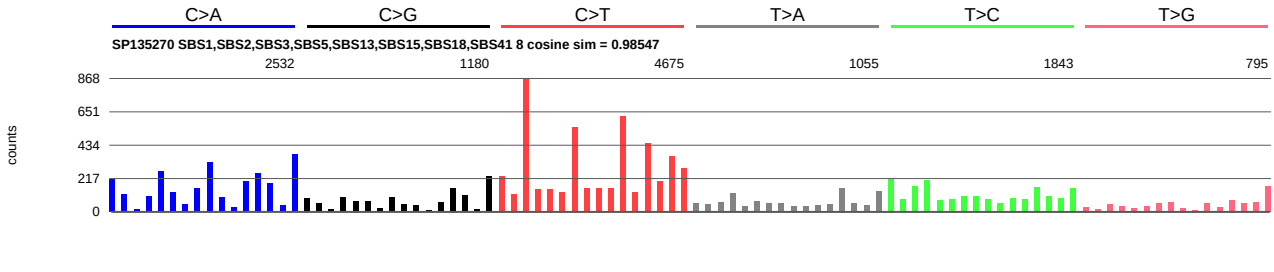
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98547
2672
1057
4693
926
1720
1017
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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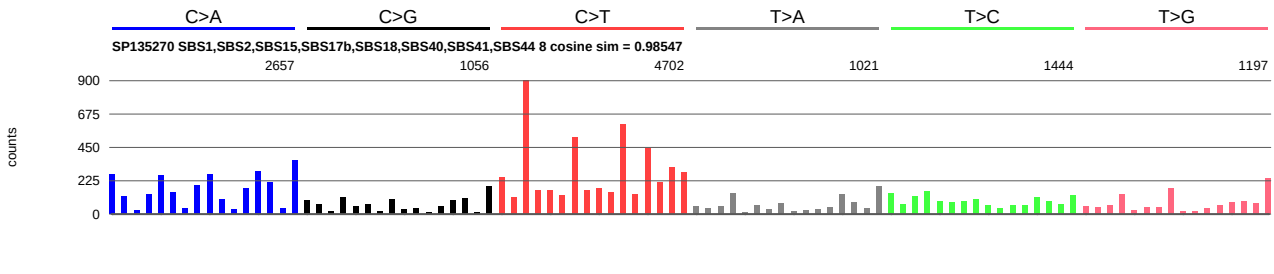
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS41 8 cosine sim = 0.98547
2532
1180
4675
1055
1843
795
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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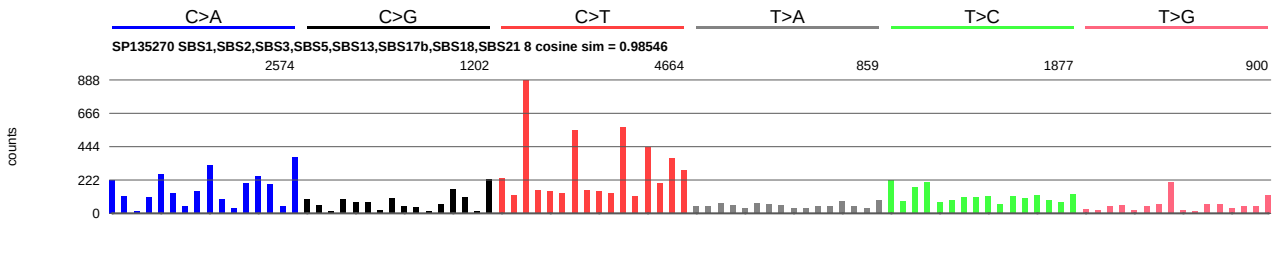
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98547
2657
1056
4702
1021
1444
1197
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1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C>G
C>T
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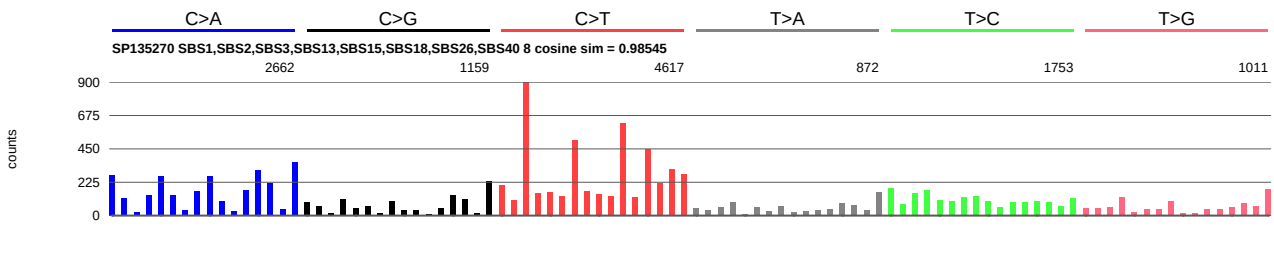
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 8 cosine sim = 0.98546
2574
1202
4664
859
1877
900
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS26,SBS40 8 cosine sim = 0.98545
2662
1159
4617
872
1753
1011
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
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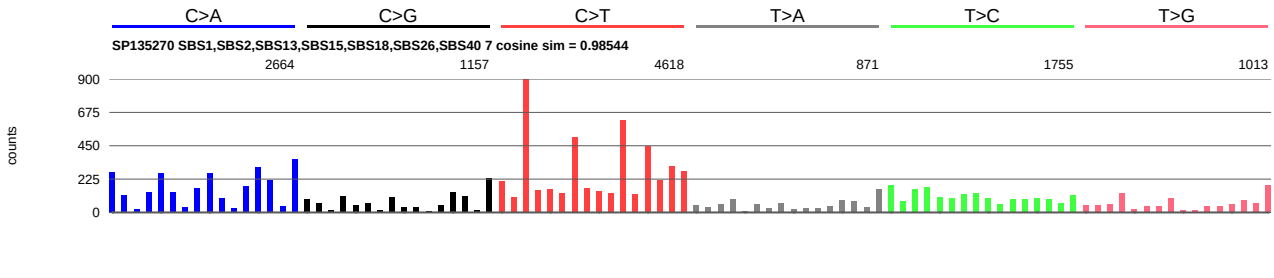
SBS2
SBS1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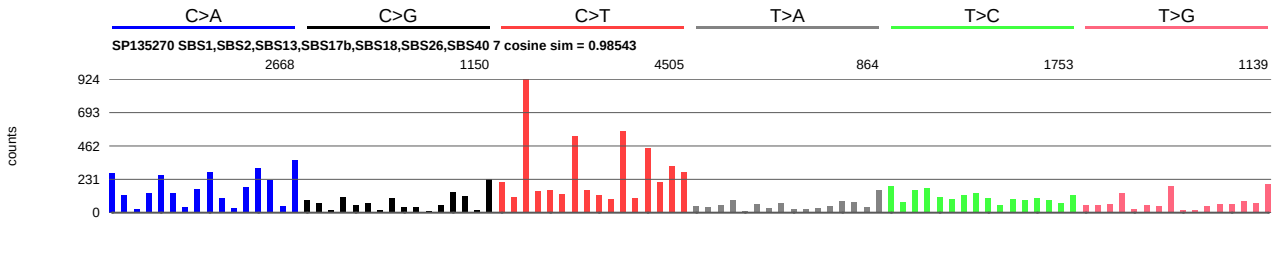
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40 7 cosine sim = 0.98544
2664
1157
4618
871
1755
1013
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
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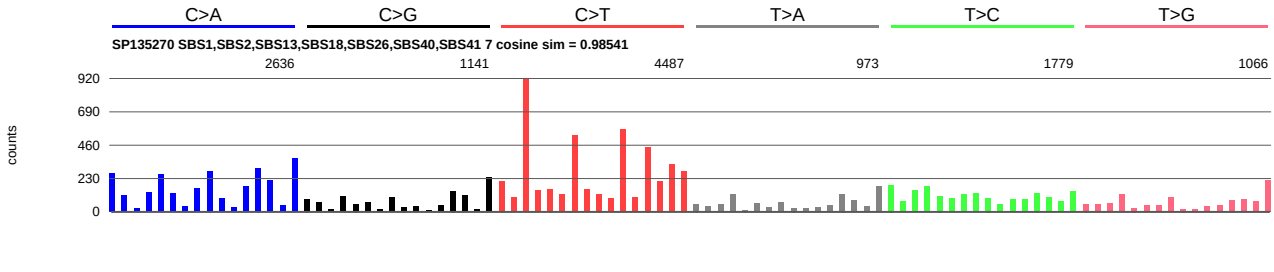
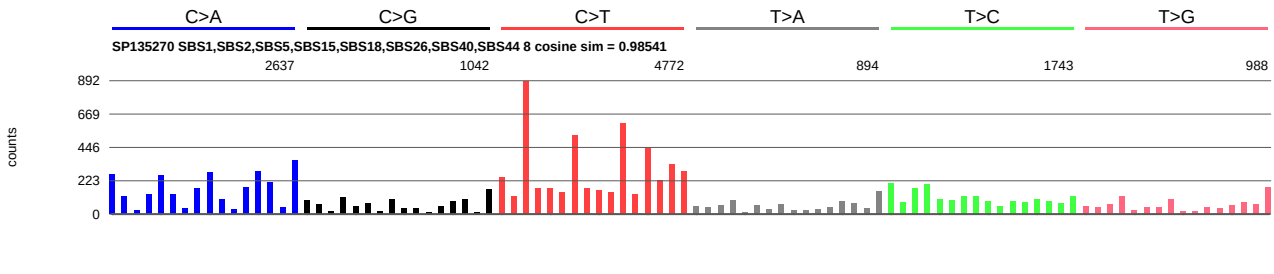
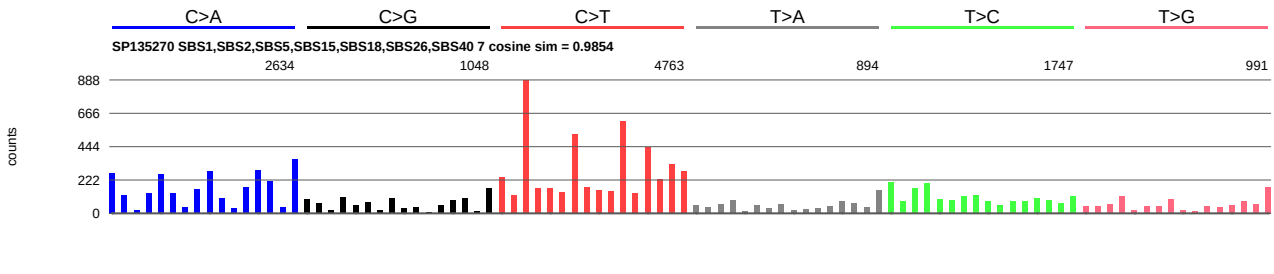
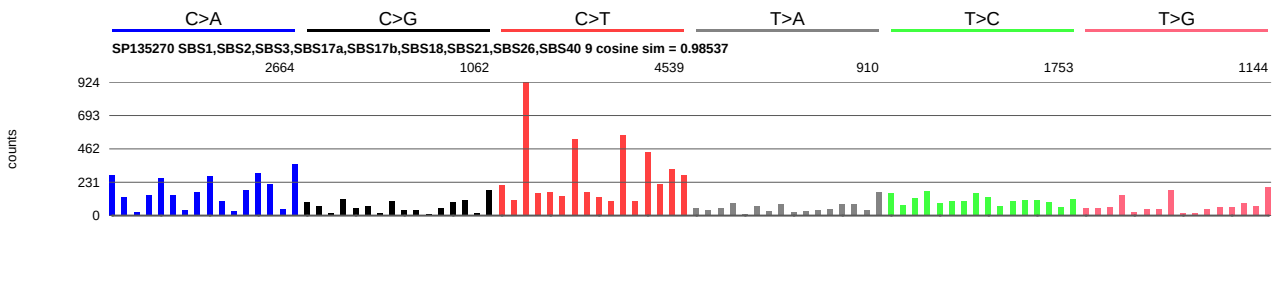
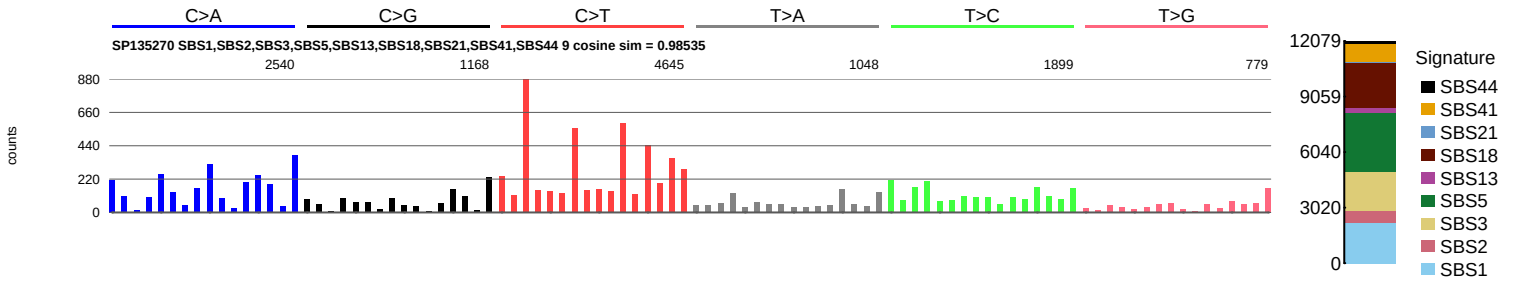
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98543
2668
1150
4505
864
1753
1139
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1

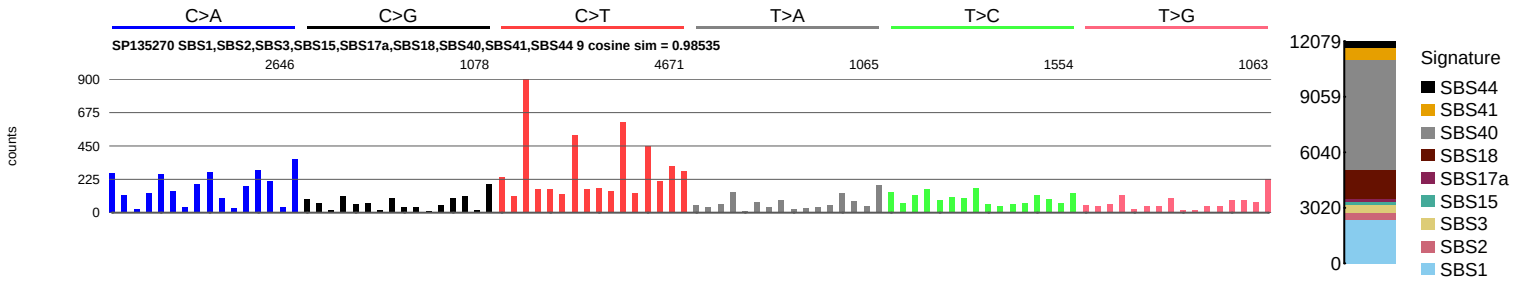

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98541
2636
1141
4487
973
1779
1066
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1

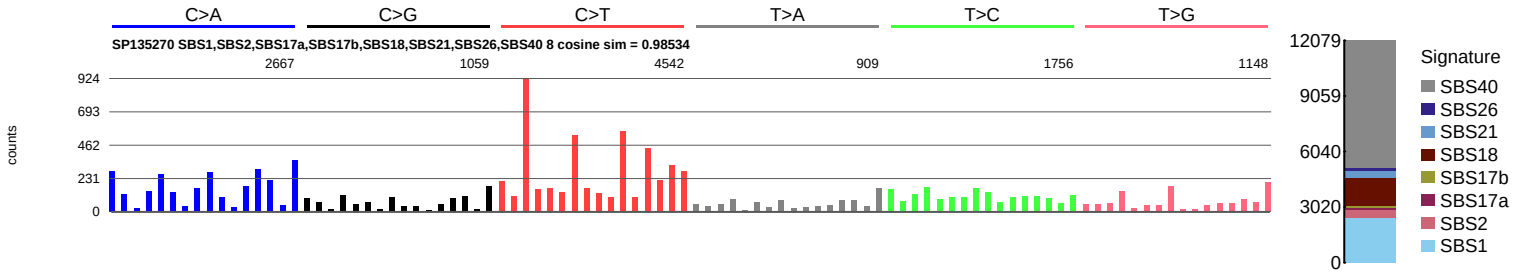

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98541
2637
1042
4772
894
1743
988
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS5
SBS2
SBS1

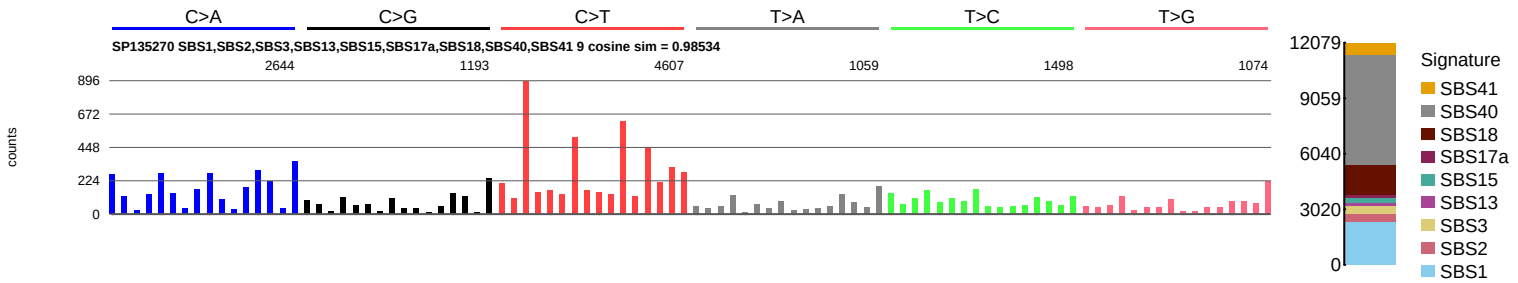

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40 7 cosine sim = 0.9854
2634
1048
4763
894
1747
991
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98537
2664
1062
4539
910
1753
1144
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98535
2540
1168
4645
1048
1899
779
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98535
2646
1078
4671
1065
1554
1063
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98534
2667
1059
4542
909
1756
1148
924
693
462
231
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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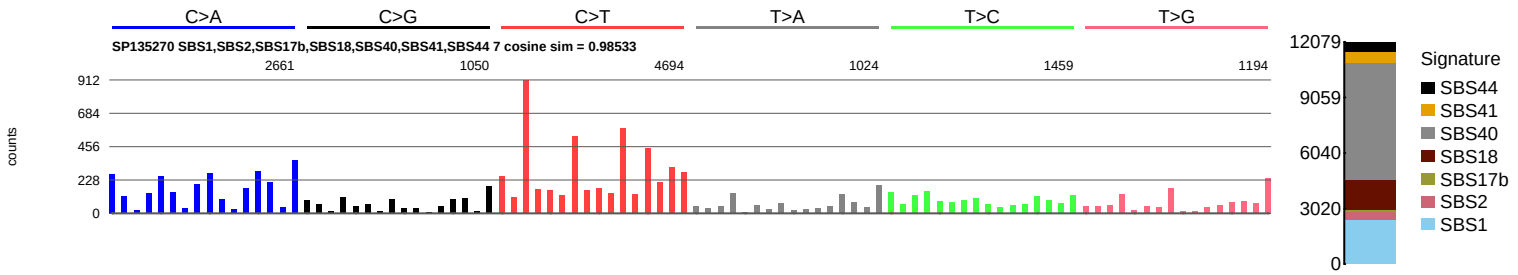
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98534
2644
1193
4607
1059
1498
1074
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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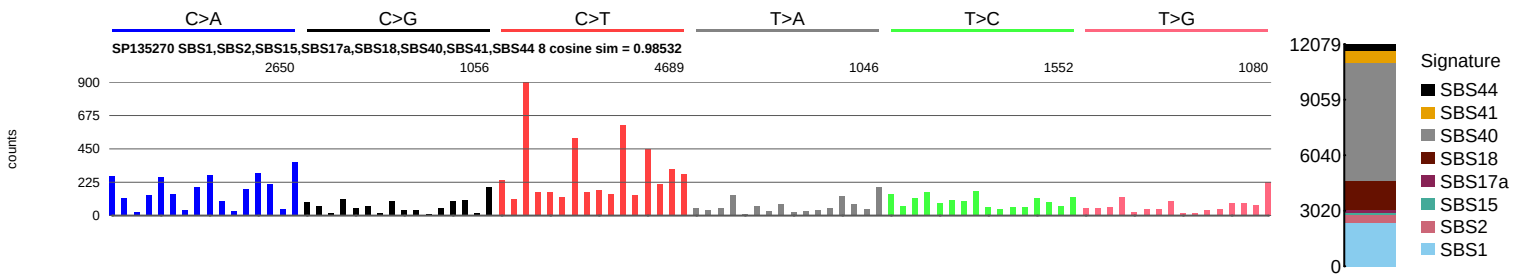
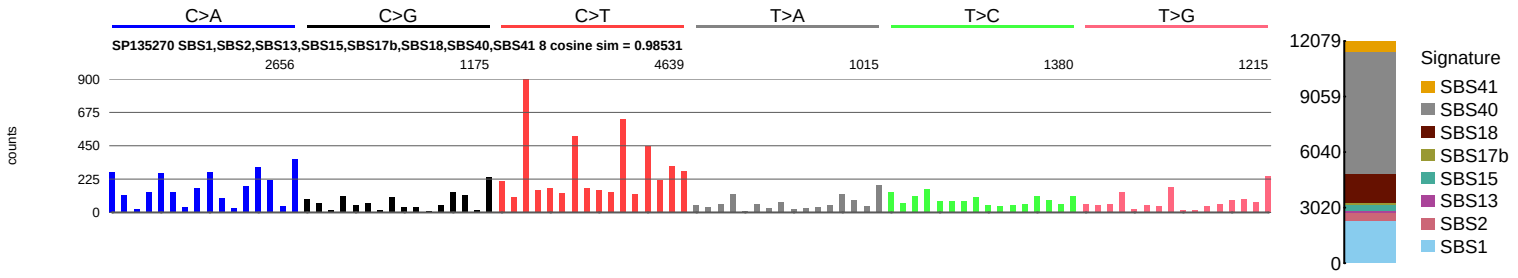
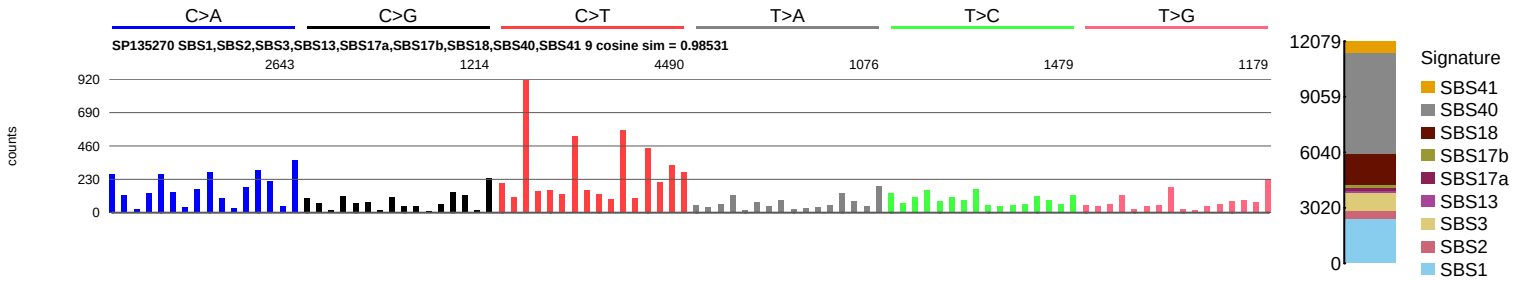
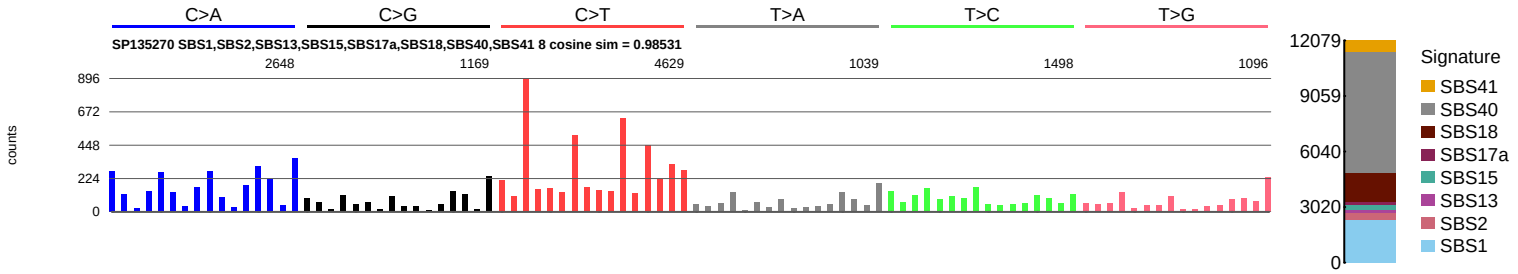
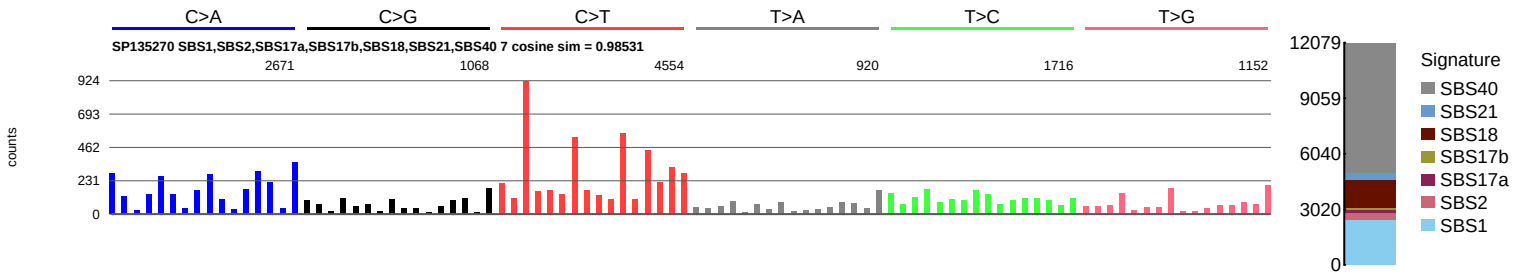
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98533
2661
1050
4694
1024
1459
1194
912
684
456
228
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS2
SBS1

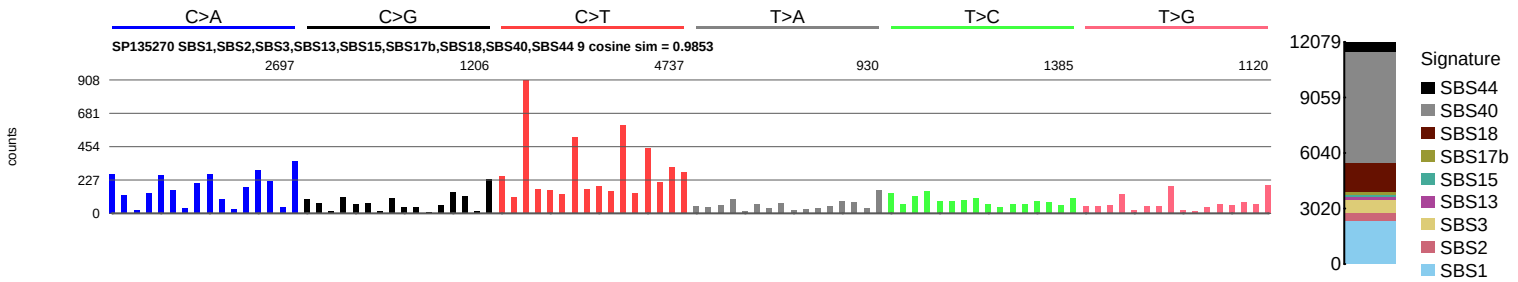

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98532
2650
1056
4689
1046
1552
1080
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1

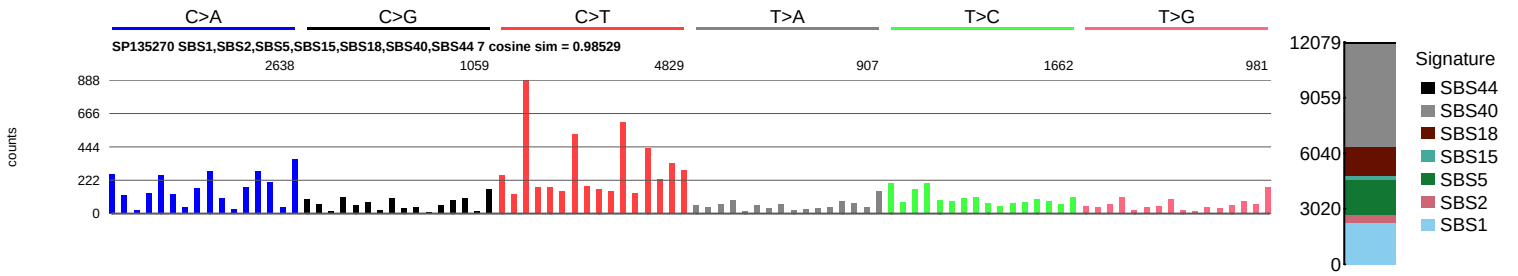

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98531
2656
1175
4639
1015
1380
1215
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

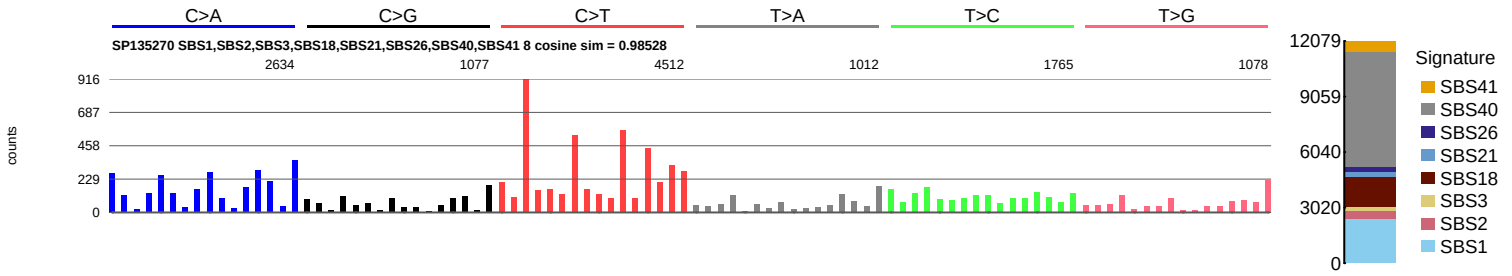

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98531
2643
1214
4490
1076
1479
1179
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

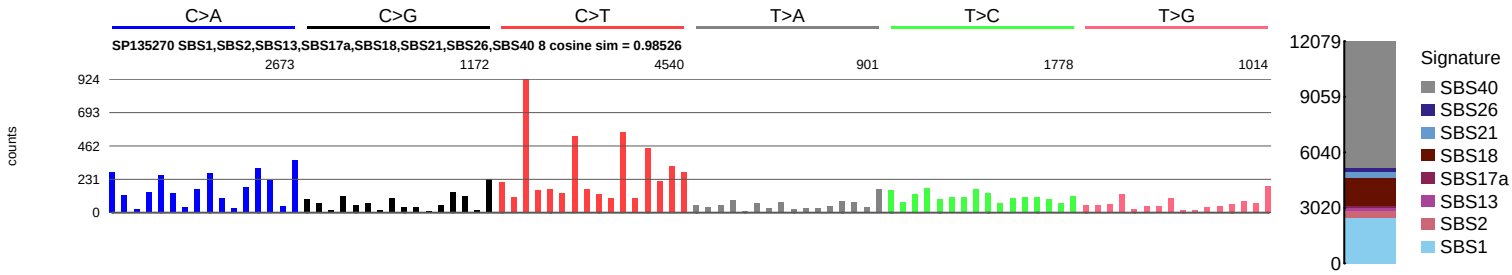

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98531
2648
1169
4629
1039
1498
1096
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98531
2671
1068
4554
920
1716
1152
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.9853
2697
1206
4737
930
1385
1120
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40,SBS44 7 cosine sim = 0.98529
2638
1059
4829
907
1662
981
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98528
2634
1077
4512
1012
1765
1078
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS3
SBS2
SBS1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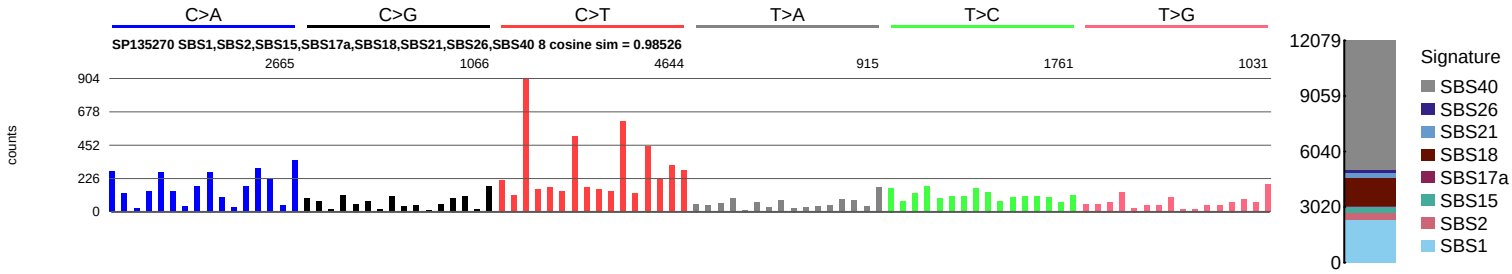
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98526
2673
1172
4540
901
1778
1014
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1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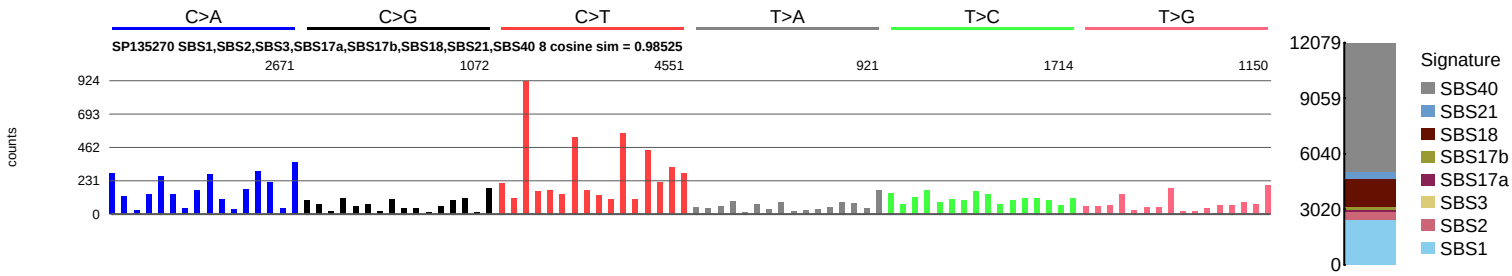
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98526
2665
1066
4644
915
1761
1031
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1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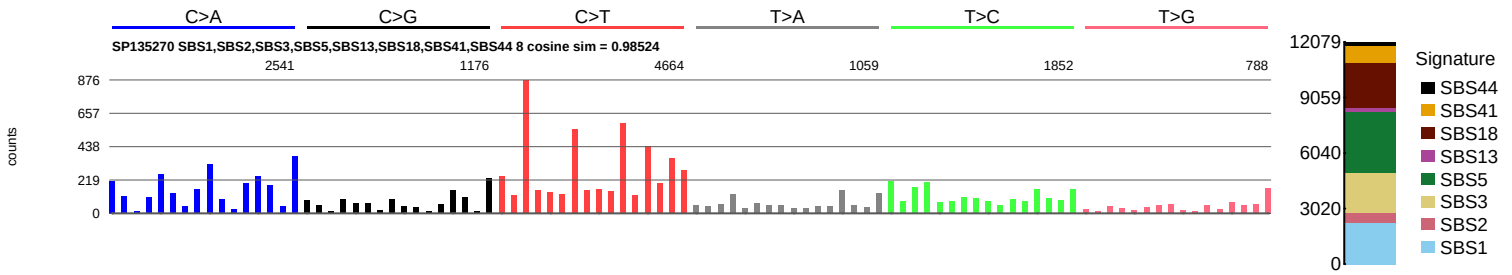
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98525
2671
1072
4551
921
1714
1150
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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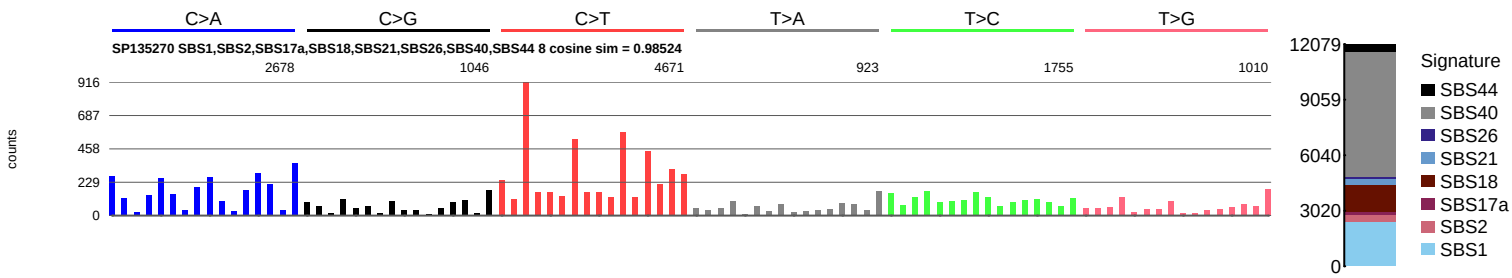
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS41,SBS44 8 cosine sim = 0.98524
2541
1176
4664
1059
1852
788
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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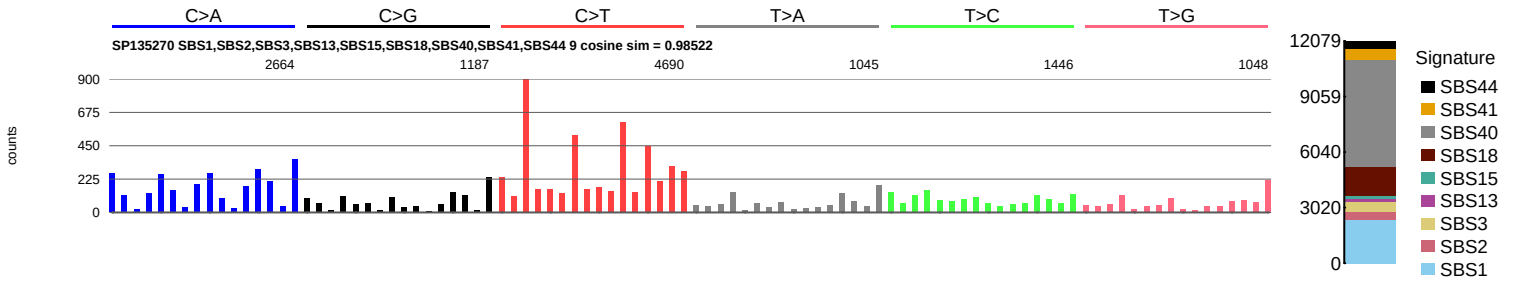
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98524
2678
1046
4671
923
1755
1010
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS2
SBS1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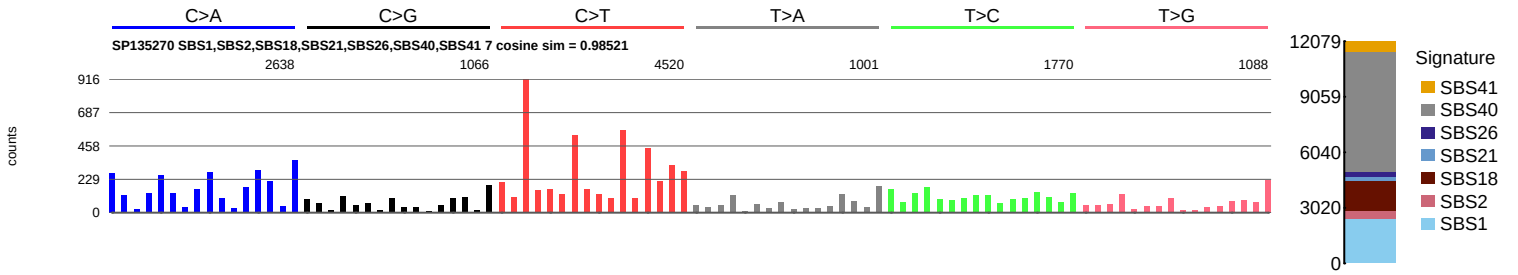
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98522
2664
1187
4690
1045
1446
1048
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
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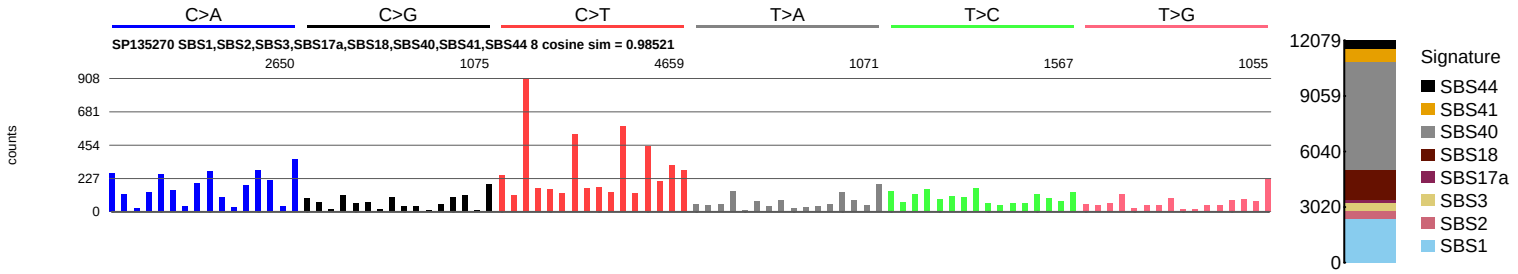
T>G
SP135270 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98521
2638
1066
4520
1001
1770
1088
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1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T>A
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T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98521
2650
1075
4659
1071
1567
1055
counts
908
681
454
227
0
12079
9059
6040
3020
0
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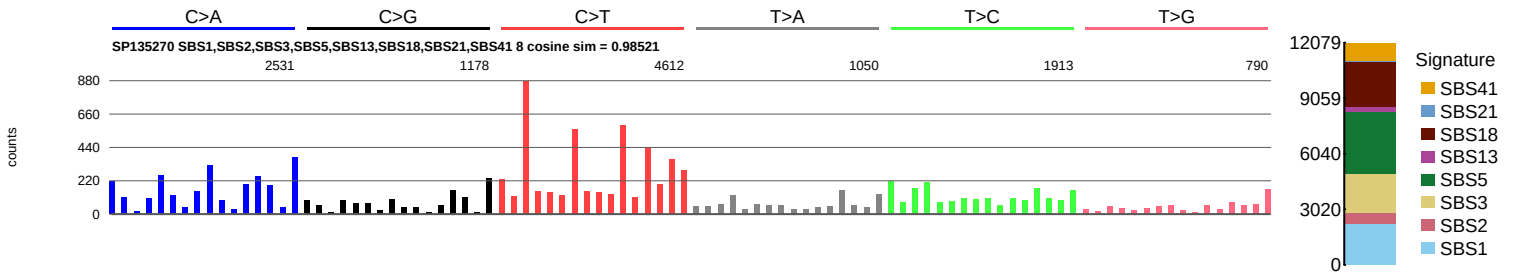
SBS2
SBS1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C>T
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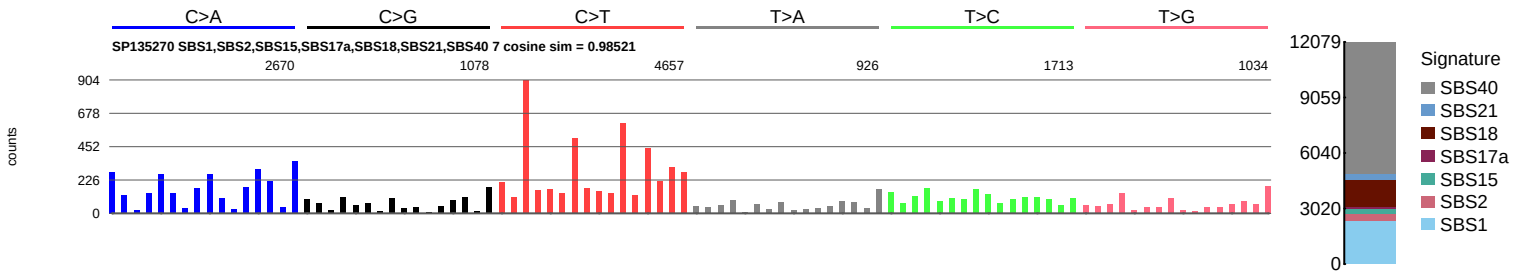
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS41 8 cosine sim = 0.98521
2531
1178
4612
1050
1913
790
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1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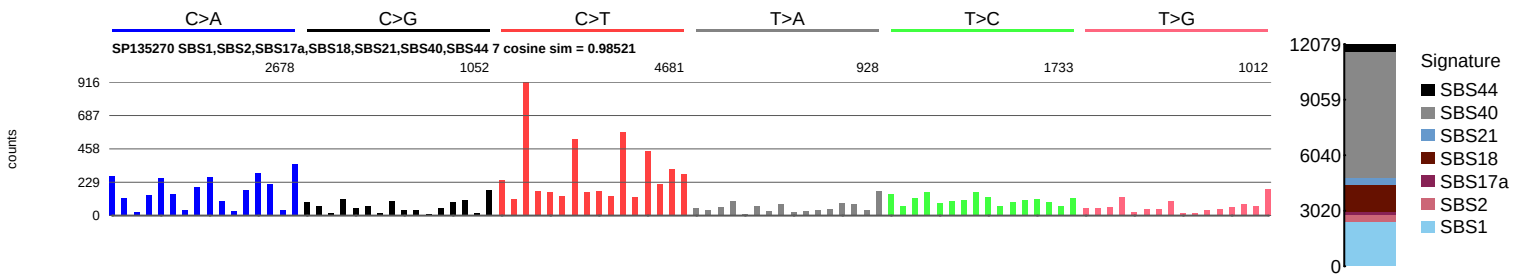
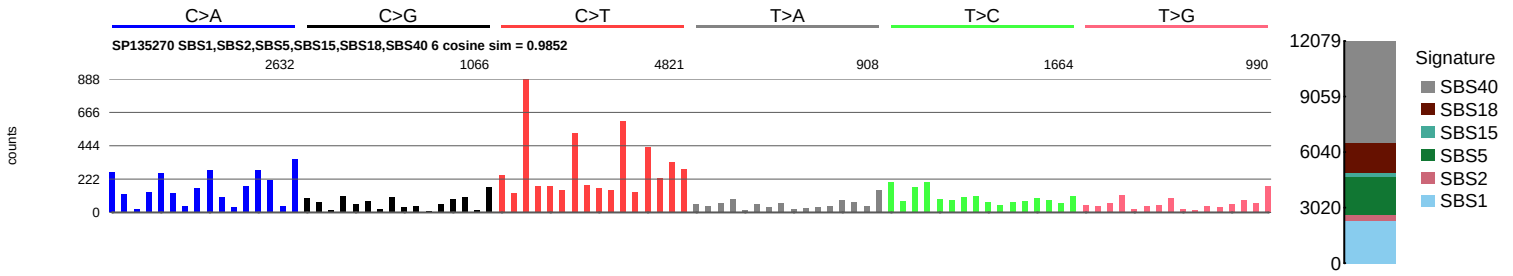
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98521
2670
1078
4657
926
1713
1034
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98521
2678
1052
4681
928
1733
1012
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
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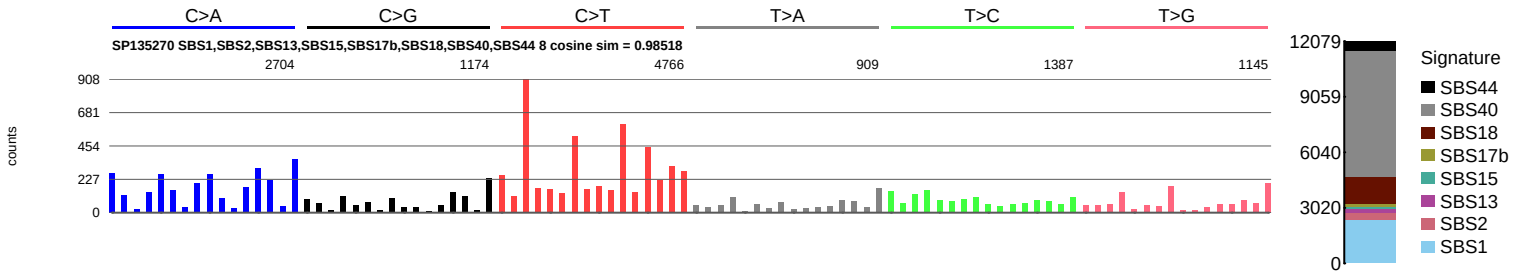
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40 6 cosine sim = 0.9852
2632
1066
4821
908
1664
990
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98518
2704
1174
4766
909
1387
1145
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
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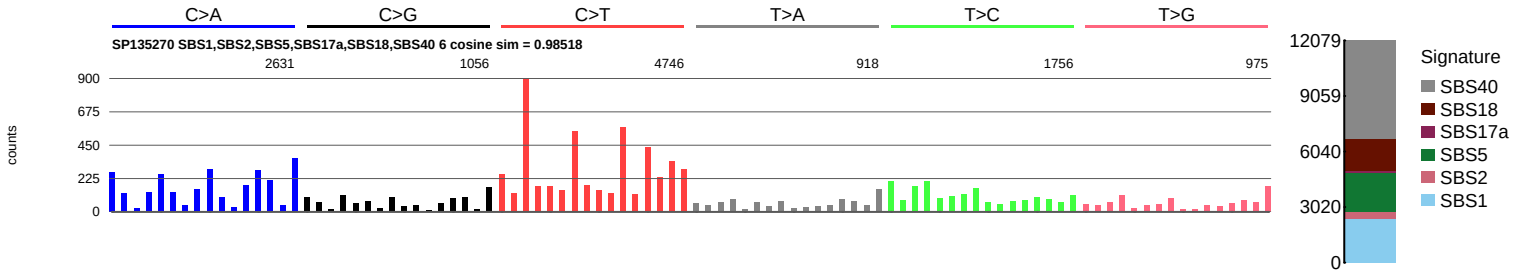
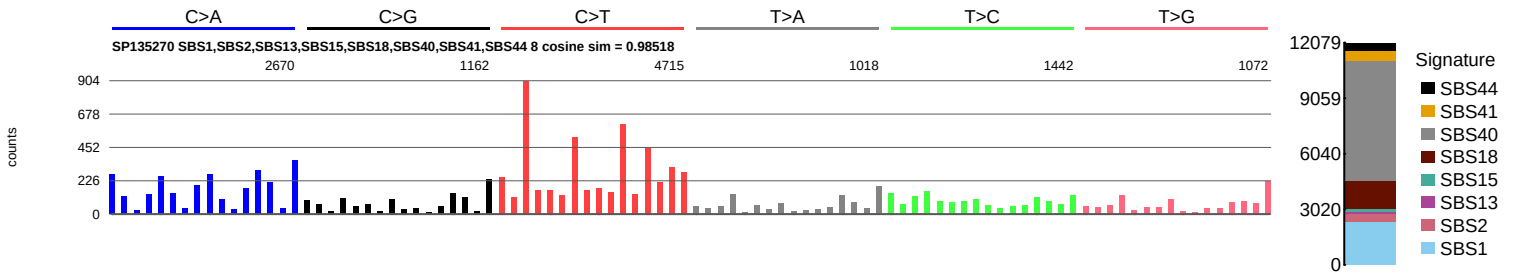
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40 6 cosine sim = 0.98518
2631
1056
4746
918
1756
975
900
675
450
225
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1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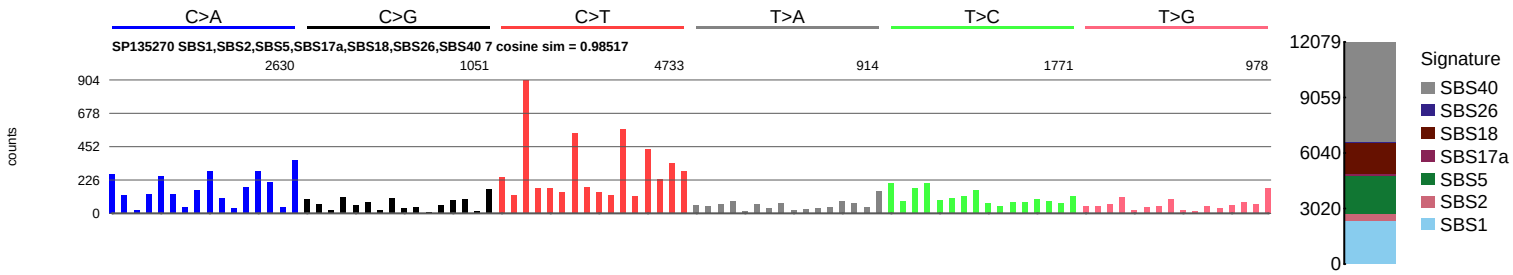
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98518
2670
1162
4715
1018
1442
1072
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS13
SBS2
SBS1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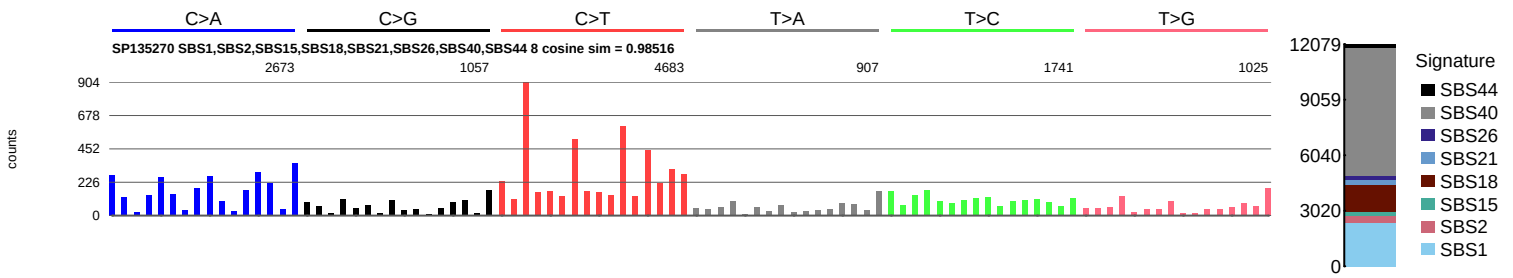
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98517
2630
1051
4733
914
1771
978
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS2
SBS1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C>G
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T>C
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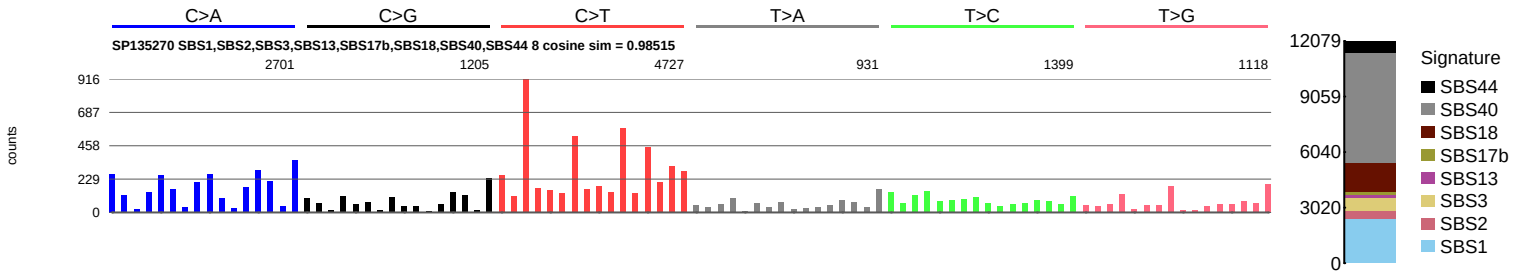
SP135270 SBS1,SBS2,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98516
2673
1057
4683
907
1741
1025
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS15
SBS2
SBS1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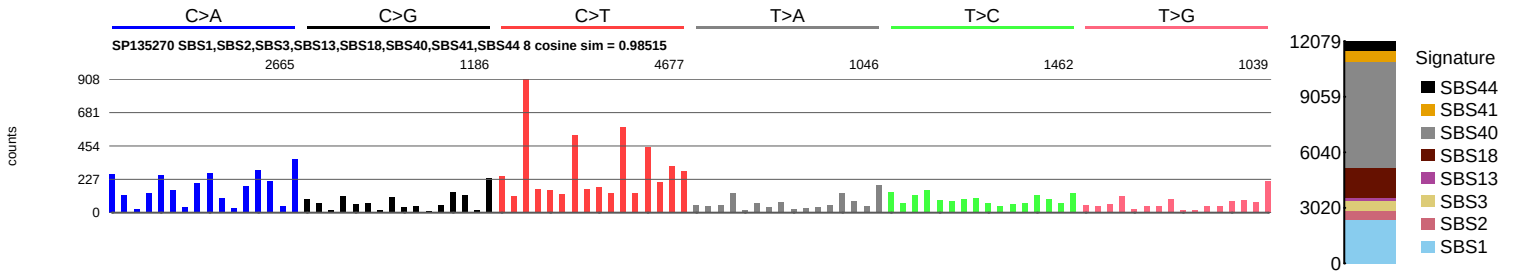
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98515
2701
1205
4727
931
1399
1118
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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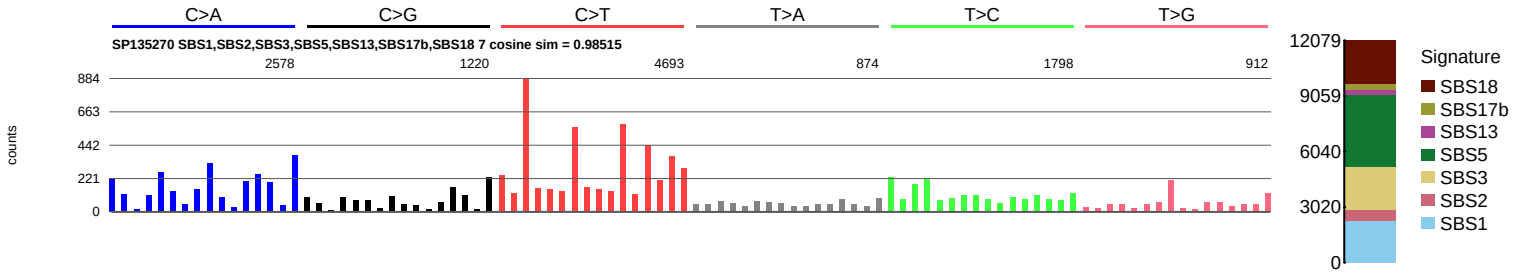
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98515
2665
1186
4677
1046
1462
1039
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18 7 cosine sim = 0.98515
2578
1220
4693
874
1798
912
884
663
442
221
0
counts
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS13
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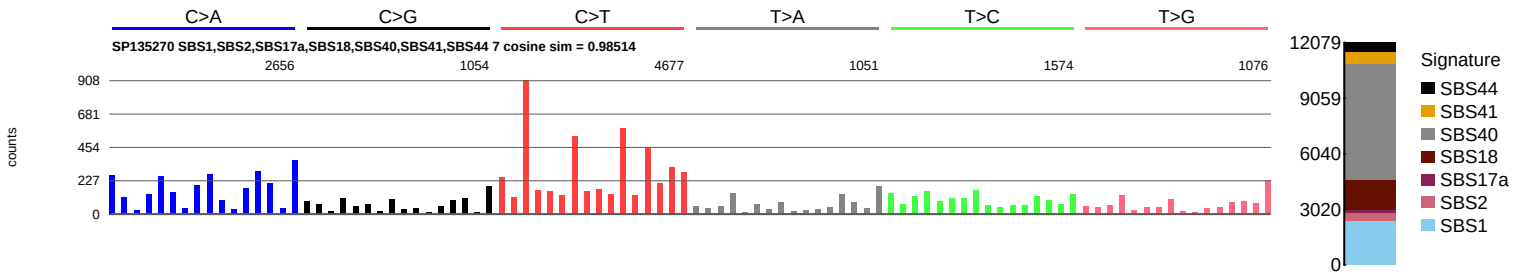
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98514
2656
1054
4677
1051
1574
1076
908
681
454
227
0
counts
12079
9059
6040
3020
0
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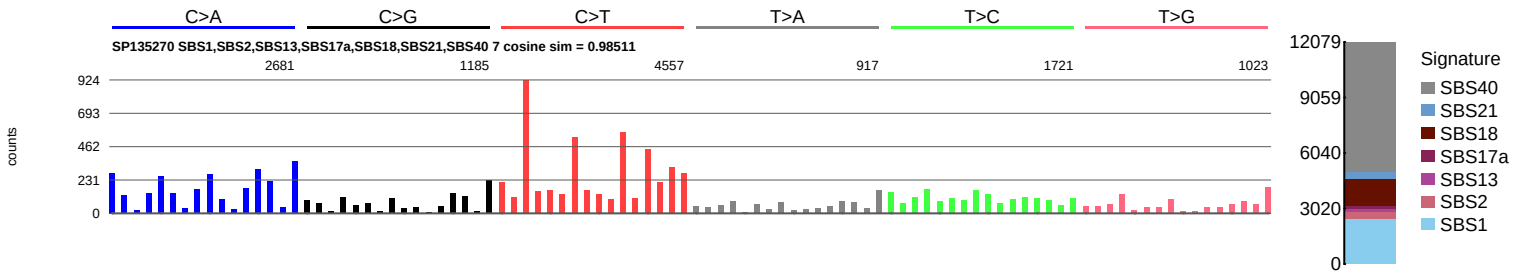
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98511
2681
1185
4557
917
1721
1023
924
693
462
231
0
counts
12079
9059
6040
3020
0
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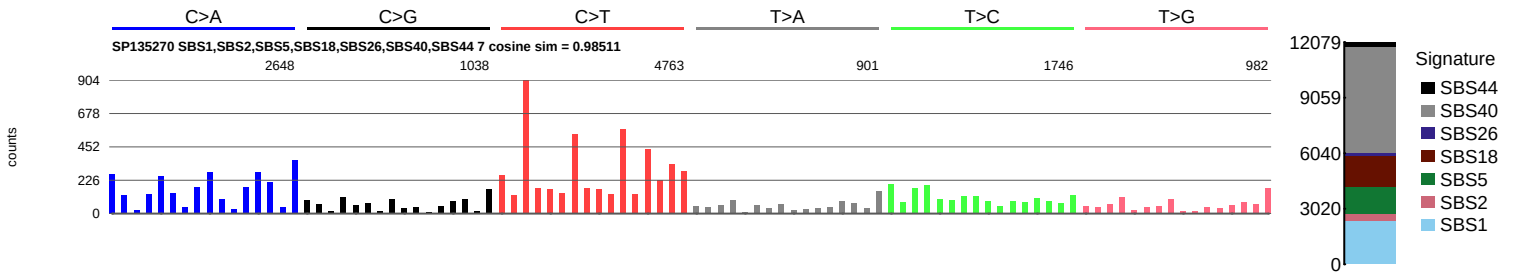
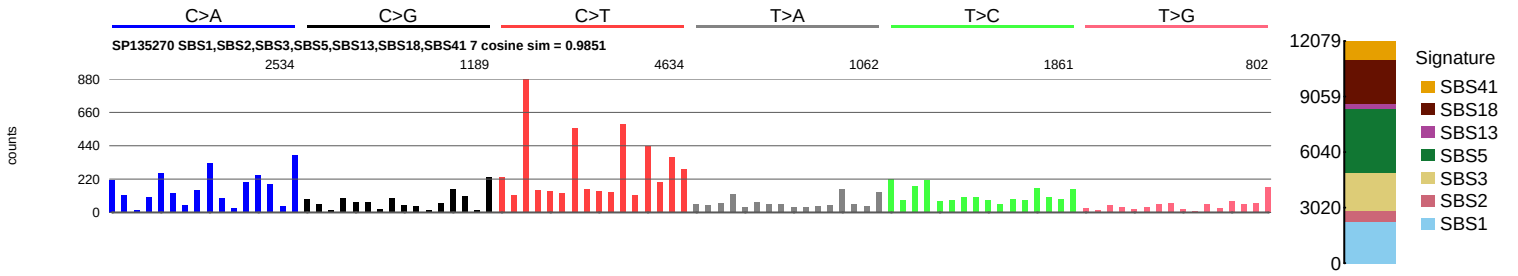
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98511
2648
1038
4763
901
1746
982
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS41 7 cosine sim = 0.9851
2534
1189
4634
1062
1861
802
880
660
440
220
0
counts
12079
9059
6040
3020
0
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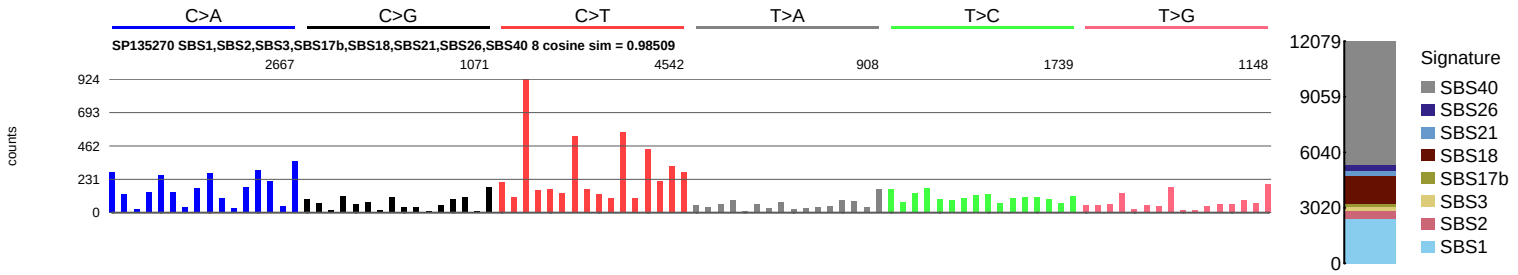
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98509
2667
1071
4542
908
1739
1148
924
693
462
231
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
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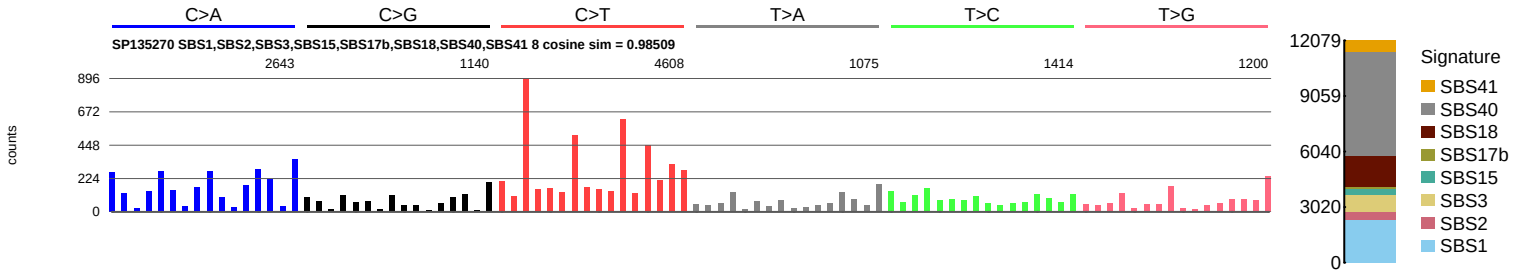
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98509
2643
1140
4608
1075
1414
1200
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
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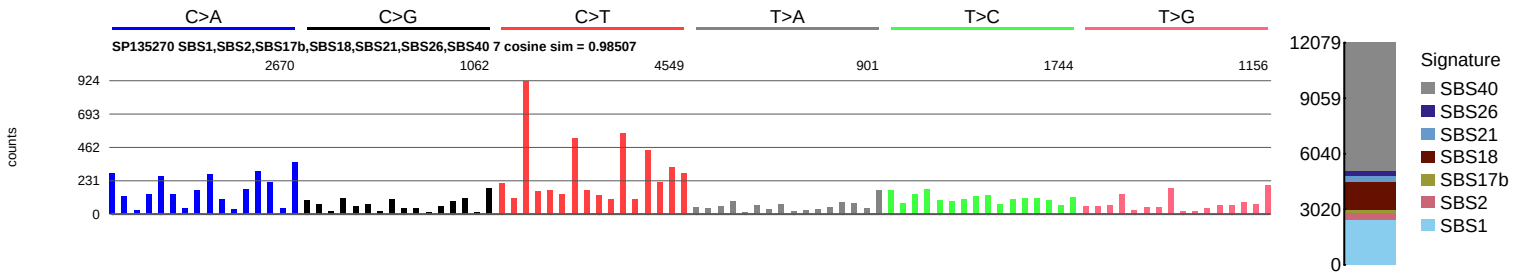
SBS2
SBS1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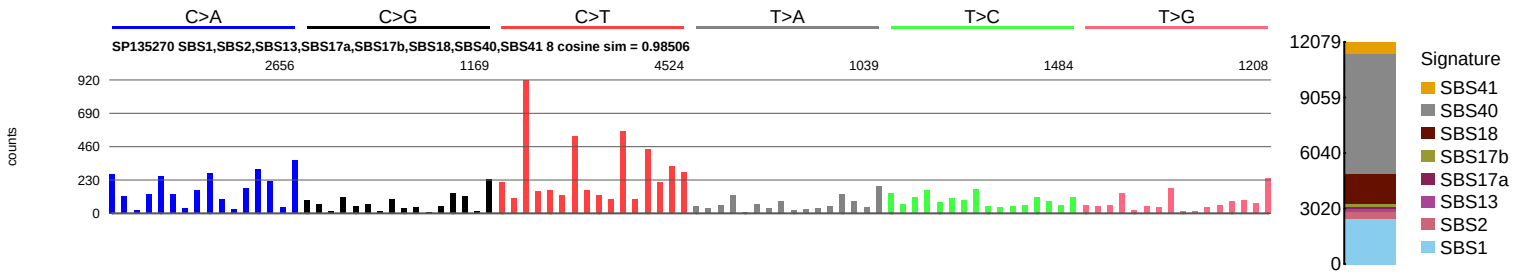
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98507
2670
1062
4549
901
1744
1156
924
693
462
231
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1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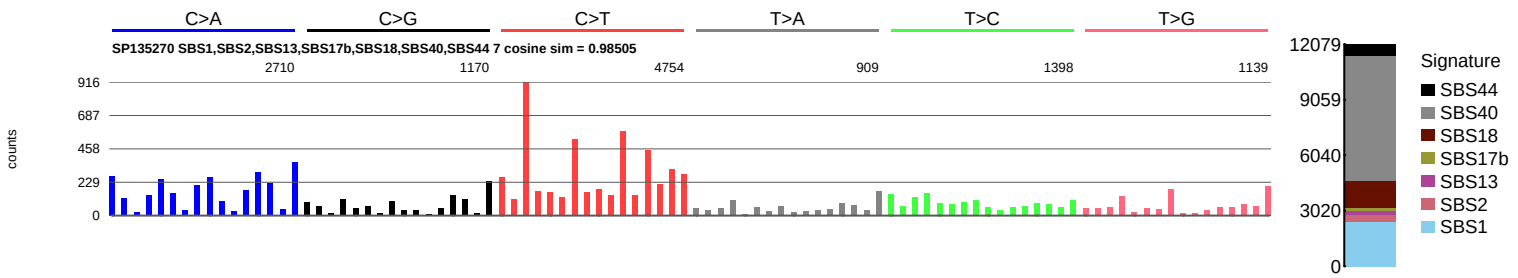
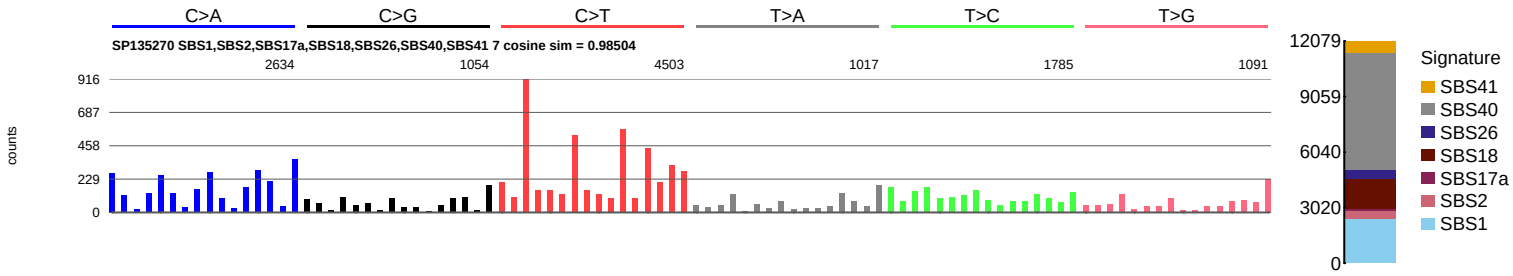
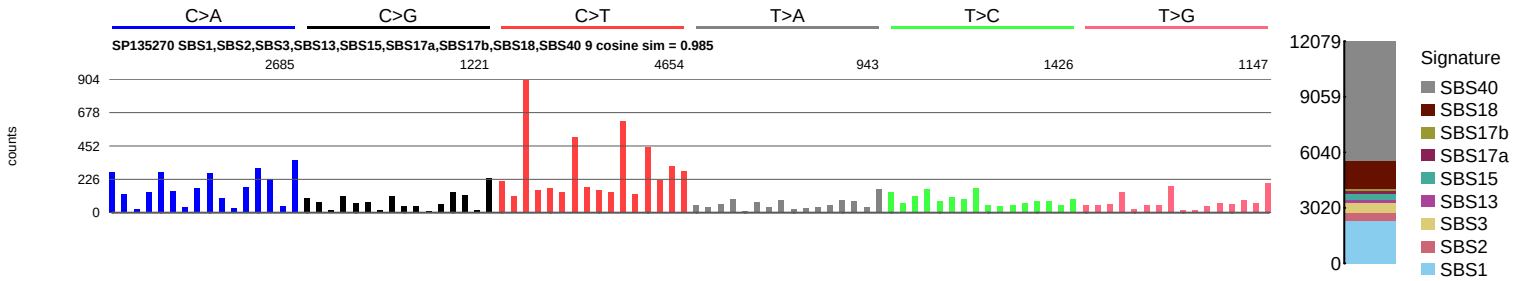
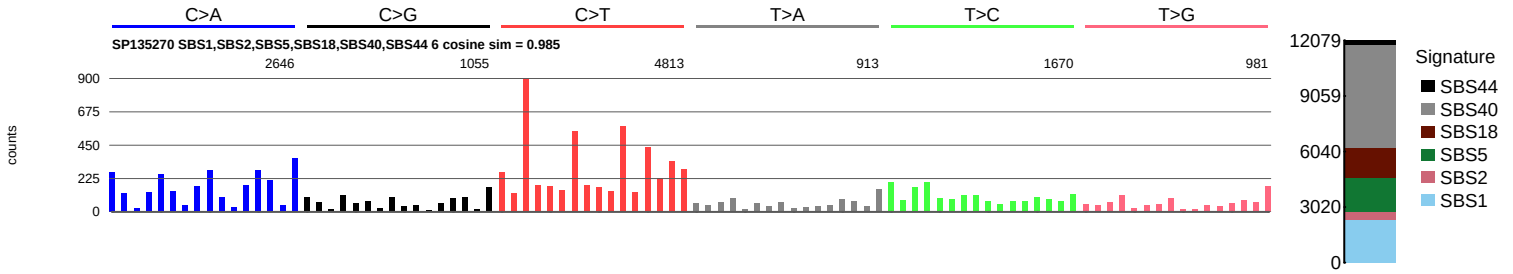
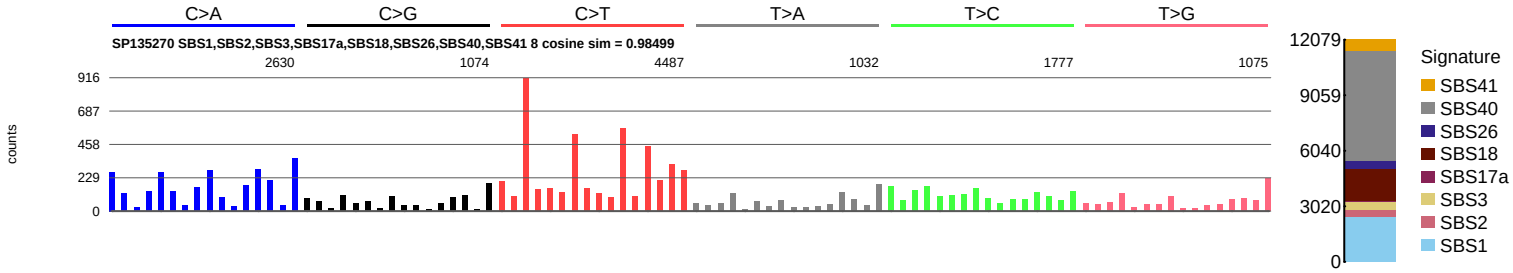
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98506
2656
1169
4524
1039
1484
1208
920
690
460
230
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

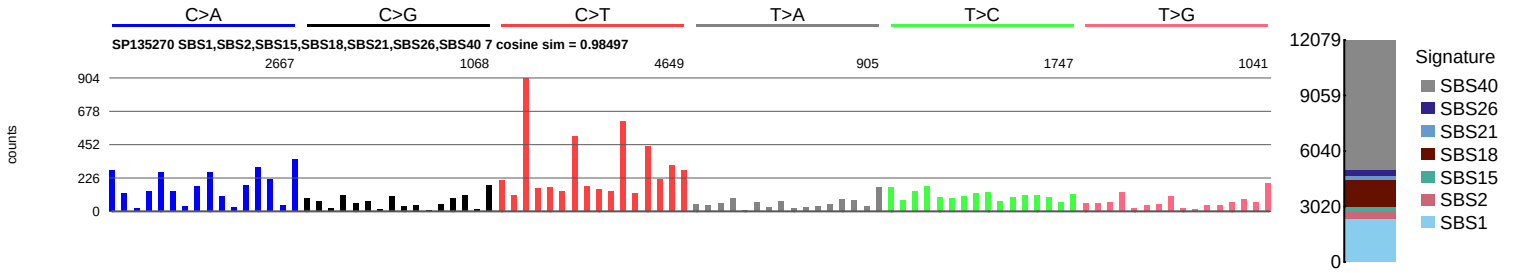

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98505
2710
1170
4754
909
1398
1139
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1

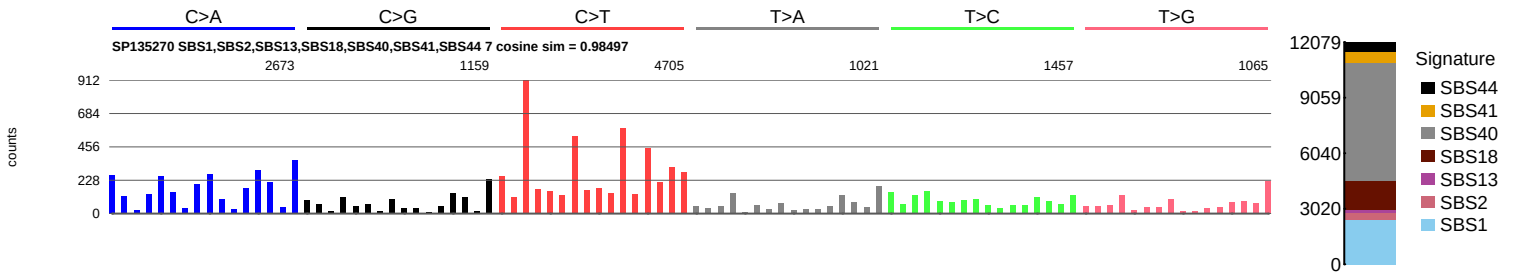

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98504
2634
1054
4503
1017
1785
1091
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1

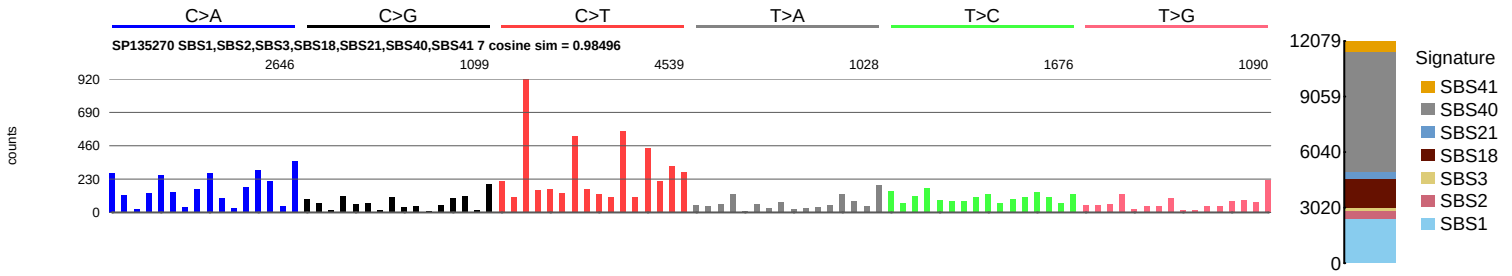

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.985
2685
1221
4654
943
1426
1147
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

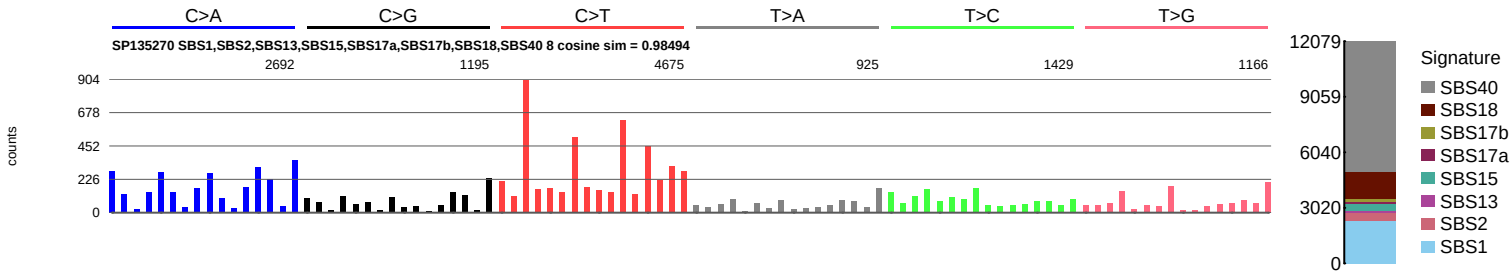

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS40,SBS44 6 cosine sim = 0.985
2646
1055
4813
913
1670
981
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS5
SBS2
SBS1

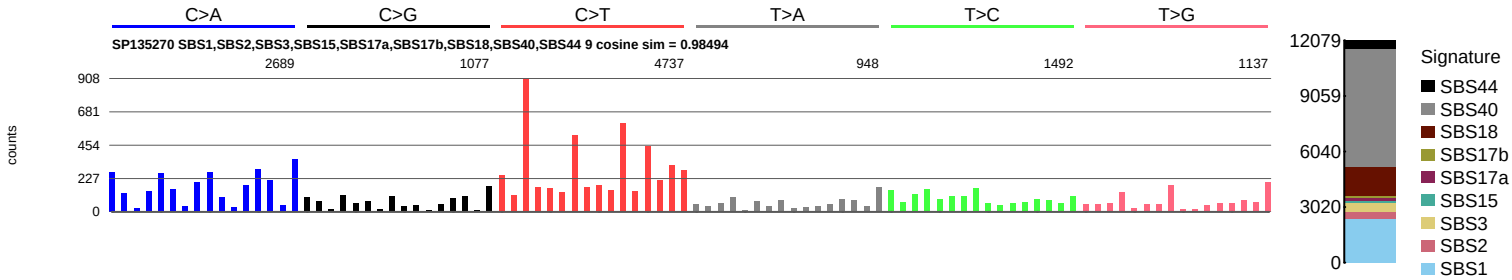

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98499
2630
1074
4487
1032
1777
1075
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS3
SBS2
SBS1

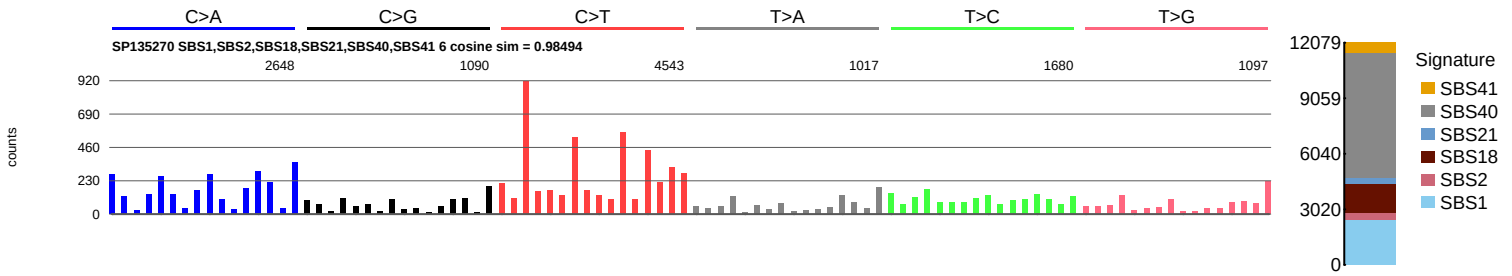

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98497
2667
1068
4649
905
1747
1041
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS2
SBS1

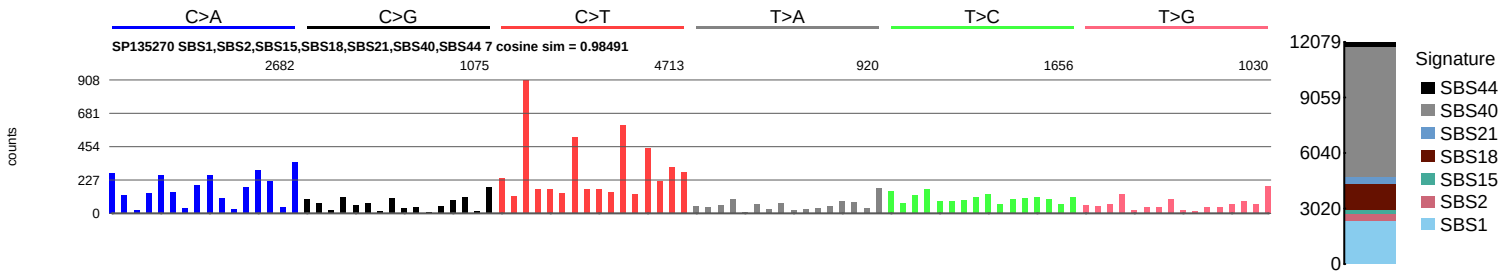

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98497
2673
1159
4705
1021
1457
1065
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS2
SBS1

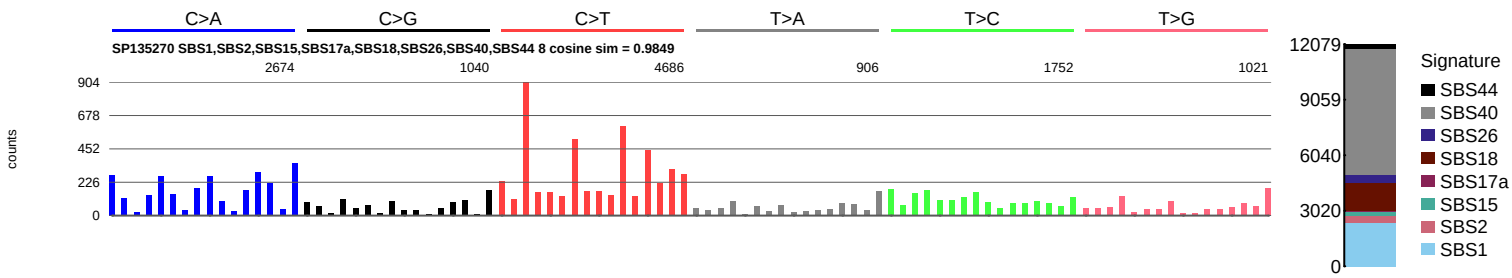

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98496
2646
1099
4539
1028
1676
1090
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS3
SBS2
SBS1

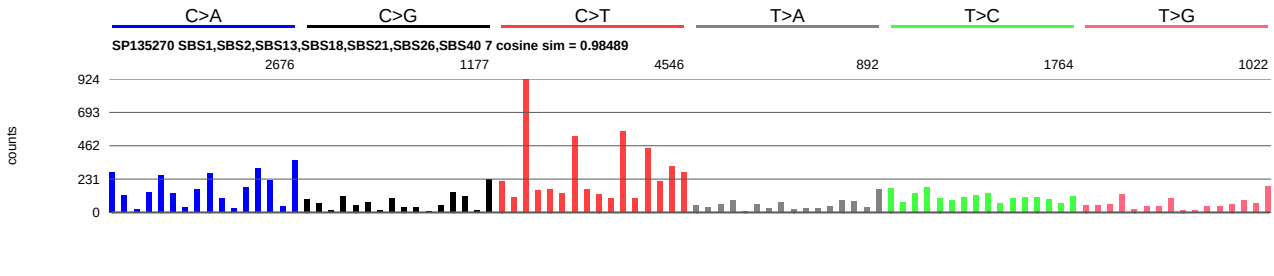

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98494
2692
1195
4675
925
1429
1166
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

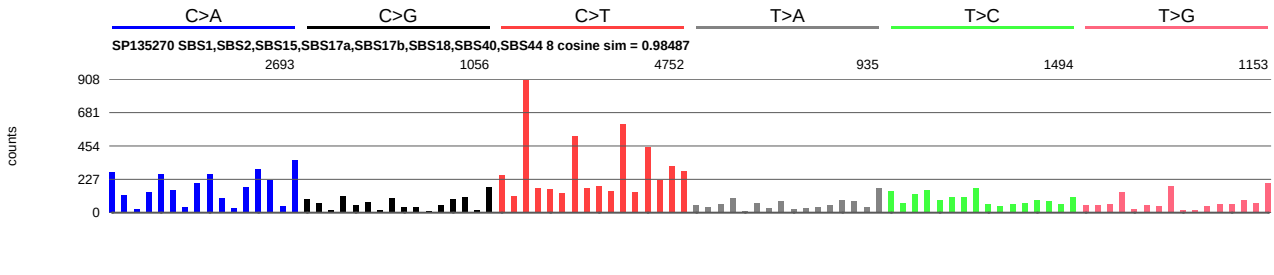

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98494
2689
1077
4737
948
1492
1137
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

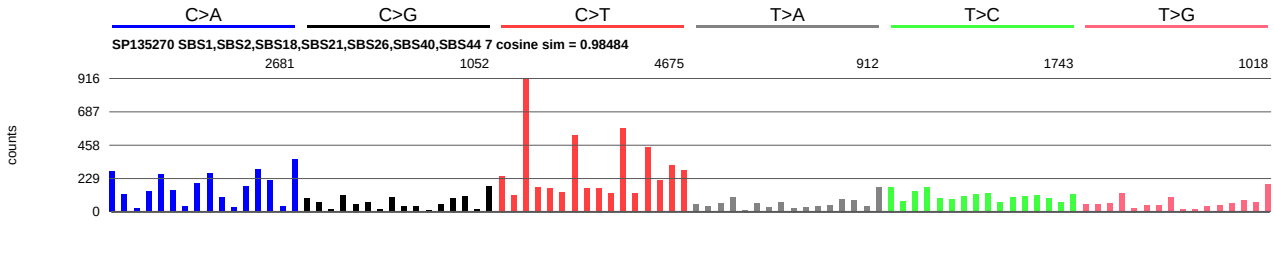

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.98494
2648
1090
4543
1017
1680
1097
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS2
SBS1

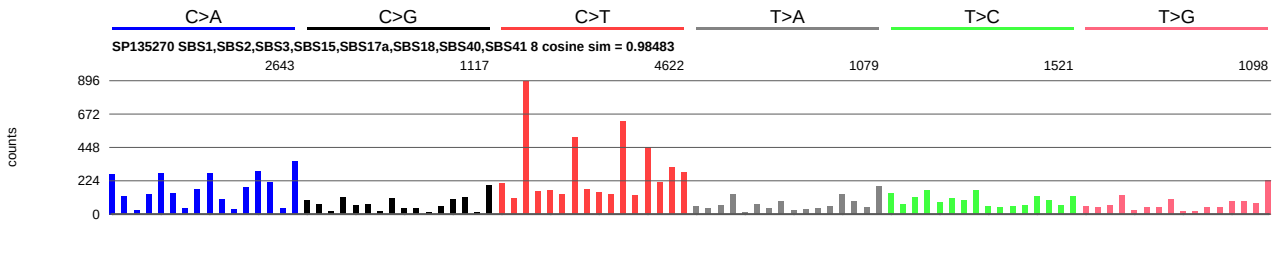

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98491
2682
1075
4713
920
1656
1030
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS2
SBS1

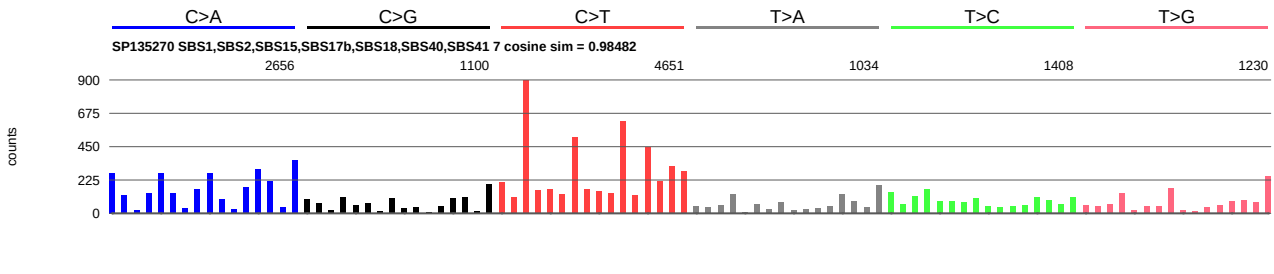

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.9849
2674
1040
4686
906
1752
1021
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1

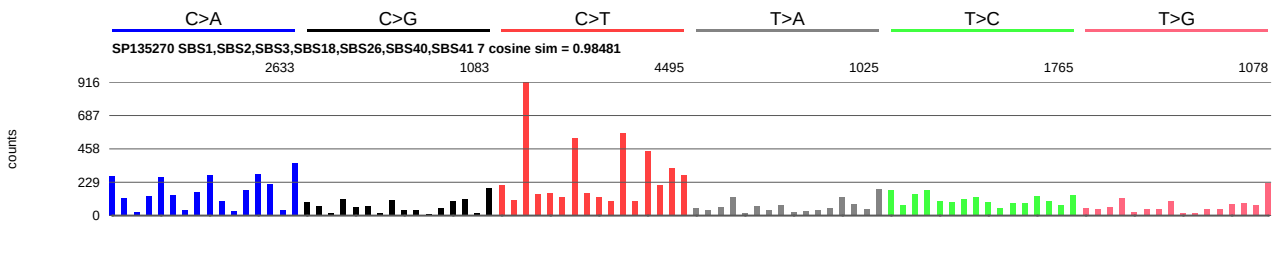

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98489
2676
1177
4546
892
1764
1022
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1

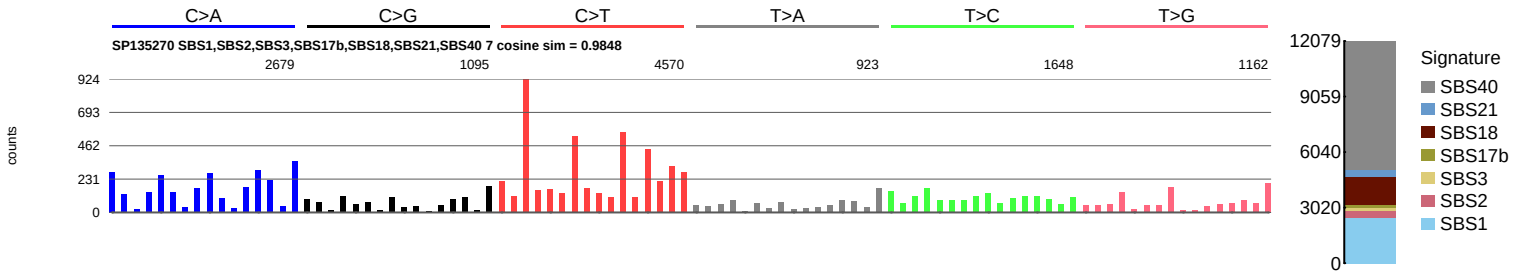

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98487
2693
1056
4752
935
1494
1153
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

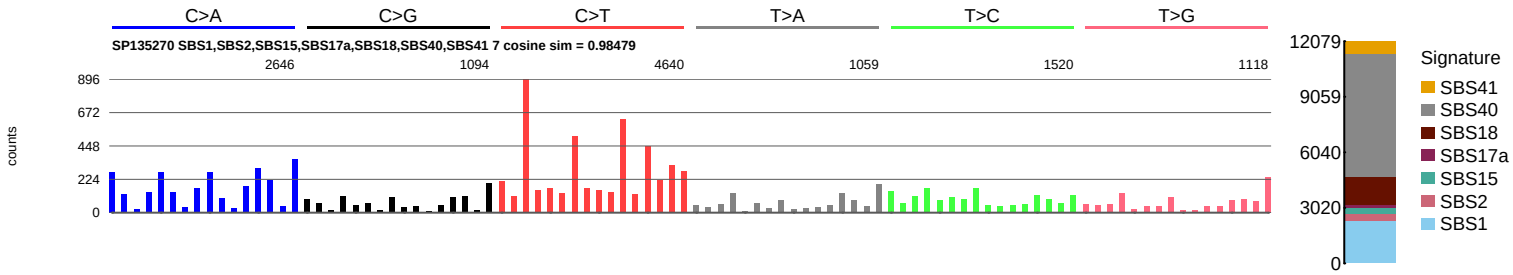

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98484
2681
1052
4675
912
1743
1018
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1

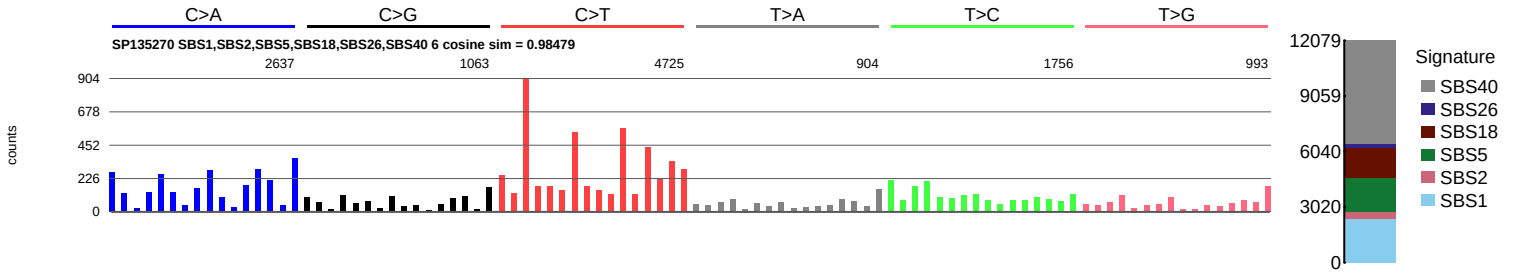

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98483
2643
1117
4622
1079
1521
1098
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1

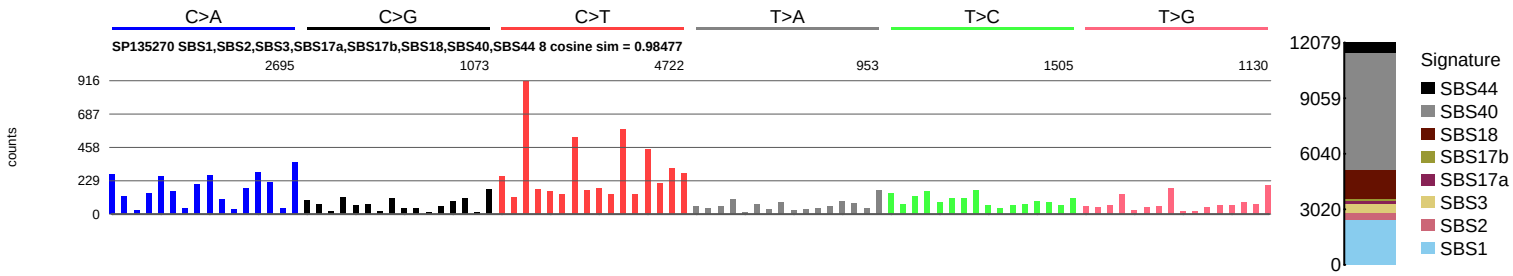

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98482
2656
1100
4651
1034
1408
1230
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1

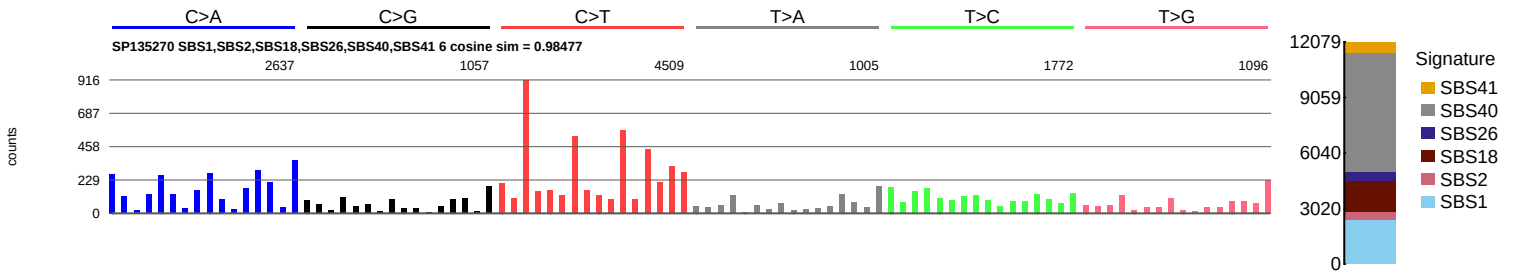

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98481
2633
1083
4495
1025
1765
1078
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.9848
2679
1095
4570
923
1648
1162
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98479
2646
1094
4640
1059
1520
1118
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40 6 cosine sim = 0.98479
2637
1063
4725
904
1756
993
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98477
2695
1073
4722
953
1505
1130
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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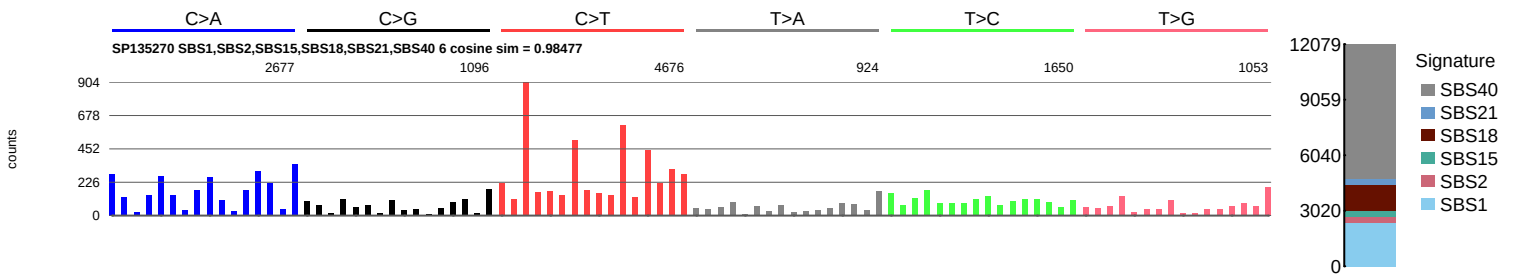
SP135270 SBS1,SBS2,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98477
2637
1057
4509
1005
1772
1096
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS2
SBS1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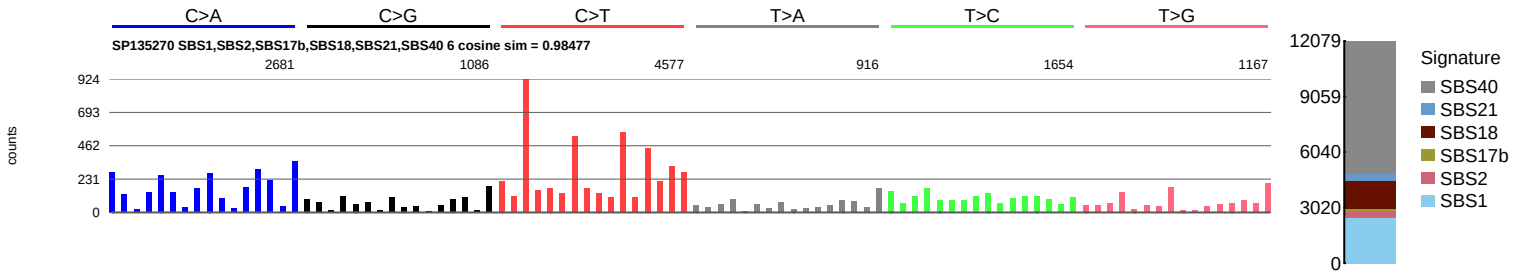
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40 6 cosine sim = 0.98477
2677
1096
4676
924
1650
1053
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS2
SBS1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T>C
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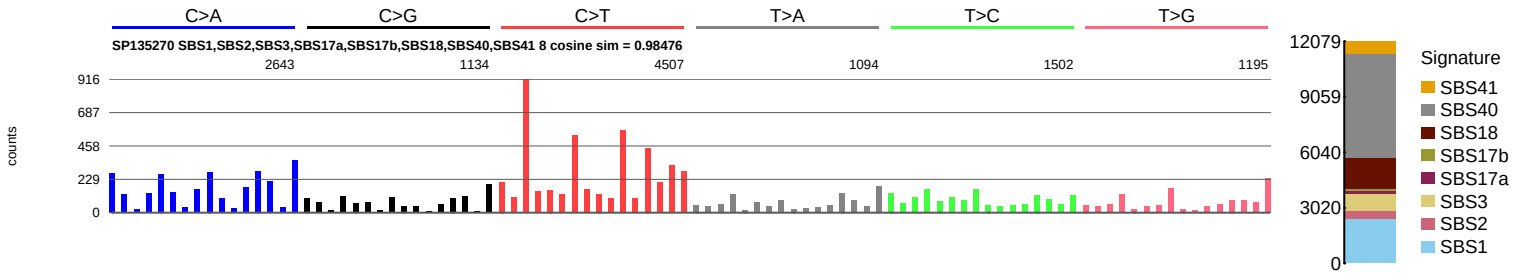
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98477
2681
1086
4577
916
1654
1167
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1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C>G
C>T
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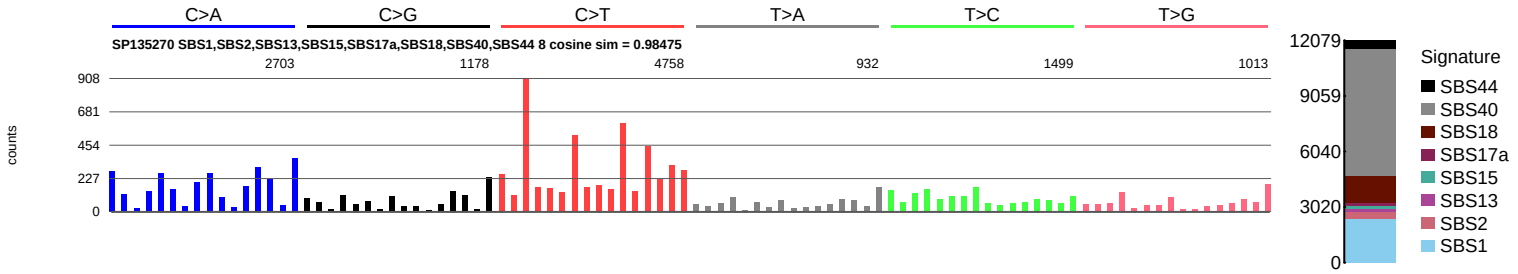
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98476
2643
1134
4507
1094
1502
1195
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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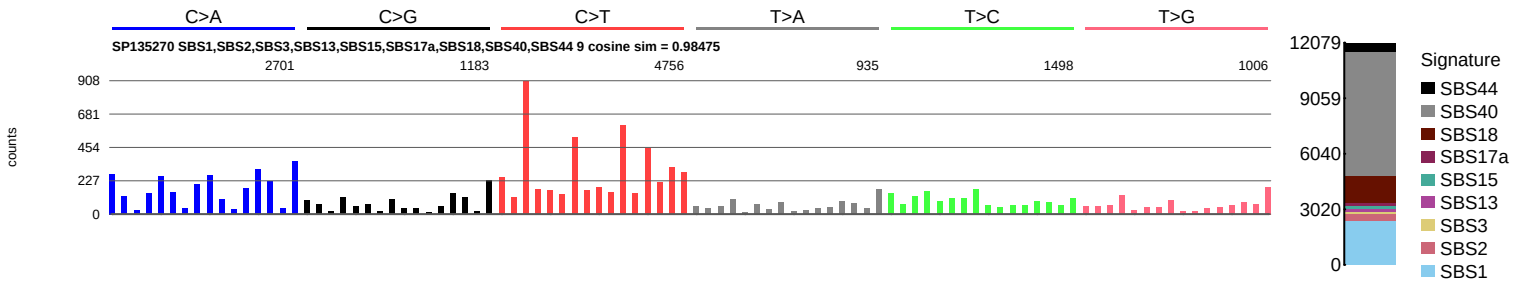
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98475
2703
1178
4758
932
1499
1013
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1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C>T
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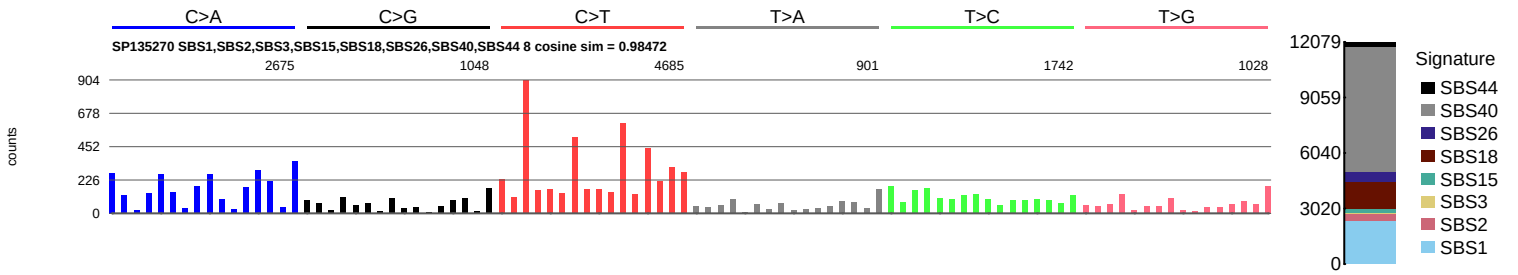
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98475
2701
1183
4756
935
1498
1006
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
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T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98472
2675
1048
4685
901
1742
1028
counts
904
678
452
226
0
12079
9059
6040
3020
0
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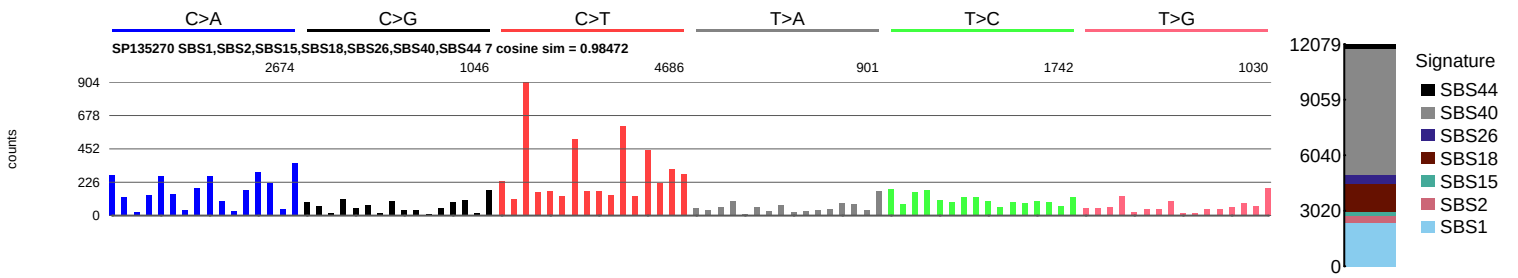
SBS2
SBS1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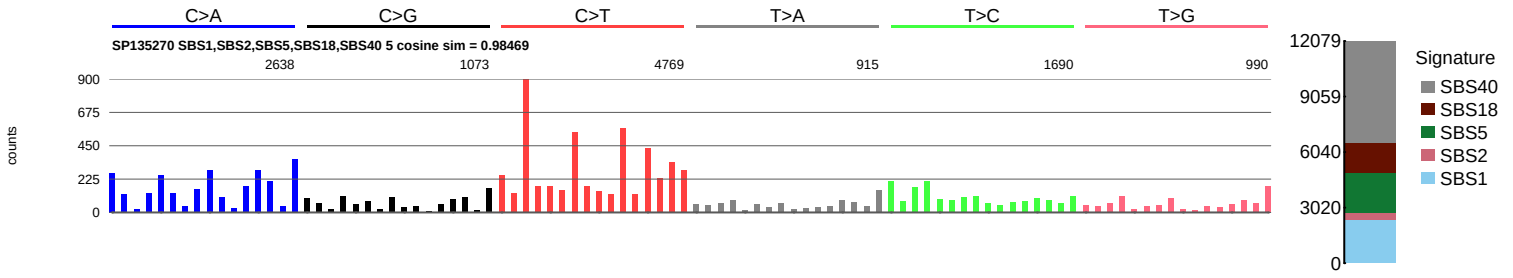
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98472
2674
1046
4686
901
1742
1030
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1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T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS18,SBS40 5 cosine sim = 0.98469
2638
1073
4769
915
1690
990
counts
900
675
450
225
0
12079
9059
6040
3020
0
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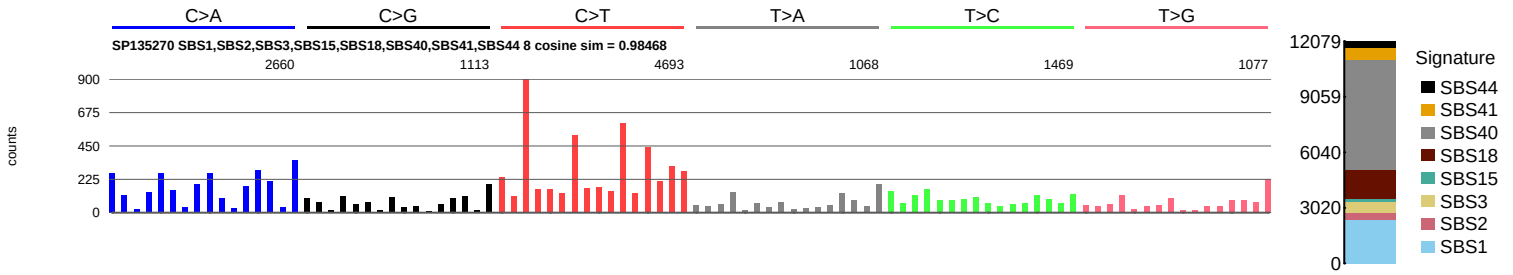
SBS5
SBS2
SBS1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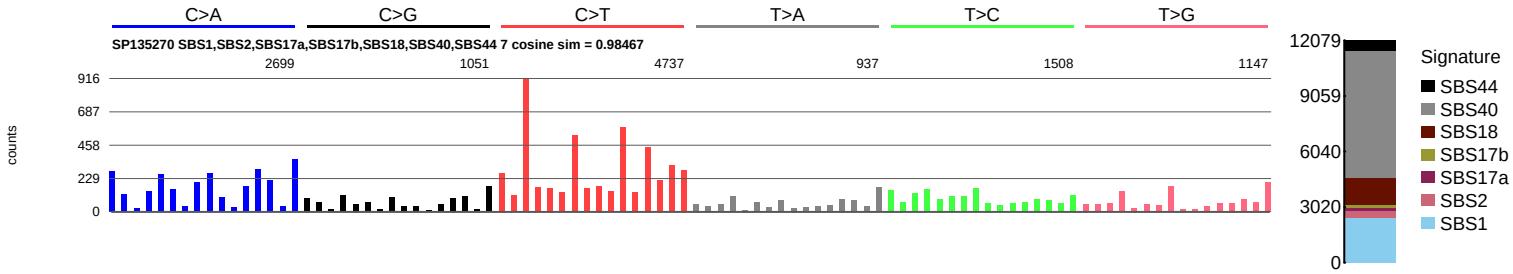
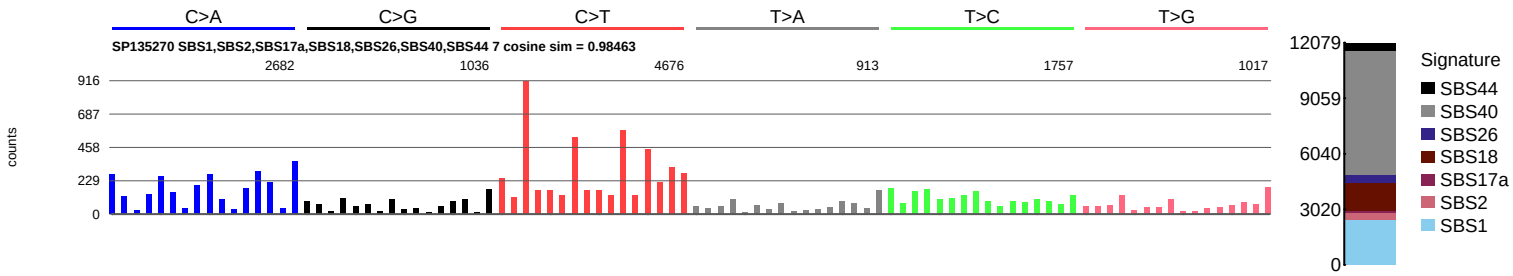
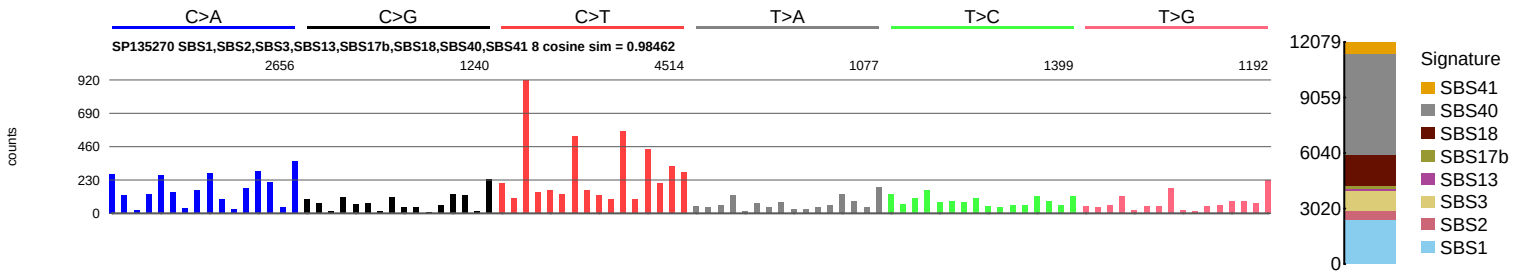
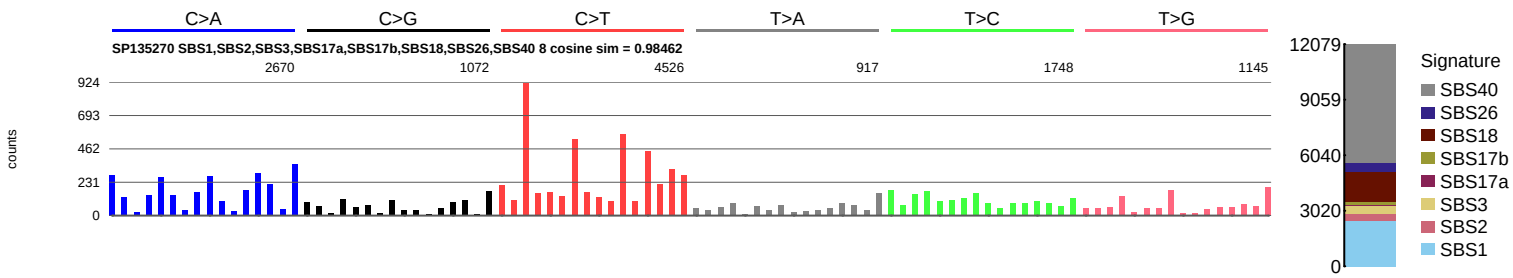
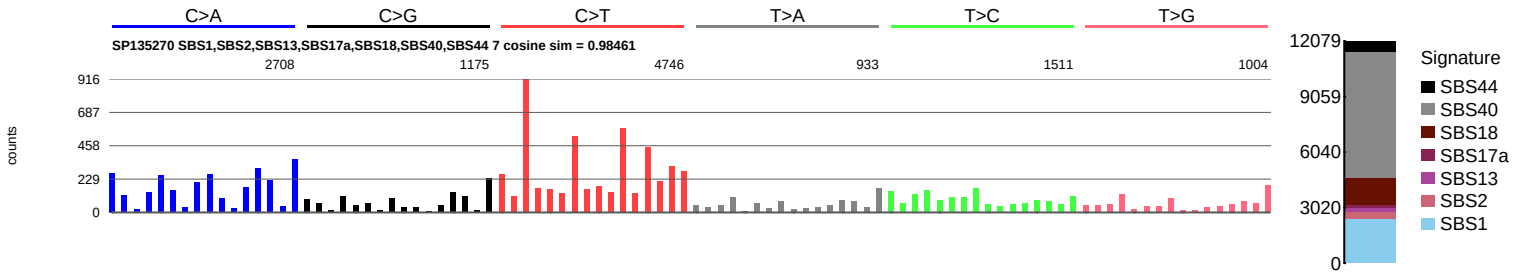
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98468
2660
1113
4693
1068
1469
1077
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS3
SBS2
SBS1

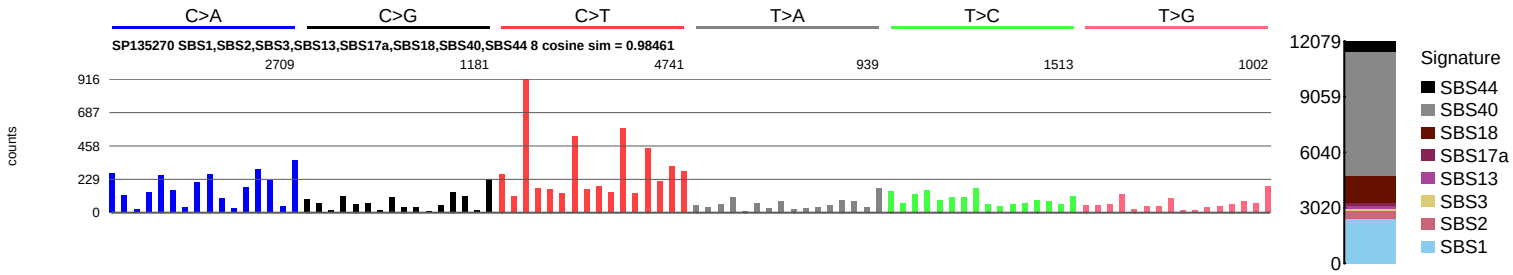

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98467
2699
1051
4737
937
1508
1147
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98463
2682
1036
4676
913
1757
1017
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98462
2656
1240
4514
1077
1399
1192
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98462
2670
1072
4526
917
1748
1145
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.98461
2708
1175
4746
933
1511
1004
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1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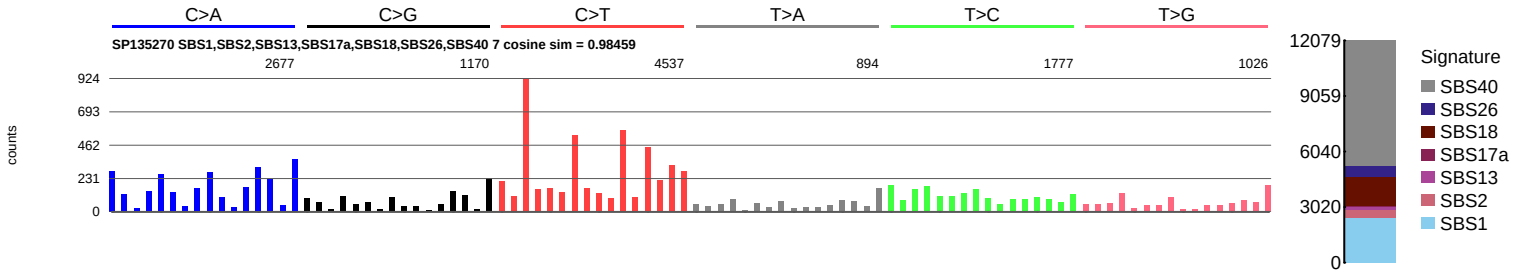
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98461
2709
1181
4741
939
1513
1002
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1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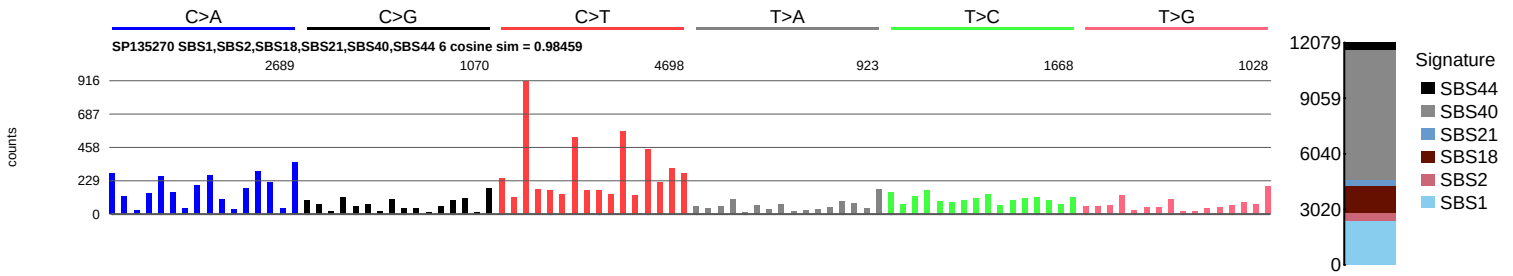
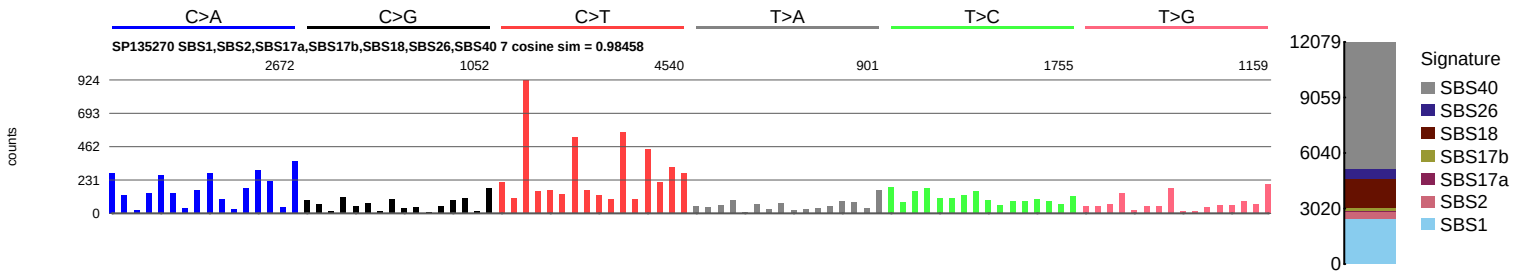
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98459
2677
1170
4537
894
1777
1026
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS21,SBS40,SBS44 6 cosine sim = 0.98459
2689
1070
4698
923
1668
1028
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98458
2672
1052
4540
901
1755
1159
counts
924
693
462
231
0
12079
9059
6040
3020
0
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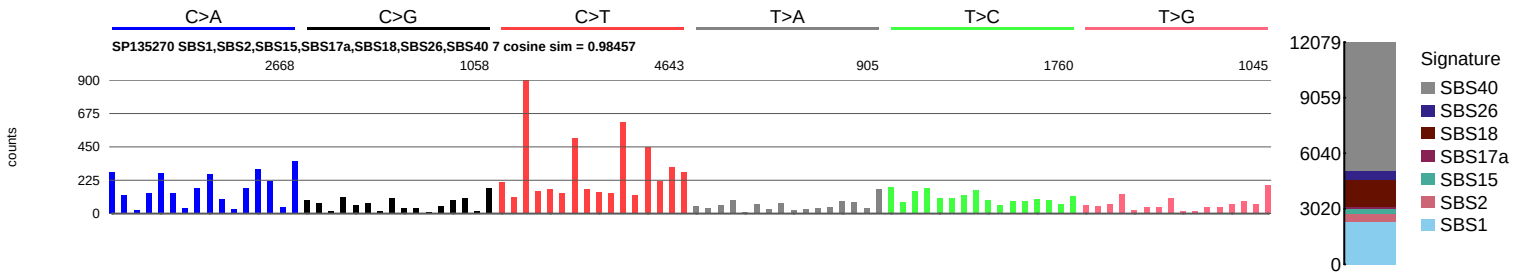
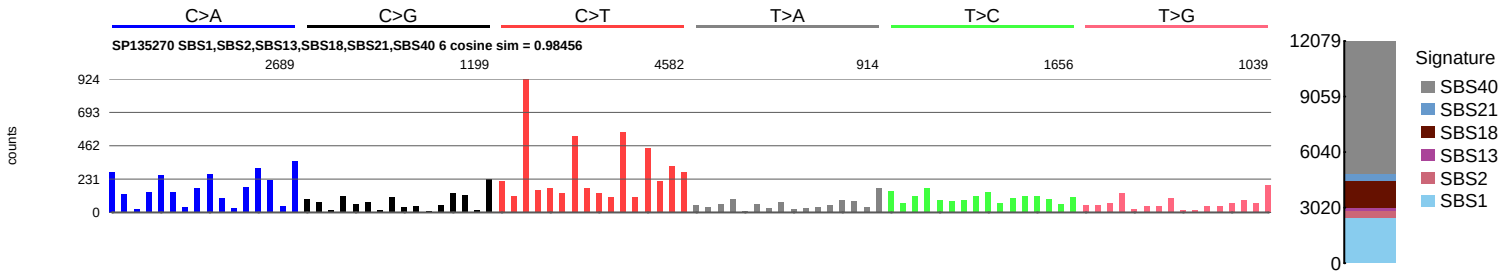
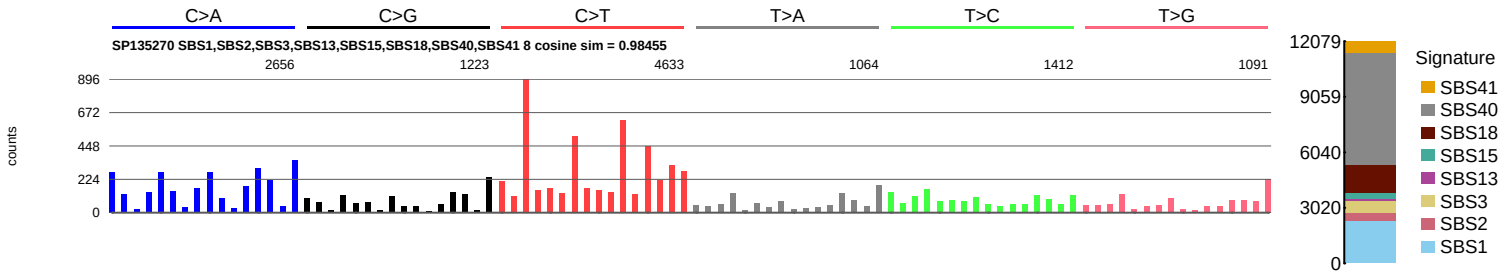
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98457
2668
1058
4643
905
1760
1045
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS21,SBS40 6 cosine sim = 0.98456
2689
1199
4582
914
1656
1039
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40,SBS41 8 cosine sim = 0.98455
2656
1223
4633
1064
1412
1091
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
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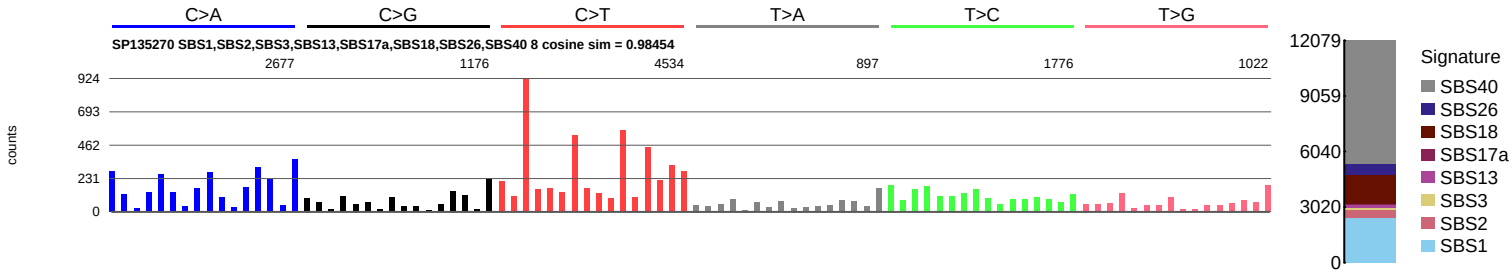
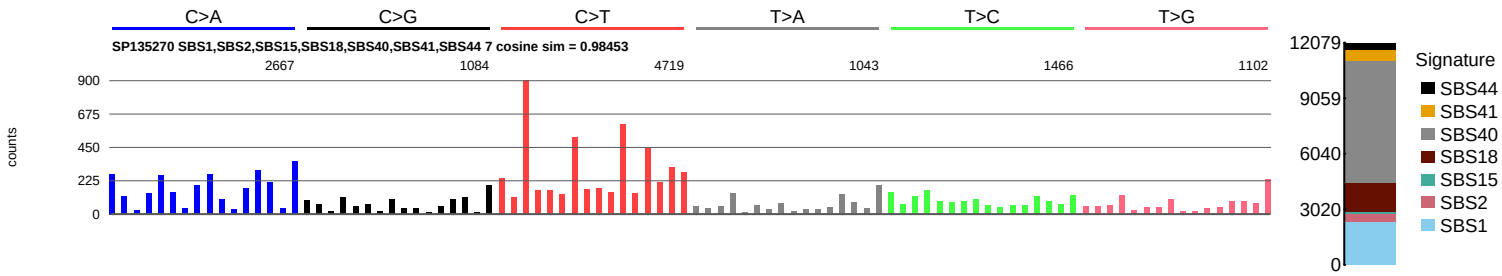
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98454
2677
1176
4534
897
1776
1022
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98453
2667
1084
4719
1043
1466
1102
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
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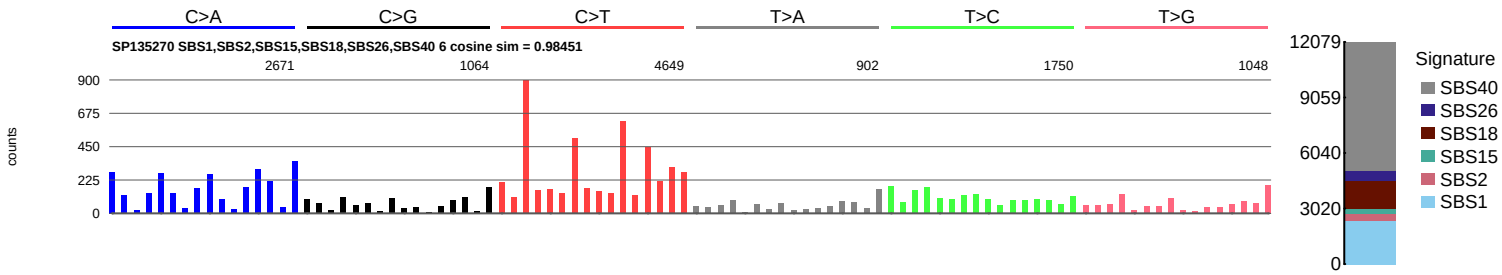
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40 6 cosine sim = 0.98451
2671
1064
4649
902
1750
1048
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
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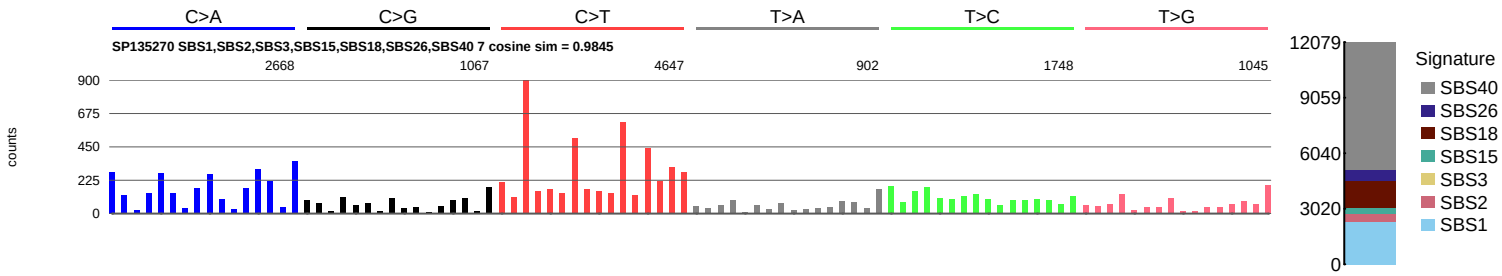
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS40 7 cosine sim = 0.9845
2668
1067
4647
902
1748
1045
counts
900
675
450
225
0
12079
9059
6040
3020
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SBS40
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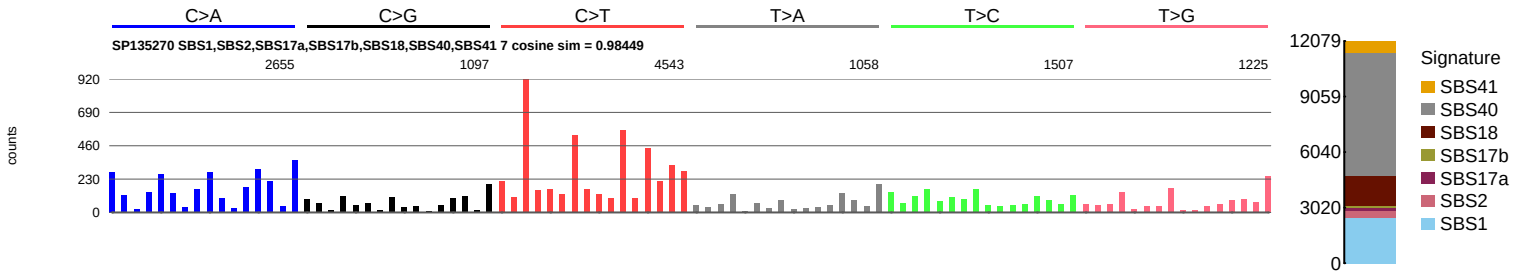
SBS2
SBS1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C>G
C>T
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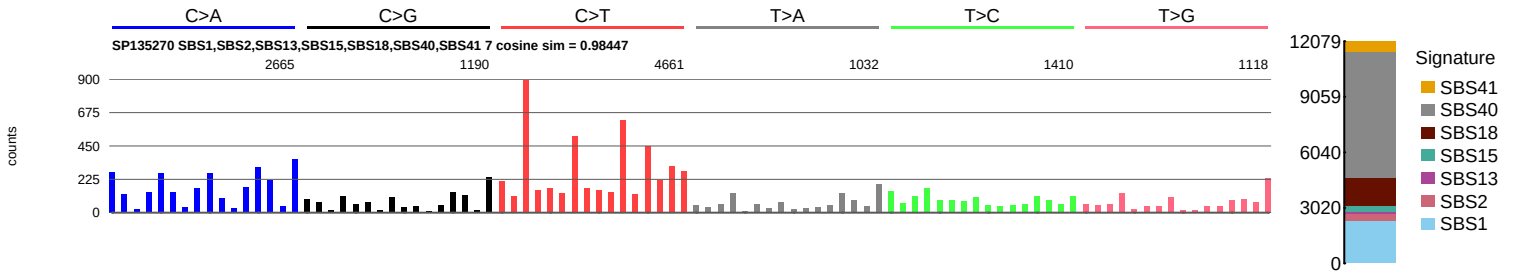
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98449
2655
1097
4543
1058
1507
1225
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
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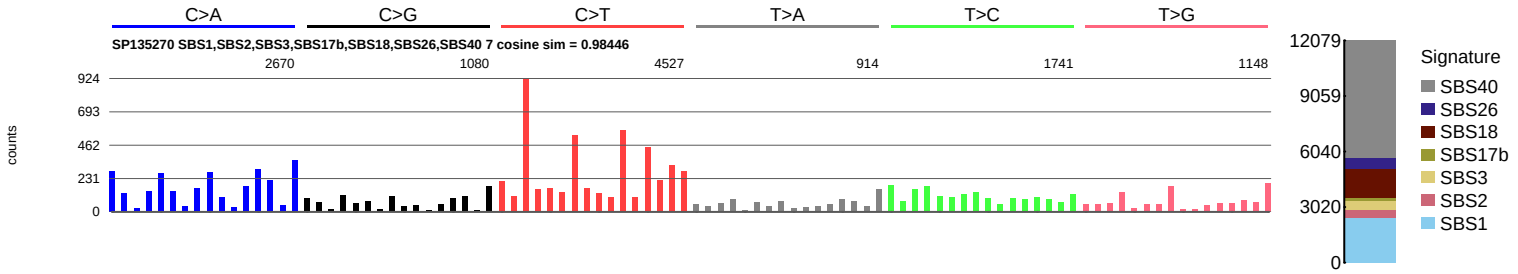
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.98447
2665
1190
4661
1032
1410
1118
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
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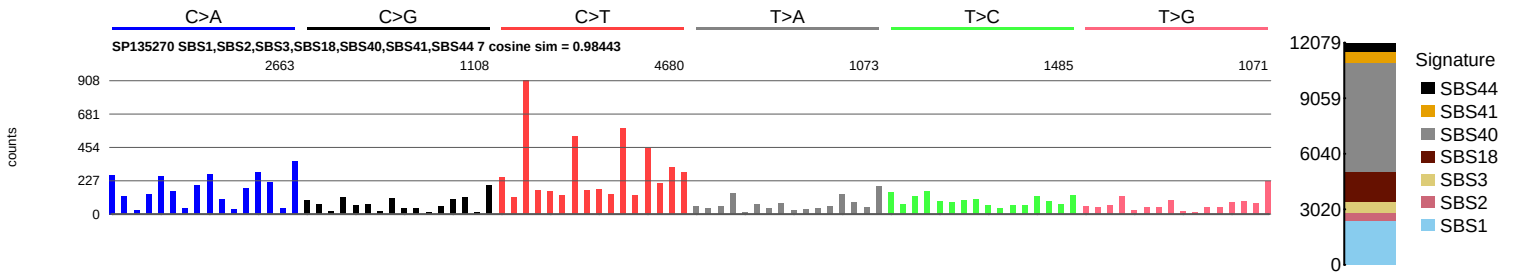
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98446
2670
1080
4527
914
1741
1148
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
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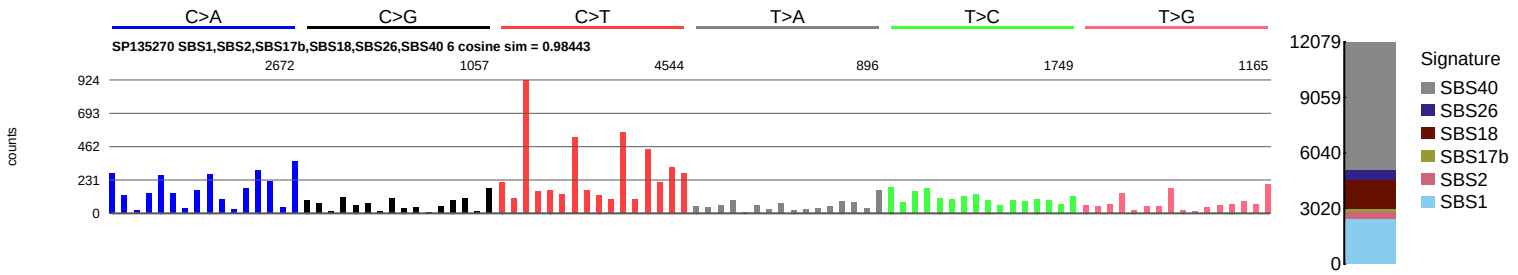
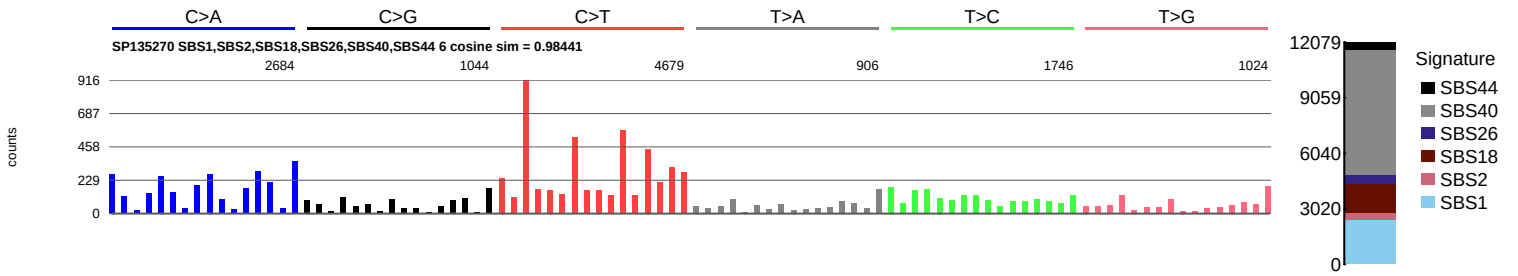
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98443
2663
1108
4680
1073
1485
1071
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98443
2672
1057
4544
896
1749
1165
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98441
2684
1044
4679
906
1746
1024
counts
916
687
458
229
0
12079
9059
6040
3020
0
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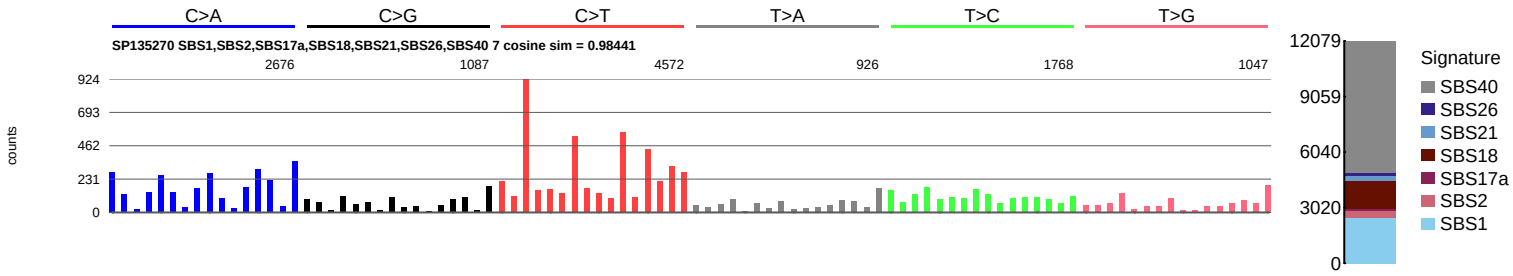
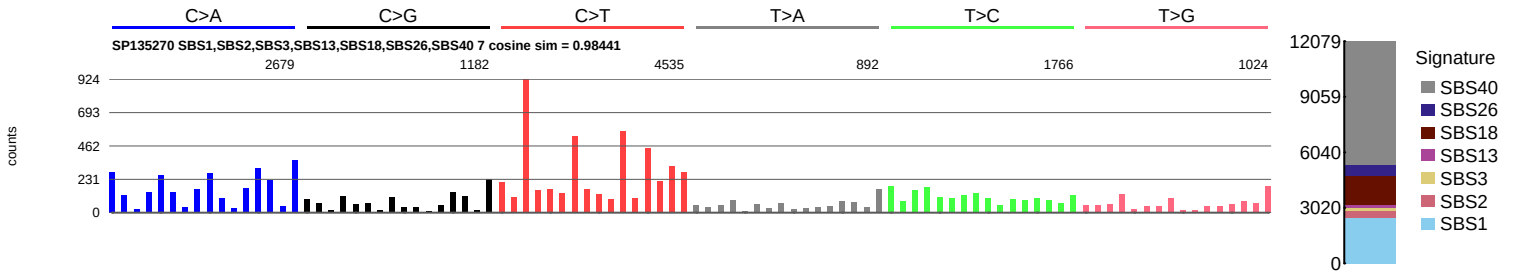
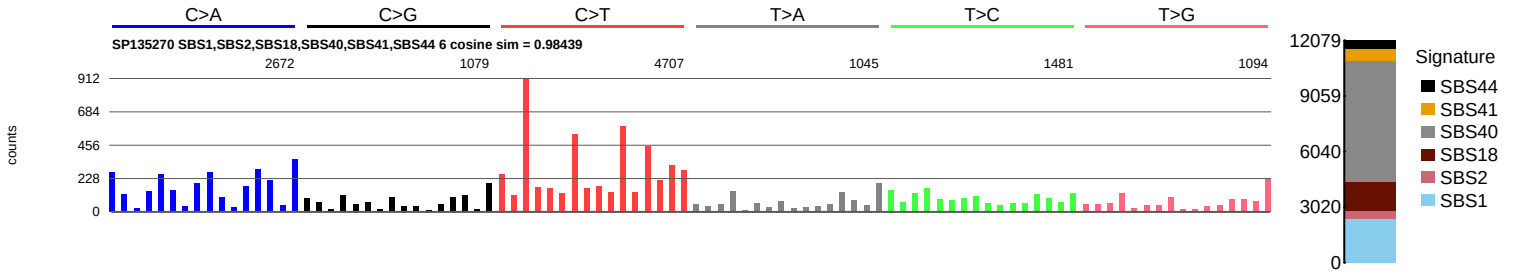
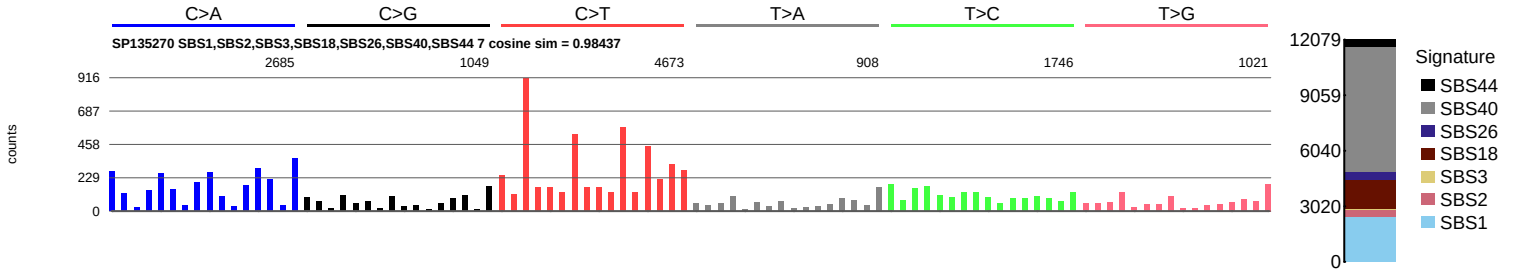
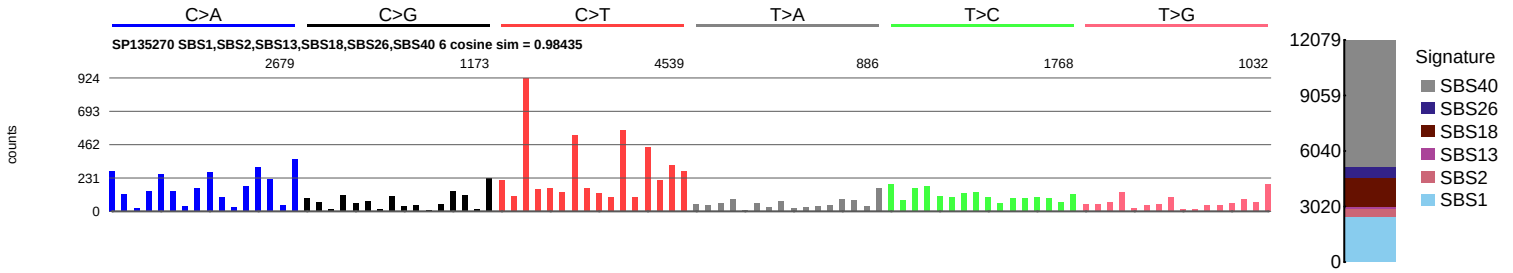
SBS44
SBS40
SBS26
SBS18
SBS2
SBS1

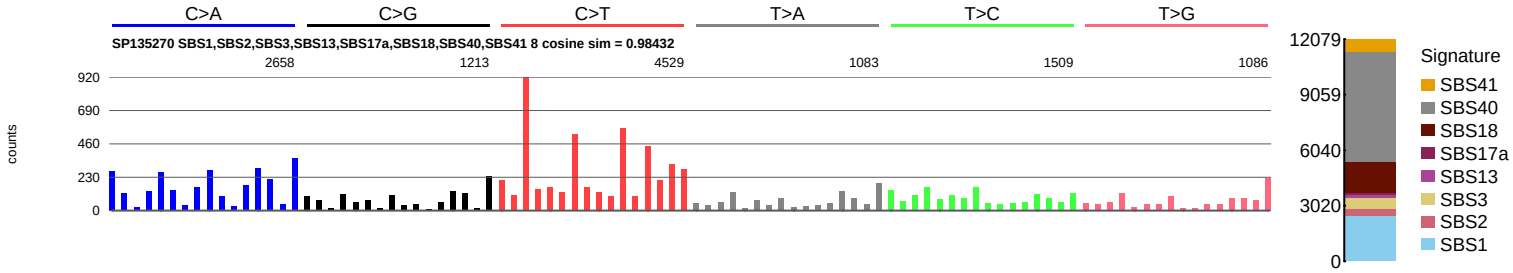

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98441
2676
1087
4572
926
1768
1047
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS26,SBS40 7 cosine sim = 0.98441
2679
1182
4535
892
1766
1024
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS40,SBS41,SBS44 6 cosine sim = 0.98439
2672
1079
4707
1045
1481
1094
counts
912
684
456
228
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98437
2685
1049
4673
908
1746
1021
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40 6 cosine sim = 0.98435
2679
1173
4539
886
1768
1032
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1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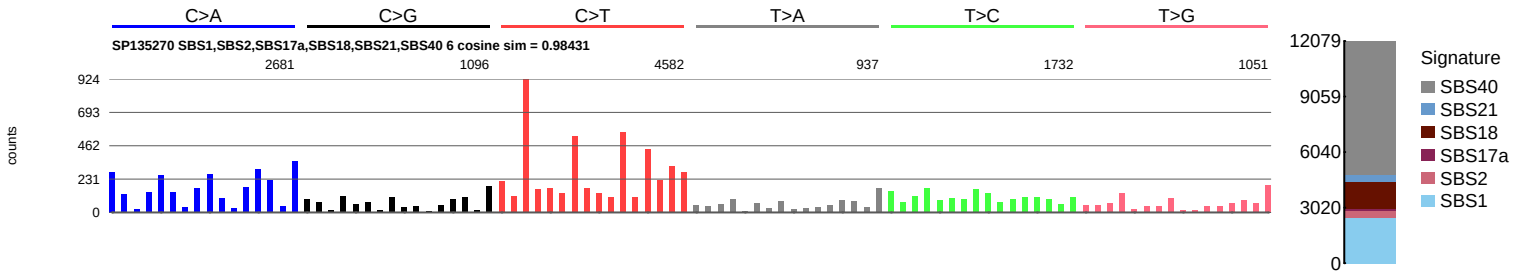
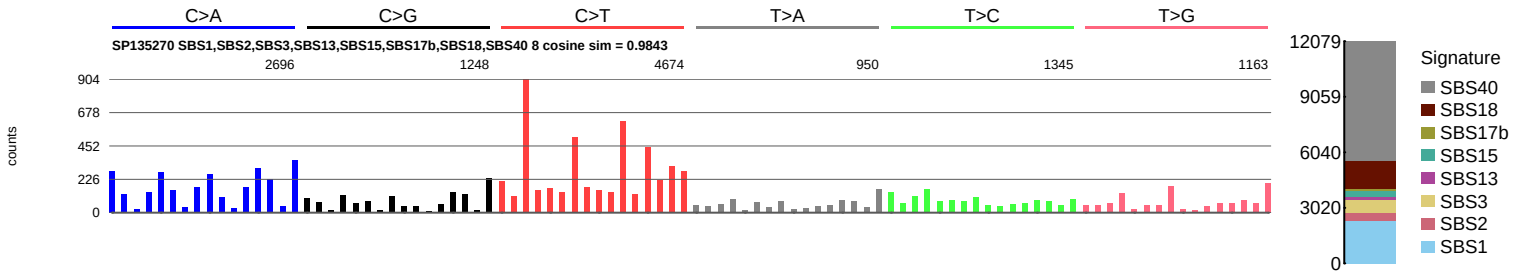
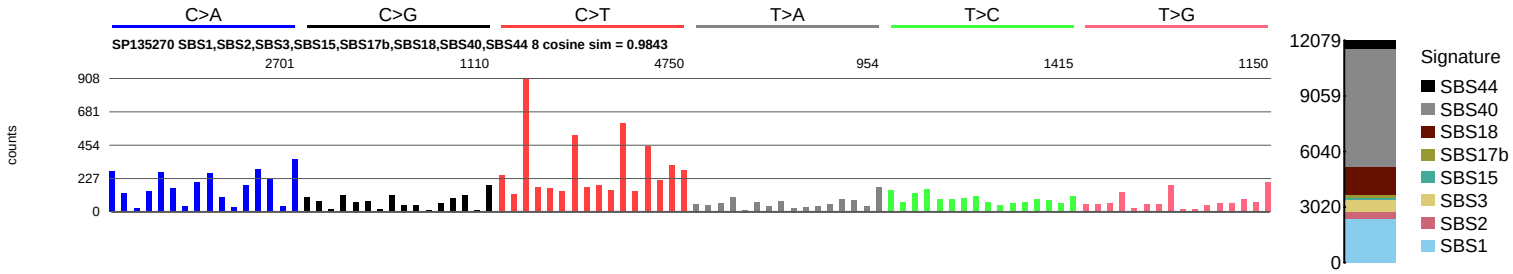
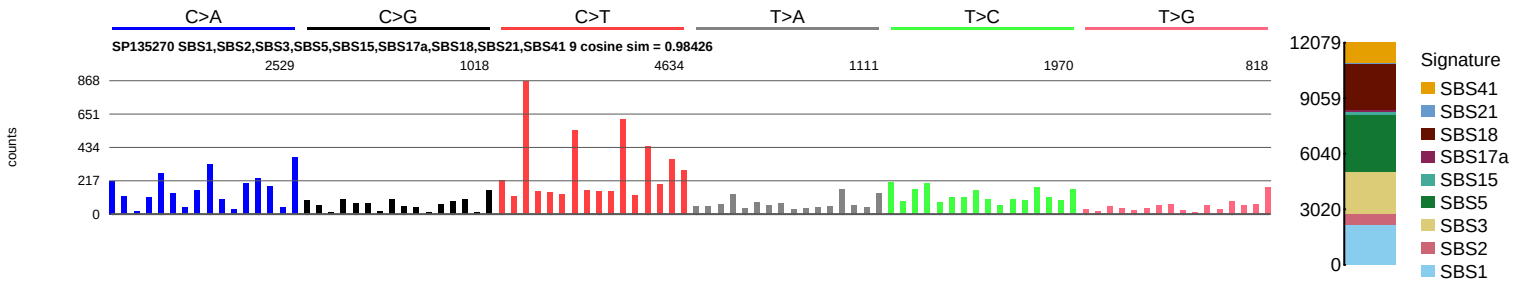
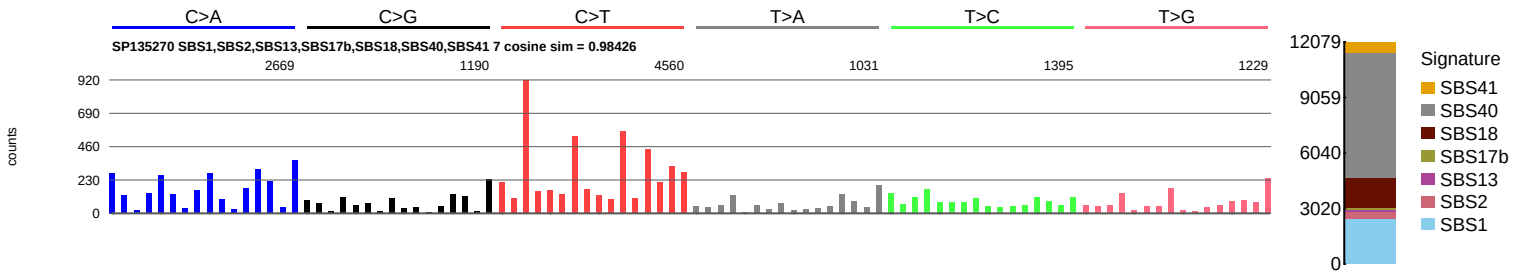
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98432
2658
1213
4529
1083
1509
1086
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

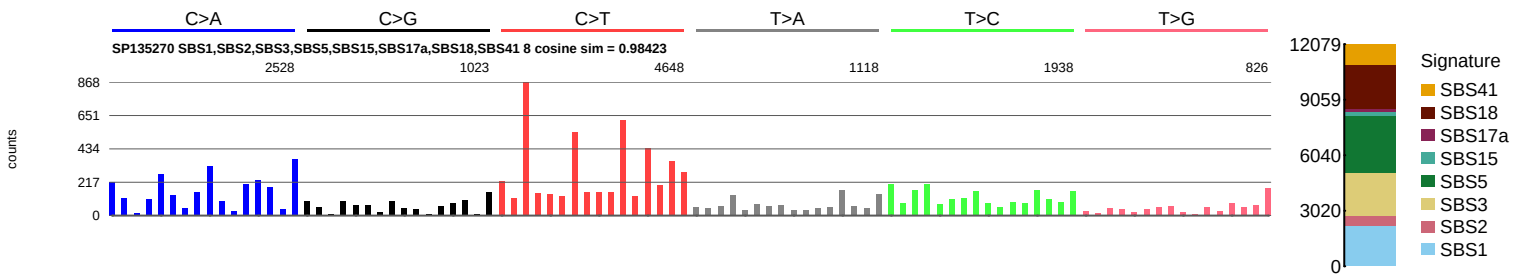

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40 6 cosine sim = 0.98431
2681
1096
4582
937
1732
1051
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1

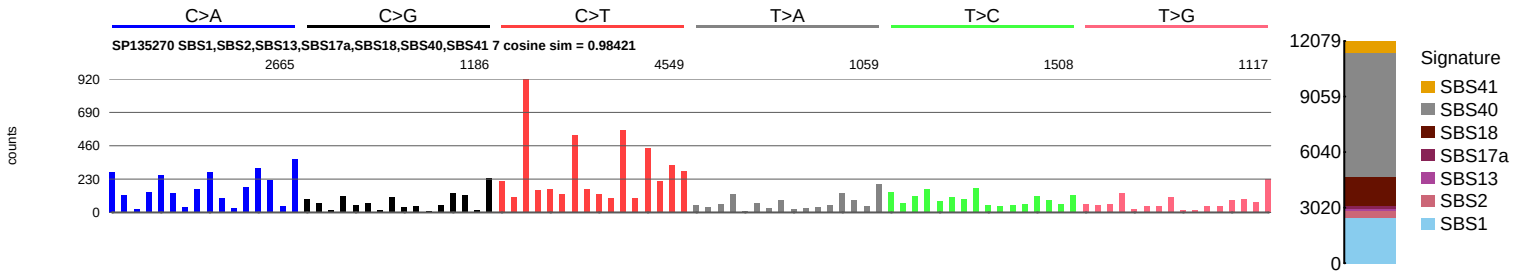

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.9843
2696
1248
4674
950
1345
1163
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

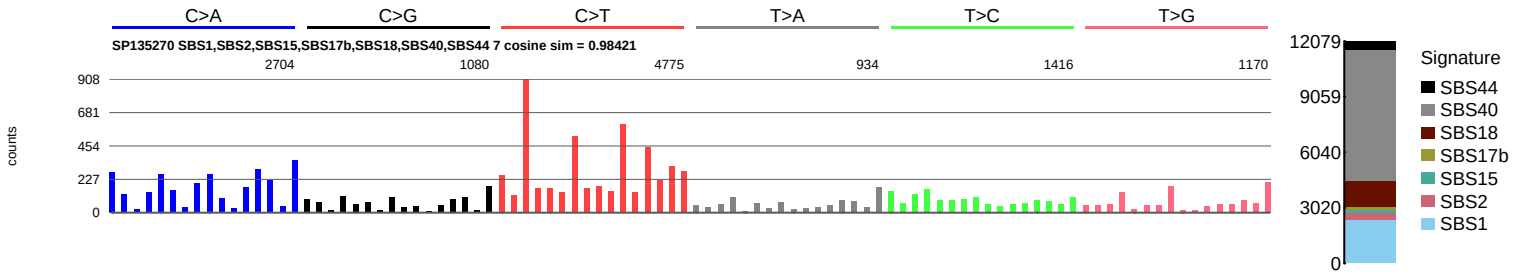

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.9843
2701
1110
4750
954
1415
1150
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

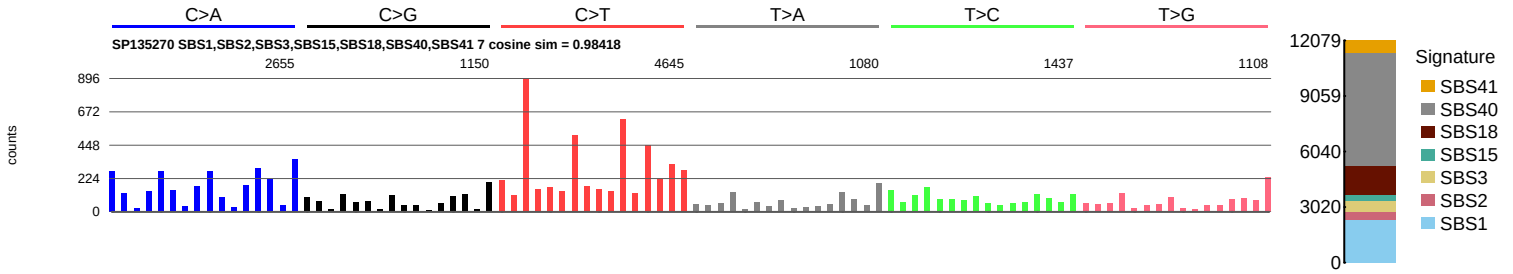

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS41 9 cosine sim = 0.98426
2529
1018
4634
1111
1970
818
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

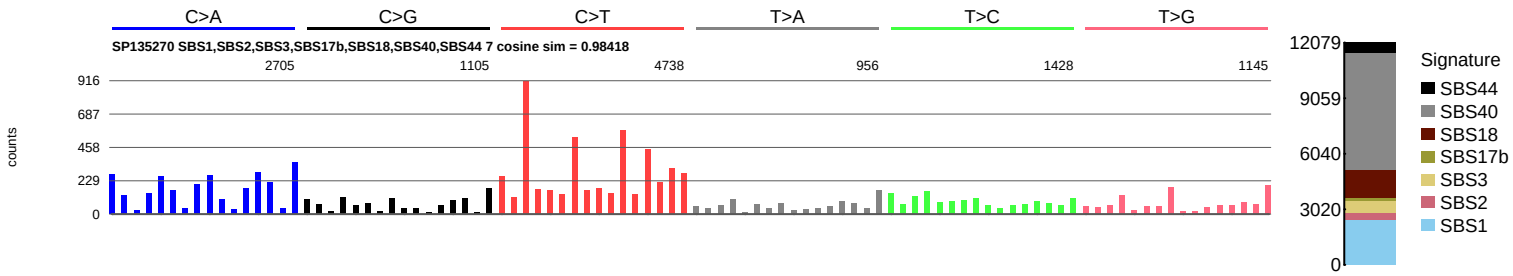

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98426
2669
1190
4560
1031
1395
1229
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1

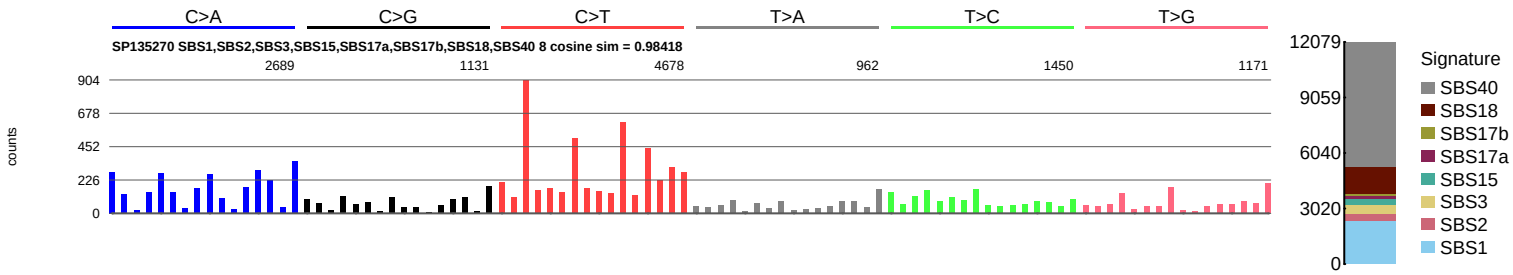

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS41 8 cosine sim = 0.98423
2528
1023
4648
1118
1938
826
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98421
2665
1186
4549
1059
1508
1117
920
690
460
230
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98421
2704
1080
4775
934
1416
1170
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.98418
2655
1150
4645
1080
1437
1108
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98418
2705
1105
4738
956
1428
1145
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98418
2689
1131
4678
962
1450
1171
904
678
452
226
0
counts
12079
9059
6040
3020
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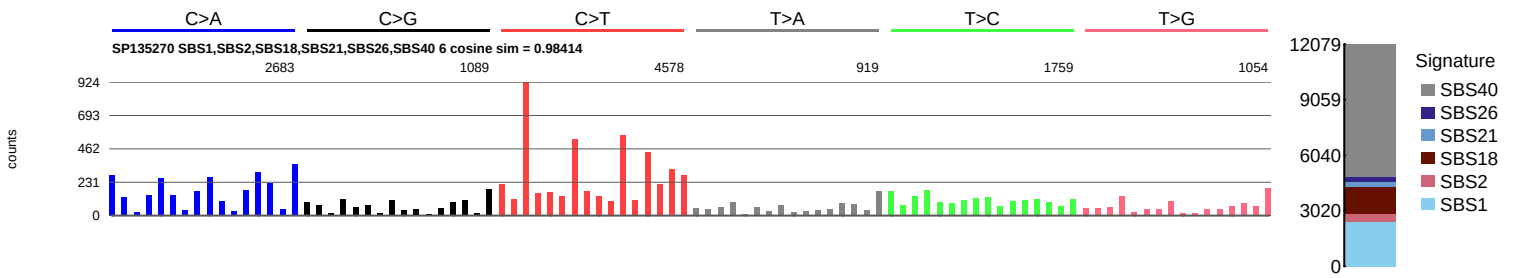
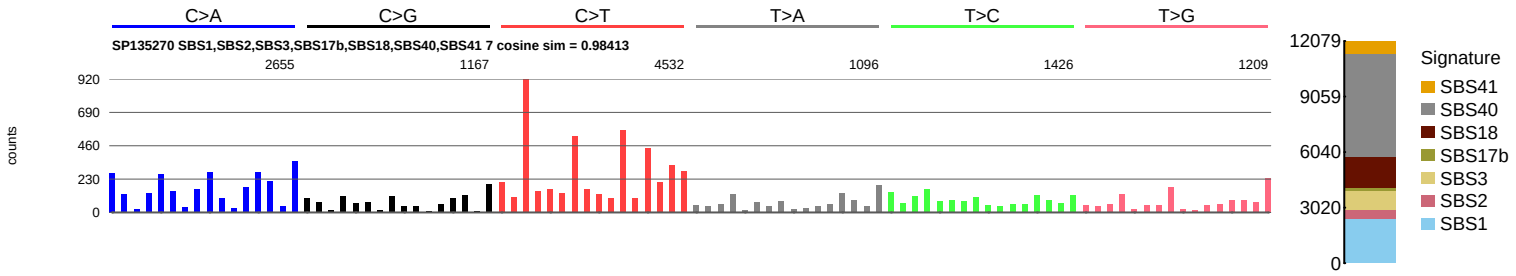
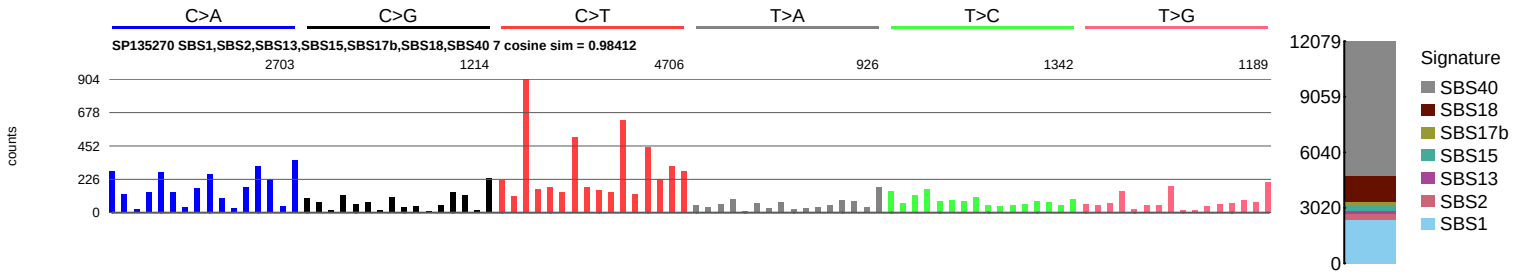
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40 6 cosine sim = 0.98414
2683
1089
4578
919
1759
1054
924
693
462
231
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98413
2655
1167
4532
1096
1426
1209
920
690
460
230
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
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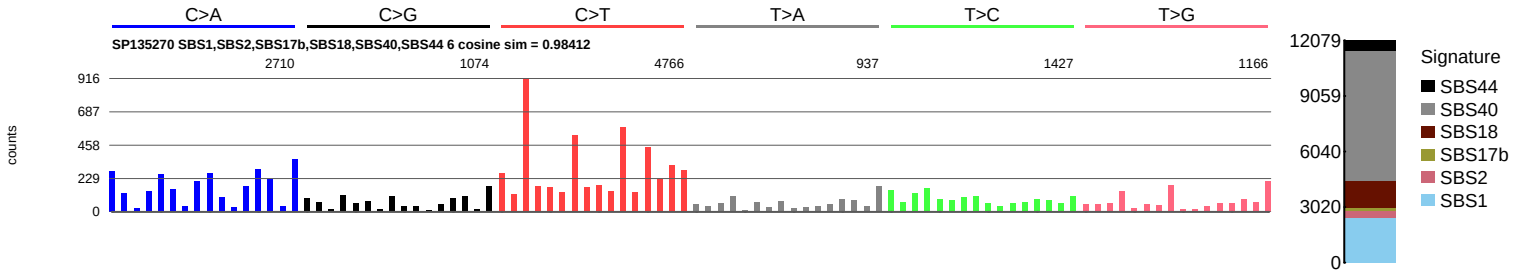
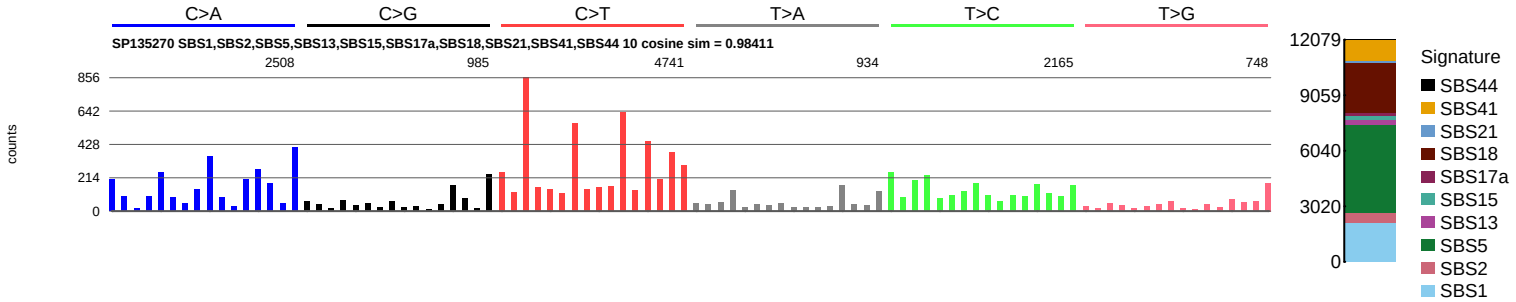
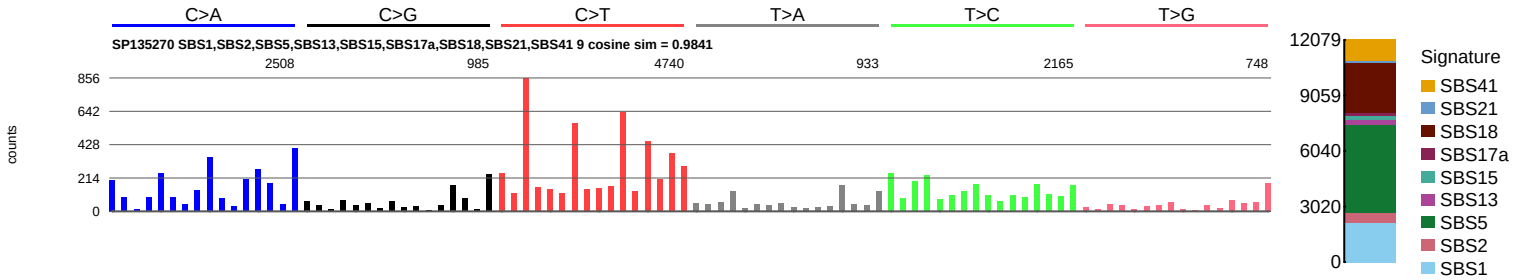
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98412
2703
1214
4706
926
1342
1189
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98412
2710
1074
4766
937
1427
1166
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98411
2508
985
4741
934
2165
748
856
642
428
214
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41 9 cosine sim = 0.9841
2508
985
4740
933
2165
748
856
642
428
214
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17a
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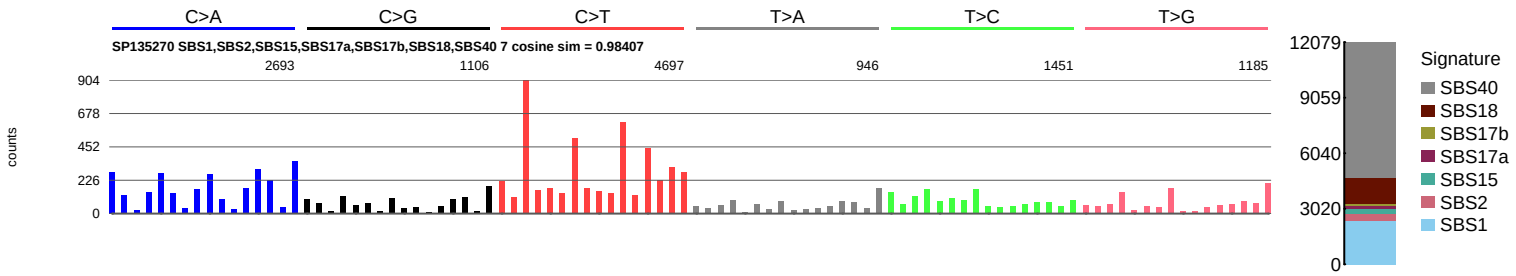
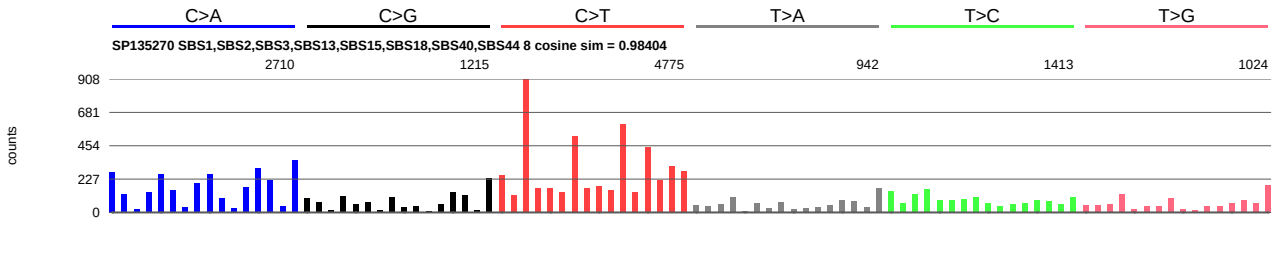
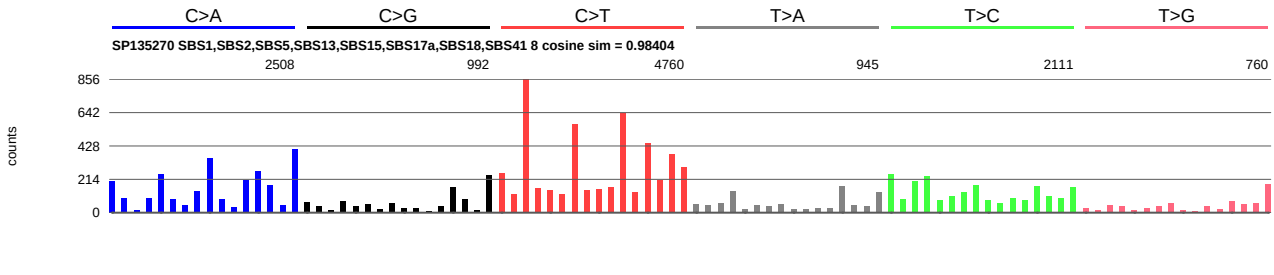
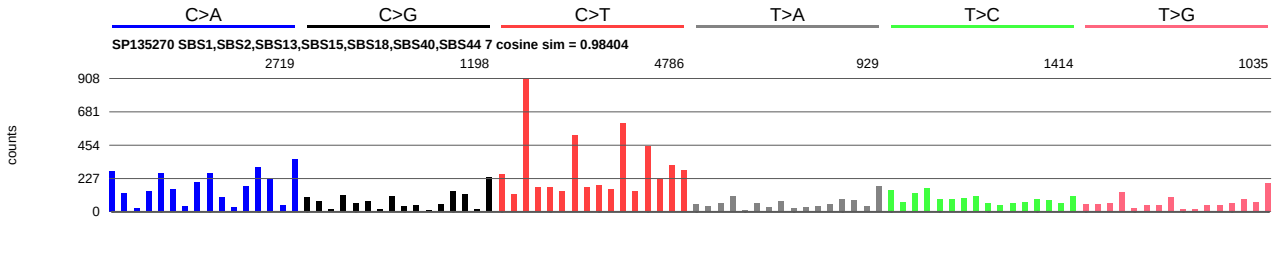
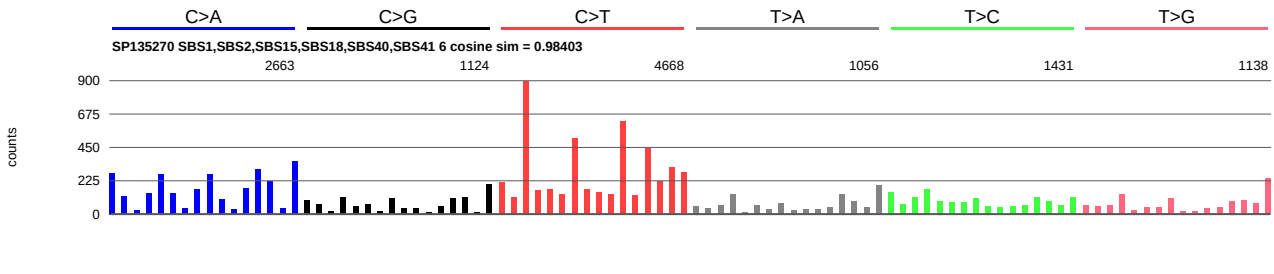
SBS2
SBS1

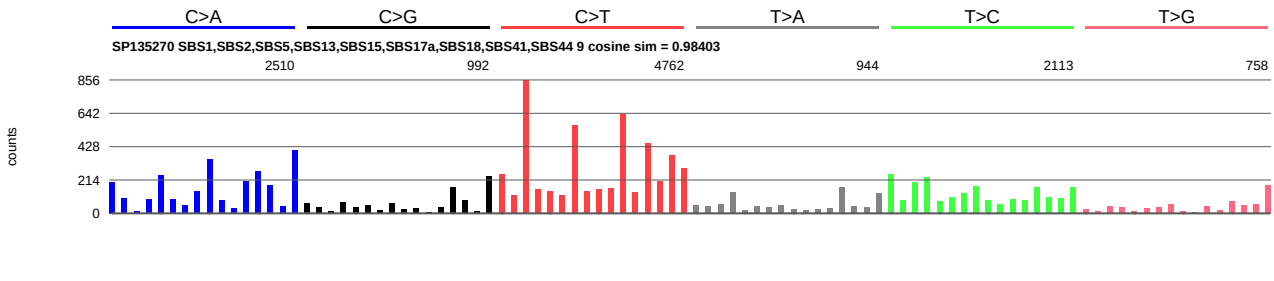

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98407
2693
1106
4697
946
1451
1185
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

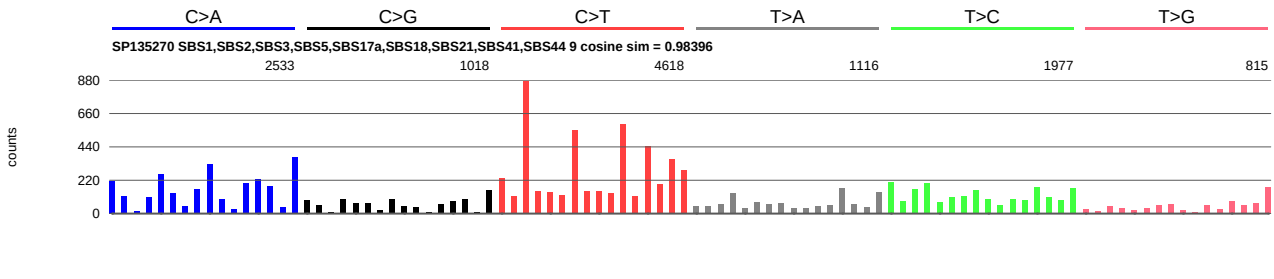

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40,SBS44 8 cosine sim = 0.98404
2710
1215
4775
942
1413
1024
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1

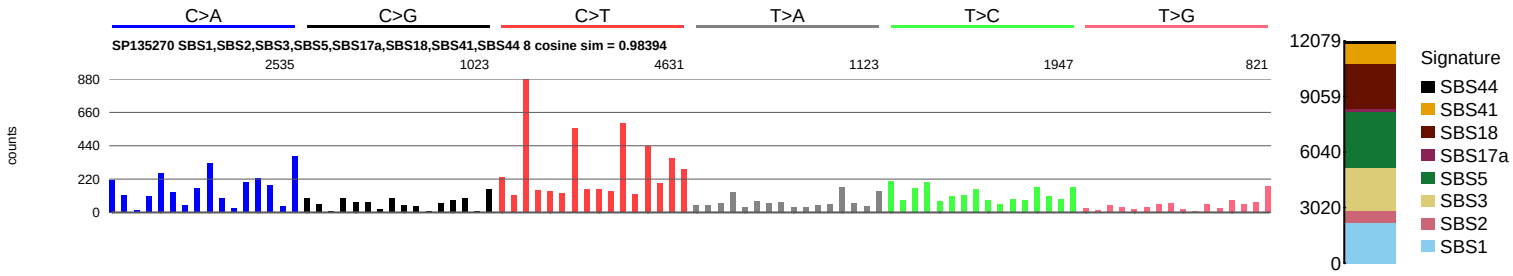

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 8 cosine sim = 0.98404
2508
992
4760
945
2111
760
counts
856
642
428
214
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

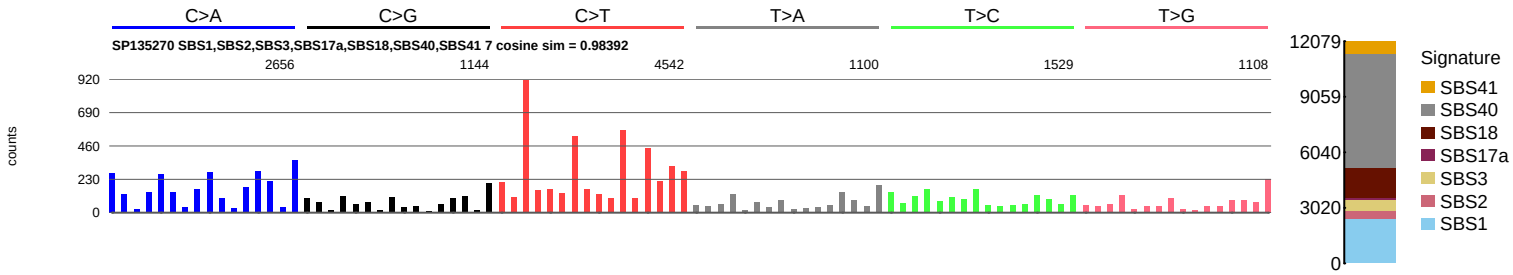

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40,SBS44 7 cosine sim = 0.98404
2719
1198
4786
929
1414
1035
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS13
SBS2
SBS1

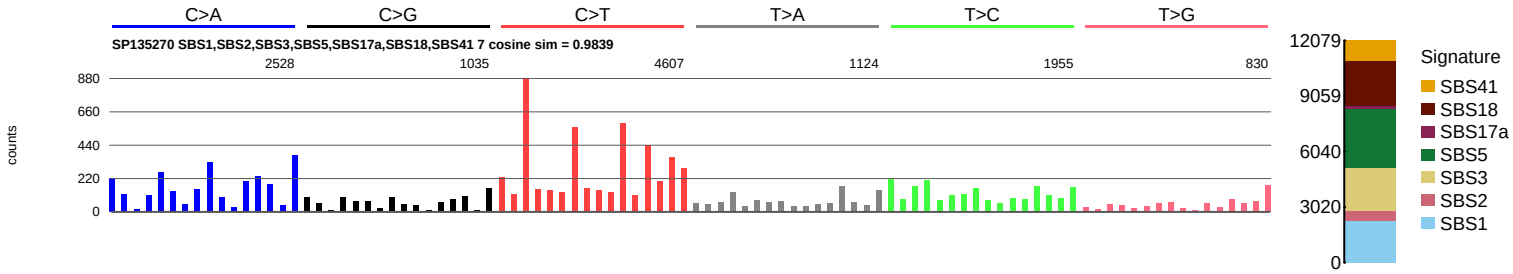

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS40,SBS41 6 cosine sim = 0.98403
2663
1124
4668
1056
1431
1138
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS2
SBS1

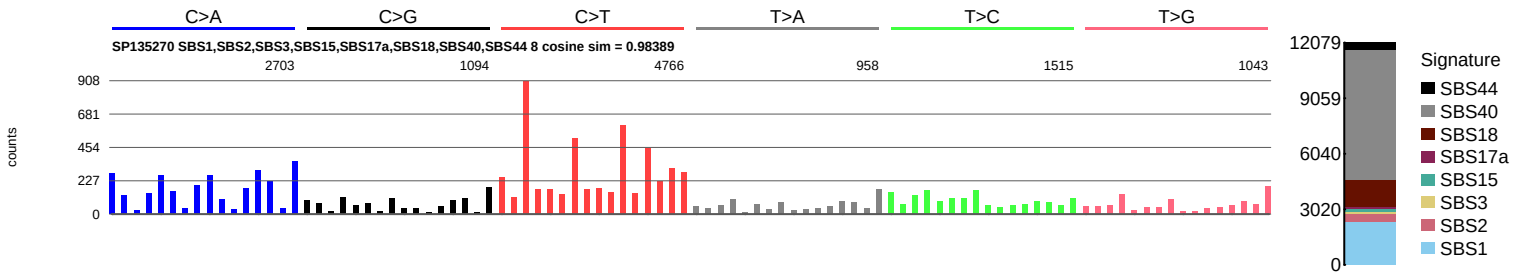

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.98403
2510
992
4762
944
2113
758
counts
856
642
428
214
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

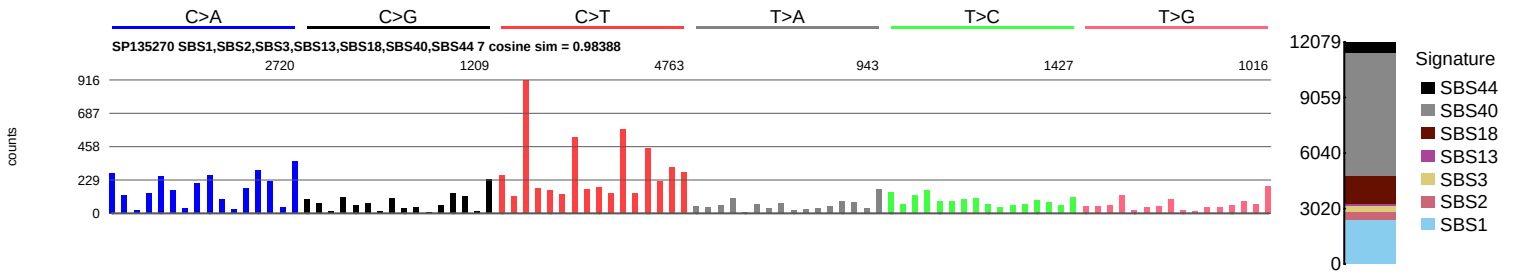

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98396
2533
1018
4618
1116
1977
815
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1

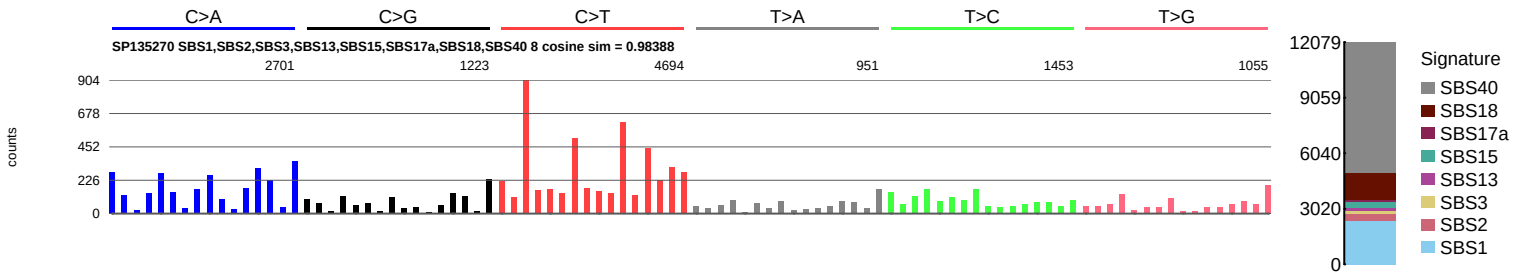

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS41,SBS44 8 cosine sim = 0.98394
2535
1023
4631
1123
1947
821
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1

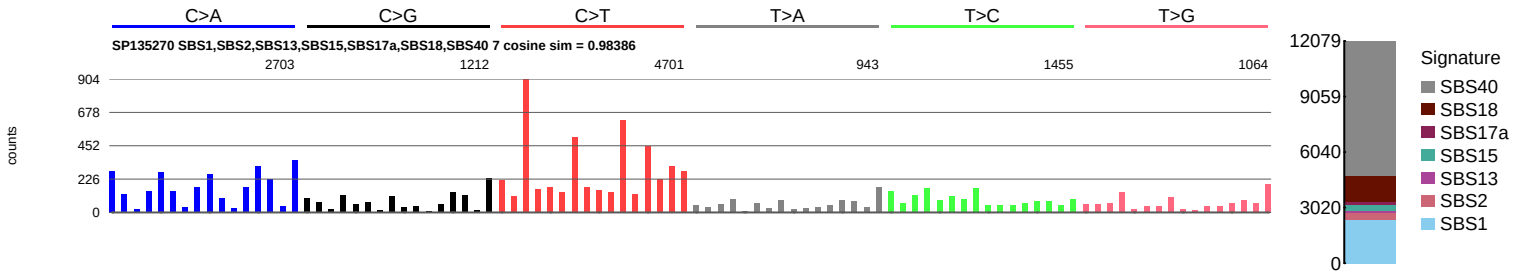

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98392
2656
1144
4542
1100
1529
1108
920
690
460
230
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS3
SBS2
SBS1

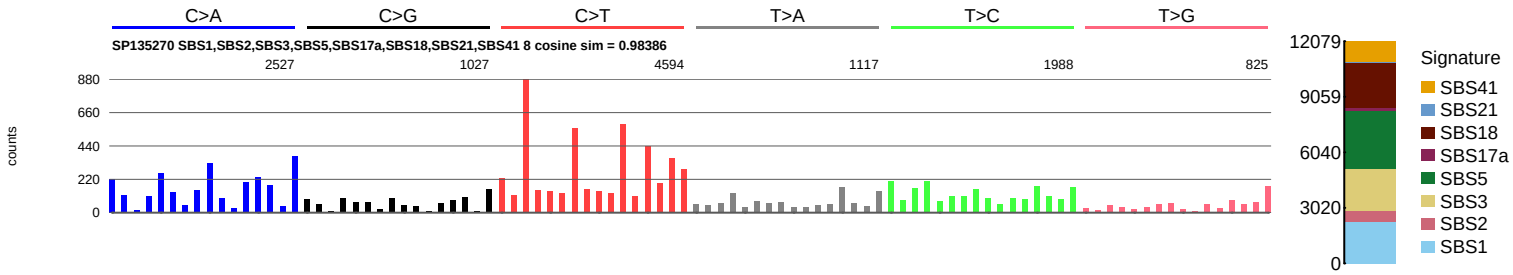

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS41 7 cosine sim = 0.9839
2528
1035
4607
1124
1955
830
880
660
440
220
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1

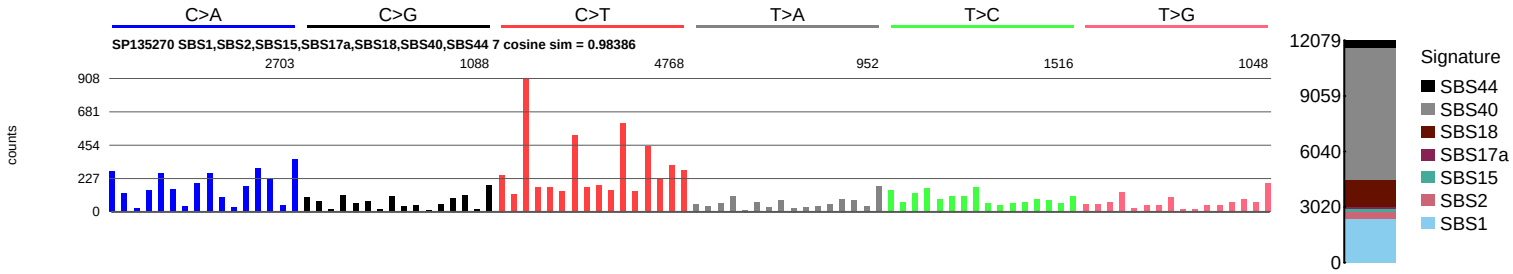

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98389
2703
1094
4766
958
1515
1043
908
681
454
227
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1

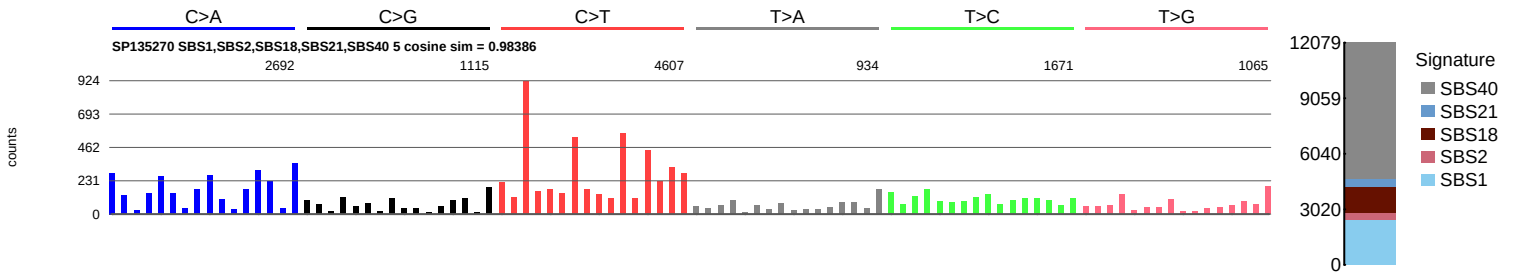

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40,SBS44 7 cosine sim = 0.98388
2720
1209
4763
943
1427
1016
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1

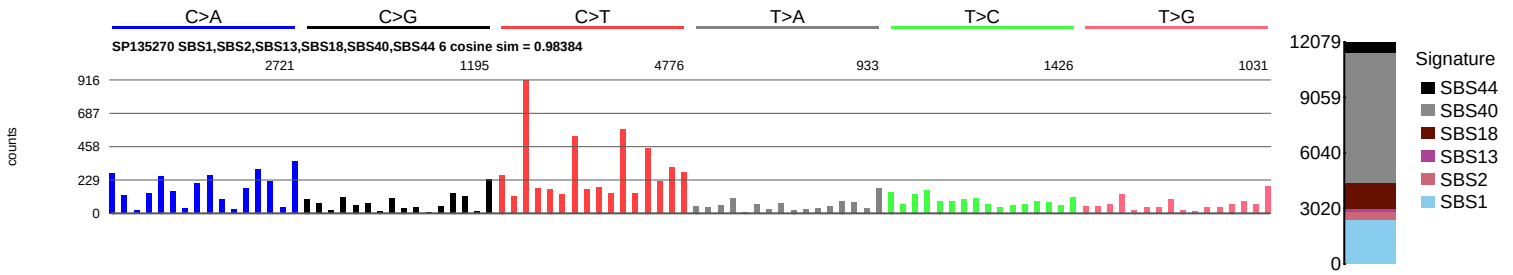

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS40 8 cosine sim = 0.98388
2701
1223
4694
951
1453
1055
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

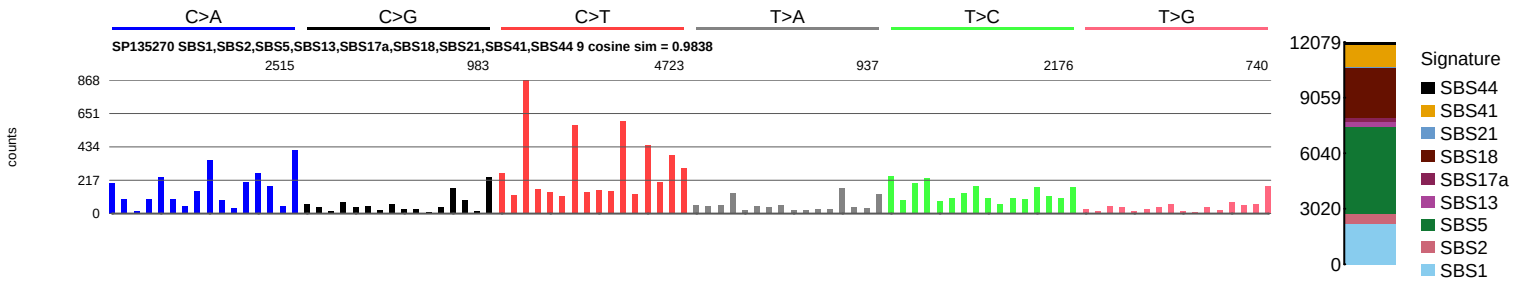

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS40 7 cosine sim = 0.98386
2703
1212
4701
943
1455
1064
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

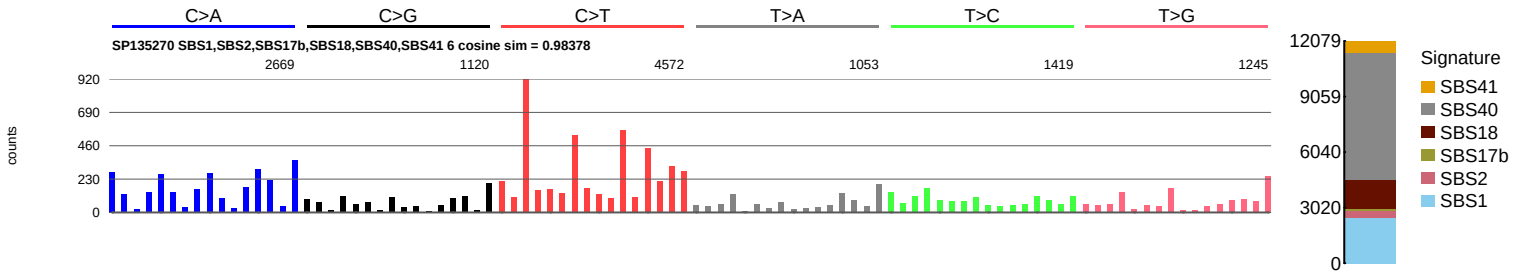

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS21,SBS41 8 cosine sim = 0.98386
2527
1027
4594
1117
1988
825
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1

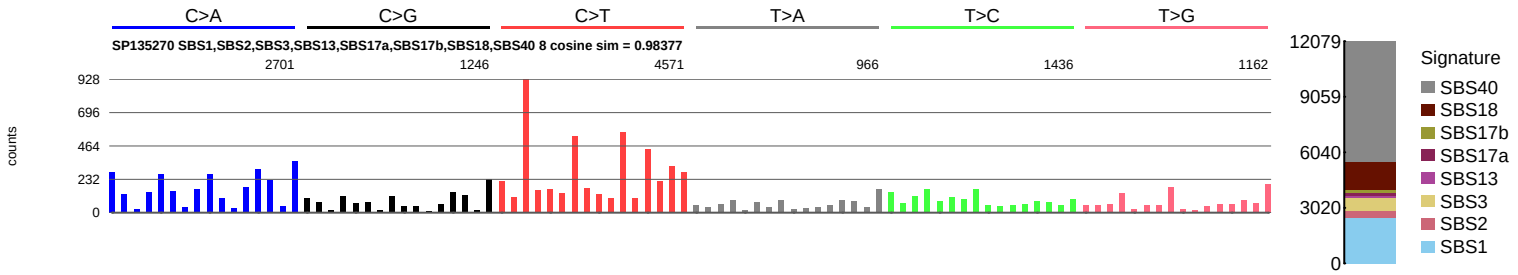

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.98386
2703
1088
4768
952
1516
1048
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1

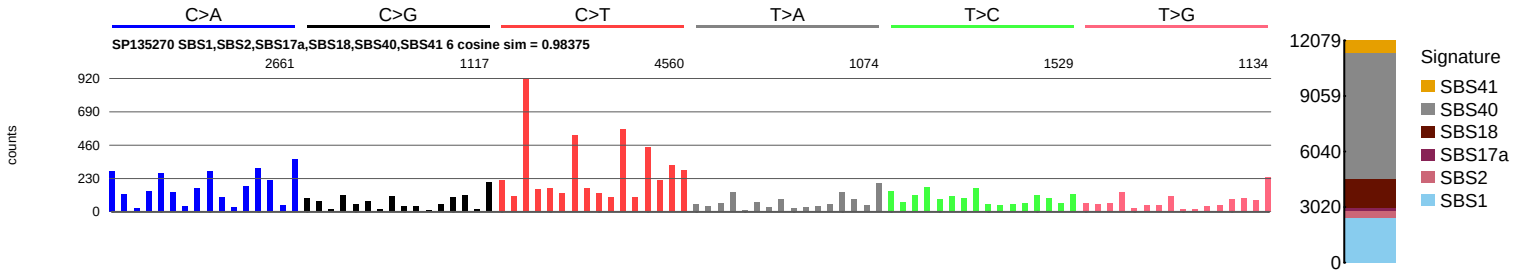

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS21,SBS40 5 cosine sim = 0.98386
2692
1115
4607
934
1671
1065
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS2
SBS1

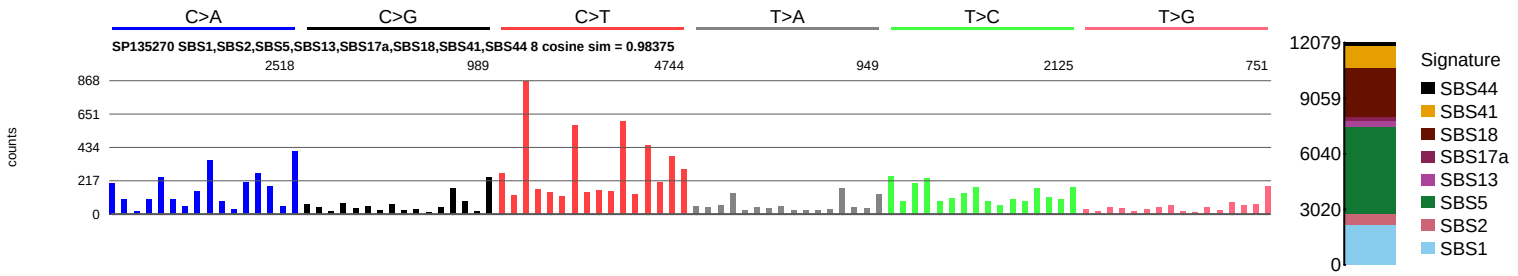

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS40,SBS44 6 cosine sim = 0.98384
2721
1195
4776
933
1426
1031
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9838
2515
983
4723
937
2176
740
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98378
2669
1120
4572
1053
1419
1245
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98377
2701
1246
4571
966
1436
1162
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS40,SBS41 6 cosine sim = 0.98375
2661
1117
4560
1074
1529
1134
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS2
SBS1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C>G
C>T
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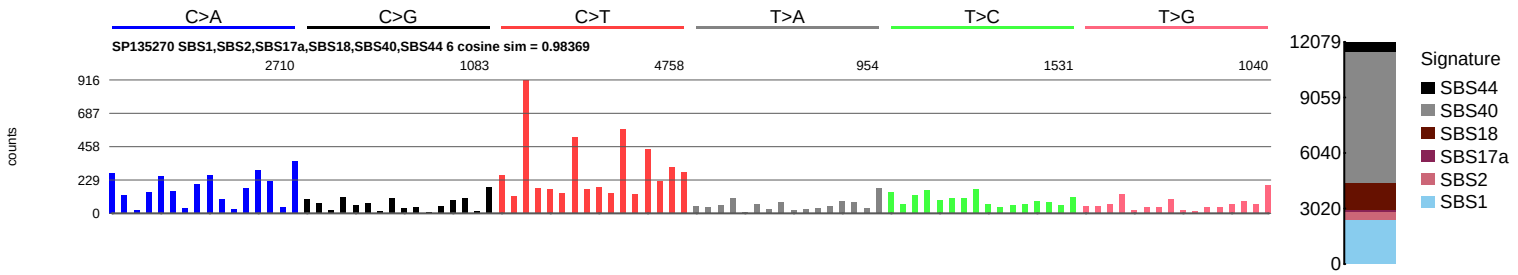
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS41,SBS44 8 cosine sim = 0.98375
2518
989
4744
949
2125
751
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
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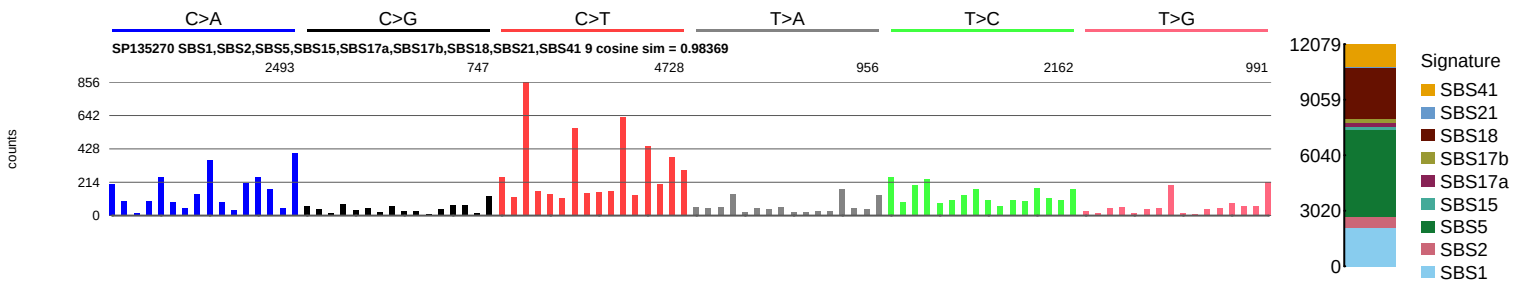
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS40,SBS44 6 cosine sim = 0.98369
2710
1083
4758
954
1531
1040
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS2
SBS1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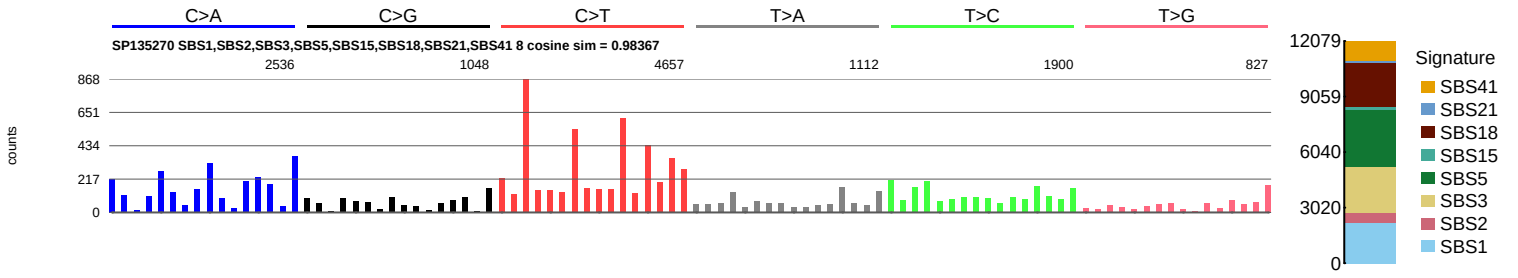
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98369
2493
747
4728
956
2162
991
counts
856
642
428
214
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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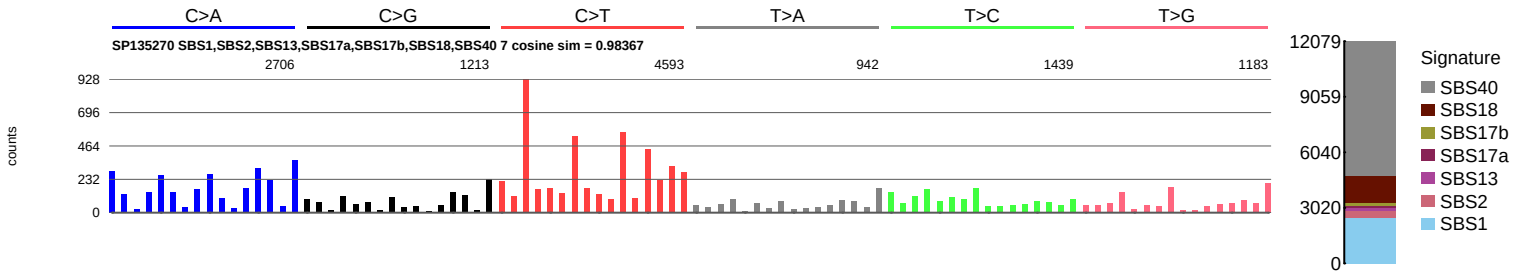
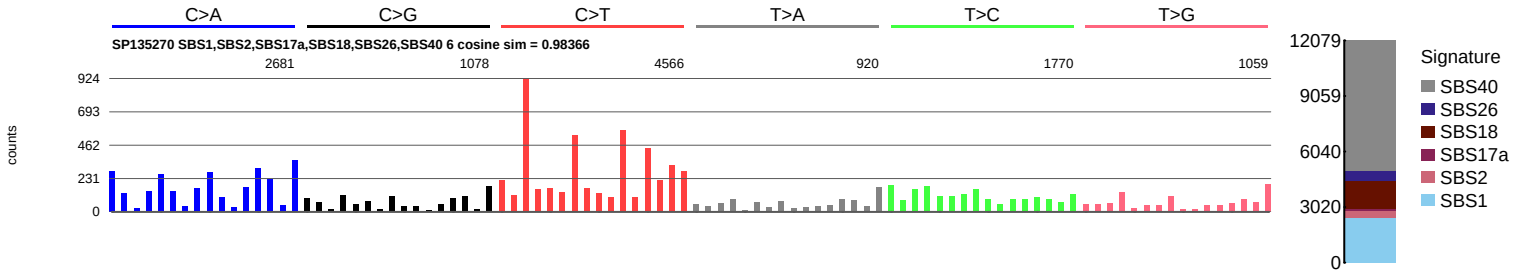
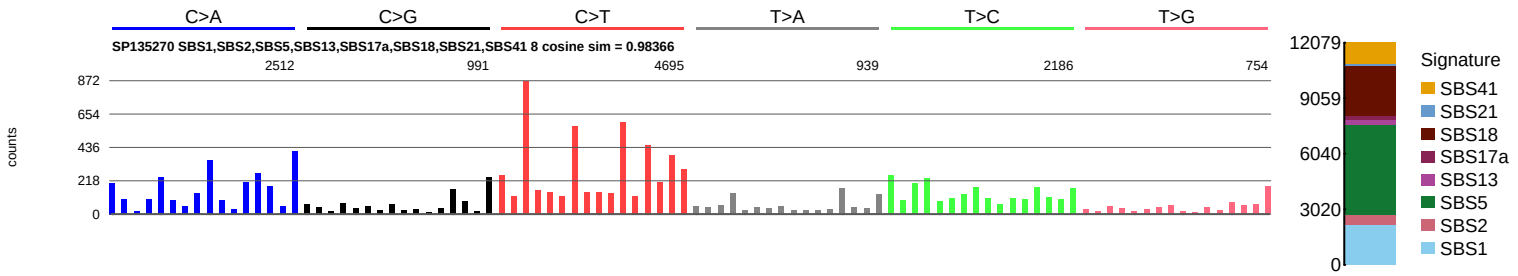
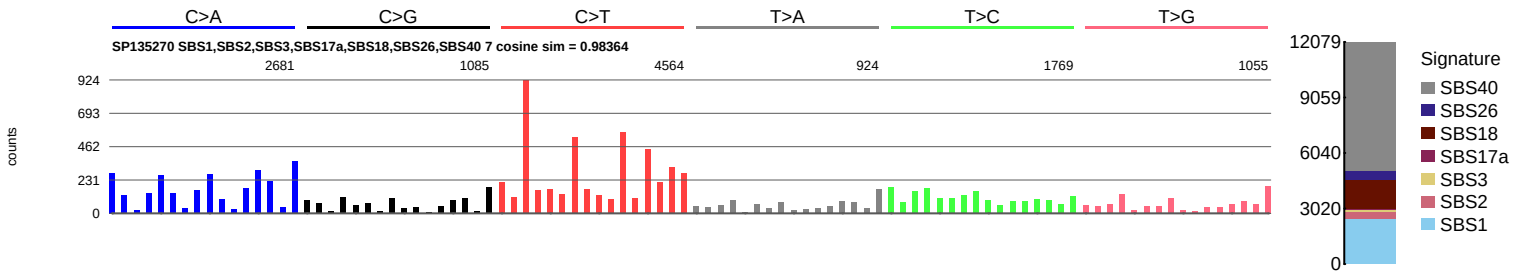
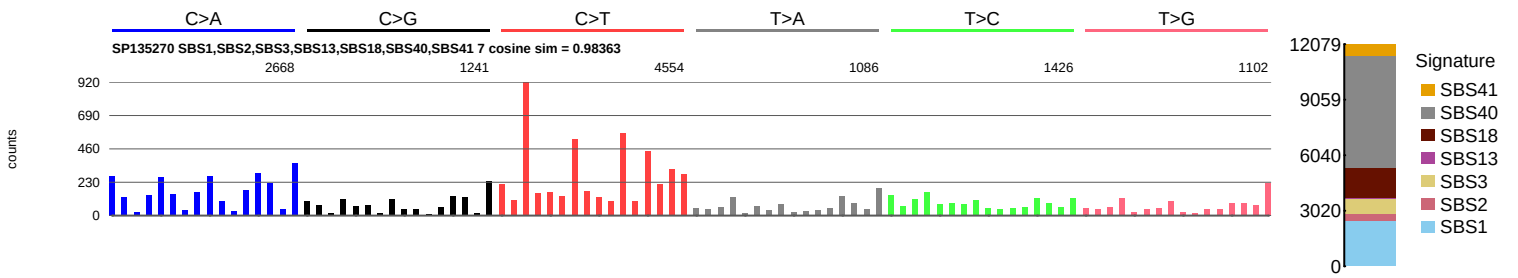
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS21,SBS41 8 cosine sim = 0.98367
2536
1048
4657
1112
1900
827
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1

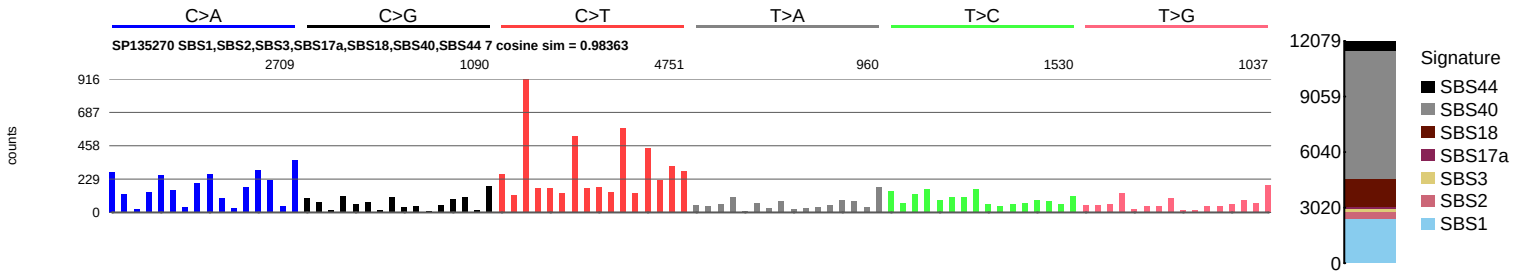

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98367
2706
1213
4593
942
1439
1183
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

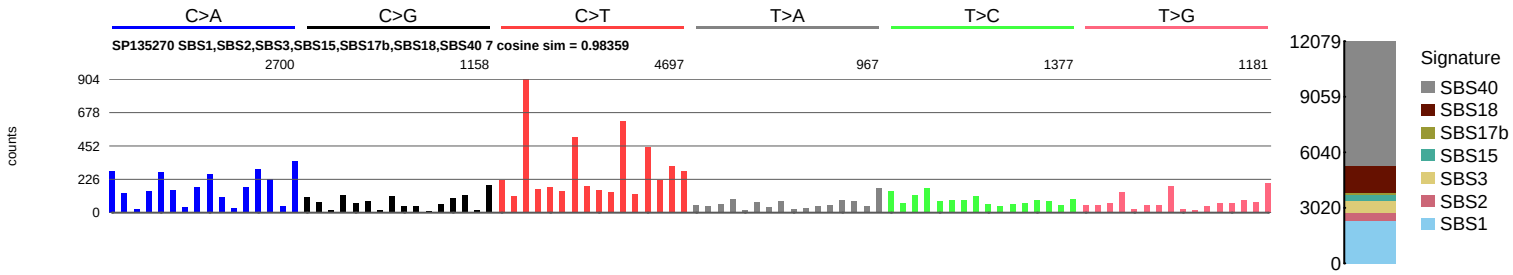

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40 6 cosine sim = 0.98366
2681
1078
4566
920
1770
1059
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1

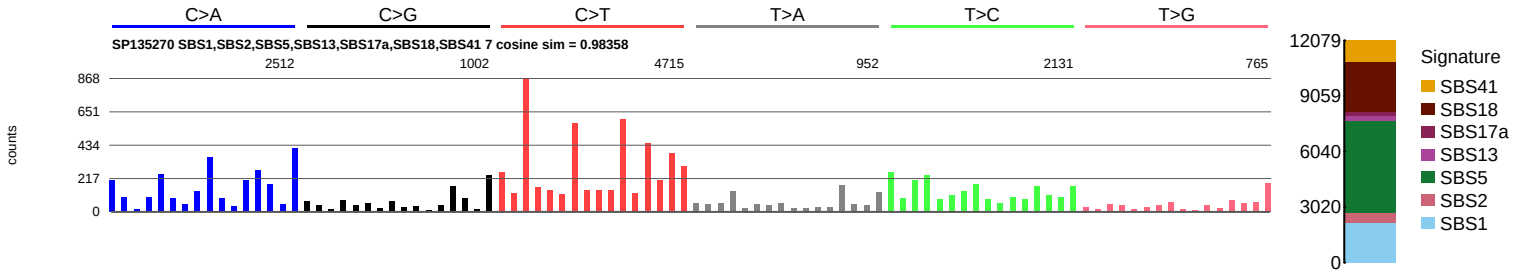

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS41 8 cosine sim = 0.98366
2512
991
4695
939
2186
754
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

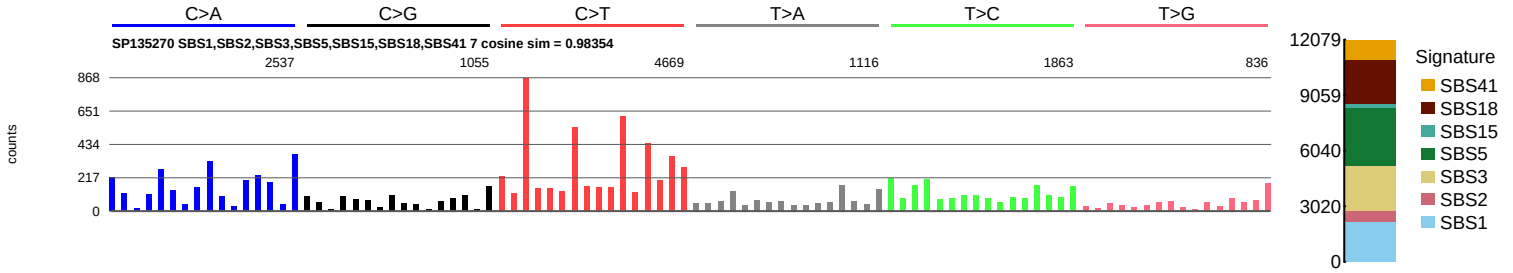

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98364
2681
1085
4564
924
1769
1055
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS3
SBS2
SBS1

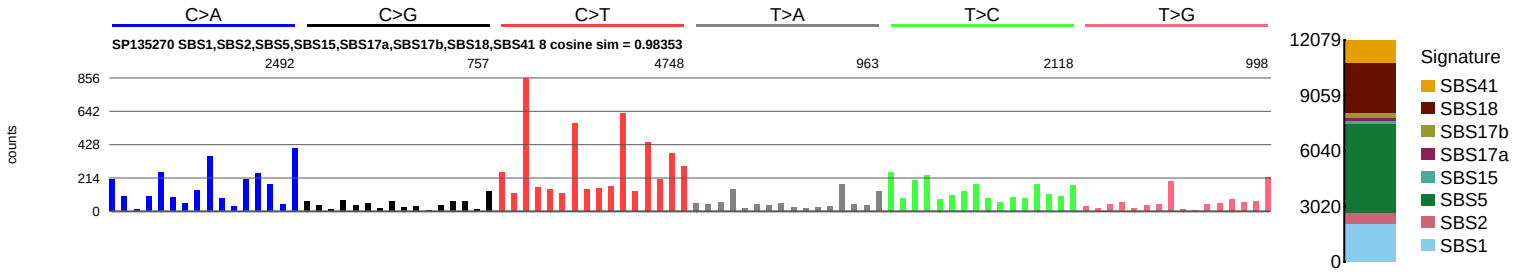

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40,SBS41 7 cosine sim = 0.98363
2668
1241
4554
1086
1426
1102
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1

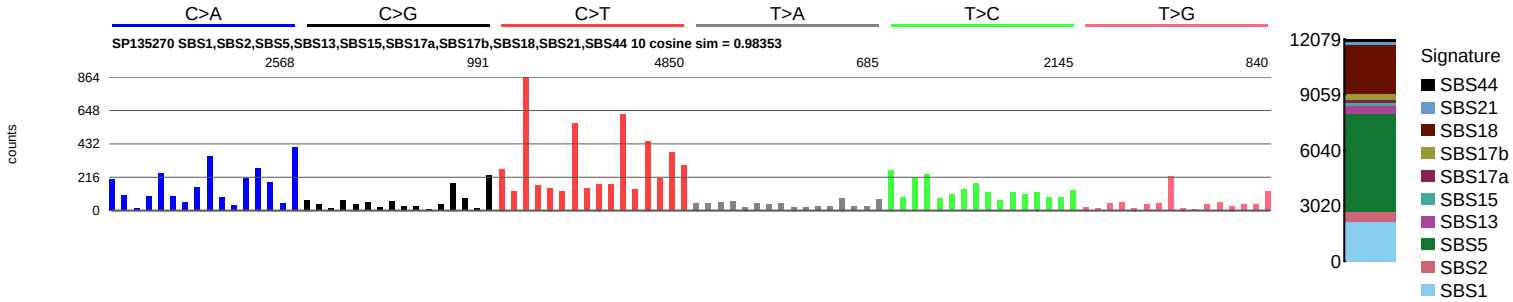

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.98363
2709
1090
4751
960
1530
1037
916
687
458
229
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS3
SBS2
SBS1

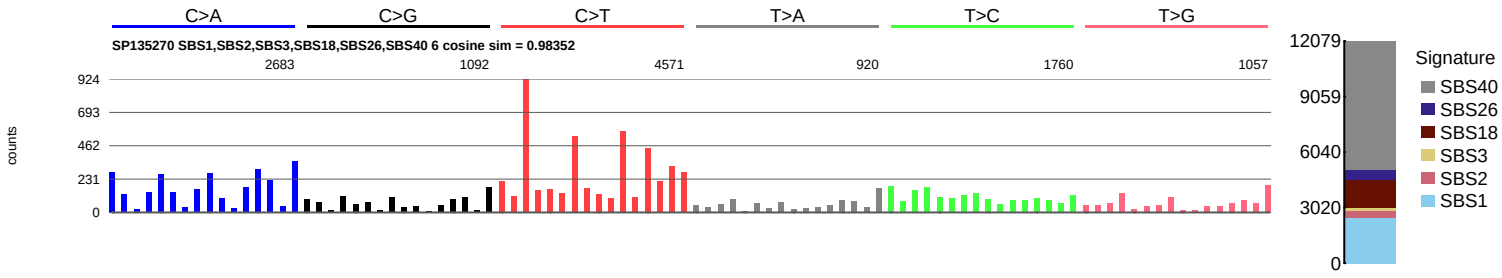

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98359
2700
1158
4697
967
1377
1181
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

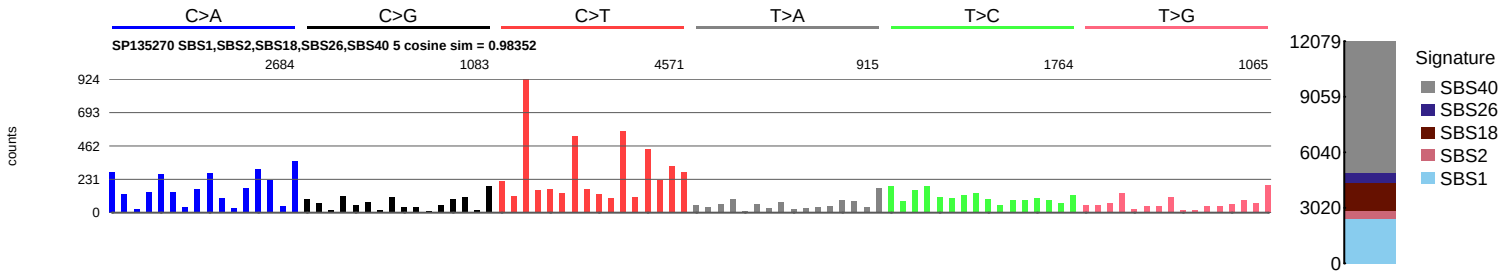

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS41 7 cosine sim = 0.98358
2512
1002
4715
952
2131
765
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

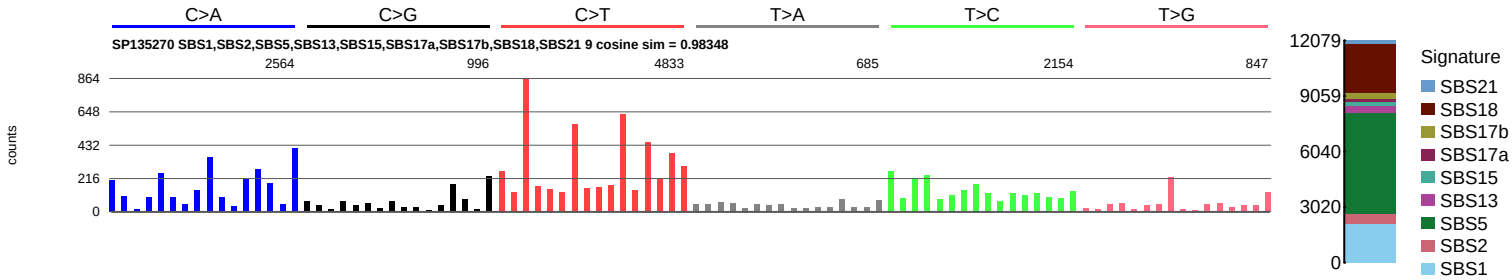

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS41 7 cosine sim = 0.98354
2537
1055
4669
1116
1863
836
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1

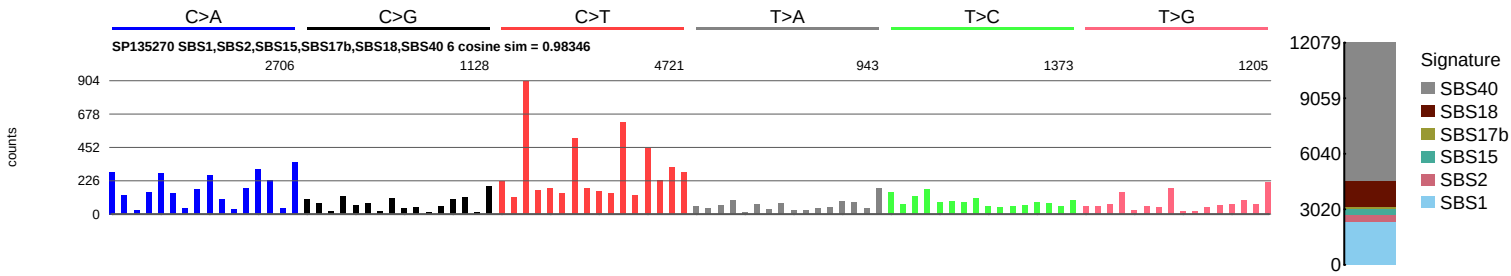

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98353
2492
757
4748
963
2118
998
856
642
428
214
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

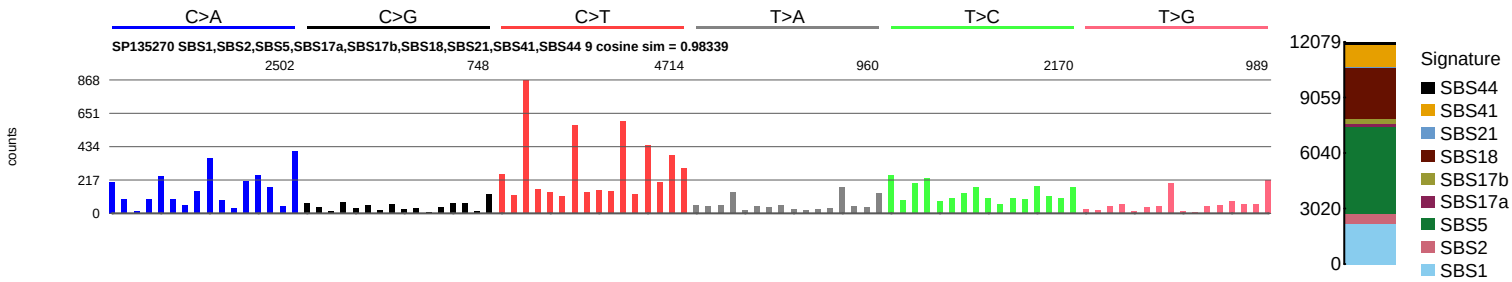

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98353
2568
991
4850
685
2145
840
864
648
432
216
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

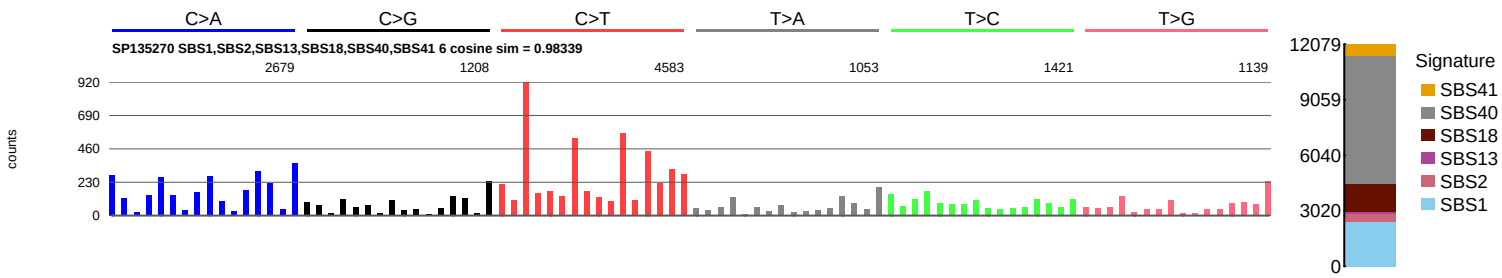

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS26,SBS40 6 cosine sim = 0.98352
2683
1092
4571
920
1760
1057
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS3
SBS2
SBS1

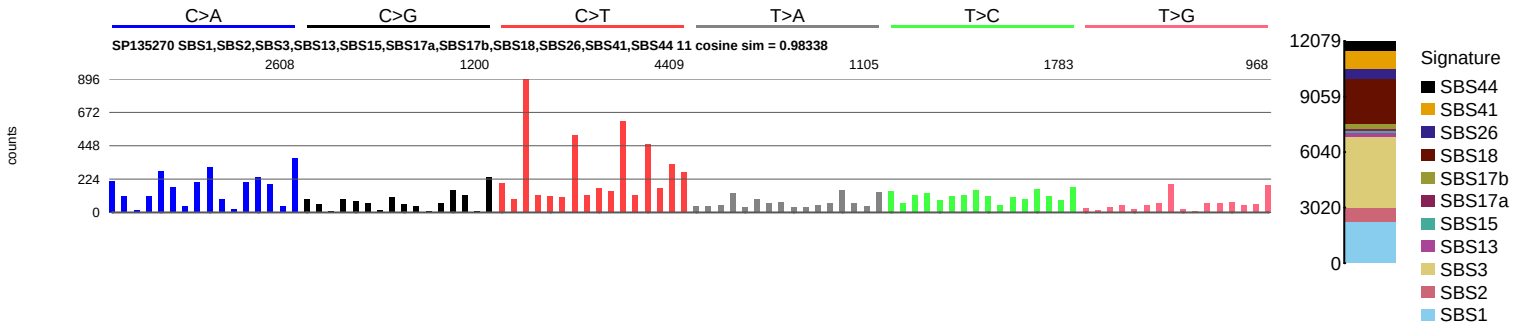

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS26,SBS40 5 cosine sim = 0.98352
2684
1083
4571
915
1764
1065
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS26
SBS18
SBS2
SBS1

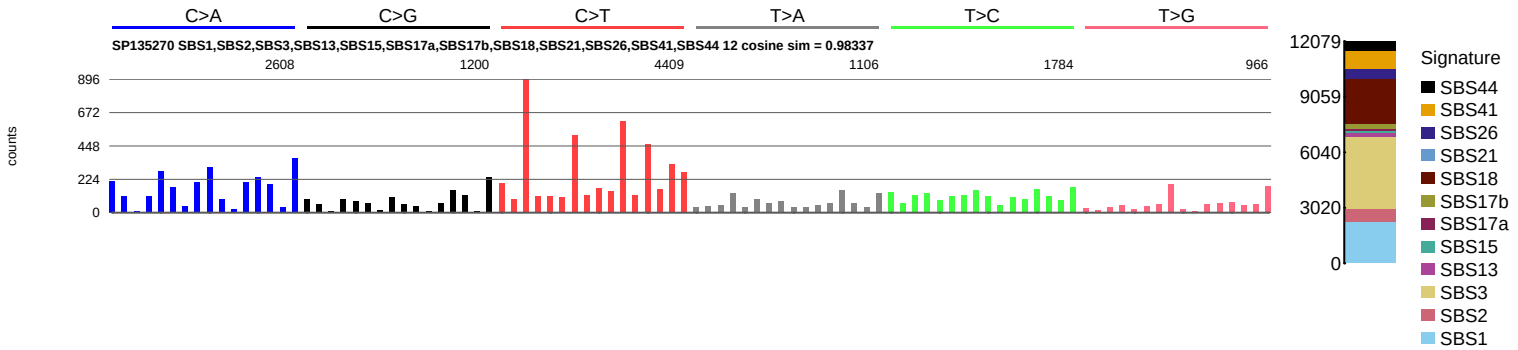

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98348
2564
996
4833
685
2154
847
counts
864
648
432
216
0
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

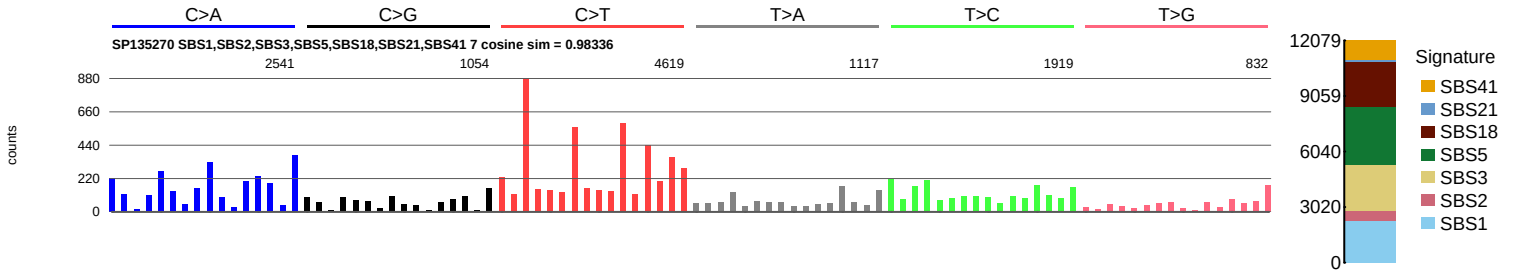

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40 6 cosine sim = 0.98346
2706
1128
4721
943
1373
1205
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1

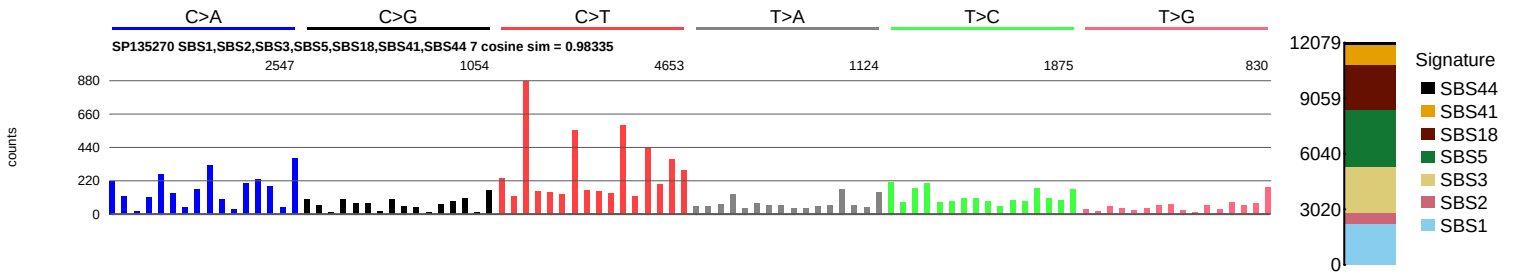

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98339
2502
748
4714
960
2170
989
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

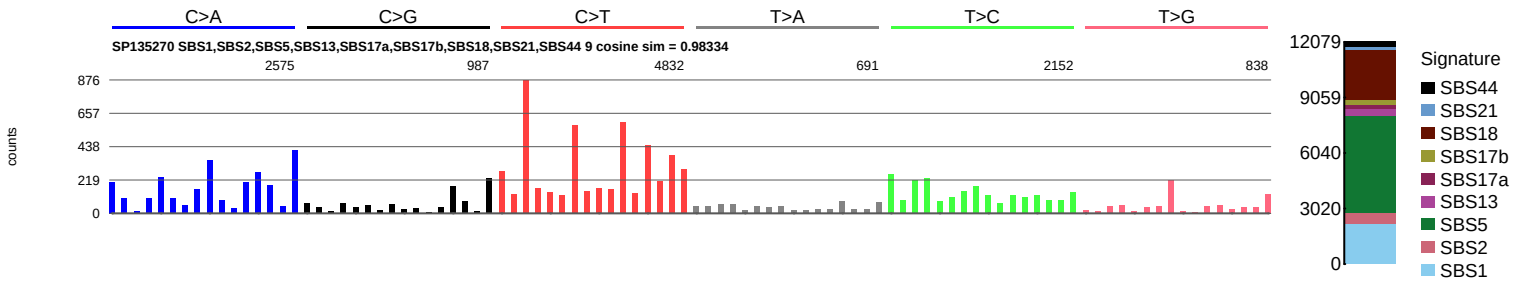

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS18,SBS40,SBS41 6 cosine sim = 0.98339
2679
1208
4583
1053
1421
1139
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98338
2608
1200
4409
1105
1783
968
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 12 cosine sim = 0.98337
2608
1200
4409
1106
1784
966
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS18,SBS21,SBS41 7 cosine sim = 0.98336
2541
1054
4619
1117
1919
832
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS18,SBS41,SBS44 7 cosine sim = 0.98335
2547
1054
4653
1124
1875
830
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS5
SBS3
SBS2
SBS1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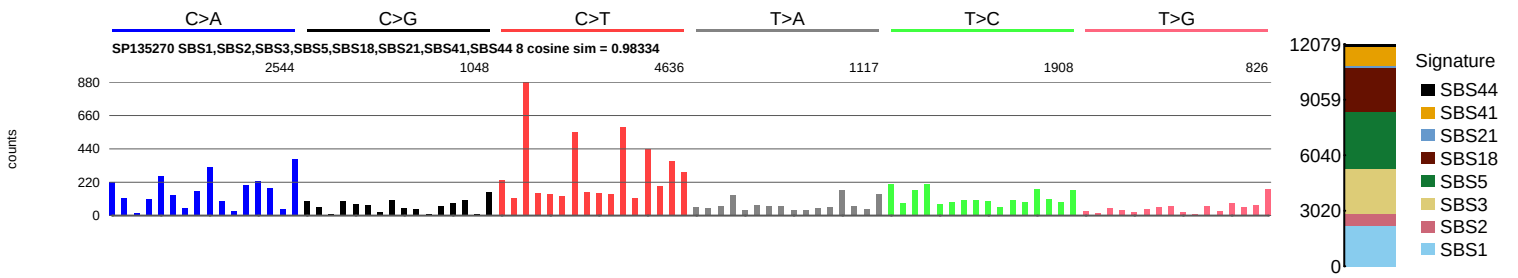
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98334
2575
987
4832
691
2152
838
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
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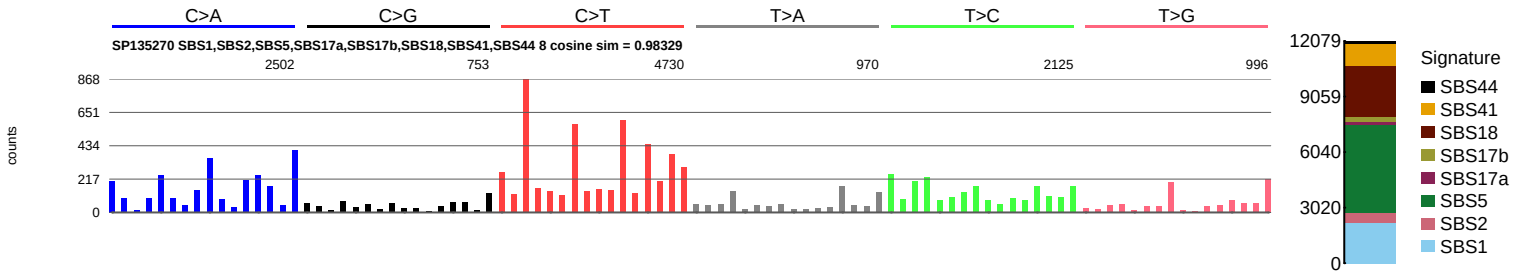
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98334
2544
1048
4636
1117
1908
826
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS5
SBS3
SBS2
SBS1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C>T
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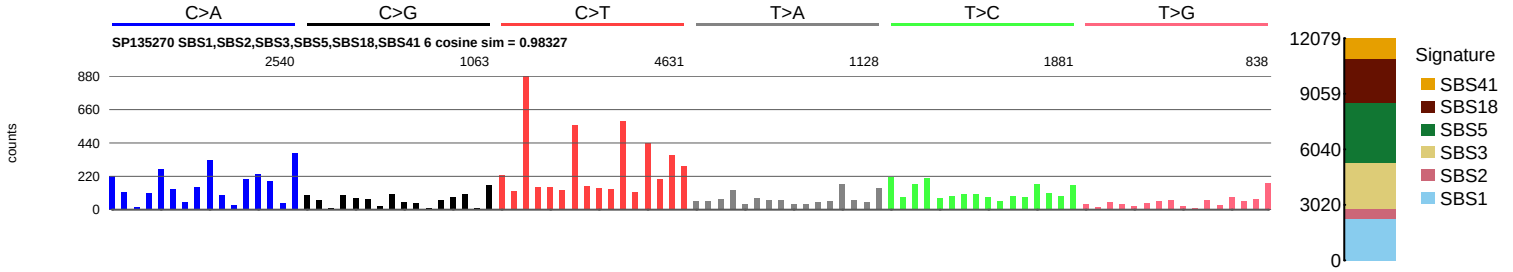
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98329
2502
753
4730
970
2125
996
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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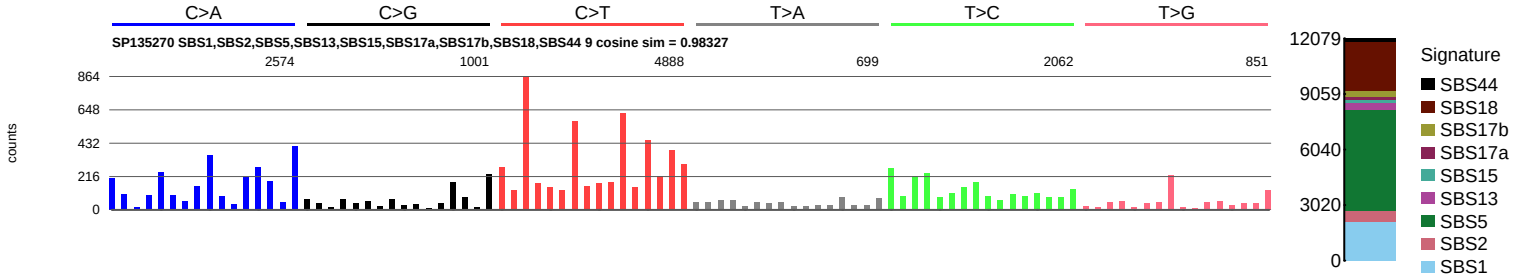
SP135270 SBS1,SBS2,SBS3,SBS5,SBS18,SBS41 6 cosine sim = 0.98327
2540
1063
4631
1128
1881
838
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS5
SBS3
SBS2
SBS1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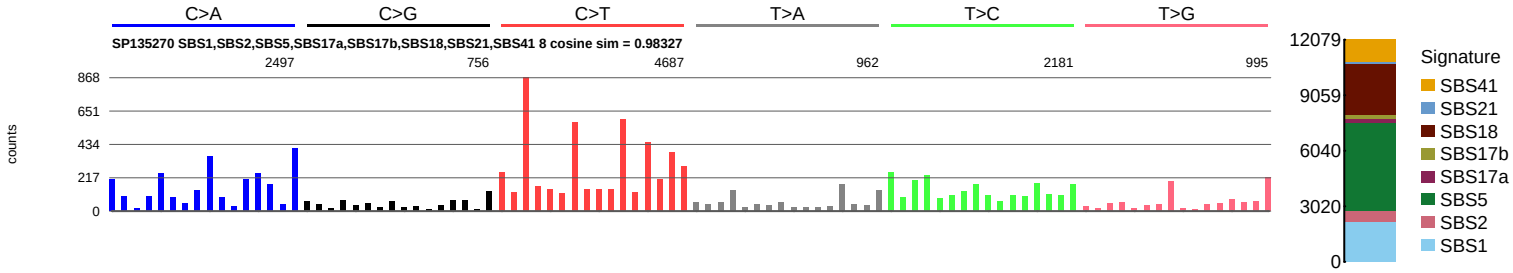
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98327
2574
1001
4888
699
2062
851
counts
864
648
432
216
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98327
2497
756
4687
962
2181
995
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
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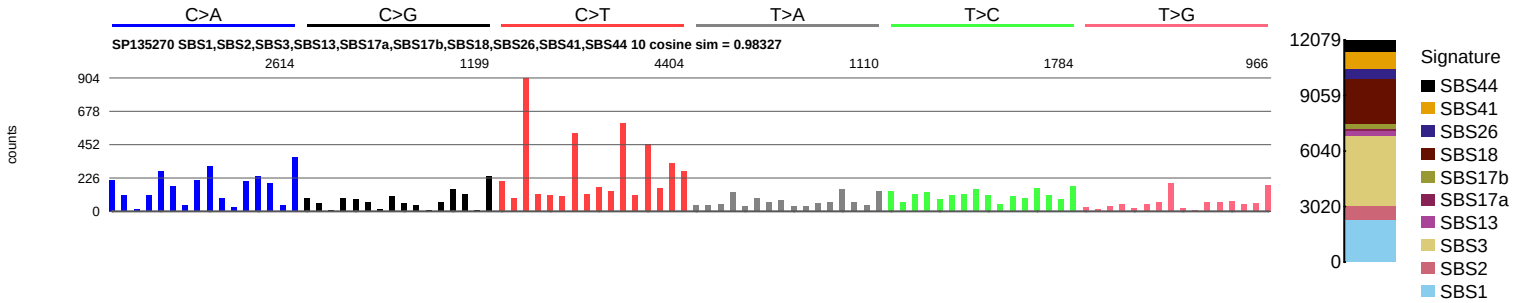
SBS5
SBS2
SBS1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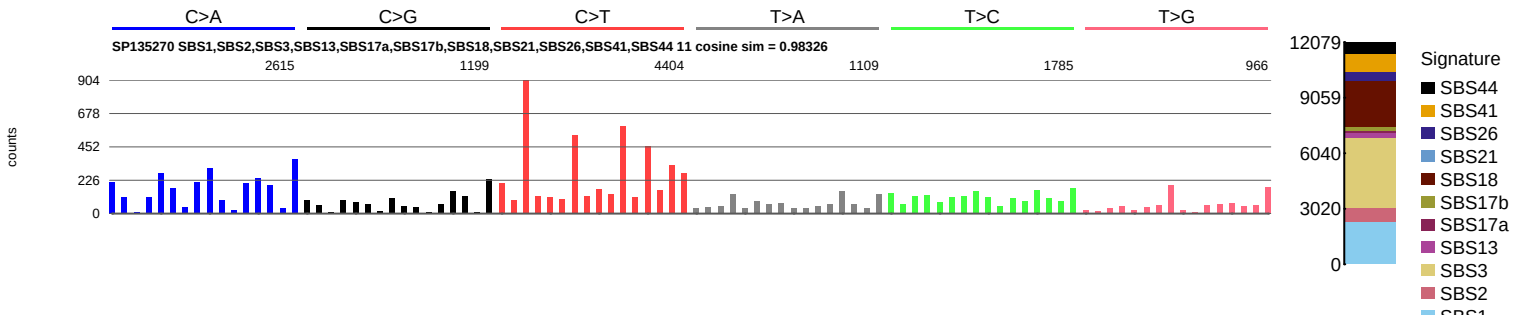
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98327
2614
1199
4404
1110
1784
966
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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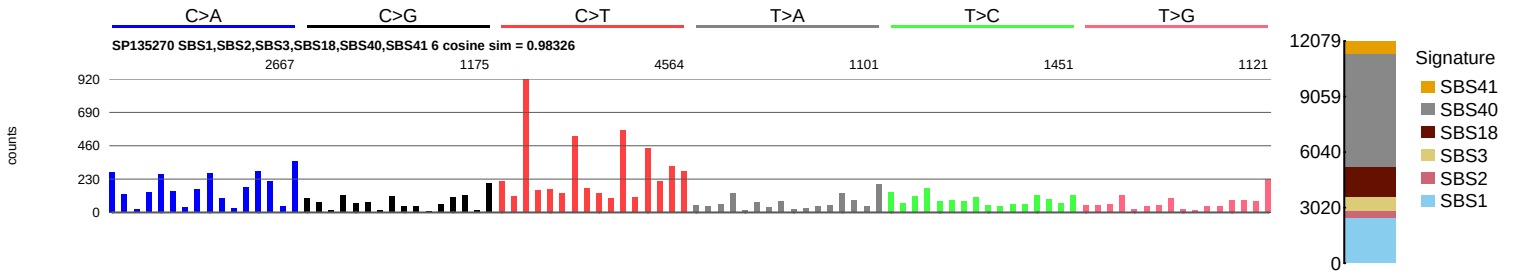
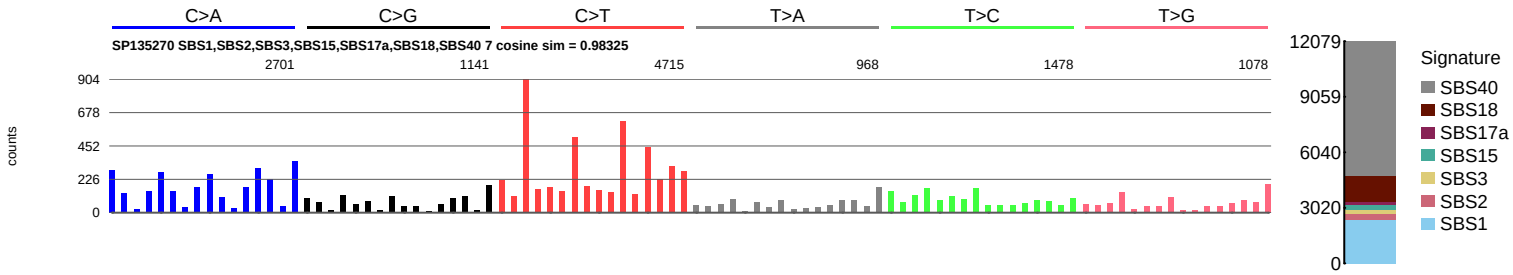
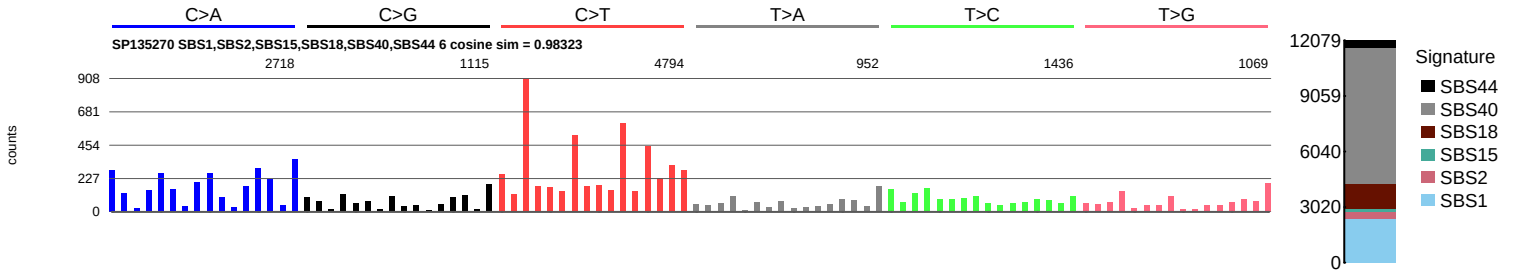
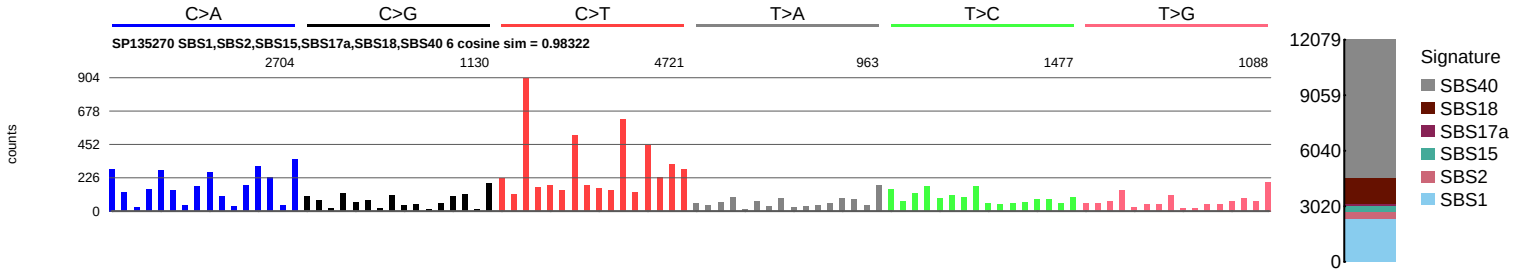
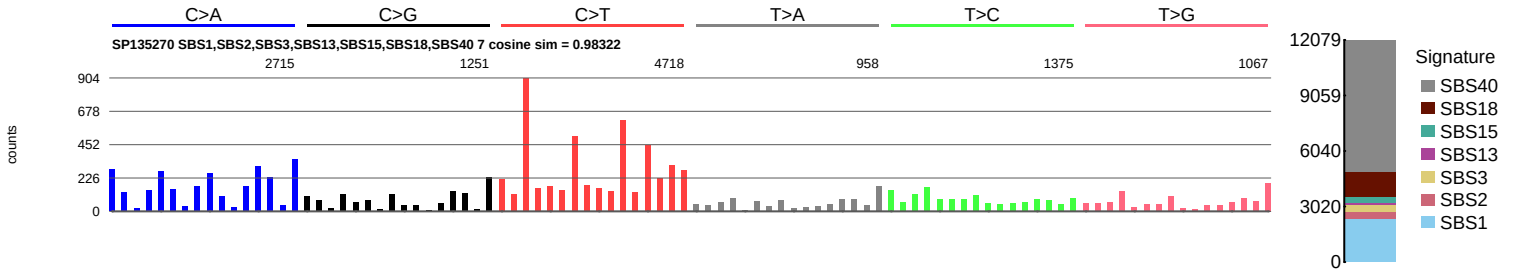
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98326
2615
1199
4404
1109
1785
966
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

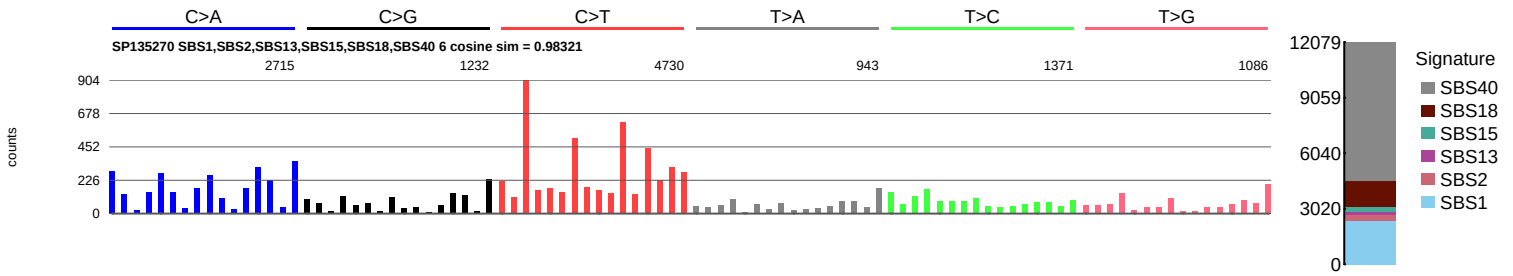

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS40,SBS41 6 cosine sim = 0.98326
2667
1175
4564
1101
1451
1121
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS3
SBS2
SBS1

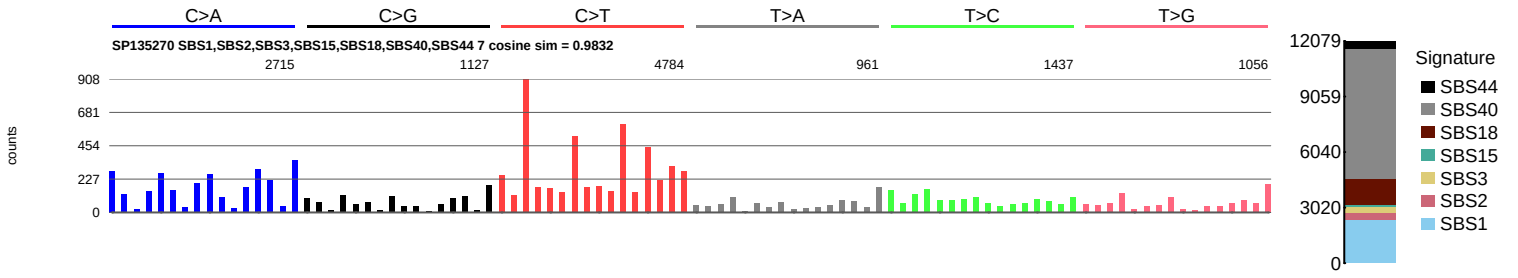

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS40 7 cosine sim = 0.98325
2701
1141
4715
968
1478
1078
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1

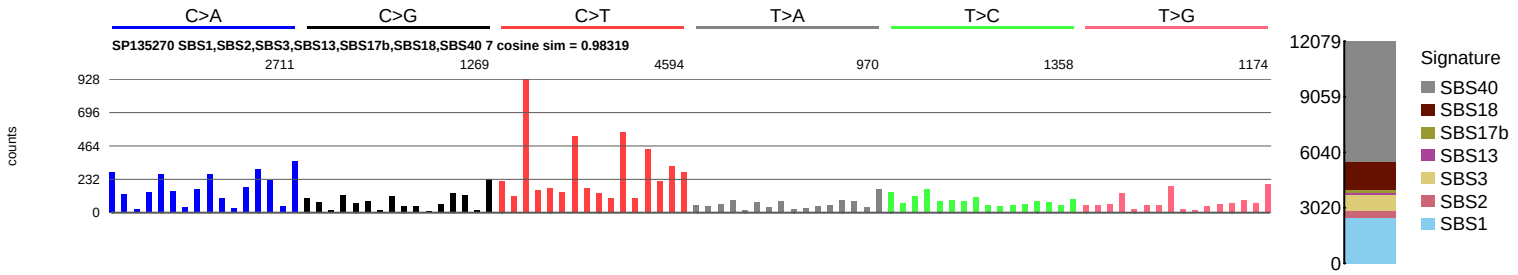

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS40,SBS44 6 cosine sim = 0.98323
2718
1115
4794
952
1436
1069
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40 6 cosine sim = 0.98322
2704
1130
4721
963
1477
1088
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS18,SBS40 7 cosine sim = 0.98322
2715
1251
4718
958
1375
1067
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS15,SBS18,SBS40 6 cosine sim = 0.98321
2715
1232
4730
943
1371
1086
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS40,SBS44 7 cosine sim = 0.9832
2715
1127
4784
961
1437
1056
counts
908
681
454
227
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS3
SBS2
SBS1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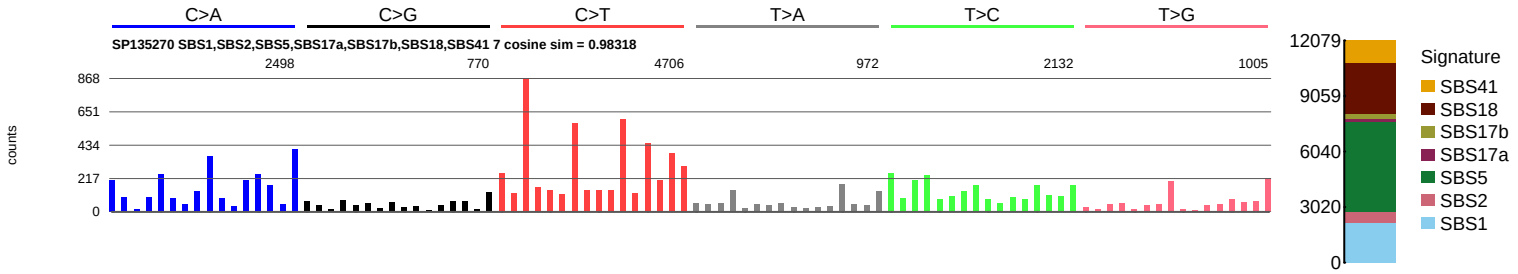
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS40 7 cosine sim = 0.98319
2711
1269
4594
970
1358
1174
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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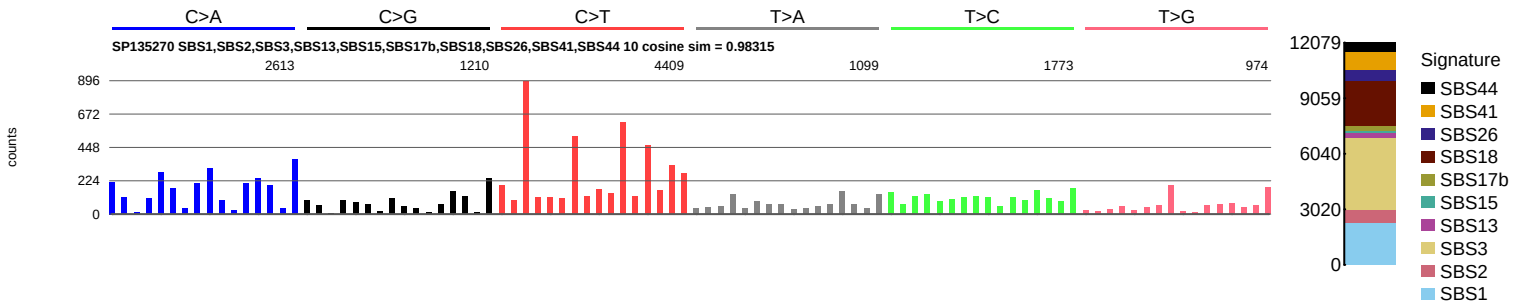
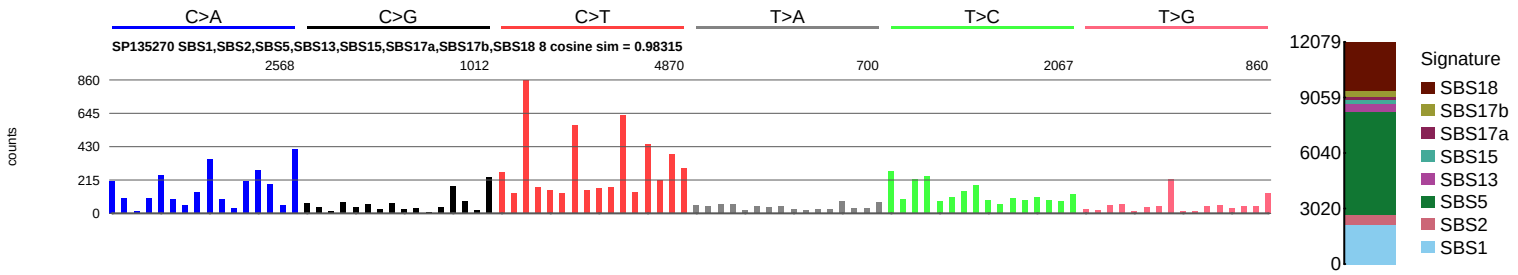
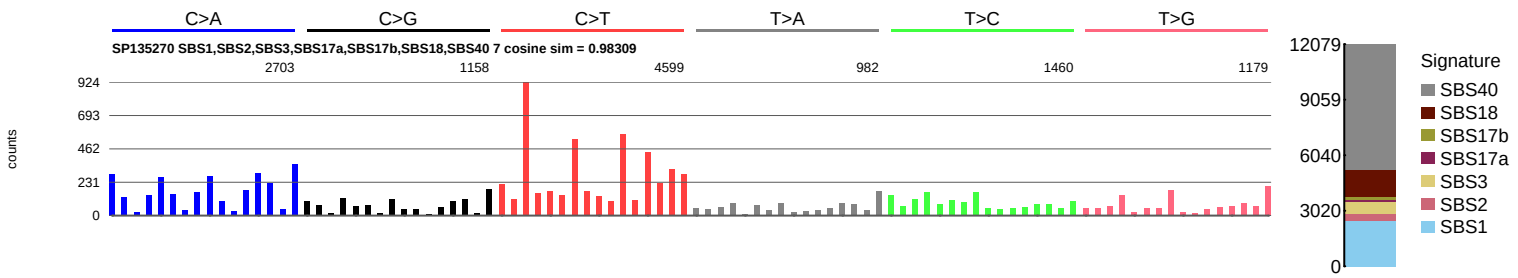
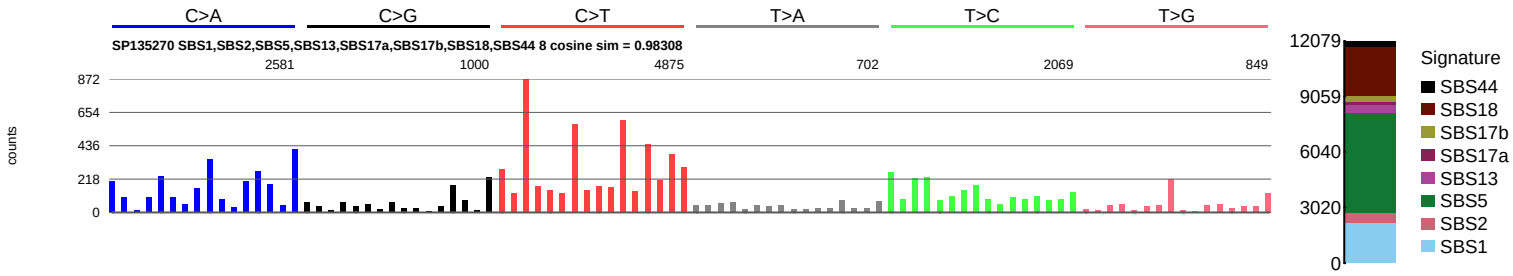
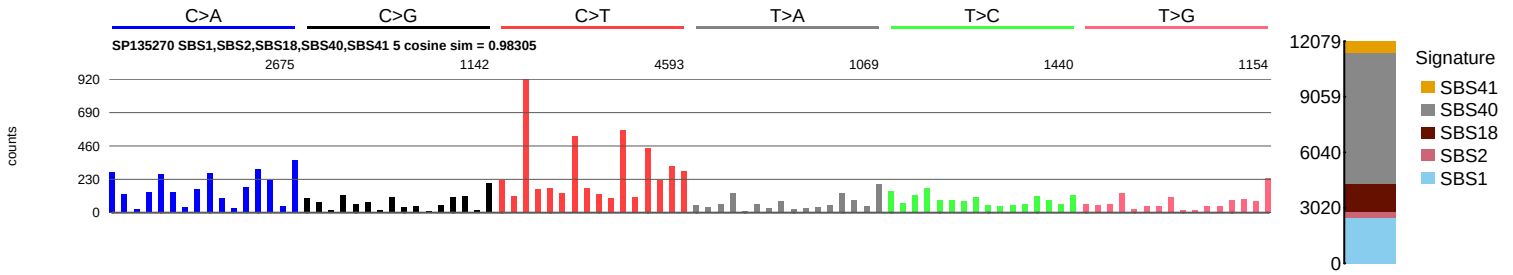
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98318
2498
770
4706
972
2132
1005
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

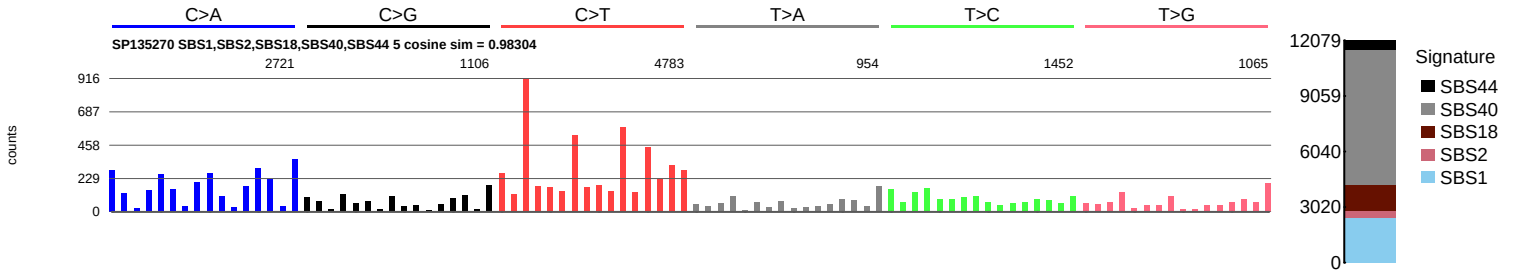

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98315
2613
1210
4409
1099
1773
974
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

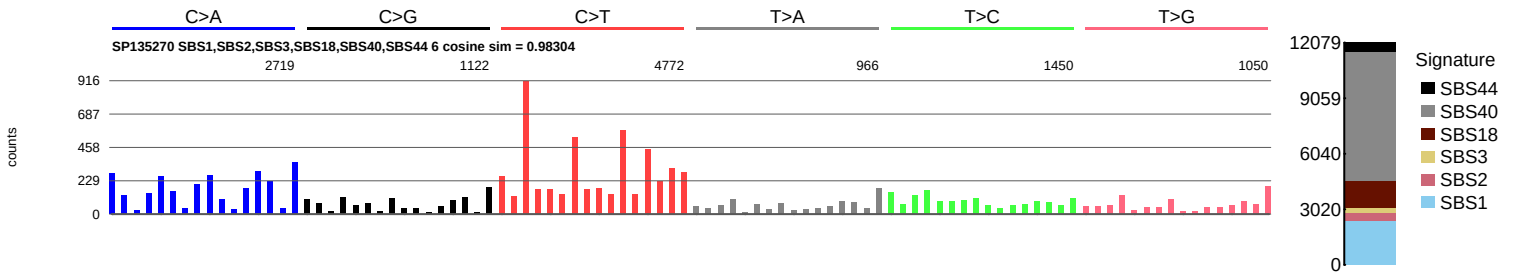

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98315
2568
1012
4870
700
2067
860
counts
860
645
430
215
0
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

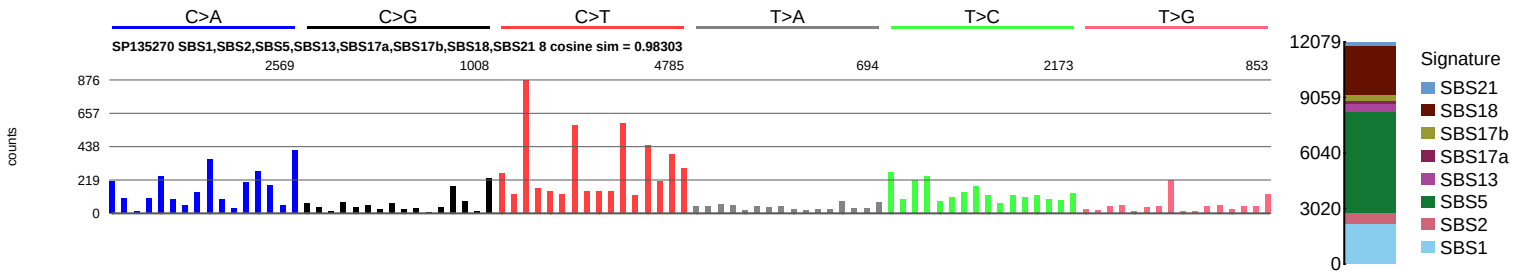

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98309
2703
1158
4599
982
1460
1179
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

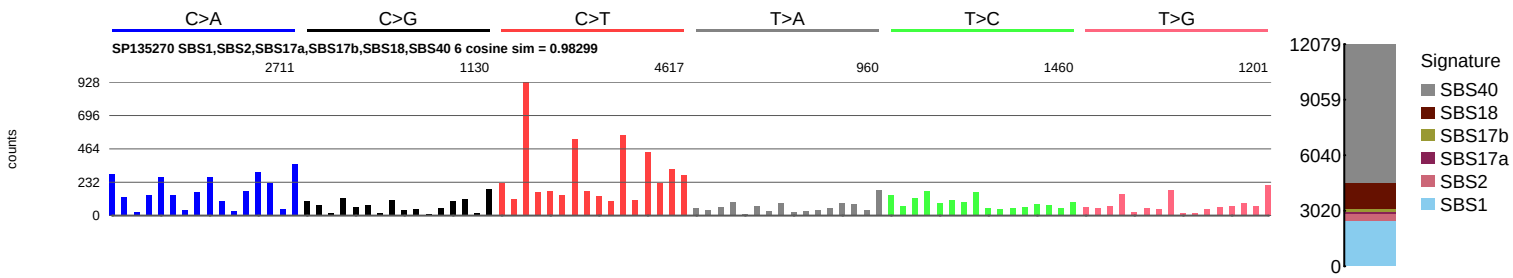

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98308
2581
1000
4875
702
2069
849
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

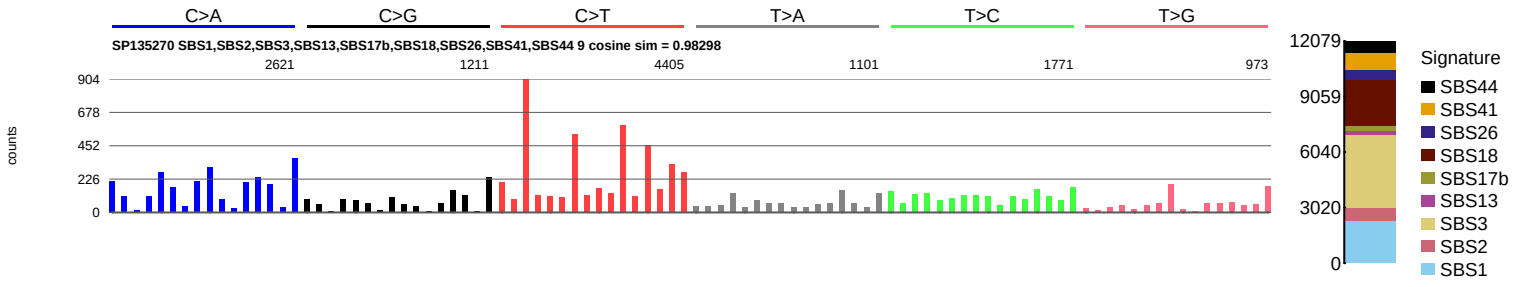

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS40,SBS41 5 cosine sim = 0.98305
2675
1142
4593
1069
1440
1154
counts
920
690
460
230
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS2
SBS1

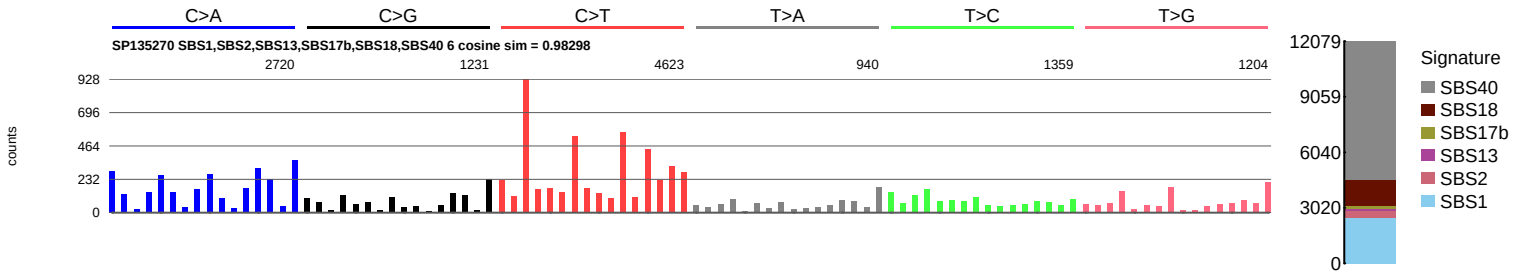

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS40,SBS44 5 cosine sim = 0.98304
2721
1106
4783
954
1452
1065
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS2
SBS1

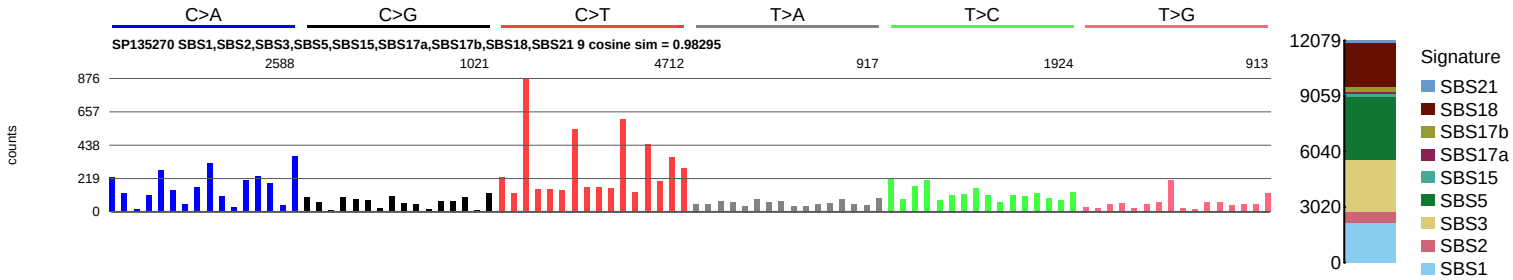

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS40,SBS44 6 cosine sim = 0.98304
2719
1122
4772
966
1450
1050
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS44
SBS40
SBS18
SBS3
SBS2
SBS1

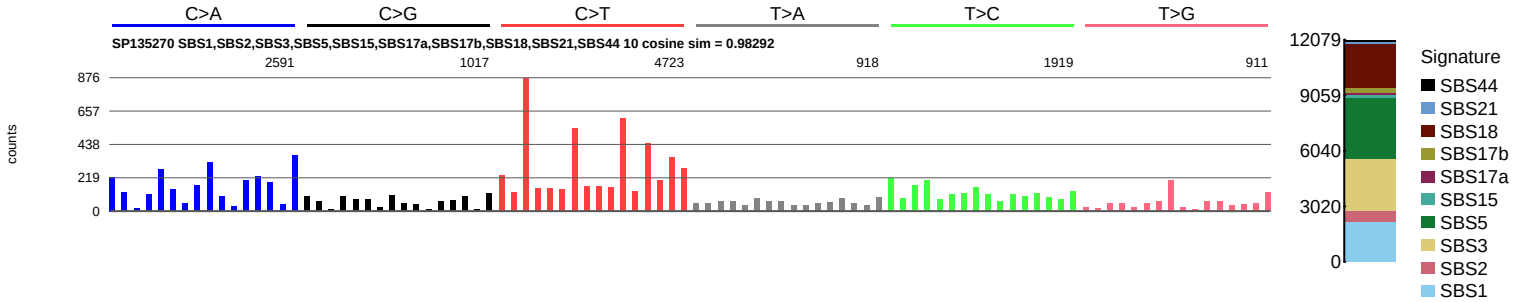

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98303
2569
1008
4785
694
2173
853
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

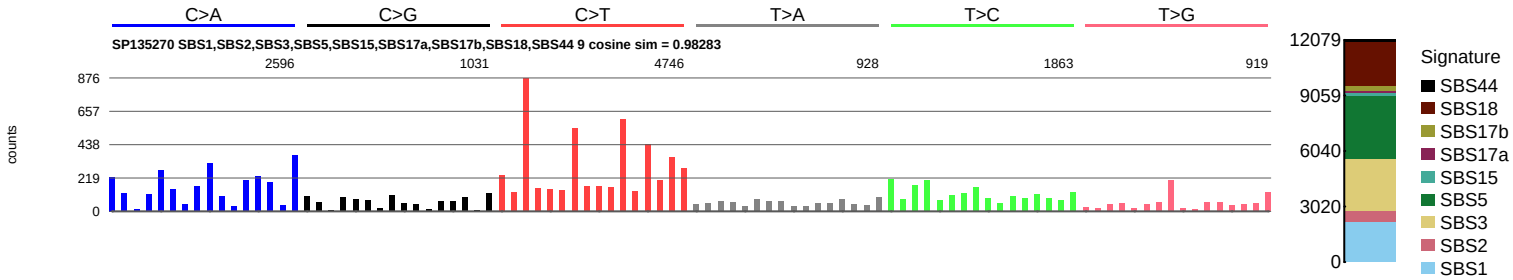

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98299
2711
1130
4617
960
1460
1201
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

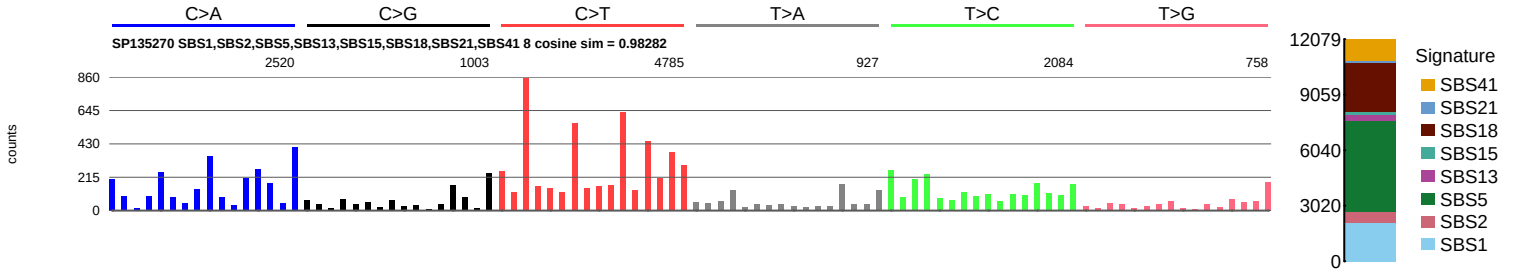

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98298
2621
1211
4405
1101
1771
973
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1

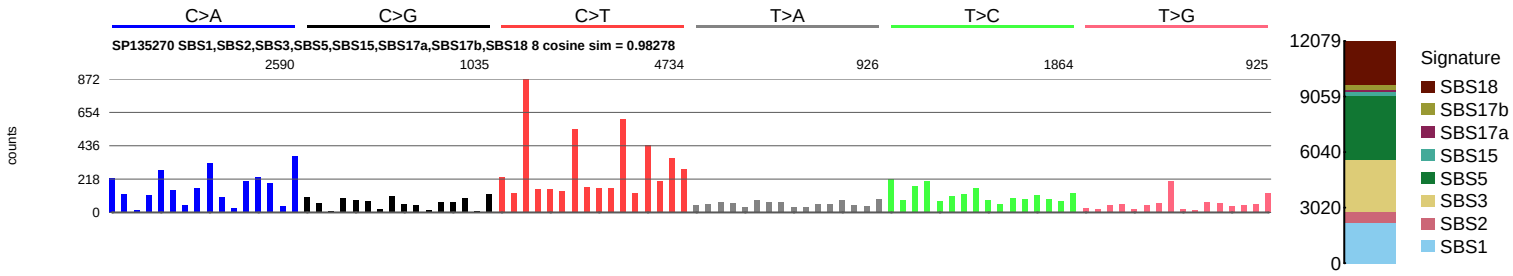

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40 6 cosine sim = 0.98298
2720
1231
4623
940
1359
1204
928
696
464
232
0
counts
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1

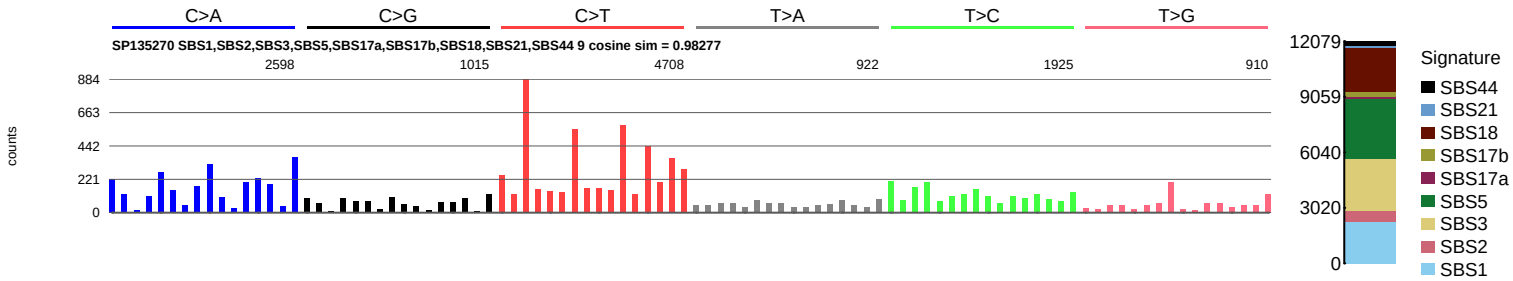

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98295
2588
1021
4712
917
1924
913
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

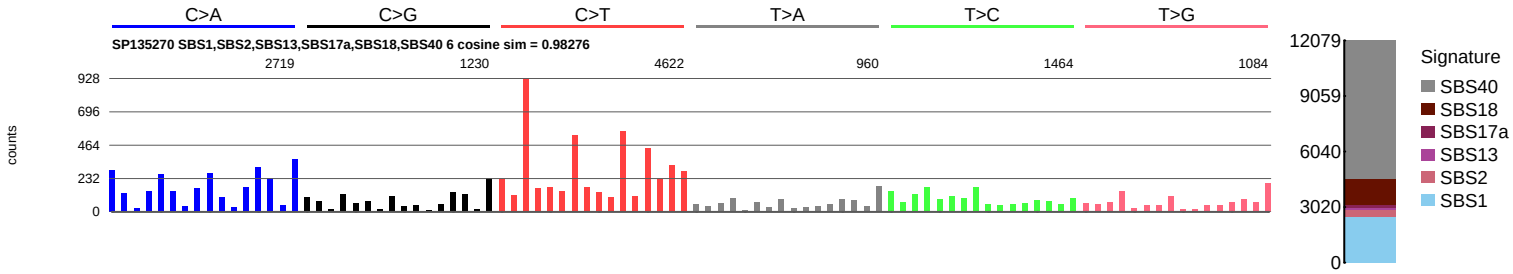

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98292
2591
1017
4723
918
1919
911
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

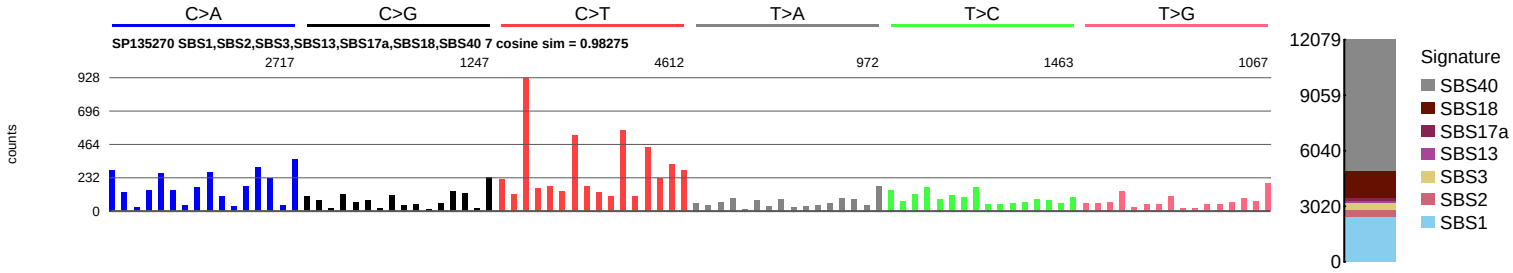

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98283
2596
1031
4746
928
1863
919
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

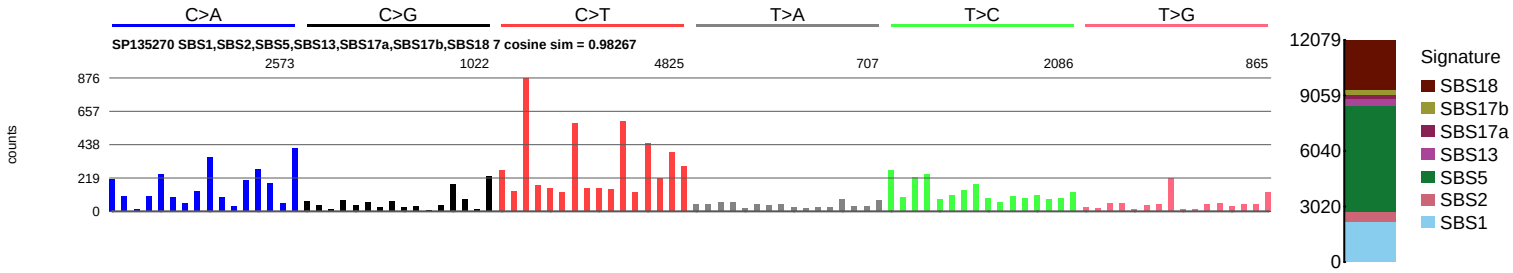

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS41 8 cosine sim = 0.98282
2520
1003
4785
927
2084
758
860
645
430
215
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1

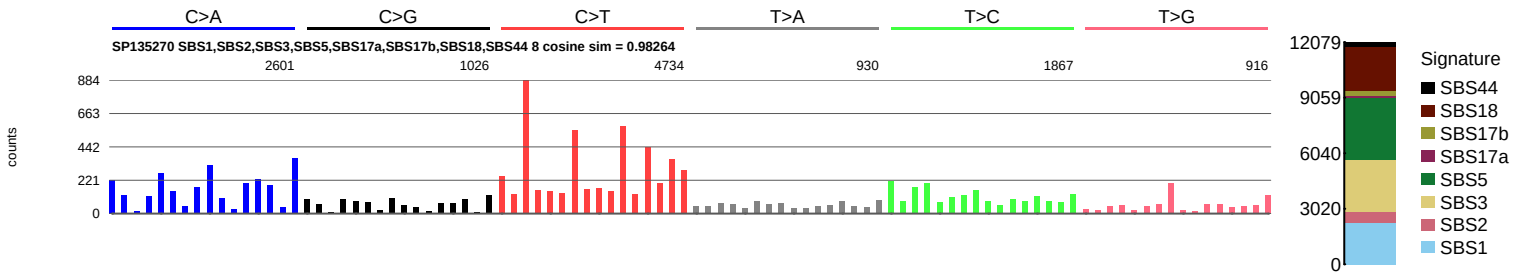

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98278
2590
1035
4734
926
1864
925
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

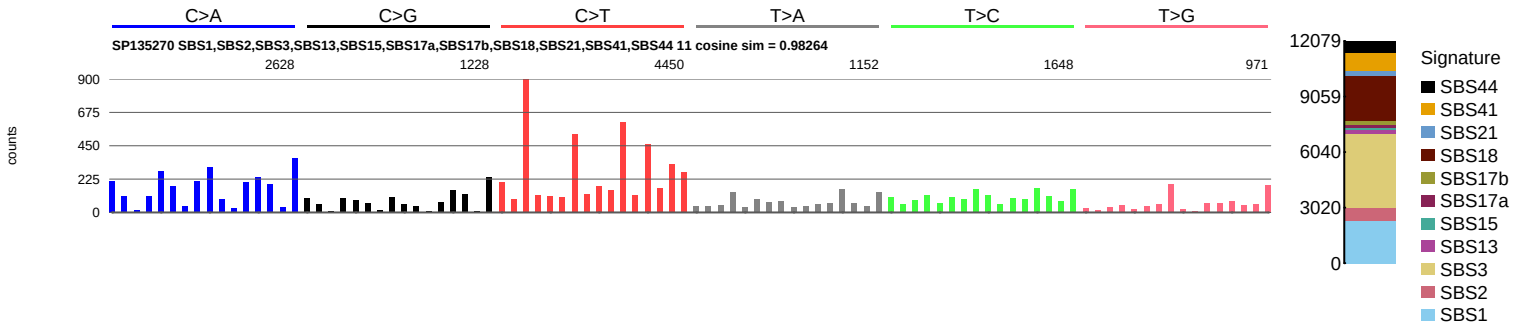

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98277
2598
1015
4708
922
1925
910
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

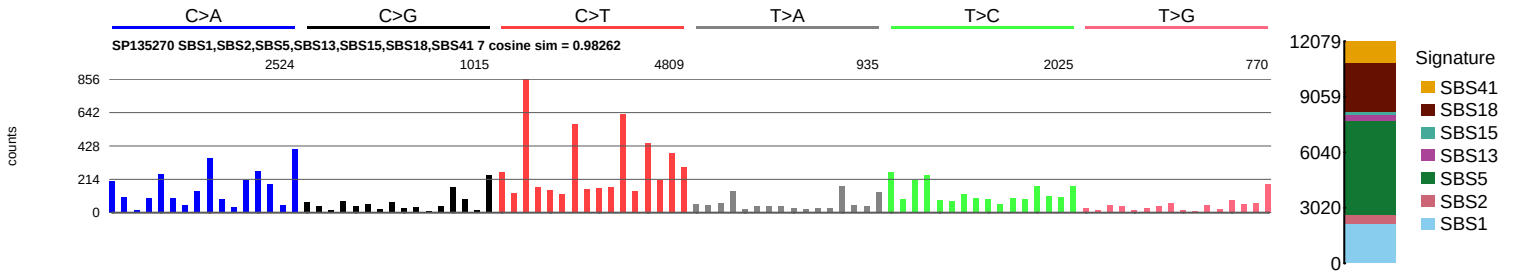

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40 6 cosine sim = 0.98276
2719
1230
4622
960
1464
1084
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1

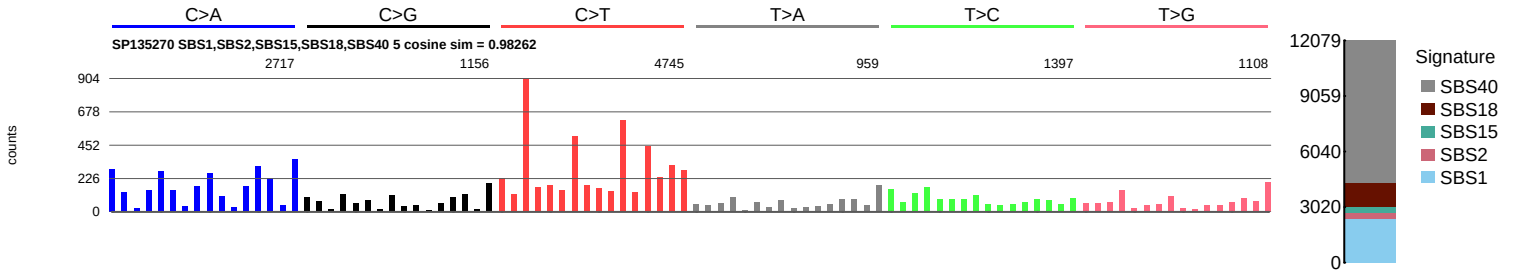

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS40 7 cosine sim = 0.98275
2717
1247
4612
972
1463
1067
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1

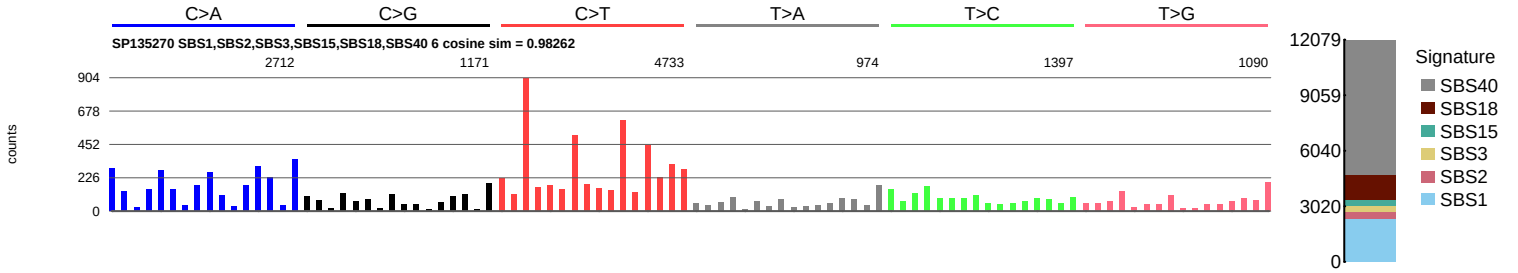

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98267
2573
1022
4825
707
2086
865
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

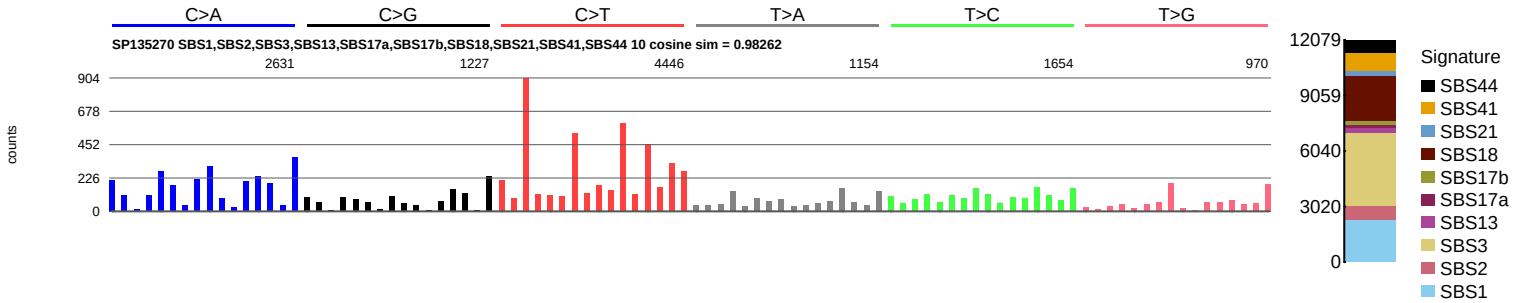

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98264
2601
1026
4734
930
1867
916
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

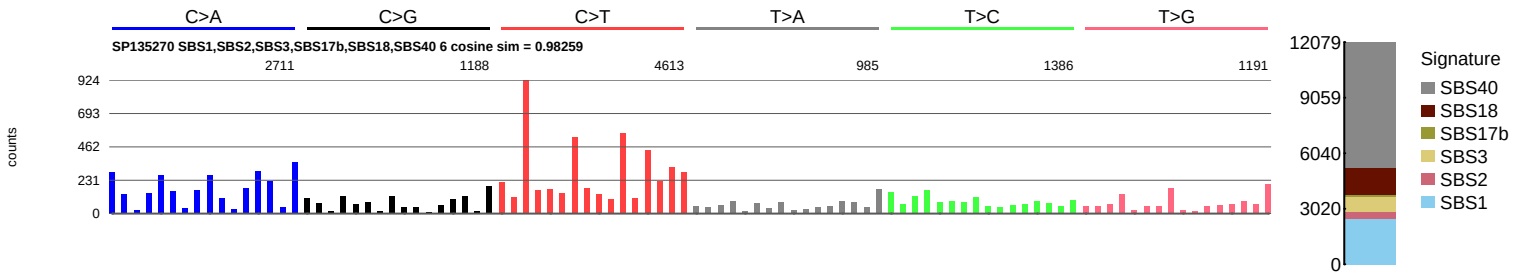

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98264
2628
1228
4450
1152
1648
971
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

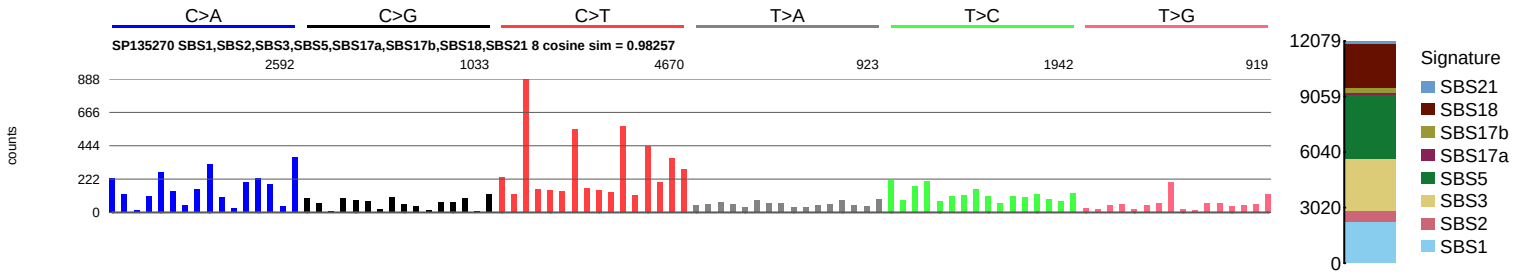

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS41 7 cosine sim = 0.98262
2524
1015
4809
935
2025
770
counts
856
642
428
214
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1

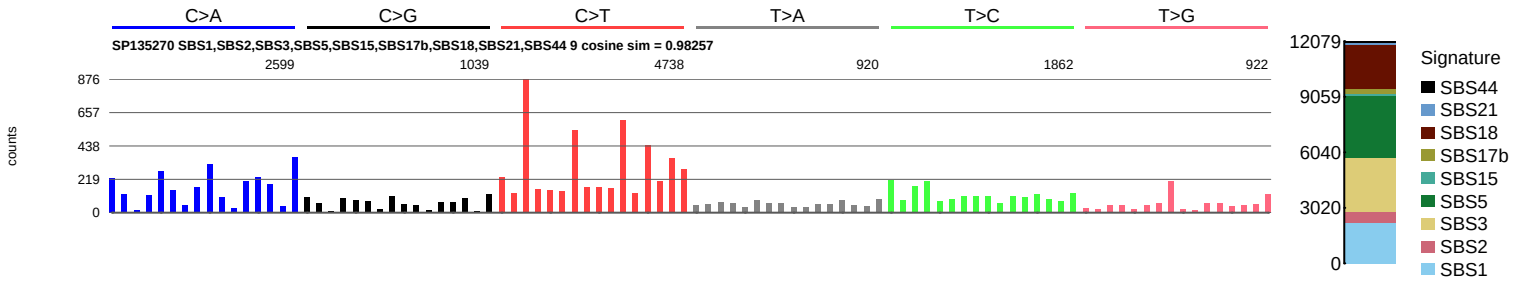

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS15,SBS18,SBS40 5 cosine sim = 0.98262
2717
1156
4745
959
1397
1108
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS15
SBS2
SBS1

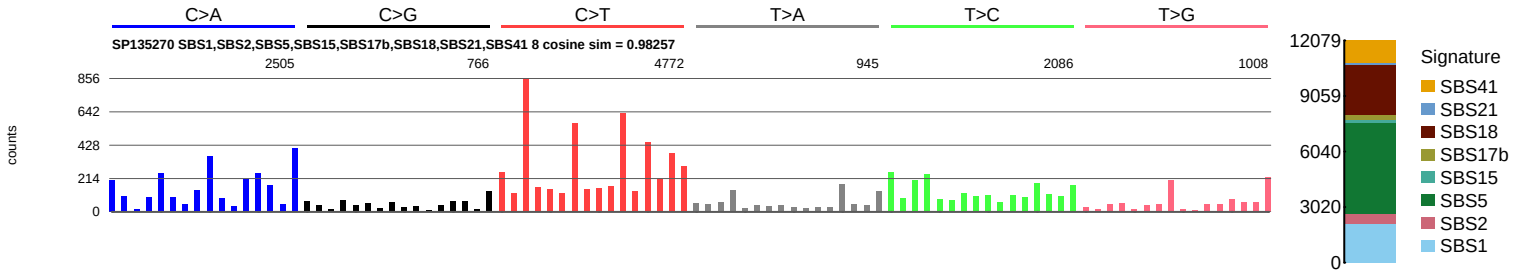

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS18,SBS40 6 cosine sim = 0.98262
2712
1171
4733
974
1397
1090
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS15
SBS3
SBS2
SBS1

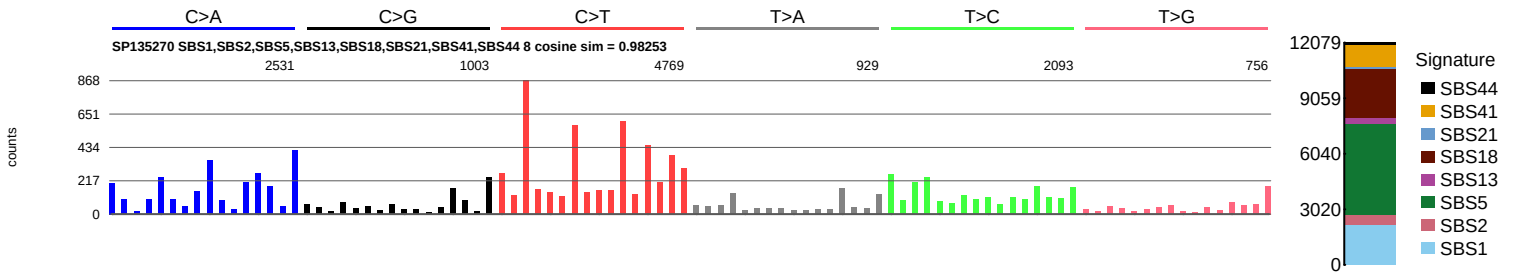

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98262
2631
1227
4446
1154
1654
970
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

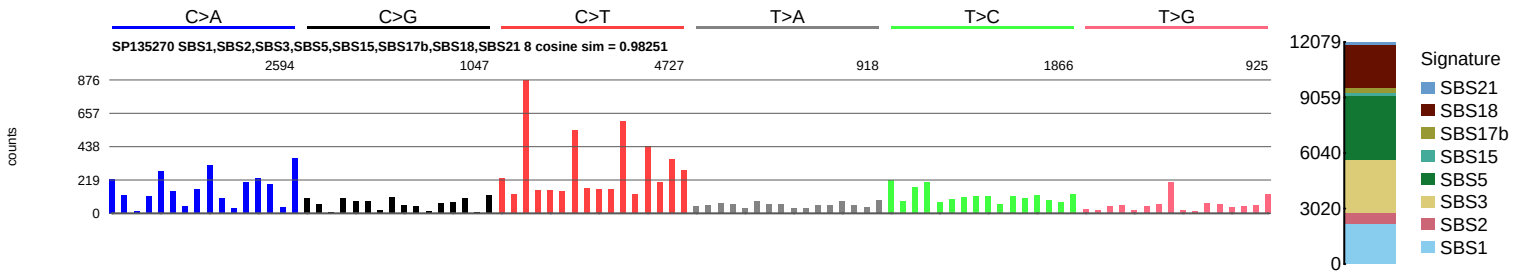

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40 6 cosine sim = 0.98259
2711
1188
4613
985
1386
1191
counts
924
693
462
231
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS3
SBS2
SBS1

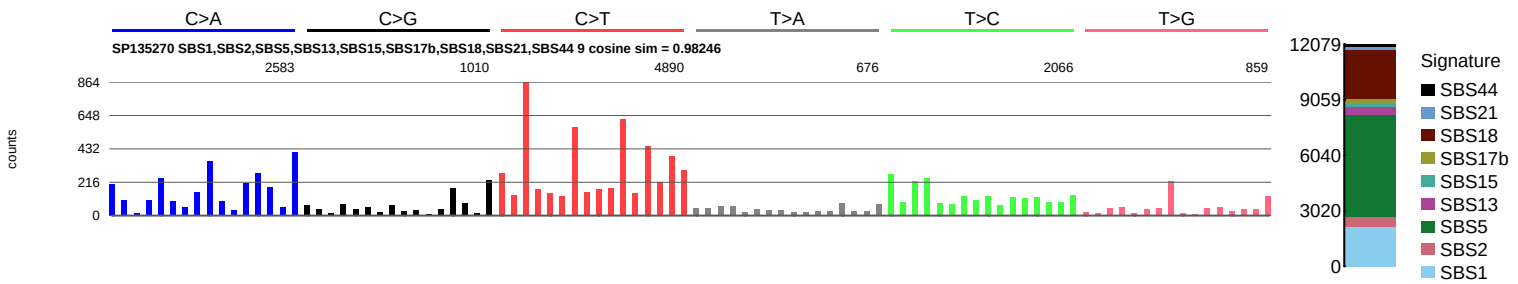

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98257
2592
1033
4670
923
1942
919
888
666
444
222
0
counts
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

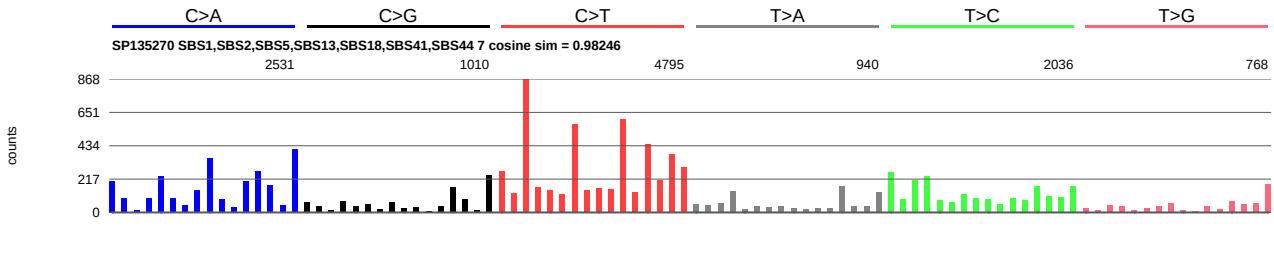

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98257
2599
1039
4738
920
1862
922
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98257
2505
766
4772
945
2086
1008
856
642
428
214
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98253
2531
1003
4769
929
2093
756
868
651
434
217
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98251
2594
1047
4727
918
1866
925
876
657
438
219
0
counts
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98246
2583
1010
4890
676
2066
859
864
648
432
216
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS41,SBS44 7 cosine sim = 0.98246
2531
1010
4795
940
2036
768
counts
868
651
434
217
0
12079
9059
6040
3020
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Signature
SBS44
SBS41
SBS18
SBS13
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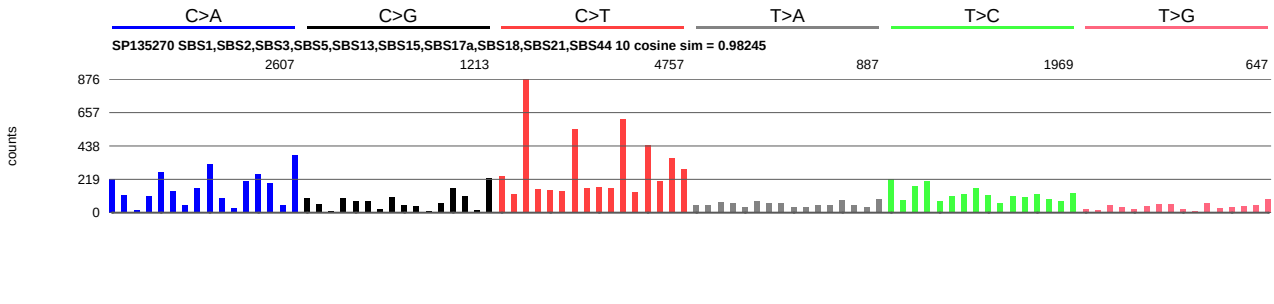
SBS2
SBS1


C>A
C>G
C>T
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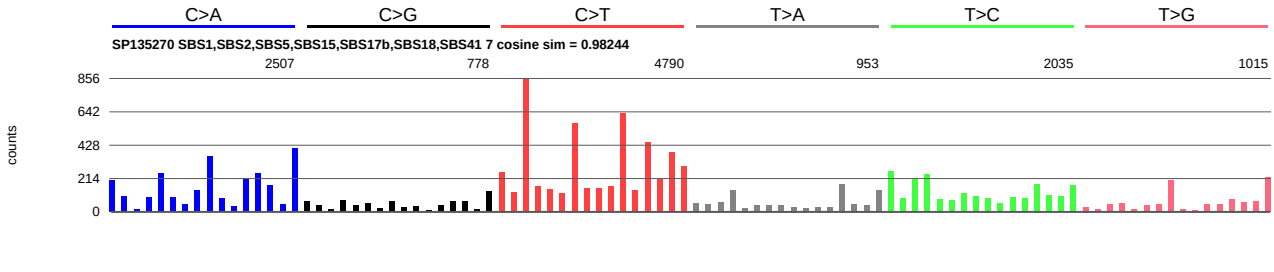
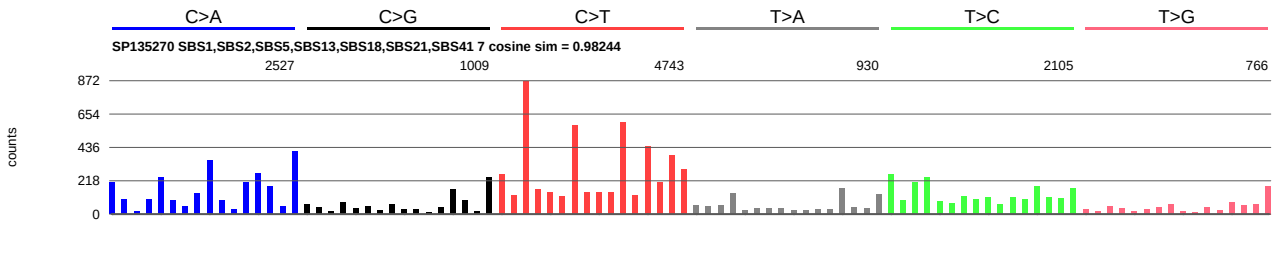
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS44 10 cosine sim = 0.98245
2607
1213
4757
887
1969
647
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS41 7 cosine sim = 0.98244
2507
778
4790
953
2035
1015
counts
856
642
428
214
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS41 7 cosine sim = 0.98244
2527
1009
4743
930
2105
766
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
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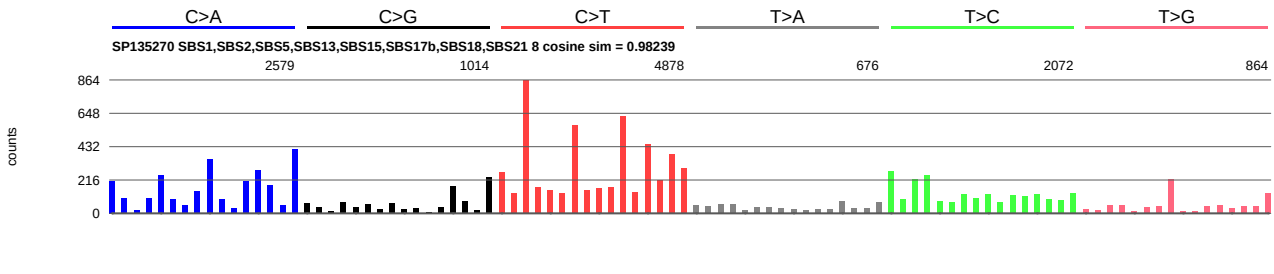
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98239
2579
1014
4878
676
2072
864
counts
864
648
432
216
0
12079
9059
6040
3020
0
Signature
SBS21
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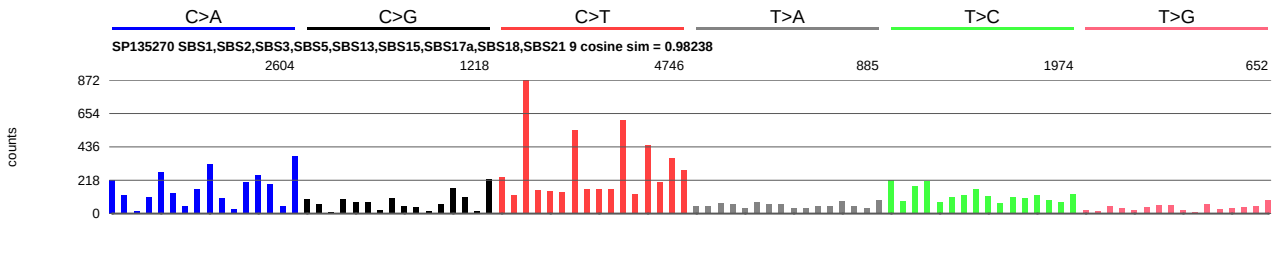
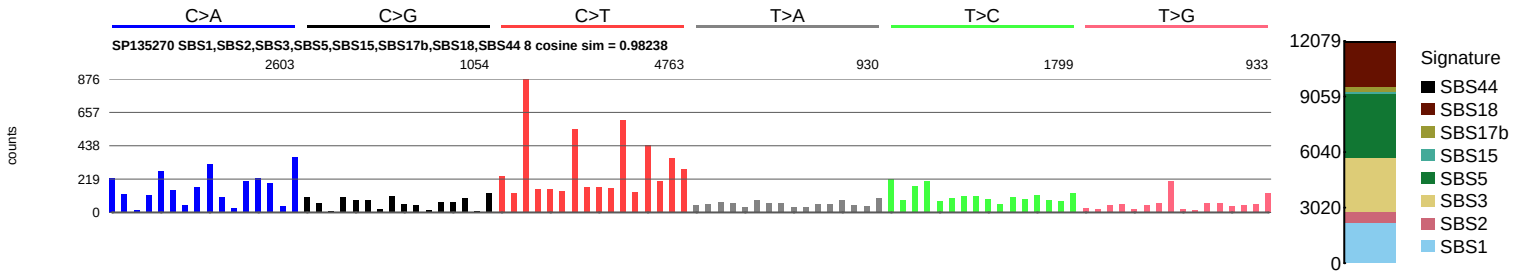
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21 9 cosine sim = 0.98238
2604
1218
4746
885
1974
652
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS44 8 cosine sim = 0.98238
2603
1054
4763
930
1799
933
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
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SBS17b
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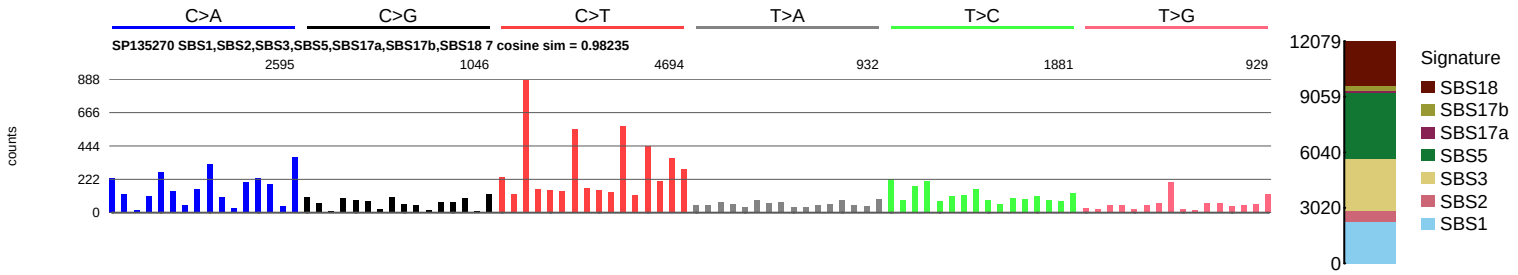
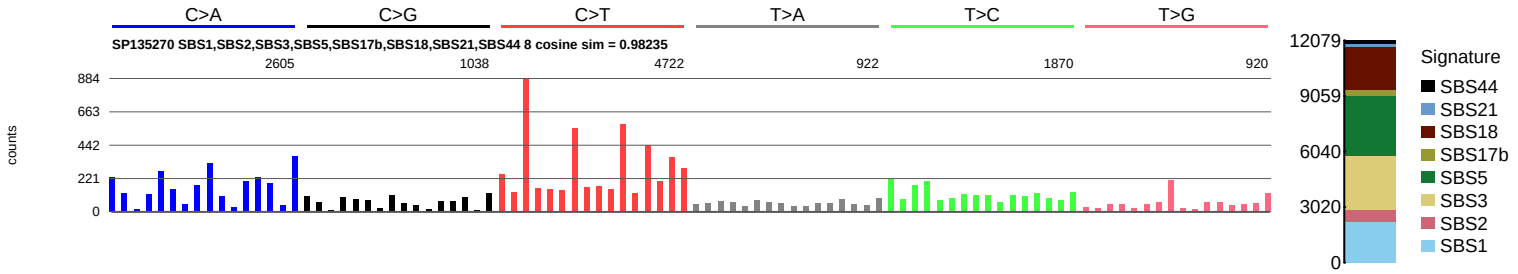
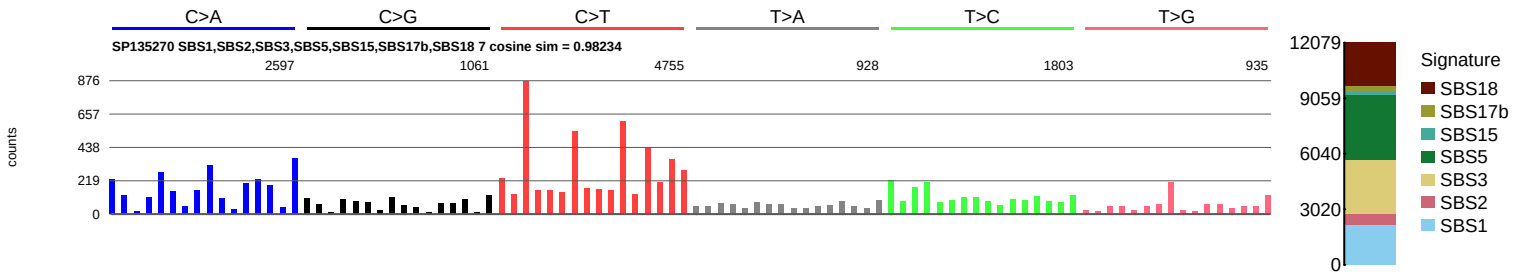
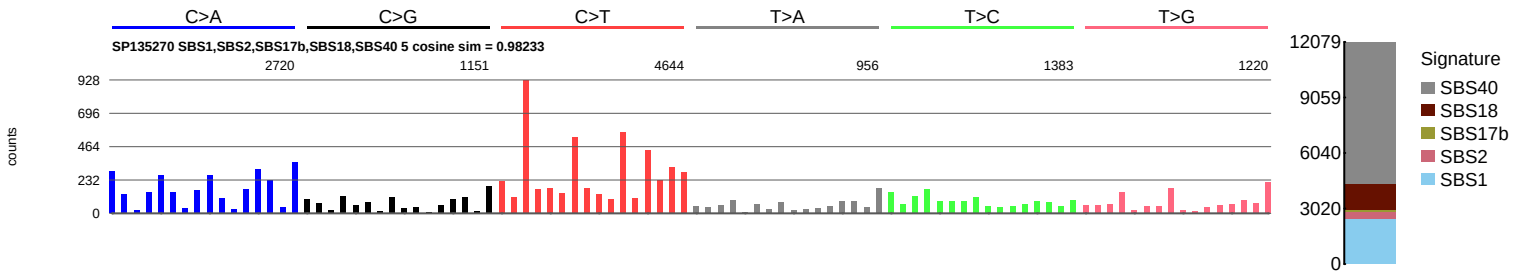
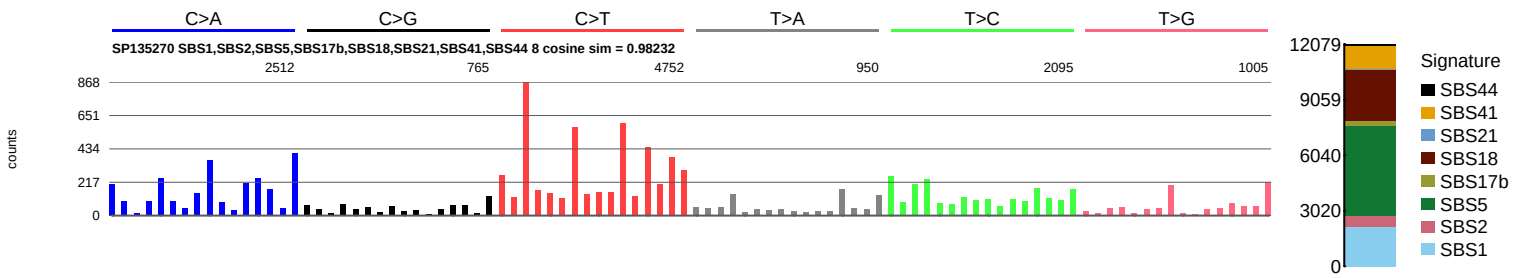
SBS2
SBS1

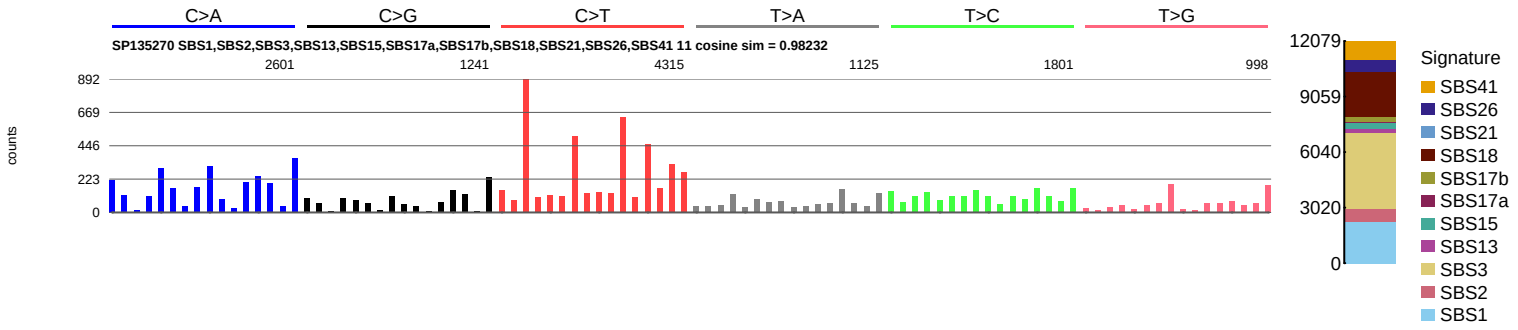

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98235
2595
1046
4694
932
1881
929
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

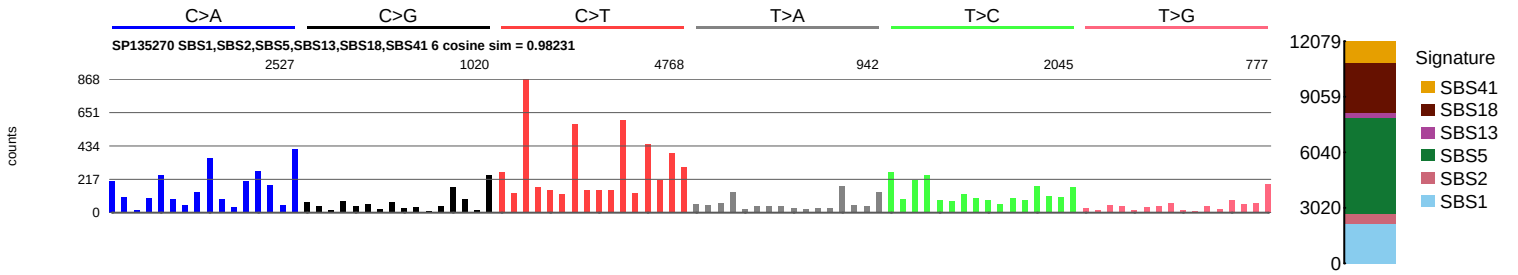

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98235
2605
1038
4722
922
1870
920
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

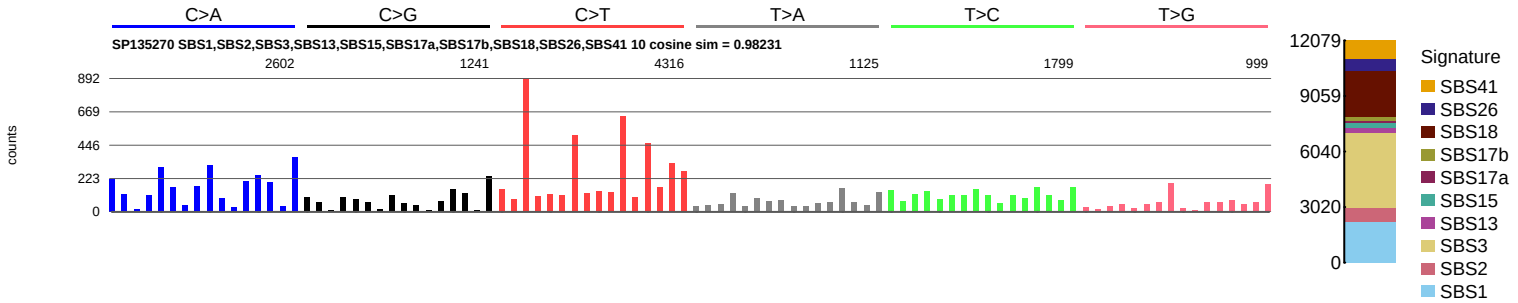

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18 7 cosine sim = 0.98234
2597
1061
4755
928
1803
935
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

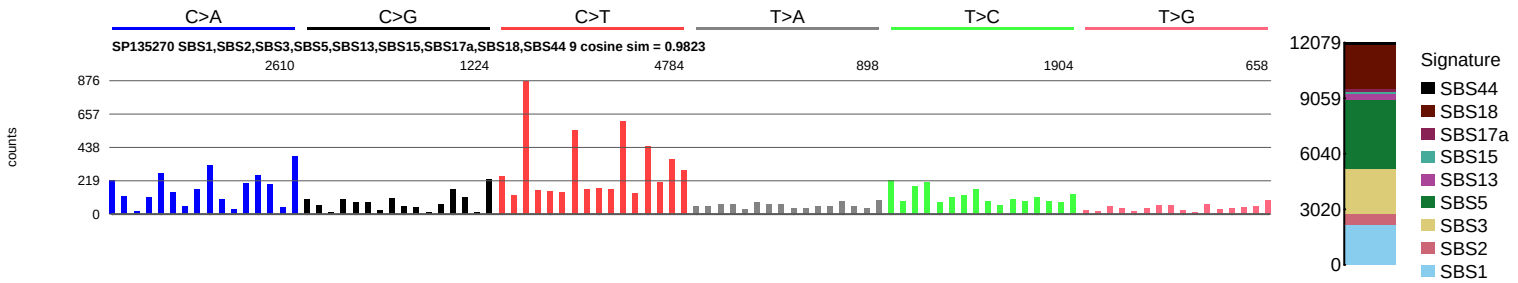

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17b,SBS18,SBS40 5 cosine sim = 0.98233
2720
1151
4644
956
1383
1220
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17b
SBS2
SBS1

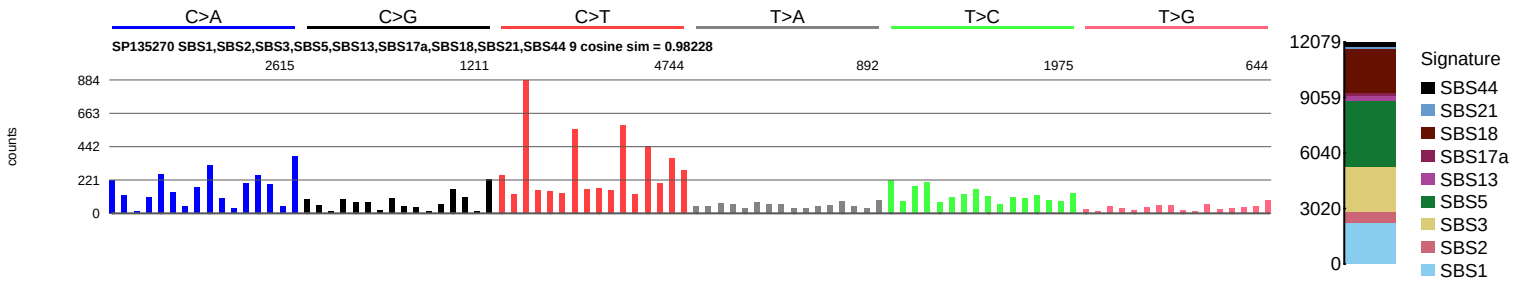

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98232
2512
765
4752
950
2095
1005
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

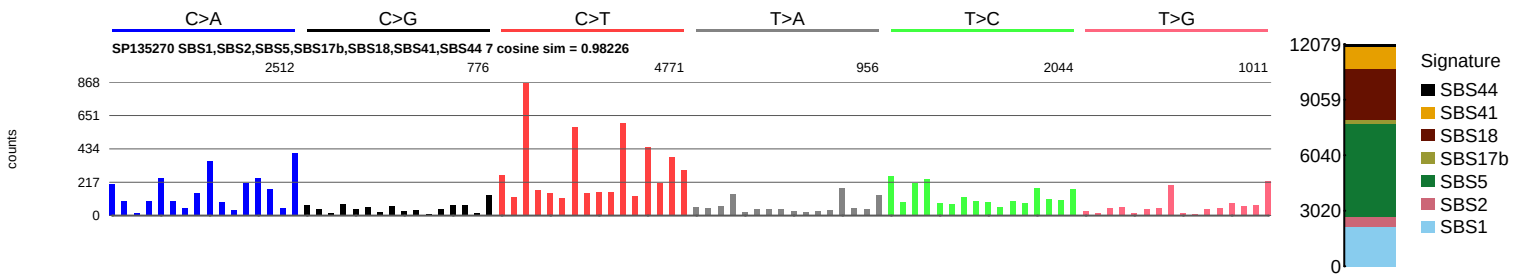

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 11 cosine sim = 0.98232
2601
1241
4315
1125
1801
998
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

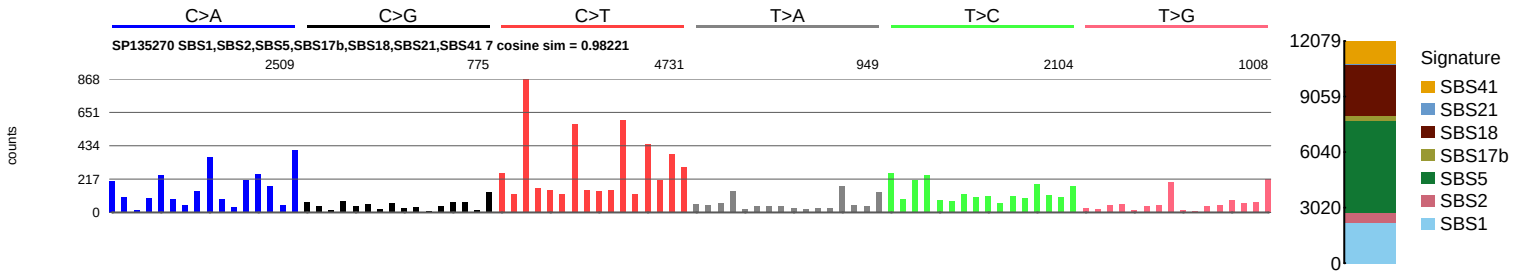

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS18,SBS41 6 cosine sim = 0.98231
2527
1020
4768
942
2045
777
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS13
SBS5
SBS2
SBS1

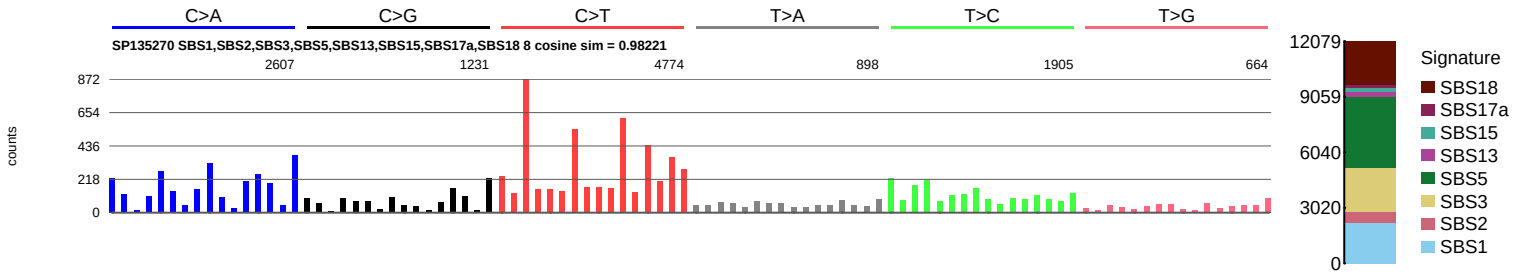

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98231
2602
1241
4316
1125
1799
999
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS44 9 cosine sim = 0.9823
2610
1224
4784
898
1904
658
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS44 9 cosine sim = 0.98228
2615
1211
4744
892
1975
644
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98226
2512
776
4771
956
2044
1011
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98221
2509
775
4731
949
2104
1008
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18 8 cosine sim = 0.98221
2607
1231
4774
898
1905
664
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS18
SBS17a
SBS15
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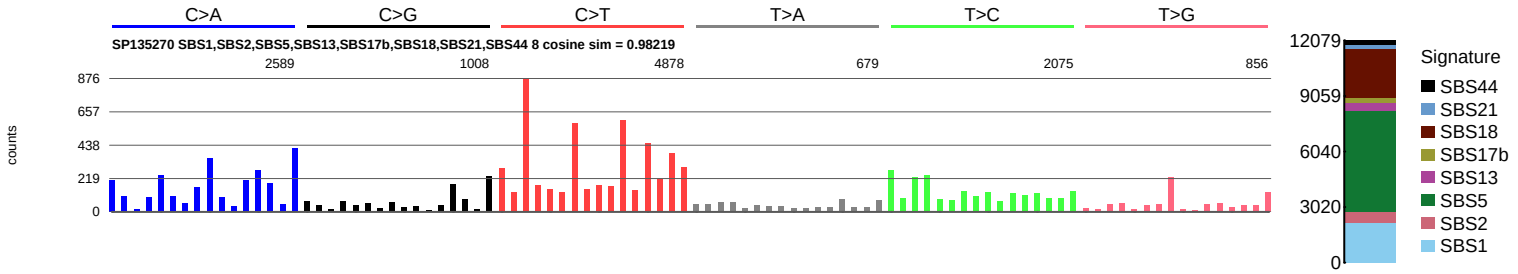
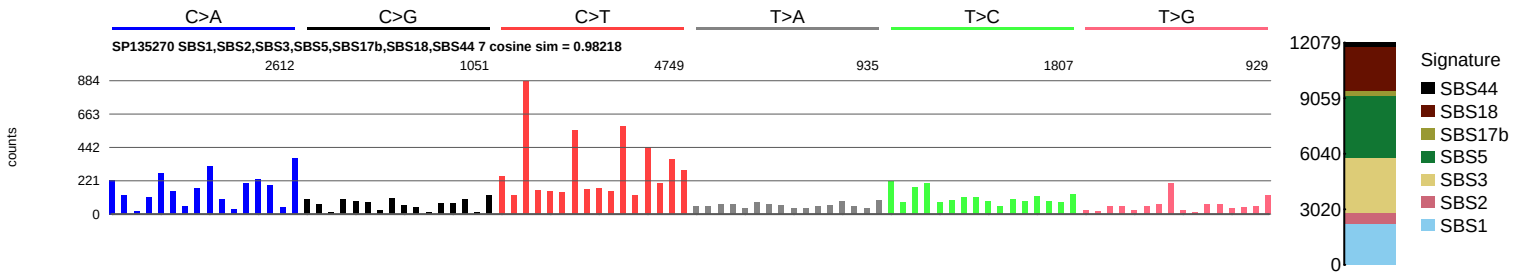
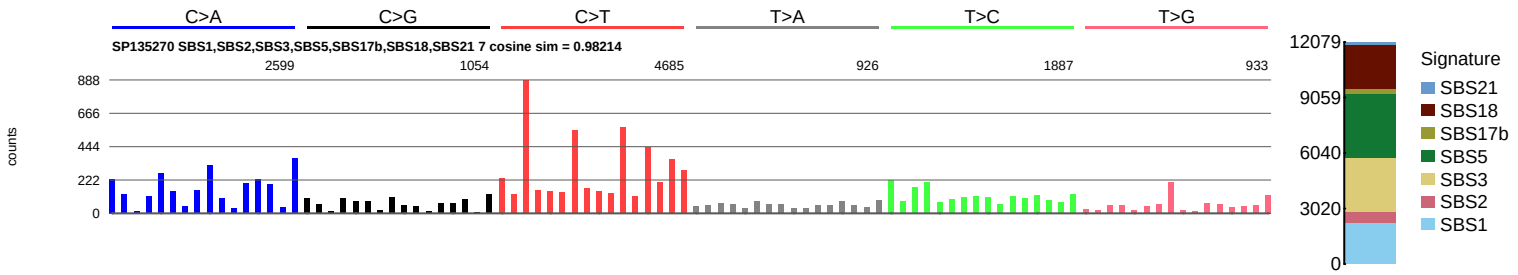
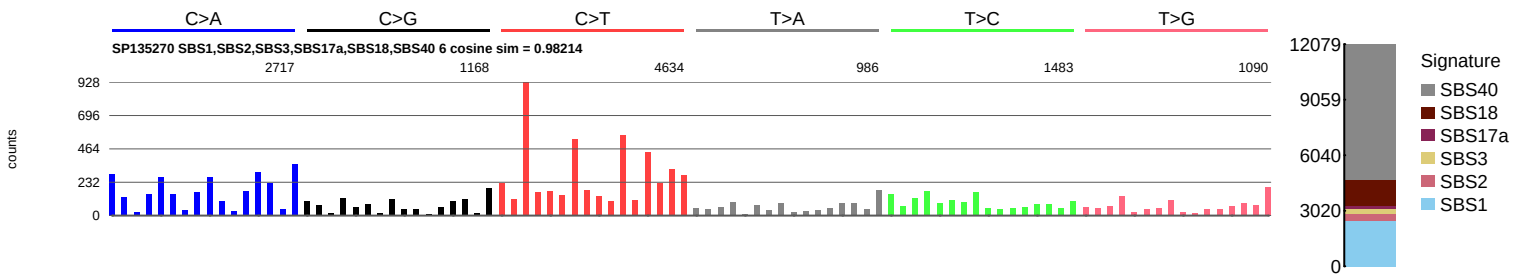
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98219
2589
1008
4878
679
2075
856
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS44 7 cosine sim = 0.98218
2612
1051
4749
935
1807
929
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21 7 cosine sim = 0.98214
2599
1054
4685
926
1887
933
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS40 6 cosine sim = 0.98214
2717
1168
4634
986
1483
1090
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
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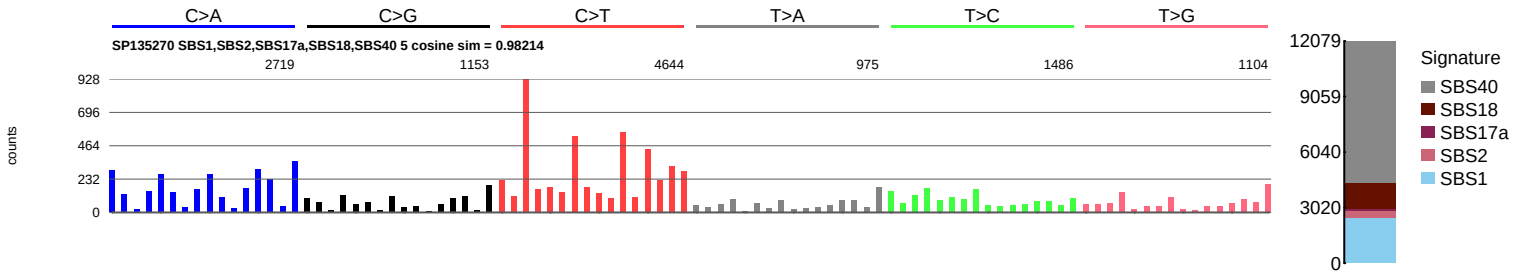
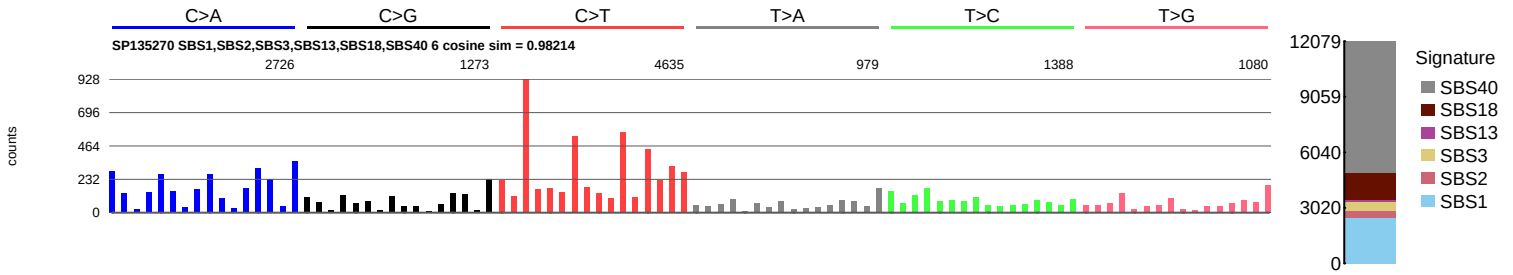
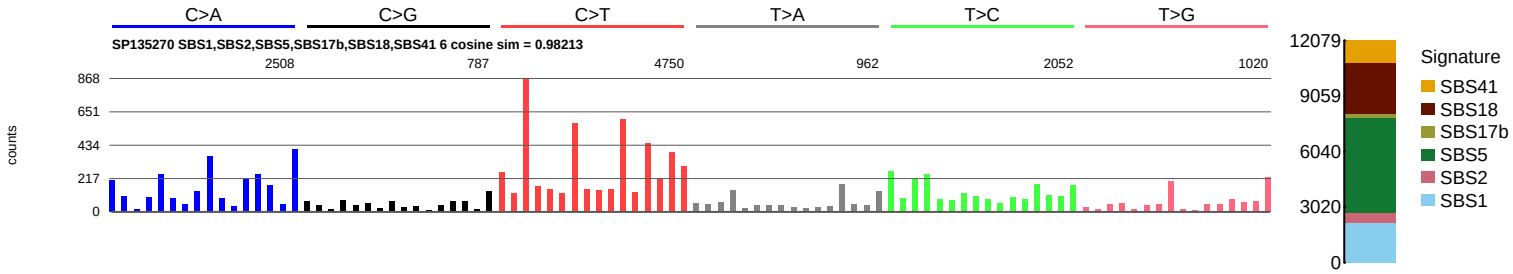
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS17a,SBS18,SBS40 5 cosine sim = 0.98214
2719
1153
4644
975
1486
1104
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS18,SBS40 6 cosine sim = 0.98214
2726
1273
4635
979
1388
1080
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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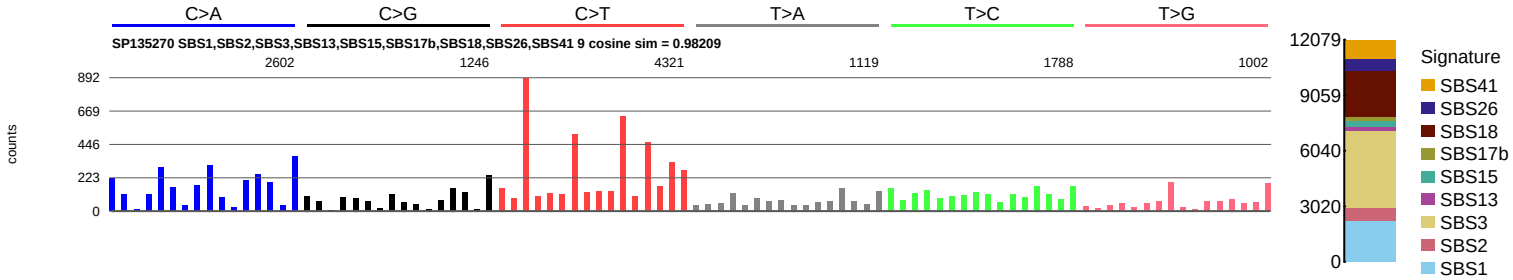
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS41 6 cosine sim = 0.98213
2508
787
4750
962
2052
1020
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS2
SBS1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C>G
C>T
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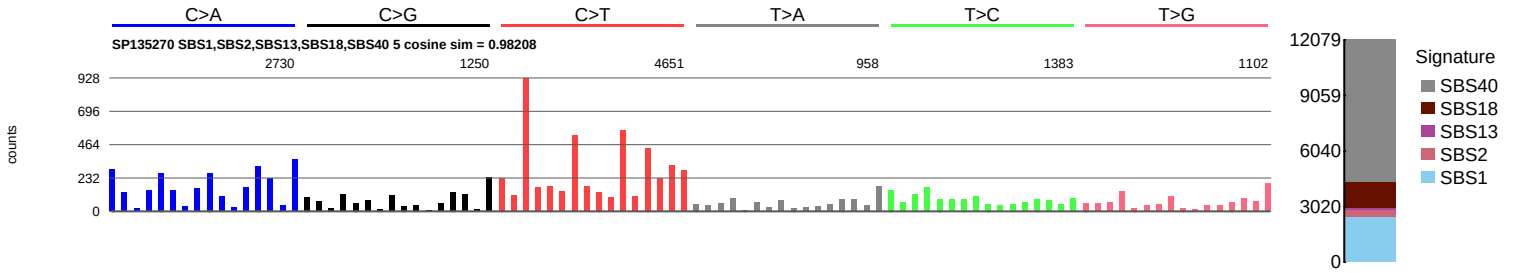
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98209
2602
1246
4321
1119
1788
1002
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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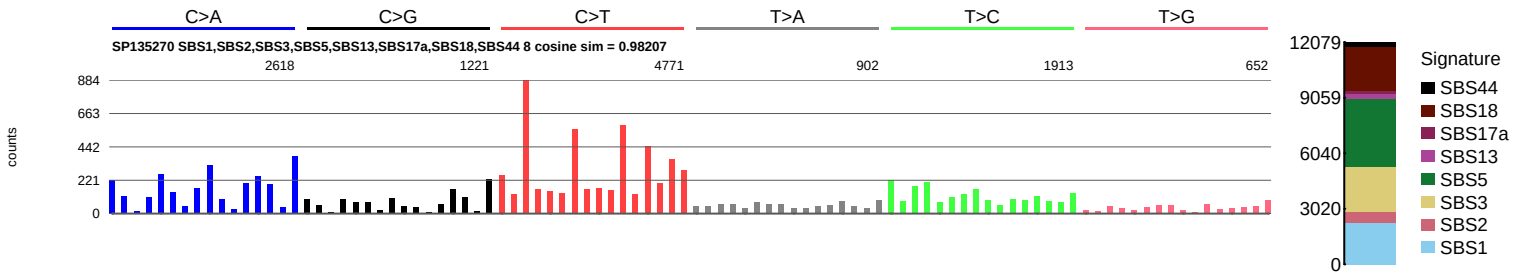
SP135270 SBS1,SBS2,SBS13,SBS18,SBS40 5 cosine sim = 0.98208
2730
1250
4651
958
1383
1102
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS13
SBS2
SBS1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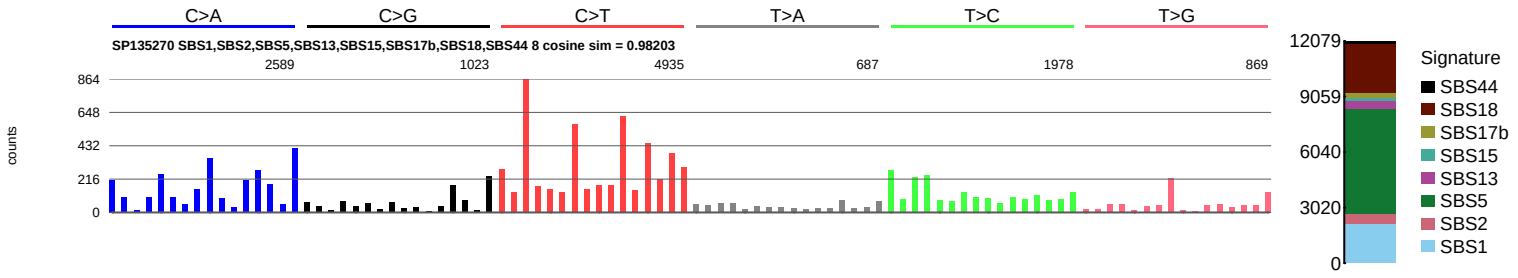
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS44 8 cosine sim = 0.98207
2618
1221
4771
902
1913
652
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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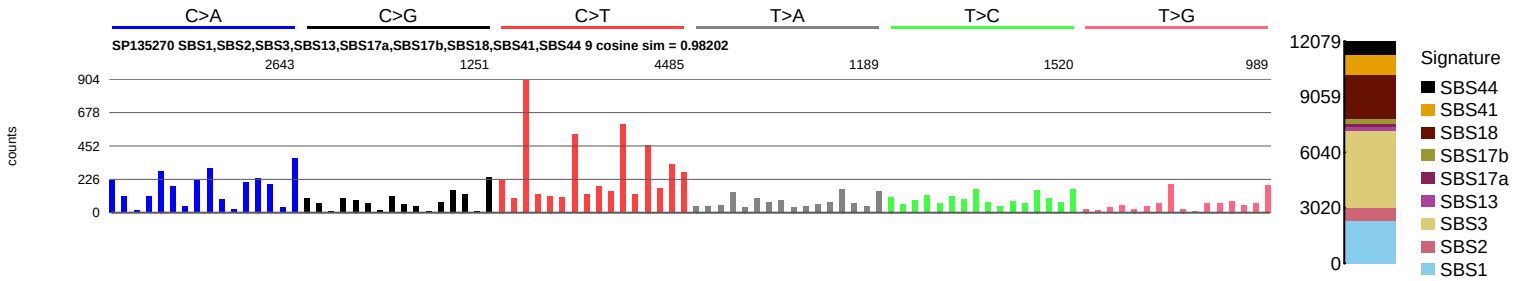
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 8 cosine sim = 0.98203
2589
1023
4935
687
1978
869
counts
864
648
432
216
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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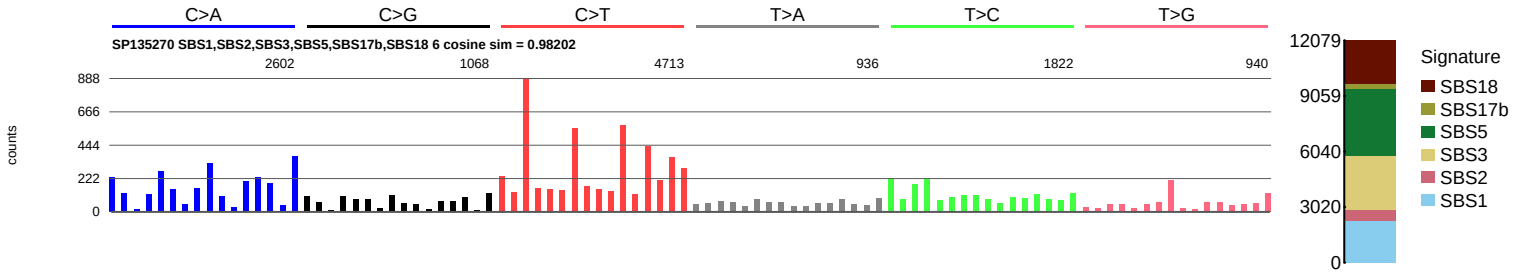
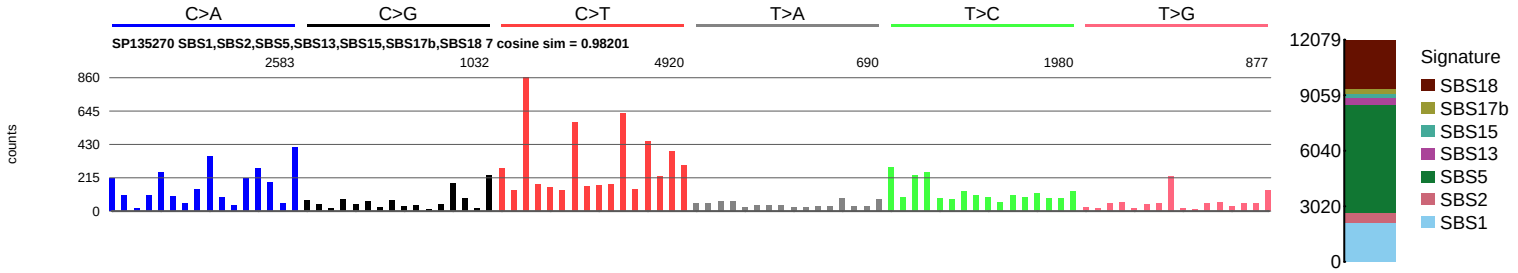
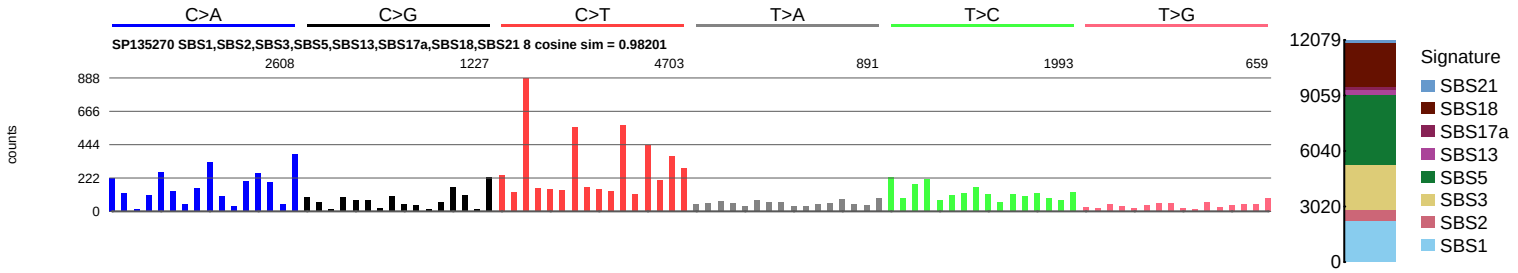
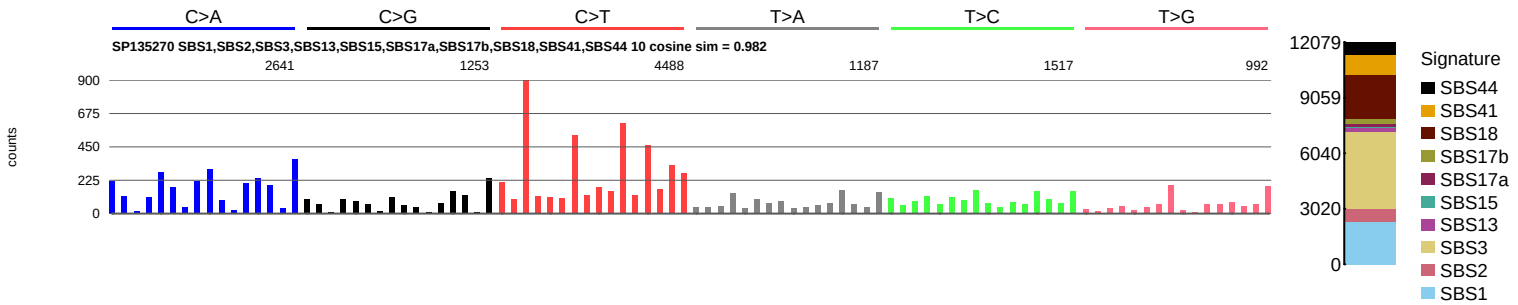
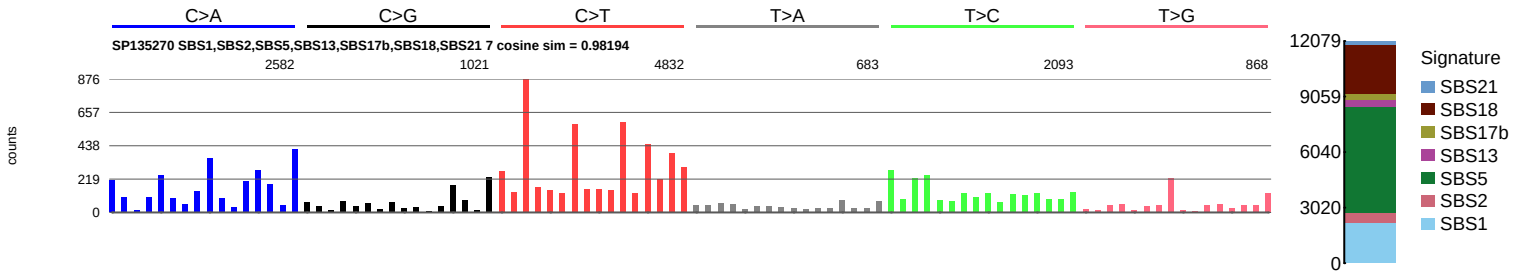
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98202
2643
1251
4485
1189
1520
989
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

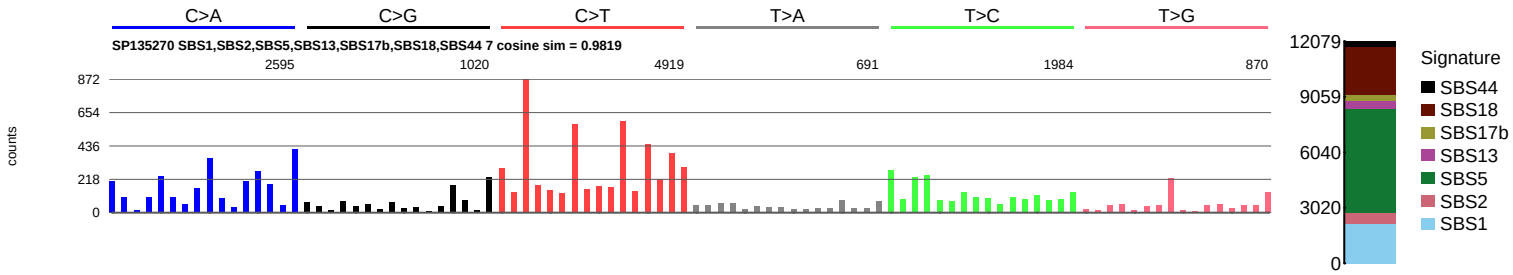

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18 6 cosine sim = 0.98202
2602
1068
4713
936
1822
940
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

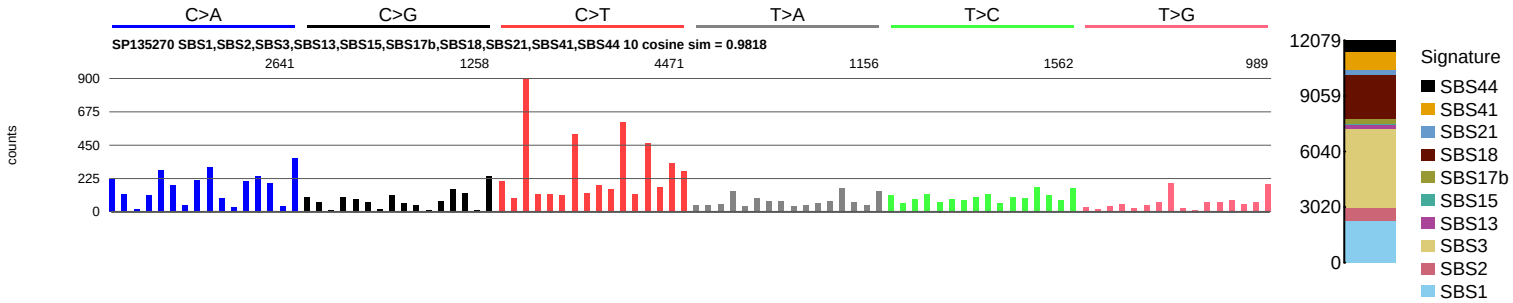

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18 7 cosine sim = 0.98201
2583
1032
4920
690
1980
877
counts
860
645
430
215
0
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

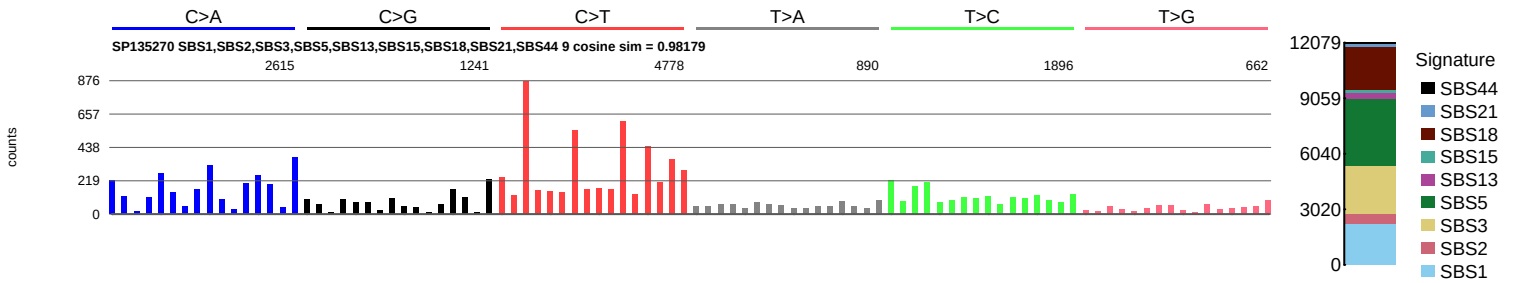

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21 8 cosine sim = 0.98201
2608
1227
4703
891
1993
659
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

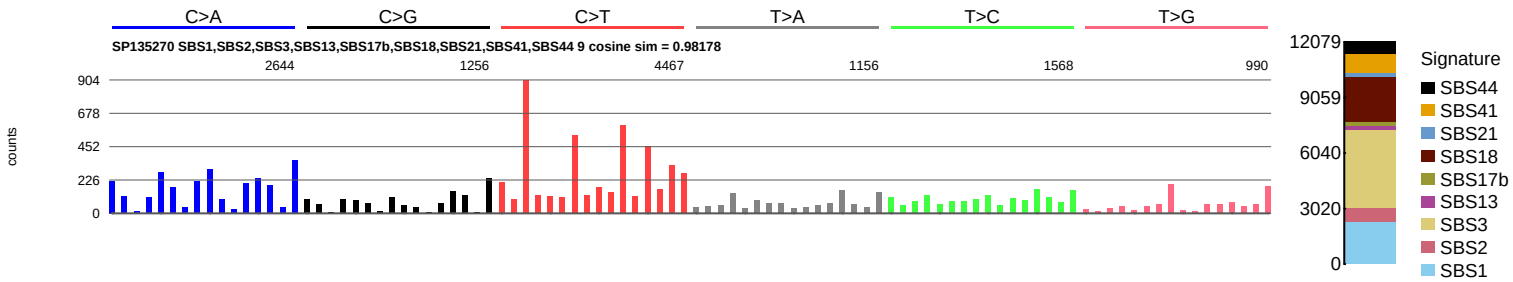

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.982
2641
1253
4488
1187
1517
992
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

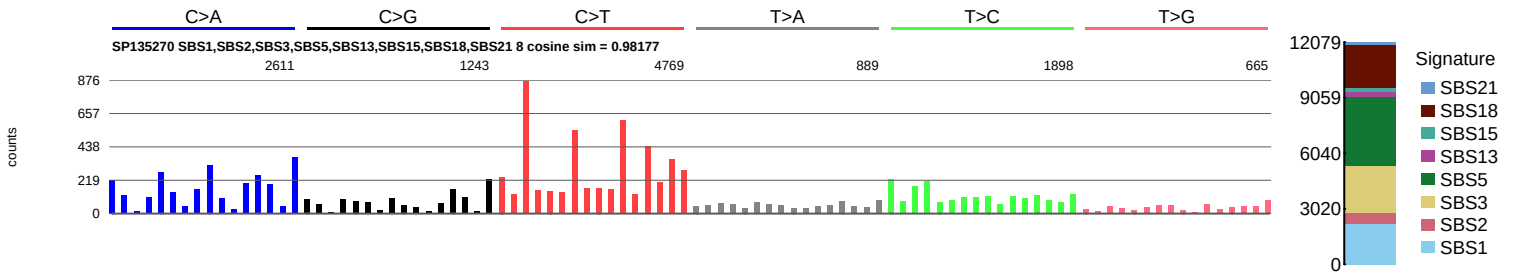

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21 7 cosine sim = 0.98194
2582
1021
4832
683
2093
868
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

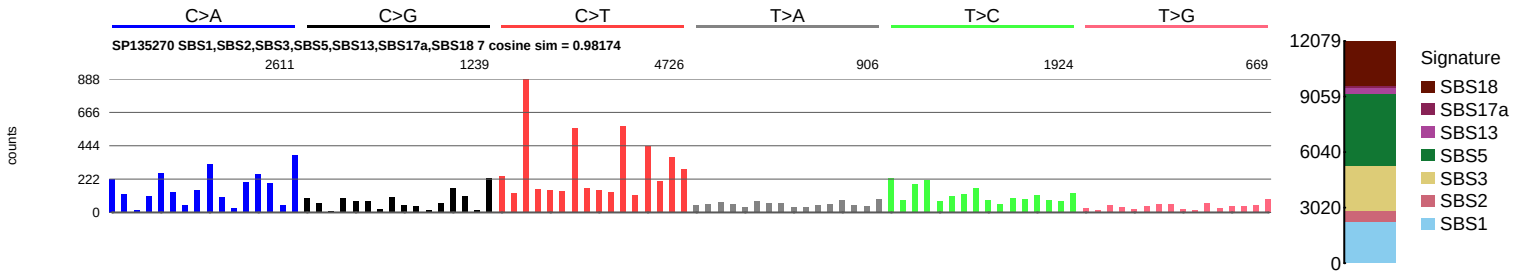

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS44 7 cosine sim = 0.9819
2595
1020
4919
691
1984
870
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

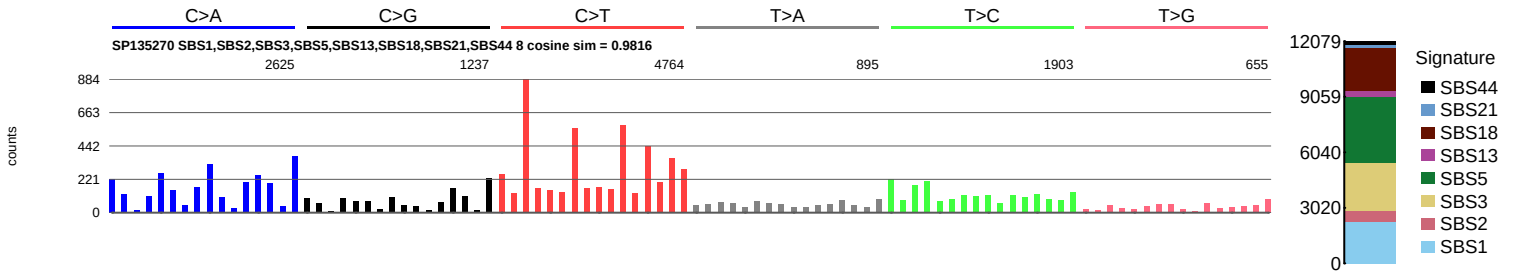

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9818
2641
1258
4471
1156
1562
989
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

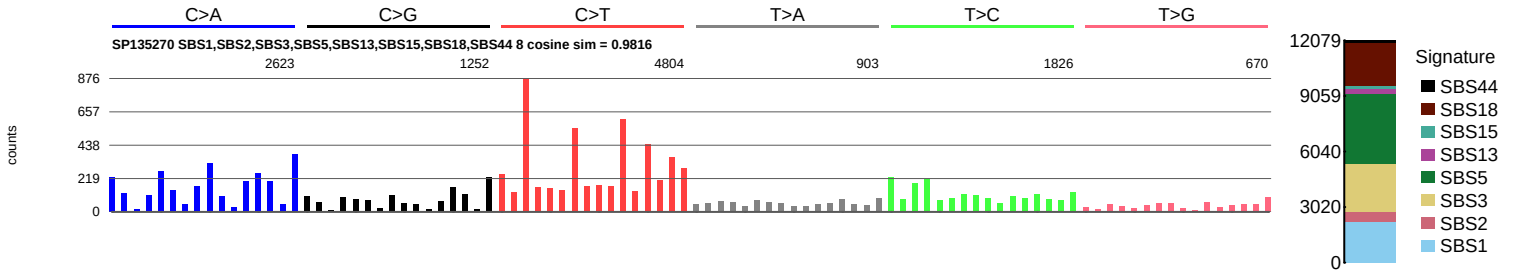

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS44 9 cosine sim = 0.98179
2615
1241
4778
890
1896
662
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98178
2644
1256
4467
1156
1568
990
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21 8 cosine sim = 0.98177
2611
1243
4769
889
1898
665
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18 7 cosine sim = 0.98174
2611
1239
4726
906
1924
669
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS44 8 cosine sim = 0.9816
2625
1237
4764
895
1903
655
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS44 8 cosine sim = 0.9816
2623
1252
4804
903
1826
670
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
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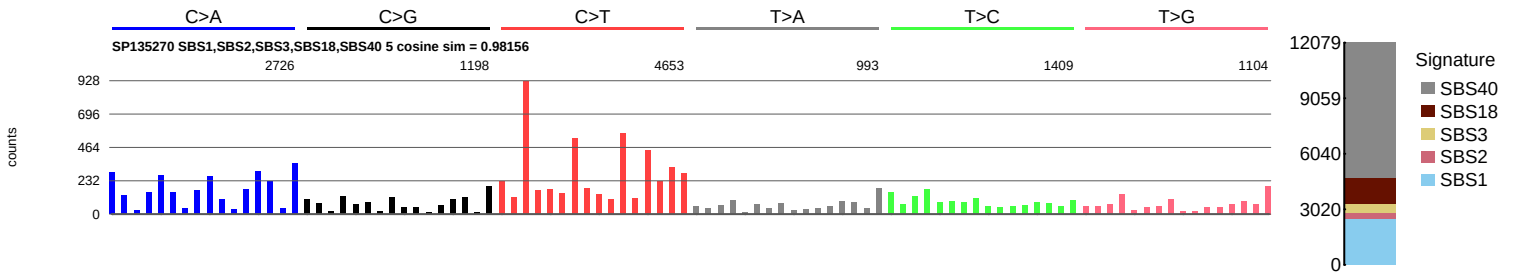
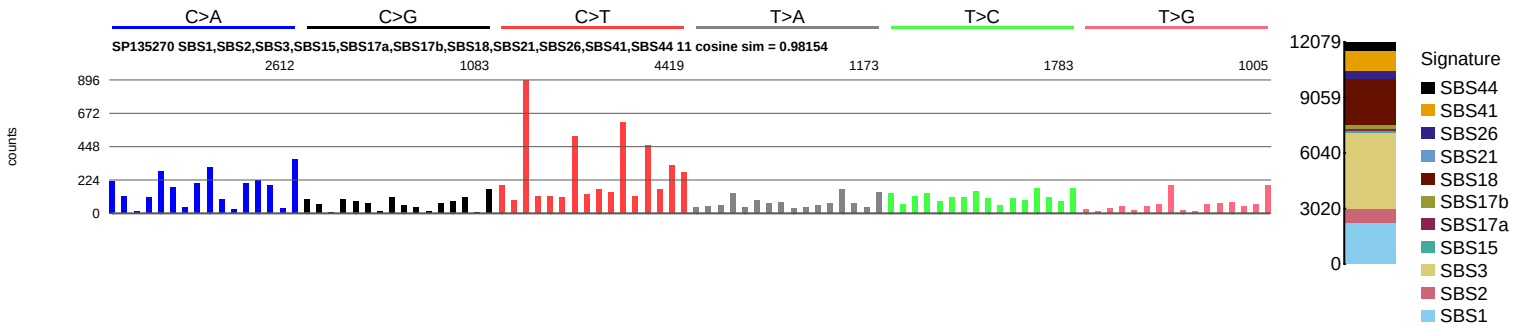
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS18,SBS40 5 cosine sim = 0.98156
2726
1198
4653
993
1409
1104
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98154
2612
1083
4419
1173
1783
1005
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
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SBS26
SBS21
SBS18
SBS17b
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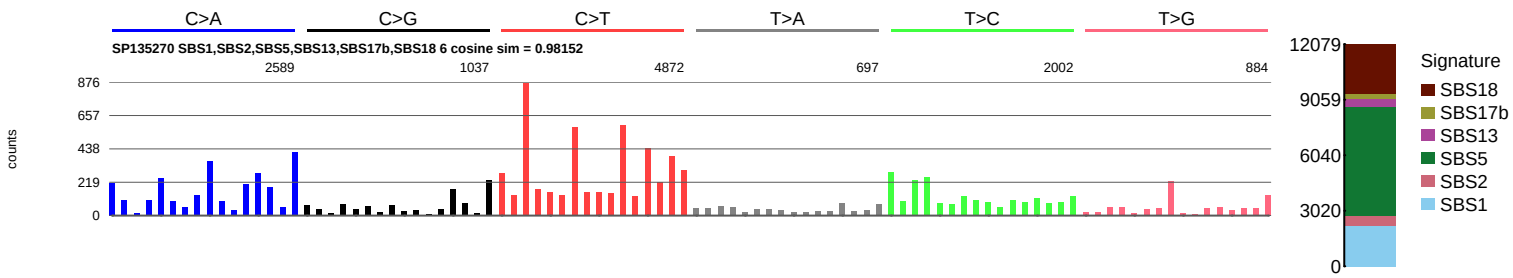
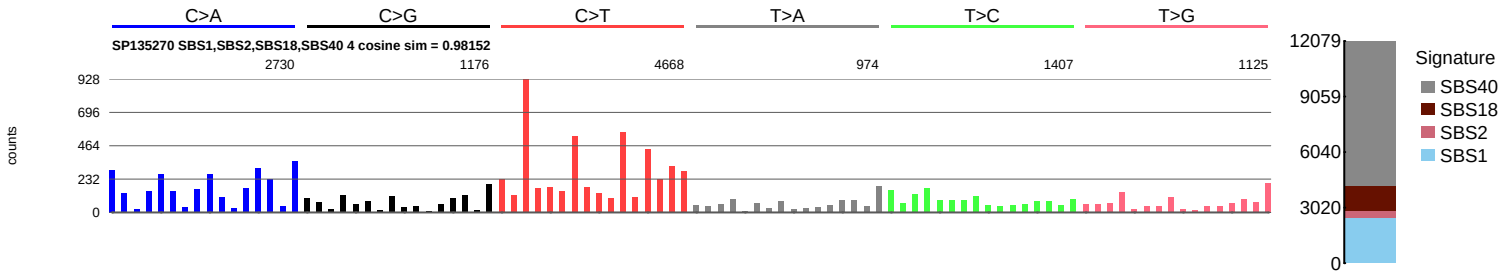
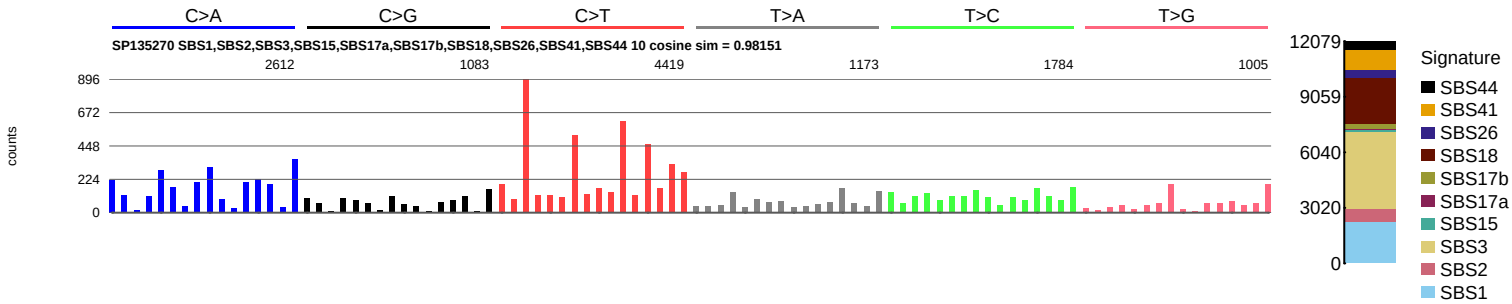
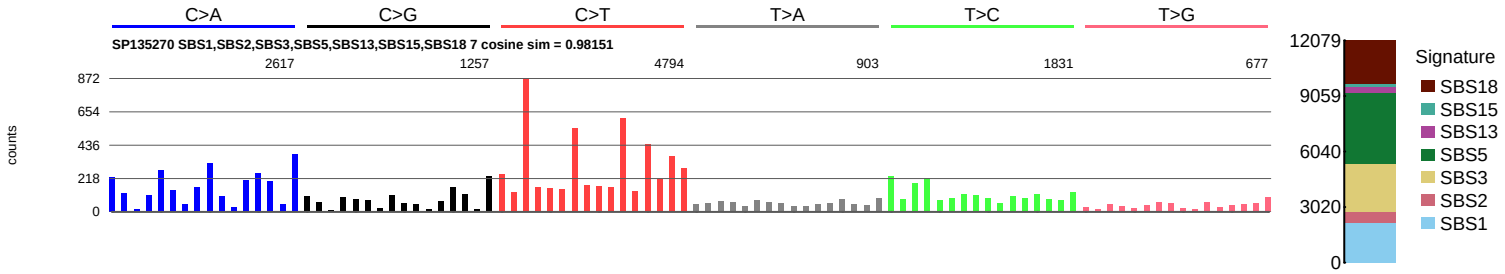
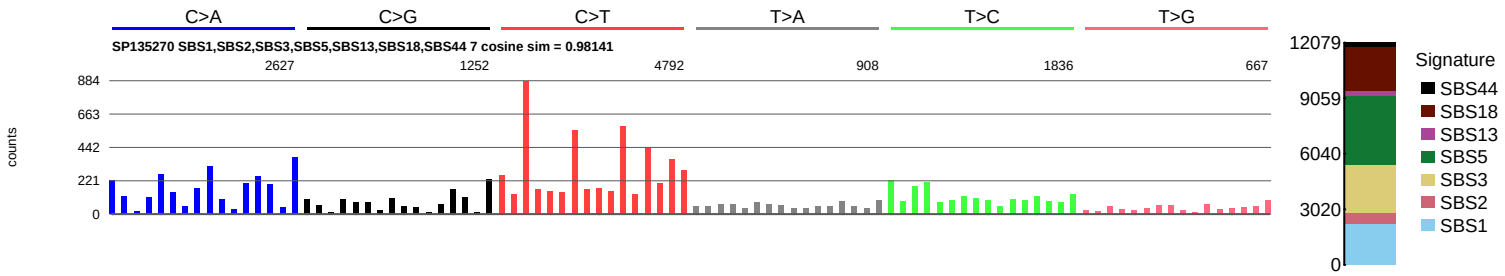
SBS3
SBS2
SBS1

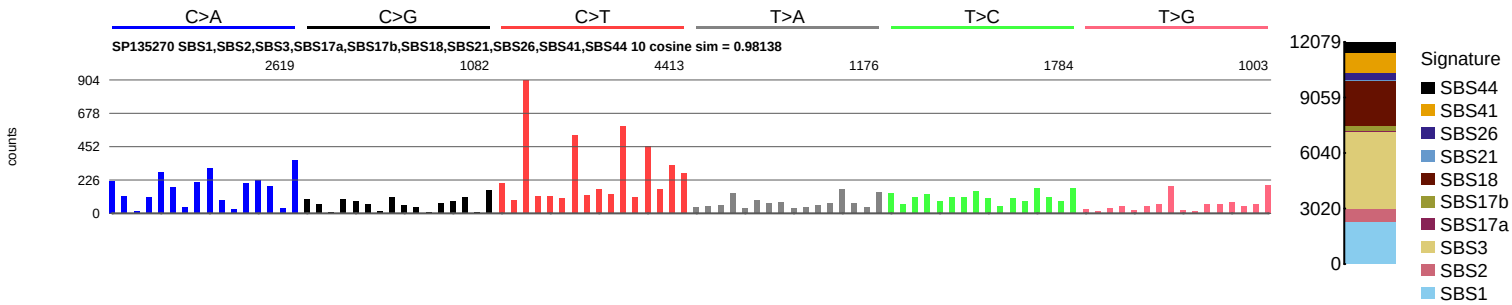

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18 6 cosine sim = 0.98152
2589
1037
4872
697
2002
884
counts
876
657
438
219
0
12079
9059
6040
3020
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS18,SBS40 4 cosine sim = 0.98152
2730
1176
4668
974
1407
1125
counts
928
696
464
232
0
12079
9059
6040
3020
0
Signature
SBS40
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98151
2612
1083
4419
1173
1784
1005
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18 7 cosine sim = 0.98151
2617
1257
4794
903
1831
677
counts
872
654
436
218
0
12079
9059
6040
3020
0
Signature
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS44 7 cosine sim = 0.98141
2627
1252
4792
908
1836
667
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS44
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1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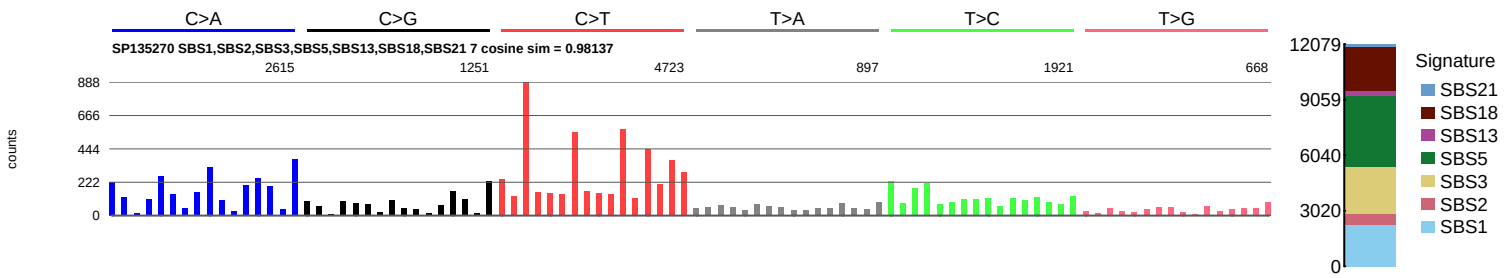
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98138
2619
1082
4413
1176
1784
1003
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21 7 cosine sim = 0.98137
2615
1251
4723
897
1921
668
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS21
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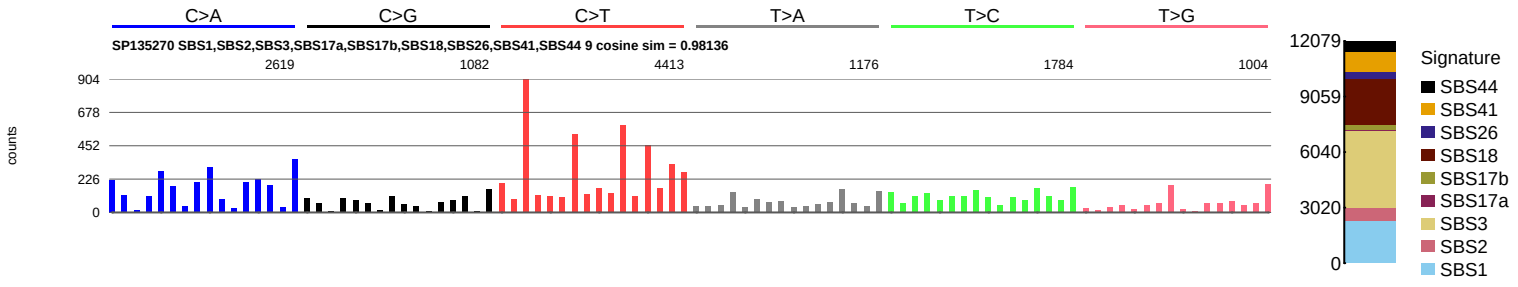
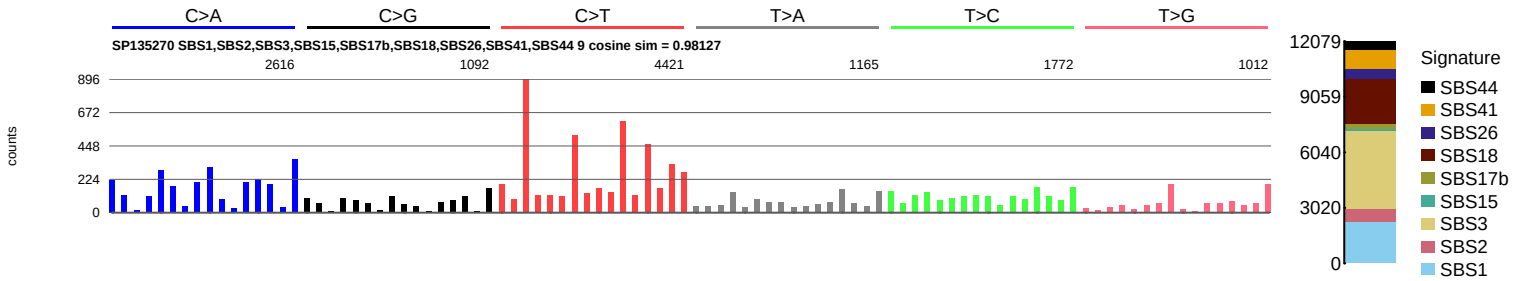
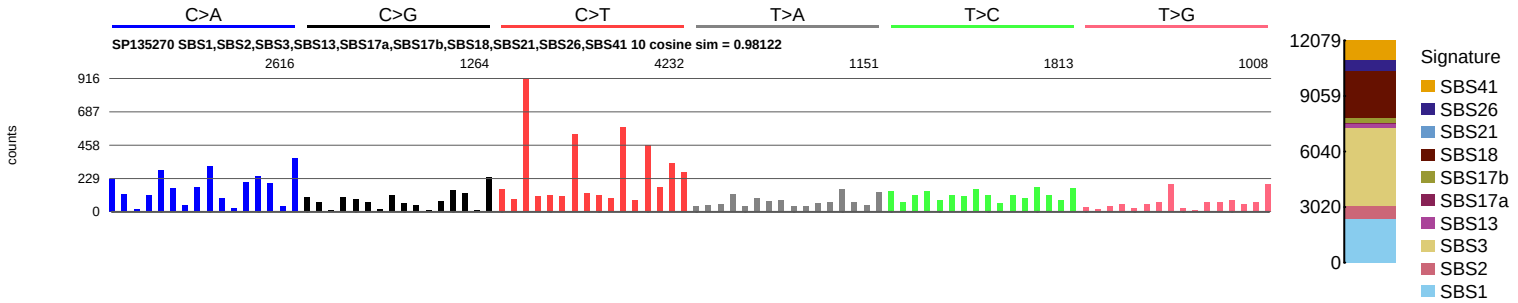
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98136
2619
1082
4413
1176
1784
1004
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98127
2616
1092
4421
1165
1772
1012
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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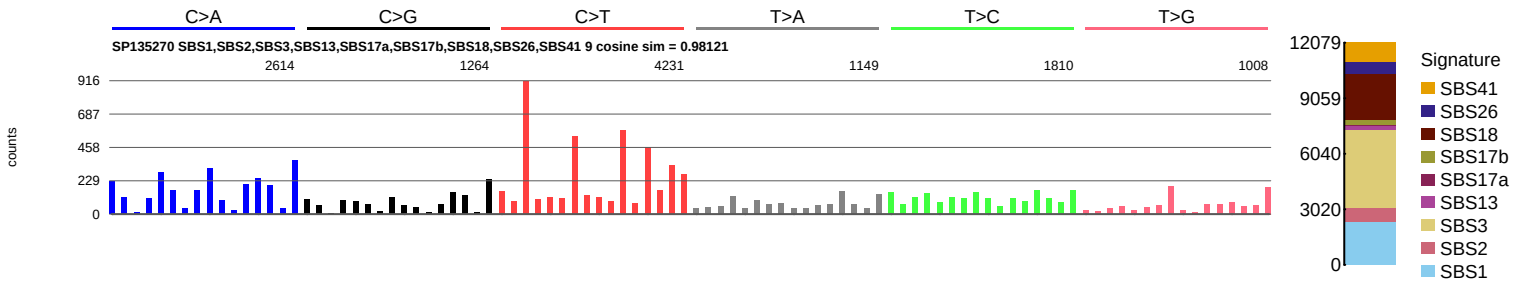
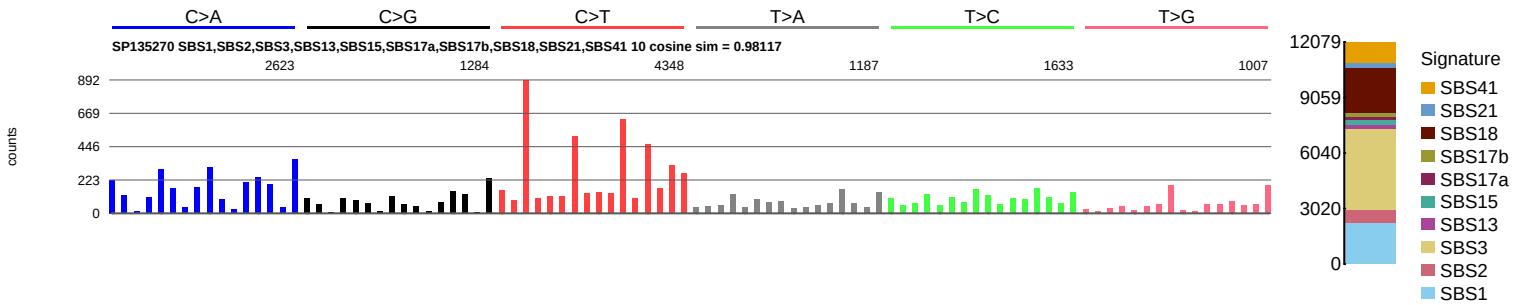
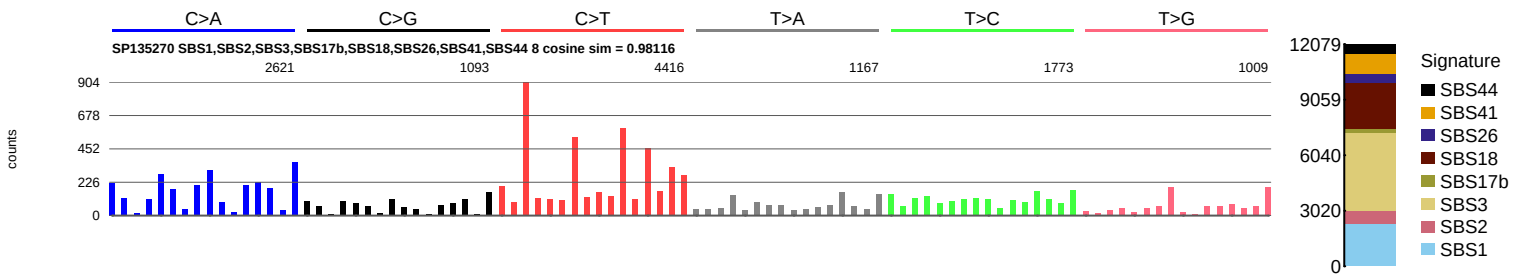
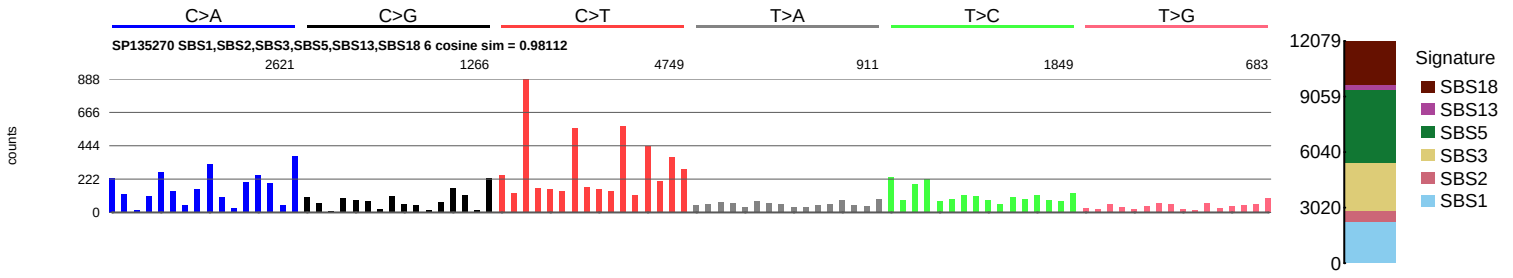
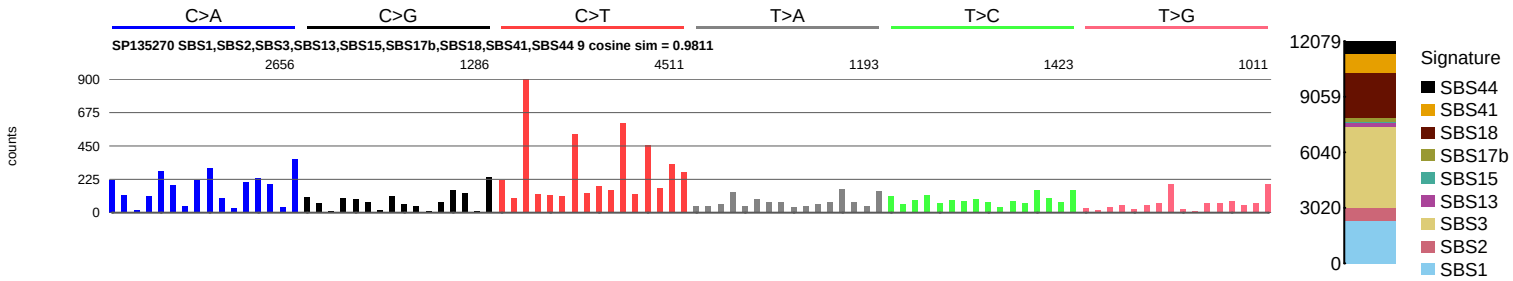
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98122
2616
1264
4232
1151
1813
1008
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

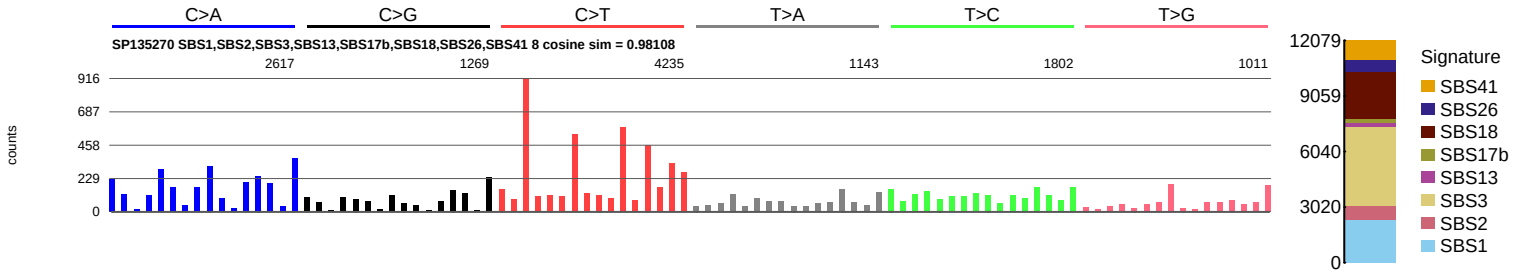

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98121
2614
1264
4231
1149
1810
1008
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

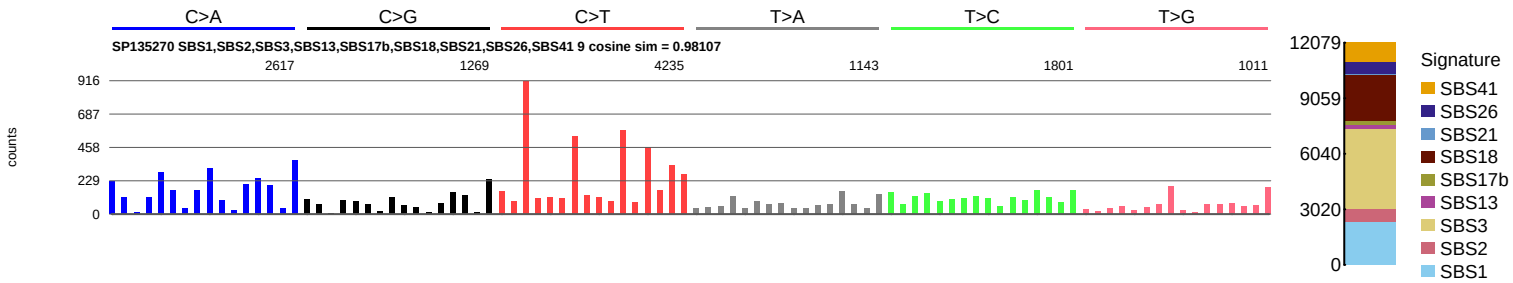

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98117
2623
1284
4348
1187
1633
1007
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98116
2621
1093
4416
1167
1773
1009
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18 6 cosine sim = 0.98112
2621
1266
4749
911
1849
683
counts
888
666
444
222
0
12079
9059
6040
3020
0
Signature
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.9811
2656
1286
4511
1193
1423
1011
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98108
2617
1269
4235
1143
1802
1011
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98107
2617
1269
4235
1143
1801
1011
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS26
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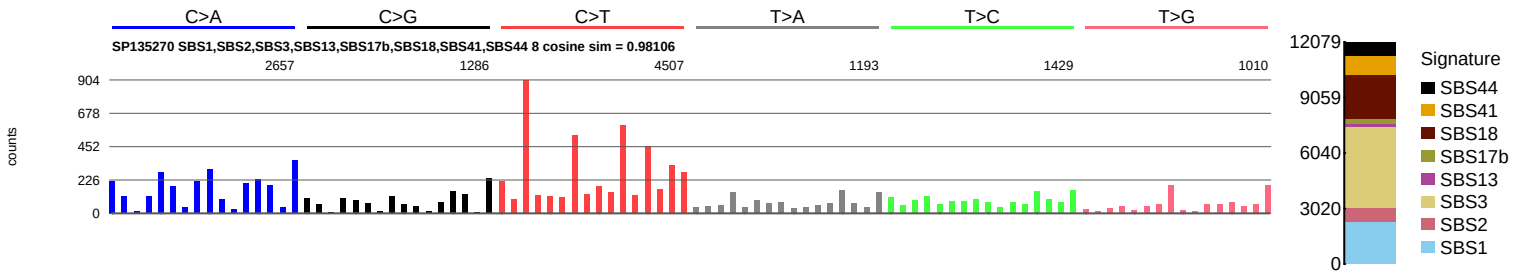
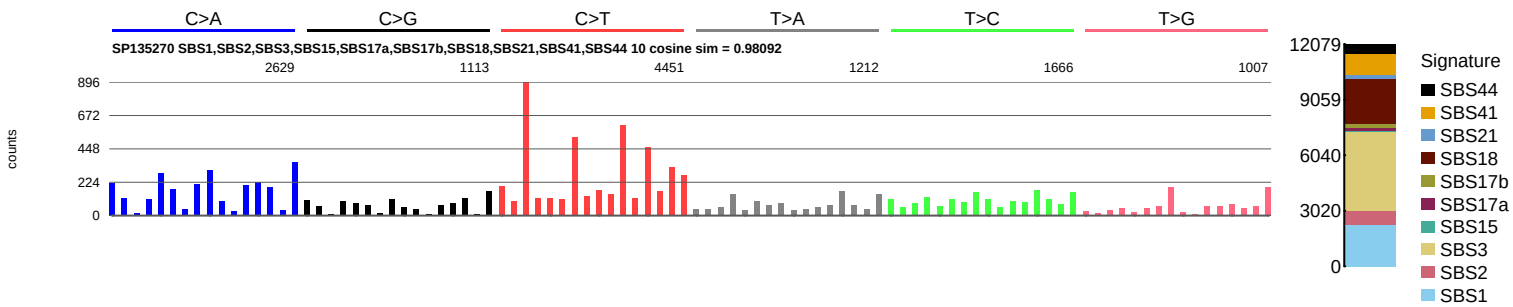
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98106
2657
1286
4507
1193
1429
1010
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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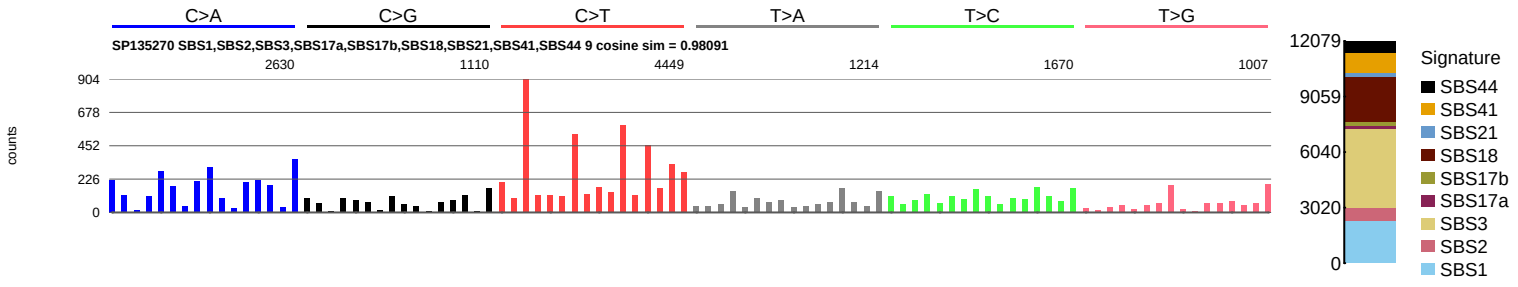
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98092
2629
1113
4451
1212
1666
1007
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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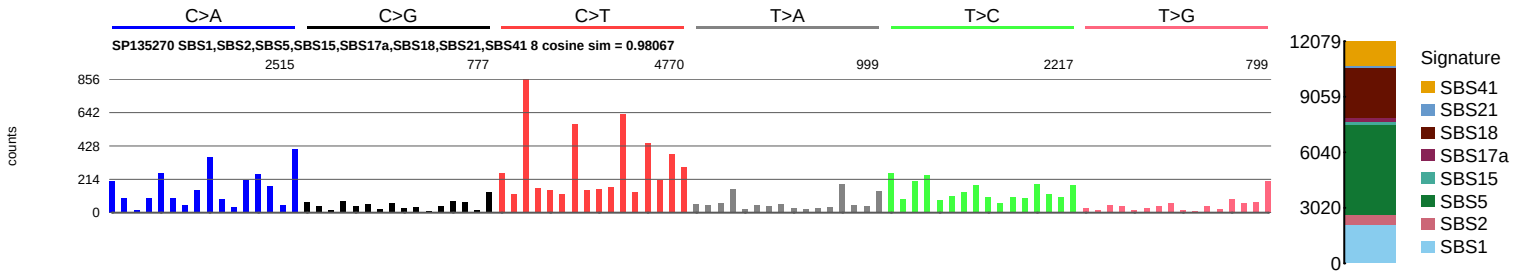
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98091
2630
1110
4449
1214
1670
1007
904
678
452
226
0
counts
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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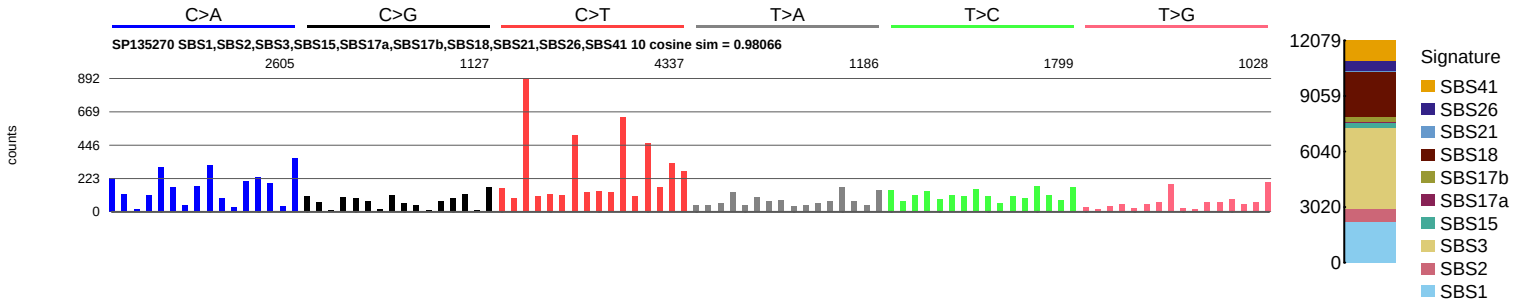
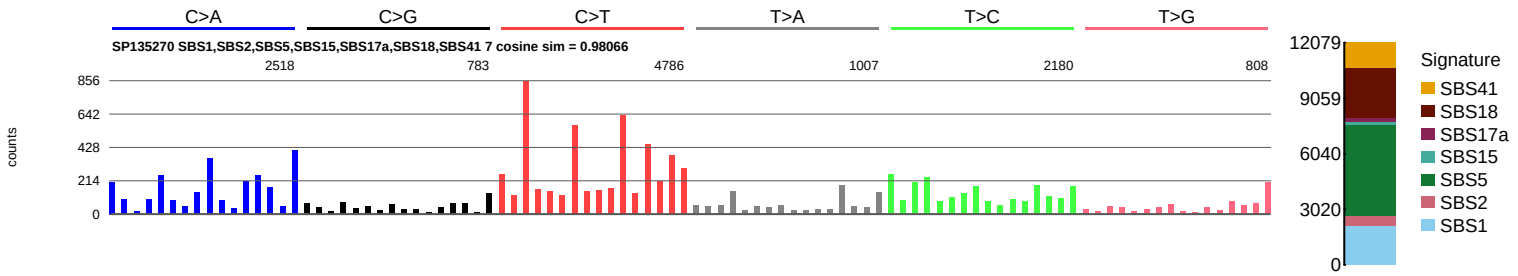
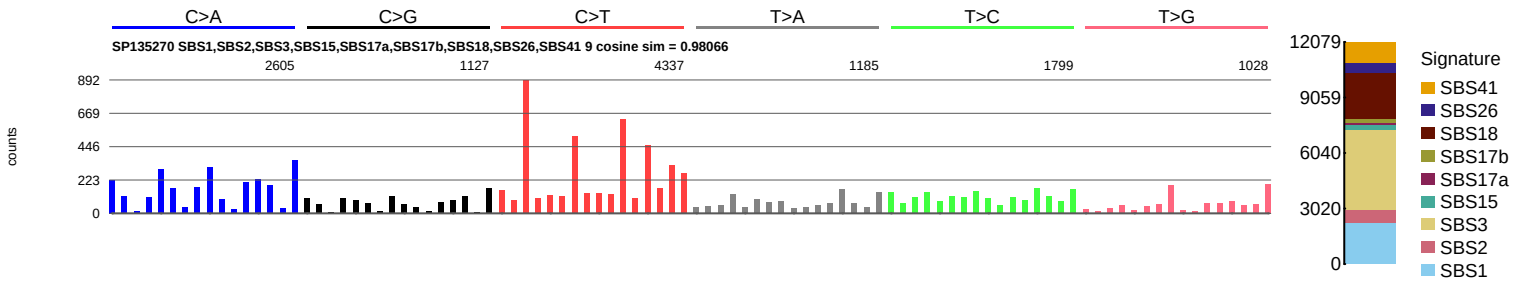
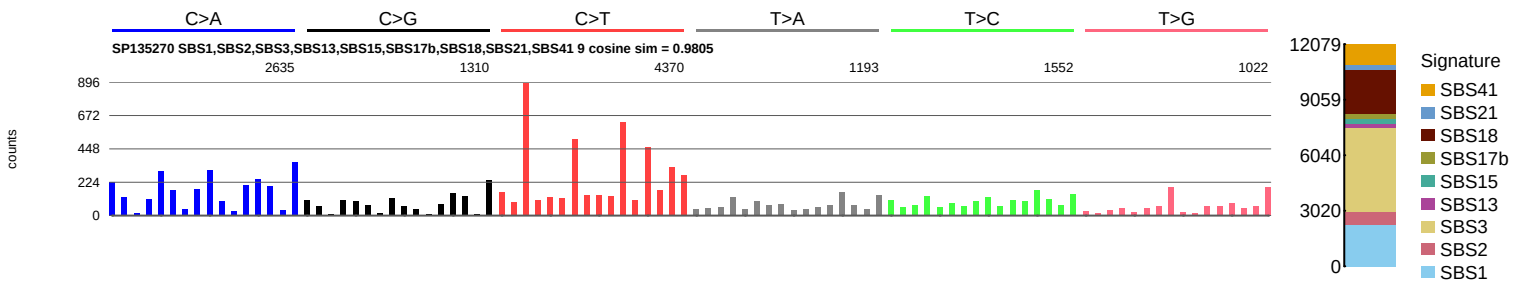
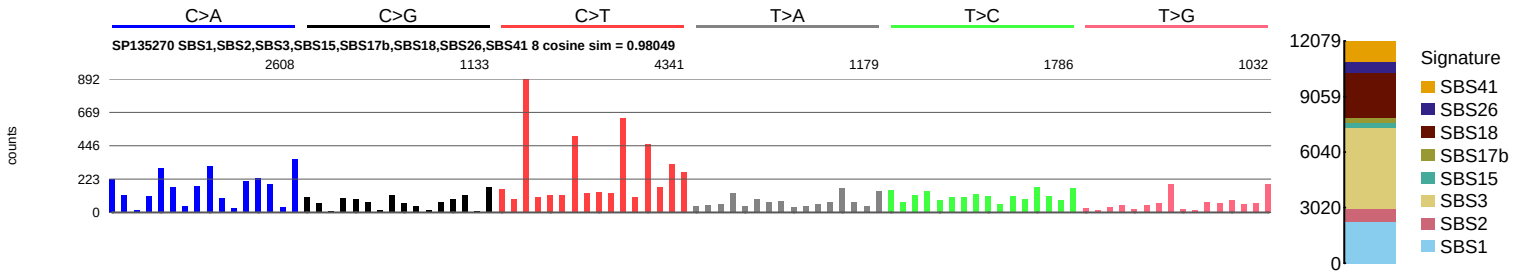
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS41 8 cosine sim = 0.98067
2515
777
4770
999
2217
799
856
642
428
214
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1

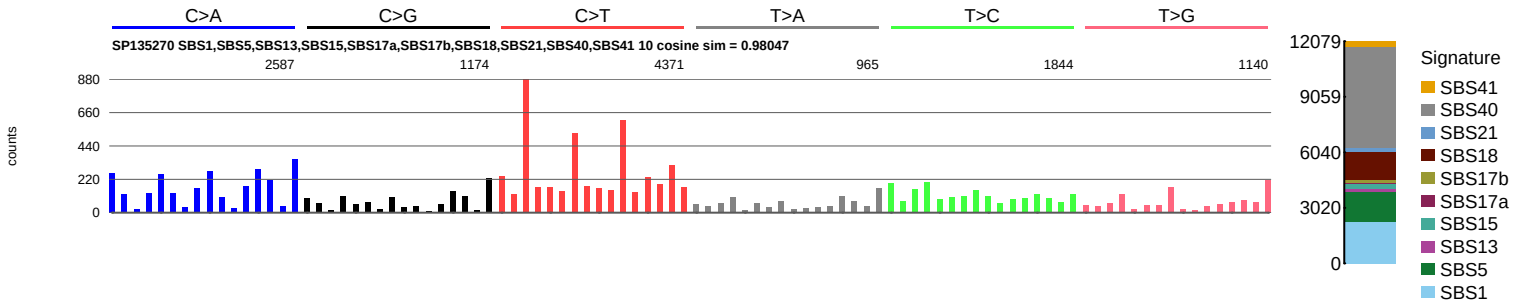

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98066
2605
1127
4337
1186
1799
1028
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

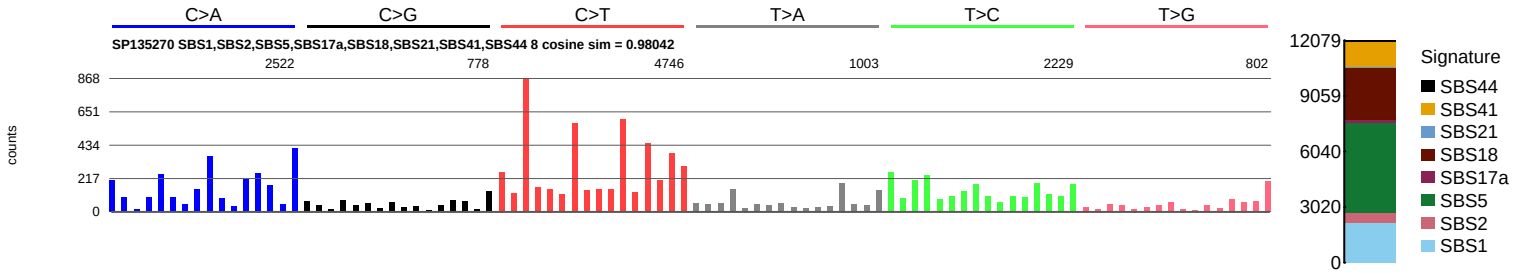

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS41 7 cosine sim = 0.98066
2518
783
4786
1007
2180
808
856
642
428
214
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1

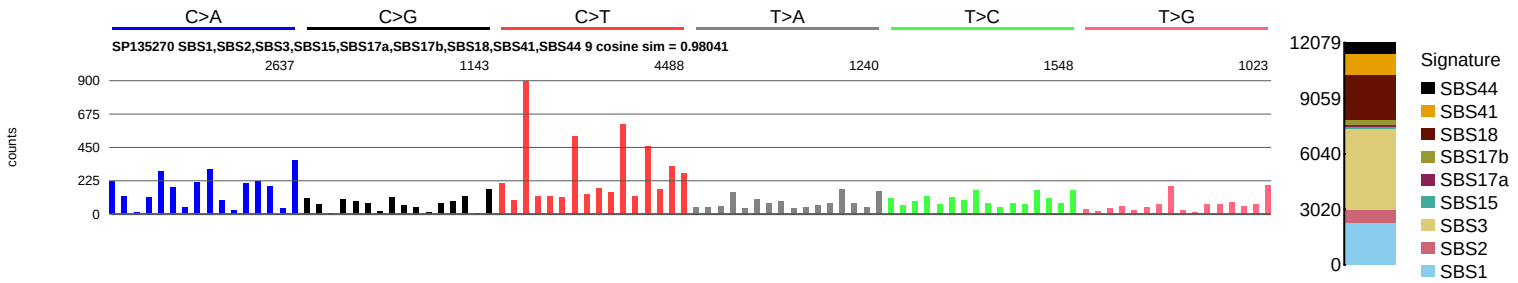

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98066
2605
1127
4337
1185
1799
1028
892
669
446
223
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

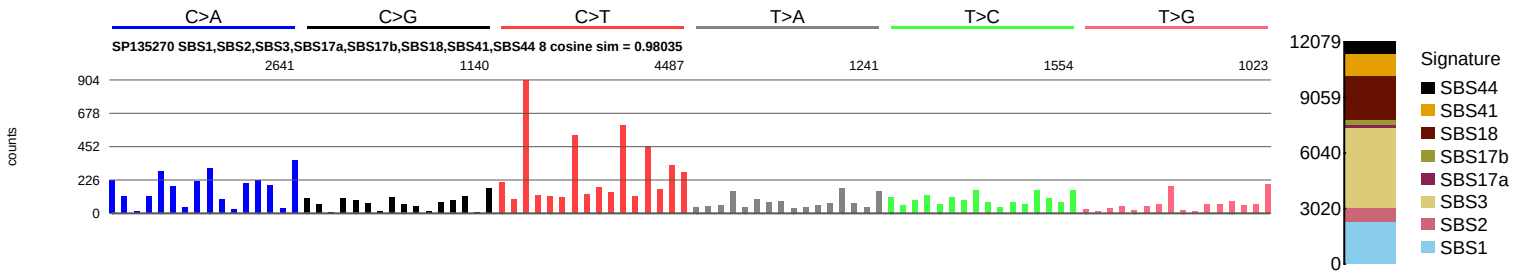

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9805
2635
1310
4370
1193
1552
1022
896
672
448
224
0
counts
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

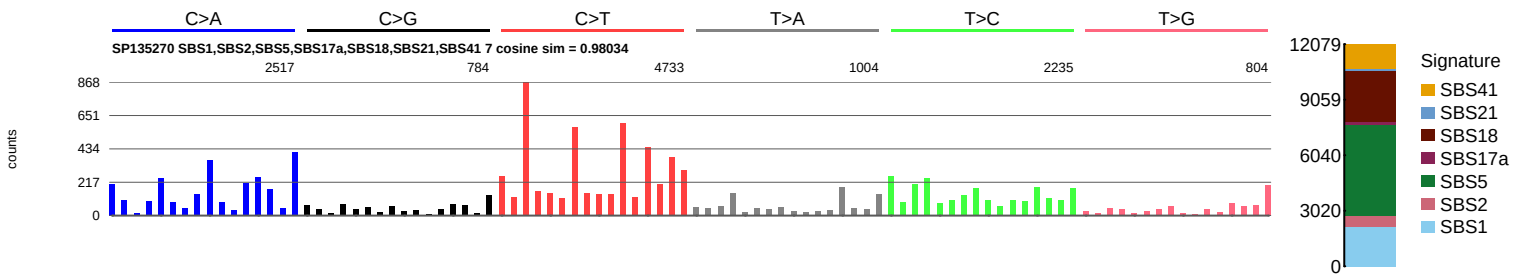

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98049
2608
1133
4341
1179
1786
1032
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

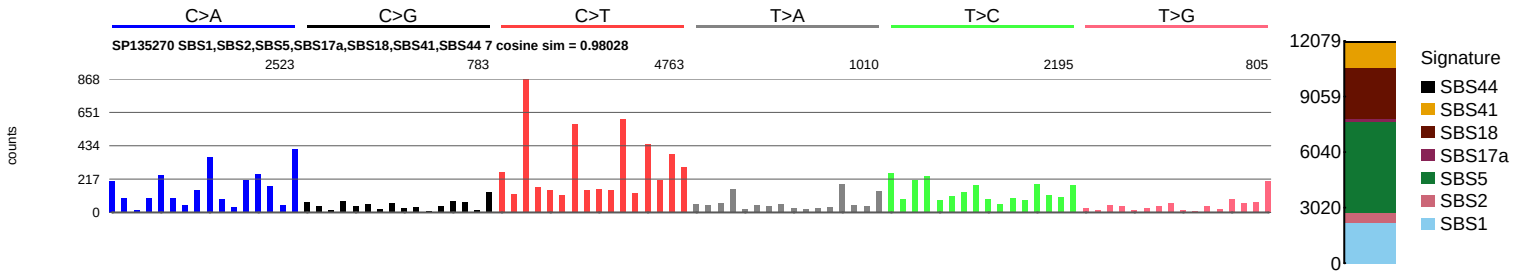

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98047
2587
1174
4371
965
1844
1140
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

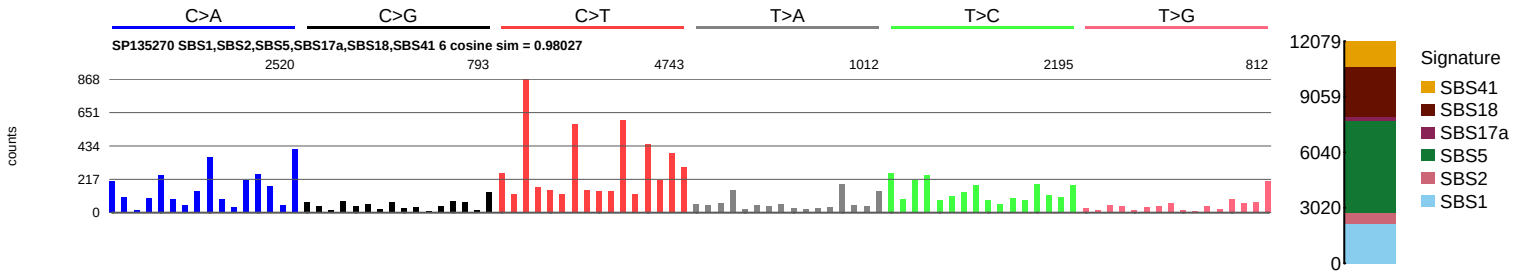

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98042
2522
778
4746
1003
2229
802
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1

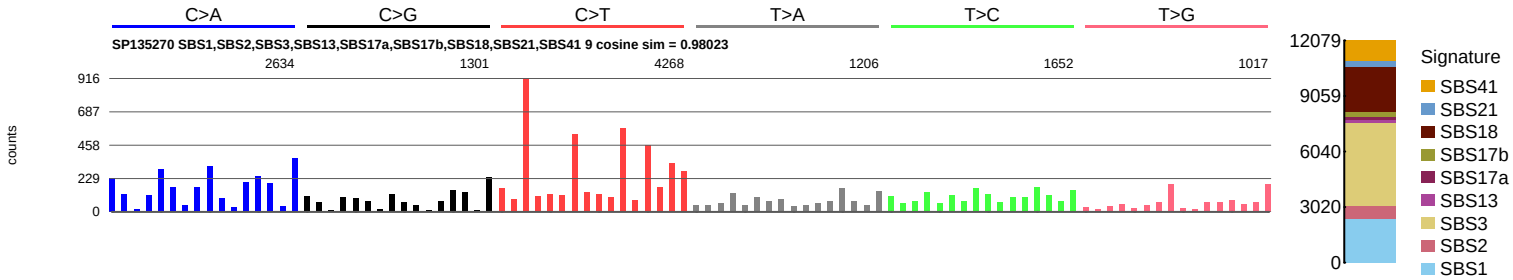

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98041
2637
1143
4488
1240
1548
1023
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

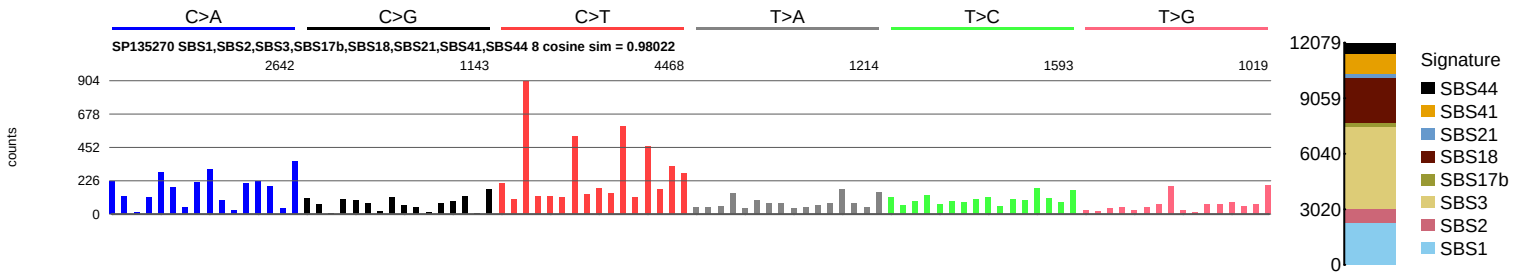

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98035
2641
1140
4487
1241
1554
1023
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

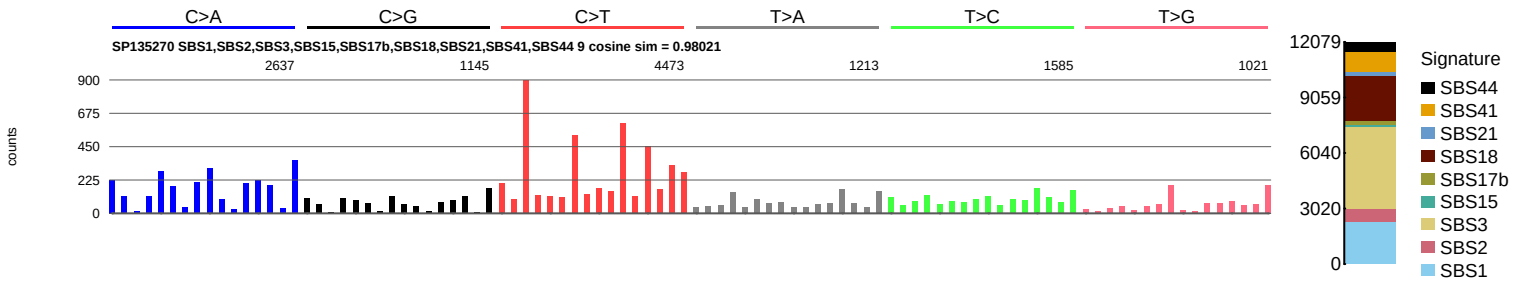

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS41 7 cosine sim = 0.98034
2517
784
4733
1004
2235
804
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1

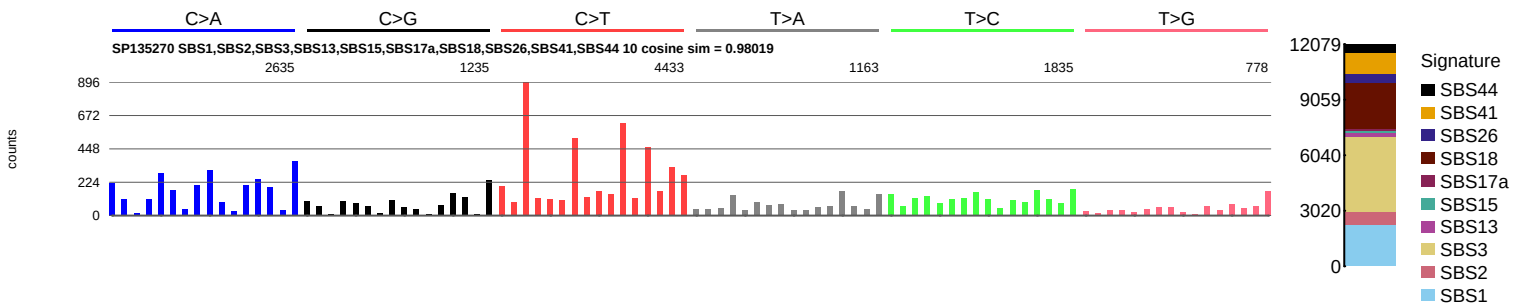

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS41,SBS44 7 cosine sim = 0.98028
2523
783
4763
1010
2195
805
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS5
SBS2
SBS1

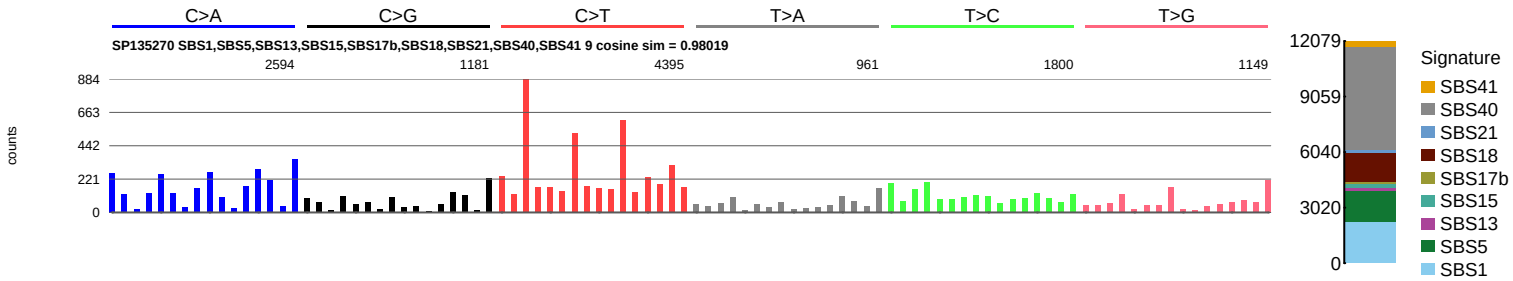

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS41 6 cosine sim = 0.98027
2520
793
4743
1012
2195
812
counts
868
651
434
217
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17a
SBS5
SBS2
SBS1

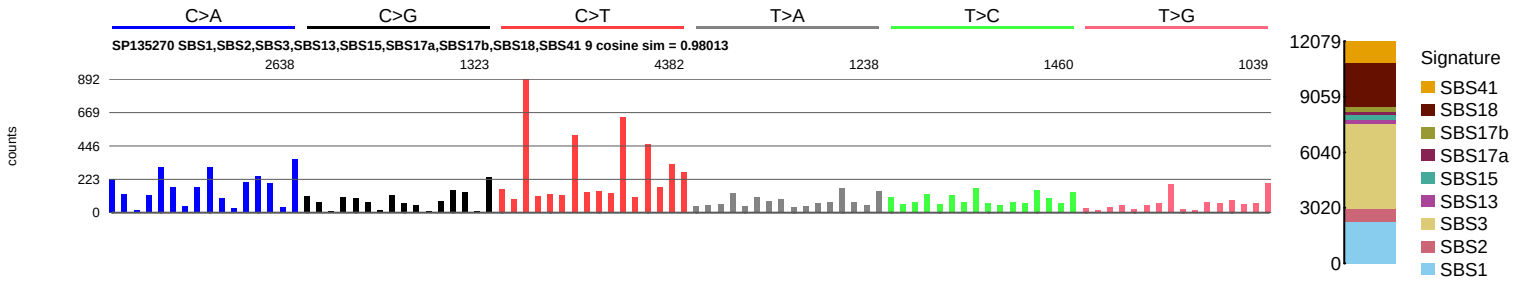

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98023
2634
1301
4268
1206
1652
1017
counts
916
687
458
229
0
12079
9059
6040
3020
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

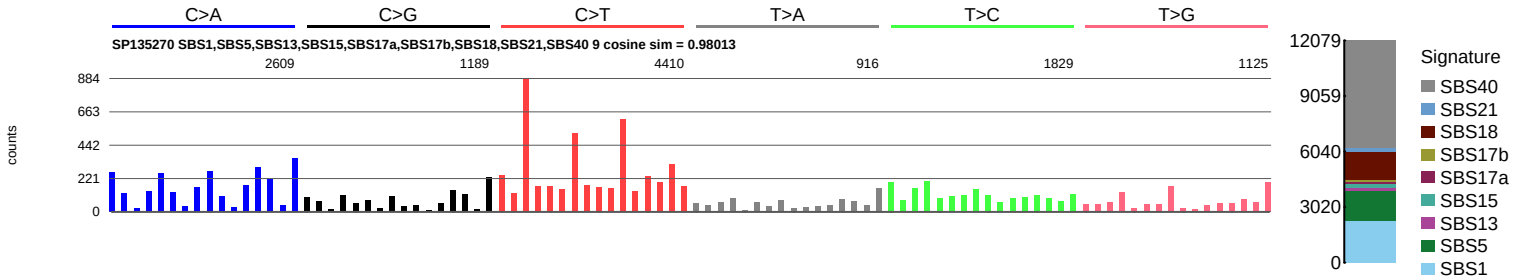

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98022
2642
1143
4468
1214
1593
1019
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1

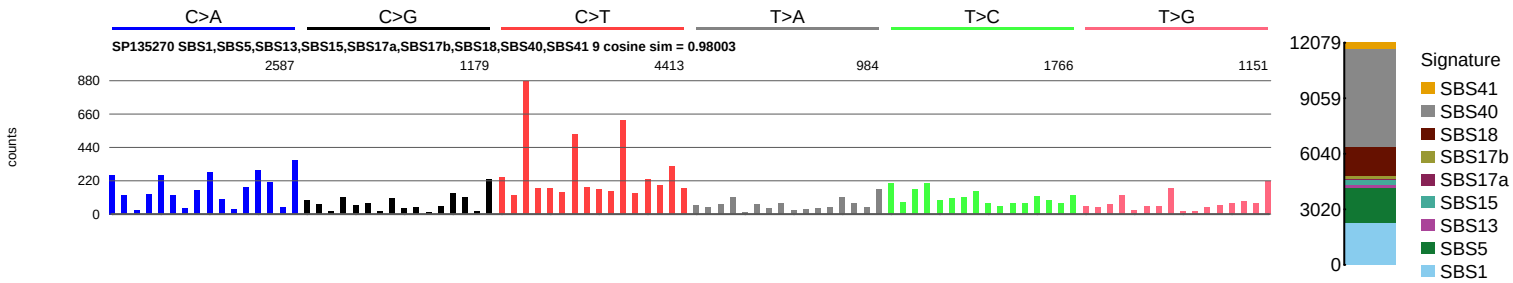

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98021
2637
1145
4473
1213
1585
1021
counts
900
675
450
225
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

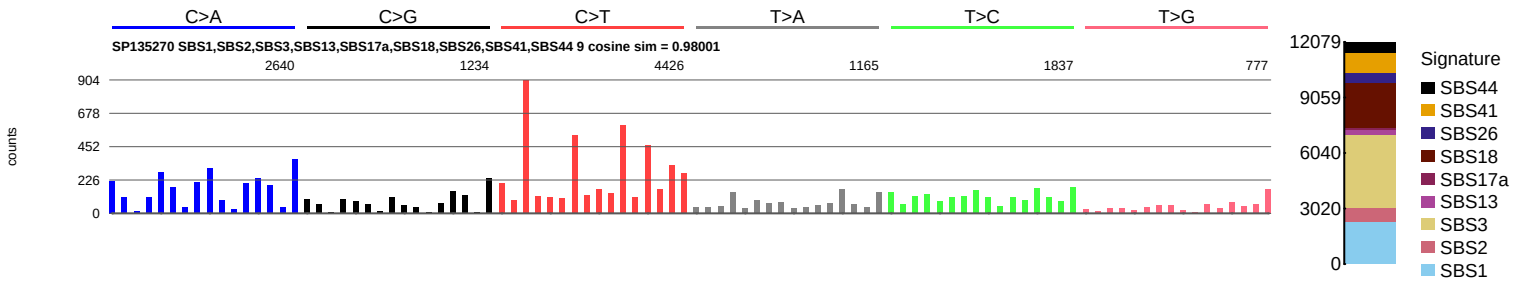

C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98019
2635
1235
4433
1163
1835
778
counts
896
672
448
224
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98019
2594
1181
4395
961
1800
1149
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98013
2638
1323
4382
1238
1460
1039
counts
892
669
446
223
0
12079
9059
6040
3020
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98013
2609
1189
4410
916
1829
1125
counts
884
663
442
221
0
12079
9059
6040
3020
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98003
2587
1179
4413
984
1766
1151
counts
880
660
440
220
0
12079
9059
6040
3020
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135270 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98001
2640
1234
4426
1165
1837
777
counts
904
678
452
226
0
12079
9059
6040
3020
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS13
SBS3
SBS2
SBS1